II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 2

Oncology, DSD, Gender & Ageing

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 24	Basic Principles of Immunology and Immunotherapy in Urologic Oncology
Ch 25	Molecular Genetics and Cancer Biology
Ch 55	Management of Abnormalities of the External Genitalia in Males
Ch 58	Management of Abnormalities of the Genitalia in Girls
Ch 59	Differences of Sexual Development
Ch 60	Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations
Ch 65	Adolescent and Transitional Urology
Ch 120	Aging and Geriatric Urology
Ch 126	Management and Considerations in Transgender and Nonbinary Individuals
Ch 127	Male-to-Female Gender-Affirming Surgery: Vaginoplasty
Ch 128	Special Urologic Considerations in Transmasculine Individuals
★	Putting it together

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · Innate and adaptive immunity

P2 Describe use and principles of immunology and Immunotherapy in Urological Oncology — innate/adaptive arms, TLR/PAMP, two signals, Th1 vs Th2, checkpoints, adaptive immune resistance, three Es

Not in the paper answer:

- Th1 cytokines (IL-2, TNF-α, IFN-γ) mean **BCG success**; Th2 cytokines (IL-4, IL-5, IL-10, IL-13) mean **BCG failure**.

ALGORITHM

1. Danger signal (TNF-α, TLR ligand) reaches a DC.
2. DC stops antigen uptake; raises MHC and B7-1/B7-2.
3. DC enters afferent lymphatics (CCL21 sensed by CCR7).
4. In the node it activates CD4, CD8 and B cells.

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; raised by IFN	Antigen-presenting cells (APCs)
Peptide	~9 amino acids	13–25 amino acids

2 · Immune editing and escape

- Escape follows equilibrium, as antigen and MHC fall and Tregs rise.
- In adaptive immune resistance, CD8-derived IFN-γ induces tumour PD-L1, which switches off those PD-1-positive CD8 cells.

KEY POINT

PD-L1 is mostly induced by the immune attack, so it marks a suppressed response, not a fixed trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating
CTLA-4	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
LAG-3	MHC class II	Inhibitory

3 · Inflammation and genitourinary (GU) tumours

	Bladder	Kidney	Prostate
Does inflammation cause cancer?	Strong: schistosomiasis, chronic UTI, catheters, cyclophosphamide	Weak	Probable: proliferative inflammatory atrophy beside HGPIN
CD8 infiltration	Better outcome in MIBC	**Worse outcome — renal paradox**	May be suppressive

	Bladder	Kidney	Prostate
Dominant restraint	PD-L1, rising with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

4 · BCG in bladder cancer

ALGORITHM

1. Instil 1–5 × 10^8 bacilli.
2. A fraction adheres via **fibronectin attachment protein**.
3. Urothelial TLR2/TLR4 release IL-8 and TNF-α.
4. Neutrophil wave (hours), then DC activation.
5. CD4 Th1 response in nodes (weeks), then memory.

- Memory explains durable remission in ~**50%** of NMIBC.
- **Nadofaragene firadenovec**, an adenovirus carrying the IFN-α2b gene with **Syn3**, is given 3-monthly for up to 12 months (CIS CR 53.4%). It was the FDA's **first gene therapy** (2022), for BCG-unresponsive NMIBC with CIS.

5 · Cancer vaccines

P4 Rationale of using cancer vaccines in Urology — antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC

- Sipuleucel-T, the first FDA-approved solid-tumour vaccine, prolongs OS by **+4.1 months** with **no effect on time to progression**.

6 · Checkpoint blockade

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052) requires cisplatin-ineligibility with CPS **≥10**, or platinum-ineligibility; with CPS <10 it was worse than carboplatin.
- Atezolizumab and durvalumab were withdrawn, and adjuvant atezolizumab (IMvigor010) was negative.
- **Enfortumab vedotin (EV)**, an anti-nectin-4 antibody–drug conjugate, gave a 68% response rate with pembrolizumab when cisplatin-ineligible (EV-103).
- †In EV-302, EV plus pembrolizumab roughly doubled PFS and OS versus platinum and is now first line for all.

Kidney

- CTLA-4 on Tregs and PD-1 on CD8 TILs make dual blockade synergistic.
- VEGF blocks DC maturation and T-cell entry, so TKIs complement immunotherapy.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- In mCRPC, tumours are rarely PD-L1-positive; pembrolizumab gives a PSA response under 10%, and ipilimumab and IMbassador250 gave no OS benefit.
- The exception is **MSI-H/dMMR** tumours (1.2–12%), treated with pembrolizumab second line or later; its 2017 approval was the **first based on a biomarker, not a site**.
- †Immune-related adverse events of grade ≥2 (such as colitis or pneumonitis) need drug hold and corticosteroids; endocrinopathies need lifelong replacement.

> **KEY POINT**
> Successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, VEGF inhibition) or removes a brake (PD-1, CTLA-4); it fails where the brake is not limiting, as in Treg-dominated prostate cancer.

7 · Landmark trials

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only phase III-based second-line approval
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	DFS 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, RCC after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab for 1 year, high-risk clear-cell RCC	Recurrence or death reduced by 32%	Adjuvant standard

25 Molecular Genetics and Cancer Biology

1 · Hallmarks and basic molecular genetics

- **Local invasion** is the one hallmark that uniquely separates malignant from benign.
- **Alternative splicing** makes ligand-independent AR variants (AR-V7), causing castration resistance.
- **Somatic** testing finds targetable mutations; **germline** testing finds hereditary syndromes for surveillance and family cascade testing.

2 · Gene alterations: tumour suppressors and oncogenes

Oncogenes need one altered allele, suppressors usually two.

P1 Oncogenes — gain of function/one allele vs suppressor loss/both alleles; mutation, amplification, fusion (TMPRSS2–ERG ~half, MYC, MET); Knudson two-hit

Not in the paper answer:

- Two-hit exceptions are **dominant negative** mutation (mutant TP53) and **haploinsufficiency** (p27/CDKN1B).

3 · Cell cycle

- The cycle takes ~24 h, and cyclin-dependent kinases (CDKs) work only with their phase-specific **cyclin**.

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins, so the nuclear envelope breaks down

Checkpoint	Trigger	Mechanism
G1/S	Mitogens, nutrients, O_2, DNA damage	p21 (CDKN1A) and 14-3-3σ inactivate G1 CDKs; TP53 triggers apoptosis if damage is unrepaired
G2/M	DNA damage (radiation)	ATM activates CHEK1/CHEK2, which sequester Cdc25; TP53 induces p21 and GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex; failure causes non-disjunction and **aneuploidy** (near-universal in solid tumours)

- CDK inhibitors are **Cip/Kip** (p21, p27, p57), acting broadly, and **INK4** (p15, p16, p18, p19), acting on CDK4/6.

RB1 — guardian of G1/S

ALGORITHM

1. Hypophosphorylated RB1 binds E2F and HDACs, silencing S-phase genes.
2. Cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1.
3. E2F is released and the restriction point is passed (irreversible).

- **TP53**, activated by DNA damage or oncogene stress, causes arrest, repair, senescence or apoptosis. **MDM2** is its negative regulator; TP53 is mutated in over half of cancers.

> **KEY POINT**
> Cancer must break both RB1 (entry) and TP53 (integrity), so both pathways are lost in almost every tumour.

4 · DNA damage and repair

Pathway	Lesion	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky adducts, cross-links (UV, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
Base excision repair (BER)	Single-base damage, single-strand breaks	MUTYH, OGG1, PARP1/2	OGG1 variants linked to RCC
Mismatch repair (MMR)	Mispairs, indel loops at repeats	MSH2/MSH6, MLH1/PMS2	Mutation rate raised ~100-fold; Lynch
Homologous recombination (HR)	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate
Non-homologous end joining (NHEJ)	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- Unrepaired DSBs (marked by **γH2AX** foci) cause apoptosis, the basis of radiotherapy. **HR** copies the sister chromatid and is precise; **NHEJ** is template-free and error-prone.

> **KEY POINT**
> Repair defects are targets: HR loss means PARP inhibitor sensitivity, MMR loss means immunotherapy, and ERCC2 loss means cisplatin sensitivity.

5 · Apoptosis

- In apoptosis, initiator caspases 8, 9, 10 activate executioners 3, 6, 7.

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, lost survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak release cytochrome c; Apaf-1 apoptosome activates caspase-9	FADD recruits procaspase-8, giving caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; IAPs	FLIP
TP53	Dependent (transactivates Bax)	**Independent**, so a target when TP53 is mutated

- Bid links the two pathways. TP53 loss causes chemoradiotherapy resistance.

6 · Senescence, telomeres, autophagy

P3 Telomerase — ~100 bp lost/division, two barriers, then telomerase immortalisation; normal tissue expression, ALT (ATRX/DAXX); short telomeres in premalignant lesions, predict prostate recurrence

Not in the paper answer:

- Telomerase is high in **85–90%** of tumours.
- **Senescent** cells are viable but permanently arrested; ADT induces senescence in some prostate cancers.

- **Autophagy** suppresses early tumours but sustains established ones.

7 · Epigenetics

- Epigenetic change alters expression without sequence change and is **reversible**. DNA methylation at CpG islands is maintained by DNMT1.
- Promoter hypermethylation silences suppressors as a "second hit": **GSTP1 in >90% of prostate cancer** (commonest somatic change), RASSF1A (up to 97% of bladder cancer) and p16. Global hypomethylation causes genomic instability.
- **H3K27me3 (written by EZH2)** represses genes; EZH2 is raised in mCRPC and predicts PSA recurrence.
- MicroRNA (18–25 nt) silences **many** mRNAs (**miR-371a-3p** in testis cancer) and siRNA **one**; long ncRNA HOTAIR drives castration resistance.

8 · Hypoxia, VHL and angiogenesis

ALGORITHM

1. In normoxia, HIF-α is prolyl-hydroxylated.
2. VHL (an E3 ubiquitin ligase) binds it for proteasomal degradation.
3. With hypoxia or VHL loss, HIF-α accumulates and dimerises with HIF-1β.
4. This switches on VEGF, angiopoietin and glycolytic genes.

- The **Warburg effect** (aerobic glycolysis) is best seen in clear cell RCC.

KEY POINT

VHL loss means permanent pseudohypoxia, driving angiogenesis, Warburg metabolism and immunosuppression; hence clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Lineage plasticity** under driver targeting gives neuroendocrine prostate cancer.

9 · GU tumour genomics

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half)
SPOP	Commonest mutated suppressor in primary tumours
TP53	Commonest alteration overall (more in advanced disease)
RB1, PTEN, TP53	Loss of ≥2 means aggressive variant, overlapping neuroendocrine
BRCA1/2	Germline plus somatic hit: give **PARP inhibitor**
ATM	Little PARP benefit
CDK12	Tandem duplications; poor response to hormones, chemo, PARP
MMR deficiency	Microsatellite instability (MSI): give immunotherapy
AR amplification	Late, under AR-targeted pressure; causes ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** plus VHL, PBRM1, BAP1, SETD2	VEGF tyrosine kinase inhibitors (TKIs); belzutifan (HIF-2α)
Papillary (15–20%)	MET (7q); FH loss in subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mTOR pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1-positive, so immunotherapy-responsive

- In clear cell RCC, VHL is mutated in 60–70%. **PBRM1** (~40%) means a better outcome and **BAP1** or **SETD2** (~10% each) a worse one.

Bladder (urothelial carcinoma, UC)

- UC has a high (APOBEC) mutation load, with TP53 most frequent.
- Of six consensus subtypes, **basal/squamous is largest (35%)**, luminal papillary 24%, neuroendocrine-like smallest 3%.
- **FGFR3** is altered in 15–20% of advanced UC and up to 37% of upper tract UC. These T-cell-poor tumours respond to **erdafitinib**, so test all advanced UC. HER2 is altered in 9–12% of MIBC.

Testis

- Testis cancer shows **12p gain** (†isochromosome 12p) and KIT/KRAS in seminoma; miR-371a-3p has >90% sensitivity and specificity.

Hereditary syndromes

P2 P1 Hereditary urological malignancies / Familial malignancies in urology — VHL, HPRC (MET), HLRCC (FH), BHD (FLCN), BRCA2 > BRCA1, Lynch, HOXB13

Not in the paper answer:

- VHL carries a >70% lifetime risk of clear cell RCC (belzutifan). HLRCC gives papillary type 2 RCC, Lynch gives UC and prostate cancer, and HOXB13 carriers need earlier PSA screening.

> **KEY POINT**
> The germline syndrome names the somatic driver (VHL: clear cell; MET: papillary; FH: type 2; BRCA: prostate), and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

10 · Clinical NGS and biomarkers

- **Tumour-only** panels cannot separate somatic from germline variants.
- **Variant allele frequency (VAF)** depends on zygosity, copy number and purity; a heterozygous variant at 60% purity gives 30%. VAF of ~50% or ~100% suggests a **germline** variant, and single-digit VAF in cell-free DNA suggests **clonal haematopoiesis**.
- Circulating tumour DNA after treatment means minimal residual disease.
- A predictive marker needs a **biomarker–treatment interaction test** in a randomised trial; crossing lines mean predictive, parallel lines prognostic.

- Absolute benefit scales with baseline risk: a 50% relative reduction gives 15% absolute benefit at 30% risk but 1.5% at 3%.
- **HSD3B1 (1245A>C)** raises DHT synthesis, causing ADT resistance but abiraterone sensitivity.

11 · Landmark trials

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-altered mCRPC	Better radiographic PFS and overall survival, driven by BRCA1/2	Germline and somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after immunotherapy	Longer overall survival	FGFR testing of all advanced UC

55 Management of Abnormalities of the External Genitalia in Males

1 · Normal development and penile size

- Male differentiation (weeks 9–13) needs Leydig testosterone, 5α-reductase conversion to DHT and a functional androgen receptor; any break causes undervirilisation.
- The genital tubercle forms the glans, urethral folds the shaft, and labioscrotal swellings the scrotum.
- **Stretched penile length (SPL)** runs from pubis (fat pad compressed) to glans tip. At term it is **3.5 ± 0.7 cm**; in adults 13.3 ± 1.6 cm.
- The micropenis cut-off is −2.5 SD: **1.9 cm** from term to 5 months, 1.5 cm at 30 weeks, 2.0 cm at 34 weeks, ≈ 3.7–3.9 cm at 5–11 years and 9.3 cm in adults. Compare with the cut-off, not the mean.
- Minipuberty (first 3 months) gives a second phase of penile growth. In Tanner staging, testicular change comes first (stage 2); semen analysis is done at Tanner 5.

2 · Prepuce: phimosis and paraphimosis

- **Physiological phimosis** is normal at birth: ≈ **90%** are **retractable by 3 years**, and < 1% remain phimotic at 17 years. Forceful retraction causes cicatricial phimosis.
- Treat persistent phimosis, balanitis, BXO or UTI.
- **Topical steroid**: betamethasone 0.05% twice daily for 30 days (or thrice daily for 21 days). About 1 in 6 fail.
- A retractile prepuce halves recurrent febrile UTI risk. The commonest acquired "penile lump" is **entrapped smegma**.
- **Paraphimosis** causes oedema within hours and may progress to gangrene.

ALGORITHM

1. Compress the glans with distal traction; this succeeds in most.
2. If it fails, reduce oedema: iced glove 5 min, granulated sugar 1–2 h, or punctures.
3. If still irreducible, do a dorsal slit, then circumcision.

3 · Circumcision

- The AAP (2012) found the benefits insufficient to recommend routine neonatal circumcision, but enough to justify access and payment.
- Before using a device (Gomco, Mogen, Plastibell), inspect the meatus to exclude hypospadias and megameatus. Older children need freehand surgery under GA.
- Analgesia is mandatory. **Dorsal penile nerve block is superior to EMLA**, whose prilocaine can cause methaemoglobinaemia.

Defer / avoid neonatal circumcision	Reason
Hypospadias, curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction
Large hydrocele / inguinal hernia	Skin pushed forward, causing secondary phimosis or a trapped penis

Defer / avoid neonatal circumcision	Reason
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
NICU setting	More complications

- **Benefits**: penile cancer occurs almost only in uncircumcised men. Febrile infant UTI falls, but **111** circumcisions prevent one UTI.
- There is ≈ **60%** protection against HIV acquisition in men, but no reduction in transmission to partners. HSV-2 and high-risk HPV are also lower, and no sexual dysfunction is proven.

Complications

P1 Circumcision undone — 0.2–5% (10–20× after infancy); bleeding commonest; adhesions/skin bridges (steroids don't help); trapped penis; revision under GA

Not in the paper answer:

- An amputated glans should be re-attached within **8 h**.
- Degloving heals by re-epithelialisation, with **no immediate suturing or graft**.
- A skin bridge can cause chordee or torsion, so excise it.

> **KEY POINT**
> Late sequelae are problems of skin quantity (too much removed: trapped penis, chordee, torsion; too little: redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis and BXO

P2 P1 Management of Urethral meatal stenosis / Lichen sclerosus (BXO) — normal calibre by age, post-circumcision meatitis, upward stream, ventral meatotomy; BXO features, steroid, circumcision, buccal mucosa

Not in the paper answer:

- Meatal stenosis occurs almost exclusively in **circumcised** boys.
- BXO in boys is rare under 5 years. Steroids help only mild preputial disease, so **circumcision is preferred**, with or without meatoplasty.
- About 1 in 5 with meatal BXO need further meatal surgery.

4 · Inconspicuous penis versus micropenis

> **KEY POINT**
> **Stretched length is the single discriminator.** An inconspicuous penis (buried, webbed, trapped) has normal SPL hidden by skin, fat or scar, so treatment is anatomical. A micropenis is small but normally formed, so it needs endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital	Inelastic dartos; skin not anchored to deep fascia	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss; liposuction only in adolescents
Trapped (post-circumcision)	Cicatrix embeds penis in fat	Dilation or betamethasone with retraction; then repair under GA at ≥ 6 months

- If retracting skin lateral to the shaft exposes the glans easily, the shaft is normal.

- **Repair** excises the dysgenic dartos bands, fixes dorsal subcutaneous tissue to pubic fascia and scrotal tissue to the ventral base, and covers with unfurled prepuce, Z-plasty, flap or graft.
- **Webbed penis** (penoscrotal fusion) is repaired by transverse incision with vertical closure, or Byars' flap. Avoid neonatal circumcision.

Micropenis

- Micropenis is normally formed but ≥ 2.5 SD below the mean (< **1.9 cm** at term). Testes are often small or undescended.
- A defect before week 12 (placental hCG) gives a small penis with hypospadias; gonadotrophin deficiency after 14 weeks gives a normally formed micropenis.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism, midline brain defects	Low LH and FSH
Hypergonadotrophic	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

1. History, examination, SPL and **karyotype**.
2. Refer to a paediatric endocrinologist; measure testosterone before and after hCG, with or without a GnRH test.
3. Screen the anterior pituitary (glucose, Na/K, cortisol, thyroid); then MRI brain.
4. Test **end-organ response: intramuscular testosterone enanthate × 3 months**. Short courses spare height; prolonged ones advance bone age. Repeat at puberty.

- Most achieve male identity, standing voiding and satisfactory sexual function.

5 · Curvature (chordee without hypospadias) and torsion

P1 Congenital Penile curvature — Devine–Horton types, artificial erection, plication/septotomy/graft

Not in the paper answer:

- Repair if sexual dysfunction is expected, with concurrent penile surgery, or if growth will worsen it.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; dorsal hood; a prominent frenulum causes distal chordee	Deglove; excise fibrous tissue superficial to Buck's; Byars' flaps; if still curved, dorsal plication, Nesbit or corporal rotation; if the urethra is short, reconstruct it; if frenular, frenulotomy
Dorsal	Isolated, epispadias	Ventral plication; protect dorsal neurovascular bundles
Lateral	Corporal asymmetry; straight when flaccid, noticed on erection	Plicate corpus **opposite** maximal curvature

- Work superficial to deep. An **artificial erection** locates the apex and confirms correction.
- **Penile torsion** is usually **counter-clockwise (to the left)**, < 60° and skin-level. If < 90°, deglove and rotate; if ≥ 90°, also divide basal bands and place non-absorbable anchoring sutures at the corporal base.

6 · Abnormal number, masses and lymphoedema

Feature	Aphallia	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, normal scrotum and testes	1 in 5 million; from accessory to complete duplication
Associations	Cryptorchidism, VUR, renal agenesis, horseshoe kidney, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI; then neophallus or reassignment (reassigned may develop male identity)	Renal US, VCUG, penile US/MRI; individualised

- In aphallia the urethra most often opens **postsphincteric** (anal verge), with the best survival.
- Penile cysts are commonest and are excised; **juvenile xanthogranuloma is observed for 1 year**.
- Lymphoedema may be primary (Milroy, Meige, praecox) or acquired (anogenital Crohn disease); surgery is excisional, not drainage. Haemangiomas involute, while vascular malformations persist and are resected.

7 · Urethral anomalies

P2 Megameatus intact prepuce / Anterior urethral valve — MIP: wide meatus under complete prepuce, found at circumcision, pyramid repair; AUV: bulbar, diverticulum lip, VCUG, endoscopic ablation

Not in the paper answer:

- If MIP is recognised during circumcision, stop and preserve the prepuce.
- **Urethral duplication** is usually sagittal (≈ 94%), and the **ventral channel is the functional one**. The **Effmann** classification is type I incomplete (IA distal, IB proximal, blind), type II complete (IIA two meatuses, including IIA2 "Y-type" perineal; IIB one meatus), and type III with bladder duplication.
- Duplication presents with double stream, incontinence or UTI and is evaluated by VCUG, RUG and cystoscopy. **Never use the hypoplastic dorsal channel as the main urethra**.
- **Congenital urethral fistula** is subcoronal with a normal meatus and is closed in layers.
- **Cowper syringocele** (Maizels types: simple, perforated, imperforate, ruptured) needs endoscopic unroofing if symptomatic.
- **Megalourethra** is **scaphoid** (commoner; spongiosum deficient) or **fusiform** (rarer; cavernosa also deficient, often lethal). Both go with prune-belly and are repaired by reduction urethroplasty.

8 · Priapism

P3 Priapism: Diagnosis and Management — ischaemic vs non-ischaemic vs stuttering, blood gas, aspiration/phenylephrine, distal then proximal shunts, embolisation

Type (children)	Cause	Treatment
Ischaemic	**Sickle cell** (commonest), leukaemia	Hydration, O_2, alkalinisation, analgesia, with or without transfusion; then aspiration with or without phenylephrine; then shunt
Non-ischaemic	Straddle injury causing an arteriocavernous fistula	Observe; if persistent, superselective embolisation
Stuttering	Sickle cell	Bedtime pseudoephedrine; oral β-agonist; long-term PDE5 inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	None — resolves in 2–6 days

- Corporal blood gas decides urgency: ischaemic blood is dark, hypoxic and acidotic in a rigid, painful penis, while arterial blood is bright red in a non-rigid, painless one. Counsel about fibrosis and ED.

9 · Scrotal anomalies

- **Penoscrotal transposition** is failed inferomedial migration of the labioscrotal swellings. It goes with proximal hypospadias, and in the complete form **up to 75%** have upper-tract anomalies, so do renal US and VCUG.
- Repair it with the hypospadias by **rotation-flap scrotoplasty**.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds	Proximal hypospadias	Scrotoplasty with hypospadias repair
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism; perineal lipoma usually isolated	Upper-tract US; scrotoplasty and orchidopexy at 6–12 months
Hypoplasia / agenesis	Under-developed / absent	Undescended testis (UDT); agenesis — all XY, bilateral UDT	—
Scrotoschisis	Rupture, extruded testis	Meconium peritonitis	Explore, orchidopexy, close

10 · Inguinal hernia and hydrocele

*All reflect incomplete closure of the **processus vaginalis**.*

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; contents enter	Repair soon after diagnosis
Communicating hydrocele	Patent, fluid only	Repair if > 1–2 y, enlarging or fluctuating
Hydrocele of cord	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia affects 1–5% of children and more with prematurity (≈ 30% if < 1 kg); right to left is 3:1. Bilateral risk is higher with prematurity, age under 1 year, cryptorchidism, ascites and VP shunts.
- An **intermittent bulge** means a patent processus, so operate. Incarceration needs immediate exploration, and reducible hernias elective repair within weeks; in preterm infants repair before discharge.

ALGORITHM

1. Skin-crease incision; open the external oblique, sparing the ilioinguinal nerve (not opened in young infants, Mitchell-Banks).
2. Split the cremaster and find the sac anteromedially; separate it from the vas and vessels.
3. Perform **high ligation** at the internal ring and tighten a wide ring.
4. Finally, **confirm the testis is scrotal**.

- Recurrence is 0.5–1% (3–6% after incarceration), from a missed or torn sac or a missed direct hernia. Other complications are iatrogenic cryptorchidism, atrophy and vas injury.
- Laparoscopic repair now matches open repair and closes the contralateral ring.
- Contralateral patency is ≈ 40% in infants, but a metachronous hernia develops in only ≈ **7%**, so routine contralateral exploration is not advised.

11 · Acute scrotum

Only cord torsion is time-critical.

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; vomiting common	Sudden or insidious; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present
Signs	High-riding, horizontal lie	**Blue dot**, upper-pole nodule	Tender epididymis; pyuria 20–40%
Colour Doppler US (CDUS)	Absent/reduced flow; **whirlpool/snail sign**	Avascular upper-pole lesion, hyperaemia	Increased epididymal flow
Management	**Emergency exploration**	Ice, NSAIDs, rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- The **bell-clapper** deformity is found in ≈ 12% at autopsy. Torsion affects ≈ 1 in 4000, is familial and is commoner if cryptorchid. Delay to presentation is the main cause of testis loss.
- A cremasteric reflex is present in up to 10% of torsions, so it does not exclude torsion.
- **TWIST score**: †swelling 2, hard testis 2, absent reflex 1, nausea/vomiting 1, high-riding 1. A score ≥ 5 is high risk, so go to theatre without imaging; ≤ 2 is low risk.
- On CDUS, absent flow is specific but only 63–90% sensitive. **Do not delay surgery for imaging when suspicion is high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	**≈ 5%**	20%	40%	60%	80%	90%

ALGORITHM

1. If torsion is suspected, explore the scrotum.
2. Detorse and apply warm packs. **Meanwhile fix the contralateral testis with non-absorbable suture**.
3. Reassess. If viable, then fix in a dartos pouch or with non-absorbable sutures. If non-viable, then do an orchidectomy; offer a prosthesis later.

- Manual detorsion fails in ≈ 1/3, and orchidopexy is still needed after success.
- Salvaged testes may still atrophy, with semen defects from ischaemia–reperfusion.
- **Intermittent torsion** precedes acute torsion in 30–50%, so offer **elective bilateral orchidopexy**.

Perinatal (extravaginal) torsion

- Cord and tunica twist together before scrotal fixation, giving a hard painless testis. Salvage is rare and it is bilateral in 5–22%; explore urgently if a normal postnatal exam was documented.

Appendage torsion, epididymitis, others

- Explore appendage torsion only if cord torsion is not excluded or pain is prolonged.
- Treat epididymitis with ice, NSAIDs and elevation, adding antibiotics if pyuria. A prepubertal boy with a positive culture needs renal US and VCUG.
- Idiopathic scrotal oedema shows the **fountain sign** and is treated with an antihistamine.

12 · Adolescent varicocele

P1 P3 Varicocele – Pathophysiology / Varicocele: controversies — left-sided anatomy, nutcracker, grading, right/non-decompressing: image for mass; adolescent: observe, repair for asymmetry

Not in the paper answer:

- Prevalence is ≈ **15%**; it appears after 10 years and peaks at Tanner 3.
- Measure testicular volume by **US (L × W × H × 0.71)**, serially. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Catch-up growth may be spontaneous, so **observation is the default**.

KEY POINT

Indications for repair: left testis > **20%** smaller (or bilateral hypotrophy), pain, abnormal semen (Tanner 5 / ≥ 18 y), or intratesticular varicocele with asymmetry. Total volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence

- Hydroceles may appear 2 or more years after repair, and up to half need intervention.

13 · Epididymal and vasal anomalies

- Spermatoceles are postpubertal, contain sperm and are mostly observed.
- Bilateral absent vas suggests a **CFTR** mutation; unilateral absence suggests **renal anomalies**.

58 Management of Abnormalities of the Genitalia in Girls

1 · Embryology that explains the anomalies

- The **perineal body** splits the cloacal sphincter, so urogenital diaphragm and external anal sphincter are **both pudendal-innervated**.
- Müllerian ducts arise lateral to wolffian ducts, so a **müllerian anomaly travels with an ipsilateral renal anomaly**. Fused ducts form uterus, cervix and upper two-thirds of vagina; **sinovaginal bulbs** form the distal vagina.
- **Caudal regression** (weeks 4–5 insult) is linked to maternal diabetes; cranial extension gives VATER or MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · Evaluation and classification

- The **pull-down manoeuvre** (labia drawn caudally and laterally) shows the distal vagina. Discharge or bleeding needs vaginoscopy under anaesthesia.
- Ultrasonography (US) shows anatomy and upper tracts; **magnetic resonance imaging (MRI) is the gold standard** for müllerian anatomy.

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	Mayer–Rokitansky–Küster–Hauser (MRKH)
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, Herlyn–Werner–Wunderlich syndrome (HWWS)

KEY POINT

Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · Clitoris

- **Clitoromegaly** is commonest from **congenital adrenal hyperplasia (CAH)**, so check electrolytes, 17-hydroxyprogesterone and karyotype (Ch 59). A large hood with normal corpora is **pseudo-enlargement**; a **diminutive clitoris** suggests complete androgen insensitivity syndrome (CAIS).

4 · Vestibule: the interlabial mass

- **Urethral prolapse** affects prepubertal African-ancestry girls: a **doughnut with central meatus** needing no imaging. Sitz baths and topical steroid or oestrogen come first, then **circumferential excision**.
- A **Skene cyst** gives a **displaced meatus, eccentric stream** and usually ruptures.
- A **Gartner cyst / covered ectopic ureter** on the anteromedial vaginal wall is incised; upper-pole heminephrectomy follows if wetting or infection persists.
- A **prolapsed ureterocele** gives an infant retention and a pink-to-dusky mass; decompress, then reduce.

- An **ectopic ureter** below the sphincter causes **daytime wetting despite normal voiding**.

> **KEY POINT**
> Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Any cyst of urethral or vaginal origin needs a **renal US, always**.

- A **labial swelling** is commonest an inguinal hernia; **childhood asymmetric labium majus enlargement (CALME)** is physiological and is observed.

5 · Vagina: obstruction and absence

Block level and uterus presence decide the operation.

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate); bluish bulge and amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through with or without flaps
MRKH	Bulbs/plate; absent	Hymenal fringe and pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy and neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

- **Imperforate hymen** is the commonest obstructive anomaly; aspiration causes ascending infection.
- **Transverse septum** (~1 in 70,000): delay causes endometriosis, and **MRI distinguishes a high septum from cervical agenesis**. **Z-plasty** makes the scar longitudinal; the main complication is **stenosis**, and a **hollow mould** is used when a uterus is present.
- **Distal atresia**: the haematocolpos acts as a tissue expander; drain, then pull through.

MRKH (müllerian agenesis)

- MRKH (~1 in 5000) is 46,XX with **functioning ovaries**, the second commonest primary amenorrhoea after gonadal dysgenesis.
- Renal anomaly occurs in ≈ **one-third**, skeletal in 10–20%. **Type A** is symmetrical without extra-müllerian anomaly; **Type B** is asymmetrical with renal, Klippel–Feil or MURCS anomalies and needs full renal and skeletal work-up.

6 · Vaginal replacement

- **Dilation is first-line** (Frank; Ingram); surgery is no better. **Vecchietti** uses an olive with laparoscopic traction.
- **McIndoe** places a split-thickness graft over a stent; it stenoses, so dilation is mandatory. Alternatives include **buccal mucosa** and **robotic peritoneal vaginoplasty**.
- **Sigmoid** is the default bowel; excise ~3 cm of intervening bowel to avoid overlapping suture lines and fix it against prolapse. **Ileum** stenoses more and is kept for an irradiated pelvis or

absent colon.

- Bowel risks discharge and **adenocarcinoma**; skin risks stenosis and **squamous carcinoma**. Every neovagina needs **annual examination**.

> **KEY POINT**
> **Cervical atresia**: a perineum-to-uterus anastomosis loses the endocervical barrier and risks life-threatening ascending infection. Hysterectomy with neovagina is often best; otherwise use endometrial suppression and IVF with transmyometrial transfer.

7 · Lateral fusion anomalies

HWWS girls menstruate, so obstruction is missed.

- **Didelphys** has a septate vagina in up to 75%; excise it only for symptoms. Incise a **septate uterus** only for recurrent pregnancy loss.
- **HWWS** is didelphys with a unilateral imperforate hemivagina: pain **without amenorrhoea**, a lateral bluish bulge and **ipsilateral renal agenesis**. Treat by wide septal incision.

8 · Acquired disorders

- **Vaginal rhabdomyosarcoma** is a grapelike bleeding mass under 2 years with the best prognosis of genital primaries. Chemotherapy is primary; IRS trials cut surgery from **100% to 13%**.
- **Labial adhesions** are the commonest interlabial abnormality (peak 13–23 months); **80% resolve in 1 year**, so treat only symptoms. Use topical oestrogen or betamethasone 0.05% before separation.

P1 Female circumcision — FGM ~200 million, WHO types I–IV (III = infibulation), complications, midline deinfibulation

- **Condylomata** suggest abuse, though perinatal transmission occurs; over 50% resolve.
- An **inguinal hernia** may present CAIS (~1–2.5%); the **round ligament is absent in genetic males**. If CAIS is found, preserve the testes.

59 Differences of Sexual Development

1 · Definition and nomenclature

- In a **difference of sex development (DSD)** chromosomal, gonadal or anatomical sex is atypical. The **2006 Chicago consensus** replaced "intersex" and "hermaphrodite" with karyotype-based terms.

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not androgen insensitivity syndrome, AIS) MCQ

- Offer endocrine and genetic testing for **proximal hypospadias even with descended testes**.

2 · Normal sex development

- **SRY** upregulates **SOX9**, making Sertoli cells. Genes acting elsewhere explain syndromic DSD: **WT1** (Wilms tumour, nephropathy), **SF1** and **DAX1** (adrenal failure), SOX9 (campomelic dysplasia).
- **Sertoli anti-Müllerian hormone (AMH)** from 7–8 weeks makes Müllerian ducts regress by ≈ 10 weeks. **Leydig testosterone** from ≈ 9 weeks builds the Wolffian ducts.
- **5α-reductase 2 makes dihydrotestosterone (DHT)**, the active androgen for prostate, urogenital sinus and external genitalia.

> **KEY POINT**
> The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. A uterus in 46,XY means Sertoli (AMH) failure, that is dysgenesis, not a synthesis or receptor defect.

3 · Classification (2006 / Chicago)

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY mixed gonadal dysgenesis (MGD); 46,XX/46,XY chimerism
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: congenital adrenal hyperplasia (CAH: 21-hydroxylase, 11β-hydroxylase, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, drugs
46,XX — other	Cloacal anomalies; Mayer–Rokitansky–Küster–Hauser (MRKH); vaginal atresia
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular; testis regression

Category	Conditions
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	Complete and partial AIS (CAIS, PAIS)
46,XY — other	Persistent Müllerian duct syndrome (PMDS); vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

- 3β-HSD and P450 oxidoreductase deficiency virilise 46,XX yet undervirilise 46,XY.

4 · 46,XX DSD

CAH is commonest and can kill a newborn through salt loss.

CAH — 21-hydroxylase deficiency

- CAH is the commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-hydroxylase deficiency (21-OHD) is **95%** of CAH; it is autosomal recessive, ≈ 1 in 15,000.
- 46,XX infants have normal ovaries and Müllerian structures, often with a **common urogenital sinus**; some look like a boy with **bilateral impalpable gonads**. Virilisation is graded **Prader I–V**.
- **17-hydroxyprogesterone (17-OHP) is raised**; measure it after **36 h**.
- A **salt-wasting crisis** (low sodium, high potassium, acidosis, shock) usually comes no earlier than **day 5–7**.
- Treat with glucocorticoid, mineralocorticoid and salt (hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone, NaCl in infancy; double-to-triple the dose in illness or surgery).
- Affected boys look normal, so **newborn screening** matters most.
- Males may get **testicular adrenal rest tumours**, which are benign and regress with more glucocorticoid but obstruct tubules, so bank sperm.
- Forms are **classic salt-wasting**, **simple virilising** and **nonclassic** (20–50% enzyme activity; hirsutism, oligomenorrhoea; may need no treatment).

Enzyme	Share	BP / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; low BP, high K^+	Normal	High 17-OHP
11β-hydroxylase	≈ 5%	Hypertension, low K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	High dehydroepiandrosterone (DHEA)
17α-hydroxylase	Rare	Hypertension, low K^+	Female/atypical; delayed puberty	Low androgens

Other 46,XX DSD

P1 P2 46-XX Male — SRY translocated to X; testes, no Müllerian structures; azoospermia (no AZF)

- Aromatase deficiency virilises mother and infant; maternal androgen exposure ends at birth, so no postnatal therapy is needed.

5 · 46,XY DSD

A uterus separates dysgenesis from synthesis or receptor defects.

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Streaks, female genitalia, uterus; absent puberty; tumour 30–45%, so prophylactic gonadectomy; oestrogen; donor-ovum pregnancy
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian structures. WT1: Denys–Drash (congenital nephrotic syndrome, Wilms 90%) or Frasier (late focal segmental glomerulosclerosis, gonadoblastoma 60%, no Wilms)
CAIS	X-linked; female genitalia, short blind vagina, no uterus; breasts by aromatisation; ≈ 50% have inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia; identity mostly male; pubertal breast growth
5α-reductase 2	Autosomal recessive; atypical/female at birth, blind pouch; **virilises at puberty**, fertility reported; testosterone:DHT **>20:1**; "guevedoces"
17β-HSD3	High androstenedione:testosterone ratio; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH or AMH-receptor mutation (autosomal recessive); male genitalia with uterus and tubes; found at hernia repair or orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester: male genitalia, impalpable testes, no Müllerian structures; undetectable testosterone/AMH, high gonadotropins at mini-puberty; testosterone from adolescence

- PAIS raised male may respond to high-dose testosterone; raised female, use a GnRH agonist or gonadectomy, then oestrogen. In CAIS, vaginal dilation usually suffices.
- **PMDS (Clarnette)**: type 1 is bilateral intra-abdominal (60–70%); type 2, testis in sac or scrotum with contralateral hernia (20–30%); type 3, transverse ectopia (10%).
- PMDS needs orchidopexy. The vasa hug the uterus, so Müllerian malignancy risk (3–8%) is weighed against careful removal sparing the vasa.

6 · Sex chromosome DSD and ovotesticular DSD

P4 P3 Klinefelter's syndrome / Mixed gonadal dysgenesis — 47,XXY features, hormones, micro-TESE; MGD 45,X/46,XY asymmetry, second commonest atypical genitalia, tumour 15–35%, remove streaks, Turner screening

Not in the paper answer:

- **Turner with Y** material: tumour 12–20%; the 2017 guideline advises gonadectomy at diagnosis, on low-level evidence.
- **Turner**: coarctation, bicuspid valve, horseshoe kidney (renal anomaly ⅓–½) and ovarian insufficiency ≈ 95%, so give lifelong cardiac and renal screening, growth hormone and oestrogen.
- **Ovotesticular DSD** has ovarian and testicular tissue; most are 46,XX, with a sinus and uterus. A **vas always accompanies a testis**, a tube an ovary.
- The ovary is usually normal and the testis dysgenetic; tumour risk is ≈ 3%. Biopsy both poles of an ovotestis and defer irreversible steps.

7 · Evaluation of the newborn with atypical genitalia

P3 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — history, exam, gonads as key sign, karyotype/FISH, 17-OHP, 1-week hormone table, karyotype-based algorithm, sex of rearing, surgery timing, gonadal risk

Not in the paper answer:

- Late presenters include CAIS, Swyer, MRKH, Turner with Y, Klinefelter and 5α-reductase deficiency; after puberty, high LH and testosterone with female genitalia means CAIS.

KEY POINT

Emergency: atypical genitalia with bilateral impalpable gonads is salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP, and watch for crisis from the end of week 1. Designation and surgery are not emergencies.

- A multidisciplinary team (MDT) reviews within **24–48 h**. A urologist without a team transfers early.
- Count palpable gonads (ovaries do not descend): none suggests CAH; one, MGD, ovotesticular DSD, PAIS or partial dysgenesis; two, 46,XY DSD.
- Hypospadias with undescended testis (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if meatus proximal.

ALGORITHM

1. Send karyotype (≥45 cells, for mosaicism; FISH for X/Y in hours), electrolytes, and 17-OHP at ≈ 36 h.
2. Pelvic ultrasonography for uterus and gonads.
3. At ≈ 1 week: LH, FSH, testosterone, oestrogen, androstenedione, DHT, AMH, with or without hCG stimulation.
4. Genetic panel, microarray or exome.
5. If still unclear, then MRI, genitography, endoscopy, laparoscopy with or without biopsy (**no gonadectomy during diagnosis**).

- Normal AMH means functioning Sertoli (or granulosa) cells; high LH and FSH mean gonadal failure.
- A precise diagnosis is nearly always made in 46,XX DSD but in **<50%** of 46,XY.

8 · Gender assignment and MDT

- The **Money (1975)** model of early designation, early surgery and secrecy failed; shared decisions replaced it.
- Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male trend: PAIS, 5α-reductase, 17β-HSD3, partial dysgenesis.
- Dysphoria is highest in 5α-reductase deficiency raised female (≈ half) and lowest in 46,XX CAH raised female.
- Micropenis (<2.5 SD; <2 cm at 0–5 months) needs an endocrine work-up; female reassignment is no longer recommended.

9 · Surgery

- **Feminising** surgery (Ch 60): clitoral surgery is irreversible and infant benefit is unproven. Ideally defer it until the individual can take part.
- **Masculinising** surgery (proximal hypospadias, orchidopexy) has high complications but is revisable, so infant repair is often chosen.
- In MRKH, dilation comes first. Uterine remnants are removed or suppressed, **never joined to a neovagina** (no cervix).

10 · Gonadal tumour risk

Risk rises with Y material (gonadoblastoma locus, TSPY), dysgenesis and intra-abdominal position, and falls with a scrotal gonad.

Risk	Condition	Germ cell tumour	Management
High	Dysgenesis with Y, intra-abdominal (Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (debated)
	17β-HSD3	28%	Monitor or gonadectomy
	PAIS / dysgenesis, scrotal	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove discordant testicular tissue
?None	5α-reductase; Leydig hypoplasia	Unknown	Unresolved

11 · Ethics

- Respect autonomy, be transparent, and avoid irreversible procedures before consent unless necessary. Psychology is integral.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

1 · Principles

P3 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — classification, work-up, CAH care, team, sex of rearing, gonadal risk

Not in the paper answer:

- Endocrine Society 2018: families with experienced surgeons decide. Never use a flap for a high confluence. Introitoplasty is needed after puberty in 25–100%.

2 · Urogenital sinus and cloaca — classification

In a **urogenital sinus** the vagina still communicates with the urinary tract (usually the mid-distal urethra), and both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly congenital adrenal hyperplasia (CAH)
Pure urogenital sinus	Normal external genitalia; behaves like a cloaca
Persistent cloaca	Rectum also joins, so there is **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH causes half of all newborn atypical genitalia and **90%** of 46,XX cases. Salt wasting occurs in **67–75%**; get a karyotype within **48 h**.

> KEY POINT
> **Governing principle**: **true urethral length** (bladder neck to confluence) decides the operation far more than common-channel length, in sinus and cloaca. Confluence level is a continuum.

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants). Virilisation lengthens the **vagina-to-meatus** distance, so a "high" CAH confluence is an illusion.

3 · Clinical evaluation

- The **first threat is metabolic**: monitor electrolytes for salt wasting before reconstruction. **Hypertension** suggests 11β-hydroxylase deficiency.
- A **suprapubic mass** (bladder, **hydrometrocolpos**, or both) is often the only early sign of a pure sinus.
- The **Huffman index** (glans width × length) is normally **<3.5 mm²**; **>10 mm²** is a concern.
- **Hydrocolpos in cloaca** forms because urine refluxes into poorly draining vagina(s). It is commoner with a common channel **>3 cm**, duplication and posterior cloaca.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, ectopia	Majority
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on magnetic resonance imaging (MRI) ≤ 43%
VACTERL; gut	Tracheo-oesophageal fistula; malrotation, duodenal atresia	Correct malrotation at colostomy

KEY POINT

Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). The number of perineal orifices tells you which world you are in.

4 · Imaging and endoscopy

CAH needs little imaging; a cloaca needs a full 3-D map.

Modality	CAH sinus	Pure sinus / cloaca
Ultrasonography (US)	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as a computed tomography (CT) or MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- On a **genitogram**, a **cervical impression** at the vaginal dome means female internal organs.
- Do a **gonadal biopsy** only if it would change the gender of rearing, using a **deep longitudinal incision** to reach testicular tissue within an ovotestis.
- **Prenatal cloaca** shows fetal ascites, bilobed pelvic cysts and hydronephrosis. **Always do echocardiography.**
- In duplicated vaginas the rectal fistula most often opens **within the vaginal septum**. At **early endoscopy**, **empty the vagina first**.

Common channel (Peña)	<3 cm	>3 cm
Anomalies; hydrocolpos	Fewer	More (nearly all have urological defects)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often

5 · DSD reconstruction — timing and consent

No trial compares early with late surgery.

- **For early surgery** gives pliable tissue; advocates seek a moratorium until consent.
- **Clitoroplasty** is only for severe hypertrophy (**Prader III–IV**) and is the most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral/sinus tissue is usable; maternal oestrogen thickens tissue; a planned one-stage repair does better
After puberty	Infant repairs stenose; avoids dilation; allows consent; delay for the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

KEY POINT

Without comparative data the decision is ethical, not evidential, and rests with parents and team. Endocrine Society CAH guideline (2018): families with experienced surgeons decide; the 2006 consensus limits surgery to Prader III–IV.

Body	Position
Lawson Wilkins and European paediatric endocrine societies' CAH group (LWPES/ESPE, 2002)	Surgery at 2–6 months; not between 12 months and adolescence
International intersex surgical society (ISHID) poll	Most prefer single-stage before 2 years
American Medical Association ethics council	Rejected a moratorium after 2 years' review
Advocacy groups; some US state bills	Moratorium until consent; bills not enacted

6 · Operative principles

Operate in this order: **clitoroplasty, then vaginoplasty, then labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving the neurovascular bundle	Basis of later techniques
Kogan (1983)	Subtunical excision, lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares the dorsal bundle
Pippi Salle (2007)	Corpora disassembled and buried in the labia	Reversible if male identity emerges

- The **Fortunoff (1964)** posterior perineal flap, now **omega-shaped**, underlies almost every repair. **Hendren and Crawford (1969)** showed that the **confluence level determines the type of vaginoplasty**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, flap inset; confluence not moved	**Never for high confluence**: vaginal voiding, infection, incontinence

Vaginoplasty	Indication	Principle	Pitfall
Pull-through	Very high	Vagina separated; sinus becomes urethra	If the vagina falls short, add flaps
Replacement	Absent vagina	See Ch 58	—

KEY POINT

Ask: **must the confluence be moved?** A flap brings skin to a low vagina; a pull-through takes a high vagina off the urethra. A flap on a high confluence is the classic error.

7 · Low confluence repair

ALGORITHM

1. Ensure metabolic stability; give **stress-dose steroids in CAH** and antibiotics.
2. Do endoscopy; place a **Fogarty catheter in the vagina** and a Foley in the bladder.
3. Prepare the whole lower body for supine–prone rotation.

ALGORITHM

1. Place a glans traction suture.
2. Incise around the glans, keeping the inner prepuce for the hood (**second only to the glans in sensitivity**).
3. Raise the omega flap and make Y-incisions on the labia majora.
4. Deglove to the bifurcation, keeping the ventral plate intact.

- **Ventral subtunical excision**: incise Buck fascia on the **ventral-most** corpora and remove the erectile tissue, leaving the tunics and dorsal bundle untouched.

ALGORITHM

1. Flap vaginoplasty: raise the omega flap with its fat.
2. Free the posterior sinus off the rectum (**hardest step**).
3. Open the posterior sinus in the midline and **continue to normal-calibre vagina**.
4. Inset the flap.

Labioplasty: split clitoral skin forms the **labia minora**; the labia majora are moved down by **YV-plasty**.

KEY POINT

Two fixed points: never touch the dorsal neurovascular bundle; always open back to normal-calibre vagina.

8 · High confluence: pull-through

- Only ≈ **5%** of CAH cases are high; they are commoner in pure sinus.
- The **critical step** is separating the anterior vaginal wall from urethra and bladder neck, which risks sphincter and nerves.
- **ASTRA** (anterior sagittal transanorectal approach) is done prone, dividing the anterior rectal wall: excellent exposure, no colostomy, continence unaffected.

ALGORITHM

1. Position prone; incise in the midline between sinus and rectum.
2. Divide the sinus up to normal vagina.
3. Free the anterior wall off the urethra **under direct vision**.
4. Close the sinus as the urethra in 2–3 layers; mobilise the vagina.
5. Turn supine for flaps and labioplasty.

Err on the side of the vagina; if it is short, add skin flaps.

9 · Total and partial urogenital mobilisation

- **Total urogenital mobilisation (TUM, Peña 1997)** mobilises the whole sinus en bloc to the perineum without separation.
- **Never discard the sinus**: split **ventrally** it becomes the vestibule; split **dorsally** it becomes the anterior vaginal wall.

	TUM	Partial (PUM, Rink)
Extent	Retropubic and suprapubic	Stops at the **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · Outcomes of sinus/DSD surgery

- After infant vaginoplasty, **secondary introitoplasty** is needed after puberty in **25–100%**, because the orifice does not grow.
- **Azziz (1986)** MCQ reported poor coitus rates, but these reflect old **cut-backs**.
- **Confluence level, number of procedures and surgeon expertise** matter more than age; no data favour postpubertal surgery.
- **Elmasri meta-analysis (2018)**: female identity ≈ 89%; vaginal stenosis **27%**; only half had comfortable intercourse.

Two-procedure dilemma	Infancy	After puberty
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations, so the key factor is a **surgeon or team with a special interest**.

11 · Cloaca — initial management

- **Hendren's rule**: never do a separate rectal pull-through; repair rectum, vagina and urethra in one stage.

ALGORITHM

1. Decompress the bowel.
2. Decompress the genitourinary tract.
3. Correct nephron-destructive anomalies.
4. Then do the definitive repair.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon available for vaginal replacement	Absorbs urine, causing acidosis
Descending, divided (Levitt, Peña)	Less resorbing mucosa; enough distal colon; less prolapse	Leave a long sigmoid

- Urine absorbed by colon causes **hyperchloraemic acidosis**. **Preserve the appendix** for a later Malone antegrade continence enema (MACE).
- **Urinary decompression**: **CIC is first line**, then a cloacal cut-back; vaginostomy or vesicostomy only if CIC fails.
- **Kidneys**: structural anomalies occur in **60–83%**; chronic kidney disease (CKD) in **75%** and end-stage renal disease in **17%**. Urodynamics are abnormal in **90%**.

12 · Cloaca — definitive repair

Repair at **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Never transabdominal TUM (ischaemia)

KEY POINT

Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags the bladder neck below the continence mechanism and causes leakage.

ALGORITHM

1. **PSARVUP (Peña; posterior sagittal anorectovaginourethroplasty)**: place catheters in each channel; position **prone jackknife**.
2. Mark the anal site with a muscle stimulator; incise in the midline.
3. Open the rectal fistula and mobilise the rectum.
4. Do TUM or vaginal separation.
5. Close the urethra in 2–3 layers with a **fat flap**; rebuild the perineal body; centre the rectum in the sphincter.

- It is **better to enter the vagina than the rectum or urethra**; separate it from the urethra by **sharp dissection, not cautery**.
- Aftercare: Foley for 2 weeks, rectal dilation from 2–3 weeks and colostomy closure at ≈ 3 months. Almost one-third empty poorly, so start CIC if in doubt.

13 · Cloaca — results and follow-up

- Outcome depends mostly on the **spine** and **common-channel length**.
- **Urinary**: about half void spontaneously. Peña: CIC in **28%** (channel <3 cm) vs **78%** (>3 cm). Review yearly for life.

- **Renal** function is preserved by aggressive bladder management. Faecal continence needs a bowel programme, then **MACE**.
- Vaginal stenosis or obstructed menstruation occurs in **36–41%**. Pregnancy is possible, with **caesarean preferred**.

ALGORITHM

1. Do US **6–9 months after thelarche**.
2. Repeat every 6–9 months until menses are normal.

65 Adolescent and Transitional Urology

1 · Definitions

- **Adolescent urology** bridges care for complex congenital conditions; the adult urologist lacks **whole-patient knowledge** (prior operations, altered anatomy, families).
- **Transition** (Got Transition) is the planned move from child/family-centred to adult/patient-centred care: **planning, transfer and integration**. Transfer is only one step.

> **KEY POINT**
> Renal function and HbA1c are expected to worsen, so **stability and retention in follow-up** are success.

2 · Who needs transition

Category	Examples	Implication
1. Complex congenital	Exstrophy, posterior urethral valves, complex hypospadias, spina bifida (SB)	Lifelong care
2. Treated, now quiescent	Vesicoureteric reflux	Early-adult review; may discharge with open return
3. Cured, with sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Failed transition (loss to follow-up, reliance on the emergency department) raises morbidity and mortality because **preventive surveillance is lost**.

3 · The process

"Three Ps"	Content	Success measure
Preparation	Independence from 12–14 years; Transition Readiness Assessment Questionnaire (TRAQ)	All families involved
Plan	Follow-up schedule; self-care (catheterisation, irrigation, medications)	Shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- The **paediatric provider must initiate**, and never during a crisis; **pregnancy** is the exception and needs expedited transfer with co-management.
- Parents fear losing control, so give the patient **private time** (sexual function) and then ask what may be shared.
- Begin at about **12 years** (AAP) and complete by the late teens or early 20s; age **26** is the latest (US parental insurance ends). If independence is unachievable, plan custodianship.

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning	14–18

Got Transition core element	Age (years)
Transfer of care	18–21
Transition complete (integration confirmed)	18–23

> **KEY POINT**
> Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · Routes and models of care

Patients arrive by transitioned referral (ideal), referral without transition, a new adolescent problem, or an **emergency after discharge**, which is the hardest and often the commonest.

Model	Strength	Drawback
1a Lifelong paediatric urologist	Continuity; suits limited life expectancy	Patient outlives the provider
1b Paediatric urologist in adult department	Continuity and adult expertise	Cost, privileges, geography
2 Transfer to an adolescent/adult specialist	Specialist coordinates the MDT	Cross-institution communication
3 Direct transfer, no plan	The default	Invisible until an emergency

Lifelong and multidisciplinary need are **independent**: exstrophy may need only a urologist, whereas SB needs the largest MDT. **Nursing** is the backbone.

5 · Development and consent

- Avoid genital surgery at **3–6 years**; the 1975 AAP panel favoured genital surgery at **<6 months**.
- Augmentation waits until the child can catheterise and wants dryness.
- **Emerging adults (18–25 years)** consent for themselves unless they lack capacity.
- Patients with **SB** usually have normal intelligence but weak **working memory and executive function**.
- **Medical PTSD** causes avoidance and procedural panic; CBT may help.

6 · Setting up a service

- The core MDT includes paediatric and adult urologists, specialist nurses, a care coordinator, nephrologist and psychologist.
- Review consent **with the family, then with the patient alone**; standing orders (for example, urine cultures) avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity with congenital conditions (biggest)**; finance	Partnerships, networks, training

120 Aging and Geriatric Urology

1 · Concepts and biology

- **Chronological age is a poor descriptor**; physiological and functional age (function, frailty, geriatric syndromes) matter. The term "geriatrics" dates from Nascher (1909) MCQ .
- **Cellular senescence** drives ageing, either by telomere shortening (**Hayflick limit**, 1961) MCQ or by stress; its secretory phenotype (SASP) causes chronic inflammation.

2 · Clinical evaluation

- **Homeostenosis** is narrowed physiological reserve, so recovery after a stressor is poorer.

Tier	Content	Significance
Basic activities of daily living (ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental ADLs (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Independent living

- A simple screen is **whether the patient can bathe independently**.
- Age is seldom an independent **surgical risk**; predictors are **longer operating time, poor ASA score and urgent surgery**. Optimise rather than "clear" the patient.

> **ALGORITHM**
> 1. Screen for frailty before elective major surgery (SPAQI).
> 2. If positive, do a diagnostic frailty assessment.
> 3. Then do a comprehensive geriatric assessment (CGA).

- The **ACS NSQIP/AGS checklist** has 13 items, only 2 of which are organ systems; the geriatrician-led **CGA** adds physical performance, albumin, creatinine, BNP and FEV1.

3 · Geriatric syndromes

- Geriatric syndromes feed one another, so treating one helps others. **Frailty** is decreased reserve causing vulnerability, distinct from comorbidity and disability.

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**
Validity	Does not predict postoperative outcome	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm often comes via the **prescribing cascade**, in which a side effect is treated with another drug.

Drug	Concern and action
Antimuscarinics (in dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention. Avoid combinations; no tricyclics in BPH

Drug	Concern and action
NSAIDs	GI bleeding; renal and heart failure at CrCl <30 mL/min
Systemic oestrogen; testosterone	Cancer, cardiovascular harm. **Vaginal oestrogen safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), co-trimoxazole	Warfarin potentiation: monitor INR; dose-reduce at CrCl <30
Benzodiazepines	Cognition, gait, delirium. Avoid with opioids or ≥2 other CNS drugs
Nitrofurantoin	Listed at CrCl <30 or chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia: treat nocturia's cause; check sodium

KEY POINT

Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- GSM causes dryness, dyspareunia and UTI **plus urgency and urgency incontinence**. Sex-steroid loss drives OAB in both sexes, including men on ADT or 5-ARIs.
- Treat with **vaginal oestrogen**. Prior breast cancer is not an absolute bar.
- A **urethral caruncle** causes painless spotting, so exclude gynaecological cancer.

4 · Incontinence management

- Incontinence is common but not normal; behavioural therapy comes first (see Voiding Dysfunction book, Ch 110). †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.

P2 P3 Overactive bladder and its drug therapy; OAB Pharmacotherapy — trospium quaternary, M3 selectivity, tolterodine QT, mirabegron BP/metoprolol, Beers avoid in dementia

Not in the paper answer:

- Beers lists **oxybutynin and fesoterodine** as "avoid" in dementia, although the paper suggests fesoterodine for the elderly; trospium has the least central penetration.
- Fewer than half persist with **antimuscarinics** at 6 months; check the post-void residual after starting.

Nursing homes

- Catheter colonisation is **98%**, so treating asymptomatic bacteriuria (ASB) causes harm.

Loeb — catheter	Loeb — no catheter	NHSN
≥1 of: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever plus new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 CFU, no other source, and clinical signs

KEY POINT

A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI and asymptomatic bacteriuria

- **A causal link between confusion and UTI is not supported** (see Infections book, Ch 26).
- ASB affects up to 22% of older women and about 10% of older men and needs no antibiotics. USPSTF gives grade D against screening; IDSA advises treatment **only in pregnancy and before urological instrumentation**.

6 · Cancer, end of life

- **Prostate**: USPSTF grade D, so no PSA screening over 70. With life expectancy **<10 years**, plan non-curative care.
- **Bladder**: restaging TURBT for T1 is often omitted in the elderly yet improves survival. BCG is less effective (immunosenescence). Cystectomy at ≥75 gives comparable results with careful care.
- **Palliative care is not withdrawal**; it means aggressive symptom control after a goals-of-care discussion.
- The Age-Friendly "4 Ms" are Mobility, Medication, Mentation and what Matters; urologists must report **elder mistreatment**.

> **KEY POINT**
> Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · Landmark trials

Trial	Design	Result	Take-home
ASPREE	RCT: aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Effective prevention includes vitamin D with calcium, antihypertensives, statins and exercise.

126 Management and Considerations in Transgender and Nonbinary Individuals

1 · Definitions and terminology

- **Transgender and gender-diverse (TGD)** people have a gender identity that differs from their sex assigned at birth. Assigned sex, gender identity, gender expression and sexual orientation are **independent axes**.

Term	Meaning
AFAB / AMAB	Assigned female/male at birth
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary	Neither male nor female
Gender dysphoria	Distress from identity–assigned sex discrepancy
Gender incongruence	Diagnostic term: marked, persistent incompatibility

Not all TGD people have dysphoria; some need only social transition, others gender-affirming hormone therapy (GAHT) and/or surgery.

2 · Diagnosis and assessment

	DSM-5 (2013)	ICD-11 (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced "gender identity disorder")
Criteria	Marked incongruence for ≥ **6 months** , significant distress or impairment	Marked, persistent incongruence; desire for treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

ICD-11 keeps a code for access but drops the mental-disorder label; **severe distress is not a prerequisite**.

WPATH Standards of Care 8 (SOC-8, 2022)

- Treat only marked, sustained incongruence, after excluding other causes and optimising coexisting conditions.
- Assess **capacity to consent**, and give **fertility counselling before any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy, **consider ≥6 months of hormones** unless not indicated.

> **KEY POINT**
> SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling stay firm because consequences are irreversible.

3 · Gender-affirming hormone therapy (GAHT)

GAHT aims for **physiological** levels of the affirmed gender in patients with persistent dysphoria, capacity, majority age and controlled comorbidities.

Direction	Agent	Dose
Transfeminine — oestrogen	Oral oestradiol	2–6 mg/day
	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate IM	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine — antiandrogen	Spironolactone	100–300 mg/day
	Cyproterone acetate	25–50 mg/day
	GnRH agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine — testosterone	Enanthate/cypionate	100–200 mg 2-weekly
	Undecanoate	1000 mg 12-weekly (rare in the US: oil embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

KEY POINT

Why the asymmetry? Oestrogen alone does not suppress testosterone to the female range, so transfeminine therapy needs an antiandrogen too; testosterone suppresses the ovarian axis on its own.

Transfeminine

- **Ethinyl oestradiol is not recommended** (about 20-fold thrombosis). Use 17β-oestradiol, target **100–200 pg/mL**, and **use transdermal if VTE risk** (age >45 or previous VTE).
- **Spironolactone** is commonest in the US and **cyproterone** in Europe; GnRH agonists and bicalutamide are alternatives.
- **Irreversible effects are breast development and reduced fertility.** Risks include VTE, CVD, hypertriglyceridaemia, liver dysfunction and osteoporosis.
- Monitor levels 3-monthly in year 1, then 1–2 times a year, aiming for testosterone **<50 ng/dL**; check potassium and creatinine on spironolactone.
- **After gonadectomy, reduce oestrogen by 50–75%** (titrate to LH, FSH and bone density) and **stop the antiandrogen**.

Transmasculine

- Contraindications are pregnancy, unstable coronary disease and untreated polycythaemia; oestrogen-sensitive cancer needs oncology input. **Contraception is essential** because testosterone is teratogenic.
- Target testosterone **300–1000 ng/dL**; persistent menses respond to a progestin or endometrial ablation.
- **Irreversible effects are clitoral growth, facial hair, voice change and male-pattern baldness.**
- The main risk is **erythrocytosis**, so monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2 times a year.
- **After gonadectomy, the testosterone dose is not reduced**. **Nonbinary** regimens are individualised (for example, **nandrolone** for muscle with little hair).

4 · Urological considerations

Lower urinary tract symptoms have five contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction and psychosocial factors**.

- **Testosterone-induced vaginal atrophy** (dryness, bleeding, urgency) is treated with topical vaginal oestrogen.
- **After vaginoplasty**, about a third report altered voiding (**nocturia is commonest**); meatal stenosis, stricture and fistula also occur (see Ch 127).
- **After phalloplasty**, post-void dribble comes from pooling at the native urethra–pars fixa junction; stricture and fistula are far commoner (see Ch 128).
- Treat **pelvic floor** dysfunction before surgery; this eases neovaginal dilation.
- **Tucking** causes stasis, prostatitis-like symptoms, epididymo-orchitis and UTI. Exclude atrophy, tucking and pelvic floor causes before blaming the operation.

5 · Fertility preservation

Counsel **before hormones and/or surgery**, as WPATH, ACOG, ASRM and the Endocrine Society advise.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation**; hold GAHT for ≥ **3 months** before collection	**Oocyte or embryo cryopreservation** after ovarian stimulation
Prepubertal, or cannot pause hormones	Testicular tissue cryopreservation (experimental)	Ovarian tissue; no hormone pause needed
Obstacles	Ejaculation distress: vibratory stimulation, electroejaculation or surgical retrieval	Bleeding returns; transvaginal scans cause distress

Off testosterone, menses return in 3–6 months, and most conceive within 6 months.

6 · Cancer screening — screen the organ, not the gender

- **Prostate (transfeminine)**: rare but may be aggressive. Offer shared decision-making on PSA at 55–69. **On GAHT, the upper limit is 1 ng/mL, or a rise of ≥ 0.3 ng/mL from nadir**, †because androgen suppression lowers PSA. After vaginoplasty, feel the prostate through the **anterior neovaginal wall**.
- **Testis**: self-examination only.
- **Cervix (transmasculine)**: cytology 3-yearly at 21–29 years; at 30–65, cytology 3-yearly or hrHPV testing 5-yearly; stop after 65. Use a small speculum and **vaginal oestradiol for 1–2 weeks before**.

7 · Affirming practice

- Keep a **detailed organ inventory** and use inclusive intake forms (name, pronouns).
- **Trauma-informed care** (SAMHSA) uses the "Four Rs": **Realise, Recognise, Respond, Resist retraumatisation**.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

1 · Aim and preparation

Many failures start with a patient who cannot dilate.

- **Mental health** input judges **surgical readiness**. Dysphoria-driven depression should **improve** after surgery, so it is no reason to defer; the key risk is **failure to dilate**.

Older criteria	WPATH Standards of Care 8 (SOC8)
Two assessments	**One** (inter-assessor agreement is high)
Mental health professional only	Any licensed clinician ≥ master's degree competent in transgender health
Hormones before gonadectomy	**6 months** suffices
Comorbidity "well managed"	Risks and benefits discussed
12 months' "real-life experience"	**Removed**: dangerous for those facing violence

Capacity is procedure-specific; if doubtful, a psychologist plans support.

> **KEY POINT**
> SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

- Stay within **1 hour's drive** of hospital for the first **3 weeks**.
- **BMI** mainly determines wound healing; no cut-off is proven, but many require **≤35 kg/m²**, and a higher BMI may suit vulvoplasty.
- **Permanent preoperative hair removal** of the scrotal–perineal graft is needed, because canal hair causes infection and hairballs.
- **Continue oestrogen perioperatively**; a 2022 meta-analysis found no added VTE risk.

Orchiectomy

- It may be the sole operation or a first stage, and afterwards spironolactone is stopped. Offer **fertility counselling and sperm banking to all**.
- Use a **vertical midline scrotal incision**; an inguinal incision may compromise the penile flap.

> **KEY POINT**
> Plan every orchiectomy as though vaginoplasty will follow: use a midline scrotal incision and leave cord fat for labial bulk.

2 · Penile inversion vaginoplasty (PIV)

P2 P4 Penile inversion vaginoplasty technique — homologous tissue table, operative steps, dilation, complication rates

Not in the paper answer:

- Pad the calves in lithotomy (**common peroneal nerve**) and give **tranexamic acid 1 g IV at the start**, as bleeding is front-loaded.

ALGORITHM

1. Foley and drain; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion).
2. Harvest scrotal graft and small posterior rhomboid flap (relieves introital tension); defat, tubularise.
3. Orchiectomy, leaving cord fat; penile disassembly; corporal excision and clitoroplasty.
4. Urethroplasty, then canal dissection; line canal, labiaplasty, pack.

- **Disassembly**: the neurovascular bundle is best **left on the dorsal tunica**, and **all erectile tissue is excised**.
- **Urethroplasty**: excise bulbospongiosus **only after canal dissection**, as it covers a rectal injury, and spatulate downward for a seated stream.
- **Canal**: use repeated digital rectal examination, **preserve Denonvilliers fascia** and close it if breached. **Packing** ensures graft take.
- Keep the labia majora incision **≥1 cm** from the groin crease to avoid dehiscence.

Vulvoplasty (zero-depth)

Vulvoplasty suits those not wanting receptive intercourse and avoids **lifelong dilation**, but is not freely reversible. **Prior radical prostatectomy favours vulvoplasty**.

3 · Abdominoperineal and intestinal options

- **Peritoneal (Davydov-type)**: robotic dissection cut rectal injury from **~7%** to about 2.3%. The **rectum may mimic seminal vesicles**, and peritoneal defects must be **closed** to prevent hernia.
- Flaps are either **two-flap** (bladder and sigmoid peritoneum) or **pull-through** (abdominal wall peritoneum).
- **Intestinal** vaginoplasty suits severe penoscrotal hypoplasia or revision stenosis; IBD, adhesions, irradiation and short gut contraindicate it. **Dilation is still required.**

KEY POINT

Lining ladder: penile and scrotal skin first, then peritoneum, then bowel. Each adds depth but imports donor pathology; none removes the duty to dilate.

4 · Complications

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care; granulation: curette or cautery plus steroid
Rectovaginal fistula (most devastating)	**0.8%** primary, 6.3% revision	Intra-op: watertight closure and bulbospongiosus flap, with or without diversion. Established: stop dilating and revise later, **or** delay closure
Urethrovaginal fistula (0.8–3.9%)	Missed injury, distal stenosis	Distal: often left. Bladder-neck (incontinence): transvaginal repair
Urethral stricture (1–6%)	Spraying, UTI, retention	Exclude pelvic floor dysfunction; dilate if short, else meatal spatulation
Urethral bulge / upward meatus	Under-debulked spongiosum	Debulk, fix meatus downward

Complication	Cause / key fact	Management
Vaginal stenosis	Poor dilation, scar, graft loss, pelvic floor dysfunction	Pelvic floor physiotherapy **first**; then scar excision and graft, or peritoneal/bowel augmentation
Prolapse (4–10%)	Poor graft take or fixation	Early repack; peritoneal fixation acts as a colpopexy

KEY POINT

Three failure modes: violated plane (fistulas), tissue not excised (bulge), graft not taken or maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient, so dilation is part of the operation.

5 · Outcomes and follow-up

- **Median time to orgasm is about 180 days**. Depth is 12–16 cm, and self-lubrication is not the norm.
- Stay is about **5 days**; on **day 5 the pack and catheter come out** and dilation is taught.
- Follow-up is weekly for 3 weeks, then at 6 weeks, 3, 6 and 12 months, then annually. †The prostate is retained; survey it via the anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if stable

KEY POINT

The body treats the canal as a defect to close, so stenosis is a lifelong risk. Any narrowing means more dilation, contacting the surgeon and physiotherapy.

128 Special Urologic Considerations in Transmasculine Individuals

P4 Enlist the steps in female to male gender reassignment surgery — staged sequence, vaginectomy, neourethra parts, metoidioplasty vs phalloplasty, scrotoplasty, prostheses, complication sites

Not in the paper answer:

- Haemorrhage is the commonest vaginectomy complication, and a two-stage repair is standard for long strictures.

1 · Scope

- Goals are a male appearance, **voiding standing**, sensation and penetration (see Ch 126).
- **Early puberty blockade has a surgical price**: the vagina is harder to remove and small labia majora give less scrotal tissue.

2 · Preoperative care and consent

- **Urological screening is mandatory** (uroflow, PVR, renal ultrasound), and voiding dysfunction, stones and hydronephrosis are treated first.
- SOC7 requires ≥**12 months** of testosterone before genital surgery (†SOC8: 6 months) for **clitoral hypertrophy**, which plateaus at about 2 years.

Agent	Helps	Hinders
Testosterone	Clitoral hypertrophy aids metoidioplasty	Vaginal atrophy: harder vaginectomy
GnRH agonist	Prevents female characteristics	Small labia majora: less scrotoplasty tissue

Consent covers fistula, stricture, vaginectomy bleeding and organ injury, voiding dysfunction, UTI and implant problems.

3 · Removal of female structures

- At hysterectomy the ureter lies beneath the uterine artery. †If the cervix is retained, screening continues.
- **Vaginectomy** risks haemorrhage, then bladder, then rectal injury. **Without vaginectomy there is no urethral lengthening** because of fistula risk.

ALGORITHM

1. Hydrodissect; circumferential introital incision.
2. Blunt rectovaginal plane; urethrovaginal plane **sharply, with no diathermy** (protects sphincter), then bladder plane.
3. If bladder injured: close, with or without cystoscopy.
4. Free levators, control paravaginal plexus; approximate levators.
5. Bulbospongiosus over urethra; layered closure.

Timing	Verdict
All-in-one	More major complications with phalloplasty; none extra with metoidioplasty
Early vaginectomy, phalloplasty later	Keeps blood loss away from free-flap perfusion

Timing	Verdict
Two-stage (preferred)	Mastectomy, hysterectomy–oophorectomy; then vaginectomy and genital reconstruction

4 · Genital reconstruction

- The fixed (perineal) and phallic neourethra both lack spongiosum. One **dorsal clitoral nerve** is coapted for sensation.
- **Bulbospongiosus over the fixed part** reduces fistulas. Scrotoplasty flaps run on the **external pudendal artery**.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · Postoperative care and catheterisation

- The neourethral catheter comes out at **14–21 days**; the suprapubic catheter is clamped until voiding is normal.
- Catheterisation is hard because the curve into the native urethra is abrupt.

> **ALGORITHM**
>
> 1. One gentle Coudé attempt, then one stylet-guided attempt.
> 2. If any difficulty, stop: flexible cystoscopy, guidewire, open-tip catheter.
> 3. In an emergency, place a **suprapubic catheter**.

6 · Urethral complications

- **Fistula** is commonest at the phallic base, and small ones close with diversion. If one persists >**3 months**, excise and close in layers with fat interposition.
- **Stricture** is commonest at the **fixed–phallic anastomosis**. A plateau uroflow prompts urethrography; treat after **≥6 months**.

Option	Use
Dilation	Low-grade meatal stenosis
Internal urethrotomy	Low success; never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative
Excision and primary anastomosis	Short, near-obliterative anastomotic
Free graft	Needs flap support or dorsal inlay
Two-stage (standard)	Long/complex; retubularise ≥6 months later
Flap / perineal urethrostomy	Recurrent, or no further reconstruction wanted

- **Hair** is prevented with a hairless flap or laser.

- **Post-void dribbling** is near-universal; if severe, exclude sacculation or a vaginal-remnant diverticulum.

> **KEY POINT**
> The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · Prostheses

- Insert an erectile prosthesis ≥ **1 year** after phalloplasty, once there is **protective sensation** and a stable urethra.
- Infection or erosion needs urgent explantation, with reimplantation ≥6 months later.

★ Putting it together

A · Newborn with atypical genitalia

ALGORITHM

1. **Emergency is adrenal:** CAH is commonest (half of all; 90% of 46,XX). Salt wasting (low Na, high K, shock) appears at day 5–7 (see Ch 59).
2. **Tests:** karyotype (SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound.
3. **Team:** †defer sex assignment until MDT evaluation.
4. **Gonads:** dysgenetic with Y (30–45% tumour risk): prophylactic gonadectomy.

B · Urogenital sinus and cloaca

KEY POINT

True urethral length outweighs common-channel length. Low confluence needs a flap; high needs pull-through or urogenital mobilisation. Clitoroplasty only for Prader III–IV, sparing the dorsal bundle. Cloaca: colostomy, decompress hydrocolpos, imaging with endoscopy, then posterior sagittal repair; seek VACTERL and tethered cord (see Ch 60).

C · Absent vagina

Cause	Key feature	Management
Mayer–Rokitansky–Küster–Hauser	46,XX, no uterus; †image kidneys	Dilation first; else McIndoe, bowel, buccal or peritoneal
Complete androgen insensitivity	46,XY, testes, no uterus	Dilation usually suffices
Septum / imperforate hymen	Uterus present, haematocolpos	Hymen: incision, not aspiration; septum: excision

D · Immunotherapy and hereditary cancer

ALGORITHM

1. **Checkpoints:** PD-L1 silences PD-1-positive T cells; CTLA-4 acts at priming. †Immune adverse events grade ≥2: hold drug, give steroids; endocrinopathies need lifelong replacement (see Ch 24).
2. **Hereditary:** VHL (clear cell; belzutifan), HLRCC (aggressive papillary), Birt–Hogg–Dubé (hybrid oncocytic), Lynch (upper-tract urothelial).
3. BRCA prostate cancer: PARP inhibitor. FGFR-altered urothelial cancer: erdafitinib (see Ch 25).

E · Older patient before surgery

KEY POINT

Function, not age, predicts outcome. Screen frailty, then do a comprehensive geriatric assessment (cognition, delirium, nutrition, falls, polypharmacy). Avoid anticholinergics (prefer a β3-agonist). For new incontinence, seek transient causes first (†DIAPPERS) (see Ch 120).

F · Transgender and nonbinary patients

ALGORITHM

1. Screen the organs present: retained prostate (†PSA lower on oestrogen) and retained cervix (see Ch 126).
2. Vaginoplasty: stenosis and rectovaginal fistula (most devastating) (see Ch 127).
3. Phalloplasty: fistula and stricture, worse without vaginectomy (see Ch 128).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 3

Infections & Inflammation

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 26	Infections of the Urinary Tract
Ch 27	Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain
Ch 28	Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders
Ch 29	Sexually Transmitted Diseases
Ch 30	Cutaneous Diseases of the External Genitalia
Ch 31	Tuberculosis and Parasitic Infections of the Genitourinary Tract
★	Putting it together: infection decisions

26 Infections of the Urinary Tract

1 · Definitions and classification

The label decides treatment length and whether to investigate.

- **Urinary tract infection (UTI)** is the urothelial inflammatory response to bacterial invasion, usually with bacteriuria and pyuria. Urine is **not sterile**.
- **Bacteriuria** and **pyuria** are not the same as UTI; bacteriuria without pyuria is colonisation. **Sterile pyuria**: look for tuberculosis (TB) see Ch 31, stone, urothelial injury and interstitial nephritis.
- **Cystitis** causes dysuria, frequency and urgency; mimics include urethritis, vaginitis, interstitial cystitis see Ch 28 and bladder carcinoma.
- **Acute pyelonephritis** is a clinical diagnosis: chills, fever and flank pain with bacteriuria and pyuria. Flank symptoms may be absent in spinal cord injury (SCI).
- **Chronic pyelonephritis** shows a focal coarse cortical scar over a **distorted calyx**.

	Uncomplicated UTI	Complicated UTI
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Often multidrug-resistant (MDR), so culture is mandatory

- **Complicating factors** are an anatomical or functional abnormality (obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO]), **male sex**, pregnancy, diabetes mellitus (DM), immunosuppression, catheter, hospital acquisition and symptoms >7 days. MCQ Male sex alone makes a UTI complicated.

Term	Definition	Discriminator
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared
Reinfection	New inoculation (~ **70%** of recurrences, often different organism)	Cleared, then new event
Bacterial persistence (relapse)	Same organism from a focus (stone, prostate)	Focus remains; recurs quickly

- **Prophylaxis** prevents reinfection of a sterilised tract; **suppression** prevents growth from an ineradicable focus.
- **Catheter-associated UTI (CAUTI)** is the commonest cause of secondary bacteraemia and ~**80%** of nosocomial UTIs.
- Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · Pathogenesis

Route	Frequency	Features
Ascending	Commonest	Bowel flora colonise the introitus and climb; reflux not required
Haematogenous	Uncommon	S. aureus, Candida; enhanced by obstruction
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- **E. coli** causes ~**85%** of community and ~50% of hospital UTIs, but less than half of complicated UTI and urosepsis.
- **S. saprophyticus** causes ~10% of cystitis in young women; complicated UTIs involve Klebsiella, Pseudomonas and E. faecalis.
- **ST131 (O25b:H4)** is an MDR E. coli clone carrying extended-spectrum β-lactamase (ESBL), with fluoroquinolone resistance as its hallmark. MCQ Enterococcus is a Gram-positive coccus, not an Enterobacteriaceae.

Virulence — fimbriae

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	fim; **FimH** tip adhesin	pap operon on a pathogenicity island
Receptor	Mannosylated **uroplakins Ia/Ib**	Glycolipid (P blood-group) receptors in kidney
Tropism	Bladder, causing cystitis	Kidney, causing **pyelonephritis** (~ **90%** of pyelonephritis strains)

- Adherence is the essential first step. In **phase variation** bacteria switch between piliated and non-piliated states.

Intracellular cycle (why cystitis recurs)

ALGORITHM

1. FimH binds uroplakin and the bacterium invades the umbrella cell (15 min–1 h).
2. It multiplies into an **intracellular bacterial community (IBC)**, which behaves like a biofilm (4–16 h).
3. Bacteria filament, escape and reinvade neighbouring cells (16–24 h).
4. Infected cells exfoliate, exposing deeper layers.
5. **Quiescent intracellular reservoirs (QIRs)** form (from ~2 weeks).

- QIRs are silent between attacks; urothelial disruption reactivates them, giving same-strain recurrence without a urological focus.

Host defences and susceptibility

- Lactobacilli are the introital barrier, disturbed by antimicrobials, spermicides and oestrogen loss.
- Bladder emptying is the key defence; **obstruction** makes infection life-threatening. In adults, vesicoureteral reflux (VUR) alone is harmless without high pressure or stasis.
- **DM** brings more abscess, emphysematous infection and papillary necrosis.

Renal papillary necrosis

- The main causes are **DM, obstruction with pyelonephritis, analgesic abuse and sickle cell disease**. A sloughed papilla obstructs the ureter; treat with antibiotics and early relief of obstruction.

3 · Evaluation and diagnosis

A fistula presents with pneumaturia or faecaluria, and an abscess with indolent fever and a flank mass.

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues / causes	E. coli	Discharge, odour, new partner; HSV, gonorrhoea, Chlamydia, Trichomonas, yeast, bacterial vaginosis	Discharge; initial-stream pyuria in men; N. gonorrhoeae, Chlamydia, HSV see Ch 29

- Dysuria with frequency and no vaginal discharge gives a ~**90%** probability of UTI; dysuria is the best single symptom.
- **Rule**: painless gross haematuria, or microhaematuria persisting after the infection clears, needs full upper and lower tract evaluation.
- Midstream urine is the default; a catheter sample is more accurate. With an indwelling catheter, replace it and sample the new one.

Urinalysis and culture

- **Leucocyte esterase** indicates pyuria. **Nitrite** is made by Enterobacteriaceae and is negative with Gram-positives and Pseudomonas; it is the best rule-in test. Dipsticks are better at **ruling out**.
- On microscopy, pyuria is ≥**5–10 WBC/high-power field (HPF)**, and squamous cells mean contamination. A positive culture without pyuria should be repeated on a catheter sample.

Threshold	Use
≥10^5 colony-forming units (CFU)/mL	Kass (originally pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheter specimens

- **Stone culture** predicts post-procedure sepsis better than bladder urine.

Imaging

- Imaging asks only whether there is obstruction, a collection to drain or a focus of persistence; most UTIs need none.
- **Indications in pyelonephritis**: possible obstruction, risk of papillary necrosis, prior reconstruction, **no response after 72 h**, DM, polycystic kidneys on dialysis, NGB and unusual organisms. Also image most men, compromised hosts and rapid recurrences.

Question	Best test
Stone	Non-contrast computed tomography (CT)
Abscess, acute focal bacterial nephritis (AFBN), drainage planning	Contrast CT
Hydronephrosis, pyonephrosis, post-void residual (PVR)	Ultrasound (US)
Extrarenal extension	Magnetic resonance imaging (MRI)
Urethral diverticulum	Pelvic MRI with contrast

Question	Best test
VUR; non-refluxing stump	Voiding cystourethrogram (VCUG)
Split function, scarring	Dimercaptosuccinic acid (DMSA) scan

Correctable causes of bacterial persistence (Box 26.6) include infection stones, chronic bacterial prostatitis, an infected atrophic kidney, ectopic ureter, foreign body, urethral diverticulum, an infected ureteric stump and fistula. Antibiotics cannot sterilise these, so **cure comes only by removal or drainage.**

4 · Antimicrobial principles

- In cystitis the **urinary** drug level matters; in parenchymal infection or bacteraemia, **serum and tissue** levels matter. Renal insufficiency and obstruction lower urine levels.
- **Collateral damage** means resistance, MDR colonisation, C. difficile and candidiasis, caused mainly by broad cephalosporins and quinolones.

P1 Antimicrobial resistance; ESBL producing enterobacterial UTI — intrinsic / chromosomal / plasmid resistance; ESBL, AmpC, KPC and their treatment options

Not in the paper answer:

- Prior quinolone use is the strongest determinant of quinolone resistance, and plasmid resistance spreads among bowel flora.
- **Stewardship** means narrow spectrum, the shortest course, culture before treating recurrent or complicated UTI, and local antibiograms.

Agents

- Fosfomycin acts on the cell wall at a different step from β-lactams, so cross-resistance is rare. Enterococci scavenge folate, so TMP-SMX fails even when "sensitive".
- **Anti-pseudomonal** agents are antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides and fluoroquinolones.
- **Enterococcal** cover comes from aminopenicillins, piperacillin, nitrofurantoin and fosfomycin, and **never from cephalosporins**.
- Ertapenem misses Pseudomonas and Enterococcus.

Agent	Key points
Nitrofurantoin	Acts in urine only, so it is useless for pyelonephritis, prostatitis and bacteraemia. Toxicity includes pulmonary fibrosis and neuropathy. Beers 2015: avoid if creatinine clearance (CrCl) < **30 mL/min**; no long-term suppression in the elderly.
TMP-SMX	Good tissue levels; misses Enterococcus and Pseudomonas; potentiates warfarin.
Fosfomycin	3 g single dose; active against ESBL producers and vancomycin-resistant enterococci, so it is an oral option for MDR cystitis.
Fluoroquinolones	Broad, including Pseudomonas. Adjust at CrCl <30. FDA warnings include tendon rupture and aortic dissection; **2016: reserve for when there is no alternative in uncomplicated UTI**. Avoid in children and pregnancy.
Cephalosporins	Poor against enterococci; use in cystitis only if first-line agents cannot be used.
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once daily 5–7 mg/kg; a nomogram level 6–14 h after the dose sets the interval at 24, 36 or 48 h. Monitor oto- and nephrotoxicity.

Agent	Key points
Aztreonam	Gram-negative spectrum without nephrotoxicity; useful in penicillin allergy and renal impairment.
Pivmecillinam	~2% E. coli resistance; Nordic first-line; not in USA.

KEY POINT

Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis; broad tissue-penetrating agents are kept for invasive infection. Resistance is made in the bowel, so breadth, duration and repetition matter more than MIC.

C. difficile infection (CDI)

- The usual culprits are cephalosporins, clindamycin, carbapenems, quinolones and TMP-SMX. Diagnosis needs ≥3 unformed stools in 24 h and toxigenic C. difficile or its toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin, with or without **bezlotoxumab** (anti-toxin B antibody)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily and IV metronidazole

5 · Asymptomatic bacteriuria

Treating ASB usually harms; know the few groups who need it.

Group (Infectious Diseases Society of America [IDSA])	Specimen	Threshold
Women	**Two** consecutive voided, same strain	≥10^5 CFU/mL
Men	One voided	≥10^5 CFU/mL
Either sex	One catheterised	≥**10^2** CFU/mL

- Prevalence is 1–5% in premenopausal women and **100%** with a long-term catheter. Untreated ASB is harmless; treating it brings more symptomatic UTI and MDR.

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; elderly (community or institution); SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (including arthroplasty and cardiac)**. No consensus: neutropenia.	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (transurethral resection [TUR], ureteroscopy, percutaneous nephrolithotomy [PCNL])

KEY POINT

ASB is treated only where bacteria are about to reach blood or tissue: pregnancy (hydroureter leading to pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · Cystitis

Acute uncomplicated cystitis in women

- Diagnose on acute dysuria with a change in voiding. Fever, flank pain or no oral route should be treated as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance > **20%** or used in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Ecology and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

A **3-day** course beats a single dose and equals 7 days (7 if complicating factors). No post-treatment culture if asymptomatic.

Cystitis in men

- Male cystitis is complicated and about half have an abnormality, so evaluate and consider prostatitis see Ch 27.
- In afebrile men, **7 days is as good as 14 days** (Drekonja). The EAU suggests TMP-SMX 160/800 twice daily × 7 d, or a quinolone guided by local susceptibility.

Complicated UTI

Category (Canadian Urological Association [CUA])	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, pelviureteric junction (PUJ) obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

Always culture and correct the abnormality. Treat **10–14 days** by culture, IV to oral once afebrile.

Emphysematous cystitis

- Bladder-wall gas, typically in an elderly diabetic woman; fever occurs in only 30–50%. **CT confirms** it.
- ~**90%** settle with broad Gram-negative cover, catheter drainage and glucose control.

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection, new resistant species	New species; suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn	Same species, still **sensitive**	Renal function and imaging
8. Non-compliance, Munchausen	—	—

7 · Recurrent UTI

Most recurrences are reinfections, so the work is prevention.

- Recurrent UTI means **2 in 6 months or ≥3 in 12 months**, each culture-documented. Reinfection is a different organism, or the same one >2 weeks after treatment.
- Risk factors include intercourse, spermicide, menopause, incontinence, cystocele and raised PVR.
- Evaluate with history, pelvic examination, **culture of each episode** and PVR. **Imaging and cystoscopy are not routine** (AUA, EAU, CUA).
- **Indications for imaging with or without cystoscopy (CUA)**: prior urinary surgery or trauma; stones; gross or persisting microhaematuria; obstructive symptoms or high PVR; urea-splitters; prior pelvic malignancy; persistence despite sensitivity-based therapy; DM or immunocompromise; pneumaturia, faecaluria or anaerobes; and repeated pyelonephritis. MCQ Recurrence frequency alone is **not** an indication.
- **Struvite** stones form when urease (Proteus mirabilis) produces magnesium ammonium phosphate, with bacteria inside the stone. The AUA advises complete removal in fit patients; unfit patients get acetohydroxamic acid and suppression.

Prevention (women, AUA 2019)

ALGORITHM

1. Confirm recurrent UTI with symptoms and culture.
2. If the same organism returns at short intervals, then evaluate for persistence (upper tract imaging and cystoscopy) and remove the nidus.
3. If it is reinfection, then start behavioural measures: ≥1.5 L water a day, frequent and postcoital voiding, stop spermicide.
4. Add cranberry ≥36 mg proanthocyanidin (PAC).
5. If UTIs follow intercourse, then give a **postcoital antibiotic**. If postmenopausal, then give **vaginal oestrogen**. If neither, then consider **methenamine hippurate** (normal renal function).
6. Use continuous antibiotic prophylaxis last.

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	RCT in premenopausal women: 1.7 vs 3.2 episodes/yr
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm). **Oral oestrogen is ineffective.**
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable; safe with warfarin
Methenamine hippurate 1 g twice daily	**Formaldehyde in acid urine (pH <6)**	Non-inferior to daily prophylaxis; fails in NGB. Contraindicated if CrCl <50.
D-mannose	Inhibits **FimH**	Cochrane 2022 inconclusive
Immunostimulants (OM-89, Urovac, Uromune MV140)	Bacterial lysates	Some benefit; small biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Works only while taken; MDR	AUA "may prescribe"; EAU: last resort
Postcoital single dose	Effective; preferred for coital UTI	Acceptable
Self-start	Old data; overuse	Prefer analgesic, sample and observe, then culture-directed treatment

KEY POINT

Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigate only for features of persistence or complication; prevention targets flora, pH, adhesion and washout.

8 · Kidney infections

Acute pyelonephritis

- **White cell casts** prove a renal origin. Take blood cultures if toxic, admitted or pregnant; the usual cause is P-fimbriated E. coli.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤**10%** ; levofloxacin 750 mg daily; TMP-SMX double-strength twice daily only if known sensitive. With or without one IV dose of ceftriaxone 1 g or an aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin and gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci: ampicillin–sulbactam with or without an aminoglycoside.	10–14 total
Pregnancy	Ampicillin and gentamicin, aztreonam or ceftriaxone, then oral cephalexin. **No quinolone.**	10–14

- **Obstruction** needs urgent decompression by **stent or percutaneous nephrostomy**.

ALGORITHM

1. If there is no sepsis or vomiting, then send a culture, with or without US, and treat as an outpatient with oral drugs for 7–10 d.
2. If there is sepsis, then take blood and urine cultures, image, and admit for IV treatment.
3. If improved by **72 h**, then step down to oral treatment.
4. If not improved, then repeat cultures and get a **CT**, and drain any obstruction or abscess.

KEY POINT

Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal bacterial nephritis (AFBN)

- AFBN is a lobar infiltrate between pyelonephritis and abscess. On contrast CT it is wedge-shaped with no wall or liquefaction, whereas an abscess is round with a liquid centre and a **ring sign**. Give IV antibiotics for ≥7 days, then 7 days oral.

Emphysematous pyelonephritis (EPN)

P1 P3 EPN. Prognostic factors & conserv. mgt.; Prognostic indicators in emphysematous pyelonephritis — definition, CO_2 mechanism, CT choice, Wan patterns, Huang–Tseng risk factors, drainage-first management

Not in the paper answer:

- EPN is mostly seen in **diabetic women**; non-diabetics are usually obstructed, with less extensive disease.

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	50–60% mortality
Fluid with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang–Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics and percutaneous catheter drainage (PCD) or stent
2	Parenchyma, no extrarenal extension	Antibiotics and PCD
3A	Perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 or PCD fails
3B	Pararenal space	
4	Bilateral or solitary kidney	PCD (nephron-sparing)

† The risk factors are thrombocytopenia, acute kidney injury (AKI), altered consciousness and shock.

KEY POINT

Fluid is a good sign and streaky gas a bad one: a contained collection can be drained; diffuse gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal and perinephric abscess

P1 Renal and perinephric abscess — haematogenous vs ascending, perinephric causes, insidious course, CT, empirical antibiotics, size-based drainage (≤3 / 3–5 / >5 cm), nephrectomy

Not in the paper answer:

- A renal abscess ≤ **3 cm** is treated with antibiotics; one > **5 cm** needs percutaneous drainage, as does any size in an immunocompromised or non-responding patient.
- A perinephric abscess has a lower drainage threshold (>3 cm); only a third are diagnosed at presentation, and delay causes nearly all deaths.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis** is infected hydronephrosis with suppurative destruction. Patients are very ill, **bacteriuria may be absent** if the ureter is blocked, and US shows a fluid–debris level.
- Give antibiotics with **immediate drainage** (stent or nephrostomy); nephrectomy if non-functioning.

XGP and malakoplakia

P1 P2 Xanthogranulomatous pyelonephritis; Malakoplakia — stone–obstruction–Proteus triad, bear-paw CT, RCC mimic, complete nephrectomy; Michaelis–Gutmann bodies, intracellular antibiotics + bethanechol/vitamin C

Not in the paper answer:

- In xanthogranulomatous pyelonephritis (XGP), urine is sterile in about a third but tissue culture is positive.

	XGP	Malakoplakia
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	Proteus, E. coli	E. coli
Laterality	Unilateral	Uni-/multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics with or without nephrectomy

> **KEY POINT**
> Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics; pyonephrosis and large or obstructed abscesses need drainage; when the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · Fournier gangrene and periurethral abscess

Skin hides the true spread; only early radical debridement saves lives.

- **Fournier gangrene** is necrotising fasciitis of the genitalia, perineum or perianal region. A source is found in ~ **95%**, and the male:female ratio is ~10:1.
- Sources are **colorectal**, **urogenital** (stricture, instrumentation, catheter) and **dermal**. Risk factors are **DM**, obesity, alcoholism and immunosuppression.
- Synergistic organisms cause obliterative endarteritis, spreading along Buck's, dartos, Colles' and Scarpa's fascia.
- Pain out of proportion comes first, then crepitus and gangrene. **The skin underestimates fascial extent**, and CT must not delay surgery.

Score	Components	Use
Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥ **6** suspicious; ≥ **8** strongly predictive

Score	Components	Use
Fournier's Gangrene Severity Index (FGSI) †	Nine vital signs and blood values, each scored 0–4	>9 predicts mortality

ALGORITHM

1. On suspicion, resuscitate and start broad IV antibiotics (piperacillin–tazobactam; or a carbapenem or vancomycin with clindamycin or metronidazole).
2. Perform **immediate radical debridement** back to bleeding fascia and leave the wound open.
3. Take the patient back for a **re-look at 24–48 h**.
4. If the source is urethral, then add suprapubic diversion; if rectal, then a colostomy.
5. Consider hyperbaric oxygen as an adjunct, then reconstruct.

- **Orchidectomy is almost never needed**, because the testicular blood supply is independent of the scrotal skin. Mortality is ~20–40%.
- A **periurethral abscess** needs drainage, suprapubic diversion and antibiotics.

10 · Urosepsis †

Term (Sepsis-3)	Definition
Sepsis	Organ dysfunction from a dysregulated response to infection: rise in Sequential Organ Failure Assessment (SOFA) score ≥ **2**
Quick SOFA (qSOFA)	≥2 of: respiratory rate ≥22/min, altered mentation, systolic BP ≤100 mmHg
Septic shock	Vasopressors for mean arterial pressure (MAP) ≥65 mmHg **and** lactate >2 mmol/L despite fluids

ALGORITHM

1. Hour-1 bundle: measure lactate and take blood and urine cultures.
2. Give a broad IV antibiotic within 1 h.
3. If hypotensive or lactate ≥4, then give 30 mL/kg crystalloid.
4. If still hypotensive, then start noradrenaline for MAP ≥65.
5. Achieve **source control**: decompress with a stent or nephrostomy; no definitive stone surgery now.
6. De-escalate antibiotics by culture.

† Not in the source notes. Manipulating infected stones releases endotoxin, so drain first and operate later.

11 · Special populations

Pregnancy

P1 P3 Bacteriuria in pregnancy; Baceriuria and use of antibiotic policy in pregnancy — prevalence unchanged but ascends, culture screen at 12–16 weeks, 4–7 day course, safe/avoid drug table, pyelonephritis and suppression

Not in the paper answer:

- GFR rises by 30–50%, so creatinine falls to about **0.5 mg/dL**, and a "normal" 0.8–0.9 may mean impairment.

- Severe or MDR pyelonephritis needs a carbapenem.

Elderly

- **Do not screen for or treat ASB**; bacteriuria with confusion is not UTI. The **absence** of pyuria **excludes** bacteriuria.
- The **revised McGeer criteria** require **localising** signs and ≥10^5 CFU/mL of ≤2 species in non-catheterised residents; catheterised residents may qualify on **systemic** signs.
- Treat for 3–6 days if uncomplicated and 10–14 if febrile.

Catheter-associated UTI

P1 Catheter Associated Urinary Tract Infections (CAUTI) — definition, 5–10% per catheter-day, extra-/intraluminal routes, biofilm, prevention bundle, change catheter, treat only symptomatic

Not in the paper answer:

- Catheters in place >30 days carry 4–5 species, and silicone is no better than latex.

Spinal cord injury

- All UTIs in SCI are complicated, and UTI is the commonest cause of fever. Symptoms are atypical: fever, spasticity, leakage, autonomic dysreflexia † and cloudy urine.
- **Clean intermittent catheterisation (CIC)** has the fewest long-term complications and an indwelling urethral catheter the most.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	≥10^2 CFU/mL
Catheter-free men (clean voided)	≥10^4 CFU/mL

- Do not treat ASB. Treat afebrile UTI for 5–7 days; febrile UTI needs admission and **14 days** parenteral-first treatment. Change the catheter.
- Prophylaxis does not work in NGB; for recurrent UTI, get imaging and urodynamics (UDS) and lower storage pressure.
- A catheter for >**10 years** raises the risk of bladder cancer, so these patients need surveillance.

Diabetes, renal transplant

- In DM, do not treat ASB, but keep a low threshold to image pyelonephritis.
- After transplant, do not screen for ASB beyond 1–2 months. Treat uncomplicated UTI for 5–10 days and pyelonephritis for 14–21 days. Stents raise UTI by about half, so remove them within **4 weeks**.

12 · Fungal UTI

P1 Fungal UTI — Candida risk factors, asymptomatic candiduria (remove catheter; treat only neutropenia, VLBW infants, pre-procedure), fluconazole 2 weeks, amphotericin for resistant species, fungus ball drainage, IDSA vs AUA pre-procedure

Not in the paper answer:

- AUA BPS 2019 advises antifungal prophylaxis for high-pressure endoscopy (TURBT, TURP, therapeutic ureteroscopy) and open entry into the tract.

13 · Perioperative antimicrobial prophylaxis

One well-timed dose suffices; more only adds resistance.

Principle (AUA BPS 2019)	Rule
Indication	Any procedure breaching a tissue barrier, weighing host defence and consequences
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min
Duration	**Single dose**; ≤ **24 h** only for PCNL, prostheses, special cases
Agent	Local flora per antibiogram
Known infection	Sterilise urine before elective surgery (this is treatment); infected urine causes bacteraemia in ~half

- Host risk factors include age, smoking, steroids, immunodeficiency, hardware and infected stones.

Wound class	Examples	Antimicrobial prophylaxis (AP)
I clean	Scrotal/inguinal, retroperitoneal lymph node dissection (RPLND), nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; anaerobic cover
IV dirty	Abscess, open trauma	Treatment, tailor to cultures

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—
Cystoscopy with biopsy/fulguration	Host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
TURP / TURBT, laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin, with or without aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single
PCNL	All	1st/2nd-gen cephalosporin; aminoglycoside and metronidazole	≤24 h
Shock-wave lithotripsy (SWL)	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract	All	Cefazolin	Single

Procedure	AP?	Agent	Duration
Colon segment	All	Cefazolin and metronidazole / cefoxitin / ertapenem, plus oral neomycin with erythromycin or metronidazole	Single
Artificial urinary sphincter (AUS), penile prosthesis, sacral neuromodulation	All	Aminoglycoside and cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- **Transrectal ultrasound (TRUS) biopsy** causes sepsis in 0.3–3.1%, mainly from quinolone-resistant E. coli. Use augmented or **targeted** prophylaxis; the AUA advises alternatives if local resistance is >20%.
- **UDS (SUFU 2017)**: give AP for neurogenic lower urinary tract dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter or recent joint implant.
- **Catheter or stent removal**: give a single dose only if risk factors are present.
- **Endocarditis** (AHA): no AP solely to prevent it; with a high-risk valve and colonised urine, sterilise the urine first.
- **Orthopaedic hardware**: only a joint replaced <2 years ago or a high-risk host needs AP for high-bacteraemia procedures.

> **KEY POINT**
> Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit; otherwise give one well-timed, flora-matched dose. Anything longer is treatment.

14 · Landmark trials

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen first-line after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment increased recurrence and resistant E. coli	Do not treat ASB; E. faecalis may protect
ALTAR (methenamine)	Methenamine vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

1 · Prostatitis: definitions and classification

Only categories I and II are infections; most men have category III.

- The Drach (1978) scheme, built on Meares–Stamey cultures, was replaced by the MCQ National Institutes of Health (**NIH, 1999**) classification.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute prostatic urinary tract infection (UTI), often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI, **same organism**; well between episodes
IIIA	Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory	Leucocytes in expressed prostatic secretion (EPS), post-massage urine (VB3) or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory	Incidental on biopsy or semen analysis

- I and II are **5–10%**; III is **90–95%**. **IIIA vs IIIB has no clinical significance**, and the prostate may not be the pain source.
- **Category III (NIH CPCRN)** needs pelvic pain for **≥3 of the previous 6 months**, excluding uropathogens, urogenital cancer, prior radiation, stricture and neurological bladder disease.
- Histological inflammation is near-universal (**>70%** at autopsy), so it cannot define disease; **Nickel grading** records its location, extent and grade.
- **Granulomatous prostatitis** classically follows **intravesical bacillus Calmette–Guérin (BCG)**, giving a **transient benign prostate-specific antigen (PSA) rise**.

2 · Category I: acute bacterial prostatitis

- Spread is mainly **ascending urethral**. Risks include catheters, instrumentation or biopsy, obstruction and insertive anal intercourse.
- **E. coli causes 65–80%**; Pseudomonas is commoner after instrumentation. After biopsy, organisms are often fluoroquinolone (FQ)-resistant or extended-spectrum β-lactamase (ESBL) producers.

> **KEY POINT**
> **Why young healthy men get ABP:** the organism, not the host — virulent uropathogenic E. coli that forms **biofilm** more often than cystitis strains; biofilm blocks antibiotics, so prostatic infection needs longer, well-penetrating therapy.

- Symptoms are frequency, dysuria, perineal pain, fever and rigors, with possible retention or sepsis. In febrile male UTI, **>90%** show raised PSA or prostate volume.

Evaluation

- First ask whether it was **spontaneous or after manipulation or biopsy**, as this decides empirical therapy.
- The prostate is tender and boggy; examine **gently and never massage**. Send midstream and **blood cultures**, and measure residual urine by **ultrasound (US)**.
- **No PSA** acutely; if raised, follow it to normal (up to **6 months**). Use transrectal ultrasound (TRUS) or CT only for suspected abscess, and **never biopsy**.

Treatment

- **Outpatient care only if** the man is not systemically ill, tolerates oral drugs and is not in retention.
- Systemic illness (**EAU**): amoxicillin or a 2nd-generation cephalosporin with an aminoglycoside, or a 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%** and fully oral outpatient care, or β-lactam anaphylaxis.
- **Febrile prostatitis after biopsy** (AUA White Paper): **no FQ or trimethoprim–sulfamethoxazole (TMP-SMX)**; use a **carbapenem**, amikacin or 2nd/3rd-generation cephalosporin.
- **Duration**: traditionally 4 weeks, but **2 weeks** of ciprofloxacin is equivalent; TB prostatitis needs ≥**6 months**. NSAIDs and α-blockers are useful adjuncts.
- **Retention**: brief urethral catheterisation is acceptable, but use a **suprapubic catheter for long-term** drainage.

Prostatic abscess

- Suspect it with high fever, immunosuppression or **no response after 48 h**. It is hypoechoic on TRUS, and **CT** shows spread.

> **ALGORITHM**
>
> 1. If the abscess is <1–2 cm, then treat with antibiotics.
> 2. If it progresses and lies peripherally, then perform US-guided percutaneous drainage (recurrence 15–33%).
> 3. If it is large or drainage fails, then perform **transurethral unroofing**, using intraoperative US to locate it.
> 4. If it extends outside the prostate, then drain it openly (rare).

3 · Category II: chronic bacterial prostatitis

Recurrent same-organism UTI in men needs a long penetrating course.

- The prostate harbours the same organism and the man is **well between episodes**, whereas CP/CPPS pain is continuous.
- Organisms are Gram-negatives and E. faecalis; biofilm explains the poor response. The role of **Chlamydia** is debated.

Localisation test	Specimens	Interpretation
Meares–Stamey 4-glass (previous)	VB1 urethra; VB2 bladder (midstream); EPS; VB3 post-massage	Prostatic if EPS/VB3 counts exceed VB1/VB2
2-glass pre/post-massage (current)	Midstream pre-massage and first 10 mL post-massage	Predicts 4-glass in **>95%**

- There is no agreed cut-off, though some use a **10-fold rise**; **do not culture semen**.
- A residual volume **>180 mL** raises infection risk. Men **<45** with a first UTI have normal imaging, but check for **stricture**.

> **KEY POINT**
> **Why fluoroquinolones?** Prostatic entry is passive (lipid solubility, ionisation, protein binding, size). FQs are lipid-soluble **zwitterions** that concentrate in alkaline CBP fluid and cover Gram-negatives, Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% with FQ)
C. trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg twice daily 10 days; azithromycin is better than ciprofloxacin
T. vaginalis	Metronidazole 500 mg three times daily 14 days

- A macrolide can be added because it **penetrates biofilm**. Refractory cases need suppression or TURP.

4 · CP/CPPS — aetiology

An insult (infection, stress, trauma) in a susceptible man starts **neurogenic inflammation**, which other factors maintain, so single treatments fail.

Domain	Key evidence
Infection	Bacterial counts are equal in patients and controls; in mice, pain persists after clearance
Inflammation	Biopsy inflammation does not correlate with symptoms; symptoms track **cytokines**
Neurological	**Central sensitisation**
Pelvic floor	Tenderness and **failure to relax**; tender in **81%**
Psychosocial	**Catastrophising and helplessness predict** pain
Endocrine	Blunted adrenocorticotropic hormone (ACTH) response

5 · CP/CPPS — clinical features and evaluation

P1 P4 Chronic Prostatitis: diagnosis, classification and management; Chronic non bacterial prostatitis — pain sites, NIH-CPSI domains, DRE/trigger points, 2-glass test, cytology, cystoscopy, UPOINT-directed multimodal therapy

Not in the paper answer:

- The **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)** † scores 0–43 (pain 0–21, urinary 0–10, quality of life 0–12), and a **≥6 points** fall is meaningful.
- **PSA is not indicated**, and a raised value must never be blamed on CP/CPPS.
- In men with bladder pain, about 42% meet interstitial cystitis criteria see Ch 28, and Hunner lesions respond to fulguration.

UPOINT phenotyping

Monotherapy fails because each man has a different mix of domains. UPOINT maps them for simultaneous treatment, and the domain count tracks severity.

Domain	Identify by	Therapy
Urinary	Lower urinary tract symptoms (LUTS), PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, stress reduction
Organ-specific	Prostate tenderness; bladder pain	†Quercetin/pollen; bladder therapy; 5α-reductase inhibitor (5-ARI) in BPH
Infection	Positive VB3 with negative VB2	Antibiotics
Neurologic/systemic	Pain beyond pelvis, irritable bowel syndrome (IBS), fibromyalgia	Tricyclics, pregabalin, acupuncture
Tenderness	Pelvic floor trigger points	Physiotherapy, myofascial release

6 · CP/CPPS — treatment

KEY POINT

Governing fact: no monotherapy has a positive definitive trial, because each tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team.

Modality	Evidence / role
Antibiotics	Quinolone trials negative. EAU: quinolone or tetracycline ≥**6 weeks** only if **treatment-naive and <1 year**. **No repeated courses without a positive culture**.
α-blockers	Trials negative; use in newly diagnosed men with voiding symptoms. Best combination: **antibiotic and α-blocker**.
Anti-inflammatories	Celecoxib helps pain only; steroids negative.
5-ARI	Dutasteride (REDUCE): ≥6-point NIH-CPSI fall in 49% vs 37%; for older men with BPH.
Neuropathic	Pregabalin missed its endpoint; amitriptyline needs ≥50 mg.
Conservative	Diet, **aerobic exercise** and cognitive therapy.
Acupuncture	Strongest conservative evidence: sham-controlled RCT, 61.5% vs 38.3% responders at 32 weeks.
Pelvic floor physiotherapy	Myofascial trigger-point release beats global massage (57% vs 21%); refer if the pelvic floor is tender.
Neuromodulation	Tibial or sacral nerve stimulation if frequency/urgency.
Surgery	No TURP or prostatectomy. **Bladder neck incision** only for proven obstruction.

7 · Acute orchitis

- **Bacterial** orchitis (age >35) is usually Gram-negative. **Mumps** is the commonest viral cause, following parotitis, with **atrophy in about half** but rare infertility. Autoimmune and vasculitic forms are treated with steroids.

- Ask about **pain duration**, because torsion is the key differential; get urine culture, STI tests and **scrotal US**. **Follow-up is mandatory** to avoid missing torsion or tumour.

8 · Acute epididymitis

Age and sexual history predict the organism.

- Epididymal pain and swelling for **<6 weeks**; infection is **ascending**.
- **In sexually active men <35, C. trachomatis and N. gonorrhoeae dominate**; **in men >35 and children, enteric organisms dominate**. Because >40% of men under 35 grow enterics, **screen all sexually active men for STI**.
- Send culture, first-void urine and STI NAAT; **Doppler US** excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** (intramuscular) single dose (1 g if ≥150 kg) and doxycycline 100 mg twice daily **10 days**
STI and enteric (insertive anal sex)	Ceftriaxone 500 mg IM and levofloxacin 500 mg once daily 10 days
Enteric only	Levofloxacin 500 mg once daily 10 days

- Give rest, elevation and NSAIDs, and treat partners. With **no improvement at 72 h**, repeat US for abscess or infarction.

9 · Chronic orchitis and epididymitis

- **Granulomatous orchitis** has infective causes (TB, brucellosis, syphilis) and an **idiopathic** form usually found after orchidectomy for suspected tumour. **Malakoplakia** shows **Michaelis–Gutmann bodies**.
- **Chronic epididymitis** (≥**3 months**) has a **Nickel classification**: inflammatory (infective, TB, **amiodarone**, idiopathic), obstructive (congenital, post-infective, post-vasectomy) and epididymalgia (normal epididymis).
- **TB epididymitis** is the commonest genital site, presenting as painless swelling with a **sinus**. Give four drugs for ≥**6 months**, and operate **only on anti-TB drugs**. **A mass not shrinking after 3 weeks needs inguinal exploration**.

10 · Chronic scrotal content pain (CSCP)

A cord block predicts who benefits from denervation surgery.

CSCP is intrascrotal pain for ≥**3 months** that interferes with life; there are no evidence-based guidelines.

- Post-procedural causes are **inguinal hernia repair** (commonest) and vasectomy (~**1 in 1000**, the consent figure). Others include varicocele, intermittent torsion and referred pain from a ureteric stone.
- **Wallerian degeneration** affects the **"trifecta nerve complex"**: cremaster, perivasal sheath and posterior lipomatous tissue. Pelvic floor dysfunction is common.
- **First decide** whether the pain is isolated or part of CP/CPPS, and get a **duplex US**.
- A **spermatic cord block** uses **20 mL 0.25% bupivacaine** at the **pubic tubercle**. **>50% relief** predicts success of microsurgical denervation of the spermatic cord (MDSC).

- Medical options are NSAIDs and **neuropathic agents**, without chronic opioids. Botulinum toxin equalled placebo in an RCT.

ALGORITHM

1. If there is an identifiable lesion, then do directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy, **microsurgical varicocelectomy**).
2. If the pain followed vasectomy, then offer **vasectomy reversal** (82–93%; wait ≥3 months).
3. If there is no lesion and the block is positive, then perform **MDSC**.
4. If the epididymis is focally affected, then perform epididymectomy.
5. Last resort: **inguinal orchidectomy**.

- **MDSC** is subinguinal and microscopic: divide the cord nerves, **preserve arteries and lymphatics** and ligate the internal spermatic veins. It cures 50–70%; consent for **testicular atrophy**. **Targeted trifecta MDSC** is equally effective and quicker.
- **Epididymectomy** succeeds in 50–75%. **Orchidectomy** is least reliable, and **inguinal beats scrotal** (75% vs 55%).

11 · Landmark trials

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point fall 49% vs 37%	5-ARI for older men with BPH
MAPP myofascial physiotherapy	Feasibility vs global massage	57% vs 21% responders	Physiotherapy for tender pelvic floor
Acupuncture vs sham	Large RCT	61.5% vs 38.3% at 32 weeks	Evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label series overestimate benefit

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

1 · Definition and nomenclature

P2 P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — AUA, ESSIC (types 1–3, A–C), NIDDK, ICD-11 definitions; IC vs BPS split; phenotypes

Not in the paper answer:

- **IC/BPS** ranges from inflammatory disease with **Hunner lesions (HLs)** to pain with a normal bladder and many non-urological complaints.
- The **AUA** defines it as bladder-related pain, pressure or discomfort with lower urinary tract symptoms (LUTS) for **>6 weeks**, with no infection or other cause. † **ESSIC (2008)** requires **>6 months** and ≥1 urinary symptom; **type 3C** is classic HL disease.
- The **NIH/NIDDK (1987)** research criteria missed **~60%** when used clinically. The current **trend** is to call HL disease "IC" and non-HL disease "BPS".

2 · Epidemiology (headline)

- The RICE study found that **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms.
- Female:male is **5:1** in clinics but **2–3:1** in populations, so men are under-diagnosed. **Ask men labelled CP/CPPS about bladder pain** see Ch 27.
- **HL prevalence** is **~20%**, rising with age.
- The course is chronic but **not progressive**, with flares, commonly after sexual activity.

> **KEY POINT**
> Uncontrolled series "work" but randomised controlled trials (RCTs) fail because of regression to the mean, placebo and extra attention. Expect improvement in the first 3–4 weeks of any treatment.

3 · Aetiology and pathophysiology

The cause is multifactorial, and each theory points to a treatment.

Theory	Mechanism / evidence	Treatment link
Glycosaminoglycan (GAG) layer / "leaky" urothelium	Urinary solutes such as potassium reach suburothelial nerves	Pentosan polysulfate (PPS), heparin, hyaluronic acid
Mast cells	Histamine release; marked in HL disease, but counts have **no diagnostic role**	Hydroxyzine, cimetidine, amitriptyline
Neurogenic	Neurogenic inflammation, pelvic cross-sensitisation and central sensitisation	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Excess of lupus, Sjögren syndrome and inflammatory bowel disease (IBD); lymphoplasmacytic infiltrate in HL	Ciclosporin A (CyA) for HL disease
Infection	No consistent organism; 18-week antibiotic RCT showed no benefit	No antibiotics without proven infection

KEY POINT

Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · Pathology: Hunner lesion vs non-Hunner phenotype

- There is **no pathognomonic histology**. Biopsy only HLs or suspicious lesions, to exclude carcinoma in situ (CIS), tuberculous and ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythema, radiating vessels, central scar; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation, dense **lymphoplasmacytic** infiltrate, increased mast cells	Near-normal; sparse inflammation
Pain source	Bladder — end-organ disease	Often regional/widespread
Treatment	Fulguration / triamcinolone; CyA; dimethyl sulfoxide (DMSO) works better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · Associated conditions

- **75–80%** have pain beyond the pelvis. Among overlapping pain conditions, **irritable bowel syndrome (IBS)** is commonest (>30%), followed by **fibromyalgia**, chronic fatigue and migraine.
- **High-tone pelvic floor dysfunction (HTPFD)** occurs in **~80%**. **Autoimmune** disease shows the largest relative excess; in lupus, think of **lupus cystitis**.

KEY POINT

Commonest is not the same as most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Widespread pain predicts failure of bladder-directed therapy.

6 · Clinical features

Pain linked to filling separates IC/BPS from overactive bladder.

- Suprapubic pain is **worse with filling and relieved by voiding**, driving day and night frequency.

Feature	Overactive bladder (OAB)	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting, pain-driven	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	Urgency incontinence	Not a feature	—
Voiding	—	Frequency, low volumes	Obstructive

7 · Diagnosis — exclusion plus localisation

Basic evaluation (AUA 2022)

- Ask about the pain–filling link, triggers, pelvic irradiation, **cyclophosphamide and ketamine**.

- Examine for vaginal and urethral pathology, do a **Q-tip test** and DRE, and **palpate the pelvic floor**.
- Test post-void residual (PVR), urinalysis and culture, with cytology if at risk. Score pain and urinary symptoms separately, for example with the O'Leary–Sant indices (trial endpoint ≥30% fall).

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynae, colon	Cystoscopy with or without biopsy; prostate-specific antigen (PSA); pelvic exam
Infection	Bacterial, atypical, **TB**, herpes; prostatitis	Culture; **sterile pyuria needs TB culture**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine causes upper-tract change
Outflow	Bladder neck obstruction, benign prostatic obstruction, retention	PVR, uroflow, (video)urodynamics
Structural/other	Stones, diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, exam, laparoscopy, colonoscopy
Neuromuscular	Pudendal entrapment; pelvic-floor myalgia	Nerve block; pelvic-floor exam

- **Ketamine cystitis** causes HL-like inflammation, fibrosis and **hydronephrosis (~30%)**. Suspect it in the **young (<30) with visible inflammation**, and treat by **cessation**.
- **HTPFD** gives pelvic pressure, obstructive voiding and dyspareunia, with tender **myofascial trigger points**. Missed HTPFD causes poor response, so refer to a pelvic-floor physiotherapist.
- **Urodynamics** are not standard. Compliance is **normal**, and detrusor overactivity **does not exclude IC/BPS**.

Cystoscopy and bladder tests

- **AUA 2022** advises office cystoscopy when **HL is suspected, especially over 50**. Flexible cystoscopy shows most HLs **without hydrodistension**; biopsy **any trigonal lesion**.
- Relief after an **anaesthetic challenge** shows bladder-centred pain and selects intravesical therapy.
- The **potassium sensitivity test** (**0.4 M KCl** vs water) is **not recommended**, because it is painful and non-specific.
- **Hydrodistension** under anaesthesia fills at **80 cm H_2O for 1–3 min** to show **glomerulations**, which are **non-specific**. Biopsy only **after** distension and only suspicious lesions.

8 · Phenotype-directed management

P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — four-phenotype table: HL, anaesthetic-responsive, pelvic floor, widespread; matched treatment

KEY POINT

MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain**; amitriptyline helped **widespread pain**. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · Treatment — AUA 2022 principles

The aim is coping and quality of life, not cure.

- There are **no tiers**, only grouped options: patients may move to any category **except major surgery** without failing "lower" ones. Stop ineffective treatment and seek other pain generators.
- The **Standard** is manual (myofascial) physiotherapy for pelvic-floor tenderness. Fulguration and/or triamcinolone for HLs is a **Recommendation**. Most drugs are **Options**. Chronic opioids are not routinely recommended.

ALGORITHM

1. If symptoms last ≥6 weeks, take a history, examine (pelvic floor, Q-tip, DRE), and check PVR, urinalysis and culture, with cytology if indicated.
2. Exclude confusable disease.
3. If HL is suspected (especially >50), then do cystoscopy. If HL is found, then fulgurate or inject triamcinolone; if refractory, give CyA.
4. If there is no HL, then match the phenotype: pelvic floor, physiotherapy; lidocaine responder, intravesical therapy; widespread pain, systemic therapy and CBT.
5. Combine behavioural, oral and intravesical treatment as needed, then BTX-A or sacral neuromodulation (SNM).
6. Only if bladder-centred and all else fails (rare), consider diversion with or without cystectomy.

10 · Treatment options (AUA 2022)

P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — options table: behavioural, physical (avoid Kegel), oral, intravesical, procedures, refractory HL, last resort

Not in the paper answer:

- SNM is favoured over BTX-A for HTPFD or obstructive voiding.

10.1 Behavioural and pelvic-floor therapy

- Common **diet triggers** are caffeine, alcohol, fizzy drinks, spicy food and citrus; an elimination diet identifies the patient's own triggers.
- Education and behavioural modification alone improved 45% in the NIH amitriptyline RCT.
- In the myofascial physiotherapy RCT, 10 sessions beat global massage, **59% vs 26%**.

10.2 Oral therapy

- **Amitriptyline**: start **10 mg at night, adding 10 mg weekly to ~50 mg**. In the NIH RCT it gave 55% vs 45% (not significant), but **77% vs 53%** if ≥50 mg was tolerated. **Contraindications** are long QT, conduction block, recent myocardial infarction and heart failure.

- **Hydroxyzine** (H1 blocker) 25–50 mg at night had a negative RCT; **cimetidine** 400 mg twice daily has one positive RCT.
- **PPS 100 mg three times daily** is the only FDA-approved oral IC drug; only **3–6%** reaches urine and it acts in **3–6 months**. Its phase IV RCT was negative.
- **PPS maculopathy** is cumulative-dose related. **FDA (2020)**: retinal examination for all **within 6 months**, then periodically.
- **Ciclosporin A** is for HL refractory to fulguration or triamcinolone, at **1–3 mg/kg/day in two doses**. Its main toxicities are **nephrotoxicity and hypertension**, so monitor blood pressure, renal function and levels.
- **Antibiotics**: the AUA advises **none long-term**, and only one empirical course if never treated.

10.3 Intravesical therapy

Intravesical therapy is for **bladder-centred** pain; a poor lidocaine response predicts failure of every instillation. It is an AUA Option.

Agent	Regimen	Key points
DMSO 50%	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved. Better in HL. Causes garlic breath and an initial flare; increases absorption of co-instilled lidocaine.
Heparin	10,000–40,000 U, 30–60 min, self-instilled	GAG mimic; benefit takes months.
Alkalinised lidocaine	200 mg with 8.4% bicarbonate daily × 5	Immediate effect, so useful for flares; tachyphylaxis.
Hyaluronic acid, chondroitin, PPS	Weekly then monthly	Weak/negative trials

- Cocktails are no better than DMSO alone.

10.4 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone, with biopsy if never biopsied. **75–90%** improve, but repeat treatment is usually needed within 1–2 years.

ALGORITHM

1. Under general anaesthesia, map the HLs before distension causes bleeding.
2. Distend mildly and biopsy suspicious or trigonal lesions.
3. Fulgurate at **low volume** (Bugbee electrode), working from the **outer margin inwards**.
4. Inject triamcinolone 40 mg/mL (or 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally.
5. Fulgurate bleeders, empty the bladder and instil lidocaine–bupivacaine.

KEY POINT

Fulgurate from the margin inward: cautery erythema and oedema spread peripherally, so the true edge is lost if you start centrally.

10.5 Hydrodistension

- AUA Option: **short (1–3 min), low-pressure (60–80 cm H_2O)** distension, with benefit fading by 6 months.
- An anaesthetic capacity **<400 mL** marks a bladder-centred phenotype, often HL.

10.6 OnabotulinumtoxinA and neuromodulation

- **BTX-A** is an Option after failed conservative, oral and intravesical therapy: **100 U** into the bladder including the **trigone**, lasting about 5–9 months. The patient **must accept clean intermittent catheterisation (CIC)**.
- **SNM (S3)** is an Option and works best with **HTPFD or levator tenderness**, but IC/BPS has high explantation rates.

	BTX-A	SNM
Favoured when	Bladder-centred pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD, CIC impossible
Main risk	Retention needing CIC	Adverse events, explantation

10.7 Major surgery — last resort

- Surgery is only for **rare (<10%), selected, bladder-centred** patients failing all else, and pain may persist after cystectomy. If pain persists after a **differential spinal block** to T10, surgery will fail.

Predicts success	Predicts failure
HLs (refractory)	No HLs
Small fibrotic bladder; anaesthetic capacity <200–400 mL	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no response
Painful filling, pain on scope touch	Urethral/neuropathic pain; HTPFD; widespread pain

- Cystectomy with a conduit leaves about three-quarters pain-free. Prefer the **simplest diversion**, as a retained bladder risks pain and pyocystis.

11 · What NOT to do (AUA)

Do not offer	Why
Long-term oral antibiotics	RCT negative; adverse events 80% vs 40%
Intravesical bacillus Calmette–Guérin (BCG)	Ineffective; DMSO superior; † BCG sepsis
High-pressure, long hydrodistension†	No benefit; rupture, sepsis †
Long-term oral glucocorticoids	No benefit; toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening	Worsens HTPFD pain
Major surgery first / non-bladder-centred pain	Irreversible; pain persists

> **KEY POINT**
> Nothing is worse than irreversible surgery followed by continued pain — every predictor of surgical success marks bladder-generated pain.

12 · Landmark trials

Trial	Design	Result	Take-home
Myofascial PT RCT (FitzGerald 2012)	10 sessions myofascial vs global massage	59% vs 26%	Guideline Standard
NIH amitriptyline RCT (Foster 2010)	EBMP with or without amitriptyline	55% vs 45% (not significant); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy works
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	Efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs by pain distribution	Hydroxyzine/PPS localised; amitriptyline widespread	Treat by phenotype

29 Sexually Transmitted Diseases

1 · Screening, prevention and partner notification

One sexually transmitted disease (STD) often hides another.

- The nucleic acid amplification test (**NAAT**) is the standard screen; men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Partner notification** covers the previous **60 days** for urethritis, chlamydia and Mycoplasma genitalium; all partners for trichomoniasis; and the previous month for scabies or lice. Abstain for 7 days and until partners are treated. Test everyone at risk for HIV and syphilis.

2 · Urethritis

- **Gonococcal** urethritis makes up 5–20%. In non-gonococcal urethritis (**NGU**), C. trachomatis is commoner than M. genitalium, which is the leading cause of **persistent or recurrent** urethritis. Up to half have no pathogen.
- **Diagnosis** needs any one: discharge; Gram stain with **≥2 white blood cells (WBC) per oil-immersion field**; positive leucocyte esterase on first-void urine; or ≥10 WBC/hpf in first-void urine.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea; a negative Gram stain does not exclude it.
- **First-void urine NAAT** is preferred; **culture** is for treatment failure and children.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (CDC 2021)	**Ceftriaxone 500 mg IM once** (1 g if ≥150 kg)	Gentamicin 240 mg IM and azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g and azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal** infection; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg bd 7 days**	Azithromycin 1 g once	No test of cure; retest at 3 months
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g, then 500 mg daily × 3, if macrolide-sensitive; or **moxifloxacin** 400 mg × 7 days if resistant or untested	Single-dose azithromycin is inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- Higher-dose **ceftriaxone monotherapy** reflects resistance; cephalosporins are safe unless a recent IgE-mediated penicillin reaction occurred. Treatment failure needs culture and susceptibility.
- **M. genitalium** has no cell wall, so Gram stain misses it; NAAT with a macrolide-resistance assay picks the second drug.
- There is no chlamydia test of cure, because nucleic acid persists ~3 weeks.

ALGORITHM

1. Persistent NGU: check adherence and look for an untreated partner.
2. Send NAAT for M. genitalium, with or without T. vaginalis.
3. If negative, then think of chronic pelvic pain syndrome.

KEY POINT

Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease or infertility. Treating the man protects his partner.

3 · Epididymitis by age

Acute means <6 weeks; chronic (≥6 weeks) is mostly tuberculosis (see Ch 27). Always exclude **torsion**. Apply urethritis criteria and NAAT; add urine culture if enterics are likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	C. trachomatis, N. gonorrhoeae	Ceftriaxone 500 mg IM and doxycycline 100 mg bd
Insertive anal sex	Above, plus enterics	Ceftriaxone and levofloxacin 500 mg daily
>35 years, obstruction or instrumentation	E. coli	Levofloxacin alone

4 · Genital ulcer disease

Herpes is commonest, then syphilis. Appearance misleads, so test **every ulcer** with syphilis serology, HSV NAAT, an H. ducreyi test if endemic, and HIV. Biopsy non-healing ulcers, because carcinoma mimics them.

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (T. pallidum)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 MU IM once
Herpes (HSV-2 more than HSV-1)	**Painful** multiple vesicles that become shallow ulcers	Tender, bilateral	NAAT of unroofed base	Aciclovir and related drugs
Chancroid (H. ducreyi)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days
Lymphogranuloma venereum, LGV (C. trachomatis L1–L3)	Small, painless, transient	Painful, matted, fistulae; groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (Klebsiella granulomatis)	**Painless, beefy red**, bleeds easily	**Absent** (pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- HSV lies latent in the **sacral ganglia** and reactivates with trauma, surgery or immunosuppression. Most transmission is from asymptomatic shedding. HSV-2 recurs more, so type it.
- Complications are aseptic meningitis and **urinary retention**; HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg tds	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode. **Episodic** therapy must start within 1 day of lesions or in the prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs a painful ulcer, a typical picture, and syphilis and HSV excluded. Aspirate or drain fluctuant buboes.
- **Donovanosis**: if antibiotics fail, biopsy, because **penile squamous cell carcinoma (SCC) mimics it**.

P1 Lymphogranuloma venereum — L1–L3; proctocolitis in MSM now commonest; groove sign; elephantiasis, rectal stricture; inguinal disease commoner in men; doxycycline 21 days

5 · Syphilis staging and serology

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre and nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	As primary
Early latent (<1 year)	Positive serology only	As primary
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if cerebrospinal fluid (CSF) normal
Neurosyphilis (any stage)	Central nervous system, eye, ear	**Aqueous penicillin G 18–24 MU/day IV 10–14 days**

- T. pallidum cannot be cultured, so **two serological tests** are used. The **nontreponemal test (NTT)** (VDRL, RPR) can be falsely positive; its titre falls with treatment, so it **monitors treatment**. The **treponemal test (TT)** (FTA-ABS, TP-PA, EIA) usually stays positive for life.
- A significant change is **fourfold (two dilutions)**. Use the same NTT each time.

ALGORITHM

1. **Reverse algorithm**
2. Start with a TT. If positive, then do an NTT.
3. If the NTT is negative, then do a second, different TT.
4. If that is positive and there is no past treatment, then treat as late latent syphilis.

- The **Jarisch–Herxheimer** reaction (fever and myalgia within 24 h) is treponemal lysis, not allergy; give NSAIDs.
- In primary or secondary syphilis, test for HIV and offer pre-exposure prophylaxis (PrEP) if negative.
- **Follow-up** serology is at 6 and 12 months (and 24 for latent disease). A fourfold rise means failure or reinfection, so retreat weekly × 3 and consider CSF.

6 · HPV and anogenital warts

- Human papillomavirus (HPV) **6 and 11** cause ~90% of warts. HPV **16 and 18** cause >70% of cervical cancer, and also penile and anal cancer.
- **Bowenoid papulosis** (HPV-16) is a high-grade squamous intraepithelial lesion. **Giant condyloma** (Buschke–Löwenstein) is a locally invasive verrucous carcinoma.
- Diagnosis is clinical. **Biopsy if** the diagnosis is uncertain, there is no response or worsening, the patient is immunocompromised, or the lesion is pigmented, indurated, fixed or ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights/week for up to 16 weeks; wash off after 6–10 h. Weakens condoms
Podofilox 0.5% (patient)	bd for 3 days, then 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	tds, not washed off. Weakens condoms. Not in HIV, herpes or pregnancy
Cryotherapy, trichloroacetic acid (TCA) 80–90% (provider)	Cryotherapy every 1–2 weeks; TCA weekly
Excision, curettage, CO_2 laser	For large warts; use smoke extraction

Change treatment if there is no response by 3 months; observation is acceptable.

- The **vaccine** is **nonavalent** (6, 11, 16, 18, 31, 33, 45, 52, 58). Give it at **11–12 years** (from 9), with catch-up to **26**, in both sexes. Starting <15 needs **2 doses**; ≥15 or immunocompromised needs **3 doses**.

7 · Molluscum and ectoparasites

- **Molluscum** contagiosum forms **umbilicated** papules that regress in 6–12 months; use cryotherapy, curettage or imiquimod. Extensive lesions suggest HIV.
- **Pubic lice**: use a **permethrin 1%** rinse for 10 min, or malathion 0.5% or ivermectin 250 µg/kg if resistant.
- **Scabies** causes nocturnal itch and burrows. Apply **permethrin 5%** to all skin, or give ivermectin 200 µg/kg repeated at 2 weeks. **Crusted** scabies (immunocompromised) needs topical treatment and repeated ivermectin, never lindane.
- Treat contacts from the past month and wash bedding.

8 · HIV for urologists

P1 Describe clinical implication of AIDS on the Genito-Urinary Tract? — staging (AIDS = CD4 <200), diagnosis, GU manifestations by site (UTI, prostate, testis, voiding, ED, HIVAN, indinavir stones, cancers), PEP 72 h/28 days, needlestick 0.3%, circumcision 50–60%

Not in the paper answer:

- Start post-exposure prophylaxis (PEP) within **72 h**, and only for mucosal, broken-skin or percutaneous contact with blood or genital fluid. Needlestick risk is **0.3%**.
- Circumcision cuts heterosexual acquisition by **50–60%** but does not protect partners. **After sexual assault** (men), give ceftriaxone 500 mg once, doxycycline 7 days, hepatitis B and HPV vaccines, and PEP within 72 h.
- PrEP is daily emtricitabine with tenofovir disoproxil fumarate, or tenofovir alafenamide in men only.

30 Cutaneous Diseases of the External Genitalia

1 · Approach to genital dermatoses

Genital rashes look alike, so biopsy early to avoid missing cancer.

- Genital skin has few reaction patterns, so mimics abound and the **biopsy threshold is low**. Ask about over-the-counter creams, and do a **full skin survey**, because extragenital lesions often give the diagnosis.
- Identify the **primary lesion**. A papule is solid and **≤0.5 cm**; larger solid lesions are nodules or plaques, and fluid-filled ones are vesicles (≤0.5 cm) or bullae. An **erosion** loses epidermis only and does not scar; an **ulcer** loses dermis too and scars.
- Bedside tests: KOH prep for hyphae or yeast, mineral oil prep for scabies, Tzanck smear for herpes, and Wood's lamp for erythrasma (coral-red) and vitiligo.

Biopsy: a 2–3 mm punch under local anaesthetic is often diagnostic; large or meatal lesions go to theatre.

2 · Principles of therapy

- Use **topical treatment first**; urologists overprescribe systemic antibiotics.
- Give **systemic antifungals** only for extensive dermatophytosis, Majocchi granuloma, severe immunocompromise or topical failure, and avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- Give systemic prednisone for ≤3 weeks as a single morning dose.
- **Topical steroids** cause atrophy in thin genital skin, and the **prepuce acts as an occlusive dressing**. The rule is low potency and a short course; lichen sclerosus (LS) is the exception.
- Narrow-band UVB or the 308-nm excimer treats psoriasis and vitiligo. **PUVA causes genital squamous cell carcinoma (SCC)**, so it is avoided.

3 · Eczematous dermatoses

Exclude Candida, herpes, impetigo and drug reaction first.

Atopic dermatitis (AD)

- AD is relapsing, flexural and intensely pruritic; **90%** present before 5 years. Use emollients, a short topical steroid, calcineurin inhibitors and a sedating antihistamine; severe disease needs **dupilumab**.

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct cytotoxicity; dose-driven	Type IV hypersensitivity
Share	~**80%**	Minority; genital ACD is rare
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest), neomycin, fragrances, latex, topical steroids
Picture	Burning, limited to the exposed zone	Demarcated, pruritic, weeping
Confirm	Exposure history	**Patch testing**

Feature	Irritant (ICD)	Allergic (ACD)
Treat	Avoidance, barrier	Remove allergen. If severe, a **3-week** prednisone taper (a short course rebounds)

Seborrhoeic dermatitis (SD)

- SD gives greasy-scaled plaques in sebaceous areas. It affects up to **83%** of patients with AIDS, so **extensive SD should prompt an HIV test**. A topical **antifungal** (ketoconazole) is the backbone.
- Lichen simplex chronicus (lichenification, in the LS differential) needs the itch–scratch cycle broken with a short potent steroid.

4 · Papulosquamous disorders

Tinea, erythrasma, secondary syphilis and neoplasia (Bowen disease, extramammary Paget disease (EPD), mycosis fungoides) mimic psoriasis, so "psoriasis" failing treatment needs KOH, syphilis serology and biopsy.

Psoriasis

- Psoriasis affects ~2% and forms sharp plaques with silvery scale; check nails, scalp, **umbilicus** and natal cleft. **Inverse psoriasis** loses its scale in folds, so use KOH to exclude tinea or Candida.

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%) continuously for **≤2 weeks**; triamcinolone 0.1% for thick plaque
Steroid-sparing	Calcipotriene, calcineurin inhibitors
Localised	308-nm excimer laser
Contraindicated	**PUVA** (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, apremilast, anti-TNF, ustekinumab, **ixekizumab (anti-IL-17), which has the best genital data**

Reactive arthritis (formerly Reiter)

- Reactive arthritis follows urethritis or enteric infection in **HLA-B27**-positive people. **Balanitis circinata** looks like psoriasis, so diagnose from **context**; it is self-limiting and needs only a low-potency steroid.

Lichen planus (LP)

- LP forms **polygonal, violaceous, flat-topped papules**; glans papules are white and reticular. Papular LP usually resolves within a year.
- **Erosive mucosal LP persists, scars and carries an SCC risk**, so review regularly and biopsy ulcers. Use an **ultrapotent** steroid (clobetasol 0.05%).

5 · Lichen sclerosus (LS) / BXO

P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — scarring disease; phimosis, then meatus, then anterior stricture; histology; HPV-independent SCC 2.3–9%; clobetasol 3 months, then maintenance; circumcision; BMG not genital skin

Not in the paper answer:

- LS is 6–10× commoner in females, and affected men are usually >60; **15–20%** have extragenital disease, and in men the **perianal skin is spared**.
- Circumcision is usually curative, but LS may recur in the scar.

ALGORITHM

1. Biopsy.
2. Give a **potent steroid (clobetasol 0.05%) for 3 months**, then taper.
3. Continue twice-weekly maintenance (or mometasone 0.1% or a calcineurin inhibitor).
4. Lifelong surveillance of non-healing fissures, ulcers, nodules.

- Recurrence is **40–50%**, so maintain proactively. Meatal disease needs meatotomy or meatoplasty.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled
Lichen sclerosus	Potent	3 months, then maintenance

KEY POINT

Steroid strength tracks the scarring and malignant stakes of the disease. LS is the single exception to the short-course rule.

6 · Vesiculobullous and drug eruptions

Feature	Erythema multiforme (EM) minor	Stevens–Johnson syndrome (SJS)	Fixed drug eruption (FDE)
Trigger	**HSV**	**Drugs**: sulfonamides, NSAIDs, β-lactams, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, carbamazepine
Lesion	Symmetric **targets**; palms, soles, mouth	Blisters and necrosis; genitals involved in ~**60%**	Solitary violaceous plaque, often on the glans
Course	Self-limited, recurs	Mortality up to ~**30%**	**Recurs at the same site within 24 h** of re-exposure
Sequelae	None	Meatal stenosis, strictures	Brown pigmentation
Treatment	**Suppressive aciclovir prevents it**	See algorithm	Stop the drug

ALGORITHM

1. **SJS**
2. Stop the offending drug.
3. Give burn-type supportive care in an intensive care or burn unit (lowers mortality).
4. Limited data support cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis or IVIG; systemic steroids are controversial.

KEY POINT

One drug class can cause a trivial FDE or a lethal SJS; extent and systemic illness separate them. Genital FDE mimics herpes; "same spot each time" is the clue.

Autoimmune blistering disease needs perilesional biopsy with direct immunofluorescence.

7 · Noninfectious genital ulcers

A genital ulcer is infectious until proven otherwise (see Ch 29).

Behçet disease (BD)

- BD is an **HLA-B51**-linked vasculitis, **more severe in males**, with oral and genital ulcers, uveitis, epididymitis and **pulmonary artery aneurysm**. Oral ulcer is most sensitive; **genital ulcer is most specific**. No confirmatory test exists. Treat with colchicine, steroids or **anti-TNF**.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; central nervous system; vascular; positive pathergy (optional)	1 each

Pyoderma gangrenosum (PG)

- PG accompanies inflammatory bowel disease and myeloproliferative disease as a painful ulcer with an **undermined violaceous border**. Biopsy the edge, treat with steroids, and avoid debridement.

8 · Infections

Balanitis and balanoposthitis

ALGORITHM

1. Remove irritants and improve hygiene.
2. Give a topical antibiotic or antifungal, then a short low-potency steroid.
3. If this fails, then **biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV).
4. If recurrent with phimosis, then **circumcision**.

Type	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny red plaque, elderly uncircumcised	Biopsy, then circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
Pseudoepitheliomatous keratotic micaceous balanitis (PEKMB)	Mica-like scaly plaque	Biopsy; excision

Soft-tissue infection

- Folliculitis needs hygiene. A furuncle needs incision and drainage with MRSA cover. Erysipelas has a sharp border and cellulitis an indistinct one; give antibiotics and **mark the border**.
- Fournier gangrene is a polymicrobial fascial infection with pain out of proportion, grey skin and crepitus; it needs emergency debridement (see Ch 26).

KEY POINT

Fournier gangrene is fascial, so skin shows a fraction of the necrosis; testes are spared. Imaging must never delay surgery. Debride to bleeding tissue, with a second look at 24–48 h.

- **Ecthyma gangrenosum** (P. aeruginosa bacteraemia in the immunocompromised) gives black eschar and needs immediate IV antipseudomonals.
- A **genital bite** needs amoxicillin–clavulanate 875/125 mg bd, irrigation and debridement.

Hidradenitis suppurativa (HS)

- HS is a **follicle-centred** disease of apocrine-bearing skin that causes abscesses, **sinus tracts** and scars, with fistulae and **SCC in chronic scars** as complications.

Hurley stage	Features	Mainstay
I	Abscess(es), no tracts or scarring	Topical clindamycin, deroofing
II	Recurrent abscesses, tracts, scarring; separated areas	Oral tetracycline, or clindamycin and rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts	Adalimumab and **wide excision**, graft or flap

- Advise stopping smoking and weight loss; **adalimumab** is given 40 mg weekly. **Repeated incision and drainage is discouraged** because the tract recurs.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Scrotum	**Involved**	**Spared**	—	May be involved
Look	**Satellite lesions**	Scaly advancing border, central clearing	Red-brown patch	Non-scaling plaque
Test	KOH: yeast	KOH: hyphae	Wood's lamp: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; test for diabetes	Topical azole or allylamine; terbinafine 250 mg/day if deep; no steroids	Clindamycin 1% or miconazole; oral erythromycin if severe	Low-potency steroid

KEY POINT

Azoles cover yeast and dermatophyte, so use them when unsure. Nystatin misses tinea; allylamines are weaker on Candida. A negative KOH from an untreated lesion means rethink the diagnosis.

Infestations

- Pubic lice and scabies both get permethrin 5%, **repeated at 1 week**; for scabies treat the whole body and the household. Ivermectin 200 µg/kg ×2 is the alternative, and lindane is disfavoured. Screen lice patients for other STDs.

9 · Premalignant and malignant lesions

P1 Premalignant conditions in urology (including premalignant lesions of penis) — PeIN HPV vs non-HPV; Queyrat, Bowen, bowenoid papulosis; Buschke–Löwenstein (no RT); glans therapy (5-FU, imiquimod, laser, resurfacing)

Not in the paper answer:

- Penile intraepithelial neoplasia is HPV-related (type 16) or non-HPV (LS, chronic LP); biopsy **several areas**.
- Verrucous carcinoma needs local excision or Mohs surgery, not primary radiotherapy (anaplastic transformation).

Lesion	Treatment
SCC in situ (SCCis) — accessible (scrotum)	Excision with a **5-mm margin** (favoured)
SCCis — glans	Mohs surgery, laser, cryotherapy; topical 5-fluorouracil (5-FU) or imiquimod (higher recurrence)
Bowenoid papulosis	Observe in a reliable young man; or 5-FU, imiquimod, cryotherapy, laser
PEKMB	Biopsy (exclude verrucous carcinoma); excision or ablation

Tumour	Key features	Treatment
Basal cell carcinoma	Rare; pearly papule	Excision or Mohs
Kaposi sarcoma	HHV-8; classic, AIDS and transplant types; a glans lesion may block the meatus	Restore immunity; excision, laser or vinblastine if solitary; radiotherapy 15–30 Gy if regional; chemotherapy if disseminated
Melanoma	<1% of penile cancers	Wide excision or partial penectomy; immunotherapy if advanced
Extramammary Paget	Red plaque at apocrine sites; CEA and EMA positive; underlying carcinoma in **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision or Mohs
Mycosis fungoides	Patches progress to plaques, then tumours; Sézary syndrome if in blood	Ultrapotent steroid, phototherapy

10 · Benign male genital lesions

Lesion	Site / look	Management
Pearly penile papules	Coronal rows of angiofibromas in 14–48% of young men, mistaken for condyloma	Reassure
Fordyce spots	Pin-sized shaft papules (ectopic sebaceous glands)	Reassure
Angiokeratoma of Fordyce	Red-purple scrotal papules; may bleed	Observe; laser if bleeding
Zoon balanitis	Shiny red glans plaque; elderly uncircumcised men	**Biopsy**, then **circumcision (curative)**
Median raphe cyst	Ventral, near the glans	Excision

> **KEY POINT**
> Zoon balanitis and erythroplasia of Queyrat are both red glans plaques in uncircumcised men: one benign and cured by circumcision, the other SCCis. A persistent red glans plaque always needs biopsy.

11 · Pigmentary disorders

- Vitiligo is accentuated by Wood's lamp; unlike LS, it has normal texture without atrophy or scarring.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

1 · Urogenital tuberculosis — organism and pathogenesis

TB seeds the urinary tract late and heals by scarring.

P3 P2 Renal and genitourinary TB: etiopathogenesis, presentation, diagnosis and management — MTBC, granuloma/IFN-γ, latency and reactivation, routes, renal sequence to autonephrectomy, site frequencies

Not in the paper answer:

- UG TB is **15–20%** of pulmonary TB in developing countries. Infection stays latent in 90–95%; lifetime reactivation is **5–15%**, so UG TB often appears without active lung disease.
- **M. bovis (and bacille Calmette–Guérin, BCG) is innately pyrazinamide-resistant**.
- Isolated urogenital (UG) TB is generally **not infectious**, except during aerosol-generating procedures.
- Ascending spread: intravesical BCG causes UG TB in **~1%**.

KEY POINT

UG TB is blood-borne and lands in the two best-perfused UG organs, kidney and epididymis. Blood-borne seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · Clinical features and pathology by site

- It is indolent, and patients are repeatedly treated for urinary tract infection (UTI). Constitutional symptoms occur in **<20%**. **Lower urinary tract symptoms dominate**; haematuria or flank pain occurs in ~⅓. **The hallmark is sterile pyuria** (>90%).

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities, then autonephrectomy (up to ⅓); "putty" kidney; end-stage renal failure ~7%
Ureter	Descending	Stricture at the vesicoureteric junction (VUJ); beaded corkscrew ureter. **Stricture obstruction is an important cause of renal failure**
Bladder	Descending	Starts at orifices; trigone spared. After ~1 year <100 mL; "thimble" <20 mL
Epididymis	10–55%; second commonest; bilateral in ~⅓	Starts in the **globus minor (tail)**; posterior scrotal sinus in up to half
Vas	Contiguous	Thickened, beaded
Testis	From epididymis	Hard tumour-like mass
Prostate	22–49%	Calcified hard gland. "Prostatitis" responds partly to quinolones, then relapses
Seminal vesicles	Via vas or ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia; infertility may be the first symptom
Urethra	1.9–4.5%	With prostatic disease; urethroscrotal fistula

Site	Frequency / route	Signature lesion
Penis	Exceedingly rare	Ulcer; papulonecrotic tuberculid

3 · Diagnosis

P3 Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management — 3–5 early-morning urines, ZN, LJ/MGIT culture (gold standard, monitors treatment), GeneXpert, LF-LAM, TST/IGRA, imaging signs, cystoscopy, algorithm

Not in the paper answer:

- KUB calcification progresses from punctate to globular to ring, and IVU may show cloverleaf calyces.

ALGORITHM

1. **Hierarchy of proof**
2. If culture or MTBC DNA is positive, then the diagnosis is made.
3. If negative, then caseating granulomas on tissue in the right context confirm it.
4. If still negative, then a compatible picture, exposure and response to empirical treatment suffice.
5. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

- Culture sensitivity is 10–90% because shedding is intermittent. The Ziehl–Neelsen (ZN) smear detects **<50%**. Acidic urine inhibits mycobacteria, so culture promptly.
- **GeneXpert** detects rifampicin resistance, a surrogate for multidrug-resistant (MDR) TB. The WHO uses it in the initial work-up of all TB, but a negative result does not exclude TB, and it **cannot monitor treatment** because dead bacilli shed DNA.
- Fine-needle aspiration cytology of an epididymal nodule is diagnostic in ~⅔. At cystoscopy ulcers mimic cancer, and a **"golf-hole" orifice** means imaging the upper tract.
- Tuberculin skin test (**TST**) cut-offs are **≥5 mm** for contacts and the immunosuppressed, **≥10 mm** for immigrants, high-risk workers and drug injectors, and **≥15 mm** with no risk factors. The interferon-γ release assay (**IGRA**) has no BCG false positives and is preferred for people from endemic countries. Both show exposure only.

Modality	Signs / role
Plain film (KUB)	Calcification in 24–44%; **lobar "putty" kidney** is pathognomonic of end-stage disease; ureteric calcification is **intraluminal** (schistosomiasis: mural)
Intravenous urography (IVU)	**Moth-eaten calyx**; **phantom calyx**; **Kerr's kink**; **pipe-stem ureter**; contracted bladder. Findings are mostly **obstructive**
CT urography	Standard; granuloma may mimic renal cell carcinoma
Ultrasound (US)	**Serial monitoring of hydronephrosis during treatment**
MRI	Diffusion-weighted imaging separates pyonephrosis from hydronephrosis

KEY POINT

Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). So obstruction is the commonest finding and must be watched during treatment.

4 · Medical treatment

- Combine drugs to prevent resistance. Check baseline liver function, renal function and HIV status, and use directly observed therapy (DOT).

ALGORITHM

1. **Standard 6-month short course (2HRZE / 4HR)**
2. Intensive, 2 months: daily INH (H), rifampicin (R), pyrazinamide (Z) and ethambutol (E); stop E once fully susceptible.
3. Continuation, 4 months: HR daily (or thrice weekly by DOT).
4. Give pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, **drug interactions**	Antiretroviral and immunosuppressant levels; rifabutin with antiretrovirals
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, hepatitis	M. bovis is resistant
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green vision)	Check vision; the **only non-hepatotoxic** first-line drug

- Some give 12–24 months for complicated disease, as UG TB relapses more.
- **Monitor** liver function, vision, urine culture and **serial US**, because healing fibrosis can worsen strictures and bladder contraction.
- **Steroids** are anecdotal in UG TB; consider them if a stricture worsens on treatment.

5 · Surgical treatment

*About half need surgery, now mostly **reconstructive**.*

- The goal is organ preservation. Indications are obstruction or abscess, a non-functioning infected kidney, resistant hypertension and reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy**, except in emergencies.

ALGORITHM

1. If uraemia or sepsis, then decompress urgently.
2. Try a retrograde double-J stent first (preferred; succeeds in < half, 41%).
3. If that fails, then try an antegrade stent.
4. If that fails, then percutaneous nephrostomy (PCN); loculated systems may need several.
5. Then repair definitively; if the kidney is unsalvageable, then nephrectomy.

- Removing a PCN without fixing the cause leaves a tuberculous cutaneous fistula.
- **Nephrectomy is established** for a non-functioning kidney with recalcitrant TB or resistant hypertension; **routine removal** of an asymptomatic non-functioning kidney is **contested**. Partial nephrectomy suits a polar lesion unresponsive to drugs.
- At nephrectomy, expect dense fibrosis and **ligate artery and vein separately** (late arteriovenous fistula).
- **Strictures** are commonest in the lower ureter, and endoscopic results are poor. Stent early for strictures arising on treatment and **image every stricture**; if worsening, give steroids,

and **if there is no improvement after 6 weeks, operate**.

Site / gap	Procedure	Key point
Pelviureteric junction (PUJ), short	Dismembered pyeloplasty; flap if longer	Harder than congenital PUJ obstruction
PUJ unreconstructable	Ureterocalicostomy (lower pole)	Cover with capsule or omentum
Upper/mid ureter	Ureteroureterostomy; Davis intubated ureterotomy	—
Lower ureter	Excise, then ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity
Long gap	Boari flap, with or without hitch	**10–15 cm**; poor in a contracted bladder
Multiple / recurrent	**Ileal ureter**	Native ureter unusable

KEY POINT

Trap: in TB the bladder may itself be contracted, ruling out psoas hitch and Boari flap. Assess capacity first; ileal ureter, with or without augmentation, is then the option.

P2 Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30- year male — assessment, augmentation vs substitution, ureteric reimplantation, bowel complications, follow-up

Not in the paper answer:

- Contracted bladder: operate for intolerable symptoms or capacity **<100 mL**, once chemotherapy controls disease; a thimble bladder needs orthotopic substitution, and the spared trigone allows supratrigonal substitution.
- **Genital** surgery is only for failed medical therapy: **epididymectomy** if the testis is spared, otherwise orchidectomy. Obstructive azoospermia is often irreversible, so offer sperm retrieval.

6 · BCG-related infection, relapse, special situations

- **BCG UG TB** (~1%) causes granulomatous prostatitis, epididymo-orchitis, renal abscess or sepsis. For organ infection give INH and rifampicin for 3–6 months, since **pyrazinamide is useless**; for sepsis give triple therapy and steroids, and stop BCG permanently.
- **Relapse** occurs in **6.3–22%**, at a mean of ~5 years, so follow up for **10 years**.

Situation	Change	Duration
MDR-TB (INH and R resistant)	Second-line; BPaL (bedaquiline, pretomanid, linezolid), not yet approved for extrapulmonary TB	> 18 months; 6–9 months with BPaL
Extensively drug-resistant (XDR) TB	MDR plus fluoroquinolone and bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	HRE and pyridoxine; Z omitted in the USA (WHO includes it); breastfeeding allowed	9 months (without Z)
HIV on effective antiretroviral therapy (ART)	Standard daily regimen; rifabutin with protease inhibitors	6 months (9 if not on ART)

Situation	Change	Duration
Renal transplant	Rifamycin with drug-level monitoring, or a rifamycin-free regimen	≥ 18 months if rifamycin-free

- **HIV** raises TB risk 15–20-fold, with more dissemination but **less fibrosis and fewer strictures**.

7 · Schistosomiasis (bilharzia)

Trapped eggs cause obstruction and squamous bladder cancer.

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical/pelvic plexus — UG tract	**Bulinus**
S. mansoni	Africa, Middle East, S America	Mesenteric — liver, gut	Biomphalaria
S. japonicum	China, SE Asia	Mesenteric — liver, gut	Oncomelania

ALGORITHM

1. **Life cycle**
2. Cercariae penetrate intact skin in fresh water and pass via the lungs to the liver, where they pair.
3. Paired adults move to the vesical plexus and live 3–5 years.
4. Eggs (**terminal spine**) cross the bladder wall into urine and hatch as miracidia.
5. Miracidia infect Bulinus, which releases cercariae.

- **Retained eggs cause the disease**; adults do not multiply in humans. Children aged 5–15 carry the heaviest burden.
- The **egg granuloma** forms polypoid masses (active), then fibroses and calcifies into **"sandy patches"**, leading to obstruction, metaplasia and cancer. Obstruction is **distal**; hydroureter may be **tonic** (reversible once relieved) or **atonic** (fibrotic).
- **Salmonella** shelters in the worm, causing chronic bacteriuria; treat both.
- **Bladder cancer** is rated **Group 1** by IARC. Patients are younger (40–50), and **SCC is 60–90%**; mass drug treatment in Egypt shifted SCC to urothelial cancer.
- Acute infection gives swimmer's itch and **Katayama fever**; chronic infection (from 8–12 weeks) gives **painless terminal haematuria**. Men may get haemospermia and infertility.

KEY POINT

Trap: a negative urine egg count does not exclude severe disease; the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Collect **09:00–15:00** (noon peak); negative early and in light or inactive disease
Rectal snip (squash prep)	Active or past infection; egg viability	Avoids bladder biopsy risks
Serology	Exposure	Stays positive, so it **never judges cure**
Circulating antigen	Live worms	Urine CCA is poor for S. haematobium
PCR	Most sensitive	Impractical in the field

- Imaging shows wall thickening, polyps and hydronephrosis on US, and a **calcified bladder ("fetal head in the pelvis")** with mural ureteric calcification on plain film.
- **Praziquantel (PZQ)** is the only WHO-listed drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart, curing 60–90%. PZQ is weak on immature worms, so repeat after several weeks.
- Active polypoid obstruction resolves with PZQ alone; late fibrotic stenosis needs surgery.

Complication	Surgery
Distal / intramural stenosis	Antireflux reimplantation; balloon dilatation (restenosis common); meatoplasty for meatal stenosis
Long / multifocal stenosis	Boari flap; ileal ureter; ileal conduit; long-term nephrostomy
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

8 · Filariasis

In India, filariasis commonly causes hydrocele and chyluria.

- W. bancrofti causes >90% and is the **only cause of genital filariasis**; Brugia affects the leg. Microfilariae are **nocturnally periodic**, and doxycycline kills the Wolbachia endosymbiont.
- **Acute adenolymphangitis** is a **retrograde** lymphangitis. **Funiculoepididymitis** is a recurrent painful cord swelling that spares the testis and is often mistaken for tumour.
- **Hydrocele** is the commonest chronic lesion; elephantiasis comes late.
- Diagnosis uses a midnight smear, the **circulating antigen** card, and the US "**filarial dance sign**".

P3 Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria — pyelolymphatic fistula; ether test, triglycerides, cystoscopic lateralisation, lymphoscintigraphy; MCT diet + DEC, then pelvic instillation sclerotherapy, then hilar lymphatic disconnection

Not in the paper answer:

- Most chyluria remits on conservative treatment; instil sclerosant on the side identified at cystoscopy.

Drug	Action	Regimen / caution
Diethylcarbamazine (DEC)	Microfilaricidal, modest on adults	Drug of choice: 6 mg/kg/day; a single-day course equals a 12-day course. **Contraindicated if onchocerciasis or Loa loa is possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg bd for 21 days; single dose in mass treatment
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with Loa
Doxycycline	Anti-Wolbachia; kills adults	200 mg daily for 4–8 weeks; improves lymphoedema and hydrocele; not in pregnancy or children

- Mass treatment uses albendazole with DEC in India, or with ivermectin where Loa or onchocerciasis is co-endemic.
- **Hydrocele** recurs after aspiration, so **excise the sac**. In funiculoepididymitis excise nodules and preserve the testis. Elephantiasis needs reduction scrotoplasty.
- **Chyluria** occurs when dilated retroperitoneal lymphatics rupture into the collecting system, giving milky urine with fat and protein loss. Exclude TB, tumour and trauma.

9 · Hydatid disease and other parasites

Never puncture a hydatid cyst blindly.

P1 Hydatid disease of kidney — kidney third site (2–3%); hydatiduria; water-lily sign, WHO stages; albendazole 400 mg bd 1–6 months; no aspiration; scolicide only if non-communicating; nephron-sparing surgery; PAIR not for kidney

Not in the paper answer:

- Pack the field with hypertonic (20%) saline swabs, and have the anaesthetist ready for anaphylaxis.
- Remove the cyst intact (pericystectomy) or do a partial nephrectomy; nephrectomy is only for a destroyed kidney.
- Enterobiasis, amoebiasis, trichomoniasis (see Ch 29) and fungal UTI (see Ch 26) are rarer genitourinary infections.

★ Putting it together: infection decisions

A · UTI: drug matches site

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g once, TMP–SMX 3 days (resistance <20%), pivmecillinam	Narrow, urine-confined
Pyelonephritis, complicated UTI, men	Quinolone or cephalosporin; carbapenem if resistant	Nitrofurantoin and fosfomycin do not reach tissue
ESBL producers	Fosfomycin or nitrofurantoin for cystitis; carbapenem if invasive	Co-resistance to quinolones
Asymptomatic bacteriuria	Treat only in pregnancy and before mucosa-breaching surgery	Otherwise harmful

B · Beyond antibiotics

ALGORITHM

1. **Image**: no response by 72 h, sepsis, obstruction, gas, febrile men.
2. **Drain** any obstructed infected kidney; treat the stone later.
3. Renal abscess: if ≤3 cm, then antibiotics; 3–5 cm, variable; if >5 cm, then percutaneous drainage.
4. **Emphysematous pyelonephritis**: gas within kidney: antibiotics and drainage; if beyond it, then nephrectomy (see Ch 26).
5. **Fournier: urgent repeated radical debridement**.

C · Recurrent UTI in women

ALGORITHM

1. Confirm each episode by culture.
2. Behaviour; vaginal oestrogen if postmenopausal.
3. Methenamine, cranberry, D-mannose.
4. Self-start or post-coital antibiotic; finally continuous prophylaxis.

D · Pelvic pain syndromes compared

	CP/CPPS	IC/BPS
Core	Male pelvic pain >3 months, no infection (NIH III)	Filling-related bladder pain, no other cause
Phenotyping	UPOINT; NIH-CPSI	Hunner vs non-Hunner
Treatment	By domain: α-blocker, pelvic-floor physiotherapy, neuromodulators, psychology	Behavioural, oral (amitriptyline, PPS), intravesical, Hunner fulguration, BTX-A; diversion last

E · GU tuberculosis in four lines

ALGORITHM

1. **Suspect:** sterile pyuria, contracted bladder, beaded vas.
2. **Confirm:** three early-morning urines for AFB, culture, GeneXpert.
3. **Treat:** 2HRZE, then 4HR; watch for ureteric stricture.
4. **Operate** after 4–6 weeks of drugs (see Ch 31).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 4

Paediatric Urology 1

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 41	Management Strategies for Vesicoureteral Reflux
Ch 49	Anomalies of the Kidney and Collecting System
Ch 50	Renal Dysgenesis and Cystic Disease of the Kidney
Ch 51	Pathophysiology of Urinary Tract Obstruction
Ch 52	Ectopic Ureter, Ureterocele and Ureteral Anomalies
Ch 53	Surgery of the Ureter in Children
Ch 56	Hypospadias
Ch 57	Etiology, Diagnosis and Management of the Undescended Testis
Ch 61	Urologic Considerations in Pediatric Renal Transplantation
Ch 63	Paediatric Urologic Oncology: Renal and Adrenal
Ch 64	Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia
★	Putting it together: common paediatric decisions

41 Management Strategies for Vesicoureteral Reflux

1 · Definition and concept

We treat the infection risk, not the reflux.

- **Vesicoureteral reflux (VUR)** is retrograde urine flow to the upper tract, linked to **acquired scarring** and **congenital dysplasia**.
- **Infection, not reflux, is the injury**, so aim for sterile urine and good emptying.

2 · Embryology and antireflux mechanism

P1 P4 Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management — Mackie–Stephens bud position (early bud gives high lateral orifice, VUR and dysplasia; late bud gives ectopia).

Passive flap valve; Paquin tunnel length to diameter **5:1** vs 1.4:1 in refluxing units.

Not in the paper answer:

- Abnormal bud position also causes hypoplasia, so some "scars" are congenital.

3 · Primary vs secondary VUR

Secondary reflux needs the bladder treated first.

P4 VUR: etiopathogenesis, recent trends in management and STING technique — primary (deficient UVJ) vs secondary (PUV, neurogenic, dysfunctional voiding, ureterocele, duplication, diverticulum); overlap; detrusor overactivity commonest urodynamic finding; infant male high voiding pressures

Not in the paper answer:

- In spina bifida, VUR occurs in up to **30%**: start early clean intermittent catheterisation (CIC) and an anticholinergic or β3-agonist.
- **Reflux at low pressure or in early filling means poor resolution.** Treating detrusor overactivity improves resolution.

> **KEY POINT**
> Primary and secondary are not exclusive: "primary" VUR often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Remove it and only true primary VUR remains.

4 · Epidemiology

- VUR is found in **~30%** of children with UTI and ~17% of those imaged for hydronephrosis. It is present in **38.6%** under 2 years versus 7.6% at 12–21 years.
- Girls are affected ~2× overall, but infant boys predominate.

5 · Grading — International Reflux Study (1981)

Grade	Ureter	Pelvis and calyces
I	Non-dilated, ureter only	Not reached
II	Not dilated	Reached, no dilatation
III	Mild–moderate dilatation	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate tortuosity	Moderate dilatation

Grade	Ureter	Pelvis and calyces
V	Gross dilatation and tortuosity	Gross dilatation; **papillary impressions lost**

- Grades I–II are non-dilating and III–V dilating. Radionuclide cystography (RNC) uses 3 grades.
- **Ureteral diameter ratio (UDR)**, the largest distal ureteric diameter divided by the L1–L3 vertebral distance, predicts resolution and breakthrough UTI (BUTI) better than grade.

6 · Diagnosis

When to image after UTI

- **AAP 2011**, febrile UTI at 2–24 months: do **ultrasound (US) first**, and a voiding cystourethrogram (VCUG) only if US is abnormal or after a **second febrile UTI**.
- **NICE**: US, with VCUG mainly for atypical or recurrent UTI in younger children.
- **Top-down**: DMSA scan first, VCUG only if abnormal. A normal DMSA leaves a **<1%** chance of high-grade VUR but misses ~1/3 of (mostly low-grade) VUR.

Cystographic methods

P1 VCUG — gold standard; gravity ≤70 cm H_2O; early study once urine sterile; cyclic VCUG +12–20%; filling-phase reflux poor prognosis; postvoid pelvic retention suggests UPJO; RNC and ceVUS alternatives

Not in the paper answer:

- Direct RNC is more sensitive for grades II–V but **misses grade I**.
- Suspect coexisting **UVJ obstruction** with a distal aperistaltic segment, dilute contrast and delayed drainage. It is a **relative contraindication to endoscopic injection**.

Upper tract

- **Renal and bladder ultrasound (RBUS)** is the serial mainstay but cannot exclude VUR.
- **DMSA** is the gold standard for cortical defects (sensitivity 98%, specificity 92%); a defect that persists is a scar.

7 · Congenital dysplasia vs acquired scar

Only the acquired scar is preventable.

P4 P1 VUR: etiopathogenesis, recent trends in management and STING technique — dysplasia (boy, IV–V, antenatal, not preventable) vs acquired scar (girl, febrile UTI, BBD, preventable); polar compound papillae, intrarenal reflux at 2 mmHg; Ransley–Risdon sterile reflux; "big bang" debated

Not in the paper answer:

- Hypertension develops in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure**; removing a small kidney or segment may.

8 · Bladder and bowel dysfunction (BBD)

BBD is the main modifiable factor.

- **Treat constipation first**; it causes sphincter co-contraction.
- BBD brings more UTI on prophylaxis and after surgery, less resolution, and **lower endoscopic success, but not open surgical success** (AUA).
- **Phimosis**: in boys with VUR, circumcision cuts the number needed to treat (NNT) to prevent one UTI from ~111 to **~4**. Topical steroid is equivalent.

9 · Natural history and resolution

Grade	Spontaneous resolution
I–II	Most; grade II 63–85%
III	~50%; ~90% of resolutions within 4 years
IV–V, bilateral III	Few (infants with IV–V resolve well in year 1)

- Resolution favours **low grade, young age, unilateral reflux, no BBD, voiding-only reflux, higher bladder volume at onset and low UDR**.

10 · Medical management — continuous antibiotic prophylaxis (CAP)

CAP failure is the main trigger for surgery.

P3 Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome — CAP agents by age with doses (TMP/amoxicillin <2 months; TMP-SMX, nitrofurantoin >2 months). Hidas risk groups, indications for correction, 8–10% febrile UTI after stopping CAP.

Not in the paper answer:

- Give a low dose **at night**; avoid TMP-SMX in neonates (bilirubin displacement).
- **If febrile UTI breaks through CAP despite good bladder and bowel habits, stop medical management and correct the VUR.**

Risk stratification (Hidas)

Factor	Low risk	High risk
Sex / age	Male; >5 y	Female; <1 y
Laterality / grade	Unilateral; I–II	Bilateral; III–V
BBD / scars	No	Yes
Presentation	Prenatal hydronephrosis	Multiple febrile UTIs

- Two-year BUTI on CAP is **8.6% low, 26% intermediate, 62.5% high** risk. About two-thirds of children are low risk.

11 · Practical algorithm (AUA-based, risk-stratified)

ALGORITHM

1. Confirm true febrile UTIs; do RBUS. If multiple febrile UTIs or upper-tract findings, then VCUG.
2. In all, treat BBD (constipation first) with timed voiding.
3. If low risk (grade I–II, no BBD, male, older), then observe with or without CAP and treat febrile UTI promptly.
4. If high risk (grade ≥III, girl, BBD, scars, infant), then give CAP.
5. If breakthrough febrile UTI, new scars, or persistent high-grade VUR, then correct (endoscopic or reimplant).

- CAP helps little at grade ≤II without BBD, but helps at ≥III, at least in girls.

12 · Surgical correction

Surgery stops reflux but does not protect kidneys better.

- Open correction succeeds in **>95%**, but neither open nor robotic surgery beats medical therapy for scarring or growth. The adult UTI risk persists.
- **Endoscopic dextranomer/hyaluronic acid (Dx/HA)**: BBD lowers success; avoid it with UVJ obstruction. Calcified mounds can **mimic ureteric stones**.
- After a cross-trigonal reimplant (Ch 53), **retrograde ureteric access is difficult**.

13 · Associated conditions

- **UPJO with VUR**: **correct the obstruction first**; repairing both at once risks the ureteric blood supply.
- In a **duplex** kidney, VUR usually enters the **lower pole** (Weigert–Meyer).
- A **ureter entering a paraureteral diverticulum** will not resolve; a diverticulum beside the orifice resolves like primary VUR.
- Contralateral VUR with a multicystic dysplastic kidney usually resolves.
- **Megacystis–megaureter** (massive VUR recycling urine into a large thin bladder) and **obstructive refluxing megaureter** are treated by reimplant, not injection or outlet surgery.

14 · Sibling screening

- Sibling prevalence is ~**30%**, higher when younger.
- The target is **renal damage, not VUR**. AUA: **VCUG for siblings with a cortical abnormality or renal asymmetry on US, or a UTI history**; otherwise observe.

15 · Follow-up and adulthood

- If scarred, check BP, proteinuria and renal function for life.
- **Pregnancy**: scars bring ~3× hypertension and **~8× pre-eclampsia**. **Scarring, not VUR, drives morbidity.**

16 · Landmark trials

Trial	Design	Result	Take-home
IRSC	<9 y, high grade; CAP vs open surgery	UTI equal (38%); surgery fewer pyelonephritis; scars, 10-y function equal	Surgery ≈ medical for scarring
Birmingham Reflux Study	High grade; medical vs surgical, 5 y	New scars equal, all within 2 y	Equal; supports "big bang"
Swedish Reflux Trial	1–2 y, grade III–IV; CAP vs endoscopic vs surveillance	Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; boys no difference	Treat girls with dilating VUR
RIVUR	607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo	Recurrent UTI **12.9% vs 23.6%**; scarring unchanged	CAP prevents UTI, not scars
PRIVENT	576 children with UTI; CAP vs placebo	Absolute UTI reduction 6% (19% to 13%)	Modest benefit
French / low-risk trials	Grade I–III; CAP vs none	No significant difference	No benefit at low risk

KEY POINT

Trials disagree because of population risk: RIVUR was diluted by low-risk children; reanalysis gives NNT 5 high-risk vs 18 low-risk. Neither CAP nor surgery prevents scarring better.

49 Anomalies of the Kidney and Collecting System

1 · Embryology that explains everything

- The ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches; its tips induce nephrons (**reciprocal induction**).
- In weeks 6–9 the kidney ascends and rotates 90° ventromedially, so the pelvis faces medially.
- Renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies; the adrenal develops independently.

> **KEY POINT**
> Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

2 · Bilateral renal agenesis (BRA)

BRA is uniformly lethal, so recognise it and counsel.

- It occurs in **≈1 in 5000** births, **75% male**. First-degree relatives carry ≈1000× risk, so **scan parents and siblings by ultrasound (US)** of any infant with unilateral agenesis (URA), BRA or dysgenesis.
- The ureters are atretic and the bladder absent or hypoplastic. The flat adrenal gives the **"lying down" sign**.
- **Potter facies** and limb deformities are **deformations** from absent amniotic fluid, not malformations. Undescended testes (43%) and gastrointestinal anomalies (≈60%) are common.
- **Pulmonary hypoplasia** is the cause of death. **Peters two-step model**: a kidney-derived "renal factor" acts before 16 wk, then amniotic fluid in the canalicular phase (16–26 wk), so fluid alone only partly restores the lungs.
- **Diagnosis**: prenatally, severe oligohydramnios, no kidneys and a small chest. In a neonate, **anuria >24 h without bladder distension** means BRA until proved otherwise; confirm with US and power Doppler.
- **Prognosis**: ≈40% are stillborn and the rest die in 24–48 h. Amnioinfusion is experimental (RAFT, §9) and needs ethics oversight with full counselling.

> **KEY POINT**
> BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD).

3 · Unilateral renal agenesis (URA) and the solitary functioning kidney (SFK)

The issues are genital anomalies and lifelong hyperfiltration.

- URA occurs in **1 in 1200**, male 1.8:1. Many apparent URAs are an **involuted multicystic dysplastic kidney (MCDK)**.

- Associations include VACTERL, MURCS, Turner, Kallmann, branchio-oto-renal and DiGeorge syndromes. **Maternal diabetes triples the risk**. An isolated preauricular pit or single umbilical artery needs no US; two or more anomalies do.
- **The ipsilateral ureter is absent in ≈60%**. The defect is in the UB, so the gonad is rarely affected.
- **Contralateral kidney**: vesicoureteral reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction (VUJ) obstruction ≈7%.
- **Genital anomalies occur in 20–40%** (females 25–50%, males 10–15%).

	Male (WD derivatives)	Female (MD derivatives)
Always present	Testis, head of epididymis	Fimbriated end of tube
Commonest	Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney**	Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia/duplication; Gartner duct cyst with blind ectopic ureter
Named syndrome	**Zinner**: seminal vesicle cyst and ipsilateral agenesis	**OHVIRA** (obstructed hemivagina and ipsilateral renal anomaly; Herlyn–Werner–Wunderlich): pubertal haematocolpos, pain
Trigger to image kidneys	Impalpable vas (infertility, herniotomy/orchidopexy)	Any uterine anomaly; pelvic US after thelarche, before menarche

- **Magee classification**: type I (before wk 4) gives a unicornuate uterus; type II (early wk 4) gives didelphys with obstructed hemivagina on the agenesis side; type III (after wk 4) gives isolated URA with normal genitalia.
- **MRKH** (46,XX müllerian agenesis): **type II has renal anomalies in 40–60%**; type I rarely does.
- **Work-up**: compensatory hypertrophy is seen from 20 wk in ≈90%. DMSA finds a missed ectopic kidney. A voiding cystourethrogram (VCUG) for contralateral VUR is **not routine**. Sport is not precluded.

Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass leads to glomerular hypertension, then albuminuria and sclerosis. Risk is highest when the SFK itself has a congenital anomaly (CAKUT), low birth weight or prematurity.
- In KIMONO, ≈⅓ had renal injury by ≈10 y, so follow-up is lifelong.

Situation	Blood pressure (BP) and first-morning urine albumin	Ultrasound	Creatinine
SFK without CAKUT	Annually	Every 5 y if glomerular filtration rate (GFR) >60 and no medication	Periodic, into adolescence
SFK with CAKUT	Twice yearly	As indicated	Periodic
GFR <60 or on antihypertensive/antiproteinuric	BP, albumin, creatinine 2–4×/year		

Hypertension or microalbuminuria (>30 mg/24 h) calls for an ACE inhibitor, salt limitation and no excess protein.

4 · Supernumerary kidney

- It has its **own capsule, collecting system and separate blood supply**, unlike a duplex kidney. About half are hydronephrotic, with obstruction, infection or stones in early adulthood.

5 · Ectopia (anomalies of ascent)

An ectopic kidney is easily mistaken for agenesis or tumour.

Simple ectopia

- It occurs in **1 in 900** at autopsy and is bilateral in ≈10%. **Pelvic is commonest**.
- The kidney is small and malrotated (pelvis anterior). **Hydronephrosis is common**, half from obstruction (mostly PUJ). **VUR occurs in 30–50%**, often into the orthotopic kidney.
- The blood supply is anomalous: **map it by CT or MR angiography (MRA) before surgery**.
- Genital anomalies occur in 15–45%. Continuous incontinence in a girl suggests an ectopic ureter from the ectopic kidney.
- **Never remove a "pelvic mass" without imaging**: solitary pelvic kidneys have been excised as tumours.

Cephalad and thoracic kidney

- Cephalad ectopia is seen with **omphalocele**. A thoracic kidney lies in the **foramen of Bochdalek**, is usually silent, and needs CT or MRU rather than thoracotomy.

6 · Fusion anomalies

Fused kidneys bring obstruction, stones and variable vessels.

Crossed renal ectopia

- The kidney lies opposite its ureteric orifice; **90% are fused**, making it the second commonest fusion anomaly. Left-to-right crossing is 3:1, and the ureteric orifice is orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed and bilateral crossed (rarest).

Crossed fused form	Key feature
Inferior ectopia (≈⅔, commonest)	Crossed upper pole fused to normal lower pole; pelves anterior
Sigmoid (S-shaped)	Second commonest; pelves face opposite ways
Lump (cake)	Lobulated pelvic mass
L-shaped (tandem)	Crossed kidney transverse, anterior to L4
Disc (pancake)	Medial fusion, reniform outline kept
Superior ectopia	Least common; crossed lower pole to normal upper pole

- An **anterior pelvis means early fusion**; a medial pelvis means late fusion. VUR occurs in ≈20%.
- **A stone in the crossed ureter hurts on the anephric side**; pyelonephritis or PUJ obstruction hurts where the kidneys lie. Obtain MRU or MRA before major surgery.

Horseshoe kidney

- It is the **commonest fusion anomaly**: **1 in 400** (0.25%), male 2:1. **Lower-pole fusion occurs in 95%**, before rotation.

- The **isthmus** lies at L3/L4, **just below the IMA**, in front of the aorta and IVC.
- The calyces **point posteriorly**, and the ureter inserts high and bends over the isthmus. **The blood supply is highly variable**, so obtain MRA or CT before surgery.
- **≈60% of Turner patients have a horseshoe kidney**; it is also associated with neural tube, anorectal and VATER anomalies.
- **PUJ obstruction occurs in 13–34%**, but many hydronephroses are not obstructed. VUR occurs in 8–32%.
- **Stones affect ≈⅓ of adults**, so still do a metabolic work-up.
- **Wilms risk is 1.76–7.9×**; preoperative chemotherapy preserved ≈75% of parenchyma (NWTS). Renal cell carcinoma is not increased.
- At least 50% are asymptomatic. Pregnancy is unaffected and the kidney is transplantable.

KEY POINT

† PUJ obstruction in horseshoe kidney: dismembered pyeloplasty with dependent anterior anastomosis; symphysiotomy not routinely needed.

7 · Malrotation and vascular anomalies

- Malrotation occurs in ≈1 in 939. **Weyrauch** types: **ventral (no rotation, commonest)**, ventromedial, dorsal (rarest) and lateral (reverse). **The vessel course is the only clue** to direction.
- Hilar fibrosis can cause intermittent PUJ obstruction and stones; suspect malrotation when a stone lies in an unusual position.

P2 Anomalies of renal artery. — multiple / aberrant / accessory terms; single artery 71–85%; end-arteries; crossing vessels in normal PUJs; aneurysm (Abeshouse, treat >2 cm, pregnancy, symptoms).
Congenital vs acquired AVF; embolisation.

Not in the paper answer:

- Treat an aneurysm for rupture, growth, pregnancy, size **>2 cm** or symptoms. An acquired AVF may close within months, so observe it.
- A congenital arteriovenous fistula presents with bruit, haematuria and high-output failure and rarely resolves.

8 · Collecting system anomalies

P1 Pelvicalyceal anomalies other than PUJ obstruction — calyceal diverticulum (types I/II, stones ≈40%, delayed contrast pooling, percutaneous ablation and fulguration), hydrocalycosis, megacalycosis (non-obstructive, 12–20 calyces, male 6:1, no surgery), infundibulopelvic stenosis (bilateral, >⅓ renal failure), extrarenal and bifid pelvis (≈10%)

Not in the paper answer:

- Extrarenal pelvis: confirm with diuretic renography before calling it obstruction.
- Infundibulopelvic stenosis: monitor renal function yearly; if it falls, give an ACE inhibitor and restrict protein.

KEY POINT

Collecting-system anomalies split on one question: is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

9 · Landmark trials

Trial	Design	Result	Take-home
RAFT (Renal Anhydramnios Fetal Therapy)	Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk	94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early	Prevents lethal lung hypoplasia but leaves neonatal ESRD — research only

50 Renal Dysgenesis and Cystic Disease of the Kidney

1 · Genetics in one table

Gene	Renal phenotype
PAX2	Renal hypodysplasia or aplasia
EYA1	Branchio-oto-renal (BOR) syndrome
HNF1B	Renal cysts and diabetes syndrome
PKD1, PKD2 (primary cilium)	Polycystic kidney disease

2 · Dysgenesis: dysplasia vs hypoplasia

Dysplasia is the leading cause of childhood renal failure.

- **Dysgenesis** (abnormal size, shape or structure) includes dysplasia and hypoplasia.
- **Dysplasia** is a **histological** diagnosis of arrested metanephric differentiation. Its sine qua non is **primitive ducts** with fibromuscular collars; cartilage may also occur. Not all cysts are dysplastic, and not all dysplasia is cystic.
- Causes are primary UB failure (syndromes such as BOR and renal–coloboma) or fetal outflow obstruction. **It is the leading cause of childhood end-stage renal disease (ESRD).** Ultrasound (US) shows a small echogenic kidney with lost corticomedullary differentiation.
- **Hypoplasia** is reduced mass from too few cells; simple hypoplasia has a normal nephron number.

Entity	Defining feature	Course / management
Oligomeganephronia	Few, hypertrophied nephrons; usually bilateral	Polyuria, polydipsia, short stature; **proteinuria first sign**; ESRD in second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, ACE inhibitor; living related donor acceptable (not familial)
Ask-Upmark (segmental hypoplasia)	Mostly acquired reflux nephropathy; deep "slit-scars"	Girls ≥10 y, **severe hypertension**; partial/total nephrectomy cures blood pressure (BP); bilateral: medical, dialysis/transplant. Reflux correction does not cure hypertension
Hypodysplasia	With ectopic orifice, megaureter, ureterocele, posterior urethral valves (PUV), prune-belly	**Dysplasia is proportional to the degree of ureteric ectopia**: lateral orifice gives reflux-associated, medial/ureterocele gives obstruction-associated dysplasia

KEY POINT

The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

3 · Cystic disease: concepts and classification

Sort first by inheritance, then by the extrarenal clue.

- Cysts arise from tubules; the **primary cilium** is central.
- A **multicystic** kidney (MCDK) is dysplastic with no normal nephrons. A **polycystic** kidney starts normal, and its nephrons are progressively replaced by cysts.

Inheritable	Non-heritable
Autosomal recessive polycystic kidney disease (ARPKD)	Multicystic dysplastic kidney (MCDK)
Autosomal dominant PKD (ADPKD)	Benign multilocular cyst (cystic nephroma)
Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant)	Simple cysts
Congenital nephrosis; familial hypoplastic glomerulocystic disease	Medullary sponge kidney; sporadic glomerulocystic disease
Syndromes: tuberous sclerosis, von Hippel–Lindau (VHL)	Acquired renal cystic disease; calyceal diverticulum

The extrarenal clue decides: hepatic fibrosis means ARPKD; berry aneurysm, ADPKD; retinitis pigmentosa, nephronophthisis; angiofibroma, tuberous sclerosis complex (TSC); haemangioblastoma, VHL.

4 · ARPKD

Huge echogenic kidneys and an always-fibrotic liver.

P3 Autosomal recessive polycystic kidney disease. — PKHD1 chr 6, fibrocystin; always congenital hepatic fibrosis; oligohydramnios, pulmonary hypoplasia, 30–50% neonatal death; homogeneously hyperechoic kidneys; scan parents; supportive care; ARegPKD early bilateral nephrectomy warning

Not in the paper answer:

- In later-onset disease, portal hypertension dominates.
- Nephrectomy costs nephron mass and brings dialysis earlier, so delay it where possible.

5 · ADPKD — paediatric focus

In children the key finding is hypertension with normal function.

P2 ADPKD (Adult polycystic kidney disease) — PKD1 (16, 85%, ESRD 50s) vs PKD2 (4, 15%, ESRD 70s); pain, haematuria ≈50%, stones 20–30%, hypertension ≈50% at 20–35 y; liver cysts, aneurysm 10–30%; Ravine criteria; BP <130/80 with ACEi/ARB, tolvaptan (TEMPO, REPRISE), lipophilic antibiotics, unroofing

Not in the paper answer:

- **Contiguous gene syndrome**: deletion of PKD1 and the adjacent TSC2 gives TSC with severe early polycystic kidneys.
- A negative US in a young person does **not** exclude PKD2, so use genetic testing after counselling when certainty matters.
- Screen for aneurysm by MR angiography only for family history or high risk. Observe <7 mm; coil or clip if >7–10 mm or symptomatic.

> **KEY POINT**
> ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

6 · Juvenile nephronophthisis (NPH) / medullary cystic kidney disease (MCKD)

A salt-wasting cause of renal failure that ultrasound often misses.

- These ciliopathies give tubular atrophy, corticomedullary cysts and interstitial fibrosis in small-to-normal kidneys. **Polyuria and polydipsia occur in >80%**, with **salt wasting**. ESRD

follows 5–10 y after presentation.

	Juvenile NPH	MCKD
Inheritance	Autosomal recessive	Autosomal dominant
ESRD	Mean 13 y; **10–20% of childhood renal failure**	Third–fourth decade
Hypertension	Absent	Present
Extrarenal	20%: **retinitis pigmentosa (Senior–Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl	None; type 2 — hyperuricaemia, gout in teenage males

- CT or MRI shows the cysts better than US. **Replace sodium, and avoid salt restriction and diuretics**. Transplant works, with no recurrence.

7 · Syndromes with renal cysts

Tuberous sclerosis complex (TSC)

- It is autosomal dominant: **TSC1 (chr 9), TSC2 (chr 16)**. Diagnosis needs 2 major features, or 1 major and 2 minor (angiomyolipoma is major).
- Kidney: cysts and **angiomyolipoma (AML) in 40–80%**, bilateral and multiple. AMLs never metastasise; the risk is haemorrhage. RCC occurs in ≈2%; suspect it in a growing fat-free lesion.
- **MRI is preferred**, because US misses 25–30% of fat-poor AMLs.
- **If an AML is >3 cm and growing, give everolimus (a mammalian target of rapamycin, mTOR, inhibitor) first line**. For an acute bleed, embolise and then give corticosteroids. Kidney-sparing surgery is second line; avoid nephrectomy. Image every 1–3 y, check BP, proteinuria and GFR yearly, and treat hypertension with an ACE inhibitor or ARB.

Von Hippel–Lindau disease

P2 VHL gene and von Hippel–Lindau disease — AD, 3p25, pVHL degrades HIF, which drives VEGF; lesion frequencies (cysts 76%, clear cell RCC, phaeo, haemangioblastoma, epididymal cystadenoma); diagnostic criteria; VHL Alliance 2020 surveillance (abdominal MRI 2-yearly from 15); nephron-sparing surgery, recurrence 75–85%, DSS 81–94%; partial adrenalectomy

Not in the paper answer:

- Partial adrenalectomy for a small phaeochromocytoma avoids lifelong steroids.

8 · Multicystic dysplastic kidney (MCDK)

The MCDK is harmless; watch the opposite kidney.

- It has non-communicating cysts, **no normal parenchyma, no function** and an atretic ureter ("bunch of grapes").
- It is the **commonest renal cystic disease of the newborn and the second commonest neonatal abdominal mass**: 1 in 1000–4000. Bilateral disease is lethal.
- **The contralateral kidney** hypertrophies and has **vesicoureteral reflux (VUR) in 18–43% and pelviureteric junction (PUJ) obstruction in 3–12%**.
- About 40% involute (later imaged as "agenesis"). Hypertension and Wilms tumour are not increased.

US feature	MCDK	Severe PUJ obstruction
Cysts	Haphazard, varied, non-communicating	Calyces around central pelvis, communicating

US feature	MCDK	Severe PUJ obstruction
Central large cyst	Absent	Present (pelvis)
Isotope uptake	None	Usually some

ALGORITHM

1. For a prenatally detected MCDK, do US on day 2–3 and at 1 month to assess the contralateral kidney and bladder.
2. If the diagnosis is in doubt, do a DMSA or MAG3 scan.
3. If there is a urinary tract infection (UTI) or an abnormal contralateral kidney or bladder, then do a voiding cystourethrogram (VCUG); otherwise do not.
4. Hand over to primary care to check BP, mass and UTI.

- **Nephrectomy** is only for mass effect, hypertension, cyst rupture or **increasing solid tissue**. Routine US surveillance has no proven benefit.

KEY POINT

MCDK removes itself; management looks to the solitary contralateral kidney and its reflux or obstruction.

9 · Benign multilocular cyst (cystic nephroma)

Imaging cannot exclude cystic Wilms tumour, so excise.

- **It is bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into the pelvis)**.
- **Excision is both diagnosis and treatment** (nephrectomy in children; partial if localised).

10 · Simple cysts in children and Bosniak

- **US criteria (all four)**: anechoic, thin smooth wall, posterior enhancement, round or oval. On CT a simple cyst is −10 to +20 HU without enhancement. A hyperdense cyst is benign if non-enhancing, homogeneous and ≤3 cm.
- **Children**: if a simple cyst is <15 mm and asymptomatic, repeat US at 2 y and discharge if stable. Continue follow-up if it is ≥15 mm or complex, or with a family history or genitourinary anomalies.

Bosniak	Features	Management
I	Simple	None
II	Fine septa, minimal calcification, hyperdense ≤3 cm	None
IIF	More septa/calcification, no measurable enhancement	Follow-up (≈11% progress)
III	Thick/irregular enhancing walls or septa	Excision (paediatric III–IV: >90% intermediate/malignant)
IV	Enhancing solid components	Excision

Treat a symptomatic benign cyst by aspiration with sclerosis or laparoscopic unroofing.

11 · Medullary sponge kidney, glomerulocystic and acquired cystic disease

P2 Medullary sponge kidney — ectatic collecting ducts, urographic triad ("brush bristles", "bouquet of flowers"), presents >20 y, colic 50–60%; hypercalciuria ⅓–½, incomplete dRTA 30–40%; potassium citrate first, thiazide, phosphate (not with urease infection)

- **Sporadic glomerulocystic disease** is a diagnosis of exclusion.

P2 Acquired renal cystic disease — ≥3 cysts/kidney in ESRD; uraemia not dialysis; 10% pre-dialysis, rising to >90% at 10 y.

Small kidneys; haemorrhage, adenomas, RCC (aggressive in native kidneys after transplant); bleeding: embolisation/nephrectomy; tumour >3 cm excise, <3 cm nephrectomy or yearly CT.

Not in the paper answer:

- Acquired renal cystic disease affects 20–45% of children on dialysis, and a retroperitoneal bleed calls for nephrectomy because an occult RCC is likely.
- **Parapelvic cysts** mimic hydronephrosis on US; delayed contrast CT shows an undilated system, and management is conservative.

12 · Landmark trials

Trial	Design	Result	Take-home
HALT-PKD	Randomised: lisinopril and telmisartan vs lisinopril; rigorous vs standard BP	Dual blockade no benefit; low BP slowed total kidney volume (TKV) growth, lowered left ventricular mass and albuminuria, GFR unchanged	Single renin–angiotensin blocker; tight BP early
TEMPO / REPRISE	Randomised, tolvaptan vs placebo	Slower TKV growth and GFR decline	First disease-modifying drug; aquaresis, hepatotoxicity
ARegPKD registry	Observational ARPKD cohort	Bilateral nephrectomy <3 months gave the worst neurological outcome	Avoid early bilateral nephrectomy

51 Pathophysiology of Urinary Tract Obstruction

1 · Definition and classification

Function, not dilation, defines obstruction.

- **Obstruction** is any impediment to flow, calyces to meatus, that injures the kidney or lower tract: **acute or chronic, partial or complete, intrinsic or extrinsic, unilateral or bilateral**.
- It affects ≈ **1 in 500** live births; high-grade obstruction carries a lifetime end-stage renal disease (ESRD) risk of up to **30%**.

	Lower urinary tract obstruction (LUTO)	Upper tract
Commonest	Posterior urethral valves (PUV) and urethral atresia ≈ 50%	Ureteropelvic junction obstruction (UPJO), 1 in 1500; M:F 2:1
Others	Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis, trisomy 18/21	Ureterovesical junction obstruction (UVJO), ectopic ureter, ureterocele, polyp, stones, crossing vessel
Outcome driver	ESRD and pulmonary hypoplasia	Unilateral function loss; contralateral kidney buffers

Presentation is prenatal hydronephrosis (mostly at 18–20 wk), or later infection, mass, pain or respiratory failure.

> **KEY POINT**
> Trap: **detrusor hypertrophy** causes UVJ obstruction by narrowing the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

2 · Haemodynamic changes

Rising pressure lowers GFR and renal blood flow (RBF), causing ischaemia. Injury begins within **2 h** of complete obstruction, worse if bilateral or solitary.

† Phase (complete)	Unilateral	Bilateral / solitary
I (0–1.5 h)	RBF rises, ureteric pressure rises (afferent dilatation: PGE2, nitric oxide)	Modest RBF rise
II (1.5–5 h)	RBF falls, pressure still high (efferent constriction)	RBF falls, pressure rises
III (> 5 h)	RBF and pressure fall (afferent constriction: angiotensin II, thromboxane)	Pressure stays high (retained solute, ANP), the basis of post-obstructive diuresis

- **Hypertension** is renin driven: bilateral ≈ **77%**, unilateral ≈ **20%**.

3 · Tubular function

- **Aquaporins and sodium transporters are downregulated**, causing a concentrating defect, salt wasting, polyuria and **distal renal tubular acidosis**.
- Potassium and other solutes are **wasted early and retained late**. FENa is < 1% in acute obstruction and > 1% with tubular injury.

4 · Pathology and fibrosis

- Fibrosis is the final common path, driven by stretch and **TGF-β/Smad**, amplified by angiotensin II.
- **Bladder** hypertrophies, compensates, then decompensates with poor compliance. Storage pressure > **40 cm H_2O** injures the kidneys.

5 · Recovery and compensatory growth

- Relief can reverse damage, but **incomplete recovery is common**.
- Compensatory growth is **glomerulomegaly, not new nephrons**.

6 · Post-obstructive diuresis (POD)

P1 Post-obstructive diuresis — unilateral ≈ **2%**, bilateral ≈ **50%**; aquaporin/transporter loss, medullary washout, retained solute; risk after valve ablation (oligohydramnios, preterm, creatinine); monitoring, partial replacement, avoid overcorrection

Not in the paper answer:

- Preterm infants have low aquaporin-2, hence their POD risk.

ALGORITHM

1. Monitor intake, output, and serum and urine electrolytes.
2. Replace losses and correct electrolytes.
3. **Do not over-replace**: it perpetuates POD.
4. Expect resolution within **24 h**; it seldom lasts > 48 h.

7 · Fetus and child

P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — UTD APD thresholds (**≥ 7 mm** at 16–27 wk, **≥ 10 mm** ≥ 28 wk), serial fetal urine (Na < 100, Cl < 90, osmolality < 200, β2-microglobulin < 6) Fetal intervention: shunt/fetal cystoscopy, amnioinfusion

Not in the paper answer:

- Second-trimester oligo-or anhydramnios carries **60–80%** mortality.
- In the Whitaker test, obstruction is a pelvic pressure > **12 cm H_2O** or a continuous rise.
- After birth, use ultrasound first, MAG3 for drainage, DMSA for scars and VCUG for outlet and reflux.

8 · Treatment principles

Drain to buy function, then reconstruct.

Option	Indication / key point
Clean intermittent catheterisation (CIC)	Neurogenic bladder; reduces UTI, stones, retention
Overnight catheter drainage	Valve bladder with nocturnal polyuria and progressive CKD
Vesicostomy	Infant with complex LUTO, unstable or unable to do CIC
Percutaneous nephrostomy	Upper tract; needs some hydronephrosis
Stent / cutaneous ureterostomy	Temporising; stent excellent for extrinsic obstruction
Pyeloplasty / reimplantation	Definitive; anastomose anterior to aberrant vessel; non-refluxing reimplant in children

- Use NSAIDs only if function is preserved. ESRD needs dialysis or transplant (Ch 61).

52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

1 · Definitions and classification

- Ectopic ureter and ureterocele are one defect of **ureterotrigonal connection** on a continuum. The appearance changes the technique, not the strategy.
- An **ectopic ureter** is a single or duplex ureter that does not enter the trigone but enters **at or distal to the bladder neck**; merely opening near it is not ectopic.
- A **ureterocele** is **cystic dilation of the distal ureter** within the bladder or spanning bladder neck and urethra. It is never entirely urethral and never attached to wolffian structures.
- In a **duplex system, both lesions always involve the upper pole** ureter.

	Ectopic ureter	Ureterocele
Distal ureter	Dilated or not, opens ectopically	Cystic dilation
Position	At/distal to bladder neck; perineum, vagina, rectum (F) or wolffian ducts (M)	Bladder or spanning bladder neck and urethra
Wolffian attachment	Yes (male)	Never
Incontinence (untreated)	Yes, in females	No

Ureterocele classification

P3 Ureterocele — Glassberg intravesical vs ectopic (most useful, predicts TUI outcome); Stephens incl. cecoureterocele; Churchill functional

Not in the paper answer:

- A **cecoureterocele** tunnels submucosally beyond the bladder neck into the urethra; it is often missed before surgery and risks incontinence.
- **Ureterocele disproportion** is a typical ureterocele with a small, dysplastic, non-dilated upper pole, so the kidney looks normal.

2 · Embryology essentials

Bud position explains upper pole obstruction and lower pole reflux.

- In a duplex, the upper pole bud arises **later and more cephalad** on the wolffian duct, so its orifice is carried **caudally** and becomes ectopic.
- **Weigert–Meyer rule**: **the upper pole orifice lies caudal (and medial) to the lower pole orifice**. So the upper pole gets ectopia or ureterocele (obstruction), and the lower pole gets a short tunnel and vesicoureteral reflux (VUR).
- Classically, a ureterocele results from failure of the **Chwalle membrane** to rupture. Renal dysplasia lies **downstream** of the connection defect.
- An ectopic ureter reaches the vagina, cervix or uterus only through the **Gartner duct**, never directly into müllerian structures.

3 · Ectopic ureter — sites and sex asymmetry

P3 Embryology and management of ectopic ureter. — female vs male sites, continuous dribbling vs never incontinent, epididymitis, Zinner syndrome, Gartner duct

Not in the paper answer:

- A **single-system ectopic ureter** may mimic an "absent kidney" on ultrasound; use contrast CT or a renal function study.
- **Bilateral single-system ectopia** (rare) gives a hypoplastic bladder, poor bladder neck and bilateral dysplasia.

> **KEY POINT**
> Why does only the girl leak? The female ureter may open distal to the external sphincter and leaks continuously; the male ureter, confined to wolffian derivatives above it, presents as infection or obstruction.

- Male sites are the bladder neck, prostatic urethra, vas, seminal vesicle and ejaculatory duct.

4 · Clinical presentation

Prenatal

- Most are found as prenatal hydronephrosis. A fetal upper pole "cyst" is upper pole hydronephrosis until proven otherwise.
- **The diagnosis is made at the bladder**, because the upper tracts look identical; a large ureterocele can mask the bladder.
- Dilation of the ipsilateral lower pole or contralateral kidney suggests reflux. Dilation of the whole kidney and ureter suggests a single system.

Incidental and infection

- A "presumed ovarian cyst" may be a dilated ureter, so include it in the differential of a cystic pelvic mass.
- Do ultrasound in every child with urosepsis. If sepsis does not respond, do an **urgent transurethral incision (TUI)** of a ureterocele, or drain an ectopic ureter (usually end ureterostomy).
- An infected ectopic ureter gives low-grade spiking fever, and a **negative culture does not exclude infection**.
- **Any young male with bacterial epididymitis needs upper tract ultrasound**; it is often recurrent before diagnosis.

Incontinence

- Suspect it in a toilet-trained girl with **continuous low-volume dampness day and night** and normal voiding. Untreated ureteroceles do not cause incontinence.
- The key question is: **can she be dry for 30–60 minutes?** With an ectopic ureter she cannot.
- The leaking moiety may be non-dilated and missed on imaging, so suspicion rests on the history. Exclude voiding postponement and constipation.

Ureterocele prolapse

- It appears as a **smooth, congested, mucosa-covered interlabial mass**, often with voiding difficulty. A lobulated grape-like mass suggests **sarcoma botryoides** instead.
- Ultrasound confirms it; VCUG shows the prolapse, lower pole reflux and the "drooping lily".

Late presentation

- Teenagers and adults present with infection or pain; an adult single-system ureterocele often contains a **stone**. Upper pole heminephrectomy is the mainstay.

5 · Evaluation

- Examine the perineum of every continuously wet girl. The ectopic orifice lies **between the urethral meatus and the vagina**.

Imaging

Test	Answers	Note
Ultrasound	Anatomy; inferred function; bladder	First always; cannot separate ectopic ureter from ureterocele above bladder
Voiding cystourethrogram (VCUG)	Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra	Optional if obstruction alone dictates incision; repeat after incision
Dimercaptosuccinic acid (DMSA)	Differential function — gold standard	Only "some vs none"
Mercaptoacetyltriglycine (MAG3) renogram	Function and drainage	Observation candidates; after 2 months
MR urography	Most detailed anatomy and function	Complex anatomy, non-dilated upper pole; needs sedation
Intravenous pyelogram (IVP)	Anatomy and qualitative function	Non-dilated ectopic ureter with incontinence

- A **ureterocele** is a thin-walled cyst within the bladder. A **pseudoureterocele** is a dilated ectopic ureter indenting the bladder; its wall is **bilaminar and thicker, and its lumen lies outside the bladder**.
- A full bladder can efface a ureterocele so it mimics an ectopic ureter, so scan through filling and voiding.
- On VCUG, the **"drooping lily"** is the lower pole pushed down and out by a dilated upper pole.
- † A single-system intravesical ureterocele gives a **"cobra head" (spring-onion)** sign on urography.
- **Preoperative function does not predict function after decompression**.

VCUG terms

Term	Meaning	Significance
Effacement	Flattens with filling	Mimics ectopic ureter
Prolapse	Moves into urethra	Outlet obstruction; predicts secondary surgery
Eversion	Diverticulum-like protrusion of bladder base	**Weak trigonal floor, so open repair likely**

Endoscopy

- **Choose the incision site** with little bladder filling, when the lowest part of the ureterocele is seen best.
- Detect any urethral extension (cecoureterocele) so the incision leaves no obstructing segment.
- A patulous bladder neck or cecoureterocele carries a later incontinence risk; consider bladder neck tightening.

6 · Management principles

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total morbidity**.
- Start antibiotic prophylaxis early in significant hydronephrosis, even without reflux.
- Relieving obstruction may **create** reflux, or may let lower pole reflux resolve. This is the central trade-off, and decisions are empirical.
- **Fetal intervention**: none for ectopic ureter; for ureterocele only if it obstructs all units (experimental).

Neonatal algorithm

ALGORITHM

1. After prenatal hydronephrosis, do postnatal ultrasound.
2. If there is significant hydroureteronephrosis, start prophylaxis.
3. Do VCUG (optional if it will not change the decision to decompress), then a renal scan after 2 months.
4. If the ureterocele is small, with minimal hydronephrosis and no obstruction, observe.
5. If obstruction is significant and parenchyma is at risk, do TUI (or cutaneous distal ureterostomy if endoscopy is impossible).
6. If sepsis does not respond, do TUI (ureterocele) or nephrostomy or end ureterostomy (ectopic ureter); elective drainage is still needed later.

- **Observation** suits selected ureteroceles: no obstruction of the lower pole or contralateral kidney and lower pole reflux ≤ grade III, a non-functioning upper pole, or no obstruction on renogram. Counsel about unpredictable acute presentation.

7 · Operative options

Transurethral incision (TUI) of ureterocele

- TUI relieves obstruction in **78–97%**. It is the choice in **sepsis** and in the **infant**.
- Make a **low transverse full-thickness incision as distal and close to the bladder floor as possible** to reduce reflux. For an ectopic ureterocele, extend it into the urethral part so no obstructing flap remains.
- The **"watering-can"** technique (multiple holmium punctures) decompresses equally with less de novo reflux.
- It is a day-case procedure. Do ultrasound and VCUG at **4–6 weeks**.
- New reflux occurs in **0–50%**. **Intravesical** ureteroceles decompress without reflux in 70–80%; **extravesical** ones need secondary surgery in up to 70%. Lower pole reflux (risk ratio 1.74) and an extravesical site (2.78) predict reoperation.

KEY POINT

Why TUI first even if half need more surgery? It is outpatient, may be definitive, defers the second operation to an older child, and decompression often avoids tapering at reimplant. It is less appealing in an older child with a massive non-functioning upper pole, who needs definitive surgery at diagnosis.

Upper pole partial nephrectomy (heminephrectomy)

- It is for a non-functioning upper pole, by open, laparoscopic or robotic approach. The main concern is **not to damage the viable lower pole**.
- Use the upper pole ureter as a handle, passing it **behind the renal vessels**, ligate the upper pole vessels and resect at the demarcation.
- **Dissect on the wall of the upper pole ureter**, and leave its back wall on the lower pole ureter near the bladder, because they share a blood supply. A small distal stump is left; a large or refluxing one is excised low, sparing the bladder neck and vas.
- Outcomes: for ectopic ureter it is usually curative. For ureterocele, new reflux appears in **15–50%** and secondary surgery is needed in **40–50%**.

Ureteroureterostomy / pyeloureterostomy

- The upper pole is joined end-to-side into the lower pole system, for functioning and increasingly non-functioning upper poles; feared hypertension has not been substantiated.
- For a **distal** repair, **stent the lower pole ureter cystoscopically first**. A **proximal** repair is preferred for a dilated upper pole or size mismatch. Robotic repair matches open results, and size mismatch is no barrier.

Lower tract reconstruction

- **Ureterocele excision with common sheath reimplantation** relieves obstruction and reflux at once, and is the fallback for persistent reflux. It risks the bladder neck, sphincter and vagina.
- Mobilise **both ureters as a unit**, excise the ureterocele, **plicate the detrusor defect** and reimplant submucosally.
- A cecoureterocele remnant can act as an obstructing **urethral windsock** flap-valve.
- Reflux persists in **5–10%**, more often after tapering.
- **Total reconstruction** suits an **older child with a massive ureterocele, non-functioning upper pole and significant lower pole reflux**.

End ureterostomy for ectopic ureter

- It is for a neonate with sepsis or massive dilation. **Do not trim the redundant ureter**, because its length is needed for reimplant. Do DMSA after a few months, then reconstruct or remove.

8 · Decision summary

Upper pole function and lower pole reflux pick the operation.

Ectopic ureter (duplex)

> **ALGORITHM**
>
> 1. If the upper pole is worth keeping and the lower pole refluxes, do a common sheath reimplant, or a lower pole reimplant with distal ureteroureterostomy.
> 2. If the upper pole is worth keeping and there is no lower pole reflux, do a proximal or distal ureteroureterostomy.
> 3. If function is unclear, or there is massive dilation or sepsis, do a temporary end ureterostomy, then DMSA, then reconstruct or remove.
> 4. If removal is chosen, do upper pole nephrectomy with ureterectomy.

Ureterocele (Table 52.1)

Procedure	Ideal indication	Advantage	Limitation
TUI	Infant; large ureterocele with VUR	Outpatient; effective; occasionally definitive	Reflux into ureterocele moiety needs bladder surgery
Upper pole nephrectomy	Older; large non-functioning upper pole; no VUR	Removes pathology; avoids bladder	Lower pole risk; may still need bladder surgery
Ureteroureterostomy / pyelostomy	Older; functioning upper pole; no VUR	Drains with little risk	Ureterocele left; may reflux
Excision and common sheath reimplant	VUR; functioning upper pole without major dilation	Removes obstruction and reflux; no renal risk	Complex; vagina/bladder neck risk; tapering

KEY POINT

VUR is the pivot: without VUR choose an upper tract solution; with VUR, a bladder-level solution, except in the infant, where TUI defers bladder surgery. Upper pole function then picks nephrectomy or ureteroureterostomy.

9 · Complications

- **Loss of lower pole function** after heminephrectomy is the most significant; prevent it by preserving the shared blood supply and using gentle traction.
- **Urinoma** occurs in up to 20% after laparoscopic or robotic cases; observe it if asymptomatic.
- **De novo reflux** after TUI or heminephrectomy is observed if there is no infection; otherwise do excision and reimplantation.
- **Voiding dysfunction** comes from poor trigonal support, which causes obstruction, poor emptying and recurrent VUR. Assess with VCUG **including lateral voiding views**, and treat by reconstruction, CIC or bulking.

10 · Other ureteral anomalies

Duplication

- It occurs in ~**1 in 125** (0.8%), F:M 1.6:1, and is mostly unilateral. **Look for contralateral duplication**: an occult second ectopic ureter explains persistent wetting.
- A **bifid ureter** is a partial duplication joining below the ureteropelvic junction (UPJ).
- **The lower pole gets single-system disease** (UPJ obstruction, VUR); **the upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Lower pole VUR is often high grade and resolves slowly; treat by common sheath reimplant, endoscopic injection or low ureteroureterostomy.

Triplication

- Smith types: 1, three separate ureters (**35%**, commonest); 2, incomplete with two orifices (21%); 3, trifid with one orifice (31%); 4, two ureters with three orifices (13%).

Valves, diverticulum, polyp

- † **Ureteral valves** are transverse mucosal folds with smooth muscle and proximal dilation; excise the segment. A ureteral **diverticulum** is usually a **blind-ending bifid ureter**.

- A **fibroepithelial polyp** is commonest at the **UPJ**, **89% male, 78% left**. At the UPJ, do pyeloplasty; elsewhere, **remove it ureteroscopically**.

Vascular position anomalies

P1 P2 IVC anomalous development and retrocaval ureter; Retroileal Ureter — persistent subcardinal vein, right only, type I fishhook vs II, CT urography, transpose anterior; retroiliac at L5–S1 with mesonephric ectopia

Not in the paper answer:

- A retrocaval ureter is always right-sided and mimics UPJ obstruction, so a simple pyeloplasty fails.

53 Surgery of the Ureter in Children

1 · Pelviureteric junction (PUJ) obstruction — definition and aetiology

P1 Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management. — Koff definition; intrinsic (infant) vs crossing vessel (older, intermittent pain) vs secondary; bilateral 10–40%; associated anomalies up to 50%; SFU grades; MAG3 T½ cut-offs; one-third need surgery

Not in the paper answer:

- Paediatric hydronephrosis is a pelvic anteroposterior diameter (APD) of **12 mm**, but hydronephrosis is not obstruction.
- With a crossing vessel, **dismembering and transposing the ureter anterior to the vessel prevents recurrence**. For secondary PUJ obstruction from high-grade reflux, avoid reimplant and pyeloplasty at one sitting; stage them.
- For **lower-moiety PUJ obstruction in a duplex**, do ureteropyelostomy or pyeloplasty; if the moiety has ≤ **5–10%** function, do partial nephrectomy.

2 · Diagnosis and indications

Decide on function trend and dilatation, not on washout alone.

- Most are found on antenatal ultrasound (US) at ≈ 20 weeks.
- Neonatal renogram washout has false positives up to **44%**, so decide on the trend in differential renal function (DRF) and on dilatation, not on half-time (T½) alone.
- **Pyeloplasty indications**: increasing APD; low or falling DRF; breakthrough infection on prophylaxis; symptoms (pain; failure to thrive in infants; haematuria; rarely hypertension).

3 · Anderson–Hynes dismembered pyeloplasty

It is the **gold standard** for any access, with success ≈ **95%**. It preserves a crossing vessel, excises the adynamic segment and allows pelvic reduction.

ALGORITHM

1. Expose the PUJ and free the lower pole, looking for an accessory vessel.
2. Use bipolar diathermy sparingly and place stay sutures.
3. Decompress the pelvis with a 21-G needle, then dismember, with or without pelvic reduction.
4. **Spatulate laterally** into healthy ureter, keeping the stenotic segment as a "no-touch" handle.
5. Bring the apex of the spatulation to the inferior pelvis and make a tension-free, watertight 6-0/5-0 absorbable anastomosis.
6. Excise the stenotic segment last, irrigate out clot and cover with perinephric fat.

- For a long gap, mobilise the whole kidney. A drain is usually unnecessary.

Alternative	Best indication	Point
Foley Y-V-plasty	High insertion, **no vessel**	V base on dependent medial pelvis, apex at PUJ
Spiral flap	Long proximal stricture, dependent PUJ	Length:width ≤ 3:1

Alternative	Best indication	Point
Vertical flap	Dependent PUJ, large box-shaped extrarenal pelvis	Flap length must match gap
Vascular hitch (Hellström)	Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex and split function; peristalsing PUJ at surgery	Vessel fixed cranially in pelvic wall; no hypertension seen
Ureterocalicostomy	Failed repair, short scarred ureter, dilated lower calyx with thin cortex	Lower-pole nephrectomy exposes calyx; spatulated ureter joined over stent

Use flaps only when a tension-free primary anastomosis is impossible. Non-dismembered repair is inferior.

> **KEY POINT**
> Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

> **KEY POINT**
> One fixed principle, many routes: dismember, excise, spatulate, anastomose without tension. Failure is managed by repeating the principle.

4 · Access

- In **infants**, open surgery is preferred; **posterior lumbotomy** suits only infants and toddlers, while the flank approach suits any age.
- Laparoscopic results equal open, and robotic surgery has a shorter learning curve.

	Transperitoneal	Retroperitoneal
For	Easier landmarks; left transmesocolic	No bowel injury; leak extraperitoneal; shorter stay
Against	Bowel risk; urinary ascites causing ileus	Learning curve; small space
Note	Preferred for horseshoe kidney, retrocaval ureter	One randomised trial favoured it; table **not flexed**

- **Endopyelotomy** does poorly long-term under 18 years; its niche is **failed pyeloplasty** without a vessel, with a double-J stent for 6 weeks.

5 · Stents, outcomes, failure

Stent	Removal	Point
Internal double-J	Second anaesthetic unless string/magnet	Catheter 24–48 h; string gives more febrile UTI (girls)
Externalised nephroureteric	7–14 days, no anaesthesia	Fewer complications, shorter stay in infants
Stentless	—	Needs flawless technique

- Stent duration of 1–4 weeks does not change outcome.
- For emergency diversion before repair, use a **nephrostomy, not a retrograde stent**, because a stent inflames the PUJ.

- Success is 90–100%; the commonest reason for conversion is bleeding.

P1 Management of failed pyeloplasty — early failure = oedema/clot: wait with or without nephrostomy; late: endopyelotomy or redo transperitoneal pyeloplasty (best); ureterocalicostomy for short scarred ureter

6 · Primary megaureter

Most resolve; the challenge is spotting the truly obstructed.

P3 Primary obst. megaureter. — ≥ 7 mm; four types; adynamic juxtavesical segment maturing craniocaudally; < 10 mm rarely needs surgery; indications incl. DRF < 40% or fall ≥ 5%; temporising options; excise–taper–5:1 reimplant; Starr/Kalicinski/Hendren; success > 90%

Not in the paper answer:

- In infancy the small bladder cannot take a safe tunnel for a massive ureter, so temporise with a **distal loop cutaneous ureterostomy** or a **refluxing reimplant**; balloon dilatation has weak evidence.
- Follow into adulthood, since some worsen at puberty.

KEY POINT

Why not rely on the renogram? Neonatal washout depends on immature function, hydration and a huge ureter's capacity, so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures.

ALGORITHM

1. Excise the stenotic distal segment and straighten the ureter.
2. Taper if needed.
3. Reimplant non-refluxing at a **5:1** tunnel length:diameter.
4. Stent with a feeding tube for 7 days or a double-J for 4 weeks.

- Plication and folding preserve the blood supply; Hendren excisional tapering risks ischaemic stricture.
- Treat postoperative obstruction with a stent first; an ischaemic stricture needs redo.

7 · Surgery for vesicoureteral reflux

KEY POINT

Principles: exclude or treat bladder causes first; handle ureter and bladder gently; mobilise without tension; keep a tunnel length:diameter of **5:1** (Paquin); ensure a straight course with the bladder empty and full.

- Techniques are intravesical or extravesical, with the tunnel suprahiatal or infrahiatal.

Technique	Approach / tunnel	Strength	Weakness
Politano–Leadbetter	Intravesical; suprahiatal neo-hiatus, vertical tunnel	Long tunnel for high grade	Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior
Glenn–Anderson	Intravesical; infrahiatal toward bladder neck	Least kinking	Short tunnel

Technique	Approach / tunnel	Strength	Weakness
Cohen	Intravesical; cross-trigonal	Commonest; success **98%**; thick bladder; with bladder-neck reconstruction	Difficult later retrograde catheterisation

Technique	Approach / tunnel	Strength	Weakness
Lich–Gregoir	Extravesical detrusorotomy ≈ 5 cm	No haematuria/spasm; short catheter	Transient voiding dysfunction/retention up to **10%**, esp. bilateral
Robotic extravesical (RALUR-EV)	Y-shaped detrusorotomy 4–5 cm	Less pain, spasm, stay	Early series: leak, obstruction, ureteric injury

Cohen — key steps

ALGORITHM

1. Open the bladder vertically and put a feeding tube in the orifice as a handle.
2. Circumscribe the orifice and free the ureter, preserving its distal blood supply, until it reaches the contralateral wall without tension.
3. Make a submucosal tunnel across the trigone, with the new orifice above the contralateral one.
4. Suture mucosa to mucosa with 6-0, with a detrusor hitch at 6 o'clock.

Lich–Gregoir — key steps

ALGORITHM

1. With the bladder one-third full, divide the obliterated umbilical artery.
2. Free the ureter **hugging it**, only as far as the UVJ.
3. Distend the bladder and mark a 5-cm tunnel; incise the detrusor down to mucosa and raise wide flaps.
4. Lay the ureter in and close the detrusor caudocranially, catching adventitia at the apex.
5. Leave a Foley catheter for 12–48 h.

KEY POINT

Why voiding dysfunction? The pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ, so stay on the ureter and stop at the UVJ.

- In the American Urological Association (AUA) meta-analysis, open success is **95.9%**; minimally invasive results are converging.
- **Follow-up** is US at 6–12 weeks. A VCUG is needed for high grade, new hydronephrosis, UTI or a dysfunctional bladder.

Complications of reimplantation

	Early	Long-term
Reflux	Ipsilateral low grade usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves, so observe; intervene for pyelonephritis. Routine bilateral repair not justified	Short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic/valve bladder): treat the bladder first (anticholinergic and clean intermittent catheterisation)

	Early	Long-term
Obstruction	Oedema, clot, haematoma, kink within 2 weeks: double-J or nephrostomy; most resolve	Ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, stenosis: balloon and stent, then re-reimplant

- Contralateral reflux risk rises with age < 6 years, capacity < 50% of expected and a refluxing duplex.
- At **reoperation**, excise ischaemic ureter back to healthy tissue; for a short ureter use a **psoas hitch**, Boari flap or transureteroureterostomy.

> **KEY POINT**
> Early complications are tissue reactions, so drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder.

Endoscopic injection

P3 Endoscopic correction of VUR (STING procedure): indications and complications — Dx/HA standard, ≈ 90% low grade; mound within 0.1–0.2 mL, volcano; never cauterise; STING vs HIT vs double HIT; late obstruction (Vantris); Swedish trial 20% recurrence at 2 years

Not in the paper answer:

- After failure, repeat the injection once or twice, then reimplant; the encapsulated implant does not make reimplantation harder.

8 · Other ureteric lesions

- † **Retrocaval ureter** is right-sided with a fish-hook shape; re-anastomose it in front of the cava.
- A **congenital mid-ureteric stricture** is often misread as megaureter; excise it and do a ureteroureterostomy.
- A **fibroepithelial polyp** (upper third, male, left) causes pain and haematuria; remove it endoscopically.
- **Retroperitoneal fibrosis** is treated with steroids, then azathioprine, then a stent, and finally ureterolysis.

56 Hypospadias

1 · Definition, diagnosis and embryology

- The **triad** is an ectopic ventral meatus, ventral curvature and a dorsal hooded foreskin (the last two not always present). Incidence is ≈ **1 in 150–300** male births.
- **Megameatus with intact prepuce (MIP)**, ≈ 5%, hides under a normal foreskin, so retract before newborn circumcision. Avoid circumcision when curvature or ventral skin deficiency is present.

P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — SRY, testosterone, 5α-reductase type 2, DHT at 8–12 weeks; endodermal two-zipper vs outdated Glenister ectodermal theory; dorsal prepuce fusion first

Not in the paper answer:

- The critical window is **weeks 8–16**. Ventral prepuce fusion awaits urethral fusion, so the foreskin is incomplete in **95%**. MCQ

> **KEY POINT**
> Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close over an open urethra, and dysplastic ventral tissue tethers the shaft into curvature.

2 · Aetiology

It is **multifactorial**, but all causes mean too little androgen action in weeks 8–16.

- Placental insufficiency and growth restriction double the risk; assisted reproduction is an independent risk.
- Twin concordance is 9–27%, and chromosomal abnormality is found in up to 30%; about 200 syndromes include it. Endocrine disruptors are unproven in humans.
- It belongs to **testicular dysgenesis syndrome**, and a shorter anogenital distance (AGD) means more severe disease.

3 · Classification and severity

- Distal forms are **70–85%**. The meatus is the least reliable marker, so **classify after degloving** by meatus, curvature and plate.

P4 Hypospadias: present concepts in management, and hypospadias fistula — GMS score table (glans, meatus, shaft; total 3–12, higher means more complications); POST low ratio favours inlay graft over plain TIP

Not in the paper answer:

- The glans–meatus–shaft (GMS) score (Merriman 2013) is scored in theatre after degloving.

4 · Associated anomalies and DSD work-up

- Hernia or hydrocele occurs in 9–16%, and undescended testis (UDT) in ≈ 7%.
- A **DSD work-up** is needed for UDT with proximal hypospadias, a perineal meatus or atypical genitalia. DSD risk is ≈ **50%** with a non-palpable vs 15% with a palpable testis, and 64% in proximal vs 7% in distal cases. Since 2006 severe hypospadias is classed as 46,XY DSD.
- Do ultrasound for Müllerian remnants, karyotype, testosterone, FSH, LH and AMH; a 46,XY karyotype does not exclude DSD (Ch 59).

- A large **prostatic utricle** causes stasis, UTI and difficult catheterisation. Routine VCUG and renal ultrasound are not needed.

> **KEY POINT**
> Hypospadias with cryptorchidism signals a systemic androgen problem, hence karyotype before surgery.

5 · Pre-operative assessment

- The goals are a straight erection, a urethra to the tip, good cosmesis and the fewest procedures. Exclude micropenis, and note that a deflected stream may justify surgery even when distal.
- **Timing**: the AAP advises 6–12 months, and the authors **6–18 months**, before genital awareness; younger repair has fewer complications.
- A glans width **<14 mm** means more complications. Give testosterone cypionate **2 mg/kg intramuscularly at 5 and 2 weeks** before surgery if the glans is <15 mm, which gains about 4 mm. Two randomised trials showed fewer complications, but meta-analyses are inconclusive; side effects resolve within 6 months.

6 · Intra-operative principles

- Use a regional block, ketorolac 0.5 mg/kg, antibiotics, and a tourniquet or epinephrine. Cystoscopy is only for severe proximal or redo cases.
- Rapidly absorbed fine sutures beat polydioxanone, whose slow absorption favours fistulae. Close tension-free with non-overlapping lines, and **never use hair-bearing skin**.

7 · Curvature

Curvature decides one-stage or staged repair.

It comes from ventral skin tethering, dysplastic dartos, a short urethra and corporal disproportion.

> **ALGORITHM**
> 1. Deglove to the penoscrotal junction and excise dysplastic dartos.
> 2. Make an **artificial erection**.
> 3. If straight, then proceed to urethroplasty.
> 4. If residual curvature is <30°, then do dorsal plication.
> 5. If >30°, then divide the urethral plate.
> 6. If still >30°, then do ventral corporal lengthening (usually staged).

- Mild is <30°, moderate 30–45° and severe ≥45°. Eyeballing underestimates curvature.
- Plication risks nerve injury, recurrence and shortening. Lengthening uses corporotomies, with or without a dermis or tunica vaginalis graft.
- Recurrence is 28% after plication vs 9% after grafting, and not dividing the plate is an independent risk, so correct fully the first time.

8 · Urethroplasty

Tubularised incised plate (TIP) repair is used for >90% of distal repairs.

Technique	Indication	Key point / pitfall
MAGPI	Glanular, mobile meatus, no curvature	Used more proximally gives a retrusive meatus
Urethromeatoplasty	Stenotic distal meatus	Heineke–Mikulicz
GAP (glans approximation)	Wide deep groove (MIP)	Two-layer neourethra
Thiersch–Duplay	Wide plate	Tubularise over 12 Fr
TIP / Snodgrass	Distal, midshaft; narrow plate; mild curvature	Midline relaxing incision; stop 2–3 mm short; meatal stenosis; inlay graft if poor plate
Mathieu (flip-flap)	Distal, shallow plate	Narrow base causes ischaemia, fistula; fewer stenoses than TIP
Onlay island flap	Plate kept but narrow	Diverticulum if too wide; less stenosis
Duckett tube	Proximal; plate divided	Higher complications; diverticulum
Two-stage Bracka	Proximal severe curvature, redo, poor plate	Stage 1 quilted preputial/buccal graft; stage 2 (≥6 months) tubularise
Two-stage Byars	Proximal severe curvature	Dorsal prepuce flaps; highest staged complications

- Staged repair (STAG) has the lowest proximal complication rate (≈ 23%), because urethroplasty over a fresh corporotomy or graft fails. Repeat the artificial erection at stage 2. Proximal TIP has been abandoned because of devascularisation and recurrent curvature.

KEY POINT

Why proximal repairs fail (Poiseuille): resistance rises with length / radius^4. A long, rigid tube ending in a non-distensible glans raises proximal pressure, causing fistula, diverticulum and stricture. 25–75% need extra procedures after planned two-stage repair.

- Glansplasty needs a tension-free **two-layer closure**, and a **barrier layer** (dartos or tunica vaginalis) reduces fistula.

9 · Post-operative care

- Leave a compressive dressing for ≈ **72 h**, and divert urine for **3–21 days** according to complexity. Short distal repairs need no stent. Give oxybutynin until the stent is out, with paracetamol and ibuprofen.

10 · Complications

Complication	Standard definition
Fistula	Urethral leak below meatus
Glans dehiscence	Wing separation; glans fusion <2 mm
Meatal stenosis	Obstructive symptoms and **<8 Fr** prepubertal / <12 Fr postpubertal
Diverticulum	Segmental sacculation on voiding
Recurrent curvature	**>30°** on erection

P4 Hypospadias: present concepts in management, and hypospadias fistula — commonest complication (<10%); distal obstruction key; wait 6–12 months; calibrate, stress test, layered closure with barrier; coronal fistula needs redo urethroplasty; recurrence ≈ 20%

Not in the paper answer:

- **Balanitis xerotica obliterans** needs complete excision and a staged graft. Recurrence is up to **40%**.
- **Meatal stenosis** is commoner after TIP than Mathieu and is treated by dilatation, then meatotomy or redo. A late **stricture** needs buccal urethroplasty, often staged.

> **KEY POINT**
> Nearly every complication traces to distal obstruction, ischaemia or tension, so repair begins by calibrating the distal urethra.

Reoperation

- Excise scar widely and transfer tissue rather than doing "small operations".
- **Buccal mucosa** is the graft of choice. Make it 10% larger, then tubularise ≥6 months later; lip mucosa suits the glans. Always do cystoscopy and an artificial erection.
- The **Cecil–Culp** procedure buries the penis in the scrotum when skin is short.

11 · Outcomes and follow-up

- A TIP meta-analysis (4,675 boys) found complications in 10.6% and reoperation in 4.5%. Distal success is 83–95%, while proximal complications are 23–68%.
- About 75% of complications occur by 3 years, so follow proximal cases **through puberty**. Recurrent UTI needs imaging for diverticulum, stricture or utricle.
- Proximal cases have worse adult sexual outcomes, and penile length is the chief complaint.

57 Etiology, Diagnosis and Management of the Undescended Testis

1 · Definitions

A retractile testis is watched; an undescended one is operated on.

P1 P3 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis; Maldescended (undescended) testes and treatment algorithm — undescended vs ectopic (SIP commonest), retractile, acquired (8–11 years), vanishing (monorchia/anorchia) vs agenesis

Term	Key point (beyond the papers)
Undescended testis (UDT)	Not in the normal scrotal position (midpoint at or below mid-scrotum). It is palpable or non-palpable; non-palpable means undescended or absent.
High scrotal testis	Stable, above the scrotal midpoint; a mixed group, not routinely called undescended.
Retractile	One that rarely stays down or rests high may progress to acquired UDT.
Recurrent	Undescended at birth, descended spontaneously, then re-ascended (10–22%, at 1–5 years).
Secondary / testicular retraction	Suprascrotal after hernia repair / after orchidopexy (iatrogenic).
Agenesis	Never formed; comes with ipsilateral Müllerian persistence (extremely rare).

2 · Embryology of descent

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — SRY/SF1, then SOX9, then Sertoli; INSL3 via RXFP2 + androgen; gubernaculum attaches to epididymis; transabdominal 10–15 weeks (outgrowth, CSL regression); scrotum by DHT at 10–12 weeks

Not in the paper answer:

- The gubernaculum does not pull the testis; it swells, then regresses.

Phase	Timing	Event	Control
Transabdominal (relative)	10–15 weeks	Gubernacular swelling (outgrowth) from 12 weeks; cranial mesonephric ligament regresses by 13 weeks	INSL3 and androgen act together
Inguinoscrotal	**20–28 weeks**	The canal widens and the processus forms; rapid passage through the canal; the gubernaculum then regresses. Most are scrotal after 27 weeks.	Androgen via the genitofemoral nerve (**GFN**), which releases calcitonin gene-related peptide (**CGRP**)

KEY POINT

Outgrowth creates the canal; regression lets the testis follow, and a defect in either gives the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

Minipuberty is a hormonal surge peaking at 1–3 months, when gonocytes become **adult dark (Ad) spermatogonia**, the stem-cell pool. This is where a positional defect becomes a germ cell deficit.

3 · Epidemiology, aetiology and associations

Spontaneous descent ends by about 6 months.

- Prevalence is **2–4% at birth** (term), falling to **1–2% at 3 months**; up to 45% in preterm boys.
- About 88% of high scrotal testes descend spontaneously, but only about half of extrascrotal ones.
- 80–85% are isolated. Risk factors are **prematurity and low birth weight**, family history, maternal diabetes and smoking. Diethylstilbestrol doubles risk.
- It is a polygenic threshold trait, and part of **testicular dysgenesis syndrome** with hypospadias, infertility and testicular germ cell tumour (TGCT).
- No hormone predicts germ cell number, so hormones are not tested routinely outside DSD and anorchia.

Associated condition	Cryptorchidism	Implication
Prune-belly syndrome	All (bilateral abdominal)	Abdominal orchidopexy
Spigelian hernia	53–80%	Absent gubernaculum/canal
Cerebral palsy	25–54%	Often bilateral
Gastroschisis	5–25%	**50–55% descend spontaneously — observe**
Omphalocele, imperforate anus, myelomeningocele, posterior urethral valves	12–33%	Caudal-field anomalies are ipsilateral in 90%
Congenital diaphragmatic hernia	7–30%	Always ipsilateral
Klinefelter (47,XXY)	> 50% of prepubertal Klinefelter boys	< 2% of non-syndromic UDT; more with hypospadias
Persistent Müllerian duct syndrome (PMDS)	Bilateral 59%; hernia uteri inguinalis; transverse ectopia 22%	AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk

Congenital vs acquired

Feature	Congenital	Acquired
Diagnosis	Birth/neonatal	Mean 8–11 years
Position	Often higher	Lower; often superficial inguinal pouch (SIP)
Processus vaginalis	Patent (87% unilateral)	Usually closed (patent 45–50%)
Epididymal anomalies	37–60%	11–31%

An acquired UDT is a never-fully-descended testis unmasked by growth, with semen results like congenital cases. Retractile testes share this pathology and ascend in about 2–3%, so they need **annual examination through childhood**.

4 · Histology, associated anomalies and fertility

Germ cells are lost from late infancy, so operate early.

- Germ cell loss begins **after infancy** and progresses; the contralateral testis is affected less.
- The UDT is smaller at birth, so the defect is intrinsic and worsened by position.

- Spermatogonia per tubule (S/T) is 2.5 at birth, with a nadir of 1.2 at 2–3 years.
- A **patent processus** is present in 87% of unilateral and 71% of bilateral cases. A UDT is at higher torsion risk.

Anomaly	Key features	Management
Polyorchidism	Usually left, scrotal in 75%; Kumar classification by drainage (A with vas, B without)	Observe a normal scrotal testis; orchidopexy if undescended with intact drainage
Splenogonadal fusion	Left side; the continuous form comes with limb defects and micrognathia	Recognise it to avoid a needless orchidectomy
Transverse testicular ectopia	Hernia with a contralateral non-palpable testis; PMDS in 20–50%	Laparoscopy; trans-septal placement if vasa fused

Trial	Design	Result	Take-home
Kollin et al.	Randomised trial: orchidopexy at 9 months vs 3 years	Early: more Sertoli and germ cells, better testicular growth (to 4 years)	Operate in first year; age at surgery, not hormones, predicts germ cells

Fertility outcome

Outcome	Former unilateral	Former bilateral
Oligospermia / azoospermia	31% / 14%	42% / 31%
Paternity	**≈ 90% of those attempting — as controls**	≈ 65% of those attempting (≈ 1/3 overall)

- In unilateral cases the normal testis carries fertility, so semen analysis overstates infertility; in bilateral cases it reflects true infertility.
- Ad spermatogonia at biopsy predict adult sperm counts; if they are absent, all counts are subnormal.
- Even after first-year surgery, 20–25% remain at infertility risk. Surgery **before 18 months** reduced assisted reproduction, and each 6-month delay added 5%.

5 · Malignancy

Orchidopexy lowers cancer risk only partly.

- TGCT risk is **2–5×** (pooled odds ratio (OR) ≈ 4). About 1% of cryptorchid men get TGCT, and 5–10% of TGCT patients had a UDT.
- In the **Swedish population study** the OR was about 2.8 if treated before puberty vs about 5.0 after. † Pettersson 2007: relative risk (RR) 2.23 if operated before 13 years vs 5.40 at 13 or older. A meta-analysis gave RR **5.8** for orchidopexy after 10–11 years.
- Untreated adults have OR ≈ 14.
- A persistently cryptorchid testis develops **seminoma** (74%), whereas tumours after orchidopexy are mostly non-seminoma (63%).
- Germ cell neoplasia in situ occurs in 2–3% of formerly cryptorchid men, so biopsy is for DSD and chromosomal cases. Biopsy at orchidopexy does not raise TGCT risk.
- Postpubertal UDT with a normal contralateral testis needs **orchidectomy** (up to age 50), with a prosthesis at the same sitting.

- **Microlithiasis** is not an independent TGCT risk, so routine ultrasound surveillance is not needed.

6 · Diagnosis

Examination decides; imaging adds nothing.

Examination — the gold standard

ALGORITHM

1. Warm room and hands. Examine supine, then cross-legged or squatting.
2. With lubricated fingers, sweep one hand from the anterior superior iliac spine (ASIS) along the canal and palpate the lowest position with the other.
3. Compress the canal and invaginate the scrotum to feel the SIP.
4. Apply sustained traction to fatigue the cremaster. If it **stays, it is retractile; if it springs back, it is undescended**.

- About **80% palpable**, and 60–70% are unilateral. At surgery a non-palpable testis is abdominal or peeping in 25–50%, vanished in 15–40% and a missed extra-abdominal testis in 10–30%.
- Examination under anaesthesia (EUA) is the final arbiter.

P3 P1 Maldescended (undescended) testes and treatment algorithm; Vanishing testes — AUA algorithm: bilateral non-palpable, then DSD; re-examine at 6 months corrected age; EUA, then yearly review / orchidopexy / exploration / laparoscopy; contralateral hypertrophy > 2 mL (specificity 100%) suggests monorchia, explore anyway

Imaging — not recommended

The AUA (2014) advises no ultrasound or MRI. Ultrasound sensitivity is only about 45%, and imaging merely delays surgery.

Bilateral non-palpable or UDT with hypospadias: DSD work-up

- A newborn with bilateral non-palpable testes needs an **urgent karyotype and hormones** to exclude **congenital adrenal hyperplasia (CAH)**, because of salt-wasting.
- Proximal hypospadias with UDT carries a chromosomal abnormality in 32–47% (see Ch 56).
- Micropenis with bilateral UDT suggests hypogonadotrophic hypogonadism or anorchia.

Marker	Anorchia	Bilateral abdominal UDT
FSH, LH	Raised	Normal
AMH, INHB (best prepubertal Sertoli markers)	Undetectable/low	Present
Basal testosterone	Low in first year only	Normal
Testosterone after hCG	No rise	Rises

hCG stimulation is no longer the test of choice, and **laparoscopy remains the confirmatory test**.

7 · Management principles and timing

- **Timing**: observe to **6 months** (corrected age), then refer; operate at **6–18 months**, preferably by 12.
- Retractile testes need no surgery but yearly examination. Acquired UDT should be operated on, especially if bilateral.

- A pubertal or postpubertal boy with a normal contralateral testis needs orchidectomy, with or without a prosthesis.

Hormonal therapy — not recommended (AUA)

- Intranasal LHRH gives descent in only about 20%, and hCG was never placebo-tested. hCG may increase germ cell apoptosis and reduce adult volume.
- Adjuvant LHRH for bilateral UDT with absent Ad spermatogonia (Hadziselimovic) has pilot data only.

> **KEY POINT**
> Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable. Descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

8 · Surgery for the palpable testis

The goal is a tension-free scrotal testis with intact vessels.

Inguinal orchidopexy — key steps

> **ALGORITHM**
> 1. EUA, skin-crease incision, check the SIP.
> 2. Open the external oblique, sparing the ilioinguinal nerve.
> 3. Free the testis; divide the gubernaculum distal to the sac (protect a long-looping vas).
> 4. **Separate the patent processus from vas and vessels without skeletonising; high-ligate it at the internal ring** (main length gain).
> 5. If short, divide lateral spermatic bands and dissect retroperitoneally; then the **Prentiss manoeuvre** (medial to the inferior epigastric vessels).
> 6. Transverse scrotal incision and **subdartos pouch**; pass the testis untwisted.
> 7. Close dartos around the cord; transalbugineal sutures are unnecessary.

- If the testis cannot reach the scrotum, stage the procedure. Orchidectomy is only for an atrophic or abnormal testis, a postpubertal boy or an insufficient vas.
- Biopsy is not routine; it is indicated in DSD and considered in bilateral UDT or after 12–13 years.

Scrotal (Bianchi) orchidopexy

The testis is delivered through a high scrotal incision, the processus ligated and the testis fixed in a subdartos pouch. It is increasingly first choice for palpable, especially low, testes.

9 · The non-palpable testis

Laparoscopic findings at the ring decide the next step.

ALGORITHM

1. EUA: if now palpable, do an inguinal or scrotal orchidopexy.
2. If **still non-palpable, do diagnostic laparoscopy** (ring patency, vessels, vas, testis).
3. If **blind-ending vessels** (with or without vas) lie above a closed ring, the testis has vanished: stop (the only unequivocal endpoint).
4. If **vessels and vas enter the ring**, explore the groin/scrotum and excise the nubbin (viable tubules 5–15%).
5. If the testis is **intra-abdominal**, do a primary laparoscopic or Fowler–Stephens (one-or two-stage) orchidopexy; orchidectomy if postpubertal or dysgenetic.
6. If it is peeping at the ring, do a laparoscopic or inguinal orchidopexy.
7. If neither vas nor vessels are seen, search the perivesical area and retroperitoneum to the kidney.

- An open ring usually, but not always, means a viable distal testis, and apparently blind-ending vessels may coexist with an ectopic testis, so keep exploring if equivocal.
- The AUA does not require fixing the contralateral solitary testis.

Laparoscopic orchidopexy

ALGORITHM

1. Empty bladder and stomach. Open (Hasson) umbilical port; CO2 at 8–12 mmHg. Inspect both rings and add working ports.
2. Divide the gubernaculum distal to the ring and use it as a handle.
3. Incise peritoneum lateral to the vessels and distal to the vas; mobilise the vessels, **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible).
4. Make a new hiatus medial to the inferior epigastric vessels.
5. Bring the testis through a trans-scrotal port to a subdartos pouch without tension.

- Ring closure is unnecessary, and the robot offers no clear advantage.

Fowler–Stephens (FS) and alternatives

P1 Methods of orchidopexy — FS on vasal/gubernacular collaterals for testes not within 2–4 cm of ring; success table: primary laparoscopic > 90–97%, Jones/Bagley ≈ 95%, FS 60–97% (one-stage 63%, two-stage 68%), Shehata 84–85% (atrophy 10%), autotransplant 96%; Dessanti PTFE traction

Not in the paper answer:

- Two-stage FS: clip the vessels high, wait about **6 months**, then mobilise on a wide peritoneal pedicle. Two-stage is slightly favoured.
- Keeping peritoneum over the vasal vessels and **sparing the gubernaculum** lowers atrophy. Avoid dividing vessels when possible, because it reduces germ cells.
- Operate on bilateral high testes about 6 months apart.

KEY POINT

Every abdominal technique answers one question: how to gain length without losing blood supply. FS sacrifices the testicular artery; traction gains length over time while keeping it, so vessel division is the last resort.

10 · Complications and follow-up

Complication	Cause / prevention	Management
Atrophy (≈ 2% modern; historically 24% abdominal/peeping vs 10% distal)	Vessel injury, skeletonisation, FS, tension	Offer a **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos
Retraction / non-scrotal (≈ 2–10%)	Inadequate retroperitoneal mobilisation, unligated processus	Redo: a prescrotal testis by the scrotal approach; otherwise a lateral approach through unscarred tissue or a fascial patch on the cord. Results equal primary surgery.
Vas injury	Long-looping vas cut with the gubernaculum	Divide the gubernaculum distal to the sac; avoid skeletonising
Torsion after orchidopexy	Rare	Prevented by subdartos fixation
Laparoscopic	Bladder/vessel injury, hypercapnia, bowel obstruction	Hasson entry, bladder drainage

Follow-up: examine at ≥ **6 months** for position and size. Counsel on fertility and cancer, and teach self-examination from puberty.

61 Urologic Considerations in Pediatric Renal Transplantation

P1 Urological evaluation and management of pediatric kidney transplant patient. — causes, five goals, pre-transplant screen (VCUG/urodynamics indications), bladder optimisation and cycling, augment segments, transplant technique, complication rates and management

Not in the paper answer:

- Graft reflux matters more than native reflux, because the kidney is solitary and immunosuppressed.

1 · Causes of ESKD in children

- The commonest cause of end-stage kidney disease (ESKD) is congenital anomalies of the kidney and urinary tract (**CAKUT**), then obstructive uropathy, mainly posterior urethral valves (**PUV**).
- Older children have FSGS or neuropathic bladder. The commonest age at transplant is **13–17 years**.

> KEY POINT
> **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

2 · Pre-transplant evaluation

- Every child needs a history and renal ultrasound, plus uroflow with post-void residual if toilet-trained.
- VCUG and urodynamics are selective: neuropathic bladder, PUV, reflux, prior surgery, febrile UTI or worsening hydronephrosis.

> ALGORITHM
> 1. Positive history: US and VCUG. If dynamics are abnormal, urodynamics.
> 2. Treat with drugs and clean intermittent catheterisation (CIC), with or without bladder cycling, then reassess.
> 3. If still unsafe, augment; then clear for transplant.

3 · Bladder management

The bladder must be safe before the graft goes in.

- The commonest pattern is **low capacity, overactivity and poor compliance**, and high pressure harms the kidney even without reflux.
- The **target** is expected capacity at detrusor pressure **< 40 cm H_2O**, with social continence.
- Use antimuscarinics, mirabegron or botulinum toxin, with CIC as needed.
- Cycle a **defunctionalised bladder** with rising saline volumes, then repeat urodynamics. **Never use the allograft to test a bladder.**
- **Augment** if the bladder cannot reach > **75%** of expected capacity at < 40 cm H_2O with CIC no more often than every **3–4 h** despite maximal therapy. Postpone this in PUV, but augment earlier in neuropathic bladder.

- A dilated ureter is the best segment. Bowel augments need lifelong malignancy surveillance.
- Site a **Mitrofanoff medial to the planned transplant incision**.
- Reconstruct **before** transplant, allowing **6–12 weeks** to heal, except when transplanting before 2 years.

4 · Native nephrectomy

P2 Pre transplant nephrectomy — default keep; indications (hypertension, nephrotic, polyuria, infection/VUR, stones, malignancy, size); retroperitoneal on PD, embolisation if high risk; leave normal ureter, remove dilated refluxing ureter

Not in the paper answer:

- Combine it with the transplant only to make room, as in a small child with huge kidneys.

5 · Transplant into the abnormal bladder

- A Gibson retroperitoneal incision is standard; a child < **15 kg** may need the aorta and vena cava through a transperitoneal approach.
- The preferred join is an **extravesical non-refluxing ureteroneocystostomy**. **Ureteroureterostomy** to the native ureter is the alternative and the salvage.
- Stent if difficult, removing the double-J at **4–6 weeks**.

6 · Urological complications

Complication	Rate	Timing / management
Ureteric stenosis	**6%**	Most within 100 days; distal, ischaemic. Focal: balloon and stent for 4–6 weeks; otherwise open repair (native ureter, bladder flap); do not delay.
Urine leak	**3%**	Early. Drain-fluid creatinine; check catheter, then US, then MAG3 scan or cystogram. Small: observe; large: drain or explore. Short necrosis: reimplant; long: native ureter, psoas hitch or flap.
Significant VUR	2%	See below
Stones	2%	Treat if > 1 cm and growing, obstructing or infected; one-stage modality

Rising creatinine with hydronephrosis

> **ALGORITHM**
>
> 1. Indwelling catheter for 1–2 weeks: if creatinine falls, the bladder is the cause.
> 2. If not, stent and keep the catheter; biopsy if rejection is suspected.
> 3. If retrograde stenting fails, nephrostomy and antegrade study.

- The graft is unusually sensitive to obstruction, and obstruction and rejection coexist, so stent at a low threshold.
- An **allograft mass is post-transplant lymphoproliferative disorder (PTLD) until proven otherwise**.
- Recurrent febrile UTI from graft reflux with an optimised bladder needs an **open or robotic reimplant**; asymptomatic reflux is observed.

7 · Outcomes

- Urological complications rarely lose the graft, but **≥ 2 UTIs** lower graft GFR at 1 and 4 years.

63 Paediatric Urologic Oncology: Renal and Adrenal

1 · Wilms tumour (WT): epidemiology and genetics

Genetics decide who is screened and who needs kidney-sparing surgery.

P2 Wilms tumour: classification, staging and management — genes (WT1 11p13, IGF2 11p15), syndrome risks, screening, anaplasia, SIOP risk groups, COG staging, COG vs SIOP, nodes, adjuvant by stage

Not in the paper answer:

- WT is **6–7%** of childhood cancers and **95%** of kidney cancers under 15 y; median age **3.5 y**, younger if bilateral or syndromic. Refer for genetics if bilateral, under 2 y, with rests, syndromic features or family history.
- WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability) is a WT1 deletion; **Denys–Drash** syndrome (DDS) is a **dominant-negative** point mutation with early end-stage renal disease. Type I tumours (WT1) are stromal with intralobar rests; Type II (11p15, Beckwith–Wiedemann syndrome, BWS) are blastemal/epithelial with perilobar rests.
- Combined 1p and 16q loss of heterozygosity (~5%) earns the intensified **regimen M**; 1q gain (25–30%) is the next stratifier.

Syndrome risk	Examples
High (> 20%)	WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy
Moderate (5–10%)	BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3)
Low	Li–Fraumeni, neurofibromatosis 1, Sotos, trisomy 18, Bloom

KEY POINT

Trap: BWS is the **commonest** predisposing syndrome but only moderate risk (21% bilateral); WAGR/DDS are rarer and highest.

ALGORITHM

1. If WT risk is > 1%, do a renal ultrasound (US) every **3–4 months** until **7 y**.
2. If a lesion looks suspicious, do computed tomography (CT) or magnetic resonance imaging (MRI).

- In BWS, hemihypertrophy and similar syndromes, add α-fetoprotein (AFP) 3-monthly to 4 y for hepatoblastoma.
- Screening lowers stage and allows nephron-sparing surgery (NSS), but survival benefit is unproven.

2 · Pathology

- Favourable histology is **triphasic**; the tumour is friable and spills easily.
- **Anaplasia** (chemoresistant): nuclei ≥ **3×** neighbours, hyperchromasia and abnormal mitoses; it is commoner in older children (~13% over 5 y).
- Nephrogenic rests are found in > 1/3 of WT kidneys and **> 90%** of bilateral cases; most regress. Perilobar rests are peripheral (BWS); intralobar rests lie anywhere (WAGR/DDS).

3 · Clinical features and work-up

- A varicocele persisting supine, or hepatomegaly, suggests inferior vena cava (IVC) thrombus. Gross haematuria needs cystoscopy at surgery.
- **Acquired von Willebrand disease** occurs in up to 8%; give desmopressin.
- Ascites beyond the cul-de-sac best predicts rupture; operate urgently only for bleeding. Imaging cannot judge nodes or separate WT from renal cell carcinoma (RCC) or clear cell sarcoma (CCSK).

4 · Staging and COG vs SIOP

COG operates first; SIOP gives chemotherapy first.

- Children's Oncology Group (COG) stages I–V are surgical–pathological. Stage III means **positive nodes, any spill**, implants, a positive margin or piecemeal removal.
- The International Society of Paediatric Oncology (SIOP) gives pre-operative vincristine (VCR) and dactinomycin (AMD) without biopsy.

> **KEY POINT**
> COG stage is surgical-pathological, so the surgeon, not the scan, decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

5 · Surgical principles

- Do a transperitoneal radical nephrectomy and inspect the liver, peritoneum and nodes.
- Sample nodes (COG **≥5 nodes**); omission is the commonest protocol violation and under-stages the tumour.
- Spill (~10%) causes local relapse; complications (~11%) are mainly haemorrhage and small-bowel obstruction.
- COG favourable histology: stage I–II get VCR and AMD; stage III–IV add doxorubicin (DOX) with flank/abdominal (and lung) radiotherapy (RT). Cyclophosphamide and etoposide are added in regimen M and for CCSK.

Bilateral WT (stage V) and NSS

P3 Bilateral Wilms tumour: treatment strategies — 5–7%; no upfront nephrectomy/biopsy; 3 drugs 6 weeks, then NSS by week 12 (NWTS-4 anaplasia); open bilateral biopsy (discordant 4/5); enucleation + flank RT; transplant after 1–2 y; AREN0534 EFS 82%, OS 95%

Not in the paper answer:

- Use frozen section and a normal-kidney margin in NSS.

> **ALGORITHM**
> 1. Synchronous bilateral WT: **no upfront nephrectomy, no biopsy**.
> 2. Give VCR, AMD and DOX for 6 weeks, then re-image.
> 3. If NSS is feasible, operate.
> 4. If not, give more chemotherapy, but do **surgery by week 12**.
> 5. If the tumour does not respond, do a **bilateral open biopsy**.

- Unilateral NSS suits syndromic children or a solitary kidney; UMBRELLA criteria are **<300 mL**, an adequate remnant and negative nodes.

IVC thrombus

- It occurs in ~6–10%. Below the hepatic veins, do upfront nephrectomy with **en bloc** thrombectomy (piecemeal means stage III).
- **At or above the hepatic veins, give pre-operative chemotherapy** to avoid bypass.

6 · Relapse and late effects

Relapse group	Definition	Salvage
Standard	Non-anaplastic, previously VCR/AMD only	70–80%
High	Non-anaplastic, after ≥ 3 drugs	40–50%
Very high	Anaplastic/blastemal	~10%

- Late effects track **RT and anthracycline**: heart failure follows DOX (rare below **150 mg/m²**). Renal failure after unilateral nephrectomy is only **0.6%** at 20 y.

7 · Other renal tumours

Tumour	Key facts	Management
Congenital mesoblastic nephroma (CMN)	Commonest **neonatal** renal tumour; ETV6–NTRK3 fusion in the cellular type	Radical nephrectomy alone; adjuvant only if incompletely resected
Clear cell sarcoma (CCSK)	3%; **bone** and brain metastases; late relapse; never bilateral	DOX-containing chemotherapy; RT even at stage I
Rhabdoid tumour (RTK)	2%; **most lethal**; SMARCB1/INI1 loss; associated brain tumours	No standard; tazemetostat on trial
Cystic nephroma / cystic partially differentiated nephroblastoma (CPDN)	Identical on imaging; CPDN has blastema in the septa; DICER1	Excision cures
RCC	Outnumbers WT above 13–14 y. **Translocation RCC (TFE3/TFEB, Xp11.2)** is commonest.	Complete resection; resistant to chemotherapy and RT
Angiomyolipoma	Tuberous sclerosis; bleeding risk if > 3 cm and growing	Mammalian target of rapamycin (mTOR) inhibitor first; embolise for bleeding

> **KEY POINT**
> Age points to diagnosis: a neonate has CMN, a toddler WT or rhabdoid, an adolescent RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

8 · Neuroblastoma

Biology, not radical surgery, decides outcome.

- It is the commonest **extracranial solid tumour** and commonest infant malignancy; median age is 19 months, 50% are adrenal and ~70% are metastatic at diagnosis.
- Paraneoplastic syndromes include watery diarrhoea from vasoactive intestinal peptide and **opsoclonus–myoclonus–ataxia**.
- Urinary vanillylmandelic acid and homovanillic acid (**VMA/HVA**) are raised in **90–95%**. **Calcification and vessel encasement** favour neuroblastoma over WT.

- 123I-meta-iodobenzylguanidine (MIBG) scanning is required for staging; MRI is best for intraspinal extension. Screening caused **overdiagnosis** without fewer deaths.

International Neuroblastoma Risk Group (INRG) stage	Definition
L1	Localised, **no** image-defined risk factor (IDRF)
L2	Localised, ≥ 1 IDRF (vessel/nerve encasement)
M	Distant metastases
MS	Age **<18 months**; skin, liver, marrow (< 10%) only; no bone (INSS 4S: < 12 months)

- INRG is imaging-based, before treatment; the International Neuroblastoma Staging System (INSS) depends on resection and is surgeon-dependent.
- **MYCN amplification** (> 10 copies; ~20%) means high risk at any stage. M at ≥ 18 months is always high risk.
- Adverse features are 1p36 deletion, 11q LOH, 17q gain and **diploidy**; favourable features are **hyperdiploidy**, TrkA expression and age < 18 months.

> **KEY POINT**
> Trap: diploid is bad, hyperdiploid good. Stage 4S/MS has OS 80–88% with frequent regression; infants < 2 months may die of massive hepatomegaly or coagulopathy.

- **Low risk**: resection alone (stage 1 > 90% disease-free); chemotherapy only for symptoms, progression or adverse biology. Never remove the kidney or vital structures.
- **4S**: resecting the primary is not mandatory. **Intermediate**: chemotherapy, then resection.
- **High risk**: induction chemotherapy, delayed surgery after 3–4 courses, myeloablative consolidation with autologous stem-cell rescue, RT 15–30 Gy, and **13-cis-retinoic acid with anti-GD2 antibody**.
- In stage 4, radical resection improves local control but not survival.
- **Cord compression**: give chemotherapy first; laminectomy only for progressive deficit, as it causes scoliosis.

> **KEY POINT**
> Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection, so vital structures are never traded for margin.

9 · Adrenocortical tumours

P4 Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — bimodal (children < 10 y, 90% functional); Li–Fraumeni/BWS; open en bloc, intact capsule; mitotane + EDP

Not in the paper answer:

- Offer **TP53** testing to every child. Androgen excess causing **virilisation** is the commonest presentation.
- Work-up includes DHEA-S, testosterone, 17-hydroxyprogesterone, cortisol tests, a **urinary steroid profile**, metanephrines, CT/MRI and chest CT.
- Weight > 200 g and spill are adverse. Give **perioperative steroids** for cortisol-secreting tumours.

KEY POINT

Why open? Cure needs complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

10 · Landmark trials

Trial	Design	Result	Take-home
AREN0532/0533	1p/16q LOH treated with regimen M	Stage III–IV event-free survival (EFS) ~90% vs 61%	Biology-driven intensification
AREN0533 (lung)	Lung nodule response at week 6	RT omitted, EFS and overall survival (OS) preserved	Response-adapted RT omission
AREN0534	Bilateral WT: 3-drug, surgery by week 12	EFS 82%, OS 95% (vs 56%/81% in NWTS-5)	Fixed clock, maximise NSS
COG P9641	Low-risk neuroblastoma, surgery alone	EFS ~89%, OS ~97%	Chemo only if symptomatic
COG anti-GD2	Anti-GD2 + cytokines + retinoid vs retinoid	Antibody superior	Immunotherapy standard in high risk
German NB97	Extent of resection, stage 4 > 18 months	No EFS/OS benefit	Radical resection not essential

64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

1 · Rhabdomyosarcoma (RMS): biology

Fusion status, not the microscope, now sets risk.

- RMS is the **commonest childhood sarcoma**; 20–30% are genitourinary (GU), at bladder/prostate (BP), paratesticular (PT) or vulvovaginal (VV) sites. GU-RMS 5-y survival is **>80%**.
- Syndromic cases (Li–Fraumeni, DICER1, neurofibromatosis 1, Costello) are usually embryonal; diagnosis under 3 y should prompt a germline search.

	Embryonal (ERMS)	Alveolar (ARMS)
Site	GU (commonest); the **botryoid** variant protrudes from bladder or vagina	Trunk, extremities, perineum
Age	Younger (~70% < 10 y)	Older, adolescents
Fusion	Usually negative	~80% **PAX3/7–FOXO1** [t(2;13), t(1;13)]
Prognosis	Better	Fusion-positive worse; fusion-negative ARMS behaves like ERMS

KEY POINT

Fusion status predicts outcome better than histology, so trials assign risk by **fusion status**.

2 · Presentation and work-up

- BP tumours cause haematuria or retention; PT tumours are a painless mass separate from the testis; VV tumours cause bloody discharge or a prolapsing mass.

ALGORITHM

1. Start with US.
2. CT or MRI of the primary and its nodal basin: BP, pelvic and retroperitoneal nodes up to the renal vessels; PT, retroperitoneal; VV, inguinal, pelvic and retroperitoneal.
3. **Chest CT** (lung is the commonest metastatic site).
4. Bone scan and bilateral marrow sampling; omit these in fusion-negative ERMS without nodal or lung disease.

3 · Stage, group and risk

The **stage** is pretreatment tumour–node–metastasis plus site. The **clinical group** records what the surgeon left. The **risk group** adds fusion status and age.

Children's Oncology Group (COG) stage	Site	Size / nodes
1	Favourable: **paratesticular, female GU**, orbit, head–neck	Any, M0
2	Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk	≤ 5 cm, N0/NX

Children's Oncology Group (COG) stage	Site	Size / nodes
3	Unfavourable	≤ 5 cm N1, or > 5 cm any N
4	Any	M1

N1 means a node ≥ 1 cm in short axis. BP-RMS is never below stage 2.

Clinical group	Definition
I	Localised, completely resected, nodes negative
IIa / IIb / IIc	IIa: microscopic residual. IIb: involved nodes resected, most distal node negative. IIc: microscopic residual or most distal node positive.
III	Gross residual or **biopsy only**
IV	Distant metastases
Risk	Criteria
Low	Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II
Intermediate	Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative < 10 y
High	Stage 4 fusion-positive, or ≥ 10 y

KEY POINT

An organ-preserving **biopsy** places BP-RMS in group III, which means **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age < 1 y or > 10 y is independently adverse. Trap: most distal node positive = IIc, not IIb.

4 · Treatment principles

Chemotherapy does most of the work; surgery aims to save the organ.

- **Chemotherapy is universal**; local control is by radiotherapy (RT) and/or surgery. Complete resection (except PT) is done only if the organ is preserved.
- Primary exenteration fell from almost all cases in IRS-I to **13%** in IRS-IV, with overall survival unchanged.
- **Pretreatment re-excision** (before chemotherapy, without organ loss) improves GU survival and can change the group. **Delayed primary excision** cannot change the group and is done only if a negative-margin resection is likely.

Bladder/prostate RMS

- Biopsy is preferably **cystoscopic**; relieve obstruction with a **urethral catheter** first and avoid a suprapubic tube (seeding).
- An **end-of-therapy residual mass** is usually stroma or mature rhabdomyoblasts, so observe it if stable. Viable tumour after chemoradiation may need cystectomy.
- Only ~40% of preserved bladders are "normal", so do **urodynamics at 6 months** and renal US yearly for ≥ 10 y.

Paratesticular RMS

- Do a **radical inguinal orchidectomy**; after a scrotal approach, excise the remaining cord inguinally.
- Do an **ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or nodes are radiologically N1.

> **KEY POINT**
> Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

Female GU, perineal, infants

- Vaginoscopy and biopsy; chemotherapy with or without brachytherapy controls most, and exenteration is reserved for unresponsive disease.
- Perineal tumours are mostly fusion-positive alveolar, with 5-y OS ~50%. RT causes most late effects, including bladder and sexual dysfunction, infertility and second cancers.

5 · Other bladder tumours

- **Urothelial** tumours in children are mostly low grade; treat by transurethral resection and follow with US. A recurrence should prompt a search for a syndrome (Lynch).
- **Spina bifida/exstrophy**: lifetime bladder cancer risk is **2–4%**, aggressive, and **not increased by augmentation**; evaluate haematuria promptly.

Benign lesion	Key point	Treatment
Inflammatory myofibroblastic tumour	ALK rearrangement; intermediate malignancy	Complete resection
Nephrogenic adenoma	Follows trauma, infection, surgery or transplant	TURBT and treat the cause; recurs
Eosinophilic cystitis	Biopsy essential; blood eosinophilia often absent	Steroids, antihistamines
Haemangioma	Severe haematuria; Klippel–Trenaunay, Sturge–Weber	Propranolol, laser; partial cystectomy

6 · Testicular tumours

Prepubertal tumours are mostly benign, so the testis can often be spared.

- Before puberty, **teratoma** is the commonest tumour (benign) and **yolk sac tumour (YST)** the commonest malignant one.
- Work-up is Doppler US with LDH, **α-fetoprotein (AFP)** and β-hCG; an **"onion-skin" appearance suggests epidermoid cyst**.

	AFP	β-hCG
Source	**YST invariably**	Embryonal carcinoma, mixed tumours; rare before puberty
Half-life	**5 days**	**24 h**
Caveat	Normally high in infancy; reaches adult level (< 10 ng/mL) by **6–9 months**	—

ALGORITHM

1. Use an inguinal approach.
2. If markers are normal and US looks benign, do a **testis-sparing partial orchidectomy** with frozen section.
3. If benign, close.
4. If frozen section shows germ cell malignancy, or AFP is raised, do a radical inguinal orchidectomy.

- Prepubertal teratoma needs partial orchidectomy and **no follow-up**; postpubertal teratoma is malignant and needs radical orchidectomy.
- **YST**: radical inguinal orchidectomy; **90%** are stage I, managed by **surveillance without RPLND or chemotherapy**. Stage III–IV disease gets cisplatin, etoposide and bleomycin.
- **Stromal** tumours (Leydig, Sertoli, juvenile granulosa) are treated by partial orchidectomy if markers are normal. Testicular adrenal rests in congenital adrenal hyperplasia respond to more steroid.

Disorders of sex development (DSD)

Tumour risk rises with Y material in dysgenetic gonads: high in gonadal dysgenesis, Frasier, Denys–Drash and partial androgen insensitivity (PAIS); intermediate in complete androgen insensitivity (CAIS); low without Y. Gonadoblastoma is premalignant.

KEY POINT

Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

7 · Other pelvic and genital tumours

- **Sacrococcygeal teratoma** causes neurogenic bladder in 36–51%, so follow with renal–bladder US and urodynamics for higher types. The **Currarino triad** also causes neurogenic bladder.
- A Sertoli–Leydig ovarian tumour should prompt **DICER1 screening**.
- **Urethral prolapse** mimics RMS in prepubertal girls; treat with topical oestrogen.

★ Putting it together: common paediatric decisions

A · VUR after febrile UTI

ALGORITHM

1. Do renal–bladder US; add a voiding cystourethrogram (VCUG) if UTIs recur or US is abnormal.
2. Treat bladder and bowel dysfunction (BBD) and constipation first.
3. Low risk (grade I–II, older boy) is observed; high risk (grade ≥ III, girl, BBD, scars, infant) gets continuous antibiotic prophylaxis.
4. Breakthrough UTI or new scars: inject or reimplant (see Ch 41, 53).

B · Dilated upper tract

Condition	Key test	Operate when	Operation
Pelviureteric junction (PUJ) obstruction	MAG3 diuretic renogram	Function falling or < 40%, symptoms	Anderson–Hynes pyeloplasty
Primary megaureter	US, MAG3, VCUG	Falling function, infection; most resolve	Tapered reimplant
Ureterocele / ectopic ureter	US, VCUG, functional scan	Obstruction, infection, incontinence	Incision, heminephrectomy or ureteroureterostomy (see Ch 52)
Multicystic dysplastic kidney	US and DMSA scan	Rarely; most involute	Observe

C · Undescended testis

ALGORITHM

1. **Palpable**: orchidopexy at **6–18 months**; no imaging.
2. **Non-palpable**: examination under anaesthesia, then laparoscopy to find and fix the testis or confirm it is absent.
3. **Bilateral non-palpable, or with hypospadias**: DSD work-up first (see Ch 59).

D · Acute scrotum

Never let a scan delay exploring torsion.

	Testicular torsion	Appendage torsion	Epididymitis
Typical	Sudden pain, high-riding testis	Upper-pole tenderness, "blue dot"	Gradual onset
Action	**Explore at once**; fix both sides, remove if non-viable	Analgesia, rest	Culture; antibiotics if infected

E · Newborn with atypical genitalia

KEY POINT

Bilateral impalpable gonads with atypical genitalia = salt-wasting congenital adrenal hyperplasia until proven otherwise. Check electrolytes, 17-hydroxyprogesterone and karyotype; crisis comes at days 5–7.

F · Wilms tumour vs neuroblastoma

	Wilms tumour	Neuroblastoma
Age	Median 3.5 years	Median ~18 months
Imaging	Intrarenal claw sign; IVC extension	Calcification, vessel encasement; crosses the midline
Markers	WT1, 11p15; WAGR, Denys–Drash	VMA/HVA raised in 90–95%; MYCN amplification = high risk
Strategy	COG: nephrectomy first; SIOP: chemotherapy first	By INRG risk group

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 5

Paediatric Urology 2

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 32	Embryology of the Human Genitourinary Tract
Ch 33	Urologic Aspects of Pediatric Nephrology
Ch 34	Perinatal Urology
Ch 35	Urologic Evaluation of the Child
Ch 36	Pediatric Urogenital Imaging
Ch 37	Infection and Inflammation of the Paediatric Genitourinary Tract
Ch 38	Core Principles of Perioperative Management in Children
Ch 39	Principles of Laparoscopic and Robotic Surgery in Children
Ch 40	Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children
Ch 42	Bladder Anomalies in Children
Ch 43	Exstrophy–Epispadias Complex
Ch 44	Prune-Belly Syndrome
Ch 45	Posterior Urethral Valves
Ch 46	Neuromuscular Dysfunction of the Lower Urinary Tract in Children
Ch 47	Functional Disorders of the Bladder and Bowel in Children
Ch 48	Lower Urinary Tract Reconstruction in Children
★	Putting it together: common paediatric pathways

32 Embryology of the Human Genitourinary Tract

1 · Overview

- Development is mostly complete by the second trimester. The **indifferent stage** lasts to about **8–9 wk**; then androgens drive the sexes apart.

2 · Kidney and ureter

Bud position decides kidney quality and orifice site.

Three successive kidneys

Kidney	Features
†Pronephros	Cervical, week 4; non-functional and regresses
Mesonephros	†Transient, but its **Wolffian duct** persists; tubules become efferent ductules (male) or epoöphoron (female)
Metanephros (permanent)	†Week 5; ureteric bud and metanephric blastema form the nephrons

Reciprocal induction †

- The **ureteric bud** grows from the Wolffian duct into the metanephric mesenchyme and forms the ureter and collecting system.
- Mesenchymal **GDNF** binds **RET** on the bud to drive branching. **WT1** makes the mesenchyme competent, and loss of **PAX2** causes agenesis or hypoplasia (renal-coloboma syndrome).
- Nephrogenesis ends at about 34–36 wk; no nephrons form after birth.

> **KEY POINT**
> One defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder.

Ascent and rotation †

- The kidney ascends in weeks 6–9 and its hilum rotates medially; failure gives a pelvic or malrotated kidney. Fused lower poles make a **horseshoe kidney**, trapped under the inferior mesenteric artery.

3 · Cloaca, bladder, trigone and urethra

P2 Embryogenesis of urinary bladder — urorectal septum, vesicourethral canal vs UGS, wall layers, trigone (apoptosis, mostly bladder muscle), urachus, Weigert–Meyer

Not in the paper answer:

- Anorectal malformations arise from the **cloacal membrane**, not the urorectal septum (weeks 5–6).

Weigert–Meyer rule (duplex kidneys)

P3 Embryology and management of ectopic ureter. — bud too low gives lateral refluxing lower pole; too high gives caudal/medial ectopic upper pole; male vs female ectopic sites and sphincter

Not in the paper answer:

- Male ectopia stays above the external sphincter, so there is no continuous wetting; †female ectopia below it causes continuous dribbling with normal voids.

- With an **ectopic vas**, the vas joins the ureter and both drain into the prostatic urethra.

4 · Gonads

Two testicular hormones explain every disorder of sex development (DSD).

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — indifferent gonad, PGC migration, SRY then Sertoli cords, rete/efferent ductules, AMH/testosterone/INSL3 table, remnants

Not in the paper answer:

- **SRY** makes Sertoli cells enclose primordial germ cells in cords at about **6 wk**; Leydig cells appear from 8 wk.
- **Müllerian-inhibiting substance (MIS, AMH)** from Sertoli cells makes the Müllerian ducts regress at **8–10 wk**; without it, the uterus and tubes persist.
- **Testosterone** from Leydig cells turns the Wolffian duct into epididymis, vas and seminal vesicle.

Ovary — the default

- Without SRY there is no MIS or androgen, so development is female.

> **KEY POINT**
> Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every DSD is a combination of the two.

5 · Gonadal descent

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — CSL + gubernaculum; transabdominal (10–15 wk: androgen, INSL3) and inguinoscrotal (7th month: genitofemoral nerve CGRP) phases; female ligaments

6 · Genital ducts

Male accessory glands

- The **prostate** buds at **10 wk** from urogenital sinus (UGS) epithelium, mostly lateral to the **verumontanum**.
- Androgen acts via the **androgen receptor in mesenchyme**; stromal type 2 5α-reductase makes **dihydrotestosterone (DHT), which has about 10× the AR affinity** of testosterone. Its deficiency therefore gives a rudimentary prostate and severe hypospadias.

Female tract

- The Müllerian ducts use the **Wolffian duct as a guidewire**, which explains combined Müllerian and renal anomalies.
- The **entire vaginal epithelium and hymen are UGS-derived** (Bulmer view). Female Wolffian remnants include the **Gartner duct**.

7 · External genitalia

P3 Embryology of the penis and management of proximal hypospadias (including Koyanagi repair) — anogenital distance first, germ layers, double zipper (opening androgen-independent, closing dependent), glanular canalisation, timetable, 12 o'clock nerve-free zone

Not in the paper answer:

- The tunica is thickest at the 12 o'clock nerve-free zone, the dorsal plication site for mild to moderate curvature.
- The shaft is fused by **16 wk**. In females the folds do not fuse, leaving an open vestibule.

8 · Developmental step and resulting anomaly

Step that fails	Anomaly
†No bud or no induction (GDNF–RET, PAX2)	Renal agenesis (absent hemitrigone)
†Abnormal bud–blastema contact	Dysplasia, hypoplasia, MCDK
†Early bifurcation or two buds	Bifid ureter or complete duplication
Bud too low / too high	VUR / ectopic ureter
†Ascent or rotation fails	Pelvic, malrotated or horseshoe kidney
Cloacal membrane or septation	Anorectal malformation; †cloacal anomalies, exstrophy
Allantois persists	Urachal remnants
Fold fusion arrests	Hypospadias
Urethral plate splits	Urethral duplication
Androgen excess in 46,XX	Virilisation (CAH)
Cranial suspensory ligament retained	Intra-abdominal testis
Gubernacular attachment abnormal	Canalicular or ectopic testis
Gubernacular migration fails	Congenital cryptorchidism
Processus fails to close / to involute	Hydrocele, hernia / ascending testis
One Müllerian duct absent	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum persists	Uterine obstruction or duplication
Vaginal plate fails to canalise	Transverse vaginal septum
Inferior end fails to open	Imperforate hymen, the commonest vaginal obstruction
Müllerian and Wolffian both abnormal	OHVIRA

9 · Clinical correlations

- **Hypospadias** (about 1 in 250) is an **arrest, not a deformation**.
- **CAH** may make a 46,XX infant look fully male. Suspect it with **impalpable gonads and hyperpigmentation**, since it mimics hypospadias with bilateral impalpable testes.
- A **streak gonad** makes no MIS or testosterone, so the child is female; give hormones and **remove the streak if 46,XY**, because of gonadoblastoma risk.
- **Persistent Müllerian duct syndrome** is 46,XY with uterus and tubes, usually from an MIS-receptor defect, found at exploration for bilateral abdominal testes.
- **OHVIRA** presents after menarche with cyclic pain or pyocolpos. The kidney is often **dysplastic with an ectopic ureter into the obstructed hemivagina**; if missed, leakage persists after septum resection.

KEY POINT

Homologues: Littre and Skene; Cowper and Bartholin; spongiosum and vestibular bulb; gubernaculum and the ovarian plus round ligaments; labioscrotal swellings and labia majora. Traps: the vagina is UGS-derived; the appendix testis is Müllerian and the appendix epididymis Wolffian.

33 Urologic Aspects of Pediatric Nephrology

1 · Renal function in the child

- CAKUT is the **commonest cause of paediatric end-stage kidney disease (ESKD)**.
- GFR reaches the adult level by **2 years**. Creatinine in the first 48 h is **maternal**; by day 7 it is the neonate's own (**<0.5 mg/dL** at term).
- †Schwartz: estimated GFR = 0.413 × height (cm) / creatinine (mg/dL).

> **KEY POINT**
> The neonatal kidney is "hormonally deaf": maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate and poor concentration reflect immaturity, not disease.

2 · Haematuria

- Microscopic haematuria is **>5 red blood cells (RBCs)/high-power field**; it affects about 1% of school children and mostly resolves.
- **Fresh microscopy is critical**: red urine is non-glomerular, while brown urine with RBC casts is glomerular. A positive dipstick without RBCs means haemoglobin or myoglobin.
- **UTI is the commonest cause of gross haematuria**. Trauma, pain or clots need urological work-up, often CT; suspected GN needs complement, ASO and nephrology referral.
- In isolated microscopic haematuria, **VCUG and cystoscopy are not indicated**.

3 · Proteinuria

- Normal urine protein-to-creatinine ratio (UPCR) is **<0.2 mg/mg** (<0.7 in newborns).
- **Transient** (fever, exercise) and **orthostatic** proteinuria are benign; **fixed** proteinuria needs full evaluation with or without biopsy.

4 · Glomerular disease

Under 1 year causes are genetic; at 1–10 years think of minimal change, post-streptococcal GN, HSP and HUS; from 7 years, lupus, IgA, MPGN and Alport.

Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** with hypoalbuminaemia and hyperlipidaemia.
- In primary disease give **steroids first, without biopsy**, since minimal change is 90% under 10 years and mostly steroid-sensitive.

Category	Definition	Action
Steroid-sensitive	Remission within 8 weeks	~50% relapse
Frequently relapsing	≥2 relapses in 6 months or ≥4 in 12	Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors
Steroid-dependent	2 consecutive relapses on steroids	
Steroid-resistant	No remission after 6–8 weeks	Biopsy; usually FSGS, which recurs after transplant

Nephritic syndrome

Complement sorts the causes:

C3	C4	Diagnoses
Low	Normal	Post-infectious GN, C3 glomerulopathy
Low	Low	Lupus nephritis, MPGN
Normal	Normal	IgA, HSP, Alport, ANCA GN, anti-GBM

- In **post-streptococcal GN**, C3 normalises by **8–12 weeks**; if it stays low, biopsy.
- **IgA nephropathy** gives painless gross haematuria **during** a respiratory infection. **Alport** adds sensorineural deafness.
- **HUS** is the commonest cause of severe AKI in a young child; atypical (complement) HUS gets eculizumab.

5 · Tubulopathies

P2 Renal tubular acidosis — types 1/2/4 compared: defect, urine pH, K^+, stones, causes (obstruction gives type 4), treatment

Not in the paper answer:

- Type 4 RTA (**hyperkalaemia**) from obstruction improves after urological correction.

6 · Cystic disease

- **MCDK** has no function, and bilateral disease is lethal. In MCDK, VUR or PUJ obstruction occurs in up to 25%, so do a VCUG, especially with hydroureteronephrosis. Follow with US and BP until involution, then yearly.
- **ARPKD** causes enlarged kidneys and hepatic fibrosis, with >75% ESKD by 5 years. Nephronophthisis causes polyuria and ESKD by adolescence.

7 · Hypertension

- Hypertension is repeated BP **>95th percentile** for age, sex and height.
- Infants have coarctation or renovascular disease, young children Wilms or neuroblastoma, school-age children **renal parenchymal disease including reflux nephropathy**, and obese adolescents essential hypertension.
- Drugs are for stage 2, symptoms, end-organ damage or a secondary cause. Target <90th percentile, or **<50th** in CKD.

8 · Acute kidney injury

KDIGO stage	Creatinine	Urine output
I	Rise ≥0.3 mg/dL in 48 h or 1.5–1.9× baseline	<0.5 mL/kg/h for 6–12 h
II	2.0–2.9×	<0.5 for ≥12 h
III	≥3× or ≥4.0 mg/dL or clearance <35 (<18 years)	<0.5 for ≥24 h or <0.3 for 12 h

- Creatinine lags GFR by 24–72 h. FeNa **<1%** means prerenal and >3% intrinsic.
- Dialysis is for diuretic-resistant overload, refractory hyperkalaemia or acidosis, or uraemia.

9 · Chronic kidney disease

- CKD stages by GFR are I ≥90, II 60–89, III 30–59, IV 15–29 and V **<15** or dialysis.
- Complications include growth failure from growth hormone resistance, bone disease, anaemia and acidosis.
- **Do not restrict protein**; strict BP control is the only proven way to slow progression.

10 · Dialysis and transplantation

P1 Urological evaluation and management of pediatric kidney transplant patient. — lower-tract evaluation and reconstruction before transplant, native nephrectomy indications, ureteric techniques, complications

Not in the paper answer:

- Peritoneal dialysis is first choice if **<9 years / <20 kg**; omphalocele, gastroschisis, diaphragmatic hernia and exstrophy contraindicate it.
- Living donors give less delayed graft function (5% vs 15%) and a longer graft half-life (20 vs 12.5 years).
- Many centres await **≥10 kg and 65 cm** before transplant because of graft thrombosis.

34 Perinatal Urology

1 · Fetal renal development and prenatal imaging

Amniotic fluid is fetal urine, so obstruction threatens the lungs.

P3 Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO) — UTD 1–3% of pregnancies, second after cardiac; 18–20 wk scan; APD, SFU and UTD grading; A1/A2–3 and P1–P3 tables; causes with incidence; postnatal plan

Not in the paper answer:

- The metanephric kidney appears at **day 28** and urine production starts at **14 weeks**. After **~16 weeks** amniotic fluid is mainly urine, so obstruction causes oligohydramnios and pulmonary hypoplasia.
- The amniotic fluid index (AFI) is normally 8–18 cm; oligohydramnios is **<5–6 cm**.
- Ultrasound severity tracks PUJ obstruction and posterior urethral valves (PUV), but not VUR or duplex. Fetal MRI is used only for anatomy, such as cloaca or exstrophy.

> **KEY POINT**
> Prenatal detection is for triage and counselling; there is no proof it improves outcome.

2 · Grading systems

- **Anteroposterior pelvic diameter (APD)** is measured transversely at the hilum and is **meaningless without its gestational age**. Severe is >10 mm in the 2nd trimester and >15 mm in the 3rd.
- Postnatal pathology rises from about 12% (mild) to 88% (severe), but **VUR shows no trend**.
- **SFU** grading (postnatal): 1, sinus barely split; 2, major calyces; 3, minor calyces with parenchyma spared; 4, parenchymal thinning. All SFU 3 and about 70% of SFU 4 kidneys keep DRF >40%.
- Pyramidal thickness **≤3 mm** predicts pyeloplasty and reduced function.
- The **urinary tract dilation (UTD)** classification uses six parameters plus oligohydramnios and covers the whole tract, pre- and postnatally.

3 · Prenatal follow-up

> **ALGORITHM**
> 1. Normal APD ceiling, all else normal: **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**.
> 2. If A1 (APD 4 to <7 mm at 16–27 wk, or 7 to <10 mm at ≥28 wk), repeat US once after 32 weeks.
> 3. If A2–3 (any abnormal parameter), scan every 4 weeks until delivery.
> 4. In both, do a postnatal US after 48 h.

- Exstrophy, cloaca, ARPKD, outlet obstruction and myelomeningocele need a multidisciplinary prenatal consult.

4 · Fetal LUTO and oligohydramnios

Fetal intervention saves lungs, not kidneys.

- The picture is a male fetus with bilateral hydroureteronephrosis, a non-cycling bladder, the **"keyhole" sign** and oligohydramnios. Only fetal cystoscopy can separate PUV, urethral atresia and prune-belly syndrome (PBS).
- Prognosis is poor with oligohydramnios **before 27 weeks**, cortical cysts or echogenicity.
- Fetal urine biomarkers are **not validated** (European consensus 2022). †Classic favourable values were Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L and β2-microglobulin <4 mg/L.

Ruano stage	Fluid / kidneys	Urine biochemistry	Intervention
I — mild	Normal fluid and echogenicity	Favourable	Not indicated
II — severe, function preserved	Oligo/anhydramnios; hyperechoic; no cysts	Favourable within 3 samples	**Indicated** — for lungs and kidneys
III — severe, function abnormal	Anhydramnios; hyperechoic, with or without cysts and dysplasia	Unfavourable after 3 samples	Maybe — for lungs only

Options

- Options are early delivery, vesicoamniotic shunt (VAS), fetal cystoscopy and serial amnioinfusion; the window is **18–24 weeks**.
- Complications occur in about **44%**, most often shunt failure, followed by preterm labour and membrane rupture.

Feature	VAS	Fetal cystoscopy
Diagnoses cause	No	Yes
Mechanism	Bladder-to-amniotic bypass	Laser valve ablation
Specific complications	Blockage, migration, bowel herniation	Urethral fistula (≤10%)
Survival / renal function (PUV)	Improved / not improved	Improved / short-term only

- After VAS, over half have renal insufficiency and a third need renal replacement therapy.

> **KEY POINT**
> Counsel: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent renal replacement, because dysplasia is often established first.

5 · Postnatal imaging

- Do the **first US after 48 h**, because neonatal dehydration under-reads dilation.
- Postnatal P1 (APD 10 to <15 mm, central calyces), P2 (≥15 mm, peripheral calyces or abnormal ureter) and P3 (abnormal parenchyma or bladder) roughly equal SFU 1–2, 3 and 4. **Classify by the highest-risk feature**.
- **VCUG** finds VUR in up to 31%; order it by **UTI risk** (females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder).
- Contrast-enhanced voiding urosonography avoids radiation but shows the urethra poorly.
- **MAG3** is preferred over DTPA. Consider surgery if **DRF <40%**, T½ >20 min or retention persists upright. Pitfalls are dehydration and poorly timed furosemide.
- **DMSA** shows scars, dysplasia and duplex upper-pole function; **MRU** shows ectopic ureters and crossing vessels but needs sedation.

6 · Postnatal management, UTI and prophylaxis

- **P1** needs only a US at 3–6 months; **P2** needs CAP **if the ureter is >7 mm**; **P3** needs CAP, VCUG and MAG3 if dilation persists.
- In P3 without VUR, give CAP until pyeloplasty or **1 year**, as about 90% of UTIs occur in year 1.
- UTI occurs in 8–22%; agreed risks are **female sex, high-grade dilation and intact foreskin**. Circumcision has a number needed to treat of **10**.
- Evidence for CAP is weak. Use amoxicillin or a cephalosporin from birth; nitrofurantoin or trimethoprim only from **≥8 weeks**.

ALGORITHM

1. US after 48 h: if normal, the dilation was transient.
2. If P1, repeat US at 3–6 months.
3. If P2, give CAP if the ureter is >7 mm; VCUG and MAG3 at discretion.
4. If P3, give CAP, do a VCUG and repeat US at about 1 month; if dilation persists, do a MAG3.
5. If bilateral severe dilation, treat as outlet obstruction: decompress, CAP, urgent evaluation.

7 · Specific antenatal diagnoses

- **Transient** dilation (50–70%) is commonest, and mild dilation resolves in 90–98%, mostly by 2–4 years.

PUJ obstruction pattern

- It is the commonest **surgical** cause. Third-trimester APD **15 mm** has about 80% sensitivity and specificity for pyeloplasty.
- Serial US suffices if P2, APD <20 mm and pyramidal thickness >3 mm, since DRF >40% is likely.
- Operate for rising dilation, DRF <40% with poor drainage, falling DRF, upright retention or symptoms.

Vesicoureteral reflux

- Prenatally detected VUR is typically male, high-grade and bilateral, with congenital dysplasia; VUR after febrile UTI is typically female, lower-grade and acquired. Screen girls with high-grade UTD and uncircumcised boys.

Megaureter / ureterovesical junction obstruction

- Most resolve; surgery is predicted by a ureter **>13 mm** and DRF <30%.
- BAPU: a prenatal ureter **>7 mm after 30 weeks** needs postnatal US, CAP and VCUG.
- Operate for DRF <40%, febrile UTI or pain. **Under 1 year, temporise** with ureterostomy or a refluxing reimplant, and reimplant after 1 year.

KEY POINT

In PUJ, VUR and megaureter, dilation alone is not an indication: operate for function loss, infection or progression; grade sets surveillance.

LUTO conditions

- **PUV** is the commonest LUTO. **PBS** combines megacystis, a deficient abdominal wall and impalpable testes.
- **Urethral atresia** means a catheter will not pass; it is managed by vesicostomy, then progressive dilation of the anterior urethra (PADUA). Massive reflux and MMIHS also cause megacystis.

> **KEY POINT**
> Bilateral dilation with normal fluid means urine still leaves the fetus (reflux, megacystis); with low fluid it means true outlet obstruction.

Neonatal management of LUTO

> **ALGORITHM**
> 1. Have the team at delivery and resuscitate the lungs first.
> 2. Pass a 5 Fr feeding tube, with a finger in the rectum to guide it.
> 3. Confirm position on a plain film (KUB).
> 4. If it fails, place a US-guided suprapubic catheter.

- **Do not use a balloon catheter**, which obstructs the posterior urethra and ureteric orifices. Judge decompression by the **creatinine nadir**, and warn of post-obstructive diuresis.
- Oliguria after decompression suggests a thick bladder obstructing the ureters; treat with antimuscarinics.

Multicystic dysplastic kidney

- VUR occurs in about 20%. There is **no Wilms association**, so no routine nephrectomy; follow with US and blood pressure.

Duplex, ureterocele, ectopic ureter

- By **Meyer–Weigert**, the upper-pole orifice is inferomedial and obstructs, while the lower-pole orifice is superolateral and refluxes.
- Suspect a duplex when the ureter is more dilated than the pelvis.
- Endoscopic puncture of a ureterocele is increasingly first-line, and mandatory for outlet obstruction or sepsis. High-grade VUR usually needs ureterocele excision and common-sheath reimplant.

Exstrophy

- Both forms show a non-visualised bladder, low umbilicus and pubic diastasis. **Cloacal exstrophy (OEIS)** adds omphalocele, imperforate anus and spinal anomalies.

Cloaca, urogenital sinus, CAH

- A **cloaca** shows a single channel, a pelvic cyst and calcified meconium; a **urogenital sinus** has no bowel obstruction and normal external genitalia.
- **CAH** (21-hydroxylase, autosomal recessive) is 75% salt-wasting. Only **1 in 8** at-risk pregnancies is an affected girl, so prenatal treatment is controversial.

Myelomeningocele

Fetal closure before 26 weeks reduces shunting and improves motor outcome, but gives **no bladder benefit**.

8 · Neonatal urological emergencies

Urinary ascites and retention

- Urine causes up to 30% of neonatal ascites, and **PUV is the commonest cause**.
- Spontaneous bladder rupture heals with a catheter, but injury from an umbilical catheter needs repair.
- Retention >24 h suggests PUV or atresia in boys, and urogenital sinus or ureterocele in girls.

Abdominal masses

- **Congenital mesoblastic nephroma** is the commonest solid renal mass under 6 months.
- **Adrenal haemorrhage** has no Doppler flow, shrinks on serial US and calcifies like an eggshell, whereas **neuroblastoma** is vascular and persists. Haemorrhage may track down the processus and mimic an acute scrotum.

Anorectal malformation and ambiguous genitalia

- Anorectal malformation has genitourinary or spinal anomalies in up to **75%**, so get renal and spinal US.
- In ambiguous genitalia, defer gender assignment and **exclude salt-wasting first**; the commonest cause is CAH.

Interlabial masses

Mass	Position	Management
Imperforate hymen	Midline bulging membrane	Hymenal incision
Urethral prolapse	Midline collar at meatus	Conservative
Prolapsed ureterocele	Off-midline from meatus; hymen intact	US and VCUG
Paraurethral cyst	Off-midline, displaces meatus	Most resolve

Gross haematuria

First exclude an oestrogen-withdrawal bleed and urate crystals, then get a Doppler US.

Feature	Renal vein thrombosis	Renal artery thrombosis	Cortical necrosis
Risk	Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia	Umbilical artery catheter	Hypoperfusion
Presentation	**Triad**: haematuria, flank mass, thrombocytopenia	Hypertension, heart failure	Renal failure
Treatment	Hydration, with or without anticoagulation	Remove catheter, with or without thrombolysis	Supportive

Scrotal masses and torsion

- **Prenatal torsion** gives a hard, non-tender testis at birth with almost no salvage, so do urgent contralateral orchidopexy.
- **Postnatal torsion** (first 30 days) has salvage up to 44%, so explore immediately and fix the other side.

9 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT, VAS vs conservative in fetal LUTO; 31 randomised, stopped early	Fewer perinatal deaths; no renal benefit; 44% complications	Shunting saves lungs, not kidneys
Great Ormond Street pyeloplasty RCT	APD ≥15 mm, DRF >40%: pyeloplasty vs observation	Most observed kidneys stable or improved; a minority (APD 20–40 mm) lost function	Observation safe with monitoring
MOMS†	RCT, prenatal vs postnatal myelomeningocele closure	Less shunting, better motor outcome; no urodynamic benefit	Fetal closure is not for the bladder

35 Urologic Evaluation of the Child

1 · Triage by acuity

Triage asks: **what is lost by waiting** — an organ, a life, or nothing?

Tier	Conditions
Emergent	Acute scrotum; acute abdomen; AKI; obstruction with UTI and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, exstrophy, renal vein thrombosis
Urgent (24 h)	Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal or testicular tumour; spina bifida
Semiurgent (1–3 wk)	After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea
Routine	Hydrocele; circumcision; incontinence or enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele

KEY POINT

Context shifts the tier: a hernia is emergent if incarcerated but semiurgent if reducible; hydronephrosis is urgent if bilateral or in a solitary kidney.

2 · History

- For hydronephrosis, ask about prenatal history, progression and **amniotic fluid**, a proxy for fetal renal function.
- Obtain **consent from the guardian and assent from the child**; adolescents get a private history.

3 · Examination

Children hide shock until late.

- Minimum systolic BP is **≥60** in neonates, ≥70 at 1 month–1 year, **70 + 2×age** at 1–10 years and ≥90 above 10 years.
- **Shock shows first as tachycardia and tachypnoea; hypotension means decompensation.**

P1 Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis — examining for undescended testis: warm room, sweep from ASIS, cross-legged/squat, ectopic sites; retractile; no imaging

Not in the paper answer:

- **Micropenis** is a stretched length below mean − 2.5 SD, which is **1.9 cm** at term.
- In **Tanner staging**, the first male sign is testicular enlargement (stage II), not hair.
- A **simple sacral dimple** (<5 mm, within the cleft, <25 mm above the anus) needs no imaging; image for a lipoma, tag, deep dimple or deviated cleft.

4 · Urine and blood tests

- In febrile children under 24 months, use a **catheter or suprapubic aspirate**; **bag specimens have high false-positive rates** and help only if negative.

- The AAP defines UTI as pyuria and/or bacteriuria with **≥50,000 colony-forming units (CFU)/mL** (catheter or aspirate) in a symptomatic child.

Test	Threshold
Microscopic haematuria	**>5 red cells (RBC)/high-power field** (adult ≥3), on >2 occasions
Calcium/creatinine ratio (hypercalciuria)	>0.4 at age 2–5 years; >0.2 above 5 years
Normal protein/creatinine ratio	<0.7 in the first year; <0.4 up to 4 years; <0.2 after 5 years; >3.0 is nephrotic
Specific gravity	<1.010 dilute (water excess, diabetes insipidus); >1.020 dehydration

- Always **confirm dipstick blood by microscopy**, and test proteinuria on a **first-morning** sample.

5 · Imaging

Only US and MRI avoid radiation; keep doses as low as reasonably achievable.

Modality	Answers
US	Kidney, bladder, scrotum, torsion, spine
VCUG	Reflux, urethra, genitogram
CT	Severe trauma, stones
MRI	Tumours, ectopia, duplication, Müllerian or sinus anatomy
MAG3 renogram	**Obstruction** and differential function
DMSA scan	**Scarring** or pyelonephritis, and differential function

6 · Emergent presentations

Acute scrotum

- The commonest cause is **torsion of the appendix testis**, with point tenderness in a prepubertal boy; treat it supportively.
- **Cord torsion** (mostly over 10 years) gives severe pain, vomiting, a high-riding testis and an absent cremasteric reflex.
- It is a **clinical diagnosis**: explore without imaging and fix the other side.

TWIST item	Points	Score and action
Testicular swelling	2	0: no US
Hard testis	2	1–5: scrotal US
Absent cremasteric reflex; nausea or vomiting; high-riding testis	1 each	**≥6**: explore without US

Other emergencies

- In **ambiguous genitalia**, exclude **salt-wasting CAH** (21-hydroxylase in about 95%). Crisis comes on day **4–15**, so do not discharge early; check **17-hydroxyprogesterone and electrolytes**.
- In **exstrophy**, tie the cord with silk (no clamp), cover the plate with cling film and transfer.
- **Infected obstruction** needs prompt drainage by stent or nephrostomy.

- In **paraphimosis**, exclude a **hair tourniquet**, then compress the glans under analgesia; a dorsal slit is the fallback.
- Ischaemic **priapism** is usually from **sickle cell disease**; give oxygen, fluids and analgesia.
- In **trauma**, the **mechanism** decides CT. After straddle injury consider **sexual abuse**; meatal blood needs a retrograde urethrogram, not a blind catheter.
- In **retention**, expected capacity (mL) is **(age + 2) × 30**. Try analgesia and disimpaction before a catheter.

7 · Common referral presentations

Antenatal hydronephrosis

- Scan **before discharge** if bilateral, in a solitary kidney or with suspected outlet obstruction; otherwise after **48 h**.
- A boy with bilateral dilation and a thick bladder needs a **VCUG before discharge**, catheterised meanwhile.

Haematuria

- Glomerular haematuria is tea-coloured with casts and protein; urological haematuria is bright red with clots.
- Investigate all gross haematuria with urinalysis, culture and US. Isolated microhaematuria is usually benign, so repeat it over 2–3 weeks.

Bladder bowel dysfunction and enuresis

- **Always examine the back**, and treat stepwise, starting with behavioural therapy.

Scrotal, inguinal and genital

- A communicating **hydrocele** is smaller in the morning and is treated by high ligation at the internal ring.
- An **undescended testis** with hypospadias or micropenis suggests a DSD.
- For **varicocele**, the Lambert volume is L×W×H×**0.71**; a non-decompressing or right-sided one needs a retroperitoneal US.
- Prepubertal **testicular tumours** are mostly benign, and tumour markers guide testis-sparing surgery.
- **Meatal stenosis** is treated by meatotomy under EMLA, and symptomatic **labial adhesions** by oestrogen or steroid cream.

> **KEY POINT**
> Red flags: absent cremasteric reflex with acute pain; ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension; abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft.

8 · Child comfort

- Distraction, midazolam, intranasal ketamine and EMLA cream can avoid general anaesthesia.

36 Pediatric Urogenital Imaging

1 · Principles — ALARA and the three harms

- Under **ALARA**, **every child is treated as radiation sensitive**. **Sedation risk usually exceeds contrast risk**, and ultrasonography (US) is **safe at all ages and gestations**.

> **KEY POINT**
> Every choice trades **radiation, sedation and contrast** against yield; the long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

2 · Ultrasonography

US comes first, but shows anatomy, not function.

P1 USG in neonates — technique; neonatal pyramids mimic dilation; under-read in first 48 h; SFU grades 0–4; PUV keyhole, MCDK, ARPKD, masses; normal US does not exclude VUR

Not in the paper answer:

- Fetal urine is made from **10–12 weeks**; by **20 weeks** US detects most anomalies.
- SFU grade 4 is defined by parenchymal thinning, not more dilatation.
- **Bilateral hydroureteronephrosis with a thick, poorly emptying bladder needs a VCUG** for posterior urethral valves (PUV); megaureter or ureterocele needs diuretic renography or MRU.

Cysts, masses, stones

Lesion	Side	US hallmark
MCDK (multicystic dysplastic)	Unilateral	Non-communicating random cysts, scant echogenic parenchyma
ARPKD (recessive polycystic)	Bilateral	Large homogeneously **bright** kidneys, not discrete cysts
ADPKD (dominant)	Bilateral	Parenchyma almost replaced by cysts
Multilocular cystic nephroma	Unilateral	Like MCDK, more parenchyma; infancy
Cystic Wilms	Unilateral	Overlaps; 2–4 years

- A neonatal solid mass is usually **mesoblastic nephroma**; a toddler's is Wilms. Neonatal enlargement with acute kidney injury and hypertension suggests **renal vein thrombosis**.
- **Stones** are echogenic **with** shadowing (fat is without), and Doppler **twinkling artefact** confirms them.

Infection and trauma

- In UTI, US is not routine at the start; **do US if there is failure to improve on antibiotics**. It misses low-grade VUR and scars, where DMSA is better. **Normal US neither risk-stratifies APN nor predicts VUR.**

Contrast-enhanced voiding urosonography (CeVUS)

- Microbubbles grade VUR on five grades with **no radiation**. Area under curve is **0.97**, so it is recommended for VUR.

Bladder/urethra

- The **keyhole sign** (thick bladder and dilated posterior urethra) means PUV.

- To tell clot from tumour, scan supine then prone, because a **clot moves**.

Scrotum and genitalia

- **Never delay exploration for suspected torsion**.
- The **whirlpool sign** gives torsion sensitivity ~96% vs ~76% for colour Doppler, with specificity ~99%.
- Appendage torsion is an avascular nodule with surrounding hyperaemia. **Teratoma** is the commonest prepubertal tumour; epidermoid shows **onion rings**.
- **Do not image a routine undescended testis** (AUA Choosing Wisely) except in an obese child.

3 · Voiding cystourethrography

VCUG is the reference test for reflux and urethral anatomy.

- **After UTI**, do it once the child is improving on antibiotics; a 4–6-week wait is unnecessary, though very early VCUG may overgrade VUR.
- Indications are hydronephrosis with **dilated ureter(s)**, febrile or recurrent UTI, suspected PUV, epididymitis in boys, and UTI in a circumcised boy.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN. **PIC cystography** (positional instillation of contrast at the orifice) is used when VCUG is negative but febrile UTI recurs.
- **Trauma**: with meatal blood, do a **retrograde urethrogram (RUG) before any catheter**. Capacity (mL) = **(age + 2) × 30**, max 400. A "pie-in-the-sky" bladder means pelvic fracture with extraperitoneal injury.

4 · Nuclear medicine

DMSA

- DMSA is taken up by proximal tubules, so APN shows photon-deficient areas.

Feature	Acute pyelonephritis	Scar
Contour	Preserved	Volume loss, concavity
Timing	Within days; resolves over ~5 months	Lesion at **≥6 months** = scar

- A third to a half of acute lesions scar.

MAG3 diuretic renography

P1 Radionuclide studies and diuretic renography in PUJ obstruction — MAG3 vs DTPA vs DMSA; well-tempered protocol (hydration, catheter, diuretic timing, ROI); DRF, T½ from maximal filling; biphasic curve; pitfalls

Not in the paper answer:

- Give furosemide **1 mg/kg IV** at maximal filling; if given too early, drainage is measured before the system is full.
- There is **no validated paediatric T½ cut-off**, so decide on age, function, curve, upright drainage and serial studies. A **biphasic curve** and poor upright drainage predict progression.

Radionuclide cystography and testicular scan

- **Direct** cystography is more sensitive and lower-dose than VCUG but grades coarsely, so it suits follow-up. **Indirect** cystography is insensitive and not routine.

5 · MR urography

- MRU gives anatomy and function without radiation and suits complex anomalies.
- Protocol: hydration and catheter, then **furosemide 1 mg/kg (max 20 mg)**, T2, gadolinium and dynamic T1; function is measured by **Patlak** analysis.
- It needs sedation under 6 years and is best for **bladder rhabdomyosarcoma** and **local staging of renal tumours**.

6 · CT and radiography

- CT is chosen for speed in trauma and staging, not for hydronephrosis.
- **In trauma, arterial-phase images miss collecting-system injury, so delayed excretory images are mandatory.** CT cystography separates intra- from extraperitoneal rupture.
- CT rarely changes stone care. IVP is obsolete.

7 · Choice of test by clinical question

Question	First line	Next	Avoid
Hydronephrosis — obstruction?	US	MAG3; MRU	CT, IVP
Hydroureteronephrosis, with or without thick bladder	US	VCUG	—
VUR	VCUG	CeVUS; direct cystogram; PIC if occult	Indirect cystogram
APN / scar	DMSA	—	Grey-scale US; MRI
Stones	US	Low-dose CT; KUB	Reflex CT
Renal mass	US	Contrast CT; MRI	—
Bladder mass	US (full bladder)	MRI	—
Trauma	CT with delayed phase; US	RUG; cystography	Catheter before RUG if meatal blood
Acute scrotum	Clinical, with or without Doppler/whirlpool	Scintigraphy if equivocal	Imaging delaying exploration
Undescended testis	Examination	US only if obese	Routine US, MRI
DSD	US and genitography	MRI	CT

37 Infection and Inflammation of the Paediatric Genitourinary Tract

1 · Overview and the paradigm shift

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls**, and 30–50% recur.
- Congenital dysplasia now explains more scarring than reflux does, so the approach is **selective imaging, prompt treatment and prophylaxis only for the high-risk child**.

2 · Pathogenesis and virulence

- Infection is **ascending**: gut bacteria colonise the perineum, enter the bladder, adhere to umbrella cells and may ascend. Uropathogenic E. coli (UPEC) causes most UTIs.

Factor	Mechanism	Association
Type 1 fimbriae (mannose-sensitive)	FimH binds uroplakins, allowing entry	Cystitis, asymptomatic bacteriuria (ASB)
P fimbriae (mannose-resistant)	Bind Galα1–4Galβ (P blood group)	**Pyelonephritis**
Aerobactin	Siderophore	Growth in iron-poor urine
α-Haemolysin, CNF1	Cell lysis, cytoskeletal damage	Inflammation, invasion
Capsule (K antigen)	Antiphagocytic, anticomplementary	Pyelonephritis
Biofilm, flagella	Shield; motility	Recurrence; ascent

- **Intracellular bacterial communities** (IBCs) form a **quiescent reservoir** that reseeds the lumen for months, causing same-strain recurrence despite adequate antibiotics.

> **KEY POINT**
> IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations.

3 · Host risk factors

- In the first year UTI affects boys **3.7%** vs girls 2%; by 6 years, boys 2% and girls 7%.
- **Circumcision** lowers UTI in the first 6 months **~10-fold**. The number needed to circumcise is **111** normal boys, 11 with recurrent UTI and **4** with high-grade VUR.
- The **faecal reservoir** recycles the same strain, and antibiotics such as TMP-SMX destroy protective flora.
- **VUR** is found in 1–2% of newborns and **25–40%** after a first UTI. Most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change.
- Treating **bladder and bowel dysfunction (BBD)** cuts recurrence and aids VUR resolution.
- In **neurogenic bladder**, 40–80% on CIC are bacteriuric; **ASB without VUR needs no antibiotics**, and frequent catheterisation prevents UTI.
- **Catheter-associated UTI** is the commonest nosocomial infection, so remove catheters early. Early infancy (IgG nadir) is the highest-risk period.

KEY POINT

Host factors either raise inoculum/dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lower defence (immature immunity, receptor-bearing blood groups); prevention reverses whichever applies.

4 · Classification

- UTI is **complicated** (anomaly, dysfunction, foreign body; all neonatal UTI) or uncomplicated; upper or lower; first or recurrent.
- Febrile UTI is presumed pyelonephritis, but only **50–66%** of febrile children show acute changes on DMSA scan, the gold standard.

Recurrence	Between episodes	Cause	Remedy
Unresolved bacteriuria	Never sterile	Non-compliance, malabsorption, resistance	Culture-directed retreatment
Bacterial persistence	Sterile; same organism	Nidus: stone, anomaly, papilla, catheter, stent	**Remove nidus**
Reinfection	Sterile; new organism	New ascent	Prevention

Asymptomatic bacteriuria

- Kass definition: two cultures of **≥10^5 colony-forming units (CFU)/mL** of the same organism, without symptoms.
- Infants with ASB are treated and scanned. In school-age girls without VUR or anomaly, **do not screen or treat**, because this selects resistance.

5 · Diagnosis

Clinical

- Fever is ≥38 °C. A toxic look, age <30 days and ≥39.4 °C predict bacteraemia.
- Under 2 years, risk factors are fever >40 °C, previous UTI, suprapubic tenderness and an **uncircumcised penis**. **Another fever source, such as otitis, only halves UTI risk.**

Urine collection

Method	Contamination	Role
Bag	50–60% (false-positive culture up to 75%)	**Reliable only when negative**; never culture
Clean catch	~26%	Toilet-trained; preferred by NICE
Catheter	10–15%	Sensitivity 95%, specificity 99%; **standard <2 years non-toilet-trained**
Suprapubic aspiration (SPA)	<1%	Reference; US guidance gives 80–90% success

- Collect **before antibiotics** unless delay is dangerous. For 2–24 months the AAP accepts bag urinalysis; if positive, culture a catheter or SPA sample.

Test	Sensitivity	Specificity	Use
Leukocyte esterase (LE)	~79%	~87%	Negative helps **rule out**

Test	Sensitivity	Specificity	Use
Nitrite	~50%	**~98%**	Positive **rules in**
LE or nitrite or microscopy	99.8%	70%	Best rule-out
Pyuria ≥5 WBC/HPF	~74%	~86%	No advantage over LE
Gram stain, fresh urine	~91%	92–96%	Most reliable rapid test

- **Nitrite is falsely negative** with short bladder dwell (infants), dilute urine and gram-positives (Enterococcus).
- UTI without pyuria is rare, which separates it from ASB. A threshold of **≥10 WBC/HPF** works better under 2 years.
- **Culture is essential** unless urinalysis is entirely normal.

Guideline	Specimen	Significant count (CFU/mL)
Kass	Voided	$\geq 10^5$
AAP 2011 (2–24 months)	Catheter/SPA	**≥50,000 + pyuria**
EAU	Midstream/catheter	10^3–10^4 monoculture
RCH Melbourne	SPA / catheter	Any gram-negative / $>10^3$
ISPN (Italy)	Catheter / midstream	$>10^4$ / $>10^5$

- **The cleaner the specimen, the lower the accepted count.** Mixed growth and skin flora suggest contamination.
- CRP is sensitive, not specific. **Procalcitonin ≥0.5 ng/mL** best predicts acute pyelonephritis (APN) and later scarring.

6 · Imaging after UTI

- Routine imaging has not been shown to prevent scarring or renal failure, so **order a test only if it changes management**.
- US is abnormal in ~15% after a first febrile UTI, but only 1–2% need action. It is **urgent if the child is unwell or not responding at 48 h**.
- VCUG is the gold standard for VUR, done with sterile urine in a well child; CeVUS is an alternative.
- DMSA shows APN best within **1 week**; a defect at **≥6 months** is a scar, which cannot be told from dysplasia. CT is only for occult stones or mass vs inflammation.

Guideline	Ultrasound	VCUG	Late DMSA
AAP (2–24 months)	Yes, after first febrile UTI	If US abnormal (or recurrence)	No
NICE <6 months	Yes	Atypical/recurrent	Atypical/recurrent
NICE >6 months	Atypical/recurrent	No	Atypical/recurrent
Top-down	No	If acute DMSA abnormal	If acute DMSA abnormal
RCH Melbourne	If <3 months or seriously ill	No	No

- **NICE "atypical" UTI** means seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within **48 h**, or a non-E. coli organism. Late DMSA is at 4–6 months.

- NICE and AAP miss about half of grade III–V VUR but spare most children a VCUG, because high-grade VUR declares itself at recurrence.

7 · Treatment

- **Delay matters**: new-scar odds double if antibiotics start after 72 h rather than within 48 h, so start empirically after urine collection.
- **Oral therapy equals intravenous** for non-toxic children over 2 months who tolerate oral drugs.
- **Admit** if under 1 month, toxic, dehydrated, vomiting or of doubtful compliance. Neonates need a septic work-up and IV therapy, switching to oral after ≥24 h afebrile.
- 90% are afebrile by 48 h. **If not improving at 48 h, do RBUS and broaden cover**.

Infection	Duration
Febrile UTI	**7–14 days** (shorter is inferior)
Afebrile cystitis	**2–4 days** (beats single dose; equals 7–14 days)
Lobar nephronia	≥3 weeks
Renal abscess	Antibiotics; drain if no response
Pyonephrosis	Antibiotics and drainage

- Choose by the **local antibiogram**. TMP-SMX and amoxicillin are poor empirical agents; prefer **cephalexin**, and nitrofurantoin for cystitis. **Proteus is a boys' organism**.
- **Neonates** need Enterococcus cover, so give **ampicillin with a third-generation cephalosporin or an aminoglycoside**.
- Use broad cover after previous UTI, recent antibiotics or admission, or with an anomaly. Fluoroquinolones are reserved for complicated UTI.

Agent	Dose (mg/kg/day)	Restriction
Cephalexin	50–100 in 4	—
Cefixime	8 once daily	—
Amoxicillin–clavulanate	20–40 in 3	—
TMP-SMX	6–12 (TMP) in 2	**<6 weeks/premature**: kernicterus
Nitrofurantoin	3–5 in 2	**<3 months**, GFR <50%, G6PD deficiency; **never for febrile UTI**
Ceftriaxone (IV)	75 once daily	—
Gentamicin (IV)	7.5 (once daily acceptable)	Nephro-/ototoxicity

No repeat culture is needed if the organism was sensitive.

8 · Renal scarring and sequelae

- Scars favour the **poles**, where compound papillae allow intrarenal reflux, and risk rises with VUR grade.

Pyelonephritic episode	1st	2nd	3rd	4th	5th
Scar risk	5%	10%	20%	40%	60%

- Childhood UTI causes **<1% of CKD**. Scars lead to hypertension and pre-eclampsia, so bilateral scars need lifelong checks of BP, function and proteinuria.

9 · Prevention and prophylaxis

- Recurrence is **10–30%**, highest in the first 3–6 months.
- **Continuous antibiotic prophylaxis (CAP)** is not given after a first UTI (AAP, NICE). It is for the **high-risk** child: girls with dilating VUR (≥III), BBD or recurrent febrile UTI.
- Give about **¼ of the therapeutic** dose at bedtime, using a different agent from treatment. Options are TMP-SMX (not under 6 weeks), nitrofurantoin and a cephalosporin in young infants.
- With VUR and BBD, recurrence is ~45% vs 15%; treat BBD with timed voiding and laxatives, then anticholinergics, biofeedback or CIC.
- **Circumcision** helps high-risk boys (recurrent febrile UTI, high-grade VUR, hydroureteronephrosis). Cranberry and probiotics are at most adjuncts.

10 · Special infections

P1 Xanthogranulomatous pyelonephritis. / Fungal UTI — XGP: Proteus, stone–obstruction triad, mimics tumour, nephrectomy (diffuse) vs partial (focal), hard dissection; Candida risks, C. glabrata resistance, treat >10,000–15,000 CFU/mL, fungus ball drainage + amphotericin

Not in the paper answer:

- In children, XGP has calculi in only 38–70% and presents with failure to thrive and anaemia.
- **Lobar nephronia** mimics a mass on CT and needs antibiotics for ≥3 weeks. **Pyonephrosis** must be **drained**.
- **Viral haemorrhagic cystitis** is due to adenovirus 11; **BK virus** after marrow transplant is treated with **cidofovir**.
- **Prepubertal epididymitis** is usually sterile; if culture-positive or recurrent, look for ectopic ureter or utricle. **Balanoposthitis** needs circumcision only if recurrent.

11 · Landmark trials

Trial	Design	Result	Take-home
RIVUR	607 children, VUR I–IV, TMP-SMX vs placebo, 2 years	Recurrent UTI ~halved (number needed to treat 8); **no difference in scarring**	Prevents recurrence, not scars; most benefit with BBD, febrile index UTI
Swedish Reflux Trial	Randomised, dilating VUR, 1–2 years	Prophylaxis reduced recurrence in girls	CAP for girls with dilating VUR

38 Core Principles of Perioperative Management in Children

1 · Perioperative surgical home and enhanced recovery

- **Enhanced recovery after surgery (ERAS)** combines counselling, shorter fasting, judicious IV fluid, no routine tubes or drains, opioid-sparing analgesia and early feeding and mobilisation. It shortens stay with no more complications.

2 · Preoperative assessment

- Risk is raised by upper respiratory tract infection (URTI), asthma exacerbation, dehydration, uncontrolled diabetes, prematurity and fever.
- **Preoperative anxiety** causes more pain and emergence delirium; explanation and play help.
- **Investigations are indication-driven**, with no routine bloods, chest X-ray or ECG. Check urine hCG in females of menstrual age; it can be falsely positive after augmentation cystoplasty.
- Send a **urine culture** before any procedure opening the urinary tract and reschedule until sterile. Treat asymptomatic bacteriuria before endourology.
- **URTI** leaves the airway hyperreactive for **6–8 weeks**, risking laryngospasm and bronchospasm. Postpone elective surgery at least 4, ideally 6 weeks.
- **Sepsis**: give antibiotics within **1 h** of septic shock and avoid neuraxial blocks. In **trauma**, assume a full stomach.

3 · Fasting (NPO, nil by mouth)

Minimum fast	Material
2 h	Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h)
4 h	Breast milk
6 h	Infant formula, non-human milk, light meal, tube feeds
8 h	Regular meal

- Non-human milk counts as a light meal. List the youngest first, because they dehydrate faster.
- **Diabetes**: list the child first, stop metformin 24 h before and halve the long-acting insulin the night before; a long case needs IV insulin.
- Omit ACE inhibitors that morning, stop warfarin and aspirin 7–10 days before, and never stop clonidine abruptly.

4 · Anaesthesia considerations

Neonate and preterm infant

- **Postoperative apnoea** peaks at 12–23 h, so give 23-h monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Defer elective surgery beyond these ages.

Anaesthetic neurotoxicity

- The FDA warning (2016) says repeated or lengthy (**> 3 h**) anaesthesia at **< 3 years** may affect brain development. Human trials of a single short exposure are negative (§11), so do not delay necessary surgery.

Comorbidities

- **Asthma**: give a β2-agonist before induction. **Chronic kidney disease**: avoid surgery within 4–6 h of haemodialysis and give no NSAIDs in CKD 3–4.
- **Sickle cell disease**: above all avoid dehydration, and also hypoxia, cold and tourniquets; acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: give α-blockade before β-blockade, with volume loading, and expect hypoglycaemia after resection.
- **Congenital adrenal hyperplasia**: give IV stress-dose hydrocortisone with glucose-containing fluid.

Spina bifida

- Confirm VP shunt function. **Latex allergy** can cause anaphylaxis, so care should be latex-free from birth.
- **Autonomic dysreflexia** (lesion at or above **T6**): bladder distension causes severe hypertension and bradycardia. Stop the stimulus, drain the bladder, sit the child up and give a rapid antihypertensive.

5 · Fluid and electrolyte management

Hypotonic fluids can cause fatal hyponatraemia.

- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for the first 10 kg, plus **2 mL/kg/h** for the next 10 kg, plus **1 mL/kg/h** per kg above 20. For 22 kg this is 62 mL/h.
- Add dextrose under 2–4 months; give no potassium if dialysed.

Maintenance fluid	Historical	Current
Tonicity	Hypotonic (0.45% saline)	Isotonic (0.9% saline or lactated Ringer) with dextrose
Potassium	10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg	

KEY POINT

Why isotonic? Surgical stress, pain, nausea and hypoxia trigger non-osmotic ADH release, so hypotonic fluid adds free water and causes hyponatraemia (< 135 mmol/L). A child's large brain-to-skull ratio makes the resulting encephalopathy potentially fatal.

6 · Blood loss and transfusion

- Estimated blood volume (EBV) is 80–90 mL/kg in a term neonate and 70–75 mL/kg in a child. Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit.
- Transfuse at haemoglobin **< 7 g/dL** in stable children; 10–15 mL/kg raises haemoglobin by about 2–3 g/dL.

7 · Temperature

- Infants cool quickly, causing acidosis, coagulopathy and apnoea, so use a warm theatre, forced-air warming and warmed fluids.

8 · Antibiotic prophylaxis and intraoperative crises

- **Timing**: give within **60 min** of incision; vancomycin and fluoroquinolones within **120 min**.
- Doses: cefazolin **30 mg/kg** (redose at 4 h); ampicillin 50 mg/kg (redose at 2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg and ciprofloxacin 10 mg/kg (not redosed).

	Anaphylaxis	Malignant hyperthermia (MH)
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)
First sign	Collapse, bronchospasm, hives minutes after induction	Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity
Treatment	**Adrenaline**, 100% oxygen, fluids; send tryptase	Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence)

9 · Analgesia

- Under 3 years, use the FLACC pain score.
- **Regional blocks under general anaesthesia** are safe and opioid-sparing (penile, **caudal**, epidural). Caudal 0.25% bupivacaine is 0.5–1 mL/kg; maximum bupivacaine is 2.5 mg/kg.
- Give **multimodal, scheduled** analgesia: acetaminophen 15 mg/kg 6-hourly and ibuprofen 10 mg/kg 8-hourly. Ketorolac is limited to 72–96 h, and no NSAIDs are given under 6 months.
- **Codeine is contraindicated**, because ultrarapid metabolisers make lethal amounts of morphine. Limit discharge opioids to 3–5 days.

10 · Postoperative care, VTE and day-case discharge

- **Emergence delirium** occurs at 2–8 years, within 30 min of waking; the child does not recognise parents, unlike a child in pain.
- **Venous thromboembolism** is rare before puberty; at-risk adolescents need mobilisation and mechanical prophylaxis, with LMWH if several factors are present.
- **Day-case** surgery suits ASA I–II children outside the apnoea-risk ages having a short, low-blood-loss procedure.

11 · Landmark trials

Trial	Design	Result	Take-home
GAS	Randomised; infant hernia, awake regional vs sevoflurane < 1 h	No cognitive difference at 2 and 5 years	Single short anaesthetic safe
PANDA	Sibling-matched; single exposure < 3 years	No IQ difference at 8–15 years	Repeated exposure untested
MASK	Cohort; exposure < 3 years	Negligible deficit after single exposure	Reassuring

39 Principles of Laparoscopic and Robotic Surgery in Children

1 · Evolution and indications

P3 Discuss diagnostics and operative pediatric laparoscopy in urology — evolution 1976, 1994, 1995, 2002; indications; age IAP limits; only contraindication; access; complications (right common iliac, inferior epigastric, port-site hernia 0.5–3%)

Not in the paper answer:

- The first paediatric urological laparoscopy was for non-palpable testis in **1976**. The wristed robot removed the suturing bottleneck, so **reconstruction** grew, with pyeloplasty the commonest.
- The only strict contraindication is **inability to tolerate pneumoperitoneum**; age and size are not contraindications.

2 · Physiology of pneumoperitoneum

P1 Pneumoperitoneum physiology; laparoscopic access and gases — pressure vs CO2 effects on heart, lungs, kidney, ICP;
vagal bradycardia; CO2 choice;
gas embolism signs and treatment (left lateral head-down, oxygen, aspirate)

Not in the paper answer:

- Effects depend on pressure and reverse on desufflation, so use the **lowest age-appropriate pressure**: intra-abdominal pressure (IAP) < 12 mm Hg, and about 7 mm Hg is feasible.
- Infant oliguria is transient and is not a guide to resuscitation, so give **no fluid boluses**.

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	**≤ 6**	**≤ 8**	**≤ 10**	**≤ 12 mm Hg**

- **Cardiovascular**: IAP > 10 mm Hg lowers venous return and cardiac index. Insufflation can trigger a **vagal reflex**, so desufflate and reinsufflate slowly.
- **Respiratory**: infants have a compliant chest wall and low FRC, so tidal volume falls, worse with Trendelenburg, and CO2 absorption affects them most.
- Laparoscopy is generally safe with **VP shunts**. Infants cool fast, so use warmed humidified gas.

3 · Access

- Open (Hasson) and closed (Veress) entry do not differ in injury rates (Cochrane); confirming intraperitoneal position is what protects.
- After scarring or failed umbilical entry, use **Palmer's point** (left midclavicular line, about 3 fingerbreadths below the costal margin).

4 · Ports, instruments and placement

- **3- and 5-mm** laparoscopic ports suit infants, whereas robots need **8 mm**. Robotic wristed instruments have 7 degrees of freedom against 4 for laparoscopic ones.
- Triangulation is essential in laparoscopy, while the robot tolerates linear placement. **HIdES** puts ports below the Pfannenstiel line.

5 · Approaches

	Transperitoneal	Retroperitoneal	Vesicoscopic
Space	Largest; familiar	Limited	Limited; small bladder predicts complications
Strength	Complex reconstruction	Avoids peritoneum; preserves peritoneal dialysis	Intravesical access
Limitation	Colon mobilised or mesenteric window	Unfamiliar view; peritoneal tear collapses space	Learning curve; port-site closure
Uses	Testis, kidney, ureter, bladder	Pyeloplasty, (hemi)nephrectomy, lymph node dissection	Cross-trigonal reimplantation

- In **left pyeloplasty**, a transmesenteric window avoids colonic mobilisation.

6 · Robotics in children

- **Working space** is about **1 L at 1 year vs 6 L in adults**, so arms collide; wide port spacing and bowel preparation help.
- **Limitations**: there are no 5-mm robotic ports, and single-port needs a 25-mm cannula and about **10 cm** to the target, which is prohibitive in small children.
- **Benefits** are shorter stay and less narcotic use; incisions under 15 mm contract with growth.

7 · Complications

- The major vascular injury is to the **right common iliac** artery at Veress entry; the minor one is the **inferior epigastric**, so insert ports under vision.
- Port-site hernia peaks around 1 year, so **close fascia** at every port.
- Experience is the strongest predictor, with fewer complications above 12 cases a year.

> **KEY POINT**
> **Ergonomics**: set the table at 70–80% of elbow height and the monitor at eye level. The robot offloads the arm but fixes the neck, so use the clutch to keep the elbows on the armrest.

40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

1 · Principles and terminology

- Non-invasive assessment excludes anatomical or neurological causes, then defines **storage vs emptying** dysfunction (ICCS 2014).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small is <65% EBC; large is >150% (some use >130%).

2 · History

- **Secondary enuresis** is wetting after **≥6 months** dry, so look for stressors.
- In girls, poor **toilet posture** activates the pelvic floor and causes incomplete emptying; a footstool helps.
- **Holding manoeuvres** (potty dance, squatting) suppress detrusor overactivity (DO).
- Bowel dysfunction may be the sole cause of urinary symptoms. Target a painless **Bristol 3–4 stool daily**.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month) needs **≥2 of**: ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful or hard stools; large rectal mass; stools blocking the toilet.
- **Neurological red flags** are perineal numbness, or gait or limb change with new LUTS.

ALGORITHM

1. **Bowel and bladder dysfunction (BBD)**
2. Apply Rome IV; obtain a 7-day diary, urinalysis and pelvic US.
3. **Treat the bowel first.**
4. If LUTS persist, do uroflow with EMG and start urotherapy.
5. If that fails, add an anticholinergic, then a β3-agonist; then botulinum toxin, biofeedback or neuromodulation.

- The **48-h frequency–volume chart** shows capacity and polyuria and is **mandatory** when first-line therapy fails.
- Scores: **DVSS** best tracks treatment, and **DVISS** is best for diagnosis.

3 · Physical examination

- Measure height, weight and BP at **every visit**.
- **Neurological**: include perineal sensation, anal tone and sacral reflexes (anal wink, bulbocavernosus); their absence means the S2–S4 arc is interrupted.
- **Spine**: a simple sacral dimple (<2.5 cm from anus) is low risk and gets spinal US up to 6 months. High-risk signs (hair tuft, lipoma, vascular malformation, multiple stigmata, VACTERL) need **MRI**.
- **Genitalia**: **continuous incontinence in a girl suggests an ectopic ureteric orifice.**

4 · Laboratory tests

- Do a **urinalysis for every child with LUTS**; **culture is not routine.** On CIC, culture only with fever, pain, new incontinence or spasm.

5 · Non-invasive urodynamics

P3 P1 Uroflowmetry; EMG — bell/tower/staccato/interrupted/plateau patterns and meaning; Qmax peak ≥2 s; patch EMG at 3 and 9 o'clock; lag time <2 s DO, 2–6 s normal, >6 s bladder-neck dysfunction

Not in the paper answer:

- Uroflow is valid only if toilet-trained and voiding **≥50% EBC** (replacing the adult ≥150 mL rule), with more than one curve.
- A staccato curve never reaches zero, whereas an interrupted one does.

> **KEY POINT**
> The curve reflects detrusor against outlet. Too much detrusor gives a tower; too much outlet gives a plateau; intermittent outlet activity gives staccato; intermittent detrusor effort gives an interrupted curve. Patch EMG tells which side is at fault.

- **Pelvic US**: the bladder wall is abnormal if **>3 mm full / >5 mm empty**. A rectal diameter **>30 mm** without urge means faecal loading.
- **PVR** is abnormal at 4–6 y if **>30 mL or >21%** of capacity and at 7–12 y if **>20 mL or >15%**. It is valid only **<5 min** after voiding with the bladder at 50–115% EBC, so repeat it.

6 · Invasive urodynamics

Filling pressure, not wetting, decides renal risk.

Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, transverse myelitis or spinal cord injury (SCI).
- **Severe obstruction**: posterior urethral valves (PUV) or urethral atresia.
- Non-neurogenic LUTS: study only after failed 2nd/3rd-line therapy, and only if it will change management.

Technique

- **Avoid sedation or general anaesthesia**; if unavoidable, use ketamine rather than midazolam.
- Routine prophylaxis is not needed, because post-study UTI is about **1.4%**.
- Use a **6 Fr dual-lumen** urethral and a rectal catheter, and **fill at 5–10% of EBC per minute** with warm fluid.
- **Stop filling** at strong urge, discomfort, voiding, leak greater than fill rate, Pdet persistently **>40 cmH_2O**, or volume **>150% EBC**.
- Report the first fill, which gives the highest Pdet. **Videourodynamics** adds trabeculation, VUR with its onset volume and pressure, and bladder-neck shape.

Storage phase

- **Pdet = Pves − Pabd**, and normal filling is flat. **DO** means phasic involuntary rises.
- **Compliance = ΔV/ΔPdet**: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better guide.

P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — DLPP = outlet resistance, high = upper-tract risk; ALPP = sphincter competence, low = bad; McGuire 1981

Not in the paper answer:

- A DLPP **>40 cmH₂O**, or no leak at high pressure, signals danger; the adult ALPP cut-off (<60 cmH₂O) has no paediatric norms.

> **KEY POINT**
> **McGuire (1981)**, myelodysplasia: with leak pressure >40 cmH₂O, hydronephrosis or secondary VUR developed in **85%**. Aim for the lowest achievable pressure.

Voiding phase

- **High Pdet with low Qmax means obstruction.**
- Anatomical obstruction has a **quiet EMG**; functional obstruction (DSD) has an **active EMG**.
- Normal Pdet at Qmax is about 66 cmH₂O in boys and 57 in girls; infant boys reach 118 physiologically MCQ.

> **KEY POINT**
> **Upper-tract threats**: compliance <20 (especially <10); DLPP >40 cmH₂O; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow.

- **UMPIRE** (spina bifida) studies at about **3 months**, then at 1, 2 and 3 years, then yearly to 5 y if VUR or a hostile bladder. Hydronephrosis is a late sign, so re-study on any change and before reconstruction.

7 · Evaluation in specific conditions

Condition	Typical findings	When to study
Non-neurogenic LUT dysfunction / enuresis	DO, dysfunctional voiding	Only if refractory
Spinal dysraphism	UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high**	Baseline, serial, pre-surgery
Sacral agenesis	Areflexia with weak sphincter, or DO with DSD; level does not predict	If voiding dysfunction
Anorectal malformation	Tethered cord common	Baseline about 1 y after repair
PUV	DO, then poor compliance, then myogenic failure ("valve bladder" about 15%)	Serial
Cerebral palsy	Upper motor neuron; DO, DSD rare	Only if conservative therapy fails
SCI	Spinal shock causes retention; higher lesions have more DSD	Video-urodynamics 3–6 months after injury
Transverse myelitis	DO most often; motor recovery does not predict bladder	Early, with CIC

> **KEY POINT**
> UMPIRE dropped DSD from high risk because reviewers disagreed on who had it, not because DSD is benign. Neurogenic bladder goals are normal capacity, low storage pressure, complete emptying and upper-tract protection.

8 · Landmark trials

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia urodynamic cohort	DLPP >40 cmH_2O led to upper-tract damage in 85%	40 cmH_2O = safety threshold

42 Bladder Anomalies in Children

1 · Classification and embryology

P3 Embryological development of urinary bladder and management of Hutch Diverticulum? — cloacal division weeks 4–6, UGS fates, trigone, urachus months 4–5; anomaly table; Hutch primary vs secondary, VCUG, treat outlet first, diverticulectomy + reimplant

Not in the paper answer:

- Anomalies are **structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional**; prune-belly syndrome (PBS) is both.

2 · Fetal bladder on ultrasound

- The fetal bladder empties every **15–20 min**, so if unseen, **repeat in the same sitting**. Amniotic fluid is mainly fetal urine after **16 weeks**.
- In exstrophy there is no filling with **normal amniotic fluid**.

	Agenesis	Hypoplasia	Exstrophy
US	Absent after repeat scan	Small, never reaches capacity	No filling; bladder template on abdominal wall
Core defect	Cranial UGS atrophy or no trigone	Inadequate filling or storage	Open bladder plate
Outcome	Survives only with ectopic ureteric drainage	May grow; augmentation often	Reconstruction (see Ch 43)

- **Agenesis** is very rare (about 1:600,000, mostly female); ureters drain into müllerian structures or the rectum, and it is treated by ureterosigmoidostomy or stoma.

3 · Megacystis

- In classic congenital megacystis, **massive VUR causes the dilation**; today the term means any prenatally dilated bladder.

Gestation	Normal	Abnormal
10–14 weeks	Length ≤6 mm; <10% of CRL	**≥8 mm**; ≥13% of CRL
15–40 weeks	Length about GA − 5 mm	> GA + 12 mm
Later (functional)	Empties within 45 min	No emptying in 45 min

- **Causes**: LUTO, with PUV in about 50%, urethral atresia or anterior valves; non-obstructive causes are PBS, MMIHS and cloacal anomalies. Male:female is 8:1.
- **Aneuploidy is about 1/5** (trisomy 13/18), so karyotype.
- In euploid fetuses, 7–15 mm mostly resolves, while **>15 mm** never does. **Persistent megacystis with oligohydramnios means LUTO.**

MMIHS

- MMIHS is a visceral myopathy (**ACTG2**), mostly female, with megacystis and normal or increased fluid. The bladder is acontractile and most need **lifelong CIC**; TPN complications are the leading cause of death.

KEY POINT

A female fetus with megacystis cannot have PUV, so think of MMIHS, cloacal anomaly or urethral atresia.

4 · Bladder duplication

- **Complete sagittal duplication is commonest**, with external genital duplication in up to 90% and lower GI duplication in up to 42%.

ALGORITHM

1. Early: relieve obstruction and prevent infection.
2. Later, if both sphincters are competent, then join the distal urethras.
3. If one is incompetent, then close that bladder neck and excise its urethra.
4. Join duplicated vaginas, with vulvoplasty.

- **Sphincter competence, not anatomy, decides reconstruction.**

5 · Bladder diverticula

- A primary (Hutch) diverticulum is observed if small and excised if there is VUR. For a secondary one, **treat the outlet first**.
- **Syndromic** diverticula (**Ehlers–Danlos, Williams, Menkes**) are multiple and giant; resect only for UTI or stones, as recurrence is common.

6 · Urachal anomalies

	Patent urachus	Sinus	Cyst	Vesicourachal diverticulum
Open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Presents	Urine from the umbilicus in a neonate	Draining umbilicus	Infection (S. aureus), painful mass	Incidental
Imaging	US; fistulogram or VCUG	Sinugram	US; CT/MRI if extensive	Incidental
Treat	Excise with cuff	Complete excision	Drain, then excise	Only if a narrow neck causes stones or UTI

- The differential is a patent **omphalomesenteric duct**. Incidental remnants are increasingly excised only if symptomatic.

ALGORITHM

1. If infected, drain and give antibiotics first.
2. Infraumbilical incision; circumscribe the tract, preserving the umbilicus.
3. Dissect extraperitoneally and excise with a **bladder cuff**.
4. Two-layer closure; catheter overnight.

7 · Acquired lesions

Lesion	Key features	Treatment
Haemangioma	Klippel–Trénaunay, Sturge–Weber; haematuria	Fulguration or Nd:YAG laser

Lesion	Key features	Treatment
Fibroepithelial polyp	2nd commonest paediatric bladder tumour	Endoscopic resection
Nephrogenic adenoma	Female 5:1; after infection, surgery, augmentation	TUR; **80% recur**, so surveillance cystoscopy
Cystitis cystica	Almost all female; mimics tumour	Long-term antibiotic prophylaxis
Eosinophilic cystitis	Male	**Biopsy mandatory**; steroids and antihistamines
Haemorrhagic cystitis	Cyclophosphamide or ifosfamide; BK virus, adenovirus	Mesna prevents; pentosan polysulfate

43 Exstrophy–Epispadias Complex

1 · Spectrum, epidemiology and embryology

Rupture of the cloacal membrane explains the whole spectrum.

P3 P1 Bladder and ureter development, exstrophy–epispadias complex and its management; Describe embryology of bladder… theories for EEC… skeletal abnormalities — spectrum, CBE 1 in 46,000, M:F 2.3:1, offspring 1 in 70; 22q11.2, ISL1, p63; IVF/progesterone/smoking/folate; failed mesodermal reinforcement of cloacal membrane, Marshall–Muecke and other theories; pelvis 12°/18° rotation, anterior pubis 30% short, diastasis 4.8 cm; MSRE stages, osteotomy indications, CPRE outline

Not in the paper answer:

- Spica cast fails in 66% without osteotomy, and diastasis >6 cm needs staged closure.
- Rupture timing explains the phenotypes: early, extensive rupture opens bladder **and** hindgut (cloacal exstrophy); later rupture opens the bladder only; rupture at the genital tubercle gives epispadias.

2 · Anatomy of classic bladder exstrophy

- **Pelvis**: the acetabula are retroverted and the posterior segment is normal. The resulting waddling gait and out-toeing improve with growth.
- **Pelvic floor**: the levator ani lies posterior and flattened. Continent patients have the shortest intersymphyseal distance and deepest bladder neck, hence **deep placement, osteotomy and pelvic fixation**.
- The **intrasymphyseal band** tethers the posterior vesicourethral unit to the pubic rami. It contains **no striated sphincter**, so radical incision down to the levator hiatus is essential.
- **Indirect inguinal hernia** is common, often bilateral in males, so repair it at closure.
- **Anorectum**: the anus is anterior. **Rectal prolapse before closure** resolves with closure, but prolapse **after closure** suggests outlet obstruction and needs urgent cystoscopy.
- **Male**: the **anterior corpora are about 50% shorter** and 30% wider. The prostate is never circumferential. Cavernous nerves lie laterally, so potency is retained, though retrograde ejaculation may follow BNR.
- **Female**: the vagina is short with a stenotic anterior orifice, the **cervix sits in the anterior vaginal wall** and the clitoris is bifid. A wide levator hiatus causes **uterine prolapse**, even in nulliparous women, and **osteotomy does not prevent it**.
- **Bladder**: it is **immature, not neuropathic**. The exposed plate develops polyps, squamous metaplasia and cystitis glandularis, which carry an **adenocarcinoma risk**. Compliance falls after BNR, so follow-up is lifelong.
- The ureters enter laterally without obliquity, so **vesicoureteral reflux (VUR) occurs in 100%** of closed bladders and is corrected by reimplantation at BNR.

3 · Variants

Except superior vesical fissure, variants are managed like CBE.

Variant	Feature	Management
Skin-covered (split symphysis), the commonest	Musculoskeletal defect; urinary tract intact	Formal closure at birth

Variant	Feature	Management
Duplicate exstrophy	Everted mucosal patch plus a closed pelvic bladder	Excise patch, close wall
Superior vesical fissure	Rupture only at the top; sphincter intact	Toilet-trains normally; no BNR
Pseudoexstrophy	Musculoskeletal defect only; low umbilicus	—
Epispadias with bladder prolapse	Prolapse may show only on Valsalva	Repair, with or without osteotomy

4 · Prenatal diagnosis

- Only about 25% are diagnosed prenatally. Signs are **absent bladder filling**, a low umbilicus, wide pubic rami, small genitalia and an **enlarging lower abdominal mass**. Plan **delivery at an exstrophy centre**.

5 · Newborn care and selection

A good first closure best predicts continence.

ALGORITHM

1. Tie the cord with 2-0 silk (no clamp, which traumatises mucosa).
2. Cover the plate with non-adherent plastic wrap; irrigate with saline at each nappy change.
3. Assess heart and lungs; obtain a renal US.
4. Transfer to an exstrophy centre for MDT counselling.

- **Assess** the template (≥**3 × 3 cm** is suitable), diastasis, pelvic malleability under anaesthesia and penile length. A plate that **bulges on crying** is adequate.
- A plate <3 cm, polyp-covered or fibrotic is **inadequate**. Penoscrotal duplication, hypoplastic bladder and bilateral hydronephrosis preclude primary closure.
- A small plate is closed **later, at 6–9 months, with osteotomy**, because a bad closure ruins continence. If it has not grown by 6–12 months, excise it and divert. After delayed closure a **catheterisable channel is needed far more often**.

6 · Surgical philosophies

Repair	Stages / feature	Weakness
Modern staged repair (MSRE)	Closure at birth; epispadias repair at 6–12 months; BNR with reimplant at 4–7 years	Multiple operations
Complete primary repair (CPRE, Mitchell)	Newborn closure with penile disassembly and neck tailoring	Hypospadias; glans loss; early reimplant
Kelly	Closure, then radical soft-tissue mobilisation at 3–6 months, then hypospadias repair; avoids osteotomy	Fistula, stenosis
Erlangen	Total repair at about 8 weeks without osteotomy	—
Mainz	Closure, then sigma pouch at about 2 years	Colonic malignancy

KEY POINT

Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all still need an outlet procedure. Compressing the reconstruction adds ischaemic and outlet risk without removing that need.

7 · Primary closure (MSRE stage 1)

Closure creates an incontinent epispadias with enough outlet for growth.

ALGORITHM

1. General anaesthesia with **epidural** (catheter to T10–T12).
2. Outline a 2-cm mucosal strip from the trigone to mid-shaft (boy) or vaginal orifice (girl); incise the bladder–skin junction.
3. Free the bladder from the rectus and strip peritoneum off the dome.
4. **Divide the urogenital diaphragm fibres sharply down to the levator hiatus**; free the corpora from the rami.
5. Close bladder, neck and posterior urethra; **taper over a 12-Fr sound**.
6. Place a suprapubic Malecot (4 weeks) and ureteric stents; leave the **urethra unstented**.
7. Close the pubis with a **No. 2 nylon horizontal mattress** suture, knot away from the neourethra; fashion a neoumbilicus.

- **Tapering** gives enough resistance for bladder growth and against prolapse, but not enough to harm the kidneys.
- **Ureteric stents** stay 10–14 days because oedema can obstruct the ureters; a urethral stent causes necrosis.
- The **commonest error** is inadequate urogenital diaphragm dissection, after which the unit is pulled forward and prolapses.

8 · Osteotomy and immobilisation

Osteotomy and rigid immobilisation cut closure failure several-fold.

- In a **newborn <72 h** whose malleable pubes meet easily, osteotomy is not needed. **If in doubt, osteotomise.**
- Osteotomise if closure is **>72 h** after birth, diastasis is **>4 cm**, the pelvis is stiff, the template is large, after **failed closure**, and in **cloacal exstrophy**. Diastasis >6 cm needs staged closure with a fixator.

Type	Position	Features
Combined bilateral anterior innominate and vertical iliac (preferred)	Supine	Best malleability; least recurrence
Bilateral posterior iliac (original)	Prone	Worse approximation, more recurrence
Anterior innominate alone	Supine	Less mobility

- Aftercare is **modified Buck traction for about 4 weeks with an external fixator for 4–6 weeks**. Without osteotomy, use **modified Bryant traction** (hips 90°) for 4 weeks.

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
Spica cast	47%	30%	**66%**

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
External fixation with traction	**7.5%**	8%	— (few)

- Failure falls with traction plus fixator for **>4 weeks**; osteotomy type and epidural make no difference. Complications are pin-site inflammation, transient femoral nerve palsy and delayed union.

9 · After primary closure

- At 4 weeks calibrate the outlet, scan and check the residual. Give **continuous antibiotic prophylaxis**, because all have VUR.
- Perform **yearly cystoscopy and cystogram under anaesthesia** to measure capacity. If capacity is **<30 mL at 1–2 years**, a continence procedure may not be feasible.
- If the **outlet is too tight**, escalate from dilation to incision and revision. Recurrent UTI with a distended bladder suggests **intrapubic stitch erosion**.

10 · Epispadias repair (stage 2)

Urethroplasty raises outlet resistance, so the bladder grows.

- It is done at 6–12 months, before BNR, because urethroplasty **increases capacity**. Give **testosterone enanthate 2 mg/kg intramuscularly (IM) at 5 and 2 weeks** before every urethroplasty.
- **Modified Cantwell–Ransley** (preferred): the plate is freed but its **distal 1 cm stays on the glans**; it is tubularised over 8 Fr and the **corpora are rotated over the neourethra**, which then lies ventral.
- **Chordee** is corrected by dorsomedial corporotomy with cavernocavernostomy; skin cover uses ventral preputial flaps.
- Early fistula occurs in about 1 in 6, usually at the base; **coronal fistula is rare**.
- **Mitchell–Bägli disassembly** risks plate ischaemia, and most end **hypospadiac**. Glans or corporal loss is linked to closure **without osteotomy**, so if ischaemia is seen, osteotomise immediately.
- **CPRE**: about half need reimplant in year 1, 65–80% need an outlet procedure, and most end on CIC.
- **Kelly**: unaided voided continence is under 1 in 5, with glans loss reported.
- **Combined closure and epispadias repair** suits failed or delayed closures and boys >6 months, but risks ischaemic penile loss.
- **Female closure**: bladder, urethra and vagina move deep as one unit, with **monsplasty and joining of the clitoral halves**.

11 · Bladder neck reconstruction (stage 3)

Capacity at BNR best predicts dryness.

- **Prerequisites**: capacity **≥100 mL** under anaesthesia, stable urodynamics, a motivated child of 5–7 years and a voiding programme. Reimplant earlier for pyelonephritis or worsening hydronephrosis.

ALGORITHM

1. **Modified YDL**
2. Open the bladder neck transversely.
3. Perform **Cohen cross-trigonal reimplantation**.
4. Outline a posterior mucosal strip **15 × 30 mm**; denude the lateral muscle triangles.
5. Tubularise the strip and **double-breast the muscle flaps** over it.
6. Suspend to the rectus fascia; suprapubic tube for 3 weeks, then clamp trials.

- After MSRE, about **70–75%** are dry by day (intervals >3 h); CPRE with YDL is lowest (about 56%).
- **Predictors** are template size, **successful primary closure** and, most importantly, **capacity at BNR**. Closure within 72 h or with osteotomy makes continence likely, whereas delayed closure without osteotomy gives about 10%. After two closures, fewer than 1 in 5 void continently.
- Day dryness comes within about 12 months and night dryness takes 2–3 years; **not dry by 2 years means failure**.

Category (after BNR)	Definition
Continent	Dry day and night, interval ≥3 h
Socially continent	Dry 3 h by day, occasional wet nights
Wet	Dry <3 h by day, wet at night

12 · Failures and complications

Each failed closure shrinks the bladder.

- **Failed closure** means dehiscence, **bladder prolapse**, outlet stricture or vesicocutaneous fistula. Causes are incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection and tube malfunction.
- In expert series, major dehiscence or prolapse occurs in about 3% and femoral palsy in about 2%.
- **Reclosure** waits at least 4–6 months and uses **osteotomy**, rigid immobilisation and radical urogenital diaphragm division. After secondary closure, voided continence is only about 1 in 6.
- Outlet stricture presents with UTI, an enlarging bladder, stones or rectal prolapse; obstruction for more than 6 months usually means augmentation.
- **Failed BNR** means no 3-h dry interval by 2 years. **Redo YDL** only if the neck is patulous, capacity >100 mL and urodynamics stable; otherwise do **bladder neck transection, augmentation and a Mitrofanoff channel**.
- **Genital**: urethrocutaneous fistula is commonest; glans or corporal loss may need **radial forearm phalloplasty**.

13 · Diversion and augmentation

- **Ureterosigmoidostomy** needs anal continence. It risks pyelonephritis, hyperchloraemic acidosis and **colonic adenocarcinoma about three decades later**, so **yearly colonoscopy** is required.

- If the bladder is too small, use a **non-refluxing colon conduit**. After failed BNR, do **augmentation with an appendiceal channel, with or without neck closure**; stones are the commonest complication.
- **Continent diversion** is indicated for closures that do not grow, inadequate capacity or wetness after BNR, and upper-tract decompensation.

14 · Long-term issues

- **Malignancy**: augmented exstrophy bladders develop adenocarcinoma, so survey endoscopically from 5 years after reconstruction.
- **Male**: libido, erection and orgasm are preserved. Fertility is impaired by retrograde ejaculation and oligospermia; about 1/4 achieve paternity, half via ICSI.
- **Female**: a stenotic introitus needs **perineal flap vaginoplasty**, because a cutback worsens prolapse. **Uterine prolapse** is treated by sacral suspension. **Pregnancy** ends in a **planned caesarean with a urologist present**.
- Psychosocial problems need lifelong support and transition care.

15 · Epispadias

P3 Development of the urethra, congenital urethral anomalies and management of epispadias — incidence, male/Davis classification, VUR rates, >3 cm means exstrophy pathway, Cantwell–Ransley then YDL at ~100 mL, female repair, continence males 47–82% / females 70–87%

Not in the paper answer:

- Glanular epispadias is never incontinent.

	Male	Female
Incidence	1 in 117,000	1 in 484,000
Classification	Glanular, penile, **penopubic** (about 70%, incontinent)	Davis: vesicular, subsymphyseal, retrosymphyseal (incontinent), in rising severity
VUR	30–40%	30–75%
Presentation	At birth	May present only with incontinence

ALGORITHM

1. **Female repair (6–12 months)**
2. Incise vertically through the urethral wall at 9 and 3 o'clock.
3. Excise the urethral roof (dorsal wedge taper) and close over a 10-Fr Foley.
4. Unite the clitoral halves and labia minora; bring mons fat over the suture line.
5. Catheter 5–7 days; BNR at 4–5 years.

- Treat as exstrophy (osteotomy and closure) if diastasis is >3 cm, the sphincter is scant or capacity is small.
- In **males** with a small bladder, do urethroplasty first, then BNR at about 100 mL (4–5 years).
- Female continence after BNR is about **87%**, as good as in CBE girls, because the bladder was never exposed.

16 · Cloacal exstrophy

A multisystem defect; neurological level sets disability.

- Incidence is 1 in 200,000–400,000. **OEIS** stands for Omphalocele, Exstrophy, Imperforate anus and Spinal defects. **Two hemibladders flank exstrophied ileocaecal bowel**, with a rudimentary hindgut and separated hemiphalluses.

Feature	CBE	Cloacal
Spinal defects	Uncommon	**85–100%** (lipomeningocele commonest)
Omphalocele	Rare	Near-universal
Upper-tract anomalies	Uncommon	41–60%; solitary kidney commonest
Osteotomy	Selective	Always
Continence	Mostly voiding	Mostly augmentation with CIC

- Only about 1 in 5 has a sacral skin sign, so all need spinal MRI. A 46,XY child may have an **intravesical phallus**, so biopsy before calling it aphallia.
- Prenatally, the **"elephant trunk" sign** is prolapsed ileum. **Gender** follows **karyotype**.
- **Bowel**: never discard hindgut; tubularise the caecal plate with an end colostomy.

ALGORITHM

1. Stage 1 (newborn): close the omphalocele, separate the caecal plate, join the hemibladders and form a colostomy.
2. Stage 2: bladder closure with **combined osteotomy** and fixation for 6–8 weeks.
3. At 5–7 years: YDL only if capacity >100 mL (rare); usually augmentation with a continent stoma.

- Combined osteotomy gives about 90% closure success but no continence gain; staged closure beats single-stage.
- Primary closure fails in about 31–43%. Continence relies on augmentation with CIC, and the **neurological level** determines disability.

44 Prune-Belly Syndrome

P1 P3 Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management; Prune belly syndrome — triad, 1 in 29,000–44,000, 95% male; four theories, MYOCD/FLNA; pathology by organ (dysplasia ~50%, VUR ~75%, failed pump); extra-GU rates; Woodard I–III; neonatal work-up; vesicostomy, reconstruction after 3–6 months; orchidopexy ~6 months; Randolph/Ehrlich/Monfort; creatinine <0.7 mg/dL, 30% CRF

Not in the paper answer:

- Renal failure is driven by pyelonephritis, dysplasia and polyuria; transplant outcomes are equal.
- Female PBS is ~5%, with normal gonads.

1 · Definition and aetiology

- The **triad** (Eagle–Barrett) is deficient abdominal wall muscle, bilateral intra-abdominal testes and a dilated urinary tract. Incidence is ~**1 in 29,000–44,000** and **95% male**; it is commoner in twins and African ancestry.
- **Survival follows the kidneys and lungs, not the belly**; the wall is least prognostic.
- There is no inheritance pattern; multiplex families show the smooth-muscle genes **MYOCD** and **FLNA** MCQ.
- Theories are early urethral obstruction, a lateral plate mesoderm defect, an intrinsic urinary tract defect and a yolk sac defect.

> **KEY POINT**
> Weakness of the obstruction theory: most have **no obstruction after birth**. The tract is **dilated, collagen-rich and largely non-obstructed**, and this drives management.

2 · Genitourinary pathology

- **Kidneys**: dysplasia in ~**50%**, worst with outlet obstruction without urachal decompression. Dilation does not correlate with dysplasia; **infection, not obstruction, is the main threat**.
- **Ureters**: the distal ureter is worse, so the **proximal ureter is preserved** in reconstruction. Vesicoureteric reflux (VUR) occurs in ~**75%**. The ureter is a **failed pump, not a blocked pipe**.
- **Bladder**: massive, thick but smooth, with a urachal pseudodiverticulum; the urachus is patent at birth in **25–30%**. It has normal compliance, large capacity and poor sensation; about half void normally but may deteriorate.
- **Prostatic hypoplasia** dilates the posterior urethra and causes ejaculatory failure. **Urethral atresia** is lethal unless the urachus is patent. **Megalourethra** is scaphoid (spongiosum only, milder) or fusiform (cavernosa too, more severe).
- **Testes** lie over the iliac vessels. Infant germ cells are normal, so the abdominal environment destroys later potential, hence early orchidopexy.

3 · Extragenitourinary anomalies

About **75%** have them, and nearly **half premature**. Pulmonary problems (>50% of survivors) include hypoplasia, pneumothorax and pneumonia; cardiac (~10%), gastrointestinal (≥30%, malrotation) and orthopaedic (up to 65%, talipes, hip dysplasia, knee dimples) anomalies are common.

KEY POINT

Two mechanisms: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

4 · Presentation and spectrum

- **Prenatal** appearance mimics posterior urethral valves: massive bladder and hydroureteronephrosis. Dilation does **not** predict function.

Woodard category	Kidneys / lungs	Course	Urological management
I	Dysplasia; **pulmonary hypoplasia**; oligohydramnios, urethral atresia	Stillbirth or early death (~20% perinatal mortality overall)	Supportive; simple bladder drainage
II	Full triad; minimal or unilateral dysplasia; no lung hypoplasia	Risk of progressive renal insufficiency	Individualised — **controversial group**
III	Incomplete or mild; no dysplasia; stable	Majority; good	Surveillance; surgery for UTI or deterioration; orchidopexy in year 1

- The abdominal wall is not a sorting criterion. **Pseudo-prune** (partial flaccidity, unilateral undescended testis) is diagnosed late and has higher late renal failure.

5 · Evaluation

ALGORITHM

1. Team; **heart and lungs first** (chest X-ray).
2. Creatinine trend; the day-1 value is maternal.
3. Renal–bladder ultrasound.
4. Prophylaxis and circumcision; then selective imaging.

- A voiding cystourethrogram is done only for renal insufficiency or suspected outlet obstruction, under antibiotic cover. A **DMSA** scan is done at **4–6 weeks**; MAG3 renograms are unreliable with massive dilation. Outlet obstruction is drained by a percutaneous suprapubic tube.

6 · Management philosophy

	Conservative	Early reconstruction
Premise	Dilation is non-obstructive and may improve	Stasis, reflux and infection drive renal loss
Operate when	Proven obstruction, intractable infection	Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function
Best for	Category III	Selected category II

- Delay major reconstruction to **≥3–6 months** for the lungs. Single-stage comprehensive surgery adds no morbidity.

7 · Urinary tract surgery

- **Blocksom vesicostomy** (sepsis, renal failure, atresia): make the stoma larger than usual, as it tends to stenose. Avoid proximal ureterostomy.
- **Ureteroplasty**: excise the distal ureter, taper the upper part and reimplant. Reduction cystoplasty is reserved for a large urachal diverticulum, as capacity recurs; clean intermittent catheterisation empties better. Circumcision reduces UTI.

8 · Orchidopexy

- A broad mesorchium defeats the inguinal approach. Do a **bilateral transabdominal orchidopexy at ~6 months**, often with other surgery. Later, a **staged Fowler–Stephens** procedure is preferred.

9 · Abdominoplasty

- Cosmetic gain is certain; functional gains (cough, defecation, emptying) are plausible. Operate from ~6 months with other surgery.

Technique	Incision / umbilicus	Key step / note
Randolph	Transverse, 12th rib to 12th rib	Excision of the lower wall; lateral bulge persists
Ehrlich	Vertical; umbilicus on a pedicle	Vest-over-pants advancement
Monfort	Vertical ellipse; umbilicus in situ	Central fascial plate with lateral flaps; **transperitoneal**, so ideal with reconstruction
Dénes	Fusiform	Modified Ehrlich
Furness	Midline, limited	Plication; peritoneum not opened

ALGORITHM

1. **Monfort**
2. Skin ellipse, umbilicus spared; incise the central plate at the lateral rectus borders.
3. Do the genitourinary surgery.
4. Suture plate edges to scored lateral peritoneum.
5. Overlap lateral flaps over drains.

10 · Long-term outcome

- Creatinine **<0.7 mg/dL at 1 year** predicts stable function. Up to **30%** develop chronic renal failure, driven by pyelonephritis, dysplasia and polyuria. Transplant outcomes are equal.
- **Bladder**: timed and double voiding; if these fail, use CIC.
- **Fertility**: puberty and erections are normal, but **azoospermia means no spontaneous paternity**; sperm retrieval with ICSI is used.

KEY POINT

Long-term care is a contest against **stasis**: lowering residual urine and infection protects the kidneys, while dysplasia and polyuria set a ceiling surgery cannot raise.

45 Posterior Urethral Valves

1 · Definition and classification

- **Posterior urethral valves (PUV)** are the commonest cause of bladder outlet obstruction in infant boys. Ablation relieves obstruction but **not the embryological injury** to bladder and kidney, so care is lifelong.
- Lower urinary tract obstruction (LUTO) affects ~2.2/10,000 births; PUV is commonest (~1.4/10,000), then urethral atresia and prune-belly.

P1 Embryology of Posterior Urethral Valves. — Young 1919; types I–III and origin; COPUM (one membrane torn by catheter); genetics: chr 11 duplications, CNVs (worse renal outcome), TBX5, PTK7

Not in the paper answer:

- Type III resembles a congenital stricture; it is contentious but accepted.

Young type	Anatomy	Status
I (95%)	Leaflets from verumontanum fuse anteriorly just proximal to external sphincter	Obstructing; the lesion treated
II	Folds from verumontanum to bladder neck	Non-obstructing; probably not real
III (5–10%)	Annular membrane, central aperture (persistent urogenital membrane)	Obstructing

- Copy number variant burden predicts higher nadir creatinine and CKD stage 5.

2 · Pathophysiology

- Obstruction causes detrusor hypertrophy, then **fibrosis replaces smooth muscle**. The bladder is the **driver**: renal dysfunction, reflux (VUR) and voiding dysfunction act through it.
- The voiding cystourethrogram (VCUG) triad is a **dilated posterior urethra, hypertrophied bladder neck and flattened verumontanum**.
- Ureters dilate from **pressure transmission, VUR (up to 70%) and polyuria**, then lose peristalsis, causing stasis and infection.
- Renal injury is **obstructive uropathy** (partly reversible) plus **dysplasia** (fixed at birth, which sets the ceiling). Tubular injury causes **nephrogenic polyuria** and post-obstructive diuresis.
- Angiotensin II induces TGF-β1 and fibrosis; ACE inhibitors may help selected children.

Organ / lesion	Effect	Natural history
Lung — hypoplasia	Respiratory failure	Commonest cause of neonatal death
Kidney — obstructive uropathy	Reversible insufficiency	Improves; may recur with bladder dysfunction
Kidney — dysplasia	Irreversible insufficiency	Progressive failure, hypertension
Kidney — tubular injury	Salt and water wasting	Nephrogenic diabetes insipidus; worsens with age
Bladder	Poor sensation, hypercontractility, low compliance; later myogenic failure	Lifelong, changes with age

Organ / lesion	Effect	Natural history
Ureters	Poor coaptation, stasis	Most retain hydronephrosis — **not** a sign of residual obstruction

"Pop-off" mechanisms

- **VURD** (unilateral high-grade VUR into a non-functioning dysplastic kidney) was thought protective. This is **refuted**: most with normal creatinine at 2 years lose it by 8–10 years, and urodynamics are worse. Urinoma, ascites and diverticula are debated but managed the same way.

> **KEY POINT**
> Every pop-off is false reassurance: normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

3 · Diagnosis

Antenatal

- Ultrasound shows a **thick distended bladder, bilateral hydroureteronephrosis, "keyhole sign" and oligohydramnios**; an echogenic or cystic cortex means dysplasia. It is only ~50% accurate for the cause.
- About half are diagnosed antenatally, but this has **not** improved renal outcome. A male fetus with these findings needs postnatal US and a **VCUG before discharge**.

Postnatal

- **US** comes first: the wall is thick if **>3 mm distended / >5 mm empty**.
- **VCUG** is definitive, showing a dilated elongated posterior urethra with **abrupt funnelling** at the valve; about half have high-grade VUR. Use a 5–8 Fr feeding tube and **withdraw the catheter during voiding**.
- A **MAG3** renogram is done at **2–3 months**. Creatinine in the first 48 h is maternal; follow the trend to the **nadir** in year 1.

4 · Fetal intervention

- Decisions hinge on **oligohydramnios, hydronephrosis degree and laterality, and gestational age**. Favourable features are oligo- (not an-)hydramnios, normal karyotype, no cortical cysts and favourable urine.

Fetal urine (after 20 weeks)	Favourable
Sodium / chloride	**<100 / <90 mEq/L**
Osmolality	**<200 mOsm/L**
β2-microglobulin	**<6 mg/L** (most sensitive)

- **Triple serial sampling** 48–72 h apart beats one sample, though prediction was poor on systematic review.
- A **vesicoamniotic shunt** may prevent **pulmonary hypoplasia** but gives **no renal benefit**. Fetal cystoscopic laser ablation is no better. Both carry **40–50%** pregnancy loss.

KEY POINT

Fetal intervention can rescue the lung, not the kidney: fluid deficiency is still in progress, but dysplasia is already established.

5 · Neonatal presentation and initial management

- **Pulmonary hypoplasia** is the commonest early death and the usual cause of delayed surgery.

ALGORITHM

1. Stabilise the airway.
2. Pass a **5–7 Fr feeding tube**. If it curls: coudé, curved stylet, or rectal finger.
3. Confirm on US; minimal balloon fill.
4. Creatinine, electrolytes, fluids.
5. US, then VCUG; antibiotic prophylaxis.
6. Ablate when stable.

- **Post-obstructive diuresis** needs monitoring of output, weight and electrolytes with IV replacement. **Hyponatraemia** is transient pseudohypoaldosteronism, treated with sodium.
- **Urinoma or ascites** (3–10%) resolves with bladder drainage.
- **Delayed presentation** (UTI, incontinence, renal failure) is common: over 40% present after 6 months, and late does not mean milder.

6 · Surgery

Primary valve ablation (preferred)

- Ablation restores flow and **normal bladder cycling**, superior to passive diversion. Use a **7.5–9 Fr cystoscope** with holmium laser or Bugbee.
- Incise at **5 and 7 o'clock, with or without 12 o'clock**. Valves are thin and avascular, so **incise, don't resect**.
- For a small urethra, use vesicostomy or **PADUA** (progressive urethral dilation with up-sized tubes). Leave a catheter ≥24 h and **repeat VCUG within 1 month**.

Complication	Cause / prevention	Management
Urethral stricture	Over-resection, hot loop, oversized instrument	Dilation / urethrotomy; urethroplasty if dense
Meatal injury	Resectoscope through small meatus	Smaller scope / PADUA
Residual valve	Incomplete incision	Persistent funnel on VCUG: re-ablate
Sphincter injury	Ablating distal to valves	Incontinence — precise incision

- **Bladder neck incision** gives no benefit and risks retrograde ejaculation; it is not recommended.

Vesicostomy

- It is used for a very small or premature infant, failed decompression, or persistent deterioration. It **does not defunctionalise** the bladder and matches ablation in final outcome.

P4 Vesicostomy. — Blocksom: 2 cm transverse incision midway, dome (not anterior wall) to skin to stop prolapse; indications, complications, closure

Not in the paper answer:

- The urachus is excised with a small dome cuff; using the dome keeps the posterior wall taut.

Upper tract diversion, circumcision and nephrectomy

- Upper tract diversion gives **no renal protection**; use it only for deterioration **despite full lower tract drainage**.
- UTI risk is 50–60% in PUV; circumcision cuts it by over 80%, so offer it to all.
- VURD nephroureterectomy is rare; **keep the ureter** for augmentation.

7 · VUR in PUV

- VUR occurs in 50–80% and is **secondary**. It resolves after ablation in 25–40% and does not predict renal outcome.
- Treat the **bladder**. Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder care.

> **KEY POINT**
> VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

8 · Bladder dysfunction and valve bladder syndrome

P2 Valve bladder: management — Mitchell definition; polyuria, poor sensation, residual; age pattern; continence ~5.5 y; ladder: behavioural, then oxybutynin 0.1 mg/kg bd / α-blocker, then CIC, then overnight drainage 7–10 h, then Mitrofanoff, then ureterocystoplasty

Not in the paper answer:

- Stop oxybutynin if residual and capacity rise, as myogenic failure can follow.

> **ALGORITHM**
> 1. High-pressure voiding causes fibrosis and weaker contraction.
> 2. Post-void residual (PVR) rises; nephrogenic polyuria adds volume.
> 3. Upper tracts store urine and refill the bladder, so it is never empty.
> 4. Pressure is back-transmitted; renal function declines, causing more polyuria.

- Continence comes at ~5.5 vs 2.9 years. Do **uroflow and PVR** in every toilet-trained boy; new incontinence after dryness is a red flag.
- An α-blocker suits early bladder-neck obstruction and an antimuscarinic the overactive wall; both harm once the bladder is hypocontractile. Do **videourodynamics** serially.

9 · Prognosis for renal function

- Lifetime end-stage renal disease (ESRD) is **20–50%**. Risks are dysplasia, **year-1 nadir creatinine**, recurrent UTI and bladder dysfunction.

Nadir creatinine (1 year)	ESRD risk
<0.8 mg/dL	Minimal (best single predictor)
>1.2 mg/dL	High
<0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL	**0% / 2% / 27% / 100%**

- **0.7 mg/dL is the watershed**, where risk leaps. On US, echogenicity, cysts and a smaller renal parenchymal area predict a worse outcome.

10 · CKD, ESRD and transplantation see Ch 61

- Do **videourodynamics in every candidate** and optimise with CIC or overnight drainage.
- Graft and patient survival are equal if the bladder is managed; late graft loss follows bladder dysfunction. Augmentation can be done before or after transplant.

11 · Long-term follow-up, sexual function and fertility

- Transition to adult urologists who know valves. Adult LUTS are 2–3 times commoner; erectile function and fertility are similar overall but worse with incontinence or renal insufficiency.
- Lifelong checks are blood pressure, creatinine, urine protein, US, and uroflow with PVR.

12 · Related urethral lesions

P2 Anterior urethral valve — 25–30× rarer than PUV; diverticulum; penile bulge; vesicostomy if premature, else endoscopic ablation; open repair of large diverticulum; 80% bladder dysfunction, 78% normal renal function; 25-fold risk with azotaemia+VUR+UTI

- **Urethral atresia** needs decompression to survive; treat with vesicostomy, then PADUA.
- **Urethral duplication** (Effmann I–III): the **ventral** channel has the sphincter, so it is the one moved to the glans.
- **Urethrorrhagia** in peripubertal boys resolves in up to 92%; treat voiding dysfunction and reserve cystoscopy for atypical cases.
- Anorectal malformation often has a rectourinary fistula; catheterise before PSARP.

13 · Landmark trials

Trial	Design	Result	Take-home
PLUTO	RCT: vesicoamniotic shunt vs conservative in LUTO	Trend to better 28-day survival; poor long-term survival and renal function in both	Shunt may help lungs, not kidneys
Circumcision RCT	Circumcision and prophylaxis vs prophylaxis	Febrile UTI 3% vs 49% over 2 years	Circumcise boys with PUV
Oxybutynin RCT	Oxybutynin to toilet training vs observation post-ablation	Better hydronephrosis/VUR; no change in creatinine, UTI, continence	Selective, monitored use

46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

1 · Definition and causes

- **Neurogenic lower urinary tract (LUT) dysfunction** is bladder and/or sphincter dysfunction from aberrant innervation.
- The commonest cause is **neural tube defect (NTD)**, mainly spina bifida (SB). Survival is about 90% to 10 years, so **renal failure** is a leading late morbidity; **10%** of SB deaths are urological. Clean intermittent catheterisation (CIC, Lapides 1971) was the pivotal advance.

Congenital	Acquired
Myelomeningocele; occult NTD (lipomeningocele, tethered cord); sacral agenesis; anorectal malformation (ARM)	Brain: cerebral palsy (CP), tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive surgery

2 · Myelodysplasia (NTD)

P1 Neurogenic bladder in myelodysplasia — lesion levels; detrusor × outlet patterns; EAU work-up; McGuire risk; early CIC outcomes

Not in the paper answer:

- Trospium (a quaternary amine) has few central effects.

Classification and prevention

- **Myelomeningocele** is open, commonest and most severe; **SB occulta** is closed and usually benign.
- The neural tube closes by **day 35**. **Folic acid** cuts NTDs by 50–70%: all women of child-bearing age take **400 µg/day** from 4 weeks before conception, and women with SB take **4–5 mg/day**.
- Recurrence is **5%** after one affected child. Almost all open NTDs have **Chiari II**, and most have hydrocephalus.
- **LUT function is not predictable from bony level or the legs**, so urodynamics defines every bladder.

> **KEY POINT**
> Management of Myelomeningocele Study (**MOMS**): prenatal repair (<26 weeks) reduced death or shunt need (RR 0.70) and improved motor and cognitive scores at 30 months, at the cost of prematurity, uterine-scar thinning or dehiscence, and more re-tethering.

Pathophysiology

- In **detrusor–sphincter dyssynergia (DSD)**, the sphincter fails to relax during detrusor contraction; it is diagnosed by electromyography.

> **KEY POINT**
> Every pattern is detrusor (over/under) × outlet (competent/incompetent/dyssynergic). Dangerous: **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**, where storage pressure rises and causes reflux and hydroureteronephrosis. An incompetent outlet "pops off": the child is wet but the kidneys are safe.

Newborn evaluation

ALGORITHM

1. Before discharge: renal–bladder US and post-void residual (need for CIC?).
2. After week 1: serum creatinine.
3. Once stable: urodynamics, **safe from ≈3 months**.
4. Cystography: ideally videourodynamics, otherwise VCUG.
5. Optional baseline DMSA scan (EAU).
6. US at 6, 12, 18 and 24 months, then yearly.

- Guidelines come from ICCS 2012, BAPU 2016 (most selective), SBA 2018 and EAU 2022. In the UMPIRE newborn cohort, dilating VUR was **7%**.
- **Cystatin C** beats creatinine, because low muscle mass hides a falling GFR. High-grade VUR acts as a pop-off, so urodynamic pressures underestimate risk.

Urodynamic risk factors

Category	Definition
Hostile / high	Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 cmH_2O**; NDO with DSD (DSD removed in UMPIRE revision)
Intermediate	NDO, reduced compliance, 25–39 cmH_2O
Abnormal but safe	<25 cmH_2O
Normal	Normal capacity, compliance >15 mL/cmH_2O, no NDO/DSD, minimal residual

- Poor compliance independently predicts renal injury.

Tethered cord

- Growth stretches a low-lying cord, so gait, sensation and bladder deteriorate. Suspect it with new wetting, rising pressure, new UTIs, worsening scoliosis or back pain. Do urodynamics **before untethering and 3–4 months after**.

3 · Proactive (early) management

Priorities in order are **kidneys, then continence, then UTI avoidance, then sexual function and fertility**.

- **Early CIC and an antimuscarinic** (Bauer) treat hostile dynamics. All paediatric societies endorse this; there is **no benefit if the child empties spontaneously**. If hydronephrosis persists, add overnight drainage.
- Give **no** antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic.

Outcome	Expectant	Early CIC + antimuscarinic
Hydroureteronephrosis	18–80%	0–15%
Renal scarring	27–50%	4–12%
End-stage renal disease (ESRD)	≈30%	0–1.6%
Augmentation	—	18–24% fewer

KEY POINT

Proactive won because it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure).

4 · Management ladder

ALGORITHM

1. CIC and antimuscarinic, with or without a β3-agonist.
2. If that fails: intradetrusor botulinum toxin.
3. If CIC is impossible: vesicostomy as a bridge.
4. Then augmentation, with or without a catheterisable channel and an outlet procedure see Ch 48.

Pharmacotherapy

- **Antimuscarinics** are first-line. FDA-licensed drugs are **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch or intravesical), **solifenacin** (≥2 y), **fesoterodine** (≥6 y, >25 kg) and **mirabegron** (≥3 y; may cause hypertension).
- α-Sympathomimetics for low-outlet incontinence are unlicensed and help a minority.

Botulinum toxin A

- Under general anaesthesia, inject **≈10 U/kg (max 360 U)** into about 30 sites, **sparing the trigone**. Continence is 65–87%, lasting 6–10 months, and repeats stay effective. Poor pre-existing compliance predicts poor response.

Surgery

- **Vesicostomy** is a temporary bridge when CIC is unmanageable. **Augmentation** (ileum preferred) treats the refractory bladder, with a channel or outlet procedure as needed see Ch 48. After ileum, check B12 yearly.

VUR in the neurogenic bladder

- VUR is secondary, and treating the bladder resolves 30–55%. **Avoid Credé** with a reactive sphincter, because it worsens VUR.
- Antireflux surgery is for breakthrough febrile UTI despite low pressure. At augmentation, **reimplant grade ≥III**.

5 · Bowel management and latex allergy

- Neurogenic bowel is managed stepwise with diet, laxatives and transanal irrigation, then a **Malone antegrade continence enema (ACE)**, often at augmentation.
- Latex sensitisation risks intraoperative anaphylaxis, so care is latex-free **from birth**.

6 · Sexual function and transition

- Function is **inversely proportional to neurological level**. Central precocious puberty affects about half of girls.
- **Males** have erectile dysfunction in about 75%; **sildenafil** works in about 80%. **Females** conceive in 70–80%; avoid the combined pill and IUD. Offer a structured transition.

7 · Other causes

- **Sacral agenesis** is linked to maternal diabetes; look for flat buttocks and a short gluteal cleft. Level does not predict the bladder, and about a third present late.
- **Spinal cord injury**: CIC during spinal shock; a **suprasacral lesion later gives NDO with DSD**. **Autonomic dysreflexia** (lesion **≥T6**) causes severe hypertension with bladder filling, so monitor blood pressure during urodynamics.
- **Transverse myelitis** causes acute retention in over 95%; **late CIC leads to poorer compliance**, so do early urodynamics and CIC.
- **Cerebral palsy** gives coordinated overactivity; upper-tract risk is **<5%**, so urodynamics is only for failed conservative care or recurrent UTI.
- **ARM** has a cord anomaly in 18–50%, so do spinal US before 3 months; neuropathy comes from the cord, not the PSARP. Pelvic surgery can injure pelvic nerves.

KEY POINT

Lesion level predicts pattern. **Suprapontine** (CP) gives coordinated overactivity and low risk. **Suprasacral cord** (SCI, myelitis, tethering) gives NDO with DSD and high risk. **Sacral/peripheral** gives underactivity. Myelodysplasia is mixed and unpredictable.

8 · Landmark trials

Trial	Design	Result	Take-home
MOMS	RCT, prenatal vs postnatal repair	Less shunting, better motor score; more prematurity, uterine dehiscence	Fetal repair in selected centres
McGuire 1981	Myelodysplasia cohort	DLPP >40 cmH_2O led to upper-tract damage	40 cmH_2O = safety line
Sildenafil in SB	Double-blind placebo RCT	≈80% improved erections	First-line for ED

47 Functional Disorders of the Bladder and Bowel in Children

1 · Normal bladder and bowel function

P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children — innervation, PAG–PMC control, maturation, ICCS functional types, flow patterns, management principles

Not in the paper answer:

- The International Children's Continence Society (ICCS) now uses expected bladder capacity = (age + 1) × 30 mL; the paper's (age + 2) is the older formula†.
- In **infancy**, voiding pressures are often **>100 cm H2O** and interrupted voiding is normal; both are gone by **~2 yr**. Dyscoordination later is **dysfunction, not immaturity**.
- Late toilet training is linked to incontinence, enuresis and urinary tract infection (UTI).

> **KEY POINT**
> Bladder and bowel are one unit, sharing pelvic space and afferents. A loaded rectum causes overactivity and poor emptying, so treating constipation improves lower urinary tract symptoms (LUTS).

2 · Definitions — ICCS

Bladder and bowel dysfunction (BBD) is lower urinary tract (LUT) dysfunction plus constipation and/or faecal incontinence in a neurologically normal child.

Group	Symptom	ICCS definition
Storage	Increased frequency	**≥8 voids/day**
	Decreased frequency	**≤3 voids/day**
	Urgency	Sudden immediate need to void
	Nocturia	Waking at night to void
Voiding	Hesitancy	Difficulty initiating when ready
	Straining	Valsalva to initiate and maintain
	Intermittency	Non-continuous stream
	Retention	Cannot void despite effort; distended bladder
Other	Holding manoeuvres; incomplete emptying; recurrent UTI	Incomplete-emptying sensation counts even with minimal post-void residual (PVR); recurrent UTI is itself a BBD symptom

- **Daytime urinary incontinence (DUI)** is leakage while awake at age **≥5 yr**. **Enuresis** is intermittent leakage while asleep at ≥5 yr: **monosymptomatic (MNE)** has no other LUTS, **non-monosymptomatic (NMNE)** has some.
- **Faecal incontinence** is mostly **retentive** (overflow); ~1 in 5 is non-retentive.
- † **Expected bladder capacity (EBC)** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

Daytime LUT conditions

Condition	Definition / mechanism	Key features
Overactive bladder (OAB)	Urgency, with or without frequency, nocturia, incontinence; no UTI or other pathology	Urgency incontinence, holding manoeuvres
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction **during voiding**, neurologically intact; staccato flow	Learned; high PVR, UTI, secondary vesicoureteral reflux (VUR)
Underactive bladder	Raises abdominal pressure to initiate, maintain or complete voiding	Infrequent, interrupted flow, PVR
Voiding postponement	† Habitual deferral with holding until urgency incontinence	† Low frequency; oppositional defiant disorder
Giggle incontinence	Extensive emptying during or after laughing; LUT otherwise normal	Distinguish from laugh-provoked OAB leak and vaginal reflux
Vaginal reflux	Vagina fills retrograde during voiding	Prepubertal girl; small leak just after voiding
Extraordinary daytime frequency	† ≥ hourly small voids, no incontinence or nocturia	† Benign, self-limiting; exclude UTI, diabetes
Hinman syndrome (non-neurogenic neurogenic bladder)	End stage of dysfunctional voiding	Trabeculation, VUR, hydronephrosis, renal damage, hypertension

KEY POINT

Trap: dysfunctional voiding (neurologically intact) is not detrusor–sphincter dyssynergia (DSD), which needs a neurological lesion. Urgency incontinence is not always OAB: a voiding postponer also leaks with urge, and anticholinergics help briefly, then worsen emptying.

3 · Clinical presentations and associations

- **Acute retention** (peak 2–4 yr) follows genital inflammation, instrumentation or **faecal impaction**; try **analgesia, sitz bath and/or enema before a catheter**.
- **Functional constipation** affects ~14% and is driven by a painful-stool **withholding cycle**. Exclude organic causes such as anorectal malformation (ARM), Hirschsprung disease and spinal causes.
- **ADHD, autism and intellectual disability** have the highest BBD rates and respond worse.

4 · Evaluation

*Evaluation mainly **excludes organic disease**; avoid over-investigation.*

Warning signs

Red flags are rectal bleeding, weight loss or vomiting (organic disease); poor growth or hypertension (chronic kidney disease, CKD); a lumbosacral skin lesion or abnormal leg neurology (spinal MRI); an abnormal anus (ARM); low specific gravity (SG) or glycosuria (diabetes insipidus, DI, or diabetes mellitus); and constant dribbling in a girl (ectopic ureter).

- Ask about bowels in **every** child with LUTS, and use a **3-day bladder diary** and a **bowel diary**. Bristol types 1–2 indicate constipation.

- A **gush on separating the labia** means vaginal reflux. Routine rectal examination is **not** needed.

Investigations (stepwise, non-invasive)

- **ICCS response**: complete 100%, partial 50–99%, none <50%.
- **Urinalysis**: SG >1.005 makes DI unlikely. Persistent proteinuria needs a first-morning protein:creatinine ratio (>**0.2 mg/mg** is abnormal).
- Measure **PVR** within 5 min of voiding at 50–115% EBC.
- **Uroflow** (3 flows, >50% EBC): bell is normal, tower suggests OAB, staccato dysfunctional voiding, interrupted underactive bladder, and plateau obstruction.
- Pelvic floor **EMG** active during voiding means dysfunctional voiding; an **EMG lag time <2 s** means detrusor overactivity. **Video urodynamics** is only for treatment failure or suspected obstruction or upper-tract risk.

Bowel — Rome IV functional constipation (child ≥4 yr)

ALGORITHM

1. ≥2 of, ≥1/week for ≥1 month, irritable bowel syndrome (IBS) criteria not met:
2. ≤2 defecations in toilet/week; ≥1 faecal incontinence/week; retentive posturing/withholding; painful or hard stools; large rectal faecal mass; large-diameter stools that may block the toilet.

- Pain is **not** a Rome IV criterion.
- No routine labs or **plain radiograph**. A rectal diameter **≥30 mm** on US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU).
- If refractory, use a transit study, contrast enema and **anorectal manometry** (absent RAIR needs rectal biopsy for Hirschsprung).

5 · Management of daytime incontinence

ALGORITHM

1. History, exam, urinalysis, PVR, with or without uroflow.
2. Exclude neurogenic causes, diabetes, CKD and anatomical causes.
3. Optimise neurodevelopmental disorders; **treat constipation first**.
4. If wetting is only at night, treat as MNE.
5. Otherwise, **standard urotherapy** (≥4 wk); re-evaluate.
6. Then **specific urotherapy** by diagnosis.
7. If refractory: botulinum toxin or neuromodulation.

Standard urotherapy

- Give information, fluid and diet advice and diaries. Sit with hips and knees at ~90° and feet supported, and use **timed voiding** 6–7 times a day. **Reward adherence, not dryness**. Urotherapy cures DUI in more than half.

Dysfunctional voiding

- Diagnose by **staccato flow with active pelvic floor EMG**.

Renal risk	Management
High (hydronephrosis, VUR, rising creatinine)	Video urodynamics; lower storage pressure; clean intermittent catheterisation (CIC) past the sphincter
Low	Urotherapy, double voiding, treat constipation, short-course prophylaxis for recurrent UTI

- **Biofeedback is the first-line specific therapy**. **α-blockers** (doxazosin) lower PVR. If refractory, give **intrasphincteric onabotulinumtoxinA** 50–300 units.

Overactive bladder

Use drugs only after urotherapy and bowel management.

Agent	Dose / point
Oxybutynin (non-selective)	Immediate release **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day; patch 3.9 mg/day. Only US-approved drug for paediatric OAB; causes dry mouth and constipation.
Tolterodine	1–2 mg twice daily or ER 2–4 mg; ER randomised controlled trial (RCT) negative
Solifenacin (M3-selective)	2.5–10 mg daily; positive phase III RCTs (LION, LEOPARD)
Trospium (quaternary amine)	Does not cross the blood–brain barrier
Mirabegron (β3-agonist)	25–50 mg if >35 kg; approved ≥3 yr for **neurogenic** detrusor overactivity only; safe add-on

- Check PVR and stools before and during antimuscarinics.
- **Intradetrusor onabotulinumtoxinA** (refractory): **5 units/kg, max 100 units**; it lasts ~12 months.

Neuromodulation options are parasacral TENS over S2–S3 (beat oxybutynin, but a sham RCT was negative), the weaker tibial nerve stimulation, and off-label sacral nerve stimulation (SNS), which helps bladder and bowel.

Other conditions

- **Underactive bladder**: timed and double voiding and biofeedback; CIC if PVR is high.
- **Postponement**: timed voiding and treat comorbidity; † avoid anticholinergics.
- **Giggle**: **methylphenidate** or **biofeedback**.
- **Vaginal reflux**: void with legs abducted (sit backwards on the toilet). **Extraordinary frequency**: † reassure.

6 · Nocturnal enuresis

P1 P3 Nocturnal enuresis / Enuresis — ICCS definitions, primary vs secondary (6 months dry), triad, sleep apnoea, evaluation, alarm vs desmopressin, second line

Not in the paper answer:

- Prevalence is **5–10% at 7 yr**; ~15% resolve each year, and ~2% persist into adulthood.
- The **triad** is nocturnal polyuria (relative antidiuretic hormone deficiency), reduced nocturnal capacity or overactivity, and failure to arouse.
- In **NMNE**, treat bothersome daytime LUTS and constipation first. In **MNE**, treat from **~6 yr**.

	Enuresis alarm	Desmopressin
Use	Nightly; the parent wakes the child; continue until **14 consecutive dry nights**	**0.2–0.4 mg** tablet or 120–240 µg melt ~1 h before bed; stop if no effect after ~1 week
Response	50–70%; slow	Immediate; rule of thirds; best with nocturnal polyuria
Durability	**Durable**; overlearning lowers relapse	**Not durable**; drug holidays or special nights
Safety	Nil	Hyponatraemia: † no fluids from 1 h before dose

ALGORITHM

1. MNE at ≥6 yr: start the alarm or desmopressin.
2. If it fails, check use, then switch or **combine**.
3. Re-evaluate: hidden daytime LUTS, constipation, sleep-disordered breathing, neurodevelopmental disorder; uroflow and PVR.
4. Then **add an anticholinergic** to desmopressin.
5. If refractory: imipramine, a β3-agonist or specialist drugs.

- **Imipramine** 25–50 mg is **cardiotoxic** (overdose, long QT), so do an electrocardiogram if there is syncope, palpitations or a family history of sudden death.

7 · Functional constipation — treatment

P4 Management of Defecation Disorders resulting into LUTS — why bowel causes LUTS, Rome IV, bowel first, disimpaction/maintenance PEG, biofeedback, irrigation, ACE, toxin, neuromodulation

Not in the paper answer:

- Daily enemas for 3–6 days are an alternative to PEG.

ALGORITHM

1. If impacted, **disimpact** with oral polyethylene glycol (PEG) or enemas.
2. Give education, a diary, toilet sitting and **maintenance PEG**.
3. Review at 2 wk; if failing, check adherence, dose, impaction, mental health.
4. Continue for **≥2 months**. Wean when ≥1 month symptom-free, with ≥3 stools/week and Rome IV not met.
5. Relapse: reinstate, or refer to gastroenterology.

Laxative	Dose
PEG 3350 — disimpaction	**1–1.5 g/kg/day × 3–6 days** (>3 yr)
PEG 3350 — maintenance	**0.2–0.8 g/kg/day**, start 0.4 (>6 months)
Lactulose — maintenance	1–2 g/kg once or twice daily (safe in infants)

- **Titrate** PEG: increase it if there is no stool in 24 h, and add a **stimulant** (senna, bisacodyl) if watery accidents occur. There is **no dependence**.
- Refractory cases may need biofeedback, transanal irrigation or SNS. Before an antegrade continence enema (**ACE**), do **colonic manometry**, because diffuse dysmotility may need resection (see Ch 48).

8 · Recurrent UTI and BBD

- In RIVUR/CUTIE, recurrence was **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**.

- Secondary VUR often resolves once BBD is treated.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile or recurrent UTI, and give prophylaxis during BBD treatment. The **AUA** advises prophylaxis for BBD with VUR.

9 · Landmark trials

Trial	Design	Result	Take-home
RIVUR + CUTIE (BBD analysis)	Prophylaxis vs placebo, <6 yr after febrile UTI, with or without VUR	BBD + VUR highest recurrence; BBD alone > VUR alone	Treat BBD; prophylaxis for BBD + VUR
LION / LEOPARD	Phase III placebo RCTs, solifenacin	Improved all efficacy measures; mild–moderate side effects	Best paediatric antimuscarinic evidence

48 Lower Urinary Tract Reconstruction in Children

1 · Goals and principles

- The priorities, in order, are: **protect the kidneys, then prevent urinary tract infection (UTI), then achieve continence**.
- The indication is a hostile bladder (poor compliance, high pressure, poor emptying) despite medical care, mostly in spina bifida; also exstrophy, posterior urethral valves and bilateral ectopic ureters.
- **Clean intermittent catheterisation (CIC)** is the key enabler, and the main determinant of outcome is **patient and family commitment**.

> **KEY POINT**
> Every reconstructive step trades storage for emptying. A continent reservoir that is not catheterised causes hydronephrosis, sepsis and perforation, so incontinence can be "protective". CIC and motivation are prerequisites.

2 · Patient selection and preoperative evaluation

- **Storage**: a chronic filling pressure **>40 cm H2O** causes hydroureteronephrosis, a falling glomerular filtration rate (GFR) and acquired vesicoureteral reflux (VUR). Start anticholinergics and CIC; persistent poor compliance needs **augmentation**.
- **Outlet**: resistance must exceed bladder pressure during all activity; a deficient outlet needs a procedure.
- **Emptying**: nothing predicts voiding after augmentation, so every patient practises CIC beforehand.

Urodynamic finding	What it decides
Capacity and fluoroscopic bladder shape	Length of bowel segment
Detrusor leak point pressure (DLPP), end-filling pressure, open bladder neck	Need for a bladder neck procedure
VUR with a history of UTI	Reimplantation
Sphincter electromyography (EMG)	Dyssynergia

- Fill at **≤10% of predicted capacity per minute**, because fast filling falsely lowers compliance.
- Every child has ultrasound (US). Hydronephrosis needs a renogram **with a bladder catheter** and voiding cystography or videourodynamics.
- Check electrolytes, blood urea nitrogen (BUN) and creatinine. If raised, involve nephrology and measure 24-h urine **volume**: the reservoir must hold **3–4 h** of output.
- In **renal insufficiency**, acidosis **always worsens** with ileum or colon, so consider stomach. In **liver failure**, ammonia reabsorption is dangerous.
- Low-grade secondary **reflux** usually resolves after augmentation; correct high-grade reflux at reconstruction.
- In short gut, diarrhoea or faecal incontinence, spare the ileocaecal valve; add an antegrade continence enema (ACE) if needed.

- A sensate urethra, wheelchair use or prior bladder neck repair favour an abdominal stoma. Do all steps in **one operation** where possible.
- **Sterile urine** is mandatory; bowel preparation is optional. With a ventriculoperitoneal shunt, infection is rare but shunt failure increases.

3 · Managing the ureters

- Reimplant into the native bladder, with or without a **psoas hitch** (avoid the genitofemoral and femoral nerves). A short ureter needs transureteroureterostomy.
- Into bowel: **colon** under a taenia; **stomach** submucosally; **ileum** needs a **nipple valve** (hardest).

4 · Bladder augmentation (enterocystoplasty)

P1 Instestinal segments in urology — segments compared with electrolyte disturbance; NH_4Cl absorption mechanism; other metabolic, stone, cancer complications; renal function for continent diversion

Not in the paper answer:

- Base deficit is the most sensitive marker; height can fall even without frank acidosis.
- Mucus: colon more than ileum more than stomach; stomach forms uric acid stones.

Principles

- **Bivalve the native bladder widely**; a narrow anastomosis makes a diverticulum.
- **Always detubularise** and **reconfigure into a sphere**. Intact bowel contracts at **60–100 cm H2O**, and a sphere gives the most volume per surface area.
- Length: ileum **20–40 cm**, colon **~20 cm**; err on the large side.
- Use **no permanent sutures or staples**, because they seed stones.

Segments

- **Ileocystoplasty**: take ileum at least **15–20 cm proximal to ileocaecal valve** (to protect B12 absorption) and open it antimesenterically into a U or W.
- **Gastrocystoplasty** suits **renal insufficiency with acidosis, short gut and irradiated bowel**. **Jejunum** is avoided (salt-losing acidosis).

Segment	Advantages	Disadvantages
Ileum (default)	Least pressure activity; plentiful, mobile mesentery	Hyperchloraemic acidosis; B12 loss; mucus; hard to make antireflux
Ileocaecal	Taenial reimplant; appendix; imbricated valve	Diarrhoea in neurogenic bowel; more mucus
Sigmoid	Taenial reimplant; close to bladder	Strongest contractions; most mucus; more reaugmentation
Stomach	No acidosis; least mucus; fewer struvite stones; usable in renal failure	Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria syndrome; tumours
Ureter	No mucus, no metabolic effect	Needs a massively dilated ureter of a non-functioning kidney; limited volume
Autoaugmentation (detrusorectomy)	Extraperitoneal; no bowel	Seldom adds capacity; compliance may not last

- Seromuscular and tissue-engineered augments are experimental. Technique matters more than segment, so use ileum unless there is a reason not to.

> **KEY POINT**
> Augmentation is needed less now, thanks to early anticholinergics, CIC, botulinum toxin and neuromodulation (immediate vs expectant treatment: 17% vs 41% augmented). This is not trial-proven, but it is the rationale for treating at diagnosis.

Incontinent alternatives

- **Vesicostomy** is an infant bridge for worsening hydronephrosis, febrile UTI with high-grade VUR, or high pressure. **Ileovesicostomy** suits those who cannot catheterise.

5 · Continent catheterisable channels

P2 Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction. — flap vs nipple valve, conduit options (appendix, Monti, tapered ileum, ureter, Casale), appendicovesicostomy steps, MACE, results, stenosis

Not in the paper answer:

- In a **flap valve**, a narrow tube tunnelled into a **low-pressure** reservoir is **compressed** as pressure rises; the appendix needs only **~2 cm** of tunnel. Nipple valves (Kock, Mainz) are **effaced** by pressure (about 85% vs more than 90–95% continent). A flap valve has **no pop-off**.

> **ALGORITHM**
> 1. Detach the appendix with a caecal cuff and preserve the mesoappendix.
> 2. Tunnel the **distal end** submucosally.
> 3. Bring the caecal end to skin without tension or twist, with a **skin flap** in the stoma.
> 4. Keep the channel **short and straight**; fix the bladder to the abdominal wall.
> 5. Test catheterisation after each step.
> 6. Stent with 6–12 French (Fr) for 10–14 days; the surgeon does the first catheterisation.
> 7. Then CIC at least 4-hourly by day.

- Site the stoma at the umbilicus or right lower quadrant. The appendix can be **split** for both an ACE and a channel.
- **Yang–Monti**: a 1–2 cm ileal ring is retubularised **transversely** over 12 Fr; it needs about twice the revisions of the appendix. The **Casale** bladder flap spares bowel but has high stenosis.
- **Stomal stenosis** is the commonest complication and occurs early. Stomal incontinence (rare) means a short tunnel or high reservoir pressure.
- **Ureterosigmoidostomy and Mainz II** are largely abandoned because of acidosis, pyelonephritis and cancer.

6 · Bladder outlet procedures

Outlet procedures sacrifice voiding and pop-off, so neurogenic repairs need **concurrent augmentation** and surveillance for **≥5 yr**.

Procedure	Mechanism	Voiding	Main problem / place
Fascial sling	Coaptation and elevation	Rarely adequate, so CIC	Better in girls; only predictor of success is augmentation; AUS still possible later
Artificial urinary sphincter (AUS)	Inflatable bladder-neck cuff	**Preserved**	Infection, erosion, malfunction; best if voiding
Bulking injection	Submucosal	Preserved	Poor durability
Young–Dees–Leadbetter	Tubularised trigone	Possible	Exstrophy (see Ch 43); poor in neurogenic
Kropp	Anterior bladder tube under trigone (flap valve)	Abolished	Highest continence; hard urethral CIC (boys); new VUR
Pippi Salle	Anterior flap onlaid on trigonal strip	Abolished	Easier CIC; lower night continence; fistula
Bladder neck closure	Complete separation and continent stoma	Abolished	After failed repairs; no pop-off

- The **sling** suits a marginal bladder neck in an augmented child on CIC.
- The **AUS** cuff goes at the bladder neck in girls and prepubertal boys and is left **deactivated ~6 weeks**; most neurogenic children still need CIC. New hydronephrosis means unmasked bladder hostility.
- **Kropp** and **Salle** repairs often need later augmentation. Bladder neck **closure** with **interposed omentum** keeps fistula low, and suits reliable catheterisers only.

> **KEY POINT**
> Choosing the outlet: a child who voids gets an AUS (the only option preserving voiding). A child on CIC who is being augmented gets a sling (girls) or flap-valve lengthening with a channel. After failed repairs, do bladder neck closure with a Mitrofanoff. Always re-study the bladder afterwards.

7 · Malone antegrade continence enema (MACE)

- † An appendicocaecostomy gives scheduled antegrade washouts and faecal continence in spina bifida, anorectal malformation and refractory constipation. Plan it with the urinary reconstruction.

8 · Postoperative care

- Irrigate the suprapubic tube **≥3 times daily**. Do a **cystogram at ~3 weeks**, then start CIC 2–3-hourly by day, **never beyond 4–5 h**.

9 · Complications of augmentation

*Up to ⅓ need further surgery: **bowel is not bladder**.*

Segment	Derangement	Management
Ileum, colon	**Hyperchloraemic metabolic acidosis** (overt mainly with renal impairment)	Bicarbonate; **base deficit** is most sensitive
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis	Avoid

Segment	Derangement	Management
Stomach	**Hypokalaemic hypochloraemic alkalosis** with vomiting; **haematuria–dysuria**	Fluids, H2 blockers or proton pump inhibitor
Stomach and bowel	Near neutral	Complex patients

- Acid load demineralises bone and lowers height, so treat base deficit early. Check B12 long term.
- **Bacteriuria** is expected with CIC; treat only symptomatic infection and **urea-splitters**.
- **Stones** are mostly **struvite**; prevent them with **daily irrigation ≥240 mL** saline, which also clears mucus.

Delayed spontaneous perforation

- Counsel a **~9%** risk, mostly in myelodysplasia. **Outlet procedures double the risk**, as do missed CIC and substance abuse.
- Signs are pain, distension, fever, sepsis and shoulder-tip pain; poor sensation means **late presentation** and deaths. Do a **cystogram** with a post-drain film, then a CT cystogram if negative. Treat by **immediate laparotomy and repair**; manage conservatively only if stable, sterile and contained.

Malignancy

P1 Cancer in augmented bladder — adenocarcinoma at anastomosis, earliest 4 yr, N-nitroso pathogenesis, gastrocystoplasty/exstrophy/transplant risk, SB 2–4%, nephrogenic adenoma, surveillance debate, cystectomy

- After ureterosigmoidostomy, adenocarcinoma arises at the anastomosis after **~26 yr**, from N-nitroso compounds.
- After augmentation it is reported from **4 yr**, is highest after **gastrocystoplasty**, and is aggressive. Bladder cancer in spina bifida is ~2–4%. There is no consensus on annual cystoscopy; consider it after gastrocystoplasty, in exstrophy and after transplant.

Pregnancy

- Vaginal delivery is possible; caesarean after major bladder neck repair or AUS, with a urologist present.

10 · Follow-up and transition

When	Tests
3, 6, 12 months	US; electrolytes, BUN, creatinine, urine culture (with or without urodynamics by 6 months)
Yearly, lifelong	US, chemistry (with or without blood gas, B12), CIC and irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR
Symptomatic	† Haematuria needs cystoscopy; pain or sepsis needs perforation excluded

- Late chronic kidney disease after augmentation is driven by **CIC non-compliance**, so adult urology must police CIC and the upper tracts.

11 · Landmark trials

Trial	Design	Result	Take-home
Early vs expectant neurogenic bladder care	Cohort, treated at diagnosis vs expectantly	Augmentation 17% vs 41%	Treat hostile bladder early
Daily bladder irrigation	Augmented cohort, ≥240 mL saline daily	Marked fall in stones	Routine daily irrigation
Tissue-engineered bladder	Industry multicentre trial	No gain in compliance or capacity	Bowel remains standard

★ Putting it together: common paediatric pathways

A · Antenatal urinary tract dilation (UTD)

ALGORITHM

1. **Prenatal:** A1 before 28 weeks: ultrasound (US) after 32 weeks. A2–3: US 4-weekly. Male, thick bladder, bilateral: suspect posterior urethral valves (PUV).
2. **Postnatal:** first US **after 48 hours**, except suspected PUV or bilateral severe dilation.
3. Normal (<10 mm): discharge; mild: repeat US; moderate–severe or dilated ureter: voiding cystourethrogram (VCUG), MAG3, consider prophylaxis (→ Ch 34).

B · First febrile urinary tract infection (UTI)

ALGORITHM

1. **Diagnose:** catheter or suprapubic sample if not toilet-trained; bag only rules out.
2. **Treat** within 48 h (>72 h doubles new scarring); oral if well.
3. **Image:** US all; VCUG if US abnormal, atypical or recurrent; DMSA at 4–6 months if needed.
4. **Prevent:** treat bladder and bowel dysfunction and constipation; prophylaxis if high risk (→ Ch 37, 41).

C · Newborn with PUV

ALGORITHM

1. Stabilise breathing → 5–7 Fr feeding tube → creatinine, electrolytes, fluids.
2. US, then VCUG; prophylaxis.
3. When stable, **endoscopic ablation**; if the urethra is small, vesicostomy.
4. Prognosis: first-year nadir creatinine (<0.8 mg/dL good); lifelong renal follow-up (→ Ch 45).

D · Neurogenic bladder in spina bifida

Step	What	Why
Birth	US, post-void residual, creatinine; latex-free care	Baseline
≈3 months	Urodynamics	Hostile: leak point or end-filling ≥40 cmH_2O, poor compliance, dyssynergia
Hostile bladder	Early clean intermittent catheterisation (CIC) and antimuscarinic	Kidney protection
Failing	Botulinum toxin, vesicostomy (infant), then augmentation	Safe storage (see Ch 46, 48)
Always	Bowel programme; watch for re-tethering	Safety

E · Daytime wetting and bedwetting

ALGORITHM

1. Urinalysis, residual with or without uroflow; exclude organic causes.
2. **Constipation first**, then urotherapy ≥4 weeks.
3. Antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding; toxin or neuromodulation if refractory.
4. **Monosymptomatic enuresis:** alarm or desmopressin; combine if partial (→ Ch 47).

F · Augmentation segment

Segment	Advantages	Disadvantages
Ileum	Default; compliant	Mucus, hyperchloraemic acidosis, B12 loss
Ileocaecum / colon	Appendix for channel	Diarrhoea in neurogenic bowel; acidosis
Stomach	Suits renal failure; little mucus	Haematuria–dysuria, alkalosis
Ureter	No mucus or metabolic effect	Needs a massive ureter

KEY POINT

Reconstruction trades storage for emptying: an unreliably catheterised reservoir causes sepsis and perforation, so CIC commitment comes first.

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 6

Female Urology

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 101	Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis
Ch 108	Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology
Ch 105	Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse
Ch 114	Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices
Ch 116	Retropubic Suspension Surgery for Incontinence in Females
Ch 118	Slings: Autologous, Biologic, Synthetic and Mid- Urethral
Ch 117	Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse
Ch 119	Genitourinary Complications of Mesh and Their Repair
Ch 121	Urinary Tract Fistulae
Ch 122	Bladder and Female Urethral Diverticula
Ch 77	Sexual Function and Dysfunction in the Female
★	Putting it together: female pelvic decisions

101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

1 · Perineum and external genitalia

- The meatus lies **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands** sit beside the meatus and are the prostate homologue; obstructed periurethral glands form a diverticulum or cyst.
- **Bartholin's gland** lies under the posterior end of each vestibular bulb and may form a cyst or abscess.
- The **deep perineal space** holds the urethral sphincter, urethrovaginal sphincter, compressor urethrae and deep transverse perineal; the **superficial space** (under Colles fascia) holds the superficial perineal muscles, clitoris, vestibular bulbs and Bartholin's glands.
- The perineal muscles and external anal sphincter converge at the **perineal body**, so episiotomy is angled laterally to spare the sphincter.
- The **labia majora have a dual supply** (superficial external pudendal from above, posterior labial from below), so the **Martius flap** is reliable.

2 · Bony pelvis and ligaments

- The **ischial spine** marks apical sutures.

Fixation point	Used for	Adjacent danger
Sacrospinous ligament (SSL): ischial spine to lower sacrum/coccyx	SSL fixation (apical prolapse)	Pudendal nerve/vessels near the spine; sacral plexus just posterior
Cooper's ligament (medial superior pubic ramus)	Burch colposuspension for stress urinary incontinence (SUI)	Obturator vessels; aberrant obturator
Obturator foramen; canal superolateral	Transobturator tape (TOT) trocar	Obturator nerve/vessels, causing groin pain or bleeding
Anterior longitudinal ligament (ALL) over S1	Sacrocolpopexy mesh	Presacral veins, middle sacral vessels, left common iliac vein, right ureter
Uterosacral ligament	Uterosacral (apical) suspension	Ureter (lateral, closest at the cervix); sacral nerve entrapment

- The pudendal nerve leaves through the **greater** sciatic foramen and re-enters through the **lesser** to reach Alcock's canal.

3 · Pelvic floor muscles

- The pelvic diaphragm is the **levator ani** and coccygeus; the levator is supplied directly by **S3–S5 (mainly S4)**.

P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — levator table: puborectalis, pubococcygeus, iliococcygeus; coccygeus NOT levator, lies on SSL

Not in the paper answer:

- The **levator hiatus** transmits the urethra, vagina and rectum, and the **levator plate** is a horizontal visceral shelf; injury leaves the hiatus lax, the plate tilts and **prolapse** follows.

4 · Endopelvic fascia and visceral ligaments

- The **intermediate** fascial stratum (parametrium, paracolpium, pubocervical fascia) **gives most organ support**.
- The **pubourethral ligaments** (puboprostatic homologue) and anterior vaginal wall form a mid-urethral hammock; weakness causes SUI.
- A vesicopelvic fascia defect at the arcus tendineus gives a paravaginal cystocele.
- The **cardinal (Mackenrodt's) ligament** carries the pelvic plexus, so dividing it at hysterectomy can cause bladder dysfunction.
- Distinguish the **infundibulopelvic ligament** (ovarian vessels, ureter beneath) from the **ovarian ligament** (ovary to uterus) MCQ.

DeLancey levels of vaginal support †

P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — I suspension (uterosacral–cardinal; uterine/vault prolapse, enterocele), II attachment (ATFP; anterior/posterior wall), III fusion (urethral hypermobility, distal rectocele, perineal descent)

> **KEY POINT**
> Support ligaments are connective tissue condensed around vessels and nerves, so the best anchors (uterosacral, cardinal) are exactly where sutures injure the ureter, pelvic plexus and sacral nerves. See Ch 108

5 · Urethra and continence anatomy

- The urethra is **4.7–5.1 cm** long and hangs from the posterior pubourethral ligaments on the anterior vaginal wall.
- **The mucosa and spongy vascular submucosa generate most closure pressure and are oestrogen-dependent**, which is the rationale for vaginal oestrogen.
- The **internal sphincter** is smooth and autonomic at the bladder neck; the **external sphincter** (sphincter urethrae, **compressor urethrae**, **urethrovaginal sphincter**) is striated, voluntary (pudendal) and covers the distal two-thirds.
- **Hypermobility** is one cause of SUI. An autologous pubovaginal sling sits at the **bladder neck**; a synthetic mid-urethral sling (MUS) sits at the **mid-urethra**.

6 · Bladder and ureter

The ureter is the structure most often injured.

- The **dome** is peritonealised, so intraperitoneal injury needs formal repair. Injury to the **base** causes vesicovaginal or vesicouterine fistula.
- The **pelvic ureter** crosses the iliac bifurcation, passes behind the ovary, runs **under the uterine artery** ("water under the bridge"), then crosses the anterior fornix to the bladder.

ALGORITHM

1. **Ureteric danger points**:
2. Pelvic brim, at the infundibulopelvic ligament.
3. Under the uterine artery (hysterectomy).
4. Beside the cardinal/uterosacral ligaments near the cervix.
5. Anterior vaginal fornix (vaginal surgery).
6. Sacral promontory: the right ureter (sacrocolpopexy).

7 · Nerves at surgical risk

- The **pudendal nerve (S2–S4)** crosses the SSL **close to the ischial spine** into Alcock's canal; pudendal block is given transvaginally just behind the spine.
- In SSL fixation, place sutures about 2 cm medial to the spine, through (not around) the ligament †.

Nerve	Procedure / mechanism	Deficit
Common peroneal	Lithotomy: compression at the fibular head	Foot drop
Sacral plexus (L4–S4)	SSL fixation (plexus exits just posterior to the SSL)	Posterior leg motor/sensory loss
Pudendal	SSL fixation near the spine	Perineal pain, sensory loss, sphincter dysfunction
Sacral trunks S1–S4	Uterosacral suspension (deep sutures)	Buttock/posterior thigh pain
Ilioinguinal (L1), genitofemoral (L1–L2)	Hernia repair, caesarean, hysterectomy	Mons/labial numbness or pain
Obturator (L2–L4) †	TOT trocar, obturator lymphadenectomy	Groin pain, adductor weakness
Posterior femoral cutaneous (S1–S3)	Gluteal pressure, pelvic tumour	Perineal sensory loss

KEY POINT

Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Leg pain and swelling after long lithotomy suggest **compartment syndrome**, an emergency († fasciotomy).

8 · Vessels and lymphatics

Internal iliac	Anterior division	Posterior division
Visceral	Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal	None
Parietal	Obturator, internal pudendal, inferior gluteal	Iliolumbar, lateral sacral, superior gluteal

- The **internal iliac** has rich collaterals, so ligation or embolisation is tolerated; ligating the **external iliac** threatens the limb.
- The **presacral plexus** is valveless and under high pressure, and a 1–2 mm vein can bleed more than **1 L/min**.

ALGORITHM

1. **Sacrocolpopexy**
2. Expose the ALL at the promontory over the L5/S1 disc.
3. Open the peritoneum distally.
4. Suture into the ALL over **S1** (strongest; less discitis).
5. Avoid the left common iliac vein, middle sacral vessels, presacral plexus and right ureter (each about 2.7 cm from the mid-promontory).

Organ	First nodes
Distal ⅓ urethra, lower ⅓ vagina, vulva	Inguinal, then pelvic
Proximal ⅔ urethra, upper ⅔ vagina	Iliac, obturator (pelvic)
Bladder, uterus	Perivisceral, then pelvic, presacral and great-vessel nodes
Ovary MCQ	Along the gonadal vessels to para-aortic nodes (bypasses the pelvis)

9 · Endoscopic anatomy

ALGORITHM

1. Distal urethra: glands, diverticular ostia, mesh, stones.
2. Mid-urethral coaptation (external sphincter).
3. Bladder neck coaptation.
4. Trigone and orifices (30° lens).
5. Survey the whole mucosa.

- Squamous metaplasia and cystitis cystica are benign, but biopsy if in doubt.

10 · Imaging

- **Translabial ultrasound** shows bladder-neck mobility, prolapse, and **mesh, MUS and bulking agents**.
- On **endoanal ultrasound**, an internal sphincter defect is a bright break in dark muscle; an external sphincter defect is a dark lesion in bright muscle.
- **T2-weighted MRI is best for urethral diverticulum** and separates other cysts such as Gartner, Skene and Bartholin cysts.
- **MR defecography** grades descent below the **pubococcygeal line** and is used for defaecatory dysfunction or symptoms out of proportion to examination.

108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

1 · Definitions and terminology

The type of UI decides treatment.

- **Urinary incontinence (UI)** is complaint of involuntary urine loss (ICS/IUGA, 2010).
- Symptoms are non-specific, so combine them with signs and observations (diary, urodynamics [UDS]). **Bother drives treatment**: without quality-of-life impact, none may be needed.

Type	Complaint of loss…	Urodynamic correlate / trap
Stress UI (SUI)	On effort, cough, sneeze	Urodynamic SUI: leak with abdominal pressure rise **and no detrusor contraction**
Urgency UI (UUI)	With urgency	**Detrusor overactivity (DO) not required**; DO incontinence (DOI) is a separate UDS diagnosis
Mixed UI (MUI)	With urgency and effort	Urge- or stress-predominant, or equal
Overflow UI	Frequent small-volume loss with retention	Underactivity or obstruction; high post-void residual (PVR)
Functional UI †	Impaired mobility/cognition, normal lower urinary tract	Frail elderly, long-term care
Continuous UI	Day and night, few voids	Suspect fistula
Nocturnal enuresis	During sleep	Differs from nocturnal UUI (wakes, cannot reach toilet)

- **Urgency** is a sudden compelling desire that is hard to defer (pathological), whereas **urge** is normal desire.
- **Overactive bladder (OAB)** is urgency with or without UUI, usually with frequency and nocturia; it is **defined by urgency**. Frequency ≥ **8 voids/24 h** is abnormal once polyuria is excluded.

2 · Epidemiology of female UI

- Remission reaches about 12%/year and is **lowest for MUI**.
- **Pregnancy** is the commonest cause of new UI in young women (about **40%**, mostly SUI). UI at 3 months postpartum usually persists, and pelvic floor muscle training (PFMT) during and after pregnancy reduces it.
- **Hysterectomy** can cause or cure UI; vaginal and laparoscopic routes carry more risk than open.

3 · Risk factors for female UI

Factor	Key point
Ageing	Strongest independent non-modifiable factor. SUI peaks at 40–59 years; UUI/MUI dominate after 60. Over 40% in long-term care at admission

Factor	Key point
Vaginal delivery	Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is by CS; number needed to treat (NNT) is **13** to prevent one moderate–severe UI. Elective CS protects more than CS in labour. Baby ≥4 kg: OR about 1.5
Parity	First delivery OR about 1.5, then modest rise; strongest for SUI
Obesity	Body mass index (BMI) ≥30 more than doubles risk (SUI, MUI); **reversible** with weight loss. Little effect on SUI surgery success
Menopause / oestrogen	**Systemic oestrogen increases UI** (new UI nearly doubled vs placebo). Vaginal oestrogen is not linked to SUI and treats atrophy and recurrent UTI
Race	SUI commoner in White than Black/Asian women (higher closure pressure in Black women)
Smoking	Urgency, frequency, UUI (dose-dependent)
Caffeine	UUI, MUI, OAB, but not SUI
Diabetes	Up to 2×; neuropathy and macroalbuminuria are independent factors

KEY POINT

Obstetric injury and loss of support cause **stress** UI; ageing, neurological, metabolic and lifestyle factors cause **urgency**. So the dominant type moves from stress to urge with age.

Male UI (contrast)

- In men **UUI is commoner than MUI, which is commoner than SUI**. Male SUI is nearly always iatrogenic (radical prostatectomy [RP]) or neurological, and remission reaches **40%**.
- Post-RP UI is 8–60% depending on definition; after TURP it is about 1%.

4 · Physiology of continence

- During filling, **pelvic** parasympathetic outflow is suppressed, **hypogastric** sympathetic β-action relaxes the detrusor and α-action closes the bladder neck, and the **pudendal** nerve (Onuf's nucleus) keeps the sphincter closed.
- In **women**, the continence zone is the **proximal and mid-urethra**; the bladder neck is often incompetent in continent women. The oestrogen-sensitive spongy submucosa gives up to **30%** of closure.
- **Support**: the anterior vaginal wall is the firm backing. Lax vaginal support with intact pubourethral ligaments causes **shear** in the continence zone.
- **Male vs female**: men tolerate losing one outlet unit (TURP) because the bladder neck is powerful; in women the bladder neck is weak, so external sphincter deficiency causes UI.

KEY POINT

The female outlet relies on a **single** mid-urethral sphincter against a mobile backing, so continence needs support, intrinsic urethral properties and muscle together.

5 · General pathophysiology: bladder vs outlet

Mechanism	Causes	Result
Loss of suprapontine inhibition	Stroke, multiple sclerosis (MS), Parkinson; early diabetes mellitus (DM)	Neurogenic DO, causing UUI

Mechanism	Causes	Result
Outlet obstruction	Women: after anti-incontinence surgery; men: bladder outlet obstruction (BOO)	DO with new (de novo) UUI
Detrusor underactivity	Conus/cauda lesion, neuropathy (DM, alcohol), radical hysterectomy, abdominoperineal resection (APR), opioids including tramadol	Retention, then **overflow UI**
Poor compliance	Pelvic radiotherapy	High detrusor leak point pressure (LPP), causing UI
Sphincter deficiency	RP, urethral/anti-incontinence surgery, cord injury, α-blockers, muscle relaxants	Intrinsic sphincter deficiency (ISD), causing SUI

- In **DM**, early polyuria and UUI give way to lost sensation and contractility, with poor emptying, UTI and overflow.
- In multiple system atrophy ≥20% leak after TURP.
- In post-RP UI, **ISD predominates (>70%)**, and DO or poor compliance contributes in >30%.
- Each failed operation adds fibrosis and ISD; **labour** causes fascial injury and **pudendal denervation**.

6 · Pathophysiology of female SUI

P3 P1 Female pelvic floor, continence mechanism, classification and management of SUI — theories table: pressure transmission (Enhorning), hammock (DeLancey; repositioning not essential), integral theory (Petros–Ulmsten, giving MUS), ISD (McGuire)

Not in the paper answer:

- Support does not explain everything: some women with marked hypermobility are dry, and some with a fixed urethra leak severely.

Hypermobility vs ISD

- ISD equals **type III SUI** (Blaivas) MCQ and is usually seen after previous surgery.
- A **"pipestem" urethra** is fixed and non-coapting; subtler ISD coexists with hypermobility in most SUI.
- **Valsalva leak point pressure (VLPP)** is the abdominal pressure that causes leakage. **VLPP < 60 cm H_2O means ISD**, historically treated by bulking or pubovaginal sling.

> **KEY POINT**
> **Slings work in hypermobility but Burch fails in ISD**. Almost all SUI includes some ISD, so sphincter-directed procedures (pubovaginal sling, mid-urethral sling [MUS], bulking) work in hypermobile women. Burch corrects only hypermobility, so it is weak with severe ISD and a fixed urethra.

7 · OAB pathophysiology (brief)

- Neurogenic causes (§5) and outlet obstruction (BOO, post-sling) both lead to DO and de novo urgency.
- With pelvic organ prolapse (POP), urgency is mainly due to obstruction from kinking. **OAB improves after POP repair in 60–80%**, more after anterior/apical repair, with no reliable predictor.

8 · Insensible incontinence

- It is usually ISD or DOI; if treatment fails, seek rarer causes. Post-void dribble suggests diverticulum (pelvic MRI), continuous loss with normal voids suggests ectopic ureter (upper-tract imaging), and frequent small losses suggest overflow (UDS and PVR).

9 · Pelvic organ prolapse: definition and staging

- **POP** is descent of the uterus/vault and/or vaginal walls from loss of support. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ.
- The most specific symptom is a **visible or palpable bulge**, which correlates best once past the hymen.
- Never attribute bleeding or pain to POP; relief with a **pessary trial** predicts benefit from surgery.

POP-Q stage	Most distal point on maximal strain
0	Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2)
1	> 1 cm above hymen (< −1)
2	Within 1 cm of hymen (−1 to +1)
3	> +1 but < +(tvl − 2)
4	Complete eversion: ≥ +(tvl − 2)

P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — POP-Q points: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D, gh, pb, tvl; 3 × 3 grid

Not in the paper answer:

- The stage is set by the single most distal point, and measuring right after pessary removal underestimates it.
- POP-Q is the IUGA/ICS-recommended system; Baden–Walker grades have poor interobserver reliability.

10 · POP epidemiology and risk factors

- Loss of support is seen in 43–90%, and **asymptomatic stage 1–2 is normal**; symptomatic POP affects 3–12%.
- **Anterior POP is commoner than posterior, which is commoner than apical**. Apical descent accompanies most significant anterior POP and is universal at stage 4. **So when planning cystocele repair, assess the apex and consider vault suspension.**
- POP progresses slowly; progression predictors are age >60, Ba ≥4 cm and genital hiatus (gh) ≥5 cm.

Risk factor	POP	UI (contrast)
Vaginal childbirth	Strongest; rises steeply with parity; forceps worse. Elective CS risk is about nulliparous; CS in labour about vaginal	SUI, not UUI
Age	About 40% rise per decade; surgery peaks in the 7th decade	Shift to UUI/MUI
Obesity	About 1.4–2.5-fold	>2-fold, reversible
Hysterectomy	Raises risk, most if done for POP (no vault suspension)	Cause or cure
Race	Hispanic highest, Black lowest	White highest

Risk factor	POP	UI (contrast)
Oestrogen / menopause	Not consistently associated	Oral oestrogen increases UI
Other	Family history (about 5× in siblings), connective-tissue disease, chronic constipation (2.5×)	—

- Recurrence after surgery is predicted by preoperative stage 3–4.

11 · Pathophysiology of POP

- The **levator ani** is the primary support; the uterosacral–cardinal ligaments hold the organs in position over it.

P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — DeLancey I suspension (uterine/vault, enterocele), II attachment to ATFP (lateral or central anterior defect), III fusion (urethral hypermobility, distal rectocele)

- The commonest cystocele defect at surgery is detachment of the arcus tendineus fasciae pelvis (ATFP) from the ischial spine.

ALGORITHM

1. **POP sequence**
2. Labour causes levator trauma or denervation.
3. The muscle loses tone and the urogenital hiatus widens.
4. Fascia and ligaments now carry the whole load.
5. They stretch and fail (ageing, straining, hysterectomy): POP.
6. POP widens the hiatus further.

- MRI shows levator defects in up to **20%** of primiparas and none in nulliparas; **avulsion** predicts advanced, multicompartment POP.
- Levator neuropathy follows vaginal delivery or CS in labour but almost never elective CS, so **labour itself is the injury**.
- An altered collagen type I:III ratio and Ehlers–Danlos or Marfan syndrome increase risk.

12 · POP and UI interaction

Prolapse can hide SUI until repair.

- 40–55% of women with SUI have significant anterior POP, as do about 20% with OAB.
- **Occult SUI** is SUI seen only with the prolapse reduced (up to a quarter of advanced POP).

ALGORITHM

1. A cystocele kinks the urethra, and the obstruction masks SUI.
2. POP repair relieves the kink, so SUI is unmasked.
3. Detect it by UDS or stress test **with the prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery is predicted by age >66, DM and low urethral closure pressure.
- Examine for POP in every UI evaluation, because POP missed at sling surgery means later prolapse surgery.

13 · Landmark trials

Trial	Design	Result	Take-home
CARE	Stress-continent women, abdominal sacrocolpopexy with or without Burch	Burch reduced postoperative SUI; SUI on UDS with prolapse reduced predicted it	Prophylactic Burch at sacrocolpopexy prevents de novo SUI
OPUS	Stress-continent women, vaginal POP repair with prophylactic MUS vs sham	UI at 1 year 27% vs 43% (NNT about 6); more bladder perforation, UTI, bleeding, retention	Fewer de novo SUI, more adverse events, so counsel

105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

1 · Definitions and risk factors

Anatomical prolapse is common; disease means symptoms.

- Risk factors for **urinary incontinence (UI)** are age, obesity, smoking, comorbidity and postmenopausal hormones; vaginal delivery causes stress UI (SUI).
- **Pelvic organ prolapse (POP)** is commonest anteriorly, then posteriorly, then apically. Half of parous women have anatomical POP, but disease means **symptoms attributable to it**. A **vaginal bulge** is the most specific symptom.

Category	POP risk factors
Primary only	Older age, high body mass index (BMI), high birth weight, levator defects, low education, poverty
Recurrence only	**Younger age**, preoperative Pelvic Organ Prolapse Quantification (POP-Q) stage 3–4
Both	Vaginal delivery, parity, levator hiatal ballooning
Protective (primary)	Caesarean delivery; **smoking** MCQ

- **ICS UI terms** follow the trigger: SUI, urgency UI (UUI) and mixed UI (MUI). **Insensible** UI occurs without awareness, and **occult SUI** appears only after POP reduction.

2 · History

- Ask directly and exclude reversible causes with **DIAPPERS**: Delirium, Infection, Atrophic vaginitis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI is urethral failure, UUI is bladder failure, and MUI is both**. In MUI, treat the more **bothersome** component first.
- **POP** gives a bulge and pressure that worsen later in the day. **Splinting** suggests high-grade POP. Anterior POP kinks the urethra and causes voiding difficulty; posterior POP obstructs defecation.
- SUI coexists in about **40%** of women with POP, and many expect repair to cure leakage, so counsel them.
- **Occult SUI** is found in about ⅓ before vaginal POP repair, and **de novo SUI** can appear after POP surgery; both need a shared decision on concurrent versus staged surgery.
- Ask about neurological disease, diabetes, radiation, pelvic or mesh surgery and obstetric trauma. **Local** oestrogen may improve UI and **systemic** oestrogen may worsen it.
- **Drugs**: **α-blockers worsen SUI in women**, and α-agonists, anticholinergics and opioids impair emptying. Diuretics, SGLT2 inhibitors and lithium cause polyuria. **ACE inhibitors cause cough, which worsens SUI**.

3 · Physical examination

- Atrophic (oestrogen-deficient) tissue is pale, flat, dry and without rugae.

SUI testing

- AUA/SUFU 2023 says: **objectively demonstrate SUI with a comfortably full bladder before surgery**.

P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — cough stress test (supine, then standing), reduce prolapse for occult SUI, Q-tip >30° = hypermobility

Not in the paper answer:

- The Q-tip test is now mostly for teaching and trials; **visual urethral mobility** (meatal rotation < or **>30°**) is routine.

POP examination

- Examine in lithotomy **and standing**, while straining, to see the worst prolapse.
- Use a **split Graves speculum** to isolate each compartment in turn. If the blade is expelled with posterior foreshortening, there is **level I loss, meaning vault prolapse**; if it stays, there is an isolated rectocele or enterocele.

POP-Q system

P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — POP-Q points table: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D (absent after hysterectomy), gh, pb, tvl

Not in the paper answer:

- Total vaginal length (tvl) is measured with the prolapse reduced, and the stage is set by the single most distal point.

Stage	POP-Q criteria	Baden–Walker grade (by eye)
0	Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2)	0: normal
I	Leading edge < −1 cm	1: upper vagina to midplane
II	Leading edge −1 to +1 cm	2: midplane to hymen
III	> +1 cm but < +(tvl − 2)	3: through/beyond hymen
IV	≥ +(tvl − 2): complete eversion	4: complete eversion

> **KEY POINT**
> Stage II is a 2-cm window around the hymen, and the III/IV boundary depends on tvl. Example: tvl 8, Bp +5 lies between +1 and +6, so **stage III**.

Other elements

- The **bulbocavernosus reflex** (S2–S4) is present in only about **70%** of normal women, so its absence alone is not pathological.
- For **prior mesh**, IUGA/ICS terms distinguish extrusion (through vaginal epithelium) from perforation (into bladder, urethra or another viscus).

4 · Supplemental evaluation

The uncomplicated patient needs few tests.

Urinalysis (UA) and post-void residual (PVR) are universal. For the AUA/SUFU 2023 index patient having SUI surgery, **UA, PVR and objective SUI demonstration** are required; cystoscopy, urodynamics (UDS) and imaging are not routine. In non-neurogenic OAB only UA is required (PVR optional).

P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — non-index features needing more tests: SUI not shown, neurogenic, urgency-predominant MUI, raised PVR, stage ≥3 POP without SUI,

failed surgery; questionnaires; VALUE

Not in the paper answer:

- Unexplained haematuria or pyuria also makes a patient non-index.

Symptom quantification

- A 3-day **bladder diary** is both diagnostic and therapeutic (see Ch 114), and its largest void equals functional capacity.
- Use the ICIQ-UI SF for UI and the **PFDI / PFIQ** for POP.
- **Pad tests** are mainly for research. In the ICS 1-hour test, 500 mL is drunk in <15 min, standard activities follow, and the pad is weighed.

	1-hour test	24-hour test
Normal	<1.4 g	<4.4 g
Mild	1–10 g	4–20 g
Moderate	11–50 g	21–74 g
Severe	>50 g	≥75 g

- In the **double-dye tampon test** (oral phenazopyridine and intravesical methylene blue), orange staining means a ureterovaginal and blue a vesicovaginal fistula.
- Measure **PVR** by bladder scan and repeat it. There is no agreed cut-off, but asymptomatic women are almost always <100 mL; a raised PVR changes counselling before SUI surgery or antimuscarinics.
- **UDS** are not needed in the index SUI patient (see Ch 107).
- **Cystoscopy** is not routine; use it for haematuria, urgency or irritative symptoms, recurrent UTI, prior sling or mesh, and pelvic radiation.

Imaging

Modality	Main indication
Voiding cystourethrogram	Reflux, diverticula, cystocele
Renal ultrasound	Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder
Translabial ultrasound	**Modality of choice for mesh** (CT and MRI poor); levator defects
MRI / dynamic MRI	**Urethral diverticulum (standard of care)**; complex or recurrent POP; symptoms out of proportion to findings

- A diverticulum gives the **3 Ds**: dysuria, postvoid dribbling and dyspareunia (see Ch 122).

5 · Management pathway

ALGORITHM

1. History, examination, UA and PVR.
2. Exclude DIAPPERS causes, UTI, obstruction and tumour.
3. Classify the UI and grade the bother.
4. If there are red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation or pelvic surgery, prolapse beyond the introitus, fistula), refer for specialist work-up.
5. If not, start initial management and reassess.
6. Add tests only if they change management; then surgery.

Diagnosis	Initial management
SUI	Lifestyle and behavioural therapy including pelvic floor muscle training (PFMT) (see Ch 114); † pessary; surgery if these fail
UUI / OAB	Lifestyle, behavioural therapy; antimuscarinic trial (see Ch 113)
Significant PVR	Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess
Symptomatic POP	† Observation, PFMT, pessary or surgery, chosen by bother and goals

6 · Landmark trials

Trial	Design	Result	Take-home
† VALUE	Randomised controlled trial, uncomplicated demonstrable SUI: office evaluation with or without UDS	Office evaluation alone non-inferior	Preoperative UDS not needed in index SUI

114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

1 · Definition and principles

- **Conservative treatment** is any therapy that is neither surgery nor drugs.
- **Behavioural treatment** changes behaviour or environment, or teaches skills. It is effective and **essentially risk free**, so it is first-line in all guidelines for stress (SUI), urgency (UUI) and mixed (MUI) urinary incontinence, overactive bladder (OAB) and nocturia.
- Programmes are multicomponent and individualised, combining education, pelvic floor muscle training (PFMT), bladder training (BT), biofeedback (BF), electrical stimulation (E-stim) and lifestyle advice.
- Outcome is unpredictable, so offer it to anyone willing; dementia patients benefit from caregiver-led programmes.

	Outlet-directed	**Bladder-directed**
Target	Urethral closure	Bladder function via voiding habits
Examples	PFMT; behavioural training with urge suppression	BT; delayed voiding
	Patient-dependent	**Caregiver-dependent**
Examples	BT, PFMT, behavioural training	Timed voiding, habit training, prompted voiding
Needs	Intact function, learning, motivation	Knowledgeable, motivated caregiver
Suits	Cognitively and functionally intact	Impaired, nursing home

KEY POINT

Why first-line if surgery cures SUI more often? Conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation.

2 · Assessment

- A **bladder diary for 3–7 days** records intake, voids and volumes, urgency, leaks and pads. It estimates **functional capacity**, and its **self-monitoring effect** makes it assessment, outcome measure and treatment at once.
- **Education** comes first: anatomy and the mechanism of the patient's own leakage. Improvement is gradual and needs daily practice.

3 · Lifestyle modifications

Cheap adjuncts; evidence is weak except for weight loss.

Fluid management

- **Inadequate intake is more common** than excess; self-restriction causes dehydration, urinary tract infection (UTI), constipation and concentrated irritant urine.
- Excess intake worsens frequency, so reduce it if 24-h output is >~**2100 mL**.
- **Target** a **normal range, not restriction**: **30 mL/kg/day**, minimum **1500 mL**, at set times.

- Change the **timing**, not the total: stop fluids **3–4 h before bed** for nocturia. For oedema-driven nocturia, use stockings or leg elevation in the late afternoon, or a mid-to-late afternoon loop diuretic.

Caffeine and irritants

- Caffeine is a **diuretic and irritant** that raises filling detrusor pressure and is a risk factor for detrusor overactivity (DO).
- About **250 mg/day** (≈2 cups) is associated with moderate–severe UI in men.
- Only a subset is caffeine-sensitive. Try a 3–5-day reduction, tapering with decaf to avoid withdrawal headache.

Bowel function

- Constipation and straining are linked to lower urinary tract symptoms (LUTS) through shared innervation, cross-sensitisation and straining-induced **pudendal denervation**. **Faecal impaction** can cause OAB or overflow.
- Advise fibre **with adequate fluid**, exercise, and regular evacuation after meals.

Obesity, smoking, diet

- Obesity roughly doubles UI risk. **Loss of 5–10% body weight** sustained for 12 months substantially reduces UI, mainly SUI; bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women is the **only grade A lifestyle recommendation**.
- For smoking, no UI recommendation is possible (grade D); advise cessation for general health. A fruit, vegetable and whole-grain diet helps prevent UI after menopause (B).

4 · Pelvic floor muscle training (PFMT)

Most failures come from a wrong contraction.

- PFMT, introduced by **Arnold Kegel in 1948**, is the cornerstone for SUI, UUI, MUI, urgency and frequency.
- **Mechanism**: contraction lifts the urethra and compresses it against the symphysis. Training causes striated muscle **hypertrophy**, which raises urethral pressure and improves bladder-neck support. There is also a **skill** route (the knack).

Assessment of PFM function

- A correct contraction gives **perineal elevation**; perineal descent means the patient is straining.
- On **digital vaginal exam**, levator ani lies just above the hymen at **4 and 8 o'clock**. Assess tone, symmetry, hold and quick flicks, and note **overactivity** (hypertonic, cannot relax).
- Scales: **Modified Oxford** is the most used; **Brink** scores pressure, duration and displacement.

Teaching

- Say: "Tighten as if stopping gas, pull up and in", without using buttocks or abdomen. For men: "move the penis up and down".
- The commonest errors are **co-contraction** (rectus, gluteals) and **Valsalva substitution**; counting aloud prevents breath-holding.
- **Verify correct contraction before prescribing.** Failure to isolate the muscle is probably the commonest cause of poor outcome; a pamphlet does not ensure it.

Regimen

No regimen is ideal; good results come with **45–60** contraction–relaxation pairs per day.

ALGORITHM

1. Empty the bladder.
2. **Lying**: 5 quick (1–2 s) and 5 sustained (5, building to 10 s) contractions.
3. Repeat **sitting**, then **standing** (total 30).
4. Rest as long as each contraction (work:rest **1:1 to 1:2**).
5. Do this twice daily = **60/day**.

- **The knack**: contract **just before and during** a cough, sneeze or lift. It reduces leakage immediately.

Evidence

- PFMT beats no treatment for SUI, UUI and MUI, short and long term; pure SUI does best.
- **Supervised training beats unsupervised** (grade A), and **age is no deterrent**.
- Antenatal PFMT in continent women prevents late-pregnancy and postpartum UI; give supervised postpartum PFMT if UI persists ~3 months.
- **After radical prostatectomy (RP)**, PFMT hastens continence in the first 3 months, and behavioural training helped men leaking years after RP. Adding E-stim or BF does not help.
- In **neurological OAB** (multiple sclerosis, incomplete spinal cord injury, Parkinson disease, stroke), PFMT is safe and first-line (B). PFMT adds to tolterodine for UUI.

Cones and devices

- **Weighted vaginal cones** (2 × 20 min/day) work by prompting a reflex contraction, but they **cannot be retained with POP or a wide introitus** and are no better than PFMT. Motion-sensing, app-guided devices can distinguish lift from strain and modestly improve results.

5 · Urge suppression

- A well-timed PFM contraction **reflexly inhibits the detrusor**, whereas rushing raises bladder pressure.

ALGORITHM

1. When the urge comes, do not rush.
2. Stand still or sit, and relax.
3. Do several quick PFM squeezes, then distract yourself.
4. The urge subsides like a wave; then walk normally to void.
5. Once mastered, wait 5 min after the urge, then extend.

- Used alone for UUI, it gives a mean **60–80%** reduction in incontinence, and also reduces urgency, frequency and nocturia.
- For **nocturia**, suppress the urge on waking; void if it has not passed in 1–2 min.
- It is at least as good as antimuscarinics and adds to drugs. In men on an α-blocker, adding behavioural therapy reduced nocturia more than adding an antimuscarinic.

6 · Biofeedback (BF)

- BF is a **teaching technique**, not a treatment. It shows PFM activity in real time by manometry or surface electromyography (EMG); EMG separates PFM from abdominal

activity, whereas pressure BF cannot separate lift from strain.

- BF **adds no clear benefit to supervised PFMT**. Use it for very weak PFMs, poor proprioception, an uncertain contraction, or motivation.

7 · Electrical stimulation (E-stim)

- Non-implanted stimulation (vaginal or anal probe, or surface electrodes) produces **PFM contraction** and **detrusor inhibition**.

	Long-term (chronic)	Maximal
Intensity	Below sensory threshold	Just below pain threshold
Aim	Detrusor inhibition via pudendal afferents	Urethral closure (direct and reflex striated contraction)
Dose	6–12 h/day for months	15–30 min, several times a week to twice daily

Indication	Frequency
UUI	**5–20 Hz** (low)
SUI (including post-prostatectomy)	**20–50 Hz** (higher)
MUI	~20 Hz or alternating
All	Pulse 200–1000 µs, rectangular, biphasic

- E-stim beats sham but **adds nothing to PFMT**, so PFMT is first-line (A). Its **niche** is women who **cannot contract voluntarily**.
- **Tibial nerve stimulation** for UUI or OAB: transcutaneous equals percutaneous in efficacy, with fewer adverse effects (see Ch 115).

> **KEY POINT**
> E-stim is dosed by goal. Sub-sensory, low-frequency, prolonged stimulation inhibits the detrusor; near-painful, higher-frequency, brief stimulation contracts the sphincter. Volitional PFMT does both, so E-stim earns its place only where volitional contraction is absent.

8 · Bladder training and scheduled voiding

P1 Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management — BT breaks urgency, frequent voiding, reduced capacity cycle; void by clock not urge (dissociates voiding from urgency)

Not in the paper answer:

- BT also works for SUI and MUI; it is grade A first-line in women, but evidence in men is insufficient.
- Adding behavioural training to drugs gives benefit that persists after the drug stops (UITN).

> **ALGORITHM**
> 1. Start with a diary.
> 2. Set an initial interval of ~1 h while awake (30 min if needed).
> 3. Void on waking, at each interval and at bedtime, but not between.
> 4. If leaks and urgency are controlled, increase by **15–30 min/week**.
> 5. Target an interval of **3–4 h**; review weekly with reinforcement.

- BT beats no treatment for UUI, SUI and MUI, and the Agency for Healthcare Research and Quality (AHRQ) rates it equal to PFMT.
- **Delayed voiding** has no schedule: at each urge, suppress it, **wait 5 min**, and extend gradually. BT is triggered by the clock; delayed voiding by the urge plus a wait.

Caregiver-dependent toileting: elderly, nursing homes

Programme	Patient role	Key features
Timed voiding	Passive	Fixed 2–4 h by day; dryness goal. Also for impaired sensation and infrequent voiders. ~20% of frail elders become dry, 30–50% improve; a 2–3-day trial predicts success
Habit training	Passive	Schedule individualised from the diary, shorter than the usual interval, to **pre-empt** expected leaks; suits diuretic or time-patterned UI
Prompted voiding (PV)	Responsive	**Monitoring**, then **prompting**, then **praise**; every 2–2.5 h. 25–40% of nursing-home residents respond (daytime leaks ≤1); a 3-day trial predicts
Functional incidental training	Responsive	PV plus mobility and strength exercise
Bladder training	Active	Lengthens interval; needs ability and motivation

- **Only BT lengthens the interval**; caregiver programmes aim at dryness, and PV success depends more on the caregiver than the patient.
- First exclude reversible causes (DIAPPERS) and check post-void residual (PVR) (see Ch 105).

9 · High-tone PFM dysfunction

A tight pelvic floor needs relaxing, not strengthening.

- Raised tone causes pelvic pain, hesitancy and incomplete emptying. Treat by **down-training**: contract, then relax, with work:rest **1:4 or longer**; BF is useful.

10 · Mechanical devices

Devices are **intravaginal** (support) or **intraurethral/meatal** (occlude); no device is proven superior.

Continence pessaries

- **Rationale**: the pessary **stabilises the bladder neck, lengthens the urethra and compresses it against the pubis**, like a sling. The knob of the ring sits at the urethrovesical junction.
- **Types** for SUI: incontinence **ring** (with knob), **dish** and **Hodge**. For pelvic organ prolapse (POP), support pessaries (ring, Gehrung) suit milder prolapse and space-filling ones (Gellhorn, cube, donut) suit advanced POP.
- **Fitting**: choose the largest comfortable size, one finger-breadth from the wall. The patient walks, coughs, strains and voids before leaving, with no expulsion, pain or obstruction.
- **Contraindications**: active pelvic infection, severe ulceration, silicone or latex allergy, **likely non-compliance with follow-up**.
- **Complications**: discharge and odour. Neglected use causes **erosion, impaction, vesicovaginal or rectovaginal fistula**, and bleeding.

- **Care**: follow-up every **24 weeks** is acceptable (versus 12; grade C), inspecting for erosion each time.
- About half improve. The International Consultation on Incontinence (ICI) says pessary or PFMT may be offered; pessary with PFMT beats PFMT alone for POP symptoms. See **ATLAS** under trials.

Other devices

- **Uresta** (reusable bell-shaped pessary) and **Impressa** (disposable tampon-like device) give suburethral support. Urethral inserts are balloon plugs for pure SUI, with risks of UTI, haematuria and migration.
- Use pads when treatment fails or is declined, and prevent **incontinence-associated dermatitis**.
- **An indwelling catheter is the last resort**: unmanageable retention, urine-contaminated sacral or perineal wounds, terminal care, or preference. Clean intermittent self-catheterisation (CISC) is preferred for retention.

11 · Vaginal oestrogen

- For SUI with atrophy, PFMT with vaginal oestrogen beats oestrogen alone (C). Local oestrogen improves urgency, frequency, recurrent UTI and pessary tolerance (see Ch 113).

12 · Adherence, prevention and delivery

- **Adherence caps efficacy** and declines over time; realistic expectations, follow-up, reminders and linking exercise to daily cues help.
- **Prevention in childbearing**: supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (A). Offer a postnatal programme after forceps or a baby ≥ **4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse.
- **Prevention at prostatectomy**: PFMT before surgery, with or without early postoperative PFMT, shortens time to continence.

13 · Evidence grading: key ICI recommendations

Grade (Oxford)	Meaning
A	Consistent level 1 evidence; effectively mandatory
B	Consistent level 2/3 studies or majority of randomised controlled trial (RCT) evidence
C	Level 4, or Delphi-processed expert opinion
D	**No recommendation possible** (not "weak")

ICI 2023 recommendation	Grade
Supervised PFMT first-line for UI in women of all ages	A
PFMT over E-stim as first-line	A
BT first-line for women with UI	A
Behavioural weight loss for overweight women with UI	A
Antenatal supervised PFMT in continent women to prevent UI	A
PFMT reduces POP symptoms (stage reduction uncertain)	A; B

ICI 2023 recommendation	Grade
PFMT over drugs first-line (fewer adverse effects, less dropout)	B
PFMT before surgery (surgery more effective but riskier)	B
Adding BF, cones or E-stim to PFMT: not routine	B/C
Fluid adequacy and caffeine reduction	B
Smoking cessation for UI	D

- AHRQ (2018): behavioural therapy is **more likely than anticholinergics to cure UUI** (high strength).

14 · Landmark trials

Trial	Design	Result	Take-home
First behavioural vs drug RCT	Older women, UUI: behavioural vs titrated drug	Behavioural fewer leaks, more satisfaction	Behavioural at least equal to drugs for UUI
MOTIVE	Men with OAB, no obstruction: behavioural vs oxybutynin extended release (ER)	Equivalent reduction in voids	Behavioural therapy works in men
UITN (BE-DRI)	Urgency-predominant UI: tolterodine ER with or without behavioural	Combination better satisfaction, bother; persisted after drug stopped	Add behavioural training to drugs
ATLAS	Stress-predominant UI: pessary vs behavioural vs both	3 mo: behavioural better satisfaction; combination not better; no difference at 12 mo	Pessary reasonable alternative; behavioural first
Weight-loss RCT (PRIDE)	Overweight women: 6-month intensive weight loss vs education	Greater UI reduction	Recommend 5–10% weight loss (A)
GBT	Women with UI: one 2-h bladder-health class vs none	Modestly better ICIQ-SF score to 12 months	Group classes cheap, scalable

116 Retropubic Suspension Surgery for Incontinence in Females

1 · Mechanism

- Stress urinary incontinence (SUI) arises from **urethral hypermobility** and **intrinsic sphincter deficiency (ISD)**, alone or combined.
- Suspension restores **pressure transmission** and a firm backstop for compression.

> **KEY POINT**
> Suspension corrects **hypermobility, not the sphincter**: it lifts and stabilises bladder neck and proximal urethra, so presupposes a mobile urethra. A fixed ISD urethra needs compression (sling, artificial urinary sphincter (AUS)).

2 · Patient selection

- Do a stress test supine, then standing if negative; a Q-tip angle >30° means hypermobility.
- AUA/SUFU minimum: **post-void residual (PVR) and urinalysis**; urodynamics only if leak is not shown. Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cm H_2O suggests ISD.
- **Indications** (Burch remains an AUA/SUFU option with midurethral sling (MUS), autologous sling and bulking): SUI with hypermobility, or the stress part of mixed incontinence (MUI); **concomitant abdominal surgery** (sacrocolpopexy); **difficult vaginal access**; **mesh aversion**.

Contraindications	Raise failure risk
Immobile vagina (cannot reach Cooper's)	Baseline urgency, preoperative anticholinergic
Immobile urethra (negative Q-tip ≈ **5-fold** failure): use sling or AUS	Obesity, menopause, age >60
ISD alone; "lead-pipe" (fixed, open) urethra	Prior hysterectomy or SUI surgery; advanced POP; MUI

3 · Anatomy

- **Retzius** is an avascular preperitoneal space between symphysis and bladder. **Cooper's ligament** lies on the superior pubic ramus.
- The **arcus tendineus fascia pelvis (ATFP)** runs from symphysis to ischial spine.

4 · Burch colposuspension

Paravaginal tissue is sutured to Cooper's; sutures elevate, and **fibrosis maintains** the repair.

ALGORITHM

1. Low lithotomy, prep including vagina, Foley.
2. Pfannenstiel or low midline incision.
3. Enter Retzius; sweep fat laterally.
4. Find bladder neck by Foley balloon and vaginal finger.
5. Dissect bladder off vagina **laterally**; expose Cooper's.
6. Place sutures, then cystoscopy.

P1 Burch colposuspension — 2–4 sutures (0/1-0) each side, full thickness not lumen; distal at mid-urethra, proximal at bladder neck, lateral; complication rates

Not in the paper answer:

- Tie **distal first** and **loosely** (Tanagho), leaving **≈ 2 fingerbreadths** between bladder neck and pubis; over-tightening obstructs.

KEY POINT

Danger points: never dissect midline over urethra or bladder neck (Santorini bleeding, injury, obstruction if sutures medial). Passing through Cooper's may bleed suddenly: use pressure, clips, sutures.

Approach	Bottom line
Open (Pfannenstiel)	Cure **85–90% at 1 y, ≈70% at 5 y**
Laparoscopic / robotic	Subjective cure equivalent to open to 18 months
Needle suspension (Pereyra, Stamey, Raz)	Abandoned: sutures tore through

5 · Other retropubic procedures

A higher anchor means more obstruction.

Technique	Anchor	Holds by	Obstruction
Burch	Cooper's ligament	Sutures; high V-shaped angle	Greater
Marshall–Marchetti–Krantz (MMK)	Symphyseal periosteum	Sutures; high	Greater
Paravaginal (Richardson)	ATFP	Scarring; flatter	Less
Vagino-obturator shelf (VOS)	Obturator fascia (or iliopectineal)	Scarring; flatter	Less

KEY POINT

AUS in women is second-line for ISD (recurrent or neurogenic). Place it at the bladder neck, preferably **abdominally**; transvaginal placement raises infection and erosion.

6 · Perioperative care

- **Cystoscopy is mandatory**: no suture in bladder or urethra, and bilateral efflux. If efflux is absent, release sutures.
- For incomplete emptying, use clean intermittent catheterisation (CIC), preferred to suprapubic, until PVR is consistently **<150 mL**.

7 · Outcomes and complications

- Burch roughly equals retropubic MUS but is slower and causes more voiding dysfunction.
- Pubovaginal sling (PVS) is more effective but more obstructive (SISTEr).
- Add prophylactic Burch at sacrocolpopexy in stress-continent women (CARE); with existing SUI, MUS beats Burch.
- Voiding dysfunction (2–27%) is treated by CIC, then urethrolysis (Ch 118). De novo overactive bladder (OAB, 8–27%) needs obstruction excluded.
- **Posterior prolapse** (2.5–27%) follows because anterior elevation opens the cul-de-sac.
- **Osteitis pubis** is unique to MMK (0.7–2.5%) and causes suprapubic pain at 2–12 weeks; treat with rest and analgesia.

8 · Landmark trials

Trial	Design	Result	Take-home
SISTEr	Randomised trial (RCT), 655: Burch vs autologous fascial PVS	PVS higher cure at 2 and 5 y; more urinary infection, voiding dysfunction, de novo UUI	Sling more effective, more morbid
Ward–Hilton (TVT vs colposuspension)	First RCT, 348	No difference to 5 y; TVT more perforation; Burch more voiding dysfunction, later POP	MUS equals Burch, less invasive
CARE	RCT, 322 stress-continent, sacrocolpopexy with or without Burch	Less postoperative SUI with Burch, sustained 2 y	Add prophylactic Burch at sacrocolpopexy

118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

1 · How slings work — theories of continence

The theory decides where the sling sits and how tight.

P3 Female pelvic floor, continence mechanism, classification and management of SUI — hammock (DeLancey) vs integral theory (Petros–Ulmsten: pubourethral ligaments, suburethral hammock, pubococcygeus); PVS at bladder neck, MUS loose at mid-urethra

Not in the paper answer:

- Supporting integral theory, incontinent women have ≈**40% less collagen** in skin and round ligament MCQ .
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of stress urinary incontinence (SUI).
- A pubovaginal sling (PVS) sits under **minimal tension at the bladder neck** and rebuilds the hammock. A mid-urethral sling (MUS) lies **loose under the mid-urethra** and arrests the posterior wall. Neither should obstruct at rest.

Historical milestones MCQ

- McGuire reintroduced the fascial PVS (1978). The tension-free vaginal tape (**TVT**, 1995) was the first retropubic MUS and defined the two complications **unique to MUS**: urinary tract mesh penetration and vaginal exposure. The transobturator route followed in 2001 and the single-incision mini-sling (SIMS) in 2006.

Guideline status (AUA/SUFU)

- MUS for the index patient is a **strong recommendation**, grade A. Either route is conditional, grade A; SIMS is conditional, grade B.
- Mesh-specific risks **and** non-mesh alternatives must be discussed before a synthetic MUS (Clinical Principle).

2 · Preoperative assessment and patient selection

P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — AUA/SUFU minimum evaluation (history, exam, cough stress test supine then standing, PVR, urinalysis); indications for UDS; ValUE; ISD cut-offs

Not in the paper answer:

- Q-tip movement **>30°** on straining defines hypermobility.
- Further evaluation, including urodynamic studies (UDS), is needed for unclear diagnosis, neurogenic bladder, urgency-predominant mixed urinary incontinence (MUI), high post-void residual (PVR), POP-Q stage ≥3 without SUI on reduction, or prior surgery. Cystoscopy is not part of the index evaluation.

Urodynamic studies — the controversy

- UDS may be **omitted** in the index patient with demonstrable SUI (conditional, grade B), based on **ValUE**.
- A maximal urethral closure pressure (MUCP) **<20 cm H_2O** defines type 3 SUI, that is intrinsic sphincter deficiency (**ISD**). A Valsalva leak point pressure (VLPP) **<60 cm H_2O** also means ISD; a value **>90** means little or no ISD.

- ISD is a **matter of degree** and now largely a clinical diagnosis; it matters most with a **hypomobile urethra**.
- **Detrusor overactivity (DO)** is the most consequential bladder finding and may improve or worsen, so consider treating it first. Detrusor underactivity predicts voiding difficulty but should **not exclude** surgery.
- Offer non-surgical options to all: weight loss, **pelvic floor muscle training (PFMT)** before surgery for most, and pessaries.

3 · Choosing the sling

Situation	Preferred option	Reasoning
Index SUI, mobile urethra	MUS (retropubic or transobturator); SIMS acceptable	Equal efficacy, least morbidity
ISD with mobile urethra	Retropubic preferred over transobturator	Meta-analyses favour retropubic
Fixed, immobile urethra	Retropubic MUS, autologous PVS, bulking, adjustable sling	AUA Expert Opinion; mobility predicts MUS success
Recurrent SUI after failed MUS	Retropubic MUS or autologous PVS	Retropubic better than transobturator for recurrence
Mesh complication, urethral injury, fistula, diverticulum, reconstruction	**Autologous PVS**	No mesh near a damaged urethra; adds tissue
Declines mesh	Autologous PVS or Burch see Ch 116	—
Neurogenic SUI on clean intermittent catheterisation (CIC)	Autologous PVS (may be tight)	Only once UDS confirm safe capacity and compliance
Concomitant pelvic organ prolapse (POP) surgery	MUS, PVS or Burch	Conditional, grade C

4 · Pubovaginal sling

- **Indications:** primary or recurrent SUI from **hypermobility and ISD**, stress-predominant MUI, anteroapical POP, neurogenic SUI, **urethral reconstruction** (trauma, mesh perforation, interposition at fistula or diverticulum repair), and failed Burch or MUS.
- **Autologous fascial PVS is the gold standard for all forms of SUI.** It gives **dynamic compression without obstruction**; too much tension causes retention and erosion.
- Width **<2 cm** compresses without a constricting band.

Material	Behaviour	Status
Autologous — rectus fascia, fascia lata	Best ingrowth and neovascularisation, minimal reaction; negligible erosion	**Gold standard**; cost is harvest pain and wound complications
Allograft	Processing weakens it; variable, less durable	Mostly withdrawn
Xenograft (porcine, small intestinal submucosa (SIS), bovine)	Weaker, encapsulated; cure falls with time; porcine inferior in trials	Mostly withdrawn

Material	Behaviour	Status
Synthetic at bladder neck	Urethral perforation **15×** and vaginal exposure **14×** other materials (especially woven polyester)	**No longer used at bladder neck**
Vaginal wall sling	Weak; inclusion cysts, vaginal shortening	Historical

KEY POINT

Fibrosis paradox: more fibrosis should mean better support, but the inflammation that drives it also degrades the sling and causes erosion. Autologous fascia remodels with minimal inflammation, the ideal balance.

- **Rectus or fascia lata:** strength and cure are equal. Fascia lata avoids abdominal wound problems but causes thigh pain; use it after **ventral hernia repair**.

5 · Autologous PVS — operative procedure

- **Counsel** about voiding dysfunction and de novo urgency, and teach CIC in selected women. Give one dose of cefazolin and mechanical thromboprophylaxis.

ALGORITHM

1. Through a 6–7 cm Pfannenstiel incision, mark and excise a **2 × 8 cm** rectus graft.
2. Close the fascia with No. 1 polydioxanone (PDS).
3. Clean the graft, stitch No. 1 PDS to each end and keep it in saline.

- Keep the transverse incision **≥2 cm above the symphysis** for tension-free closure. A fascia lata graft is taken over the iliotibial band, and the fascia lata is not closed.
- **Vaginal part:** make an **inverted-U incision**. With the **bladder empty**, perforate the **endopelvic fascia** and finger-dissect the retropubic space.

ALGORITHM

1. With the bladder empty, pass Stamey needles **from above**, on the back of the pubis, into the vaginal incision.
2. Perform **cystoscopy (70° lens)**. If there is an injury, repass **more laterally** and re-scope after each pass.
3. Confirm ureteric efflux, centre the graft and tag its ends above.

- Do any POP repair **before tensioning** and remove the weighted speculum.
- **Tension:** tie the sutures over the rectus with **two fingerbreadths** of slack, deliberately **loose**. Tie tighter only if lifelong CIC is planned.
- Postoperatively, do a voiding trial on day 1; if it fails, use Foley or CIC and retry within 5 days. No heavy lifting and no intercourse for **6 weeks**.

6 · PVS outcomes

PVS rarely fails for stress leakage; "failure" is usually urgency.

- Results are durable even in complex patients (radiation, spinal cord injury, trauma). Cure stays high even with MUCP <4 cm H_2O, so **severe ISD is not a contraindication**.
- **Recurrent SUI:** a repeat PVS, or a PVS after failed MUS, succeeds in most.

- The commonest "failure" is **storage symptoms or urgency urinary incontinence (UUI)**; de novo UUI occurs in 2–22%, predicted by prior POP surgery.
- **Cochrane, PVS versus MUS:** PVS is probably no better, maybe less effective; MUS is shorter, with fewer complications (except bladder perforation) and less voiding and storage dysfunction.
- A damaged urethra does worst, yet PVS still dries most. In **MUI**, PVS cure equals that of pure SUI but the urgency outcome is unpredictable.

7 · PVS complications and management

Voiding dysfunction and bladder outlet obstruction (BOO)

- Incidence is 2.5–35%, **more than Burch** (SISTEr: all outlet procedures were in the PVS arm). Retention >4 weeks is ≈8%; **permanent retention is ≤5%**.
- The usual cause is **excess tension**; poor contractility and **straining to void** add to it.
- It presents as weak stream, retention, **storage symptoms only**, recurrent urinary tract infection (UTI) or pain, even with a low PVR. Exclude new POP kinking the urethra.
- Prediction is unreliable, so a low detrusor pressure means **counselling about CIC**, not refusing surgery. Evaluate by examination, PVR, cystoscopy and selective video-UDS; a high-pressure, low-flow pattern is often **absent in women**.

KEY POINT

The key diagnostic criterion is the temporal relationship between surgery and symptoms, not a urodynamic number. Preoperative UDS does not predict obstruction (SISTEr).

ALGORITHM

1. Most transient retention resolves within **10 days**: use catheter or CIC, timed voiding, PFMT, anticholinergic.
2. Autologous obstruction often improves, so wait up to **3 months**.
3. If retention is complete, act earlier: at **<6 weeks**, loosen the sling under anaesthesia by downward urethral traction.
4. If **>6 weeks** or conservative care fails, then do sling incision or urethrolysis.

- Downward loosening is **not for synthetic slings**. Dilation risks fibrosis, and **transurethral bladder-neck incision fails** because the sling is extraluminal.

Procedure	Features
Sling incision (transvaginal)	84–100% success; least morbid, so **first choice**; ≈20% recurrent SUI, not prevented by interposition
Transvaginal urethrolysis	Least effective for PVS: cannot relieve suburethral compression
Suprameatal urethrolysis	Better access to the lateral wings
Retropubic urethrolysis	Best results; after failed transvaginal attempt

- Urethrolysis succeeds in 65–93%, with recurrent SUI in 0–19%. Failure is usually **inadequate lysis**, and a repeat relieves ≈90%.
- **Storage symptoms persist** in over half: use an antimuscarinic or β3-agonist, **sacral neuromodulation (SNM)**, or onabotulinumtoxinA 100 U.
- Recurrent SUI after lysis is treated with bulking, repeat PVS or MUS.

Perforation, exposure and other complications

- Perforation is rare with fascia. Urethral perforation needs **sling excision and repair** (with or without Martius flap); bladder erosion is removed endoscopically.
- An exposed synthetic PVS is removed, and recurrent SUI is treated with a second, **autologous** PVS.
- Mortality is ≈**3 per 10,000** anti-incontinence procedures; harvest-site **incisional hernia** occurs in ≈1%.

8 · Mid-urethral sling — mechanism and materials

- The MUS **secures the posterior urethral wall**, with dynamic **kinking**. Mesh is fixed along its **whole length**, so 50–93% stay dry after midline incision.
- Preoperative **immobility predicts failure**; the ideal is a loose sling with a mobile urethra. **Exception, SIMS:** success needs restricted mobility, so it is set tighter.

> **KEY POINT**
> **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing the mobile urethra descends onto the tape; the posterior wall is arrested, the urethra kinks and closes. A fixed urethra never meets the tape, and tightening the tape to compensate obstructs.

- The material is **type 1 (Amid) macroporous monofilament polypropylene** with pores **>75 µm**, which admits macrophages and fibrovascular ingrowth. Microporous tapes (<50 µm) had high exposure and were withdrawn.

9 · MUS operative procedures

Each route has its own danger structures.

P2 P1 Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT; TOT in SUI — TVT bottom-up steps, TOT outside-in/inside-out steps, tensioning, Cochrane complication rates

Not in the paper answer:

- Document counselling about the **permanent implant**, mesh-specific complications and non-mesh alternatives.
- Bottom-up beats top-down, with better subjective cure and fewer perforations and exposures.

> **ALGORITHM**
> 1. Infiltrate local; make two suprapubic stabs ≈2 cm from midline.
> 2. Make a 1.5 cm midline vaginal incision 1.5 cm from the meatus.
> 3. Make minimal lateral tunnels to the endopelvic fascia, **not perforating** it (unlike PVS).
> 4. The catheter guide deflects the bladder to the opposite side.
> 5. Pass the trocar through Retzius and rectus, **hugging the back of the pubis**; repeat on the other side.
> 6. Perform cystoscopy.
> 7. Tension loosely (clamp between tape and urethra), remove sheath, trim.

- Keep the trocar in **metal-to-bone contact** with the back of the pubis, so it stays extravesical and extraperitoneal.
- **Transobturator, outside-in (TOT):** hips **hyperflexed (120°)**; puncture the labiocrural fold at the adductor longus notch, pierce the obturator membrane and turn around the ramus

onto the finger. Check the vaginal sulcus for a button-hole.

- **Inside-out (TVT-O)** has the same cure, with more voiding dysfunction and fewer vaginal perforations.
- **SIMS** uses only a vaginal incision and anchors in **obturator internus**; set it **tighter**, close to the urethra.

Feature	Retropubic (TVT)	Transobturator	SIMS
Path / anchor	Retzius, rectus fascia; U-shape	Obturator foramen; flatter hammock	Obturator internus only
Cure (index SUI)	Equal to transobturator up to 5 y; best long-term data (>17 y)	Equal; slightly more repeat surgery	Non-inferior to transobturator (TVT-Secur excluded)
ISD / recurrent SUI	**Preferred**	Worse with low VLPP	Limited data
Bladder perforation	**4.5%**	**0.6%**	Rare
Vascular/visceral injury, bleeding, voiding dysfunction	More	Less	Less
Pain	Suprapubic 2.9%	**Groin 6.4%** vs 1.3% retropubic	Least
Vaginal exposure	About 2%	About 2%	Similar (excluding TVT-Secur)
Cystoscopy	Mandatory (70°)	Recommended; check the urethra	Essential (30° suffices)

10 · Intraoperative cystoscopy

P1 Management of perforated mid urethral sling — 70° lens, full bladder; bladder injury: repass laterally and continue; urethral injury: no mesh, staged repair

Not in the paper answer:

- Inspect the urethra too after transobturator and SIMS passes. A bladder injury is benign: leave a catheter overnight to 72 h.

11 · MUS outcomes

These are the figures quoted when counselling.

- Retropubic slings are the most studied (17–20 years); objective cure stays high and subjective cure is lower. Cochrane (81 trials): retropubic and transobturator cure are equal from short to long term.
- TVT equals colposuspension at 5 years, with less de novo urgency and less later POP.
- In TOMUS at 5 years, satisfaction stayed high (79% retropubic, 85% transobturator) despite falling objective success.
- Long term, **de novo urgency rises** (≈⅓ at 10–20 years); retention is 0–7%.
- In Cochrane, SIMS did worse than retropubic, driven by **TVT-Secur (withdrawn)**; others about equal transobturator, with less pain. The key long-term endpoint is **reoperation**.

Group	Key points
MUI	Urgency resolves in ≈60–75% (best when stress-predominant). De novo urgency risk rises with age, obesity, nocturia, DO and low capacity. Persistent DO: transobturator < retropubic < PVS. New storage lower urinary tract symptoms (LUTS) may mean BOO
ISD	About 50% cure with urodynamic ISD. An immobile urethra predicts failure more strongly than a low VLPP, and low VLPP is not a contraindication to retropubic MUS. Transobturator: 12× odds of persistent SUI with low VLPP
Recurrent SUI	No level 1 evidence; repeat MUS is commonest, retropubic better than transobturator. Cure ≈81–90% but below primary surgery (≈62% vs 86%); more urgency, and more bladder perforation after prior retropubic surgery
Concomitant POP	A concomitant sling lowers postoperative SUI and reoperation but raises voiding dysfunction. About 30% of SUI resolves after POP repair alone. A prophylactic sling only after counselling (OPUS)
Elderly	Safe; similar or lower cure, more urgency and voiding dysfunction. Q-tip >30°: 90% continent; <30°: 42%; <10°: 80% stay incontinent. Cure ≈64% at ≥75 y vs 89% at 55–64 y. **Frailty, not age**, predicts readmission. Do not exclude by age (AUA Expert Opinion)
Obese	Success >80% but lower (≈87% vs 66% at 5 y), worse with BMI >35; more wound infection, fewer bladder punctures

12 · MUS complications and management

KEY POINT

Concurrent or staged sling at POP repair. Concurrent: one operation and less SUI (≈39% falls to 8–19%), but more obstruction, storage symptoms and mesh events. Staged: avoids an unnecessary sling but commits a third to a second operation. Decide after testing for occult SUI with the prolapse reduced.

- Bladder trocar injury is ≈2–4%, almost all retropubic; risk rises with prior retropubic surgery and thin habitus. Vascular (≈0.07%) and bowel (≈0.04%) injuries are rare, but deaths have been reported.
- †Unexplained hypotension, falling haemoglobin, pain, ileus or sepsis after a retropubic sling needs urgent imaging or exploration.
- Bleeding occurs in 2–3%, more with retropubic slings. A retropubic haematoma is usually **asymptomatic, so observe**.
- **UTI is the commonest complication** (3–4% at 30 days). It is less after transobturator slings and marks a worse course.
- Vaginal wound infection is 0.7–2%; rarely there is abscess or **adductor myositis**, which may present years later.
- **Groin pain** after transobturator slings affects 16–26% early and ≈4% at 12 months. Treat with NSAIDs, physiotherapy and trigger-point injection; if refractory, excise, and **complete removal** relieves more than partial.

Voiding dysfunction and retention

- The tape is **too tight** or **too proximal**, or POP kinks the urethra. Obstruction is commoner after retropubic slings (TOMUS: 2.7% vs 0%). Evaluate by **temporal relation**, examination, PVR, cystoscopy and selective UDS.

ALGORITHM

1. Retention or high PVR: CIC or catheter; most resolve within 2 weeks.
2. If obstruction persists, perform a **transvaginal midline sling incision at 2–4 weeks**.
3. Beyond 3 months the tape is fixed, so excise its lateral portion.
4. Formal urethrolysis is rarely needed.

- **Urethral dilation is harmful; do not perform it**. After incision 70–100% stay continent. **Early** release gives better voiding and **less** recurrent SUI.

KEY POINT

Why is the timing different from PVS? Fascial obstruction often settles, so one waits up to 3 months. Mesh is fixed by ingrowth and rarely loosens; late release needs wider dissection and causes more recurrent SUI, so release within ≈4 weeks.

Storage symptoms, exposure and sexual function

- De novo overactive bladder (OAB) is ≈6–12%, equal for all MUS types. A clear temporal link means **partial BOO first**, and **cystoscopy is mandatory**. Treat with behavioural measures and drugs, then sling revision, SNM, percutaneous tibial nerve stimulation (PTNS) or botulinum toxin.
- Vaginal exposure is ≈2%; †it causes discharge, bleeding, dyspareunia or partner discomfort. †Small exposures get topical oestrogen with or without excision; larger ones are excised. Urinary tract mesh needs excision and repair, with no immediate new mesh see Ch 119.
- De novo dyspareunia is ≈3–14%, and sling removal may relieve it. Many women improve as **coital incontinence** stops.

13 · Adjustable slings

- **Remeex** (mesh on sutures to a suprapubic varitensor, adjusted after surgery) is a niche option for **complex SUI**. Cure is ≈85–90%, but complications reach 32–51% and there is no level 1 evidence.

14 · Assessing the failed or troublesome sling

ALGORITHM

1. Define the symptom: SUI, storage, voiding, or pain/discharge.
2. Establish the **timeline** in relation to surgery.
3. Examine for exposure, POP, urethral mobility and tenderness over the tape.
4. Check PVR and urine culture, perform **cystoscopy**, and add UDS selectively.
5. Tailor treatment to the cause.

- If SUI recurs with a mobile urethra, repeat a retropubic MUS or do a PVS; if the urethra is fixed or there is ISD, use PVS, bulking or an adjustable sling.
- If storage symptoms have a temporal link, treat obstruction first. Pain or exposure needs excision (complete for pain) see Ch 119.

15 · Mesh controversy and regulatory status

- The **2010 FDA** safety communication on mesh for POP and SUI was based on **MAUDE** database reports. Professional bodies still endorse MUS with documented counselling.

- †The 2019 FDA order stopping transvaginal mesh concerned **POP**, not MUS. The UK (2018) paused MUS use pending safeguards see Ch 119.

16 · Landmark trials

Trial	Design	Result	Take-home
ValUE	RCT, non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation	Office evaluation non-inferior at 1 y	UDS may be omitted in the index patient
SISTEr	RCT, 655: rectus PVS vs Burch	PVS higher cure (47% vs 38% at 2 y); more voiding dysfunction, UTI and UUI; all BOO surgery in PVS arm; persisted at 5 y	PVS more effective but more obstructive
Ward-Hilton	RCT: TVT vs open colposuspension	Equal to 5 y; TVT more bladder injury; Burch more morbidity and POP	MUS replaced Burch
TOMUS	RCT, 597: retropubic vs transobturator	80.8% vs 77.7% (1 y, equivalent); surgery for voiding dysfunction 2.7% vs 0%; †more groin and neurological symptoms with transobturator; 5 y 51% vs 43% (equivalence not met)	Either; retropubic slightly more durable, more voiding dysfunction
OPUS	RCT (327): POP repair with or without prophylactic retropubic MUS in stress-continent women	Incontinence at 12 months 27% vs 43%; NNT 6.3; more perforation, UTI and bleeding	Reduces SUI but adds risk, so counsel

117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

1 · Goals and principles

Prolapse is a quality-of-life problem, so operate for bother.

P2 Vaginal prolapse repair — QoL condition, first operation best chance, restorative/compensatory/obliterative, vaginal (less morbid) vs abdominal (more durable), apex is the cornerstone

Not in the paper answer:

- **Success means absence of symptoms** (no bulge), which drives satisfaction more than anatomy does.
- Obliterative surgery is contraindicated if intercourse is desired. Open and laparoscopic routes have equal complications, but laparoscopic recovery is faster.
- **The first operation is the best chance**, because planes are then lost and recurrence rises with each re-attempt. Preoperative education improves satisfaction, and declining surgery is a legitimate choice.

2 · Choice of procedure by compartment

P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — POP-Q points (A distal, B proximal, D only with cervix), DeLancey levels, operation by defect, SSLF and promontory anatomy, uterus-preserving options

Not in the paper answer:

- Enterocele (point D) is closed by culdoplasty: **McCall** vaginally, **Halban** or **Moschcowitz** abdominally.

Compartment	POP-Q point	Organ
Anterior	Aa / Ba	Urethrocele / cystocele
Middle	C / D	Cervix or cuff / enterocele
Posterior	Ap / Bp	Rectocele / enterocele; perineal body

Site	Vaginal	Abdominal
Aa — urethra	Anterior repair; bladder neck suspension; sling	Retropubic urethropexy
Ba — bladder	Anterior repair; paravaginal; colpocleisis; wedge colpectomy	Paravaginal; sacrocolpopexy
C — cervix/cuff	Uterosacral ligament suspension (USLS); iliococcygeus; sacrospinous ligament fixation (SSLF); Manchester; hysteropexy; vaginal hysterectomy; colpocleisis	Hysterectomy; USLS; sacrocolpopexy; uterine suspension
D — cul-de-sac	**McCall** culdoplasty	**Halban**, **Moschcowitz** culdoplasty
Ap — posterior	Posterior repair; site-specific repair	Colpoperineopexy

KEY POINT

Only paravaginal repair and uterosacral suspension are done by both routes. McCall is vaginal; Halban (linear sutures) and Moschcowitz (purse-string, which may angulate the ureter) are abdominal.

3 · Anterior compartment

Anterior repair has the highest recurrence.

- A defect may be **central** (midline thinning of the pubocervical fascia) or **lateral/paravaginal** (detached from the arcus tendineus fasciae pelvis, ATFP); most are combined. **Anterior repair does not suspend the apex.**

ALGORITHM

1. In lithotomy, catheterise, hydrodissect and incise in the midline from apex to bladder neck.
2. Dissect the epithelium off the pubocervical fascia laterally to the ATFP and reduce the bladder.
3. Plicate with interrupted 2-0/3-0 delayed-absorbable sutures from bladder neck to apex.
4. Trim judiciously and close; give intravenous (IV) dye, do **cystoscopy** and pack.

- Deep sutures may enter or kink the bladder or ureter, so completion cystoscopy is mandatory.
- Anterior repair is abandoned for stress urinary incontinence (SUI): AUA meta-analysis failure approached **40%**.
- Recurrence depends on the **definition**: 30–46% for "POP-Q ≥2" versus 9–23% for "beyond hymen".
- Polypropylene lowers recurrence versus native tissue but causes far more graft complications without better QoL. Hence the **withdrawal of transvaginal synthetic POP mesh** (US 2019) see Ch 119.
- A concomitant sling **halves anterior recurrence**, more than any graft.

Complication	Point / management
De novo SUI	Rises with stage: stage 2 **41%** , early 3 **53%** , advanced 3–4 **66%** ; point Ba predicts it
Detrusor overactivity	De novo in 5–7%; pre-existing overactivity resolves in up to 63%
Retention	Usually from the sling: clean intermittent catheterisation (CIC); if persistent, sling incision or urethrolysis
Bleeding (>350 mL)	Bleeding means the wrong plane; worse at redo
Bladder injury	Drain the bladder before perforating endopelvic fascia; two-layer closure; Martius if irradiated
Others	Ureteric obstruction, vesicovaginal fistula (VVF), vaginal shortening, dyspareunia

KEY POINT

Confirming ureteric efflux: indigo carmine (blue, IV), fluorescein 10% (green, IV) or phenazopyridine (orange, oral). If there is no efflux, pass a ureteric catheter or remove plication sutures. Never close without proving patency.

Paravaginal repair

- Reattach the pubocervical fascia to the ATFP with non-absorbable sutures 1 cm apart (**5–7** vaginal, **4–6** abdominal), then do cystoscopy.
- Lateral recurrence is 2–8%, but central recurrence is ~**22%**, so these are independent failures. The abdominal route is more durable.
- **Neither colporrhaphy nor paravaginal repair treats SUI.** The vaginal route has more serious complications (transfusion 12%).

4 · Occult SUI and the concomitant sling

- High-stage anterior POP kinks the urethra and masks SUI; SUI on prolapse reduction occurs in 8–66% of continent women.
- Test **with prolapse reduced** (pack, speculum or pessary); sensitivity is low and specificity high.
- **Staged** argument: sling obstruction needing intervention (8.5%) equals intervention for SUI without a sling (8.3%).
- The **OPUS calculator** uses 7 predictors (accuracy 0.72 vs 0.54 for the stress test). In **CUPIDO**, urodynamics added nothing in asymptomatic women.

KEY POINT

A sling also helps the prolapse: distal anterior support makes the cystocele repair last. **Why not everyone?** In OPUS it caused more bladder perforation, UTI, bleeding and incomplete emptying. Weigh per patient.

ALGORITHM

1. In advanced anterior POP, do a reduction stress test, with or without urodynamic studies (UDS).
2. If SUI is symptomatic, occult or urodynamic, then offer a concomitant procedure.
3. If she is continent and the test is negative, then weigh individual risk and benefit (OPUS calculator).
4. The sling is **selective, not routine**.

5 · Apical compartment

The apex is the cornerstone of prolapse repair.

Uterosacral ligament suspension (USLS)

- **The ureter lies 4 cm from the ligament at the sacrum but only 0.9 cm at the cervix**. Injury is usually **kinking**, not ligation. Sutures near the sacrum can entrap S1–S4 roots, causing buttock and posterior thigh pain.
- The **safe zone** is the intermediate portion, ~1 cm posterior, at the level of the ischial spine; pass needles **lateral to medial**.

ALGORITHM

1. After vaginal hysterectomy, place cuff stays at 3 and 9 o'clock, excise the enterocele sac, pack bowel and palpate the spines.
2. Place Allis clamps at 5 and 7 o'clock and pass **2–4** non-absorbable sutures per side.
3. Do any anterior repair or sling now.
4. Bring the sutures through the apical pubocervical and rectovaginal fascia and tie.
5. Perform **cystoscopy after tying, before trimming**. If there is no efflux, remove the most lateral ipsilateral suture first.

- Compared with SSLF, USLS keeps the midline axis, closes the cul-de-sac and risks the pudendal vessels less.
- Apical success is ~98% and anterior ~81%; ureteric kinking or injury occurs in **1–11%**.
- **OPTIMAL**: USLS equals SSLF (~65% at 2 years on a strict composite), and perioperative pelvic floor muscle training (PFMT) adds nothing.

Sacrospinous ligament fixation (SSLF)

- SSLF is extraperitoneal and preserves length. The ligament lies under coccygeus; the pudendal bundle runs round the spine and the inferior gluteal vessels lie behind.
- The **safe zone** is **1.5–3.0 cm medial to the ischial spine**, at limited depth. Too lateral risks the pudendal bundle; too medial hits weak ligament and sacral nerves.

ALGORITHM

1. Mark the apex, incise in the midline, enter the perirectal space (rectum medial) and palpate spine and ligament.
2. Place retractors and pass 2 sutures (Capio or Miya hook).
3. Complete other repairs, then attach the sutures to the apex (pulley stitch if permanent).
4. Close the vagina halfway and tie the apex flush, with **no suture bridge**.
5. Do cystoscopy and close.

- SSLF is usually right-sided, with 2 sutures 2 cm apart. The **pulley stitch** keeps the knot out of the lumen.
- Cure is ~85% and apical recurrence ~5%; stage 4 predicts recurrence.
- **Drawback:** the axis is deflected posteriorly and to the right, causing **anterior recurrence** (~18%).
- Gluteal pain occurs in **15%** and resolves in 2–3 months. Pudendal injury, rectal injury and haemorrhage can also occur.

KEY POINT

Both ligaments are safest in their **middle third** and dangerous at their bony ends: the intermediate portion for USLS, and 1.5–3 cm medial to the spine for SSLF.

Iliococcygeus suspension is for deficient uterosacrals, with sutures **1 cm distal to the ischial spine**; anterior recurrence is commonest.

Sacrocolpopexy (SCP)

- **Indications:** other defects (including SUI) repairable at one sitting, failed vaginal repair, young sexually active women, isolated vault prolapse, or wish for the most durable repair. It maximises vaginal length without distortion.

- Use **type I macroporous monofilament polypropylene**; fascia lata recurs in ~1/3 versus <1/10 with polypropylene.

ALGORITHM

1. Low lithotomy with a vaginal stent or EEA (end-to-end anastomosis) sizer; sigmoid to the left.
2. Incise the peritoneum over the promontory and along the **right** of the rectum (watch the right ureter) down to the anterior longitudinal ligament.
3. If there is a large enterocele, then add a Halban culdoplasty.
4. Dissect the bladder off the anterior vagina (6–8 cm) and the rectum off the posterior wall.
5. Attach a Y-mesh to both walls **without entering the lumen**, tension it, and fix it with 2–3 permanent sutures at **S1**.
6. **Retroperitonealise** the mesh, add an anti-incontinence procedure if indicated, and perform cystoscopy.

- Fixation too low (S3–4) risks the presacral venous plexus; too high risks the middle sacral vessels and the **left common iliac vein**.
- Avoid hypersuspension, which causes pain and de novo SUI. Sutures or tacks into the promontory can cause spondylodiscitis.
- Open, laparoscopic and robotic SCP succeed in 87–100%; the robot saves little blood but costs more.
- Failures are mainly **posterior or distal**, with almost no apical retreatment; ~20% later need vaginal repair, so counsel.

SCP complication	Rate / management
Haemorrhage (presacral)	~4%; †pressure, sutures, bone wax, sterile thumbtacks
Cystotomy / enterotomy / ureter	3% / 1.6% / 1%; bladder is the commonest robotic visceral injury
Ileus / small bowel obstruction	3.6% / 1.1%
Mesh exposure	**~10% by 7 years** (E-CARE); more with microporous mesh; excision see Ch 119

- **Hysterectomy at SCP:** total hysterectomy raises exposure, so go **supracervical** or close the cuff meticulously.
- Cochrane, vaginal apical repair vs SCP: the vaginal route gives more prolapse awareness (relative risk 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9) and dyspareunia (2.5), yet it suits many older women.

ALGORITHM

1. If she is elderly or comorbid and not sexually active, then offer colpocleisis.
2. If she is fit, has primary prolapse and prefers the vaginal route, then offer USLS or SSLF, with or without vaginal hysterectomy or hysteropexy.
3. If young or sexually active, or with recurrence, short vagina or high stage, then offer SCP (minimally invasive).

6 · Hysterectomy vs uterine preservation

- †Hysterectomy alone does not cure prolapse, so always resuspend the apex.

- Sacrospinous hysteropexy is non-inferior to vaginal hysterectomy with USLS.
- †Preservation needs a normal uterus: exclude abnormal bleeding, endometrial or cervical disease and Lynch syndrome, and keep cervical screening up to date.

7 · Obliterative surgery — colpocleisis

For frail older women it is quick, low-risk and effective.

- **Total colpocleisis** (after hysterectomy) denudes and inverts the whole vagina. **Le Fort** (uterus in place) excises anterior and posterior rectangles and leaves **lateral channels** for drainage.
- Both need **high perineorrhaphy and levator myorrhaphy** and cystoscopy. If the uterus stays, evaluate the endometrium; hysterectomy removes cancer and **pyometra** risk but raises complications.
- Spare ≥ **1.5 cm** of distal anterior wall, because urethral traction causes SUI. Add a sling for SUI or occult SUI.
- Success is ~90–98% and satisfaction >90%, but regret reaches **13%**.

8 · Posterior compartment

Bowel symptoms often persist after rectocele repair.

- Symptoms are bulge, splinting or digitation, trapping (**not faecal incontinence**) and dyspareunia. First exclude slow transit and **anismus**.
- **Benefit is predicted** by bulge, splinting and a large rectocele, but not by constipation. **Constipation often persists**.
- **The transvaginal route beats transanal** (recurrence 7% versus 66%).
- **Midline fascial plication**, with or without perineal body repair, is the International Consultation on Incontinence (ICI) procedure of choice. In **site-specific** repair the commonest defect is **transverse**.
- Perineorrhaphy should leave a smooth contour, because ridges cause dyspareunia.

ALGORITHM

1. Repair anterior and apical compartments first; hydrodissect and incise in the midline.
2. Dissect the rectovaginal fascia off the epithelium; if the hernia is large, place a purse-string at its base.
3. Plicate in the midline (2-0 absorbable), incorporate the perineal body, trim and close.

- **Levator plication has been abandoned** because dyspareunia reaches 50%. **Grafts** are not supported (complications RR 1.8).

9 · Complications across POP surgery — summary

Problem	Prevention / management
Ureteric injury (USLS highest, 1–11%)	Bites lateral to medial; cystoscopy after tying; remove the most lateral suture; stent; reimplant
Bladder/rectal injury	Correct plane, drained bladder, rectal finger; layered closure
Mesh exposure	Type I mesh, full-thickness flaps, supracervical hysterectomy; oestrogen or excision
Recurrence	Anterior after USLS/SSLF, posterior after SCP; most are asymptomatic

Problem	Prevention / management
Buttock pain	Middle-third placement; observe up to 3 months; nerve injection
De novo SUI / retention	Reduction testing; selective sling; CIC; sling incision

10 · Landmark trials

Trial	Design	Result	Take-home
CARE	RCT: SCP with or without Burch, continent women, stage 2–4	SUI 24% vs 44% at 3 months	Prophylactic Burch reduces de novo SUI
E-CARE	7-year extension	SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48%	The benefit is a delay; exposure rises with time
OPUS	RCT: vaginal POP repair with MUS vs sham	27% vs 43% at 12 months; more perforation, UTI, bleeding, retention	Selective sling
OPTIMAL	RCT: USLS vs SSLF, with or without PFMT	Equal (~65%) at 2 y; PFMT no effect	Native apical repairs are equivalent
Bilateral SSLF vs abdominal SCP	RCT	Earlier failure and 3× cystocele reoperation after SSLF	Abdominal more durable
Transvaginal vs transanal rectocele	RCT	Recurrence 7% vs 66%	Repair vaginally

119 Genitourinary Complications of Mesh and Their Repair

1 · Mesh, terminology and regulation

- Unlike the abdominal wall, the pelvic floor is mobile, contaminated and close to viscera. Mesh harms are pain, exposure, extrusion and injury to nearby structures.

Amid type	Pore	Example	Behaviour
I	Macroporous >75 µm	Knitted monofilament polypropylene	**Only recommended mesh**
II	Microporous ~10 µm	Polytetrafluoroethylene (PTFE, Gore-Tex)	Encapsulated; >30% full-length slings excised
III	Macropores in microporous base	Mersilene, Mycromesh	Extrusion ~20%
IV	Submicronic	Silicone-coated	Extrusion/sinus up to 40%

KEY POINT

Why pore size matters: bacteria (~1 µm) enter pores that macrophages (~50 µm) cannot, so infection is sealed from immunity. Large pores (~1500 µm) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. Mesh adds no cancer or autoimmunity.

P1 Management of perforated mid urethral sling — IUGA/ICS terms: exposure, extrusion, perforation; "erosion" not used; trocar injury and late perforation management

Not in the paper answer:

- IUGA/ICS **CTS (Category–Time–Site) coding**: Category 1–3 vaginal (1 no separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 bowel, 6 skin/musculoskeletal, 7 systemic; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from it, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.
- The FDA classed transvaginal POP mesh as **class III (2016)** and it stopped selling in **2019**; slings and sacrocolpopexy (**transabdominal POP mesh, class II**) are exempt and proven safe, whereas mini-slings are not.

2 · Risk factors and prevention

- **Contraindications to vaginal mesh** are simultaneous urethrovaginal fistula repair, urethral injury or mesh perforation, and diverticulectomy. With **intraoperative urethral injury, abandon mesh** and place a sling later.
- Relative risks are prior fistula or diverticulum surgery, radiation, oestrogen-deficient tissue, smoking, obesity and age extremes; in men, radiotherapy and urethrotomy.
- POP mesh risk rises with high mesh volume, **shallow vaginal dissection**, hysterectomy, diabetes and sexual activity.

3 · Evaluation

ALGORITHM

1. History and **prior operative notes**.
2. Speculum and pelvic exam: exposure, bands, point tenderness, levator hypertonicity.
3. If haematuria, recurrent UTI, storage LUTS or voiding pain, then cystourethroscopy.
4. If voiding dysfunction, then urodynamics (UDS).
5. Translabial ultrasound for a sling; **MRI** for sacrocolpopexy mesh (invisible on CT); CT for haematoma.

Time never excludes a complication; slings can expose decades later.

4 · Midurethral sling complications

Vaginal exposure

- It follows infection, haematoma, excess tension or superficial placement, and presents with bleeding, discharge, dyspareunia, partner pain or UTI.
- A small early exposure gets **topical oestrogen** (1 g cream nightly for 2 weeks, then 2–3 times weekly; or estradiol 10 µg tablet).
- **Excision**: raise a flap, expose the full width of mesh, pass an instrument behind, divide, excise at least **1 cm** around the defect and close. Avoid blind lateral dissection, which causes vulval haematoma after transobturator (TO) slings.

Pain

- Causes are shrinkage, tension, obturator neuralgia and pelvic floor hypertonia.
- **Immediate severe pain**: exclude haematoma and consider **early removal** before ingrowth.
- Delayed focal pain gets pelvic floor physical therapy, trigger point injection and oestrogen, then segment excision (60–90% improve). Diffuse late pain reflects central sensitisation, and excision may fail.
- Remove **as much vaginal mesh as possible** (81% pain-free). For TO **groin pain**, add a thigh incision to obturator externus.

Bladder perforation

- Cystoscopy with a **full bladder** detects trocar puncture. A missed one causes storage LUTS, haematuria, UTI, stones or vesicovaginal fistula (VVF).
- **Endoscopic** removal has low morbidity, but mesh stays in the wall and re-extrudes. **Open or robotic transvesical** removal excises 2 cm beyond the defect with two-layer closure and a catheter for 2–3 weeks; re-extrusion is minimal.

Urethral perforation

- Causes are trocar injury, over-tensioning and **urethral dilation** (avoid it; divide an obstructing sling). It presents with voiding pain, retention or haematuria.
- Repair: inverted-U flap, peel the mesh off, minimal urethrotomy, layered closure with or without a **Martius** flap, and non-overlapping suture lines. A stricture needs buccal urethroplasty.
- After excision, SUI occurs in **42%**, fewer with a simultaneous fascial sling.

Voiding dysfunction

- Midline **sling division** resolves retention in 95% and symptoms in 75%. Urgency incontinence persists in up to 76% and is de novo in up to 45%. Recurrent SUI (20–74%) is treated by bulking, repeat MUS or fascial sling.
- Persistent symptoms need UDS, then wider excision and urethrolysis.

> **KEY POINT**
> **How much to remove**: retention needs midline division; exposure at least 1 cm each side; pain maximal vaginal removal, rarely arms. **Never remove asymptomatic mesh**; more removal means more bleeding and more POP and SUI recurrence.

5 · Transvaginal POP mesh

- In the largest randomised trial (n = 1358), native and mesh repair were equal at 6 years (reoperation 12% vs 10%); 8% of the mesh arm needed excision.
- Exposure is 10–12% and removal 5–8%. **Contraction** (~12%) is unique to POP mesh and causes tightness and dyspareunia. Bladder perforation is 0.7–3.5%, rectal 0.15% and fistula 0.2%.
- Treat **conservatively** (observe, oestrogen) if <1 cm, mildly symptomatic and uninfected; only **size** predicts success. After office trimming, 73% later need theatre.
- **Excision**: raise thick flaps, then clamp behind the mesh and divide, which is the **most dangerous step**. Cut arms under tension rather than chasing them, then repair natively.
- Afterwards, cystocele recurs in 19%, 20% need repeat excision and 3% develop VVF.

6 · Sacrocolpopexy mesh

- Exposure is ~**10.5% at 7 years**, higher with total hysterectomy (23% vs 5%) and stage 3–4.
- Features are discharge, granulation, dyspareunia, cuff abscess, bladder or bowel perforation and **discitis**; image cross-sectionally.
- Cut an **exposed suture** below the knot in the office. Oestrogen for small exposures and vaginal excision often fail; partial colpocleisis succeeds in ~50% but is unsuitable if sexually active.
- **Abdominal removal** suits failed vaginal repair, pain, infection, discitis or ureteric complication. **Remove the entire infected mesh**, including sacral sutures, because residual mesh causes sinus, abscess and fistula; bowel injury occurs in up to 29%.

7 · Male slings

- **Advance XP** causes retention in 14–21%, usually for days; incise one arm if it persists. **Virtue** causes perineal or scrotal pain in 14–17%, chronic in 0–7%.
- **Urethral extrusion** usually follows a perforation missed without intraoperative cystoscopy. Excise lateral to the urethra and repair primarily or with buccal graft, avoiding transection.

121 Urinary Tract Fistulae

1 · Definition and universal principles

- A **urinary tract fistula** is an abnormal epithelial- or mesothelial-lined connection from the urinary tract to another cavity (bowel, vessel, genital tract) or skin.
- Causes are iatrogenic (commonest in the developed world), malignancy, inflammation, infection, ischaemia, trauma and childbirth.
- It persists because **urine keeps flowing through a tract that cannot heal**, so it epithelialises. Every principle diverts flow or restores healing, and the **first repair has the best chance**.

Management (whole course) — Table 121.1	Operation (any route) — Table 121.2
Optimise nutrition	Evaluate concomitant injury; rule out malignancy
Control infection	Adequate exposure; remove all foreign material
Continuous drainage (catheter/stent)	Tension-free closure; vascularised edges
Eliminate obstruction	Multiple, watertight, **non-overlapping** layers
Consider malignancy — **always biopsy** if any history	Consider interposition; drain after

KEY POINT

Trap: a fistula may be the only sign of recurrent cancer, so biopsy the tract after prior malignancy or radiation.

2 · Vesicovaginal fistula (VVF) — aetiology

VVF is the commonest acquired urinary fistula (**>75%** of genitourinary fistulae).

	Developing world — obstetric (>90%)	Developed world — iatrogenic
Cause	Obstructed labour; young, small pelvis, poor nutrition	Unrecognised surgical injury; **>75%** after hysterectomy
Mechanism	Fetal head impacted for days causes pressure necrosis and a large defect	Dissection, cautery, clamp crush, bladder caught in cuff closure
Extent	Trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial)	Solitary, **supratrigonal at vaginal cuff**

- **Hysterectomy** causes VVF in ≈ 0.8/1000, lowest after vaginal and highest after laparoscopic surgery. Distorted anatomy and concomitant prolapse, incontinence or mesh surgery raise risk; immediate recognition and repair prevent fistula.
- **Universal cystoscopy at hysterectomy** raises intraoperative detection but has an inconsistent effect on delayed injury. AUGS advises it with any prolapse or anti-incontinence procedure, AAGL after laparoscopic or robotic hysterectomy, and ACOG makes no recommendation after benign hysterectomy.
- **Radiation** causes obliterative endarteritis, fibrosis and necrosis. VVF follows 0.6–2% of cervical cancer radiotherapy, typically **1.5–2 years** later (up to 30); risk is dose-dependent and worse in smokers.

- Other causes are mesh, **bevacizumab** (anti-VEGF) and cancer, and rarely pessary, intrauterine device, infection or endometriosis.

3 · Classification of VVF

Of >25 systems, the WHO uses simple vs complex.

Axis	Goh	Waaldijk
Basis	Distance of distal edge from external urethral meatus	Closing (continence) mechanism and urethral involvement
Types	1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm	I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric)
Size	a <1.5 · b 1.5–3 · c >3 cm	Small <2, medium 2–3, large 4–5, extensive ≥6 cm
Tissue	i none/mild fibrosis · ii moderate/severe · iii special (radiation, ureteric, circumferential, previous repair)	—

KEY POINT

Trap: Goh type 1 is farthest from the meatus (best) and type 4 closest (worst); closer means worse closure and continence. Goh predicts closure better. Always describe size, location (trigonal vs supratrigonal), tissue loss, and ureteric and urethral involvement.

4 · Diagnosis of VVF

- The classic symptom is **continuous leakage day and night** in any position; a small tract may leak only positionally.
- **Timing points to mechanism**: a missed cystotomy leaks in the first days, thermal or crush injury at **7–10 days**, and radiation after years. Fever, flank pain or ileus suggest urinoma or ureteric injury.
- On speculum the defect is usually at the vaginal apex after hysterectomy, and vaginal fluid creatinine approximates urine creatinine. Biopsy friable tissue after cancer or radiation, and optimise oestrogen, smoking, glucose and nutrition.
- **Single dye test**: backfill ~300 mL methylene blue, insert a tampon and walk 15–60 min. Proximal stain means VVF, distal stain means urethrovaginal fistula or SUI, and dye from the cervical os means vesicouterine fistula.
- **Double dye test**: oral phenazopyridine 100–200 mg plus blue in the bladder, then a tampon. **Orange means ureterovaginal fistula (UVF)**, blue means VVF, and both means both.
- **Cystoscopy is integral** (sensitivity ~92%) and shows size, number, orifices, bladder neck, mesh and capacity, and allows biopsy. Vaginoscopy shows the vaginal side. A cystogram (lateral view) or T2 MRI helps difficult cases.
- **Upper tract imaging is mandatory**, because ureteric injury coexists in up to **12%**. Use CT urography with a delayed phase, or MR urography if contrast allergy or renal failure. **Partial ureteric obstruction with VVF means UVF until proven otherwise.** If unclear, do retrograde pyelogram and stent, or percutaneous nephrostomy (PCN).
- Urodynamics helps planning: 10–15% have SUI needing an autologous sling, and radiation may reduce capacity and compliance.

KEY POINT

Work-up order: is it urine (creatinine), where is it from (dye tests, endoscopy), and what else is damaged (hence compulsory upper tract imaging). A preoperative PCN also allows antegrade ureteroscopy to find the ureter in scar.

5 · VVF — timing, conservative care, repair

P3 P2 P1 VVF management options; early vs delayed repair; eponymous techniques — catheter trial (~15%, <1 cm, within 3 weeks, EAU ≥12 weeks), fulguration, sealants not recommended; timing by cause; vaginal vs abdominal route (~90% either); flap-splitting, Latzko, cuff excision, O'Conor, transvesical/extravesical, lap/robotic; flaps; postoperative care; diversion; outcomes

Not in the paper answer:

- Size >3 cm predicts failure, and radiation strongly predicts failure with ureteric stricture, poor compliance and sphincter deficiency, so consider diversion.
- Apical VVF (8–10 cm from introitus) lies beyond Martius reach, so use peritoneum or omentum.

Setting	Timing of repair	Rationale
Obstetric (traditional)	≥3 months, catheter meanwhile	Inflammation settles, necrosis sloughs
Iatrogenic, clean tissue	Early; European Association of Urology (EAU): no minimum delay if no infection or recent radiation	Comparable closure; less distress
Radiation-induced	≥ **12 months** after radiation ends	Late injury still evolving

- **Catheter** drainage is favourable for tracts <1 cm catheterised within **3 weeks**, and unfavourable at ≥6 weeks or with radiation or malignancy. The EAU trial lasts ≥ **12 weeks**. Sealants are possible for <1 cm in poor surgical candidates.
- **Location**: within **1 cm** of a ureter, stent it; if the ureter is involved, reimplant abdominally. Bladder neck involvement needs reconstruction, plus a pubovaginal sling if needed.
- **Route**: there are no RCTs, and first-attempt closure is ≈ **90%** either way. Transvaginal repair has less blood loss, pain and stay.
- Abdominal repair suits large fistulae (≥4 cm primary, ≥2 cm recurrent), urethral, bladder-neck or ureteric involvement, a non-capacious vagina, radiation, or no lithotomy. Vaginal repair risks shortening and dyspareunia, whereas abdominal repair spares sexual function.

ALGORITHM

1. **Transvaginal flap-splitting (Table 121.5)**
2. Cystoscopy: stent close ureters; wire through the tract.
3. Suprapubic catheter (SPC) and urethral catheter; episiotomy if narrow.
4. Foley ≥8 Fr through the fistula for traction; hydrodissect.
5. **J- or U-incision** with a 1–3 mm cuff; raise wide anterior and posterior flaps.
6. Layer 1: 3-0/4-0 delayed absorbable. Layer 2: **perpendicular** to layer 1.
7. Leak test 200–300 mL; Martius or peritoneal flap if needed.
8. Advance the opposite flap ≥2 cm beyond (layer 3); cystoscopy for efflux.

- **Postoperative drainage is the most important part**: urethral catheter, often with SPC, for ≥10–14 days. Give an anticholinergic or β3-agonist for spasms, remove stents at 4–6 weeks, and advise pelvic rest for ~6 weeks.

- **Latzko** leaves the bladder unopened and the tract unexcised: denude 1–2 cm and close in two or more layers. It succeeds in ≥90% of non-irradiated cases and suits close orifices, but shortens the vagina.

ALGORITHM

1. **O'Conor (Table 121.7)**
2. Ureteric catheters; low midline extraperitoneal approach.
3. Sagittal dome cystotomy; **bivalve** down around the fistula; excise the tract.
4. Separate bladder from vagina ≥2–3 cm beyond; close the vagina.
5. Place **omentum or peritoneum** beyond the vaginal suture line.
6. Close bladder in 1–2 layers; SPC with or without urethral catheter; fill test.
7. Drain 10–14 days, then cystogram.

- O'Conor causes more de novo urgency and LUTS from long suture lines. Transvesical repair suits low-lying fistulae, and extravesical repair suits supratrigonal fistulae. Robotic O'Conor uses omentum or peritoneum, with indocyanine green to check edge vascularity.

6 · Interposition flaps

- Flaps reinforce, fill dead space and import blood supply. Use them for **complex VVF** (radiation, recurrent, ischaemic obstetric, poor tissue); an RCT showed **no benefit of Martius** in simple VVF.
- **Martius** is supplied by the internal pudendal (inferior), external pudendal (superior) and obturator (lateral) vessels. Keep the **superior** pedicle for VVF (inferior for rectovaginal). It reaches mid-distal VVF, bladder neck and urethra, but **not the apex**.
- A vaginal peritoneal flap suits the apex, and **omentum** suits radiation, recurrent or large fistulae. Rectus, gracilis and Singapore flaps are a last resort.
- After repair, residual incontinence (predicted by Goh 3–4 and more than one repair) drives quality of life, and ICS advises physiotherapy for all obstetric fistula. Divert if the fistula is unrepairable.

7 · Ureterovaginal fistula (UVF)

P3 Urological complications in OBS-GYN and uretero-vaginal fistula — distal ureteric injury at hysterectomy, danger sites, normal voiding + leak within 4 weeks, double dye/CTU/RGP, stent within 2 weeks for 4–6 weeks, reimplant ± psoas hitch/Boari, ≥90%, follow 30 months

Not in the paper answer:

- Up to **25%** of VVF have a coexisting UVF. Prophylactic stents are selective (e.g. caesarean hysterectomy for placenta accreta).

ALGORITHM

1. Suspect UVF; exclude VVF; confirm with CTU or retrograde pyelogram.
2. Stent; remove at 4–6 weeks and reimage.
3. If resolved, stop. If persistent or stenting fails, then do ureteroneocystostomy.

KEY POINT

UVF: wet but voids normally. The ureter heals over a stent if reached early, so the detection-to-stent interval largely decides whether surgery is needed.

8 · Vesicouterine fistula (VUF)

- It follows **low transverse caesarean** and joins the uterine body above the internal os to the posterior bladder; a competent cervix **preserves continence**.
- **Youssef syndrome** is menouria, amenorrhoea, first-trimester miscarriage and continence. Bladder endometriosis is the differential of menouria.
- Dye appears from the cervical os, and cystoscopy shows a posterior supratrigonal defect; add cystogram, hysterosalpingography or MRI.
- Spontaneous closure is ~5%. After 3–4 months, separate transabdominally and close in layers with omentum or peritoneum, or do hysterectomy if the family is complete. Deliver later pregnancies by caesarean.

9 · Urethrovaginal fistula

P3 Urological complications of OBGY surgery and urethrovaginal fistula — diverticulectomy commonest Western cause (20–25%), distal vs proximal symptoms, flexible urethroscopy, ~20% separate VVF, flap repair + Martius, autologous sling never synthetic, SPC ≥14 days, closure 90–95% vs continence 50–80%

10 · Rectourethral fistula (RUF)

P3 Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management — post-prostate-cancer treatment, faecaluria/pneumaturia, biopsy, colostomy first, transperineal gracilis vs York-Mason vs abdominal, salvage diversion

Martini	Category
Stage (size)	I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage
Grade (tissue)	0 surgical injury only · 1 prior radiation/energy · 2 salvage radical prostatectomy (RP) or salvage ablation

11 · Vesicoenteric fistula (VEF)

- Colovesical is commonest. M : F is **3 : 1** because the uterus shields the bladder.
- Causes in order are **diverticulitis**, **Crohn disease** (ileovesical) and colorectal cancer (10–15% of colovesical fistulae), then bladder or prostate cancer, radiation and appendicitis.
- It causes **pneumaturia** (50–70%), faecaluria and polymicrobial UTI. **Gouverneur syndrome** is suprapubic pain, frequency, dysuria and tenesmus.

Test	Role
Poppy seed test	Highest detection of presence; cheap; no location
CT (preferred)	Air in bladder, wall thickening, adherent bowel; shows cause
Cystoscopy	Nonspecific erythema/oedema
Colonoscopy	Poor for tract; **excludes cancer**
MRI	Crohn, anatomy

- Treat conservatively if symptoms are minimal and disease benign (especially Crohn); spontaneous closure is <2%.
- **Resect the diseased bowel with the fistula** with single-stage anastomosis; use Hartmann for abscess, contamination, advanced cancer or radiation. Leave or oversew a small bladder defect, do **partial cystectomy** if malignant, and interpose omentum.

> **KEY POINT**
> VEF is a bowel disease presenting in the bladder: remove the diseased bowel, and the bladder usually heals itself.

Renocolic fistula: spontaneous ones are mostly renal (xanthogranulomatous pyelonephritis [XGP]) and need nephrectomy; small iatrogenic ones after PCNL settle with antibiotics, drainage and nil by mouth.

12 · Urocutaneous fistulae

- **Vesicocutaneous** fistula follows long-term SPC, open prostatectomy or radiation, and persists with infection, foreign body, tumour or bladder outlet obstruction (**BOO**). Drain and relieve BOO; if unhealed, do fistulectomy or partial cystectomy.
- **Urethrocutaneous** fistula is the commonest hypospadias complication (up to 20%). Wait ≥ **6 months** and exclude distal stricture. Close <1 cm locally; ≥1 cm or redo needs vascularised flaps (**tunica vaginalis** better than dartos).
- **Nephrocutaneous** fistula follows stones, XGP, prior surgery or tuberculosis. The kidney is non-functioning in ~90%, so do **nephrectomy**.

13 · Pubovesical fistula (PVF)

- A tract from bladder, prostate or urethra to the pubic symphysis causes **osteomyelitis**. It follows prostate radiotherapy plus repeated outlet procedures, median ~10 years after radiation, and also artificial urinary sphincter surgery.
- It presents with **pubic pain on walking**, abscess, urosepsis or sinus. **MRI is most sensitive and specific**; bone biopsy guides antibiotics because urine culture is unreliable.
- Conservative care usually fails. Debride bone, excise the fistula, fill with a rectus or omental flap, and give 4–6 weeks of antibiotics; if bladder or sphincter function is poor, do **cystoprostatectomy and diversion**.

14 · Urovascular fistulae

- **Arteriocalyceal** fistula follows PCNL, transplant or trauma; treat by **angioembolisation**.
- **Ureteroarterial fistula (UAF)** is an emergency. Most (85%) are secondary to long-term stents, pelvic surgery, radiation, cancer or vascular grafts. It presents with **intermittent haematuria**, and ~20% are unstable.
- Diagnose with **angiography**, provoking by stent removal only with endovascular treatment ready; a negative CT angiogram does not exclude it.
- Treat with an **endovascular stent graft** with or without internal iliac embolisation; use open surgery for infection, abscess or enteric involvement.

15 · Renal urine leak and anastomotic leak

- After **trauma**, a leak occurs in 1–7%. Use CT with a 10-min delayed phase; a non-opacified distal ureter suggests ureteropelvic junction avulsion. Most resolve, and persistent leaks need stent or nephrostomy.
- After **partial nephrectomy** (0.5–20%), a central tumour predicts leak. Drain, adding stent or nephrostomy if needed.

- **Vesicourethral anastomotic leak after RP** (Patil): I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolong the catheter, then withdraw the drain, then place stents or nephrostomies; reoperate last.

16 · Fistula after gender-affirming surgery

- After phalloplasty or metoidioplasty, urethrocutaneous fistula and stricture are common; **primary colpectomy** roughly halves fistula. Small ones close with a catheter. After transfeminine surgery, neovaginal fistula is rare and is repaired with interposition after diversion.

17 · Landmark trials

Trial	Design	Result	Take-home
Martius flap in simple obstetric VVF	RCT, repair ± Martius	No difference in closure; ~⅓ donor-site pain/numbness	Reserve Martius for complex fistula
Prophylactic ureteric stents in gynaecological surgery	RCT (+ meta-analysis)	No/small reduction in injury (number needed to treat 224)	Selective, not routine, stenting

122 Bladder and Female Urethral Diverticula

1 · Bladder diverticula — definition and pathology

P2 Bladder diverticulum. — definition, congenital vs acquired, investigations, management, diverticulectomy steps, T3a staging

Not in the paper answer:

- Diverticular cancer: radical cystectomy gives the same stage-for-stage survival; upstaging is common.
- A diverticulum is urothelium herniating through the muscularis propria; an outer pseudocapsule gives the surgical plane.
- With **no functional muscle, the sac cannot contract**, so post-void residual (PVR) is high and stasis leads to infection, stones and malignancy.
- About 90% occur in adults; M : F ≈ **9 : 1**.

Feature	Congenital	Acquired
Age / number	<10 years; solitary	Men >60; often multiple
Site	Posterolateral to ureteric orifice (~90% near ureterovesical junction [UVJ]); vesicoureteral reflux (VUR) common	Ureterovesical hiatus commonest; anywhere
Cause	Detrusor weakness at UVJ	Bladder outlet obstruction (**BOO**) or neurogenic — almost always
Wall / presentation	Smooth; acute urinary tract infection (UTI)	Trabeculated; incidental (lower urinary tract symptoms [LUTS], haematuria, UTI)
Malignancy	Not increased	Increased

- Up to 60% of congenital cases have a syndrome, neuropathic bladder or BOO, e.g. **Menkes**, **Williams**, **Ehlers-Danlos**, prune-belly or posterior urethral valves.
- Acquired cases are due to benign prostatic hyperplasia in ~**70%**; in women, causes include dysfunctional voiding, prolapse, bladder neck obstruction and post-sling obstruction.
- A **Hutch diverticulum** is paraureteric, in a neuropathic bladder with VUR.

> **KEY POINT**
> Congenital: weak wall, normal pressure. Acquired: normal wall, high pressure, so an acquired diverticulum sends the search back to the outlet.

2 · Bladder diverticula — evaluation

- Symptoms are nonspecific (incomplete emptying, double voiding).
- Do rectal examination, PSA, urinalysis, culture, **cytology**, endoscopy and upper and lower tract imaging, and seek BOO or occult neuropathy.
- Voiding cystourethrogram (**VCUG**) shows size, VUR and emptying. **Paradoxic enlargement** means the sac fills on voiding and refills the bladder afterwards.
- CT or MRI shows number, masses and relation to ureters and rectum.
- Hydroureteronephrosis (HUN) occurs in ~7% of adults, from compression, underlying dysfunction or VUR; the pelvic ureter deviates **medially**.

- Video-urodynamics (**VUDS**) shows a **pressure-sink effect**: contractility looks falsely low, and a raised PVR may be diverticular urine.
- **Cystoscopy** inspects the whole sac with cytology from it. Biopsy with extreme care, as perforation seeds tumour.
- With VUR and a congenital diverticulum, reimplant **selectively**. Treating the cause may make excision unnecessary.

KEY POINT
Two jobs: find the cause (outlet or neurogenic) and exclude cancer in a sac each test sees imperfectly, so tests are combined.

3 · Malignancy in bladder diverticula

- Prevalence is 0.8–10%, driven by stasis and inflammation. Urothelial carcinoma is **70–80%** and squamous cell carcinoma (SCC) 20–25%.
- With **no muscularis propria**, spread is early and extravesical, transurethral resection (TUR) risks perforation, and staging is inaccurate.
- **Radical cystectomy** gives the same stage-for-stage survival. Partial cystectomy or diverticulectomy suits selected low-grade, low-stage tumours.

KEY POINT
Treat a diverticular tumour as potentially invasive from the outset; missing muscle explains early spread, perforation risk and inaccurate staging.

4 · Bladder diverticula — management

Treat the outlet first; operate for complications, not size.

ALGORITHM
1. Assess with history, cystoscopy, cytology and VCUG.
2. If a lesion is suspicious, then biopsy and give cancer therapy.
3. If not, image the upper tract. If HUN or symptoms, then do urodynamics and treat BOO, then reassess.
4. If a problem from the diverticulum persists: fit → diverticulectomy; unfit → clean intermittent catheterisation (CIC).
5. If asymptomatic and uncomplicated, then survey.

- **Diverticulectomy indications**: refractory LUTS, relapsing UTI, **stones**, **carcinoma or premalignancy**, upper tract deterioration (obstruction, VUR). **Size alone is not an indication.**
- **Treat the outlet first or simultaneously**; a small sac may then empty, and concomitant prostatectomy is common.
- If observing, counsel about cancer and survey with symptoms, cytology and cystoscopy. The unfit with poor emptying need CIC or a catheter.
- **Endoscopic** treatment for the unfit resects the neck to muscle, but large sacs do poorly.
- Diverticulectomy is open, laparoscopic or robotic: use an ipsilateral ureteric catheter, dissect in the pseudocapsule, close the neck in layers, and reimplant a ureter entering the sac.

5 · Female urethral diverticulum (UD) — anatomy and aetiology

P1 Female urethral diverticula — Routh theory, posterolateral ostium, three Ds ~5%, stones 4–10%, adenocarcinoma, imaging sensitivities (MRI 97%), L/N/S/C3, diverticulectomy principles, autologous sling only, recurrence 10–22%, de novo SUI 7–16%

Not in the paper answer:

- Configuration is simple in ~73%, saddlebag in ~17% and circumferential in ~10%; ~50% are mid-urethral and 6–19% multiple.
- Carcinoma in UD is **adenocarcinoma** > urothelial > SCC. Local excision alone recurs, so use anterior exenteration or cystourethrectomy with or without radiotherapy.
- **Periurethral glands** lie posterolaterally, with ducts draining into the **distal third**; UD lie between the leaves of the urethropelvic ligament. The striated sphincter is mid-urethral, so plan repair for continence.
- In the gland theory, a gland abscess herniates into the fascia, usually **ventrally**, then ruptures into the lumen; about a third are proximal, however.

6 · UD — presentation and evaluation

- All three Ds occur in only ~**5%**. Features are irritative LUTS, pain, recurrent UTI, mass (~⅓), incontinence (~⅓), haematuria and discharge; 10–20% are asymptomatic. **Size does not track symptoms**, and UD is often misdiagnosed for years.
- In pregnancy, give antibiotics or aspirate, and repair after delivery.
- A mass lies 1–3 cm inside the introitus, and a **hard mass suggests stone or cancer**. Check atrophy, stress test and prolapse.
- **Cystourethroscopy** shows the ostium at **4 and 8 o'clock** mid-urethra in 15–89%; not seeing it must not stop work-up.
- **VUDS** shows SUI in up to 50% and separates true SUI from dribbling from the sac.
- **MRI is best** (T2 high, maps extent) but may miss small cancers. Ultrasound shows an anechoic sac, and VCUG underestimates size.

> **KEY POINT**
> UD symptoms mimic common LUTS and do not track size, so suspicion and examination make the diagnosis. Imaging confirms and **maps** it, especially extension towards the bladder neck (bladder or ureteric injury, postoperative incontinence).

7 · Differential diagnosis of periurethral masses

Lesion	Clue	Treatment
UD	Mid/proximal; communicates	Excision, reconstruction
Leiomyoma	Firm, mobile, solid; oestrogen-dependent	Transvaginal enucleation
Skene cyst/abscess	Lateral to and distorts meatus; non-communicating	Aspiration, marsupialisation, excision
Gartner duct cyst	Anterolateral; may receive ectopic ureter — image upper tract	Observe, sclerotherapy, excision
Vaginal wall cyst	Müllerian commonest	Excise if symptomatic
Urethral prolapse	Red "doughnut"; prepubertal, postmenopausal	Oestrogen; excision
Caruncle	Red, posterior meatal lip; postmenopausal	Oestrogen; excise if atypical

Lesion	Clue	Treatment
Bulking agent	Injection history	Excise if symptomatic

Skene gland adenocarcinoma may raise PSA.

KEY POINT

Two questions sort most periurethral masses: does it communicate (only UD), and where is it relative to the meatus (Skene, prolapse and caruncle lie at the meatus)? A firm, mobile, solid mass is a leiomyoma.

8 · Classification and SUI

System	Basis	Use
L/N/S/C3 (Leach)	Location, Number, Size, Configuration, Communication, Continence	Standard; not validated
Ginsburg–Genadry	Location	Distal: marsupialisation; proximal: excision
Leng–McGuire	Periurethral fascia present?	Absent = **pseudodiverticulum**, needing flap/graft

- Significant de novo stress urinary incontinence (SUI) occurs in **7–16%**; risks are size ≥30 mm, a proximal or circumferential sac and a short urethra. Most settles within 12 months.
- About 50% of preoperative SUI improves after excision, so operate only for **bothersome** SUI, using an **autologous pubovaginal sling, never synthetic**.

9 · Surgical repair of UD

- Operate for symptoms, stones or suspected cancer; treat the asymptomatic or unfit with suppressive antibiotics and postvoid stripping.
- **Spence-Duckett marsupialisation** suits only distal-third UD in women who are not sexually active.
- Principles (Box 122.2): vascularised flap, separate fascia, excise ostium and entire sac, watertight multilayer non-overlapping closure, close dead space, and **preserve or create continence**.

ALGORITHM

1. Lithotomy and Foley; suprapubic catheter (**SPC) at the start**, never the end (it would cross the repair).
2. Make an **inverted-U incision** and raise the vaginal flap off the fascia.
3. Incise the fascia **transversely** and raise fascial flaps.
4. Excise the entire sac to the ostium, including the inflamed urethral edge.
5. Close the urethra watertight over a catheter (as small as 12 Fr).
6. Close fascia **perpendicular** to it, obliterating dead space.
7. If tissue is poor or fascia attenuated, then add a **Martius** flap.
8. Advance the vaginal flap (3 layers; 4 with Martius); antibiotic pack.

- A pseudodiverticulum needs a Martius flap or fascial graft. Do a **pericatheter VCUG at 14–21 days**, repeated weekly if leaking.

10 · Complications

Complication	Incidence	Notes
Recurrent UD	**10–22%** (pooled ~10%)	Incomplete excision, dead space; circumferential, complex, proximal, multiple
Incontinence	1.7–16%	Conservative 12 months, then autologous sling
Urethrovaginal fistula	0.9–8.3%	Distal: may need no repair; proximal: repair ± Martius
Stricture	0–5%	—
Recurrent UTI	up to 31%	Investigate for recurrence
Others	—	Distal urethral necrosis, bladder/ureteric injury, dyspareunia

Redo excision gives less relief and more fistula, so image persistent symptoms for recurrence or stricture.

> **KEY POINT**
> Complete sac removal, watertight closure, off-axis fascia and obliterated dead space prevent the three failures: recurrence, fistula, incontinence.

Feature	Bladder diverticulum	Female UD
Origin	Herniation through detrusor	Infected periurethral gland ruptured into lumen
Cause to treat	BOO / neurogenic	Infection; coexisting SUI
Commonest cancer	Urothelial (then SCC)	Adenocarcinoma
Key study	Cystoscopy + cytology; VCUG	MRI
Operate for	Complications, not size	Symptoms or complications
Operation	Diverticulectomy + outlet treatment	Transvaginal excision, layered reconstruction

77 Sexual Function and Dysfunction in the Female

P3 Female sexual dysfunction — definition, prevalence, response models, DSM-5 vs ICSM/ISSWSH, causes, PLISSIT/FSFI, treatment overview by domain

Not in the paper answer:

- Flibanserin: about half respond by 8 weeks; REMS. Bremelanotide: ≤8 doses/month; hyperpigmentation. Testosterone: stop at 6 months if no benefit.
- Genital-sparing cystectomy allows sex at about 6 vs 14 weeks; combined anterior and posterior repair increases dyspareunia; TOT spares the dorsal clitoral nerve.

1 · Functional anatomy

The **distal vagina, clitoris and urethra** share blood supply and nerves and respond as one unit, the **clitoral complex**, so urethral or vaginal surgery can change sexual function.

- The **clitoris** is the penile homologue, **separate from the urethra**; its two corpora continue as crura on the ischiopubic rami.
- The glans is very sensitive but **not erectile**. The dorsal nerves lie **superficially on the dorsum**, outside the tunica, so hood or clitoral surgery can injure them.
- **Erectile** tissue is the clitoral body, crura and vestibular bulbs; labia minora, glans, periurethral tissue and vaginal walls are vascular but do not engorge.
- **Skene** glands are the prostate homologue (PSA positive) and **Bartholin** glands the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis.
- Nerve supply of the labium majus and the glans draining **directly to deep inguinal nodes** are minor exam points.

2 · Physiology of the sexual response

ALGORITHM

1. Touch on the clitoral complex travels in pudendal (somatic) afferents to **S2–S4**.
2. Parasympathetic efferents (nervi erigentes, then cavernous nerves) dilate arterioles, causing engorgement and lubrication.

The reflex has **somatic afferents and visceral efferents**. **Orgasm** is cerebral, with **sympathetic** output (T12–L1) contracting the uterus and pudendal reflexes contracting pelvic striated muscle.

KEY POINT

Pudendal injury removes the **trigger**; pelvic plexus or cavernous nerve injury removes the **response**.

ALGORITHM

1. Neuronal nitric oxide synthase (nNOS) makes nitric oxide (NO), which activates guanylate cyclase to make cyclic GMP (cGMP).
2. cGMP lowers Ca^{2+}, so smooth muscle relaxes and tissue engorges.
3. Phosphodiesterase type 5 (PDE5) degrades cGMP; vasoactive intestinal polypeptide (VIP) works with NO.

Desire balances **excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) against **inhibition** (serotonin, opioids, prolactin, progesterone); licensed HSDD

drugs act on this balance.

	Oestrogen	Testosterone
Role	Maintains the **substrate**: epithelium, collagen, lubrication, acidic pH, NOS/VIP	Drives **desire**; inactive when bound to sex hormone–binding globulin (SHBG)
Deficiency	Thin, dry, fibrotic vagina causing dyspareunia; risk rises at oestradiol **≤50 pg/mL**	Lower libido, arousal, sensation and orgasm
Treatment risk	—	Virilisation: early reversible (acne, hirsutism); late irreversible (voice, alopecia, clitoromegaly)

Model	Stages	Key idea
Masters & Johnson (1966), linear	Excitement, plateau, orgasm, resolution	**No desire phase**
Kaplan (1977), linear	Desire, arousal, orgasm	Adds desire
Whipple & Brash-McGreer, circular	Seduction, sensations, surrender, reflection	Reflection feeds next seduction
Basson (2002), circular	Neutrality, receptivity, arousal, **responsive desire**, satisfaction	Spontaneous desire **not needed**; intimacy is the motive

KEY POINT

Under Basson, lack of spontaneous desire (common after menopause) is not a dysfunction, which is why DSM-5 merged desire and arousal.

3 · Classification

DSM-5 requires **≥6 months** and **clinically significant distress**; low desire without distress is not a disorder.

Domain	DSM-IV-TR	DSM-5
Desire and arousal	HSDD; female sexual arousal disorder (FSAD); sexual aversion disorder	**Female sexual interest/arousal disorder (FSIAD)**, **≥3 of 6**: less interest; fewer fantasies; less initiation/receptivity; less pleasure, response to cues or sensation (in 75–100% of encounters). Aversion dropped
Orgasm	Female orgasmic disorder (FOD) after normal excitement	FOD: delay, infrequency, absence **or lower intensity** in **75–100%**; normal-excitement requirement removed
Pain	Dyspareunia; vaginismus	**Genito-pelvic pain/penetration disorder (GPPPD)**, ≥1 of: penetration difficulty; pain; fear of pain; pelvic floor muscle (PFM) tensing

- **ICSM 2016** keeps HSDD and FSAD **separate** and adds **persistent genital arousal disorder (PGAD)** and orgasmic illness syndrome; **ISSWSH** adds a urological **genital arousal disorder**.
- About **40%** have a lifetime sexual problem but only about **8–12%** a distressing disorder. **HSDD is commonest**, FOD second.

4 · Causes (biopsychosocial)

- **Biological**: diabetes, hypertension, coronary disease, thyroid disease, hyperprolactinaemia, renal failure, spinal cord injury (SCI), multiple sclerosis (MS), vulval dermatoses, incontinence, pregnancy, breastfeeding and menopause.
- **Psychological**: depression (bidirectional, so screen every HSDD), anxiety, body image, abuse, conflict and the **partner's erectile dysfunction or premature ejaculation**.
- **Drugs**: SSRIs, SNRIs, antipsychotics, antihypertensives, antiandrogens, GnRH agonists, opiates and alcohol. MCQ Combined hormonal contraceptives and oral oestrogens raise SHBG.
- **Genitourinary syndrome of menopause (GSM)** causes dryness, burning, dyspareunia (often **vestibular**), urgency, dysuria and recurrent UTI. Dryness causes pain, which lowers desire, so **treat the tissue first**.
- **Cancer treatment** harms by castration, anti-oestrogens (aromatase inhibitors cause severe GSM), pelvic radiotherapy (stenosis) and nerve injury.

5 · Evaluation

Women rarely volunteer sexual problems, so the **clinician must ask**, gender-neutrally.

> ALGORITHM
>
> 1. **P**ermission.
> 2. **L**imited **I**nformation: cycle, ageing, drugs.
> 3. **S**pecific **S**uggestions: lubricants, **vaginal oestrogen**, positions, vibrators.
> 4. **I**ntensive **T**herapy means **referral** (PLISSIT).

- **DSDS** (Decreased Sexual Desire Screener) is a **validated** 5-item screen; yes to items 1–4 with no contributing cause means **generalised acquired HSDD**. The ACOG BSSC-W is not validated.
- **FSFI** (19 items, 6 domains) is most used; FSDS-R measures **distress**; **PISQ-12/IR** is the **only validated** tool for prolapse and incontinence.
- History: lifelong or acquired, generalised or situational, the **order problems appeared**, drugs, surgery, LUTS, prolapse and the **partner's** function.
- Examination is essential with pain. **pH ≥4.5** without infection or semen means atrophy. Do a **Q-tip test** of the vestibule and **retract the hood** for adhesions.
- Lichen sclerosus leads to introital stenosis; lichen planus (**Wickham striae**) to vaginal stenosis.
- **Overactive** PFM causes dyspareunia and **underactive** PFM causes incontinence and prolapse; fornix tenderness suggests endometriosis.
- Labs are rarely needed. Do **not** measure a **random testosterone**: there is no female range and no level predicts HSDD.

6 · Hypoactive sexual desire disorder

- ISSWSH defines HSDD as ≥6 months of lost desire **with distress**, not due to drug, illness, pain, abuse or partner. **Exclude relationship conflict first.**
- HSDD peaks in **midlife** and is defined by distress, not hormone level.

ALGORITHM

1. Screen (DSDS), then history and examination.
2. **Correct modifiable factors**: drugs, depression, GSM, pain, partner's ED.
3. Education; sex or couples therapy.
4. If **generalised acquired** HSDD persists, then drug; review at 8 weeks.
5. **Refer** if situational HSDD fails counselling, or psychiatric illness or sexual trauma.

	Flibanserin	Bremelanotide
Mechanism	Central **5-HT1A agonist / 5-HT2A antagonist**: lowers serotonin, raises dopamine/noradrenaline (removes the brake)	Central **melanocortin agonist** releasing dopamine (presses the accelerator)
Licence	FDA: only **premenopausal** acquired generalised HSDD	
Dose	**100 mg at bedtime** daily; about half respond by 8 weeks	**1.75 mg subcutaneous** ≥45 min before sex; ≤1/24 h, **≤8/month**
Side effects	Dizziness, somnolence, nausea	Nausea, flushing, headache, **hyperpigmentation**
Cautions	**Alcohol** (hypotension, syncope); strong CYP3A4 inhibitors; REMS restricted prescribing	Uncontrolled hypertension or cardiovascular disease

- **Transdermal testosterone** (off-label) improves desire, arousal, orgasm and satisfaction in **postmenopausal HSDD**, with no cardiovascular signal but unknown long-term safety.
- The 2019 Global Consensus targets **premenopausal physiological levels**, checks levels at baseline and 3–6 weeks, and says **stop at 6 months** if there is no benefit. Bupropion (off-label) is an alternative.

7 · Arousal disorder and PGAD

- **Subjective and genital arousal correlate poorly**, so vasodilators raise flow without felt arousal. Use mindfulness-based sex-focused CBT.
- **Transdermal testosterone** has the best evidence. PDE5 inhibitors help only with neurovascular impairment (type 1 diabetes, SCI, MS, SSRI-induced). **Bupropion 150 mg twice daily** treats SSRI-induced dysfunction.
- **PGAD** is distressing genital arousal for hours to days, **not relieved by orgasm** and without desire. Causes include Tarlov cysts, pelvic varices and **stopping an SSRI**; treat with CBT, duloxetine or pregabalin.

8 · Female orgasmic disorder

- FOD is a **diagnosis of exclusion requiring distress**: exclude drugs, disease, stress and **inadequate stimulation** (including the partner's ED/PE).
- **No orgasm at coitus without clitoral stimulation is normal variation.**
- **Education is the cornerstone**, with directed masturbation, sensate focus and mindfulness. If arousal is intact, increase stimulation (vibrators; MCQ EROS clitoral vacuum is the only FDA-cleared FSD device).

ALGORITHM

1. **SSRI-induced dysfunction**: was it present before treatment?
2. If not, wait for tolerance (~4 months).
3. If it persists at 3–6 months, then lower the dose, or switch to or add **bupropion** or mirtazapine; consider sildenafil.

9 · Genito-pelvic pain / penetration disorder

- **Dyspareunia** is superficial or deep (endometriosis, prolapse, pelvic floor). **Vaginismus** is involuntary contraction preventing penetration.
- **Vulvodynia** is vulval burning pain for **≥3 months** without a cause; it is localised or generalised, provoked or unprovoked.
- **Provoked vestibulodynia (PVD)** is commonest (about **10–12%** of women). Primary PVD starts at first penetration; secondary PVD (about two-thirds) follows pain-free sex.
- The mechanism is vestibular nerve proliferation with a **hypertonic pelvic floor** and fear-avoidance.
- **Clitorodynia**: hood adhesions trap keratin and form painful pseudocysts, so **examine the glans routinely**.

ALGORITHM

1. Education and vulval care.
2. Lidocaine 5% or EMLA **10 min before intercourse**.
3. **Multimodal pelvic floor physical therapy** (myofascial release, down-training, biofeedback, graded **dilators**) with psychological/couples therapy.
4. If hypo-oestrogenic, then vestibular oestrogen.
5. If localised PVD is refractory, then **vestibulectomy**.

- **Multimodal physical therapy is standard of care** for PVD; EMG biofeedback is best if the pelvic floor is overactive.
- Oestrogen deficiency **increases** vulval nerve density, so **topical beats systemic and vestibular beats vaginal**.
- Second-line options are amitriptyline, gabapentin and intralevator botulinum toxin. **Vestibulectomy** beats non-surgical therapy in localised neuroproliferative disease.

Treatment of GSM

ALGORITHM

1. Non-hormonal lubricants/moisturisers and regular activity (NAMS first line).
2. Then **low-dose vaginal oestrogen** (minimal absorption, no progestogen) or vaginal DHEA (prasterone 6.5 mg nightly).
3. If vaginal products are impossible, then ospemifene 60 mg orally (SERM, vaginal agonist; VTE risk).
4. If vasomotor symptoms, then systemic hormone therapy.

KEY POINT

After **breast cancer**, use non-hormonal measures first; vaginal oestrogen only after oncology discussion (aromatase inhibitors).

10 · Pelvic surgery and sexual function

Surgery harms by three routes: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (distal urethra) and **vaginal shape**. Counsel **before and after**.

- **Radical cystectomy**: loss of desire is commonest, and about half can have intercourse. Preserve function by saving the bundles on the **lateral vaginal walls**, dividing the urethra at the **bladder neck**, and keeping vaginal dimensions.
- **Genital-sparing** cystectomy (if oncologically safe) lets sex resume at **~6 vs 14 weeks**; diversion type is irrelevant. **Never trade oncological control.** See Ch 101
- **Hysterectomy** for benign disease leaves function unchanged or better; after **rectal surgery** dysfunction is common, and nearly universal after abdominoperineal resection.
- **Prolapse repair**: bother and body image, not stage, predict function. Dyspareunia is commoner after vaginal than abdominal repair, and after anterior mesh than **sacrocolpopexy**.
- **Combined anterior and posterior repair increases dyspareunia**, so stage it in sexually active women. **Keep length, avoid excess skin excision, preserve the introitus.** See Ch 119
- **SUI surgery**: about a third improve and most are unchanged; **curing coital incontinence** predicts improvement. Retropubic tape (**TVT**) may denervate the vaginal wall, while transobturator tape (**TOT**) spares the dorsal clitoral nerve. Dyspareunia is a strong reason for sling removal.
- **Coital incontinence** (about 1 in 10 incontinent women): leakage at penetration is stress; at orgasm, detrusor overactivity.
- **Bladder pain syndrome causes the worst FSD**, through pain, PFM overactivity and vulvodynia. **Sacral neuromodulation** improves sexual function whatever the cause.
- **Genital cosmetic surgery** has poor evidence; screen for **body dysmorphic disorder**.

11 · Special populations

Ask about orientation and gender identity; offer fertility preservation before transition. Oestrogen risks thromboembolism; testosterone risks erythrocytosis (haematocrit **>50%**), so check haematocrit and testosterone 3-monthly in year 1, then 1–2 times a year. SCI impairs sensation, lubrication and orgasm. See Ch 126

12 · Landmark trials

Trial	Design	Result	Take-home
Flibanserin phase III	RCTs vs placebo, premenopausal	More satisfying events and desire, less distress	First FDA-approved drug for HSDD
RECONNECT (bremelanotide)	2 phase III RCTs	More desire, less distress	On-demand option for premenopausal HSDD
Bupropion for SSRI-induced dysfunction (Cochrane)	Meta-analysis	Benefit at 150 mg twice daily	Add or switch to bupropion

★ Putting it together: female pelvic decisions

A · Incontinent woman

ALGORITHM

1. **Classify** (history, diary): stress (SUI), urgency (UUI), mixed, overflow, functional. Check stress test, prolapse (POP), residual, urinalysis (Ch 105).
2. **Urodynamics** are not needed for uncomplicated demonstrable SUI (VALUE). Use them if mixed, voiding dysfunction, prior surgery, neurological or high-grade POP.
3. **Conservative first:** weight, fluids, supervised pelvic floor and bladder training, pessary, vaginal oestrogen (Ch 114).
4. If this fails, then **SUI surgery** (Ch 116, 118); for UUI, drugs, botulinum toxin, neuromodulation.

B · Choosing an SUI operation

Operation	Best for	Drawbacks
Retropubic mid-urethral sling (MUS)	Standard; intrinsic sphincter deficiency	Perforation (cystoscopy mandatory), voiding dysfunction
Transobturator MUS	Avoiding the retropubic space	Groin/thigh pain; weaker in severe SUI
Autologous pubovaginal sling	Recurrent SUI, urethral repair, mesh declined	More voiding dysfunction
Burch	Mesh declined; concurrent abdominal surgery	Posterior prolapse, de novo urgency
Bulking agent	Frail, minimal surgery	Lower, short-lived cure

C · Prolapse: which repair?

ALGORITHM

1. **POP-Q** stage II is a leading edge within 1 cm of the hymen (Ch 105).
2. Observe if asymptomatic; offer pessary or surgery if symptomatic.
3. **Support the apex**: uterosacral/sacrospinous suspension or sacrocolpopexy (most durable) (Ch 117).
4. If **frail, not sexually active**, then colpocleisis.
5. If **occult SUI**, then discuss a concomitant sling (OPUS: fewer leaks, more complications).

D · Vesicovaginal fistula

KEY POINT

The first repair has the best chance. Do a double-dye test, upper tract imaging and, if irradiated, biopsy. Post-hysterectomy fistulae are supratrigonal and mostly repaired vaginally; go abdominal for ureteric involvement, large or high fistulae, or failed vaginal repair. Use tension-free, non-overlapping layers, a vascularised flap (Martius or omentum) for poor tissue, then a cystogram (Ch 121).

E · Mesh complications

ALGORITHM

1. If suspected, examine and cystoscope.
2. If exposure is small, then oestrogen with or without trimming; otherwise excise and close.
3. If mesh is in bladder or urethra, then remove and repair without new mesh.
4. Remove only enough to solve the problem, never asymptomatic mesh (Ch 119).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 7

Voiding Dysfunction

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 102	Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis
Ch 103	Physiology and Pharmacology of the Bladder and Urethra
Ch 104	Pathophysiology and Classification of Lower Urinary Tract Dysfunction
Ch 106	Evaluation and Management of Males With Urinary Incontinence
Ch 107	Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract
Ch 109	Neuromuscular Dysfunction of the Lower Urinary Tract
Ch 110	Overactive Bladder
Ch 111	Underactive Detrusor
Ch 113	Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure
Ch 115	Electrical Stimulation and Neuromodulation in Storage and Emptying Failure
★	Putting it together: voiding dysfunction decisions

102 Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

1 · Bony pelvis

- The sacrum and two innominate bones (**ilium, ischium, pubis**, fused at the acetabulum) meet at the sacroiliac (SI) joints behind and the symphysis in front.
- The **sacrospinous** ligament divides the sciatic foramen into **greater and lesser**.
- The pelvic brim is **50–60°** to horizontal. The **arcuate line** separates false from true pelvis. The iliac crest is at **L4**.

2 · Lower abdominal wall

Fascia and extravasation

- The umbilicus is the **T10** dermatome, at L3. **Scarpa's** fascia fuses with fascia lata about 1 cm below the inguinal ligament and continues as **Colles fascia**, then dartos.
- Colles is fused to the ischiopubic rami and perineal membrane, so **anterior urethral injury** fills penis and scrotum with blood and urine as a **"butterfly" perineal haematoma**. It spares the thigh but can spread **upward** deep to Scarpa's to the flank.
- †With Buck's fascia intact the haematoma stays on the shaft; a butterfly means Buck's is breached.

Rectus sheath

	Above arcuate line	Below arcuate line
Anterior	External oblique and part of internal oblique	All three aponeuroses
Posterior	Rest of internal oblique and transversus	**Absent**: transversalis fascia and peritoneum only

- Tendinous intersections tether rectus to the anterior sheath, so it can be **divided transversely without retracting**.
- Segmental nerves enter **laterally**, so a paramedian incision lateral to rectus causes atrophy and ventral hernia.

Inguinal canal

- It is **4 cm** long and carries the **spermatic cord and ilioinguinal nerve**. The deep ring is midway between the anterior superior iliac spine and pubic tubercle, **lateral to the inferior epigastric vessels**.
- The internal oblique and transversus roof closes onto the floor as a **shutter mechanism**.
- Direct hernias are **medial** and indirect hernias **lateral** to the inferior epigastrics. At **radical orchidectomy** the cord is ligated at the internal ring, so the stump retracts retroperitoneally.

Peritoneal folds (laparoscopic map)

Fold (medial to lateral)	Contains	Significance
Median	Urachus	Bladder apex to umbilicus
Medial (paired)	Obliterated umbilical arteries	Incise **lateral** to them to reach obturator fossa and retropubic space

Fold (medial to lateral)	Contains	Significance
Lateral (paired)	Inferior epigastric vessels	Mark deep ring and hernia type

3 · Pelvic floor and fascia

- Obturator internus fascia thickens into the **tendinous arch of levator ani**, the origin of the pelvic diaphragm (levator ani and coccygeus).
- Levator ani comprises **puborectalis** (a sling behind the anorectal junction for **faecal continence**), pubococcygeus and iliococcygeus. It is tonically contracted, relaxes to void, and **Kegel exercises** increase its tone.
- **Parietal fascia** lines the walls and **visceral fascia** sheaths the organs; their lateral attachments, the **endopelvic fascia**, carry pelvic vessels and are the gateway to the prostate.
- The **arcus tendineus fascia pelvis** marks the parietal–visceral junction and anchors urethral and bladder supports.

4 · Perineum

- The **perineal membrane** spans the pubic arch, is pierced by the urethra and supports the sphincter.
- The **perineal body** is the midline keystone, blending with Denonvilliers; injury at perineal prostatectomy causes urinary and faecal incontinence.
- Perineal prostatectomy MCQ : **Young** divides the central tendon, leaving the external sphincter on the rectum; **Belt** passes **under** it.
- The **internal pudendal artery** runs in **Alcock's canal** and becomes the dorsal penile artery.
- The **pudendal nerve (S2–S4)** supplies both external sphincters, levator ani, ischio- and bulbospongiosus, ending as the dorsal penile nerve.

5 · Pelvic vessels and lymphatics

- The **external iliac** gives the inferior epigastric and deep circumflex iliac. **Accessory obturator vessels** from the inferior epigastric cross the obturator fossa and are a PLND hazard.
- **Internal iliac**: the posterior trunk is parietal (superior gluteal, ascending lumbar, lateral sacral); the anterior trunk is visceral (umbilical, obturator, middle rectal, inferior vesical, internal pudendal, inferior gluteal).
- The patent umbilical gives the **superior vesical** and artery of the vas. The **inferior vesical** (not internal pudendal) supplies the neurovascular bundle (NVB).
- The iliac veins join **behind the internal iliac artery**, a site of major PLND bleeding.
- The **dorsal venous complex (DVC)** enters under the symphysis in front of the striated sphincter; ligating it at prostatectomy can damage the sphincter.
- Lymph often **crosses sides**. External iliac nodes drain bladder and urachus; internal iliac nodes drain prostate and bladder. The **node of Cloquet** is the **distal limit of PLND**.

6 · Pelvic nerves

Nerve	Roots	Key point
Ilioinguinal	L1	Through inguinal canal to anterior scrotum

Nerve	Roots	Key point
Genitofemoral	L1–L2	On psoas; **lateral limit of PLND**; genital branch supplies cremaster
Femoral	L2–L4	Injured by psoas hitch, retractor blades, inguinal dissection
Obturator	L2–L4	Iliac bifurcation to obturator canal; PLND injury gives **weak adduction**
Pudendal	S2–S4	Perineal sensation; levator, urinary and anal sphincters
Nervi erigentes	S2–S4	Parasympathetic to bladder, rectum

KEY POINT

Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — pelvic plexus (T10–L2 + S2–S4) beside rectum at seminal vesicles gives cavernous nerves, NVB outside Denonvilliers at 5 and 7 o'clock

Not in the paper answer:

- The NVB lies at **5 and 7 o'clock** at the apex; injury at prostatectomy or rectal resection causes erectile dysfunction.

7 · Pelvic viscera

Rectum

- The rectum begins at **S3**. Opening the **rectovesical pouch** exposes the seminal vesicles in the posterior robotic approach. **Denonvilliers fascia** separates rectum from prostate.
- The distal rectum near the apex is the **commonest rectal injury site** at prostatectomy.

Bladder

- The bladder is extraperitoneal except the dome, which leads to the **urachus** (adenocarcinoma, fistula, cyst).
- Circular bladder-neck fibres form the **internal sphincter**; these **α1-adrenergic** fibres maintain continence and **antegrade ejaculation**, and their damage adds to post-prostatectomy incontinence.
- Lymph goes mainly to the **external iliac** nodes, and also to internal iliac and obturator nodes, bilaterally.

Ureterovesical junction

- The intramural ureter runs **1.5–2 cm** obliquely in **Waldeyer's sheath**; being narrow, it traps stones.
- It is a **flap valve** in which detrusor backing compresses the ureter. Weak backing causes **vesicoureteral reflux (VUR)**, and a weak hiatus allows a **Hutch diverticulum**.
- The pelvic ureter narrows at the iliac crossing and passes **behind the vas**. Its blood supply enters **laterally**, so stripping the lateral attachments causes stricture.

8 · Endoscopic anatomy

ALGORITHM

1. Under local anaesthetic, inspect bladder, orifices and trigone with a flexible cystoscope.
2. Retroflex at the neck: intravesical prostatic protrusion, hidden tumours.
3. Find the verumontanum and ejaculatory ducts; the external sphincter coapts just distal, then the bulbar urethra.

KEY POINT

The verumontanum is the sentinel at transurethral resection of the prostate (TURP): resection distal to it injures the external sphincter and causes incontinence.

9 · Radiographic anatomy

- A **phlebolith** mimics a distal ureteric or bladder stone.
- On a **cystogram**, absent lateral distension suggests an extraperitoneal pelvic haematoma or post-prostatectomy lymphocele.
- Voiding cystourethrography (**VCUG**) best shows the posterior urethra, as the sphincters relax. On retrograde urethrography (**RUG**), steady pressure overcomes the external sphincter.
- The membranous urethra is the **shortest, narrowest** segment and is not a stricture.
- Computed tomography (**CT**) needs contrast except for **stones**. In **extraperitoneal rupture**, urine lies in the retropubic and perivesical spaces.
- Malignant nodes are rounded and heterogeneous; size is most specific at ≥ **1 cm** short axis.
- **CT urogram** excretory images are taken at **7–10 min**.
- Magnetic resonance imaging (**MRI**) has the best soft-tissue contrast. A distended bladder wall over **3 mm** is thickened.

103 Physiology and Pharmacology of the Bladder and Urethra

1 · Overview

- The lower urinary tract (LUT) has two functions, **storage** (continence) and **emptying** (micturition), integrating the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethra, pelvic floor and external urethral sphincter (EUS).

> **KEY POINT**
> **Why single-organ therapy fails:** bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent modulation acts on both (reduce it for overactivity, augment it for underactivity), making it the most promising therapeutic lever.

2 · Functional anatomy

Bladder

- The **male bladder neck** is a complete smooth muscle ring with dense sympathetic supply; it closes in ejaculation, ensuring **antegrade ejaculation**. Women lack this collar.
- The **lamina propria** is the "functional centre", containing the afferent plexus, interstitial myofibroblasts and capillaries.

Stroma

- The wall is about **50% collagen, 2% elastin**, mainly types I, III and IV.
- **Poorly compliant bladders** have more connective tissue relative to muscle and a higher **type III : type I collagen** ratio, so capacity falls.

Urethra and sphincters

- The urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from the periurethral pelvic floor.
- The male EUS is a crescent in front of the prostate, becoming a horseshoe distally. The female EUS is a horseshoe over the ventral urethra, and a subepithelial vascular plexus forms a **"urethral seal"**.

Feature	Slow-twitch	Fast-twitch
Role	Sustained resting tone	Rapid extra squeeze on coughing
Fatigue / metabolism	Slow; oxidative	Rapid; anaerobic
Where	Intramural rhabdosphincter, giving **~50% of static urethral resistance**	Periurethral pelvic floor (mixed)
Rhabdosphincter mix	Male **65% slow / 35% fast**; female **87% slow / 13% fast**	

3 · Urothelium: barrier and sensor

The urothelium senses stretch; its damage drives urgency and pain.

Barrier

- **Umbrella cells** carry **uroplakin** plaques, which form the barrier and are the binding site of **type 1-fimbriated E. coli**. **Tight junctions** block paracellular flow.
- Loss of umbrella cells (infective or radiation cystitis) lets solutes leak in, so symptoms worsen with distension.

Sensor–transducer function

Urothelial cells behave like sensory neurons, carrying purinergic, adrenergic, muscarinic and TRP receptors. On stretch they release mediators onto afferents, directly or via myofibroblasts.

Mediator	Released by / when	Effect
ATP	Stretch; more with injury, inflammation, ageing	Excites suburothelial **P2X3** afferents, giving fullness and pain; P2X3-null mice are hyporeflexic
Nitric oxide (NO)	Especially inflammation	Blocks afferent Ca^{2+} channels: **afferent inhibition**
Acetylcholine (ACh), non-neuronal	Basal; more with stretch, ageing	Modulates afferents and DSM, so antimuscarinics may act partly here
Prostaglandins (PGs)	Urothelium is the main source	Detrusor regulation, cytoprotection
Inhibitory factor (unknown)	Basal	Reduces muscarinic contraction; not NO or PG

- The afferent threshold is set by the **ATP (excitatory) : NO (inhibitory) balance**. ATP is raised in detrusor overactivity (DO) and in bladder pain syndrome/interstitial cystitis (BPS/IC).
- **Suburothelial myofibroblasts** are linked by connexin 43 (**Cx43**) gap junctions and respond to ATP and ACh; their M2/M3 expression correlates with urgency in idiopathic DO.

> **KEY POINT**
> Damaged urothelium causes symptoms twice over: by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents.

4 · Detrusor smooth muscle physiology

Each contraction step is a drug target.

- Not every detrusor cell is innervated, so **gap junctions** spread excitation. Ca^{2+}–calmodulin drives **myosin phosphorylation**, and activity is mostly tonic.

ALGORITHM

1. ACh binds **M3** receptors (Gq).
2. Intracellular Ca^{2+} rises via L-type entry and phospholipase C (PLC), whose IP3 releases sarcoplasmic reticulum (SR) Ca^{2+}.
3. Ca^{2+}–calmodulin activates myosin light chain kinase (MLCK), which phosphorylates MLC20.
4. Cross-bridges cycle and the muscle contracts.
5. As Ca^{2+} falls, myosin light chain phosphatase (MLCP) dephosphorylates MLC20 and the muscle relaxes.

- Tone depends on Ca^{2+} and the **MLCK : MLCP balance**. In **calcium sensitisation**, RhoA-activated **Rho kinase** and protein kinase C (PKC) inhibit MLCP, so force rises without more Ca^{2+}.
- **L-type voltage-dependent Ca^{2+} channels (VDCC)** are the main trigger for contraction (blocked by nifedipine, diltiazem, verapamil).
- K^+ channels repolarise the cell, and **BK is the key brake**: Ca^{2+} sparks open BK, hyperpolarising the cell. BK knockout gives overactive bladder (OAB), and BK is reduced in neurogenic DO (NDO).

G protein	Receptors	Effect
Gq	M1, M3, M5 ("odd"); α1; AT II; H1; 5-HT2	IP3 and DAG release Ca^{2+} and activate PKC: contraction
Gi/o	M2, M4 ("even")	Lower cAMP
Gs	β3	Raise cAMP, activating PKA: relaxation

- **Purinergic (P2X1) contraction** is minimal normally but appears in obstructed, OAB and ageing bladders. It suppresses muscarinic coupling, explaining detrusor hyperactivity with impaired contractility (**DHIC**): urgency with incomplete emptying.
- The detrusor is deliberately **poorly coupled**, protecting filling. More Cx43 in bladder outlet obstruction (BOO) and NDO, induced by cytokines and nerve growth factor (NGF), allows whole-bladder myogenic contraction, which is DO.

KEY POINT

Motor–sensory coupling: spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and so reduce sensory input; efferent and afferent functions are inseparable.

5 · Receptors and drug targets

Receptor	Location	Function	Drug target
M3 (Gq)	Detrusor; urothelium; salivary glands	**Mediates contraction**	Antimuscarinics
M2 (Gi)	Detrusor (**numerically predominant**)	Amplifies M3 by opposing β-cAMP relaxation; the balance shifts from M3 to M2 in neurogenic and obstructed bladders	Non-selective antimuscarinics

Receptor	Location	Function	Drug target
β3 (Gs)	Detrusor (**most expressed adrenoceptor**); afferents	Relaxes during storage; increases compliance	Mirabegron, †vibegron
α1A	Urethra, prostate	Urethral closure; female resting tone	α-blockers (benign prostatic hyperplasia, BPH); α-agonists increase closure
α1D	Bladder (predominant α); spinal cord	Storage symptoms of obstruction	Tamsulosin improves storage lower urinary tract symptoms (LUTS)
α2	Prejunctional urethra; ganglia	Reduces noradrenaline release	—
P2X3	Suburothelial afferents	Senses ATP; increased in BPS/IC	AF-219 (proof-of-concept)
P2X1	Detrusor	Purinergic contraction; increased in BOO, ageing	Potential for DHIC
TRPV1	C fibres; urothelium	Senses heat >43 °C, protons, capsaicin; increased in NDO	Capsaicin and resiniferatoxin (RTX) desensitise
TRPM8	Afferents	Senses cold (bladder cooling reflex)	Experimental
NK1	Spinal cord	Afferent transmission	Aprepitant reduced urgency (trial)
Nicotinic	EUS	Sphincter contraction	—

P3 OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — tissue > subtype selectivity; darifenacin ~30-fold M3; β3 raises cAMP; dementia and antimuscarinic burden

Not in the paper answer:

- **Oxybutynin**: its metabolite **desethyloxybutynin** causes most of the dry mouth, so the controlled-release form is better tolerated.
- CNS-penetrant agents block M1, causing delirium and cognitive decline that is reversible within weeks of stopping.
- Outlet drugs are **α-agonists** (ephedrine, midodrine); systemic oestrogen may worsen urinary incontinence (UI).

P3 Duloxetin — SNRI raises Onuf's nucleus output and so sphincter tone; mild–moderate SUI; Europe, not USA

6 · Peripheral innervation

P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children — pelvic (S2–S4) voiding, hypogastric (T10–L2) storage, pudendal/Onuf's EUS; Aδ vs silent C fibres; storage spinal vs spinobulbospinal voiding reflex; PAG–PMC; central transmitters

Not in the paper answer:

- Parasympathetic outflow is **S2–S4**; sympathetic is †**T10–L2**.
- Some parasympathetic postganglionic neurons are **intramural**, so cauda equina or pelvic plexus injury **decentralises rather than denervates** the bladder.
- C fibres are mucosal, mostly **silent** and capsaicin/RTX-sensitive, and they drive a new **spinal reflex** after spinal cord injury (SCI), inflammation, BOO or ageing.

- Afferent sensitisation by ATP, NGF and inflammatory mediators leads to central sensitisation, and **cross-organ sensitisation** explains BPS/IC overlapping irritable bowel syndrome. μ-opioids raise sphincter tone, causing postoperative retention.

7 · Central control

Lesion level predicts the bladder pattern.

- The pontine micturition centre (**PMC**) is **Barrington's nucleus** ("M region"); lesions (stroke, multiple sclerosis, tumour) cause **retention**. It does **not** project to Onuf's nucleus but relaxes the sphincter **indirectly** via GABA/glycine interneurons.
- The **L-region** (ventrolateral pons) is the storage centre and projects to Onuf's nucleus.
- The periaqueductal grey (**PAG**) relays afferents to the PMC, forming a binary "on–off" switch; without higher control the bladder empties at a critical volume, as in infants.
- In the **forebrain**, the **insula** registers desire and the **prefrontal cortex** decides, keeping the reflex under **tonic suppression**. Frontal white-matter lesions cause **incontinence**.

KEY POINT

Lesion localisation: a forebrain lesion removes tonic inhibition, giving DO with preserved coordination. A pontine lesion gives retention. A suprasacral cord lesion gives DO with detrusor–sphincter dyssynergia (**DSD**). A sacral or cauda lesion gives a decentralised, acontractile bladder (see Ch 104).

8 · Storage and emptying phases

- Storage pressure stays low through intrinsic wall properties, quiet parasympathetic outflow and a **sympathetic storage reflex**, which is weak in humans.
- **Guarding reflex**: filling recruits pudendal motoneurons, raising EUS activity, and on coughing urethral pressure rises **before** pressure transmission. Its suppression in voiding is supraspinal, so SCI causes **DSD**.
- **Sphincter-to-bladder reflex**: pudendal afferents inhibit bladder preganglionic neurons, the basis of "holding on" and neuromodulation.

ALGORITHM

1. Tension-receptor (Aδ) firing exceeds threshold.
2. If the cortex permits, the PAG/PMC switch flips.
3. Sympathetic and pudendal outflow switch off; the **urethral sphincter relaxes first**, and urethral smooth muscle relaxes (NO).
4. The detrusor contracts (parasympathetic) and flow begins.
5. Urethral afferents reinforce the contraction until the bladder is empty.

- **Urethra-to-bladder reflexes**: urethral flow facilitates contraction, linking stress urinary incontinence (SUI) with detrusor instability.
- After SCI, weeks of spinal shock (retention) precede an NGF-driven **C-fibre spinal reflex**, giving NDO with DSD.
- **Pdet = Pves − Pabd**. Detrusor power (Pdet × flow, Q) is roughly constant, so voiding **Pdet reflects outlet resistance, not contractility**.
- Women with SUI void at Pdet at Qmax ≈ **19 cmH_2O**, healthy men at ≈ **60–70 cmH_2O**. Low pressure in women does not mean impaired contractility, and male nomograms are invalid in

women.

9 · Urethral closure mechanisms

Each SUI treatment targets one closure force.

- α-blockade lowers urethral pressure by about **one third**, so smooth muscle (α1A) tone matters.
- **Mucosal seal**: lamina propria vessels fill to seal the lumen, and oestrogen increases this flow.
- **Passive pressure transmission** reaches the proximal urethra; in the **hammock hypothesis (DeLancey)** it is compressed against a firm posterior fascial layer.
- **Active** guarding adds a fast-twitch reflex squeeze on coughing.

> **KEY POINT**
> Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the SUI standard, yet DeLancey's data suggest intrinsic sphincter function matters more than support (see Ch 109).

10 · Compliance

P3 Urinary bladder compliance — C = ΔV/ΔP; collagen/elastin, coiled collagen straightens; fast filling falsely low; DLPP >40 cmH_2O means upper tract damage

Not in the paper answer:

- Normal filling keeps pressure **<10 cmH_2O**. Determinants are viscoelasticity, active tone and geometry.
- †A detrusor leak point pressure (DLPP) ≥ **40 cmH_2O** in a low-compliance neurogenic bladder predicts upper tract damage (see Ch 104).

11 · Pathophysiology links

- **Idiopathic OAB**: DO is found in a minority (11% in one cohort), so symptoms reflect dysregulated sensation: more ATP/TRPV1 and Cx43, less BK, and NGF plasticity.
- **BOO** causes hypertrophy and partial denervation, which lead to **muscarinic supersensitivity** and more P2X1 and Cx43. NGF stays raised, which explains **persistent storage symptoms after prostate surgery**.
- **BPS/IC** shows more NGF, ATP/P2X3, TRPV1 and mast cells; tanezumab and intravesical RTX failed (see Ch 113).
- **Underactive bladder (UAB)** is a symptom complex, whereas detrusor underactivity is the urodynamic finding.
- A post-void residual (PVR) ≥ **300 mL** warrants evaluation, as do bothersome LUTS, hydronephrosis, renal insufficiency or culture-proven urinary tract infection (UTI); PVR <300 mL does not prove normal contractility.

12 · Ageing

- Ageing causes **hypofunction**: contractility, sensation and closure pressure fall. Capacity does not fall per se, so small capacity means coexisting DO.
- **α-contraction increases (α1D) and β-relaxation decreases**, so noradrenaline contracts the aged bladder.

- The aged CNS is prone to overactivity, giving urgency with poor emptying (DHIC). †Prefer β3 agonists or low-CNS-penetrant agents in the elderly.

13 · Neuromodulation and botulinum toxin — mechanisms

- Sacral neuromodulation (**SNM**), pudendal stimulation and posterior tibial nerve stimulation (**PTNS**) treat refractory OAB **and** non-obstructive retention by acting on **afferent axons**.
- In OAB, somatic input inhibits preganglionics and transfer to the PMC; in retention, it switches off an overactive **guarding reflex**.
- **OnabotulinumtoxinA (OBTX)**: the light chain cleaves **SNAP-25**, reducing ACh release. It also lowers sensory receptors, neuropeptides and urothelial ATP.
- RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

14 · Landmark trials

Trial	Design	Result	Take-home
ABC	RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with urgency urinary incontinence (UUI)	Equal fall in daily UUI; OBTX gave complete dryness more often	Antimuscarinic: dry mouth; OBTX: catheterisation, UTI
ROSETTA	RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years	Comparable efficacy; OBTX more often dry	OBTX more UTIs (note 200 U dose)

104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

1 · Normal function: two phases

Storage requires	Voiding requires
Low detrusor pressure (normal compliance), appropriate sensation	Detrusor contraction of adequate magnitude and duration
Outlet closed at rest and on straining	Fall in smooth and striated sphincter resistance
No detrusor overactivity (DO)	No anatomic obstruction

- **Storage is mostly passive**. More collagen (inflammation, bladder outlet obstruction [BOO], decentralisation) gives **drug-unresponsive** low compliance.
- Storage relies on α1 bladder-neck tone, β3 detrusor relaxation and the spinal **guarding reflex**; voiding uses pelvic parasympathetic outflow coordinated by the **pons**.
- **Stress continence** needs equal pressure transmission to the proximal urethra, active reflex closure and a **mucosal seal**.

2 · Pathophysiology

Every failure is of the **bladder, outlet, or both**: the basis of the **Wein functional classification**.

Storage failure

- **Bladder overactivity** means involuntary contractions, low compliance, or both.
- **Overactive bladder (OAB)** (International Continence Society, ICS) is urgency with or without urgency urinary incontinence (UUI), usually with frequency and nocturia. With UUI assume DO, as urodynamic studies often miss it.
- **Hypersensitivity** without overactivity occurs in interstitial cystitis/bladder pain syndrome (IC/BPS). **Outlet underactivity** follows denervation, surgery, trauma, ageing or loss of female support.

	Hypermobility (genuine stress urinary incontinence, SUI)	Intrinsic sphincter deficiency (ISD)
At rest	Outlet competent	Bladder neck poorly functional or open
Cause	Poor pelvic support	Prior surgery, scarring, neurological
Treatment	Correct hypermobility	Coaptation or compression

> **KEY POINT**
> Urinary incontinence can occur from ISD alone, but **never from hypermobility alone**. In **DeLancey's hammock**, a firm suburethral layer lets abdominal pressure compress the urethra.

Emptying failure

- **Bladder underactivity** is neurogenic, myogenic (overdistension, drugs) or psychogenic.

- **BOO** (ICS) is low flow and/or raised post-void residual (PVR) **with high detrusor pressure**. Male causes are prostate, bladder neck contracture and stricture; female causes are prolapse and slings.
- **Functional** obstruction includes striated sphincter dyssynergia (commonest), smooth sphincter dyssynergia (**autonomic dysreflexia**), primary bladder neck obstruction and dysfunctional voiding (**non-neurogenic**).

3 · Functional classification (Wein) — 2 × 2 table

	Because of the bladder	Because of the outlet
Failure to store	DO; low compliance; hypersensitivity; underactivity with overflow	Hypermobility; ISD. (Plus **fistula**)
Treatment	Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation	Pelvic floor training; α-agonists, duloxetine; bulking; slings; artificial urinary sphincter (AUS)
Failure to empty	Underactivity: neurogenic, myogenic, psychogenic	Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction
Treatment	Valsalva/Credé; parasympathomimetics; electrical stimulation	α-blockers, 5α-reductase inhibitors (5ARIs); prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy
Circumvent	Pads or collecting devices, antidiuretics, catheterisation, diversion	

> **KEY POINT**
> Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ An α-agonist tightens the outlet (storage); an α-blocker relaxes it (emptying). OnabotulinumtoxinA in the detrusor treats storage; in the sphincter it treats emptying.

Combined deficits (T10 paraplegia: DO and dyssynergia) are managed by external collection, or by **abolishing DO** plus clean intermittent catheterisation (CIC).

4 · Urodynamics by phase and site

Phase	Bladder	Outlet
Storage	Filling cystometry; detrusor leak point pressure (DLPP)	Urethral pressure profile; Valsalva leak point pressure (VLPP)
Voiding	Voiding cystometry (Pdet)	Fluoroscopy; sphincter electromyography (EMG)

Flowmetry and PVR assess both phases but **cannot localise** the fault.

5 · Other classifications

P2 Pathophysiology, classification and management of LUTS. — ICS storage / voiding (incl. terminal dribble) / post-micturition (post-micturition dribble) symptoms; Wein summary

Not in the paper answer:

- Chronic obstruction causes hypertrophy, collagen deposition, diverticula and DO, so intervene promptly.

ICS urodynamic classification (most precise)

Domain	Storage	Voiding
Detrusor	Normal; DO (neurogenic or idiopathic)	Normal; underactive; acontractile
Sensation, capacity, compliance	Normal, high/increased, low/reduced, absent	—
Urethra	Normal or incompetent closure	Normal; BOO, dysfunctional voiding, detrusor–sphincter dyssynergia (DSD)

- **Maximum cystometric capacity is when one can no longer delay** voiding.
- DSD is detrusor contraction with involuntary striated sphincter contraction. † **Lapides** and **Bors–Comarr** are older neurological classifications. MCQ

6 · Nocturia and neurogenic dysfunction

- **Polyuria** is >**40 mL/kg/24 h**. **Nocturnal polyuria** is a night-to-24-h volume ratio >**20%** in the **young, >33% elderly**; causes include loss of the nocturnal vasopressin rise and sleep apnoea.
- In **IC/BPS**, normally silent **C fibres** become hyperactive through central sensitisation.

Lesion	Detrusor	Sphincter
Suprapontine	DO	Synergic
Suprasacral spinal	DO	DSD, raised PVR
Sacral	Acontractile	Impaired

> **KEY POINT**
> DSD needs an intact sacral arc **disconnected from the pons**, so it is suprasacral, never suprapontine.

106 Evaluation and Management of Males With Urinary Incontinence

1 · Definition and determinants

- **Urinary incontinence (UI)** (International Continence Society, ICS) is any involuntary loss of urine. In men the **external urethral sphincter (EUS) is most important**.
- Risk factors include **age (major, independent)**, lower urinary tract symptoms (LUTS), infection, cognitive or neurological impairment, obesity, diabetes, prostatectomy and drugs.
- Reversible causes, **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- Male:female is about 1:2. **Urgency UI (UUI) predominates**, then stress UI (SUI), then mixed UI (MUI).

2 · Types

Type (ICS)	Mechanism	Male context
SUI: on effort, cough	Sphincter deficiency	Almost only **after prostatectomy**
UUI: with urgency	Detrusor overactivity (DO) overcomes a normal outlet	~**90%** have DO on urodynamics; it is "wet" overactive bladder (OAB)
MUI	DO and sphincter deficiency	Uncommon; may follow prostatectomy
Nocturnal enuresis	Wetting in sleep	New in an adult man: suspect **high-pressure chronic retention** and renal failure
Continuous	Fistula, exstrophy/epispadias, ectopic ureter	Rare
With chronic retention	Residual ≥ **300 mL**	"Overflow" is no longer an ICS term; spinal cord injury (SCI) at or above T6 risks autonomic dysreflexia

3 · Evaluation

- The history covers timing, triggers, prior prostate or pelvic surgery, neurology, bowel and sexual function, drugs and the **most bothersome symptom**.
- Examine the abdomen, genitalia, prostate (digital rectal examination, DRE), anal tone, neurology and cognition, and do a **cough test supine then standing**.
- Record **body mass index (BMI)**: weight loss may help SUI, and high abdominal pressure can overcome an artificial urinary sphincter (AUS) cuff.
- **Dipstick** all men; haematuria needs cytology, cystoscopy and upper-tract imaging.
- In a **bladder diary**, **3 days** **equal 7 days**; include a weekend.
- The International Prostate Symptom Score (IPSS) does **not measure incontinence**, so use the **ICIQ-UI Short Form** (0–21). A **pad test** is not routine and cannot separate UUI from SUI.

Severity	Pads/day	Pad weight/day
Mild	1	<200 g

Severity	Pads/day	Pad weight/day
Moderate	2–4	200–400 g
Severe	>4	>400 g

- **Flow and post-void residual (PVR)**: flow cannot separate bladder outflow obstruction (BOO) from underactivity.
- Check prostate-specific antigen (PSA) only if life expectancy is >10 years and it would change management.
- Do urodynamic studies (**UDS) only if they alter management**, i.e. before invasive treatment (ICI, EAU and AUA guidance) (see Ch 107).
- **Urodynamic stress incontinence (USI)** is leakage on effort without detrusor contraction. **Cough-associated DO incontinence** mimics SUI.

KEY POINT

Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS, usually just before surgery.

4 · Urgency urinary incontinence

ALGORITHM

1. Lifestyle, bladder training and pelvic floor muscle training (PFMT) ≥6 weeks (ideally 3 months).
2. Antimuscarinic or β3-agonist, or both.
3. Urodynamics optional; then onabotulinumtoxinA, sacral neuromodulation (SNM) or posterior tibial nerve stimulation (PTNS).
4. Last resort: augmentation or ileal conduit.

- Cut fluid by **25%** if intake is >1 L/day, and avoid caffeine.
- **Antimuscarinics** have similar efficacy, and extended-release forms are preferred. **Anticholinergic burden** for ≥3 months raises dementia risk by about 50%.
- **β3-agonists** are co-first-line with less dry mouth; with mirabegron, **monitor blood pressure**. In **cognitive impairment, use a β3-agonist or early surgery**.
- Mirabegron plus solifenacin beats escalating solifenacin. Tadalafil is the only phosphodiesterase-5 inhibitor for LUTS with erectile dysfunction.

	OnabotulinumtoxinA	SNM	PTNS
Delivery	Cystoscopic, **100 U** (200 U neurogenic DO)	S3 lead, implanted generator	Needle behind medial malleolus, 30 min weekly × 12
Strength	Repeat every 6–9 months	Most effective, durable >10 years	Least invasive; better for dry OAB
Drawback	Retention in ~**10%**, so the man must accept self-catheterisation	Most invasive, costliest	Frequent visits

- If the SNM test gives ≥**50%** improvement, a permanent generator is implanted (see Ch 115).

5 · Stress urinary incontinence

ALGORITHM

1. Supervised PFMT ≥3 months, with or without biofeedback and electrical stimulation (better started pre-op); consider duloxetine.
2. Exclude anastomotic or urethral stricture and bladder dysfunction (UDS).
3. Choose surgery by severity.

P2 Post-prostatectomy incontinence — wait 6–12 months + PFMT; duloxetine off-label; no bulking; sling mild–moderate, AUS (cuff, balloon, scrotal pump) severe/irradiated; ProACT; MASTER

Not in the paper answer:

- After radical prostatectomy wait **6–12 months** before surgery.
- The **AUS is the gold standard**, with about 80% continence (Scott 1973 MCQ), and is activated at about 6 weeks; †deactivate it before catheterisation or endoscopy to prevent erosion.
- Radiotherapy, severe leakage and prior transurethral resection worsen fixed-sling results, so choose an AUS. **ProACT** is **contraindicated after radiotherapy**.

EAU 2023 recommendation (post-prostatectomy incontinence, PPI)	Grade
Offer AUS for moderate–severe PPI	Strong
Warn of AUS/ProACT complications, mechanical failure, explantation	Strong
Inform of duloxetine side effects, off-label use	Strong
Fixed transobturator sling for mild–moderate PPI	Weak
Do not offer bulking agents; no ProACT after radiotherapy	Weak

6 · Mixed and other types

- In **MUI**, treat the most bothersome component first; DO may mask SUI, so treat DO and repeat UDS.
- In high-pressure chronic retention with enuresis, **catheterise first**, then relieve BOO.
- Enuresis with a low residual is usually OAB and is treated with an antimuscarinic or β3-agonist, with or without desmopressin.
- Postmicturition dribble is treated with **urethral milking**.
- **Red flags** needing a specialist are haematuria, pain, recurrent infection, pelvic radiotherapy, abnormal DRE and high PVR.

7 · Landmark trials

Trial	Design	Result	Take-home
MASTER	Non-inferiority randomised controlled trial (RCT), male sling vs AUS for urodynamic SUI after prostatectomy	Equivalent at 12 months; secondary outcomes favoured AUS; few completely dry	Sling is an acceptable alternative; counsel realistically

107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

1 · Definition and role

- **Urodynamics (UDS)** is the dynamic study of urine transport, storage and evacuation. Use it when simple tests fail, to **diagnose the cause of symptoms** and to assess **diseases that silently damage the upper tracts** such as spinal cord injury (SCI), multiple sclerosis (MS) and radiation cystitis.

Wein functional classification — the frame for every study

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Outlet	Failure to empty	Failure to store

- Set the questions first and aim to **reproduce the symptom**; an unrecorded abnormality is not excluded.

> KEY POINT
> **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity (DO) throughout filling; detrusor leak point pressure (DLPP) **> 40 cmH_2O**; poor emptying with high storage pressure.

2 · Preparation

- **Decide medication status in advance**: test on regular dosing to assess drug effect, or stop the drug first to uncover aetiology.
- **ICS pre-invasive work-up**: symptom and bother score, **3-day frequency–volume chart**, free flow and postvoid residual (PVR).
- **Urinalysis before every invasive study**; postpone for symptomatic urinary tract infection (UTI).

Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

The "index" patient needs **no routine prophylaxis** (level 1). Non-index patients get a **single dose within 1 h**: oral trimethoprim–sulfamethoxazole first; alternatives include a cephalosporin, amoxicillin–clavulanate, an aminoglycoside or a fluoroquinolone.

Prophylaxis YES	Prophylaxis NO
Neurogenic LUT dysfunction (NLUTD); bladder outlet obstruction (BOO) or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (especially renal transplant); chronic catheters including clean intermittent catheterisation (CIC); high-risk orthopaedic implant	Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely)

Trap: asymptomatic bacteriuria is treated here, unlike in most of urology.

3 · Non-invasive components

P2 Uroflowmetry. — Qmax, Qave, voiding efficiency; VV ≥150 mL; Siroky/Liverpool nomograms, corrected Qmax; bell, plateau (stricture), intermittent curves; cannot separate BOO from underactivity

Not in the paper answer:

- Minimum voided volume is **VV 150 mL**, because maximum flow (Qmax) is linear above it and hyperbolic below.
- Normal Qmax is > 15 mL/s in men (< 10 suggests obstruction) and > 20 mL/s in women.

4 · Equipment, transducers and quality control

- The minimum is vesical (Pves), abdominal (Pabd) and subtracted detrusor pressure (Pdet) with flow. Infuse saline or contrast, **not CO_2**.

Feature	External water-filled	Air-charged	Microtip
Reference level	**Upper border of symphysis pubis**; re-level on position change	None (no hydrostatic effect)	—
Problems	Air bubbles; motion and hydrostatic artefact; underdamped	Overdamped (signals > 3 Hz attenuated), so coughs are blunted	Expensive, reusable

- Systems are **not interchangeable** (Pdet may differ by up to 10 cmH_2O); water-filled systems underpin most nomograms.

> KEY POINT
> **Quality control** (ICS Good Urodynamic Practice): zero to atmosphere, flush air, and keep resting Pdet near 0. Do a **cough test** before filling, at intervals and before voiding; a biphasic Pdet spike means re-flush or reposition. Rectal contractions cause an unexplained fall in Pdet.

5 · Filling cystometry (CMG)

CMG shows whether storage is at safe, low pressure.

- Filling to "permission to void" gives the **maximum cystometric capacity (MCC)**. **Pdet = Pves − Pabd**; rectal Pabd is preferred.
- Standard adult filling is **30–100 mL/min**; the ICS physiological rate is < body weight (kg)/4 mL/min. Neurogenic bladders need a physiological rate and body-warm saline (EAU). DO is commoner upright.
- A **normal ("stable")** bladder keeps Pdet near zero with no involuntary contractions. **Sensation** (first sensation, first desire, strong desire, urgency) is the only subjective parameter.

Detrusor overactivity (DO)

- DO means involuntary detrusor contractions (IDC) during filling, either **idiopathic** or **neurogenic**. The label comes from the history, as tracings look identical.
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO, and some asymptomatic volunteers do show it.
- In BOO with overactive bladder (OAB), a **single terminal IDC** predicts storage-symptom resolution after transurethral resection of the prostate (TURP). In neurogenic DO, higher Pdetmax and longer IDCs predict upper-tract deterioration.

- **Cough-associated DO** mimics stress urinary incontinence (SUI) but is detrusor in origin.

Compliance

P3 Urinary bladder compliance — ΔV/ΔPdet; < 20 mL/cmH_2O poor; absolute pressure matters more; causes; pitfalls (fast fill, low-amplitude IDC, VUR/diverticulum/outlet pop-off)

Not in the paper answer:

- Storage pressure **> 40 cmH_2O** itself leads to upper-tract damage.
- Outlet pop-off hides poor compliance, which may appear only after a sling or artificial urinary sphincter (AUS), so occlude the outlet.

6 · Leak point pressures

P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — ALPP (low = weak sphincter) vs DLPP (high = upper-tract danger); technique from 150 mL, catheter out at capacity; VLPP < cough LPP; McGuire DLPP > 40 gives 85% hydronephrosis/VUR

Not in the paper answer:

Abdominal leak point pressure, ALPP (cmH_2O)	Inference (McGuire)
< 60	ISD
60–90	Equivocal
≥ 90	Little or no ISD (hypermobility)

- Intrinsic sphincter deficiency (ISD) and hypermobility coexist; ALPP poorly predicts surgical success.
- **Occult SUI** is leakage only on prolapse reduction in a continent woman with pelvic organ prolapse (POP).
- **DLPP > 40 cmH_2O** is not a validated safe cut-off (20 is more sensitive in children), so aim **as low as achievable** by lowering bladder pressure or the outlet (e.g. sphincterotomy for DESD), then repeat the UDS.

7 · Urethral pressure profilometry (UPP)

- **Maximum urethral closure pressure (MUCP)** is maximum urethral minus intravesical pressure (normal 40–60 cmH_2O in women).
- **MUCP ≤ 20 cmH_2O** means ISD ("type III" SUI) and predicts sling or suspension failure.
- The EAU advises against using UPP or leak point pressure to grade severity.

8 · Pressure–flow studies (PFS)

Only PFS tells obstruction from a weak bladder.

Pdet	Flow	Interpretation
Low/normal	High/normal	Unobstructed
High	Low	**BOO**
Low	Low	Detrusor underactivity (obstruction may be hidden)

- ICS **detrusor underactivity** is reduced contraction strength or duration, so emptying is prolonged or incomplete. **Acontractile detrusor** replaces "areflexia".

Men — nomograms and indices

Index	Formula	Cut-offs
Bladder outlet obstruction index (BOOI)	PdetQmax − 2Qmax	**> 40 obstructed · 20–40 equivocal · < 20 unobstructed**
Bladder contractility index (BCI)	PdetQmax + 5Qmax	**> 150 strong · 100–150 normal · < 100 weak**

- BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow, so a decompensated detrusor masks obstruction.
- **Stop tests** estimate isovolumetric pressure (Piso); in men Piso < 50 cmH_2O suggests underactivity.
- About 2/3 of men with benign prostatic obstruction (BPO) have DO; half to two-thirds resolve after relief.
- MCQ In the bethanechol supersensitivity test (Lapides), 2.5 mg is given subcutaneously; a rise **≥ 15 cmH_2O** at 100 mL is positive (neurogenic), but the test is unreliable.

Women

- Women void at low pressure, so the **BOOI and ICS nomogram cannot be used**. Suspect BOO after anti-incontinence surgery or with prolapse.

Method	Criterion	Comment
Cut-points	e.g. Qmax ≤ 12 mL/s and PdetQmax ≥ 25 cmH_2O (Defreitas); Chassagne ≤ 15 / ≥ 20	Anatomic obstruction only; overlap
Videourodynamics, VUDS (Nitti)	Radiographic obstruction between bladder neck and meatus with a sustained contraction of any magnitude	Localises; works despite weak detrusor
Blaivas–Groutz	Free Qmax vs Pdetmax; none/mild/moderate/severe	Two separate voids; **overdiagnoses**
Solomon–Greenwell	PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain	Includes functional obstruction

9 · EMG and sphincter coordination

- Surface perineal electrodes are standard. Normally EMG quietens and urethral pressure falls **before** Pdet rises.
- The **guarding reflex** (EMG rise during an IDC) is normal, not dyssynergia.

	DESD	Dysfunctional voiding
Cause	**Suprasacral cord lesion** (the higher, the likelier)	Learned; no neurological lesion
Behaviour	Involuntary sphincter contraction during DO or voiding	Intermittent sphincter contraction during voiding
Significance	High pressure causes poor compliance and upper-tract risk	LUT symptoms; fluoroscopy improves accuracy

- **DISD** needs a suprasacral lesion above the T10–L1 sympathetic outflow and is seen on **VUDS only**. **Primary bladder neck obstruction (PBNO)** is also invisible to EMG and needs fluoroscopy.

10 · Videourodynamics

P4 Video Urodynamics — gold standard in NLUTD (EAU); only test for bladder neck/DISD/PBNO (types I–III); VUR pop-off; Nitti criteria; technique, screen only at key moments; indications

Not in the paper answer:

- If VUDS is unavailable in NLUTD, do CMG with PFS (EAU).
- Fluoroscopy is critical for PBNO types II and III.

11 · Ambulatory urodynamics

- Natural filling shows more DO and better compliance than pump filling; use it for symptoms unexplained by conventional UDS.

12 · Indications by condition (AUA/SUFU, EAU)

Order UDS when the result will change treatment.

Female SUI

- In uncomplicated demonstrated SUI (the "index" patient) UDS may be omitted (AUA/SUFU 2023; EAU: not routinely).
- Do UDS in non-index women: prior anti-incontinence or POP surgery, symptom–examination mismatch, voiding dysfunction, significant UUI, raised PVR, unconfirmed SUI or NLUTD.

Male LUTS

- No UDS is needed to start an α-blocker for lower urinary tract symptoms (LUTS).
- AUA/SUFU: do **PFS when obstruction matters, before invasive, morbid or irreversible treatment**, because without obstruction surgery does worse.
- EAU PFS triggers before invasive treatment are prior failed treatment, inability to void > 150 mL, **Qmax > 10 mL/s**, PVR > 300 mL, and age > 80 or < 50.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**

Women with voiding symptoms; NLUTD

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) needs VUDS.
- The NLUTD goal is **low-pressure filling**, even at the cost of catheterisation; only UDS separates DO from poor compliance.
- UDS is essential before treatment and in follow-up in SCI and myelomeningocele.

13 · Landmark trials

Trial	Design	Result	Take-home
ValUE	RCT, 630 women, uncomplicated SUI: office evaluation with or without UDS	Success ~77% both arms	Office evaluation alone non-inferior before SUI surgery
Dutch embedded RCT	Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored	Equal cure	UDS discordance need not change surgery
SISTEr	Burch vs pubovaginal sling	ALPP and DO did not predict outcome	UDS adds little in stress-predominant SUI

109 Neuromuscular Dysfunction of the Lower Urinary Tract

1 · Principles and classification

The lesion level predicts the bladder and sphincter pattern.

- The International Continence Society (ICS) uses **five parameters**: sensation; detrusor (normal, overactive, areflexic, impaired); compliance; smooth sphincter (synergic or dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).

Level	Typical disorders	Detrusor	Smooth sphincter	Striated sphincter
Suprapontine	Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy	Overactive	Synergic	Synergic (with or without impaired voluntary control; PD bradykinesia)
Multiple system atrophy (MSA)	—	Overactive, later impaired	**Open**	Denervated
Suprasacral cord	Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis	Overactive	Synergic (dyssynergic if above T7)	**Dyssynergic**
Sacral cord / cauda equina	Sacral SCI, disc, myelodysplasia	Areflexic	Competent non-relaxing; may open	Fixed tone
Peripheral	Radical pelvic surgery, diabetes, tabes	Impaired / areflexic	Open (pelvic surgery)	Fixed tone / synergic

KEY POINT

Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**. An open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

2 · Lesions at or above the brainstem

Stroke

- **Acutely** there is retention ("cerebral shock"). **Chronically** there is detrusor overactivity (DO) with **synergic sphincters**.
- **True detrusor–sphincter dyssynergia (DSD) does not occur. Pseudodyssynergia** is an electromyographic (EMG) flare from voluntary attempts to inhibit a contraction.
- In men with lower urinary tract symptoms (LUTS), do a full urodynamic study (UDS) before outlet surgery.

Dementia, brain injury, tumour, others

- **Dementia**: anticholinergics worsen cognition, so a **β3 agonist is strongly preferred** (SUFU).
- **Brain injury** above the pontine micturition centre (PMC) gives DO with synergic sphincters; brainstem injury below the PMC can give DSD. Frontal **tumours** can cause retention.

- **Normal-pressure hydrocephalus** gives dementia, gait ataxia and incontinence, improved by a shunt.
- **Cerebral palsy** gives DO with synergic sphincters, but about half of adults with LUTS have poor compliance or a detrusor leak point pressure (DLPP) ≥40 cm H_2O, so evaluate fully.

Parkinson disease vs multiple system atrophy

- **PD** LUTS occur in 35–70% and are mainly storage symptoms. Sphincter **bradykinesia** mimics DSD.
- Treat with overactive bladder (OAB) drugs with low central effect, such as mirabegron or trospium; onabotulinumtoxinA, tibial nerve stimulation (PTNS) and sacral neuromodulation (SNM) also work.
- **Transurethral resection of the prostate (TURP) in PD is not contraindicated.** The old "post-TURP incontinence in PD" cases were MSA, so confirm PD, exclude MSA and do videourodynamics first.
- **MSA** affects the intermediolateral columns and **Onuf nucleus**, causing early severe LUTS and erectile dysfunction (ED).
- In MSA, DO comes early and poor emptying late. Treating DO worsens emptying, so add clean intermittent catheterisation (CIC); give desmopressin for nocturia. **Prostatectomy is hazardous.**

Feature	MSA	PD
Urinary symptoms / ED	Precede or accompany parkinsonism; ED near universal	Years after diagnosis; ED ~⅓
Dominant problem; PVR	UI; high postvoid residual (PVR)	Frequency/urgency; PVR normal
Bladder neck	**Open**	Synergic
Striated sphincter EMG	**Denervation**	Normal (bradykinesia)
Detrusor	DO, then areflexia	DO; areflexia uncommon
Outlet surgery	Worsens	Acceptable after work-up

3 · Spinal cord disease

Cord lesions add dyssynergia and silent upper-tract damage.

Multiple sclerosis

- LUTS occur in 50–90% and are the presenting complaint in up to 15%. On UDS, DO is found in ~⅔, DSD in 30–65% of those with DO, and areflexia in ~¼.
- Symptoms do not track the disease, so **baseline UDS is essential**.
- Keep management **reversible**, because the pattern changes: antimuscarinic with CIC, onabotulinumtoxinA, or SNM in selected patients. Credé and Valsalva voiding are contraindicated, and an indwelling catheter is the last resort.

Spinal cord injury

P2 Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male — spinal shock, suprasacral NDO + DSD, autonomic dysreflexia, CIC/toxin/sphincterotomy/Brindley ladder

Not in the paper answer:

- Spinal shock usually lasts **6–12 weeks**; bladder reflexes return with the lower-limb deep tendon reflexes.
- Smooth sphincter dyssynergia occurs with cord lesions at **T7–T8** (column T6) or above.
- The sacral cord begins at vertebral column T12–L1, so state column and cord level separately.

	Suprasacral (complete)	Sacral (complete)
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal or high; may fall
Smooth sphincter	Synergic (dyssynergic above T7–T8)	Competent non-relaxing; may later open
Striated sphincter	Dyssynergic	Fixed tone, no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, reduced reflexes
Management	Lower pressure (drugs or toxin) and CIC	CIC and keep compliance safe

- **Neurology does not equal urodynamics**, and thoracolumbar (T10–L2 column) lesions are least predictable, so **manage by UDS, not by level**.
- Upper-tract risks are detrusor fibrosis, leak point pressure ≥75 cm H_2O and DO for >⅓ of filling; DLPP >40 cm H_2O is the classic threshold. Damage can occur **without vesicoureteral reflux (VUR)**.

Autonomic dysreflexia (AD)

- AD occurs in SCI **above T6** and may appear years later. Triggers include bladder or rectal distension, a blocked catheter, instrumentation, stones, sores and fracture.
- Risk during UDS is highest with complete cervical lesions, DSD and poor compliance.

ALGORITHM

1. Sit the patient upright and loosen clothing.
2. **Remove the stimulus**: drain the bladder, unblock the catheter, empty the rectum.
3. If blood pressure stays high, apply nitroglycerin 2% paste above the lesion (first line).
4. If it still persists, give labetalol or intravenous nitroprusside.
5. For prophylaxis, give terazosin or prazosin (α1-blockers).
6. If intractable, consider sacral rhizotomy, sympathectomy or cordectomy.

- Nifedipine (10–20 mg sublingual) is now banned in many centres because of uncontrolled hypotension.

VUR and women with SCI

- A **chronic urethral catheter is worst** for the upper tracts. VUR artefactually enlarges capacity and lowers pressures on UDS.
- Women paraplegics do CIC, while tetraplegics often need indwelling catheters.

Other cord disorders

- **Transverse myelitis** causes retention, then DO with DSD, and is managed as SCI. **Cervical myelopathy** gives DO with or without DSD, or underactivity, so UDS guides treatment.
- In **myelomeningocele**, level correlates poorly with UDS. The typical adult has an **areflexic bladder and open bladder neck** with a fixed sphincter, so leaks at a fixed pressure; DSD occurs in 10–15%.
- Poor compliance, high pressure and DO predict renal decline. For stress UI, bulking is safer than a sling or artificial urinary sphincter (AUS) because it does not reduce compliance.

- **Tethered cord** presents after growth spurts, usually with DO. A sensory neurogenic bladder (tabes, B12) has a large capacity.

4 · Lesions distal to the cord

Root and nerve lesions give a weak bladder behind a fixed outlet.

- **Disc disease** (L4–L5, L5–S1) most often gives a **normally compliant areflexic bladder**. Do **pre-laminectomy UDS** for clinical and medicolegal reasons.
- **Cauda equina syndrome** causes saddle anaesthesia and an acontractile detrusor. **Decompression within 48 h** gives better urinary and rectal recovery.
- **Radical pelvic surgery** (abdominoperineal resection, radical hysterectomy) gives impaired contractility, fixed striated tone, an **open non-functional bladder neck** and lower compliance, so the patient leaks or retains.
- After pelvic surgery, **use CIC and wait 6–12 months**; do **no prostatectomy unless BOO is proven**.
- Herpes zoster and herpes simplex (**Elsberg syndrome**) cause transient retention.

P2 Bladder dysfunction non genitourinary systemic disease — diabetic cystopathy (reduced sensation, then raised capacity, reduced contractility and raised PVR; DO common; pressure–flow vs BOO; timed voiding), Guillain–Barré, lupus, B12

Not in the paper answer:

- **Guillain–Barré** causes retention; use CIC until recovery.

> **KEY POINT**
> Distal lesions hit the peripheral reflex arc, giving areflexia with a non-relaxing or fixed outlet. Compliance falls only when preganglionic parasympathetic fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

5 · Specific functional and miscellaneous conditions

Detrusor–sphincter dyssynergia

P1 Detrusor–external sphincter dyssynergia (DESD / DSD) — lesion pons–sacral cord, Blaivas types 1–3, pseudodyssynergia, half of untreated men deteriorate, CIC first, sphincterotomy if CIC impossible

Not in the paper answer:

- Blaivas type 1 (incomplete) can be observed unless there is VUR, hydronephrosis or AD; types 2–3 need treatment.
- Without a neurological lesion, doubt DSD (except Hinman syndrome in children).

Functional outlet disorders

- **Dysfunctional voiding** (Hinman) is striated sphincter contraction during voiding without neurological disease, proven on pressure–flow–EMG.
- **Non-neurogenic chronic retention** is PVR ≥ **300 mL** persisting ≥6 months on ≥2 occasions. High-risk patients need drainage, preferably suprapubic.
- **Primary bladder neck obstruction** affects young and middle-aged men with a normal prostate, and videourodynamics localises it. α-blockers help some, but **bladder neck incision is definitive**.
- **Female BOO** is rare. Groutz criteria are free maximum flow (Qmax) ≤12 mL/s and detrusor pressure at Qmax ≥20 cm H_2O.

- **Fowler syndrome** is retention in young women with complex repetitive discharges on needle EMG and urethral closure pressure >100 cm H_2O. **SNM is the treatment** (~70% succeed).

Other conditions

- Rarer causes include myasthenia (more incontinence after prostatectomy), Isaacs syndrome, infective myelopathies and lupus cystitis (treated with steroids). After **radiation**, early storage symptoms recover by 6 months.

6 · Evaluation and risk stratification

- Under AUA/SUFU 2021 and EAU 2022 neurogenic LUT dysfunction (NLUTD) guidelines, **the primary goal is to prevent renal damage.**
- Everyone gets history, examination (cognition, **hand dexterity**, spasticity), urinalysis, PVR and diaries.
- **Defer full assessment during spinal shock or acute brain injury**, and give no irreversible therapy then.

Risk (AUA/SUFU)	Typical	Defining findings
Low (all)	Suprapontine or uncomplicated peripheral	Spontaneous voiding, low PVR, stable; normal imaging and renal function
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage and voiding symptoms, raised PVR, recurrent UTI; not yet assessed, so do UDS, upper-tract imaging and renal function
Moderate	—	Neurogenic retention; DO with incomplete emptying; normal tracts
High (any one)	—	Poor compliance, raised storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function

KEY POINT

Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); high risk means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis.

7 · Management

Start with the simplest, most reversible step.

ALGORITHM

1. Aim for low-pressure storage with regular emptying: CIC with an antimuscarinic or β3 agonist.
2. If refractory, give intradetrusor onabotulinumtoxinA.
3. In selected patients, try neuromodulation.
4. If these fail, do augmentation or continent diversion.
5. As a last resort, do bladder neck closure with a suprapubic catheter, ileovesicostomy or incontinent diversion.

- **CIC is the gold standard**; keep volumes **< 500 mL**. Credé and Valsalva are not recommended.

- CIC beats an indwelling urethral catheter, and for a chronic catheter **suprapubic is better than urethral**.
- For refractory neurogenic DO, use **intradetrusor onabotulinumtoxinA** (~60% continue long term) and warn about CIC. Intrasphincteric toxin is not routine.
- The Brindley stimulator is used in complete SCI (see Ch 115). PTNS suits patients who void and have OAB; SNM suits NLUTD without SCI or spina bifida.
- **UTI**: treat bacteriuria only if symptomatic, correct stones and residuals, and give **no prophylaxis** for CIC or indwelling catheters.
- **Bladder cancer** incidence is similar to the general population, but it is often muscle-invasive at diagnosis. **AUA/SUFU 2021 advise no routine surveillance cystoscopy**; scope for haematuria.

8 · Follow-up

Follow-up is lifelong, because upper-tract damage is silent.

Body	Schedule
AUA/SUFU 2021 (risk-based)	Low risk: none routine. Moderate/high: yearly history, examination and renal function; upper-tract imaging every 1–2 years; repeat UDS for change; after bowel reconstruction, yearly electrolytes, acid–base and imaging
EAU	Specialist review every 1–2 years; ultrasound and residual 6-monthly; dipstick 2-monthly; yearly examination and bloods
American Paraplegia Society (SCI)	Annual for 5–10 years, then alternate years; renogram: a >20% fall in renal plasma flow means investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU)

110 Overactive Bladder

1 · Definitions and terminology

OAB is a symptom diagnosis built on urgency.

P1 P2 Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management; Overactive bladder and its drug therapy — ICS definition, OAB vs DO, afferent/urothelial/myogenic/central mechanisms, behavioural therapy, drugs, PTNS/toxin/SNM

Not in the paper answer:

- In overactive bladder (OAB), **urgency is the defining symptom**, hence "urgency" rather than "urge" incontinence.
- OAB is a symptom syndrome, whereas detrusor overactivity (DO) is a urodynamic observation that may be asymptomatic or artefact.

Term	ICS definition (all patient-reported)	
Increased daytime frequency	Patient considers they void too often by day; upper normal	7 voids
Nocturia	Waking ≥1 time to void; each void preceded and followed by sleep	
Urgency	Sudden, compelling desire to void, difficult to defer	
UUI	Involuntary leakage associated with urgency	

- **Urgency urinary incontinence (UUI) is the most bothersome lower urinary tract symptom (LUTS) in both sexes**. Frequency bothers at ≥**8** daytime voids and nocturia at ≥**2**.
- **Isolated nocturia** usually means nocturnal polyuria (night urine >33% of 24-h output in older adults) or poor sleep.
- **Mixed urinary incontinence (MUI)** is stress UI (SUI) with UUI; SUI with "OAB dry" is **not** MUI.

OAB versus bladder pain syndrome (BPS)

BPS requires bladder pain, pressure or discomfort with ≥1 urinary symptom; pain is excluded from OAB, though overlap is large.

Feature	OAB	BPS
Primary fear	**Leakage**	**Pain**
Filling sensation	Phasic urgency waves	Steady, accelerating pain
Voided volumes	Variable	More consistent
Deferral	Limited, because of leak risk	Possible, at cost of pain
Site	Perineum/penile base, urethra/vagina	Suprapubic, with or without perineal
UUI	Yes	No

2 · Epidemiology and burden

- Adult OAB affects ≈ **12–29%**, depending on the definition (EpiLUTS highest, EPIC lowest).
- Women have UUI 2–3× as often as men, so the sex gap is in **wet** OAB. Prevalence rises with age, to ~40% of men over 75.

- Fewer than half tell a clinician. About **92%** stop the first antimuscarinic within 2 years, mainly because of **unmet expectations**, then side effects.
- OAB dry progresses to UUI at ~1% a year. Among third-line options, onabotulinumtoxinA 100 U is the most cost-effective and sacral nerve stimulation (SNS) the costliest.

3 · Pathophysiology

- **DO is no longer the sine qua non**: only about half with DO have urgency, and **44–69%** with urgency show DO.
- The periaqueductal grey and pontine micturition centre are the main regulatory level. Afferent, urothelial, myogenic (exaggerated micromotions) and central mechanisms all contribute (see pointer).

> **KEY POINT**
> Final common pathway: more afferent signal reaches awareness. The bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB.

4 · Aetiology and risk factors

- Risks are **ageing** (clearest), female sex, metabolic syndrome, bowel and hormonal factors, depression and anxiety.
- **Bladder outlet obstruction (BOO)** is only an **indirect** cause in humans, since OAB is as common in women, in whom BOO is rare.

5 · Clinical assessment

- The **minimum** work-up (AUA/SUFU and EAU) is **history, examination and urinalysis**, followed by empirical treatment; the EAU adds questionnaires and a diary.
- **History** covers bother, SUI, voiding symptoms, haematuria, fluids and caffeine, occult neurology, pelvic surgery or radiotherapy, bowel function (faecal incontinence in ~**50%** of women with OAB), drugs and causes of polyuria.

Drug class	Relative contraindications
Antimuscarinics	Uncontrolled closed-angle glaucoma; cognitive impairment; retention; impaired gastric emptying
β3-adrenergic agonists	Uncontrolled hypertension; arrhythmia

- **Examination**: frailty, cognition and oedema, but the **pelvic and genital examination is most important** (prolapse, atrophy, stress test; digital rectal examination in men; perineal sensation).
- **Urinalysis is mandatory**; microhaematuria needs full evaluation.
- **Postvoid residual (PVR)** is not needed before conservative or oral therapy if uncomplicated; measure it with retention risk, voiding symptoms, prior outlet surgery or neurology.
- Use a **diary** for 3–7 days; **maximum voided volume is the functional capacity.** Questionnaires include ICIQ-OAB and OAB-q; MCQ OAB-V8 is a screening tool, not a severity score.

6 · Urodynamics (UDS)

- UDS is **not** in the initial work-up. It is **indicated** when the result decides invasive or irreversible treatment, after multiple failures, or with suspected BOO.
- DO is found in OAB wet in **>90% of men** but is absent in 30–40% of women. It correlates **best with UUI and worst with nocturia**.
- Stop antimuscarinics to show DO, and continue them to explain refractory symptoms.
- **Assess emptying**: each **10 mL** of PVR raises the risk of catheterisation after onabotulinumtoxinA by **27%**. DO persists in up to **60%** after transurethral resection of the prostate (TURP).

7 · Management

Treatment follows the patient's goals.

- OAB is benign, so **no treatment is acceptable**. Set **realistic expectations** of improvement, not cure. AUA/SUFU lines are **not a compulsory sequence**; allow **4–12 weeks** per therapy.

> **ALGORITHM**
>
> 1. Confirm OAB and exclude mimics.
> 2. Educate, then start conservative measures.
> 3. If not enough, add drugs.
> 4. If drugs fail, offer tibial nerve stimulation, sacral neuromodulation or onabotulinumtoxinA.
> 5. Rarely, do augmentation or diversion.
> 6. Reassess at every step.

- Reducing fluid by ~**25%** helps, and cutting caffeine improves urgency and frequency.
- Pelvic floor muscle training with bladder training reduces leaks by **50–80%**. **Weight loss** reduces SUI and UUI (PRIDE trial).
- Drugs (see Ch 113): response is idiosyncratic, so switch or combine before invasive work-up.
- In **dementia**, antimuscarinic use >**3 months** raises risk by 10–65%, so prefer a β3 agonist.
- In benign prostatic hyperplasia with OAB, an α1-blocker plus antimuscarinic works; retention occurs in ~3%, so take care if PVR is raised.

Third-line (see Ch 115)

Frailty predicts functional decline and 1-year morbidity of up to 50%, even after minor procedures.

	Tibial nerve stimulation	OnabotulinumtoxinA	SNS
Efficacy	60–80% respond	UUI reduced 59%; ~¼ dry	5-year success 70–80%
Drawback	12 weekly visits; response by weeks 8–10	Urinary tract infection (UTI) ~20%; catheterisation 0–12% in trials, real-world ~20% women, ≤35% men	Revision >30% at 5 years; battery
Risk	Lowest	Moderate	Highest

- **ICI pathways**: in men, treat the outlet with the bladder; in women, treat the most bothersome MUI component first; in the frail elderly, put comorbidity and carer goals first.

KEY POINT

Escalate by invasiveness and reversibility, not severity alone. OAB is benign, so patient goals, frailty and PVR decide how far to climb.

111 Underactive Detrusor

P2 P4 Management of detrusor underactivity; Detrussor underactivity. — ICS definition, causes (neurogenic/myogenic/drugs/functional), BCI, management table: CISC mainstay, α-blockers, bethanechol not routine, SNM, Brindley, outlet surgery, diversion, investigational options

Not in the paper answer:

- Poor surgical outcome: voiding pressure under 45 cm H_2O, age over 80, PVR over 1500 mL.
- Significant PVR proposal (over 40% of capacity); multisystem atrophy 52–95%.
- Sphincter toxin 50–100 U; no bladder neck incision in women.

1 · Definitions and symptoms

Drugs and surgery lower outlet resistance but cannot strengthen the detrusor.

- **Detrusor underactivity (DUA)** (International Continence Society [ICS] 2002, urodynamic) is a contraction of **reduced strength and/or duration**, causing prolonged or incomplete emptying. **No thresholds** are given. It is managed mainly by intermittent self-catheterisation (ISC).
- A **"bashful bladder"** fails to void only during the urodynamic study (UDS), unlike a truly **acontractile detrusor**.
- **Underactive bladder (UAB)** (ICS 2018) is the symptom set: slow stream, hesitancy and straining, with or without incomplete emptying. It has **no pathognomonic symptom** and overlaps with overactive bladder (OAB) and bladder outlet obstruction (BOO).
- A significant postvoid residual (PVR) is not agreed; one proposal is over **40%** of functional capacity (voided plus PVR).

Phase	Symptoms
Voiding	Weak stream, intermittency, hesitancy, straining
Post-micturition	Incomplete emptying
Storage	Frequency and nocturia, **or** reduced desire with infrequent voiding
Very large PVR	Incontinence, especially in sleep

> **KEY POINT**
> **Sensation decides the picture.** Absent sensation makes an infrequent voider, with overflow worse at night. Preserved sensation makes a frequent voider with incomplete emptying and nocturia. DUA may be asymptomatic.

2 · Epidemiology

- BOO is rare in women, so low flow with high PVR in a woman suggests DUA.
- UDS series find DUA in men under 50 in **9–28%**, men over 70 in ~**48%** and women in 12–45%.
- It peaks in institutionalised elderly people, often as **detrusor hyperactivity with impaired contractility (DHIC)**: an involuntary contraction emptying under half the bladder.
- Elderly patients often also have detrusor overactivity (DO), BOO or stress incontinence, so do **UDS before lower urinary tract symptom (LUTS) surgery in the elderly**.
- Non-neurogenic male DUA shows **minimal progression** over 10 years.

3 · Aetiopathogenesis

Confirmed causes are neurological disease and diabetes; probable ones are BOO, ageing and atherosclerosis.

Mechanism	Site	Result
Myogenic	Myocyte, extracellular matrix (ECM)	Weak contraction despite normal drive
Efferent	Sacral cord, roots, pelvic nerves	Impaired activation
Afferent	Peripheral afferents, spinal columns	**Early termination** of the voiding reflex
Brain	Pontine micturition centre (PMC), periaqueductal grey (PAG), limbic, prefrontal	Failed integration

- Ageing contributes through ECM fibrosis and falling afferent sensation; in the **"stiff balloon" hypothesis**, afferent drive fades after a small void, so the contraction stops early and leaves a PVR.
- **Drug causes are reversible**: anticholinergics (including antihistamines, antipsychotics, tricyclics, antiparkinsonian drugs) and **opioids**.
- †**Fowler syndrome** (young women, non-relaxing sphincter) responds to sacral neuromodulation.
- **BOO** causes **decompensation** in animals, but most men in acute retention have preserved detrusor function.
- **Diabetes**: early diuresis causes hypertrophy and storage symptoms; late damage impairs sensation and emptying.

Condition	DUA	Key point
Stroke	~Half retain acutely	"Cerebral shock"; chronic sequel is DO
Parkinson disease	<20%	DO commoner; DUA drug-related
Multisystem atrophy	**52–95%**	Efferent parasympathetic atrophy; mistaken for Parkinson
Multiple sclerosis	~20%	Lumbosacral plaques
Radical pelvic surgery	Hysterectomy: ~72% LUT dysfunction	Nerve-sparing reduces it; many recover by 1 year
Guillain–Barré, zoster	—	Reversible

4 · Diagnosis

Only a pressure-flow study proves a weak detrusor.

- **Invasive UDS is the only accepted method**. There are no consensus criteria, and nearly all measure strength only.
- Typically maximum flow (Qmax) and detrusor pressure at Qmax (Pdet@Qmax) are both low; isovolumetric estimates correct for their inverse relation.

Measure	Formula	Thresholds / note
Bladder contractility index (BCI) (Abrams)	Pdet@Qmax + 5Qmax	>150 strong; 100–150 normal; **<100 weak** ; male-derived
Projected isovolumetric pressure (PIP) (Schäfer)	Pdet@Qmax + K·Qmax (K 5 in men, 1 in older women)	<50 very weak; 50–100 weak
Detrusor contraction coefficient (DECO)	BCI/100	<1 weak

- †The **BOO index** is Pdet@Qmax − 2Qmax: over 40 is obstructed and under 20 unobstructed.
- If the patient fails to void on UDS, do **ambulatory UDS**: over four-fifths of apparently acontractile patients contract.

5 · Management

Safe drainage protects the kidneys; no drug reliably helps.

ALGORITHM

1. Evaluate: diary, digital rectal examination, urinalysis, flow, PVR.
2. If there is a neurological deficit, then do spinal MRI.
3. Stop offending drugs; treat constipation.
4. Timed or double voiding.
5. Lower outlet resistance where relevant.
6. If PVR is high, then ISC; if ISC is impossible, then a suprapubic catheter.
7. Selected patients: sacral neuromodulation (SNM), outlet surgery, diversion.

Aim	Routine	Experimental	Not recommended
Drainage	Timed/double voiding; catheterisation; diversion	—	Credé, Valsalva, reflex voiding
Lower outlet resistance	Pelvic floor biofeedback; α-blockers; outlet surgery	Sphincter botulinum toxin	—
Improve detrusor function	SNM; anterior root stimulator	Intravesical prostanoids; intravesical electrotherapy; myoplasty	**Muscarinic agonists; anticholinesterases**

- **Credé or Valsalva** is used only with an incompetent sphincter, as otherwise it causes reflux.
- **ISC** causes urethral bleeding in ~⅓; keep volumes below ~500 mL. Suprapubic is the best long-term indwelling route.
- **Bethanechol and distigmine** cause bronchospasm, diarrhoea, salivation and **rare fatal cardiac arrest**, and are useless with fibrosis.
- **SNM** works best with a sphincter or pelvic floor disorder and may help DHIC, where treating DO worsens DUA and vice versa (see Ch 115).
- The **Brindley** stimulator, combined with rhizotomy, is for complete spinal cord injury. Intravesical electrotherapy had a negative sham-controlled trial.

KEY POINT

Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** to bethanechol, toxin and electrotherapy.

- **Sphincter toxin** is given as 50–100 U cystoscopically (men) or periurethrally (women). About half of idiopathic cases improve; it is short-lived and unlicensed.

	DUA, low PVR, symptomatic	DUA, catheter-dependent
Reasoning	Emptying at low Pdet, so BOO is unlikely to matter	Lowering resistance may allow voiding
Evidence	Transurethral resection of the prostate (TURP) at ~11 years: no benefit over conservative care	Flow rarely normalises, but many void
Verdict	**Surgery unlikely to help**	Consider in young, fit patients

- Outcome in retention is poor with voiding pressure under **45** cm H_2O, age over **80** or PVR over **1500 mL**.
- In women, avoid bladder neck incision because of incontinence and stenosis.
- **Diversion**: a continent catheterisable stoma suits young women with idiopathic retention refractory to SNM; an ileal conduit treats renal deterioration from intramural ureteric obstruction. Reduction cystoplasty is abandoned.

113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

1 · Framework

Drugs follow the Wein scheme (see Ch 104). Most lower urinary tract (LUT) drugs fail on **selectivity**, because the same receptor in heart and vessels caps the dose.

2 · Antimuscarinics and β3-agonists

First-line overactive bladder (OAB) drugs; side effects and age drive choice.

P1 P3 Compare and contrast various detrusor relaxants in management of OAB; OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — M3 vs M2, agent table with doses, β3/cAMP, mirabegron ≥180/110 and CYP2D6 (metoprolol, desipramine, digoxin), vibegron 75 mg, dementia risk, SYNERGY/BESIDE

Not in the paper answer:

- Beers avoidance of oxybutynin; myasthenia and severe colitis as contraindications.
- MILAI II add-on to mirabegron; combination doubles constipation.
- †Antimuscarinics act mainly in **storage**, so retention is uncommon.
- **Oxybutynin** causes the most dry mouth and has the most central nervous system (CNS) penetration, so avoid it in the elderly (Beers).
- Tolterodine is 2 mg bd or extended-release (ER) 4 mg od. **Darifenacin** is most M3-selective.
- **Trospium** is a renally excreted quaternary amine. Other doses: propiverine 20 mg, imidafenacin 0.1–0.2 mg bd.
- †**Side effects by receptor**: dry mouth, constipation and blurred vision (M3); tachycardia (M2); cognition (**M1**).
- †**Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary or gastric retention, myasthenia gravis, and severe ulcerative colitis or toxic megacolon.
- †In the elderly, cumulative **anticholinergic burden** favours a β3-agonist or non-CNS-penetrant agent. Men with bladder outlet obstruction are safe if postvoid residual (PVR) is low.
- Mirabegron goes from 25 to 50 mg od; avoid it in severe uncontrolled hypertension (≥**180/110** mm Hg). Vibegron needs no titration and had no blood pressure (BP) effect.

ALGORITHM

1. †OAB: behavioural therapy, then an antimuscarinic or a β3-agonist; if elderly or at cognitive risk, then choose the β3-agonist first.
2. If the response is poor, then switch or combine.
3. If refractory, then onabotA 100 U, sacral neuromodulation (SNM) or percutaneous tibial nerve stimulation (PTNS).
4. Last resort: augmentation cystoplasty.

3 · Combination therapy

Direction	Evidence	Agents
Mirabegron added to failing antimuscarinic	SYNERGY, SYNERGY II, BESIDE, MILAI	**Solifenacin only**
Antimuscarinic added to failing mirabegron	MILAI II (52 weeks)	Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg
Vibegron plus antimuscarinic	Phase IIb (vibegron 100 plus tolterodine 4 mg)	Extrapolated; no trial after monotherapy failure

KEY POINT

In non-responders, combination **equals** solifenacin 10 mg with less dry mouth: it gains **tolerability, not efficacy**. Heart rate, BP, QT and PVR are unchanged; constipation roughly doubles.

- An **α-blocker plus desmopressin** suits refractory nocturia in benign prostatic hyperplasia (BPH), on uncontrolled evidence.
- Adding **vaginal oestrogen** is **not established**; †systemic oestrogen worsens urinary incontinence (UI).

4 · Intradetrusor onabotulinumtoxinA (onabotA)

The main third-line option.

P1 Botulinum toxin in urology: mechanism, indications and doses — SNAP-25 cleavage, 100 U/20 sites OAB (EMBARK), 200 U/30 sites NDO (DIGNITY), PVR at 2 weeks, UTI, myasthenia

Not in the paper answer:

- Repeat ≥12 weeks apart; aminoglycoside caution.
- The dose is **100 U** for idiopathic OAB and **200 U** for neurogenic detrusor overactivity (NDO).
- †Patients must accept clean intermittent catheterisation (CIC) first, especially in non-catheterising NDO.
- The commonest adverse event (AE) is urinary tract infection (UTI). The effect lasts ≈6–9 months.

5 · Nocturia: desmopressin †

P3 Desmopressin (DDAVP) — V2 agonist for nocturnal polyuria; hyponatraemia risk age ≥65; sodium at baseline, 1 week, 1 month; contraindications incl. loop diuretics

- Low-dose sublingual desmopressin is 25 µg in women and 50 µg in men; hyponatraemia risk rises at age ≥**65**.

6 · Increasing outlet resistance (SUI)

No drug is standard for stress urinary incontinence (SUI).

†Pelvic floor muscle training and surgery are first-line.

Drug	Mechanism	Status
Duloxetine †	Serotonin–noradrenaline reuptake inhibitor at **Onuf's nucleus**; raises pudendal tone	40 mg bd; nausea causes discontinuation; Europe, not USA
Phenylpropanolamine, midodrine, norepinephrine	Systemic α-agonists	Weak evidence; BP rises first; phenylpropanolamine banned (stroke)
Imipramine †	Anticholinergic plus α-like outlet effect	Mixed UI, enuresis; orthostatic hypotension, arrhythmia; caution in elderly
Clenbuterol, vaginal phenylephrine gel	β-agonist; local α-agonist	Abandoned

7 · Facilitating emptying

- †**Bethanechol** has **no convincing controlled evidence** in detrusor underactivity and causes cholinergic AEs; avoid in asthma, peptic ulcer, bradycardia and obstruction.
- †**α-blockers** help in BPH, primary bladder neck obstruction and neurogenic LUT dysfunction, and aid catheter trials.
- †Sphincteric onabotA is an option for the striated sphincter.
- †**Drugs that cause retention** include anticholinergics, pseudoephedrine, opioids, benzodiazepines, anaesthesia and epidurals; stopping them helps more than adding any drug.

8 · Failed drugs

Agent	Indication	Reason for failure
Vitamin D3, elocalcitol; calcium antagonists	Urgency UI, OAB	No urodynamic effect
K^+ channel openers; URO-902 gene therapy	OAB	Dose capped by heart; phase 1 only
RimabotulinumtoxinB	Toxin A-refractory DO	Short duration (≤3 months)
Intraprostatic onabotA, PRX302, NX-1207	BPH lower urinary tract symptoms	No better than sham injection

9 · Landmark trials

Trial	Design	Result	Take-home
SYNERGY / II	Phase III, 12 weeks / 12 months	Solifenacin 5 plus mirabegron 50 better than monotherapy for UI	Effective, safe long-term
BESIDE	Solifenacin 5 non-responders	Add-on better than solifenacin 5, equal to solifenacin 10	Alternative to dose escalation
MILAI II	52 weeks, mirabegron non-responders	Four antimuscarinic add-ons safe, effective	Add antimuscarinic to mirabegron
DIGNITY	Pooled RCTs, NDO	OnabotA 200 U equal in anticholinergic users/non-users	Efficacy independent of anticholinergic
EMBARK	RCTs, refractory OAB	OnabotA 100 U effective	Standard idiopathic OAB dose

115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

P4 P3 Neuromodulation for voiding dysfunction, including sacral neuromodulation; Sacral neuromodulation — afferent/forebrain mechanism, FDA indications table, PNE vs staged lead, S2/S3/S4 responses, ≥50% test, InSite, ROSETTA, PTNS technique, SUmiT, TTNS, implantable tibial, pudendal, DGNS, IVES, Brindley

Not in the paper answer:

- Urodynamic DO does not predict success; PNE gives false negatives.

1 · Evolution and mechanism

One device treats both overactivity and retention.

- Direct detrusor stimulation failed, so stimulation now uses an **indirect route via afferents**.
- Milestones MCQ : sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); percutaneous tibial nerve stimulation (PTNS) derives from acupuncture point **SP6**.
- **Current theory**: sacral afferents reach forebrain awareness areas, which reduces pontine micturition centre (PMC) firing in overactive bladder (OAB); in underactive bladder, raised filling awareness lets the PMC switch on.
- **Paradox**: in retention it works mainly by **striated sphincter relaxation**, not detrusor contraction.

2 · Sacral neuromodulation

- **Third-line** for idiopathic OAB (American Urological Association/Society of Urodynamics [AUA/SUFU] and European Association of Urology [EAU] agree).
- First get a frequency–volume diary and exclude anatomical causes, urinary tract infection (UTI) and a significant postvoid residual (PVR). Urodynamic detrusor overactivity (DO) **does not predict** success.
- In **non-obstructive urinary retention (NOUR)**, PVR is the only objective outcome. Fowler syndrome predicts response; age lowers the odds.

Food and Drug Administration (FDA) approved	Not approved
Urgency urinary incontinence (UUI)	Bladder pain syndrome
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder

- †Faecal incontinence is also approved.
- Test success is ≥**50%** improvement in ≥1 diary parameter against a 3–5-day baseline; in retention, catheterisation falls ≥50%, ideally to none.
- The **S3 target** is ≈9 cm above the coccyx tip, 1–2 cm lateral. S3 gives anal "bellows" and **great-toe flexion**; S2 gives calf or foot rotation; S4 is perineal only.
- Percutaneous nerve evaluation (PNE) is an office test with a temporary lead for 5–7 days; lead migration lowers conversion, and it is not used in NOUR.
- The staged approach places a permanent tined quadripolar lead under fluoroscopy, tested for up to 4 weeks, with the generator in the buttock at stage 2.

- Programming is 210 µs, 10–16 Hz, at sensory threshold. Cycling equals continuous, and **bilateral is not superior** (randomised controlled trial, RCT).
- Battery life is ≥ **10 years** (non-rechargeable) or ≥15 years (rechargeable). Devices are whole-body **MRI**-conditional at **1.5 and 3 T**.
- Complications: revision ~3% and explantation ~9% at 2 years (ROSETTA). †Site or leg pain is reprogrammed, then revised; infection needs explant.

3 · Tibial nerve stimulation

ALGORITHM

1. Supine; insert a 34-gauge needle ≈3–4 cm above the medial malleolus, behind the tibia, at 60–90°; ground pad on the same leg.
2. Stimulate at 20 Hz, 200 µs, 0–10 mA.
3. Confirm **great-toe flexion/toe fanning** or sole tingling.
4. Give **12 weekly sessions × 30 min**, then taper to maintenance.

- Indications are drug-refractory or intolerant OAB; it is safe in neurogenic OAB and in children.
- **Transcutaneous tibial nerve stimulation (TTNS)** uses home surface electrodes; it is cheaper but less effective than PTNS, and equals tolterodine with fewer side effects.
- Success is a ≥50% fall in urgency or UUI, and **relapse is likely without maintenance**. Results are better without DO.

4 · Other targets

- **Pudendal** stimulation (posterior approach) beat SNM in a blinded crossover and is salvage after failed SNM; it is not FDA approved.
- Dorsal genital nerve, sacral dermatome and intravaginal stimulation have limited roles and low-quality evidence.
- †**Intravesical stimulation** suits neurogenic underactive bladder with incomplete lesions, since afferents must be intact.

5 · Sacral anterior root stimulation (Brindley)

It restores voiding in complete spinal cord injury.

- Anterior root stimulation gives voiding, defecation and erection. It is combined with **posterior sacral rhizotomy**, which abolishes reflex incontinence and autonomic dysreflexia, raises capacity and compliance, and protects the kidneys.
- †Voiding is "post-stimulus" because the sphincter relaxes faster than the detrusor. †Rhizotomy costs reflex erection, reflex ejaculation and residual sacral sensation.

6 · Landmark trials

Trial	Design	Result	Take-home
SUmiT	RCT, PTNS vs validated sham, 12 weeks, n = 220	Responders 55% vs 21%	First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years

Trial	Design	Result	Take-home
InSite	RCT, SNM vs standard medical therapy, mild–moderate OAB	Success 61% vs 42% at 6 months	SNM beats continued drugs
ROSETTA	RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women	Toxin slightly fewer leaks at 6 months; equal at 2 years; more UTI, some clean intermittent catheterisation (CIC)	Both valid third-line; registered OAB toxin dose is 100 U

★ Putting it together: voiding dysfunction decisions

A · Functional classification as treatment map

	Bladder	Outlet
Failure to store	Overactivity, low compliance: drugs, onabotulinumtoxinA, neuromodulation, augmentation	Sphincter weakness: pelvic floor training, duloxetine, sling, artificial sphincter, bulking
Failure to empty	Underactive, acontractile: clean intermittent catheterisation (CIC); sacral neuromodulation (SNM) for non-obstructive retention	Obstruction or dyssynergia: α-blocker, outlet surgery, sphincterotomy, CIC

If neither is fixable, use a catheter or diversion (see Ch 104).

B · Lesion level and urodynamics

Lesion	Detrusor	Striated sphincter	Key risk
Above pons (stroke, Parkinson disease, dementia)	Overactive	Synergic	Urgency incontinence; low upper-tract risk
Pons–sacral cord (spinal cord injury [SCI], multiple sclerosis)	Overactive	**Dyssynergic**	High pressure damages upper tracts; autonomic dysreflexia (AD) if above T6
Sacral, cauda equina, peripheral (diabetes, pelvic surgery)	Areflexic, reduced sensation	Fixed tone	Retention, overflow

KEY POINT

A leak point or storage pressure ≥40 cmH_2O, or poor compliance, threatens the kidneys and must be treated (see Ch 107, 109).

C · OAB ladder

ALGORITHM

1. Behavioural measures, bladder training and pelvic floor training.
2. Antimuscarinic or β3-agonist (β3 in the elderly). If partial response, then combine (see Ch 113).
3. If refractory, then onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), tibial nerve stimulation or SNM (see Ch 115).
4. Last resort: augmentation or diversion.

D · SCI bladder

ALGORITHM

1. **Spinal shock**: catheter until reflexes return.
2. **Then** urodynamics; CIC plus antimuscarinic or toxin for high-pressure overactivity.
3. If CIC is impossible, then sphincterotomy or diversion; survey upper tracts lifelong.
4. **AD** (above T6): hypertension, bradycardia, headache, flushing. Sit up, remove the stimulus and give a short-acting antihypertensive (see Ch 109).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 8

Nocturia & Male Incontinence

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 112	Nocturia
Ch 123	Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings
★	Putting it together

112 Nocturia

1 · Definition and terminology

- **Nocturia** (International Continence Society, ICS, 2019) is the number of voids during the **main sleep period**, from intending to sleep until intending to rise. The period follows the **individual's own schedule**, so shift workers are included.
- The definition does **not exclude 1 void per night** and does **not incorporate bother**.
- ICS 2019 dropped "wake **to void**"; non-urological causes are **leverage for interdisciplinary care, not grounds to exclude**.
- Guidance is from the European Association of Urology (EAU) and ICS; the AUA overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

Key diary terms

Term	Definition / formula	Meaning
Actual nightly voids (ANV)	Voids in the main sleep period	Severity count
First uninterrupted sleep period (FUSP)	Hours from sleep onset to the first night void	Key sleep measure
Nocturnal urine volume (NUV)	Night voids **plus the first morning void**	Volume made during sleep
Nocturnal polyuria index (NPi)	NUV / 24-h volume	Proportion made at night
Nocturnal urine production (NUP)	NUV / hours of sleep	Hourly rate
Maximum voided volume (MVV); nocturnal MVV (NMVV)	Largest void in 24 h; largest in the sleep period	Proxies for functional capacity
Nocturia index (Ni)	NUV / MVV	>1: night output exceeds capacity
Modified Ni (MNi)	NUV / NMVV	Uses night data only
Predicted nightly voids (PNV)	Ni − 1	Voids expected if the bladder filled fully each time
Nocturnal bladder capacity index (NBCi)	ANV − PNV	>0: patient voids below capacity

Trap: the first morning void is **split**. Its volume goes into NUV, but it counts as daytime frequency, not nocturia.

2 · Why nocturia matters

- **It is common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate.
- **Bother** is reported by most people with ≥**2** voids, but **any nocturia that brings a patient to a urologist is clinically significant**.
- **Sleep**: nocturia is the **commonest cause of adult sleep disruption**. Most first voids occur within **2–3 h** of sleep onset, breaking early slow-wave (N3) sleep and causing daytime fatigue.
- **Falls and hip fracture** are linked to nocturia more than to other lower urinary tract symptoms (LUTS).

- **Systemic disease**: nocturia may herald cardiometabolic (strongest), renal, endocrine or neurological disease; ≥2 voids is linked to higher mortality.

> **KEY POINT**
> The first void breaks restorative slow-wave sleep; the symptom may be the visible edge of systemic disease.

3 · Pathophysiology — production, capacity and sleep

Remember **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD) and **M**edications.

- Every factor upsets **night production**, **functional capacity** or **sleep**; the sleep link is **bidirectional** ("convenience" voids).

Nocturnal polyuria (NP) and global polyuria

Criterion	Threshold	Comment
NPi (ICS)	> **20%** in young adults to > **33%** in older adults	Most used, but discriminates poorly
NUP (ICS alternative)	> **90 mL/h**	More specific for production-driven nocturia
Global polyuria	> **40 mL/kg/24 h** (>2.8 L at 70 kg)	Uncommon but serious; **always investigate**

- **Krimpen study**: NPi >33% was found in 92% of older men with ≥2 voids, but also in 70% of men **without** nocturia.
- Excess night production contributes to **most adult nocturia**.

Functional capacity and the mismatch

- Normal capacity is usually >300–400 mL; **150–300 mL** suggests diminished capacity, which may be **24-hour** (all voids small) or **night-specific** (partly convenience voids).
- **Ni is the strongest diary determinant of severity and of treatment response**.

> **ALGORITHM**
> 1. Worked example:
> 2. NUV is 800 mL and MVV is 200 mL, so Ni = 4.
> 3. PNV = Ni − 1 = 3.
> 4. The diary shows 6 night voids, so NBCi = 6 − 3 = **3**: the patient voids well below capacity.

Investigate diminished night capacity when NBCi is > **1.3**; a value > **2** is strongly associated with severe nocturia.

4 · The bladder diary

EAU (strong): use a diary for female LUTS, and for male LUTS with storage symptoms or nocturia.

ALGORITHM

1. Give the patient a measuring container.
2. Record time and volume of every void and every drink.
3. Mark bedtime (intending to sleep) and rising time.
4. Grade urgency and note leakage for each void.

- International Consultation on Incontinence (ICI) urgency codes: **0** no need (social void), **1** normal desire, **2** urgency that passed, **3** urgency but dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; questionnaires overestimate it. **FUSP is the most important sleep measure**.
- Duration: EAU advises **3 days** (Male guideline: at least 3); because of "diary fatigue", fewer than half complete it.
- A **nocturnal-only diary** lets MNi substitute for Ni but **cannot diagnose global polyuria**; EAU does not recommend it for women.

KEY POINT

Every cause works through production against capacity, gated by sleep; their ratio (Ni) predicts severity better than either alone.

5 · Evaluation — the Sleep CALM work-up

Do a basic work-up in all, then tailored tests and referral. The **TANGO** questionnaire screens all domains.

Factor	History / examination	Tests (basic, then ancillary)
Sleep	Habits, apnoea, restless legs, pain, mood	STOP-BANG and sleep diary; then polysomnography, sleep specialist
Comorbidities	Blood pressure, volume overload, neurological signs	Metabolic panel; then HbA1c, ECG, echo, natriuretic peptide, urine albumin, calcium, thyroid, water deprivation
Actions	Fluid, alcohol, caffeine, sodium; body mass index (BMI)	Fluid intake diary
LUTD	LUTS, prior surgery; abdominal, genital, rectal examination	Urinalysis, post-void residual (PVR); then prostate-specific antigen (PSA), uroflowmetry, cystoscopy
Medications	Reconcile drugs and **dose timing**	—

- **Trap**: urology is one row of five; first-line is to **optimise every Sleep CALM abnormality**.

6 · Sleep disorders

Obstructive sleep apnoea (OSA) is the best-described link; frequent nocturia predicts severe OSA.

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Production (main)	Hypoxia causes pulmonary vasoconstriction, raising right atrial transmural pressure. This releases **atrial natriuretic peptide (ANP)**, causing natriuresis.	Lowers night production

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Capacity	Intermittent hypoxia causes oxidative stress and detrusor instability	Less detrusor overactivity
Sleep	More light sleep, less slow-wave sleep	Increases slow-wave sleep

- CPAP reduces nocturia by up to ≈**2 voids** in moderate–severe OSA, among the **largest effects of any intervention**. EAU Female (strong): refer suspected OSA to a sleep clinic.
- **STOP-BANG** scores Snoring, Tired, Observed apnoea, Pressure, BMI >**35**, Age >**50**, Neck >**40 cm** and Gender male. A score ≥**3** means high risk; the tool is highly sensitive.
- Ask the **primary motive for getting up** (urge, or already awake). Melatonin and insomnia therapy may help.

> **KEY POINT**
> OSA raises production, lowers capacity and fragments sleep — hence one of the largest nocturia responses to treatment.

7 · Comorbidities

- Comorbidities act mainly through **excess night production**. Nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary, diabetic or early kidney disease.
- EAU: consider systemic causes in all; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload** (e.g. heart failure): leg fluid is mobilised on lying down and excreted at night.
- **Hypertension**: blood pressure normally dips **10–20%** in sleep. **Non-dipping**, from renin–angiotensin–aldosterone dysregulation, causes **nocturnal natriuresis**.

Disorders of water homeostasis

Arginine vasopressin (AVP) acts on collecting-duct **V2 receptors**; defects give **24-hour polyuria**.

Condition	Defect	Causes
Central diabetes insipidus (DI)	Reduced AVP secretion	Brain tumour, haemorrhage, trauma, autoimmune
Nephrogenic DI	Kidney insensitive to AVP	Drugs, electrolytes, renal failure
Primary polydipsia	AVP suppressed by high intake	Idiopathic; psychiatric

Arginine-stimulated **copeptin** appears superior to water deprivation. Always evaluate global polyuria.

> **KEY POINT**
> Comorbidity acts through the kidney (third-spaced fluid, nocturnal natriuresis, failed antidiuresis). Global polyuria and non-bothersome nocturia both warrant looking beyond urology.

8 · Actions — lifestyle and behaviour

- **Timing matters more than amount**. Water causes diuresis within ≈1 h; caffeine and alcohol within 2–3 h, and they disturb sleep.
- Restrict fluid, caffeine and alcohol from **3–4 h** before sleep; older adults should drink more by day.
- Also: **sodium** restriction, weight loss, exercise, **compression stockings**, bladder training, pelvic floor muscle training (PFMT), treat constipation.

9 · Lower urinary tract dysfunction

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), underactivity, neurogenic bladder and menopause syndrome See Voiding Dysfunction book, Ch 110.
- **Isolated nocturia is not a LUTS**, so it is not expected to respond to bladder or prostate drugs.
- Do urinalysis in all, and **PVR in both sexes**. Check PSA only if it changes management. †Urodynamics are not routine See Voiding Dysfunction book, Ch 107.

Benign prostatic obstruction

- In benign prostatic hyperplasia (BPH) trials, **α-blockers** carry most of the nocturia effect; 5α-reductase inhibitors (5ARIs) alone are weakest. Gains are fractions of a void with a large placebo response: **statistical, not clinical, significance** See Voiding Dysfunction book, Ch 113.
- **Surgery**: TURP beat microwave therapy, α-blockers and watchful waiting; **counsel** that nocturia may respond sub-optimally.

Overactive bladder

- Tolterodine was negative for night voids but reduced urge-related voids; **fesoterodine** (nocturnal urgency endpoint) was positive.
- Patients with a **raised NBCi** respond best to an evening dose. β3-agonists give a marginal benefit (≈0.1–0.2 voids over placebo).
- **Behavioural therapy** comes first and matches or beats drugs. Antimuscarinics are on the **Beers Criteria** for people ≥65 years, the core nocturia population.

Therapy	EAU Male LUTS	EAU Female LUTS
α1-blocker	Offer for nocturia with LUTS (weak)	—
5ARI	Offer with LUTS and prostate > **40 mL** (weak)	—
Phosphodiesterase-5 (PDE5) inhibitor	**Do not offer**	—
Antimuscarinic	Offer with OAB (weak)	Offer after counselling (**strong**)
PFMT; vaginal oestrogen	—	Offer PFMT (**strong**); offer oestrogen (weak)

KEY POINT

LUT drugs act on capacity and outflow, but most nocturia is production-driven — effects are statistically significant, clinically small. Treat the mechanism, not the organ.

10 · Medications causing nocturia

Mechanism	Drugs
Increased urine output	Diuretics; SSRIs (block antidiuretic hormone); **calcium channel blockers** (oedema, raised ANP); tetracycline; lithium (reduces aquaporin-2); **SGLT-2 inhibitors** (glycosuria)
Insomnia / central nervous system effects	Stimulants; α- and β-blockers; salbutamol, theophylline; corticosteroids, thyroxine; SSRIs; phenytoin
Bladder toxicity	Ketamine; tiaprofenic acid; cyclophosphamide

Adjust dose timing first, or substitute another drug.

11 · Management algorithm and refractory nocturia

ALGORITHM

1. Take a history, examine, and obtain a diary and basic tests.
2. Identify the Sleep CALM factors.
3. Optimise each factor, with multidisciplinary help.
4. Add conservative measures.
5. If nocturia stays refractory, then consider an antidiuretic or a timed diuretic in selected patients.
6. Repeat the diary at every step.

- Treat interacting causes together (EAU **"Shared Care Pathway"**). **Target night volume even when capacity is small**.

Desmopressin

P3 Desmopressin (DDAVP) — selective V2 agonist, no pressor effect; hyponatraemia risk factors (age ≥65, Na <135, CrCl <60, high dose), most in first 2–3 weeks; EAU <65 / low dose >65; contraindications

Not in the paper answer:

- Desmopressin is the **only US-approved nocturia drug**, for **≥2 voids per night due to NP** (level 1b); some NP patients have a **blunted nocturnal AVP rise**.
- †Also contraindicated with systemic or inhaled glucocorticoids and estimated glomerular filtration rate (eGFR) <50 mL/min.

	Nasal spray	Orally disintegrating tablet (ODT)
Dose	1.66 mcg if <65 with no risk factors; **0.83 mcg** if ≥65 or at risk	**Sex-specific**: women **25 mcg**; men **50 mcg** (75 mcg added hyponatraemia, not efficacy)
Timing	≈30 min before bed	Sublingual, ≈60 min before bed

- **Women are more sensitive**, because the V2 receptor gene is on the **X chromosome**.
- †Advise minimal fluid from 1 h before until 8 h after the dose.
- Hyponatraemia is defined as sodium < **130 mmol/L**; most cases occur within **3 weeks**.
- **Sodium monitoring** (EAU): baseline, **day 3**, day 7, 1 month, then periodically (US label: baseline, within 7 days and ≈30 days). EAU Female: avoid if baseline sodium is low.
- A **low dose** still works and avoids antidiuresis outlasting sleep. **Age 65 is the pivot.**

Timed diuretics

- Furosemide acts for 6–8 h. Evening dosing **causes** NP, whereas **morning or early-afternoon** dosing treats it (level 1b).
- Small trials (bumetanide 4–6 h, or furosemide 40 mg 6 h before bed) cut voids; monitor sodium.
- **Trap**: afternoon furosemide with bedtime desmopressin worked in a trial, but loop diuretics are a **contraindication** to desmopressin in US labelling.

> **KEY POINT**
> Refractory nocturia: move urine production out of the night. The price is sodium balance — dose, age and monitoring define safe use.

12 · Landmark trials

Trial	Design	Result	Take-home
MTOPS	Doxazosin, finasteride, combination, placebo	Doxazosin and combination beat placebo; finasteride did not	α-blocker drives benefit
VA Cooperative	Terazosin, finasteride, combination, placebo	Only terazosin separated from placebo	5ARI alone does not help
CombAT	Tamsulosin, dutasteride, combination	Combination −0.5 vs −0.3/−0.4 voids	Significant but small
Fesoterodine RCT	Nocturnal urgency primary endpoint	Positive	Helps urgency-driven night voids
Nasal spray phase III	0.83/1.66 mcg vs placebo	Both beat placebo	Basis of approval
ODT sex-specific RCTs	Women 25; men 50/75 mcg	Positive; 75 mcg caused more hyponatraemia	Sex-specific dosing

123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

1 · Pathophysiology and device mechanisms

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters. Radical prostatectomy (RP), the commonest cause, removes the internal sphincter, leaving the rhabdosphincter.
- Other causes: radiotherapy (RT), pelvic fracture, neuropathy, vesicourethral anastomotic stenosis (VUAS) treatment. See Voiding Dysfunction book, Ch 106.

Device	Mechanism	Site	Adjustable	Needs residual sphincter
Artificial urinary sphincter (AUS, AMS 800)	Circumferential compression, cycled by the patient	Bulbar urethra or bladder neck	No	No
Advance / Advance XP	**Repositions** and lengthens the urethra; no compression	Bulbar, transobturator	No	**Yes**
Virtue (quadratic)	Prepubic compression and elevation	Bulbar, four arms	No	Less dependent
ATOMS (adjustable transobturator male system)	Cushion compression	Bulbar, transobturator	Yes, via scrotal port	Less dependent
ProACT	Two para-urethral balloons	Bladder neck	Yes, via scrotal ports	—

- The **AUS** has a cuff, a scrotal pump and a pressure-regulating balloon (PRB) that sets cuff pressure. The cuff refills passively over minutes.
- A transobturator sling raises leak point pressure **without obstruction** by elevating the bulb and rhabdosphincter.

> **KEY POINT**
> Every device compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but compress — only the Advance depends absolutely on a coapting residual sphincter.

2 · Indications and device selection

- Indication: **irreversible sphincter incompetence** with bothersome incontinence (UI) after conservative care. Do not delay surgery beyond **6 months** for severe SUI.
- 24-hour pad weight grades severity: mild **<150 g**, moderate **<400 g**. AUS is preferred if >250 g or after RT (MASTER trial See Ch 106).

Stratum (AUA/SUFU)	Recommendation
Severe UI or **any RT**	AUS first-line; sling only with counselling

Stratum (AUA/SUFU)	Recommendation
Moderate, no RT	AUS or male sling; adjustable balloons in specialised centres
Mild, no RT	Continue pelvic floor muscle training (PFMT), sling, possibly AUS

- Counsel: ~1 thin pad/day, AUS lifespan **7–10 years**; failed slings mostly go to AUS.
- **A sling needs** mild–moderate SUI, a contractile bladder (if hypocontractile, choose AUS), a non-irradiated urethra and a coapting sphincter.
- **An AUS needs** manual dexterity and cognition, a healthy urethra and adequate bladder compliance.

Absolute — need an alternative	Relative
Low compliance threatening the kidneys; low capacity; urethra too poor (alternatives: clamp, catheter, bladder neck closure, diversion)	Frequent transurethral work (bladder cancer, refractory VUAS, radiation cystitis), because it risks erosion and infection

- Radiation cystitis needs hyperbaric oxygen and **6 months** free of bleeding before AUS.

> **KEY POINT**
> Each device's contraindication mirrors the other's requirement.

3 · The high-risk ("fragile") urethra

- Strong erosion risk factors are **prior RT, prior erosion or infection, and urethroplasty** (worse than RT, as transection divides the bulbourethral arteries).
- After a prior erosion, 5-year repeat erosion approaches **50%**.
- Countermeasures: optimise medically, use a **transcorporal (TC) cuff**, a PRB of **51–60 cm H2O**, a cuff 0.5 cm larger, and nightly deactivation.

> **KEY POINT**
> Erosion = spongiosal ischaemia under constant pressure. Countermeasures add tissue (TC tunica) or lower pressure — never squeeze harder.

4 · Operative technique

- Sterile urine; give IV antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg plus a first/second-generation cephalosporin or vancomycin.
- Place the cuff on the **proximal bulbar urethra, proximal to corporal fusion**, where spongiosum is thickest.

> **ALGORITHM**
> 1. Midline perineal incision; divide bulbospongiosus.
> 2. Incise Buck fascia and make a 2-cm tunnel around the urethra.
> 3. Measure and place the cuff (usually 4.0–4.5 cm).
> 4. Place the PRB preperitoneally and fill it with **23 mL**.
> 5. Place the pump subdartos in the anterior scrotum.
> 6. Cycle the device; do urethroscopy.
> 7. Leave it **deactivated for 4–6 weeks**.

- Less tight cuffs continent better; a **3.5-cm cuff** raises erosion after RT. †Deactivate the cuff before any later catheterisation or endoscopy.

PRB (cm H2O)	Use
51–60	High-risk bulbar (RT, prior erosion, atrophy); trades continence for safety
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Tandem cuff**: a second cuff ≥1.5–2 cm distal; not first-line, because the distal cuff erodes more.
- **Transcorporal cuff** (Webster 2002): ventral tunica stays on the dorsal urethra and adds bulk. Use it for a urethra <3.5 cm, revisions, prior erosion or urethroplasty; erosion benefit is unproven.
- **Bladder neck AUS** suits an intact prostate (neuropathic bladder) and is **contraindicated after RP**; cuff ≥8 cm, PRB 71–80.

ALGORITHM

1. **Advance XP**:
2. Perineal incision; open bulbospongiosus.
3. Divide the perineal body to release the bulb.
4. Pass the helical trocar outside-in below the adductor longus tendon.
5. Fix the mesh to the spongiosum.
6. Tension under cystoscopy until the mucosa coapts (bulb moves 3–4 cm proximally).

- **ATOMS** leaves bulbospongiosus **intact** and is adjusted in clinic via the scrotal port; **ProACT** balloons are placed at the bladder neck under fluoroscopy.

5 · AUS complications and management

ALGORITHM

1. **Retention**:
2. Confirm the cuff is deactivated.
3. Insert a small catheter for 24–48 h.
4. If the second trial fails, then place a **suprapubic cystostomy**, avoiding the PRB.
5. If retention lasts weeks, then the cuff is undersized: replace it.

- **Infection** occurs in **1–3%** (up to 10% if high-risk), usually staphylococci; **scrotal pain is the key early sign**. If antibiotics fail, explant all; salvage only if not eroded, septic or immunosuppressed.
- **Erosion** occurs in **2–15%** (up to 35% with several risk factors). Remove **all components**. If <50% of the circumference is lost, leave a catheter ~3 weeks; if >50%, do in situ repair or anastomosis.
- Reimplant after **3–6 months** at a **new site**, using a TC cuff with or without a low-pressure PRB.
- **Atrophy** causes a **gradual** recurrent leak. Downsize, relocate (TC cuff distally) or add a tandem cuff; **never simply raise PRB pressure**.
- **Mechanical failure**: within 3 years, replace the failed part; **after 3 years, replace the entire device**.

Recurrent leak after AUS	Clue, then how to confirm
Deactivated	Sudden; reactivate
Fluid leak	Sudden; PRB shrinks on imaging
Atrophy	Gradual; PRB volume normal (>20 mL)
Erosion / detrusor overactivity	Cystoscopy / urodynamics

KEY POINT

Sudden = device, gradual = urethra. Manage every complication to keep a salvageable urethra for the next cuff.

6 · Sling complications and failure

Device	Commonest	Erosion / explant
Transobturator	Transient groin pain; retention or urgency 4–10%	0–0.4%
ATOMS	More complications	Erosion/infection ~10%; explant 10–16%
ProACT	**Perforation** (~5%); migration; revision 22%	~4%

- Cystoscope every worsening sling. After a failed sling the **AUS is the gold standard**, with results as good as primary AUS.

7 · Special circumstances

- A stricture at the cuff may herald erosion.
- With a penile prosthesis, synchronous and staged placement do equally well; if staged, place the AUS first. **Climacturia** responds to a sling or AUS, adding a **mini-jupette** graft if there is erectile dysfunction.

8 · Outcomes

P2 P3 Post-prostatectomy incontinence; SUI: evaluation in females, role of UDS, and surgical therapy in males — AUS social continence 55–77%, satisfaction ~90%, revision 10–29% at 3 years; Advance cure ~60%; ATOMS ~two-thirds dry

Not in the paper answer:

- Cure clusters at 55–65% for all devices (ProACT ~60–67%); Advance XP results are poorer after RT and in severe UI.

★ Putting it together

A · Nocturia: three diary questions, by danger

ALGORITHM

1. **Global polyuria?** If >40 mL/kg/24 h, then investigate for diabetes, diabetes insipidus or polydipsia.
2. **Nocturnal polyuria?** If night fraction >20% (young) or >33% (older), then seek heart failure, sleep apnoea, oedema or evening fluids; consider desmopressin with sodium checks.
3. **Reduced night capacity?** If NBCi >1.3, then treat overactive bladder, obstruction or residual (See Ch 112).

B · Post-prostatectomy incontinence

KEY POINT

Pelvic floor training first; operate on severe leakage by ~6 months. Pad weight: mild <150 g, moderate <400 g. A sling suits mild–moderate leakage, a contractile detrusor and no radiotherapy. Leakage >250 g, radiotherapy or a failed sling needs a sphincter. For erosion risk, add tissue (transcorporal cuff) or lower pressure; never tighten. Remove eroded or infected devices (See Ch 123).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 9

Andrology & Sexual Medicine

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 66	Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System
Ch 67	Male Reproductive Physiology
Ch 68	Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome
Ch 69	Male Infertility
Ch 70	Surgical Management of Male Infertility
Ch 71	Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction
Ch 72	Evaluation and Management of Erectile Dysfunction
Ch 73	Priapism
Ch 74	Disorders of Male Orgasm and Ejaculation
Ch 75	Surgery for Erectile Dysfunction
Ch 76	Diagnosis and Management of Peyronie Disease
★	Putting it together: andrology decisions

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

1 · Scrotum

Each layer is a surgical plane from the abdominal wall.

Layer (superficial to deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele collects between its layers; visceral layer fused to albuginea

- The anterior wall carries the external pudendal vessels and the ilioinguinal and genitofemoral nerves; the posterior wall carries the perineal vessels and nerve. Nothing crosses the raphe.

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — scrotum to ipsilateral superficial inguinal; testis to para-aortic/interaortocaval; cauda to external iliac

> **KEY POINT**
> Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum. †So scrotal violation in testis cancer adds the inguinal nodes to the at-risk field.

2 · Testis

Structure

- Volume is **15–25 mL**, length 4.5–5.1 cm. The capsule is visceral tunica vaginalis, **tunica albuginea** (with smooth muscle) and tunica vasculosa.
- The **mediastinum testis** is the posterior echogenic band on ultrasound (US). The appendix testis is a **müllerian** remnant.
- About 250 lobules hold **600–1200 seminiferous tubules**, which drain to the rete testis.
- The interstitium is **20–30%** of volume and holds the **Leydig cells**. **Sertoli** tight junctions form the blood–testis barrier, aided by endothelium and peritubular myoid cells (see Ch 67).

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral, lost after vasectomy
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Collaterals may save the testis after testicular artery ligation, but atrophy or azoospermia can occur.

KEY POINT

Surgical rules: no traction suture through the lower pole. Biopsy the **medial or lateral upper pole**; the midportion has fewest vessels. After vasectomy the deferential collateral may be gone, so preserve the testicular artery in later cord surgery and counsel about devascularisation.

Veins — pampiniform plexus

The plexus cools the artery by **countercurrent heat exchange** to **2–4 °C** **below rectal** temperature. A varicocele disrupts this.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) on the right; left renal vein on the left
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

P1 Varicocele – Pathophysiology — left vein into renal vein (**~90% of varicoceles left**), oblique right entry valve-like; middle/posterior groups cause recurrence

Not in the paper answer:

- **High (retroperitoneal) ligation** can take artery and vein, since collaterals suffice; **inguinal or subinguinal** ligation takes veins only, because arterial ligation there risks devascularisation.

Lymphatics and nerves

- Dividing lymphatics at varicocelectomy causes a **hydrocele**; microsurgery preserves them.
- Supply is autonomic, with **no somatic innervation**; the genital branch of the genitofemoral nerve supplies the tunica vaginalis.
- The **"trifecta" nerve complex** (perivasal, posterior periarterial/lipomatous, intracremasteric) is the target of microsurgical cord denervation for chronic orchialgia.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz with Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord or inguinal disease; air in Fournier gangrene
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- versus extratesticular; nonpalpable testis (MR angiography)

- If torsion is suspected, **explore**; Doppler must not delay surgery.
- Varicocele is a **clinical diagnosis**. Use US (standing, Valsalva) only if examination is equivocal.

3 · Epididymis

- Caput, corpus and cauda lie on the posterolateral testis. **Principal** cells form the blood–epididymis barrier, and **clear** cells acidify the lumen to keep stored sperm quiescent.
- The testicular artery supplies caput and corpus, and the deferential artery the cauda. Nerves and smooth muscle increase toward the cauda, which shifts from storage to propulsion.

- Wolffian (mesonephric) remnants include the **appendix epididymis**; müllerian (paramesonephric) remnants are the **appendix testis** and prostatic utricle.

4 · Vas deferens

- It runs **posterior** to the cord vessels through the inguinal canal, crosses **anterior to the ureter**, forms the ampulla and joins the seminal vesicle.
- It is **30–35 cm** long with a **0.2–0.7 mm** lumen and has **three muscle layers** (longitudinal, circular, longitudinal).
- About half the cord nerves surround the vas: sympathetic fibres drive emission, sensory fibres cause chronic scrotal pain.
- A **vasogram** can cause obstruction, so do it **only at reconstruction**.

5 · Seminal vesicles and ejaculatory ducts

- Behind the bladder base, they join the ampulla as ejaculatory ducts opening on the verumontanum at **5 and 7 o'clock**.
- They bud from the mesonephric duct alongside the ureteric bud, so **renal anomalies coexist**.
- The fluid is ~70% of semen and alkaline. It carries **fructose**, prostaglandins and **semenogelin**, which forms the coagulum later liquefied by prostate-specific antigen (PSA).
- On **transrectal ultrasound (TRUS)**, a post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** means obstruction. **MRI** is the modality of choice; CT is for stones.

> **KEY POINT**
> A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · Prostate

P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — McNeal zones (peripheral **70%** , transition gives BPH), fasciae, arteries, DVC, lymph, cavernous nerves

Not in the paper answer:

- TRUS volume is length × width × height × **0.52**, which underestimates. The normal gland is **~25 mL**.
- **Multiparametric MRI** is scored by PI-RADS v2.1; targeted biopsy finds less insignificant cancer.

7 · Urethra

P1 Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias — segments (membranous **2–2.5 cm** ; Cowper 3/9 o'clock), epithelium, sphincter (Onuf), blood supply

8 · Penis

Structure

- There are paired **corpora cavernosa** (crura on the pubic arch) and a **corpus spongiosum** (bulb, urethra, glans).

- The septum is incomplete distally, so the corpora act as **one unit** (†one-sided injection fills both).
- The **tunica albuginea** (2–3 mm) has an inner **circular** and an outer **longitudinal** layer; it is thin ventrally.
- **Buck fascia** encloses the corpora and splits around the spongiosum. From midline outward it holds the **deep dorsal vein, then arteries, then nerves**.
- Sinusoidal smooth muscle is **~45%** of volume. Endothelial nitric oxide synthase (NOS) sustains erection.

Arteries

The internal pudendal artery runs through Alcock canal to become the **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally; helicine arteries fill the sinusoids (erection)
Dorsal	Under Buck fascia to the glans; circumflex branches

Veins, nerves

- The superficial dorsal vein drains skin; the **deep dorsal vein** (under Buck fascia) drains to the periprostatic plexus.
- In the distal two-thirds, sinusoids drain via the subtunical plexus and **emissary veins**; the proximal third drains via cavernous and crural veins.
- The **dorsal nerve** (pudendal, **S2–S4**) gives sensation; the **cavernous nerves** (pelvic plexus) give erection.

> **KEY POINT**
> Erection is veno-occlusive: filled sinusoids compress subtunical and emissary veins against the inelastic tunica. †Tunical or smooth-muscle damage defeats this and causes venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

MRI shows fracture, prosthesis malposition and tumour.

67 Male Reproductive Physiology

1 · Hypothalamic–pituitary–gonadal axis

Every hormonal work-up reads this axis.

	Peptide hormones	Steroid hormones
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol
Storage	Secretory granules	Not stored; secretion equals production
Receptor	Cell surface, acting via cyclic AMP (cAMP)	Intracellular, altering transcription; carried bound in plasma

Hypothalamus

- Gonadotrophin-releasing hormone (GnRH) is a decapeptide whose only function is LH/FSH release. Half-life is **5–7 min**.
- Its rhythms are pulsatile (every **90–120 min**), circadian (testosterone peaks early morning) and seasonal.
- **Pulsatility is essential**: a continuous GnRH agonist (leuprolide) desensitises the pituitary, giving castrate testosterone.
- In **Kallmann syndrome** GnRH neurones fail to migrate, which causes hypogonadotrophic hypogonadism. †Anosmia coexists because the neurones migrate with the olfactory placode.

GnRH modulation	Factors
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH and FSH share an **α subunit** with TSH and hCG and have a specific **β subunit**.
- **Hyperprolactinaemia abolishes gonadotrophin pulsatility**, so testosterone falls. Low prolactin is not pathological.

	LH	FSH
Pulses	8–16/24 h; follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol to pregnenolone to testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitative output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) and **inhibin B**

Testicular hormones and feedback

- Testosterone output is ~**5 mg/day**. 5α-reductase converts it to the more potent **dihydrotestosterone (DHT**), needed in most peripheral tissues; aromatase converts it to **oestradiol**, which feeds back especially on FSH.
- FSH makes Sertoli cells produce androgen-binding protein (ABP), transferrin, lactate and inhibin. **Inhibin** suppresses FSH; **activin** stimulates it.

> **KEY POINT**
> There are two loops: LH with Leydig cells (testosterone), and FSH with Sertoli cells (inhibin, oestradiol). So **LH reports Leydig function, while FSH and inhibin B report the tubules**. High FSH with low inhibin B means spermatogenic failure.

Development and ageing

- **SRY** (Yp) starts the testis. **DAX1** and **WNT4** are "antitestis" genes; the female pathway is not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct forms internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent
Target tissue	DHT	External genitalia
Sertoli	Müllerian-inhibiting substance (MIS/AMH)	Müllerian regression; keeps germ cells out of meiosis

- Puberty (~12 years) is triggered by kisspeptin, leptin and melatonin.
- **Ageing**: testosterone is ~one-third lower by the 7th decade and FSH ~threefold higher. SHBG rises and feedback is blunted, so failure is both peripheral and central.

2 · Testis

Architecture and blood flow

- It has **600–1200 tubules** and ~**700 million Leydig cells** in young adults.
- The cord holds one artery in ~half of men, so at microsurgical varicocelectomy ligating a single artery can cause atrophy.
- **Countercurrent exchange** cools arterial blood **2–4 °C** below core and recycles testosterone. It is lost in varicocele and cryptorchidism.

Leydig cells and testosterone synthesis

- **Rate-limiting step**: LH raises cAMP, and **steroidogenic acute regulatory (StAR) protein** moves cholesterol into the inner mitochondrial membrane to become pregnenolone.
- Further enzymes (3β-HSD, 17α-hydroxylase/17,20-lyase, 17β-HSD) make testosterone; their mutations cause 46,XY disorders of androgen biosynthesis.
- †In blood, ~44% of testosterone is bound tightly to SHBG, ~54% loosely to albumin and ~2% is free. Free plus albumin-bound is "bioavailable".

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak in 2nd–3rd decade, then decline	Maintains androgen-dependent organs

Sertoli cells and blood-testis barrier

P3 P1 Spermatogenesis; Male reproductive immunology — Sertoli nurse functions, ABP (intratubular androgen ~ **50-fold** serum), basal/adluminal compartments, anatomical + functional barrier, no antibodies before puberty, chemotherapy sanctuary

Not in the paper answer:

- **Inhibin B** marks Sertoli function; Sertoli tight junctions are the body's strongest, yet open to let spermatocytes pass.

3 · Spermatogenesis

A semen test reflects events about two months earlier.

P3 Spermatogenesis — **~64 days** (42–76); proliferation (Ad, Ap every 16 days, B), then meiosis, then spermiogenesis (acrosome from Golgi, flagellum from centriole), then spermiation; six stages, helical wave; cytoplasmic bridges

Not in the paper answer:

- Output is ~ **123 million/day**, about 1000 per heartbeat.
- †Spermatogenesis needs high intratesticular testosterone, so exogenous testosterone suppresses it.

Stem cells

- Failed migration of primordial germ cells causes extragonadal germ cell tumours or Sertoli cell–only testes.
- "Selfish spermatogonia" (FGFR2/FGFR3 mutations) cause paternal-age disorders and spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- **Defective recombination** occurs in ~half of maturation arrest. Altered crossover causes aneuploidy, so IVF/ICSI offspring have more chromosomal abnormalities.

Genetics

Y microdeletions of the azoospermia factor (AZF) region occur in **15%** of azoospermic and 6–8% of severely oligospermic men, the commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

Paternal age matters because spermatogonia divide ~23 times a year, raising de novo dominant mutations (achondroplasia, Apert, Marfan), autism, schizophrenia and **DNA fragmentation**.

4 · Epididymis

- The 3–4 m duct runs from caput through corpus to cauda. Testicular sperm are **immotile and cannot fertilise**, except by ICSI.
- **Transport** takes **2–12 days**, driven mainly by spontaneous rhythmic contraction.

- **Storage**: the cauda holds about half the sperm; prolonged storage increases DNA fragmentation.
- **Maturation** gives progressive motility and zona binding. **Fertility is reached mostly by the late corpus or early cauda**.

> **KEY POINT**
> Pregnancy occurs even after vasoepididymostomy at the efferent ductules, but **the more distal the anastomosis, the better the fertility**. After vasectomy for >15 years, motility is lower after reversal because of epididymal dysfunction.

- The epididymis is **androgen dependent**, mainly on DHT, which explains impaired semen on 5α-reductase inhibitors.
- **Heat** abolishes storage, and **denervation** (spinal cord injury, diabetes) impairs transport.

5 · Vas deferens, seminal vesicles, ejaculatory ducts

Semen chemistry shows whether these organs are present and open.

- With a **muscle:lumen ratio of ~10:1** the vas is the most muscular hollow viscus.
- The **seminal vesicle** gives ~**80%** of seminal volume in the late fractions. Its fluid is alkaline and carries **fructose** and prostaglandins; **semenogelin I** forms the coagulum.
- The **ejaculatory duct** has no sphincter; reflux is prevented by its **acute insertion angle**.
- Obstruction may be **physical** or **functional** (vesicle myopathy from diabetes or drugs), and static TRUS may not tell them apart.
- An **ejaculate with pH <7.2, low volume and no fructose** means absent or blocked seminal vesicles, as in congenital bilateral absence of the vas (CBAVD) or ejaculatory duct obstruction.

6 · Spermatozoon, capacitation and fertilisation

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in a helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- The axoneme is **"9 + 2"**, and **dynein** slides the doublets. Inner dynein arm mutants have no motility; outer arm mutants have reduced motility.
- **Paternal mitochondria are degraded** in the egg, so mitochondrial inheritance is maternal.
- **Capacitation** in the female tract gives hyperactivated motility and readiness for the **acrosome reaction**.
- Binding to zona **ZP3** triggers the acrosome reaction. Sperm **phospholipase C zeta (PLCζ)** triggers calcium oscillations, which activate the oocyte.

Sperm defects

- Morphology is graded by strict (Kruger) criteria. Head defects rarely reflect DNA quality, except **pinhead** and **multiple-tailed** sperm.
- In **globozoospermia** (round head, no acrosome) fertilisation fails even with ICSI, because of PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (multiple morphological abnormalities of the flagella; fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on electron microscopy (EM)	Short, bent, coiled or absent flagella
Cause	Recessive; respiratory disease, dextrocardia (Kartagener)	Genetic; also varicocele, reactive oxygen species (ROS)
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · Ejaculation physiology †

Each cause of ejaculatory failure hits a specific phase.

ALGORITHM

1. **Emission** (sympathetic, T10–L2 via the hypogastric nerve; α1): vas, seminal vesicle and prostate contract and deliver semen to the posterior urethra.
2. The **bladder neck closes**. If it fails to close, then ejaculation is retrograde.
3. **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions, with a relaxed external sphincter, give an antegrade ejaculate.

P3 Retrograde ejaculation — failed bladder neck closure (α-blocker, diabetes, bladder neck surgery, RPLND) gives retrograde; loss of emission gives anejaculation

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

1 · The men's health centre

The urologist is often the only doctor a man sees.

- Men avoid preventive care but seek help for erectile dysfunction (ED), infertility and voiding, so the urologist is the **front door** to a multidisciplinary centre and triages the whole man.

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular ("rule out hypogonadism"); preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon screening; disease management	Gastroenterology (GI), cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition	Geriatrics, neurology, audiology, ophthalmology

2 · The longevity gap

- Male life expectancy was **76.1** versus 81.1 years for females (2016).
- Males have higher mortality in **12 of 15** leading causes; the largest gap is **suicide (>3-fold)**.
- Prostate cancer is the only US malignancy with rising stage IV at diagnosis, so discuss screening.

3 · Urologic screening

- Screen sexual and voiding complaints together, with PSA and urinalysis per guidelines.

Questionnaire	Screens
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Androgen-deficiency symptoms
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms (LUTS)

4 · Urologic disorders of focus

Testosterone deficiency (TD)

P4 Male menopause (andropause, late-onset hypogonadism) — low T on more than one occasion + symptoms; AUA **<300 ng/dL** on 2 morning samples; LH separates primary/secondary, prolactin if LH low; TRT contraindications; monitor T, haematocrit (**>54%** hold), PSA at 3–6 months then yearly

Not in the paper answer:

- For haematocrit, the AUA withholds TRT at >50% and the Endocrine Society at >54%; then reduce the dose or phlebotomise.
- **Randomised trials show TRT benefit** for obesity, metabolic syndrome, diabetes, muscle and bone, but **TRT is not a treatment for depression**.
- For men on chronic opioids, the Endocrine Society advises screening for TD and considering TRT. Exclude secondary causes before starting TRT.

KEY POINT

† **Fertility caution**: exogenous T suppresses LH/FSH and intratesticular T, which causes oligo- or azoospermia. Men wanting children should avoid TRT and use clomiphene, human chorionic gonadotrophin (hCG) or aromatase inhibitors (see Ch 69).

† TRT	Points
Contraindications	Near-term fertility wish; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe obstructive sleep apnoea (OSA); uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)
Formulations	Gels, injections, pellets, each with its own monitoring plan

Erectile dysfunction

- ED affects >50% of men aged 40–70.
- **Every man with ED, of any age, is at cardiovascular disease (CVD) risk**. †Penile arteries are small, so ED precedes coronary events by ~2–5 years.
- Treatment steps from lifestyle to PDE5 inhibitors (never with nitrates), then injection, intraurethral or vacuum therapy, then prosthesis, which has the highest satisfaction. Platelet-rich plasma and shock wave therapy remain in trials.

BPH and infertility

- Refer BPH when medication fails, is not tolerated or is not wanted.
- Infertility is no pregnancy after 1 year; a male factor is present in ~half (see Ch 69).

5 · Cardiovascular prevention

- Only **lowering low-density lipoprotein cholesterol (LDL-C)** benefits primary prevention; raising HDL or lowering triglycerides does not. Screen lipids from age 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin with or without a nonstatin
LDL-C 70–189, 10-year risk ≥ **7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or PCSK9 inhibitor

- USPSTF, NICE and ESC start statins at ≥10% risk.
- A **coronary artery calcium (CAC) score** is most useful at **intermediate risk**; zero means <1% atherosclerosis.

Blood pressure (BP) (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Elevated or stage 1 BP gets lifestyle change. Drugs are for stage 2, or stage 1 with age ≥65, T2DM, CVD, chronic kidney disease or 10-year risk ≥10%.

6 · Obesity and metabolic syndrome

Fat lowers testosterone, and low testosterone adds fat.

- Body mass index (BMI): overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40.
- Advise weight loss at BMI ≥30, or 25–29.9 with CVD risk, OSA or osteoarthritis; GLP-1 agonists are approved.
- †Metabolic syndrome (ATP III) needs ≥3 of: waist >102 cm, triglycerides ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, glucose ≥100 mg/dL. Adipose aromatase raises oestradiol, which lowers LH, so low T and metabolic syndrome drive each other.

7 · GI, skin and mental health

- Evaluate **constipation** in every man with UTI or retention. Colorectal screening runs from **45–75** years.
- Iron deficiency anaemia carries a >30-fold GI malignancy risk and needs investigation.
- Psoriasis is linked to MI, and suicide is much higher in men.

69 Male Infertility

1 · Definition and overview

Half of infertility involves the man, so always examine him.

- Infertility is no pregnancy after **1 year** of unprotected intercourse. It affects ~15% of couples; male and female factors contribute equally.
- Most conceptions occur in the first 3–6 cycles, so evaluation may start after 6 months.
- Idiopathic infertility is the largest group. **Varicocele is the commonest identifiable cause**, then obstruction.

† Classification	Examples
Pre-testicular (endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, congenital adrenal hyperplasia (CAH)
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemotherapy/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), congenital bilateral absence of the vas (CBAVD), ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · History

P2 Discuss the role of urologist in reproductive endocrinology and male infertility practice — history, examination, endocrine tests (LH localises, FSH/inhibin B = germ-cell mass, prolactin), HH/hyperprolactinaemia/CAH treatment, surgical roles, ART choice, testis cancer in azoospermia

Not in the paper answer:

- The **female partner's age is the most important determinant**; fecundity falls after 35.
- Paternal age ≥40 needs counselling about offspring bipolar disorder, schizophrenia, autism and ADHD (ASRM/AUA 2021).
- Semen quality peaks after **1–2 days' abstinence**, so advise daily intercourse around ovulation; most lubricants impair sperm.

Gonadotoxins and drugs

> KEY POINT
> **Exogenous testosterone (T) is a contraceptive**. It is aromatised to oestradiol (E2), which feeds back on the pituitary and lowers LH. Intratesticular T then falls, and spermatogenesis falls with it. Recovery usually takes ≥4 months after stopping (after ≤2 years' use); after longer use it is unknown.

- Other harmful drugs include antiandrogens, spironolactone, corticosteroids, anabolic steroids and some antiretrovirals. Cannabis, alcohol and smoking also impair semen.

Infection and childhood conditions

- Infection injures either **anatomically** (duct scarring) or **functionally** (leucocytes, reactive oxygen species [ROS]).
- After torsion, about half have later spermatogenic effects.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.**
- The American Urological Association (AUA) advises orchidopexy within **18 months**, because germ cells are lost from 15–18 months. Later surgery before 10 years still improves sperm retrieval.

3 · Physical examination

Testis size and the vasa sort most azoospermic men.

- Examine the man standing. An engorged epididymis suggests obstruction, and testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound (US)	Most accurate; added value unclear

- Unilateral vasal absence needs renal US; bilateral absence needs CFTR testing and renal US.
- Palpable seminal vesicles suggest ejaculatory duct obstruction (EDO).

Varicocele grade	Definition
Subclinical	Seen only on Doppler; **do not treat**
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · Endocrine evaluation

ALGORITHM

1. Take a morning sample for total T, sex hormone-binding globulin (SHBG) and albumin, and calculate bioavailable T.
2. Measure LH and follicle-stimulating hormone (FSH), then E2.
3. Measure prolactin only if indicated.

- Adequate total T is ≥**280–300 ng/dL**. Low T is common even with density >20 million/mL, so assess every infertile man.
- Low SHBG (obesity, diabetes) makes total T under-read; high SHBG (ageing, cirrhosis) makes it over-read.
- A T (ng/dL) to E2 (pg/mL) ratio **<10:1** means aromatase excess.
- Prolactin of 20–50 µg/L needs no further work-up; a significant rise needs pituitary MRI.

5 · Semen analysis

P4 Semen analysis nomenclature — collection, ≥2 samples 2–3 weeks apart; WHO 2021 5th centiles (volume **1.4** mL, **16** million/mL, 39 million, 42/30%, **4** % strict forms, 54% vitality); terms; only azoospermia definitive; HOS test

Not in the paper answer:

- **Low volume (<1.0 mL)** suggests EDO, CBAVD, hypoandrogenism or retrograde ejaculation. Do a **postejaculatory urinalysis**; if there are no sperm in urine, do transrectal ultrasound (TRUS).
- †Sperm DNA fragmentation testing is not routine; use it for recurrent pregnancy loss or failed assisted reproduction (ART).

6 · Genetic testing

ALGORITHM

1. If non-obstructive azoospermia (NOA) or density **<5 million/mL**, then do karyotype and Y-microdeletion testing (ASRM/AUA 2021).
2. If CBAVD (or idiopathic EDO), then test CFTR in the man **and his partner**.

P3 Genetic causes of male infertility. Enumerate and classify. — AZFa (USP9Y, DDX3Y)/AZFb complete deletion: no retrieval; AZFc (DAZ): sperm often found, sons inherit; CFTR; Kallmann; androgen receptor; globozoospermia, AURKC, PCD

Not in the paper answer:

- AZFc deletion with oligozoospermia declines progressively, so **cryopreserve sperm** early.

7 · Imaging

- Do not screen, because incidental microlithiasis causes harm. Varicocele is clinical.

P3 Ejacultory duct obstruction — TRUS vesicle AP **>1.5 cm** with or without midline cyst (low specificity); aspiration ≥3 sperm/HPF; vesiculography, chromotubation; <5% of azoospermia; idiopathic EDO CFTR in ~half; TURED with TRUS guidance

8 · Azoospermia: obstructive (OA) vs non-obstructive (NOA)

This split decides between reconstruction and sperm extraction.

	OA	NOA (spermatogenic dysfunction)
FSH	**≤7.6 IU/L**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction (see Ch 70) or sperm retrieval	Genetic tests; microdissection testicular sperm extraction (micro-TESE) with intracytoplasmic sperm injection (ICSI)

ALGORITHM

1. If the vasa are absent, then test CFTR in both partners.
2. If the vasa are present and volume is <1 mL, then do postejaculatory urinalysis. Sperm in urine means retrograde ejaculation; no sperm means TRUS for EDO.
3. If volume is normal, then use FSH and testis size to separate OA from NOA.

9 · Specific causes and treatment

Klinefelter syndrome (47,XXY)

P4 Klinefelter's syndrome. — commonest genetic cause; 1 in 500–1000; non-mosaic 80–90%; small firm testes, raised FSH/LH, low T; ejaculated sperm ~8%; micro-TESE ~ **50%** ; mediastinal GCT

- **Trap**: exogenous T suppresses residual spermatogenesis, so extract sperm before starting T. If fertility is wanted, use aromatase inhibitors or human chorionic gonadotrophin (hCG).

Hypogonadotropic hypogonadism — most treatable cause

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR)/complete HH	Low LH, low FSH	hCG with or without recombinant FSH (rFSH) or human menopausal gonadotrophin (hMG)
Isolated LH deficiency ("fertile eunuch")	Low LH, normal FSH	hCG
Isolated FSH deficiency	Low FSH	FSH
Hypothalamic GnRH deficiency	Low GnRH	Pulsatile subcutaneous GnRH every 2 h, for ≥6 months
Partial HH (common)	LH inappropriately low for T	Selective oestrogen receptor modulator (SERM) or aromatase inhibitor

- Give hCG 1500–5000 IU 2–3 times weekly (max 10,000/week), titrated to T; add hMG or rFSH 75 IU 2–3 times weekly. After androgen abuse, allow 1–2 years.
- Treat prolactinoma with cabergoline, and CAH with corticosteroids.

Empirical medical therapy

Drug	Dose	Notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks E2 feedback, so LH and FSH rise; off-label; no change in prostate-specific antigen (PSA) or haematocrit; target T 600–800 ng/dL
Anastrozole / letrozole	Anastrozole 1 mg daily	Raises T and lowers E2; use when T:E2 <10:1 (obesity); caution with bone density

Primary failure and androgen resistance

- High LH with low T has no accepted therapy; exogenous T is contraindicated, so offer sperm extraction.
- Androgen receptor defects give high T, LH and E2 and are treated with ICSI.

CBAVD

P3 CBAVD — genital form of CF; mild CFTR alleles; normal spermatogenesis, low-volume azoospermia; partner carrier ~1 in 20; renal agenesis ~1 in 10; retrieval + ICSI

Varicocele

P3 Varicocele: controversies and present status of treatment — prevalence, left predominance, right-sided: exclude tumour; heat/ROS/DNA injury; repair palpable + abnormal semen, never subclinical; may down-stage ART

- Varicocele affects ~15–20% of all men versus 35–50% of infertile men (see Ch 70).

Antisperm antibodies and structural defects

- For antisperm antibodies, washing fails and steroid trials conflict, so use **ICSI**.
- Globozoospermia needs ICSI with or without oocyte activation, and ICSI is not recommended for macrocephalic multiflagellate sperm (AURKC).

10 · Lifestyle and obesity

- Obesity raises E2 and lowers LH and T, and worsens ART outcomes, so advise weight loss. Stop smoking, cannabis, excess alcohol and anabolic steroids, and avoid heat.

11 · Assisted reproduction

Sperm count decides the technique.

- †Choose by total motile sperm count (TMSC): IUI if processed TMSC is ≳5–10 million; IVF below that or after failed IUI; ICSI for severe oligozoospermia, retrieved sperm, antisperm antibodies or prior fertilisation failure.
- †Retrieval in OA is by percutaneous or microsurgical epididymal sperm aspiration (PESA/MESA) or testicular aspiration; in NOA it is micro-TESE.

70 Surgical Management of Male Infertility

1 · Principles

- In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI) treats even non-obstructive azoospermia (NOA), but it is costly, burdens the female partner and bypasses natural genetic barriers.
- Treating the man (reconstruction for obstructive azoospermia (OA), varicocelectomy) is safest and cheapest, and moves the couple **down the ladder**: donor sperm, then IVF/ICSI with testicular sperm, then IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI), then natural conception.

2 · Surgical anatomy

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (from testicular a.) supply the caput; inferior epididymal (from deferential a.) supplies the cauda; interconnected
Vas	Abdominal end: deferential only. Testicular end: deferential and inferior epididymal. **No supply from cremasteric or cord vessels**

- The testicular artery **exceeds deferential and cremasteric combined**; after vasectomy, preserve it at any later surgery.
- The epididymis can be dissected off the testis if the **superior epididymal artery is intact**; this gains **4–6 cm**.
- If the vas is blocked and devascularised at two sites, the segment between fibroses. So **never do two simultaneous vasovasostomies on one vas**; excise it and do one tension-free anastomosis.
- **Excurrent ducts**: 7–11 efferent ducts enter the caput, but a **single tubule** runs from the caput–corpus junction to the vas. So injury distal to the caput (even **one stitch**) obstructs the whole side.

3 · Diagnostic testis biopsy

P1 Testicular biopsy and TESA (testicular sperm aspiration) — indications (azoospermia, normal testes, palpable vasa, normal FSH); diagnostic + therapeutic; open = gold standard; never formalin; percutaneous 14 G / 23 G

Not in the paper answer:

- Biopsy is usually bilateral and planned as testicular sperm extraction (TESE), with an andrology laboratory on site for wet examination and freezing.

ALGORITHM

1. Give a cord block; choose the larger testis (opposite a varicocele).
2. Make a 2 cm incision, open the tunica vaginalis, place stay sutures.
3. Incise the albuginea 3–4 mm, using bipolar on crossing vessels.
4. Take a pea-sized sample with iris scissors (no forceps).
5. Do a touch prep, then a wet prep. If negative, then biopsy the other side.
6. Close the tunica vaginalis with continuous 5-0; no antibiotics.

- A non-absorbable suture marks a positive site for future TESE.

Complication	Point
Haematoma — most common	Prevent with magnification, bipolar and continuous tunica vaginalis closure
Epididymal biopsy — most serious	If that tubule held sperm, obstruction is now certain

4 · Vasography

- **Absolute indications (all three)**: azoospermia, complete spermatogenesis on biopsy and at least one palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, and low volume with poor motility (partial ejaculatory duct obstruction (EDO)).
- Do it **only at planned reconstruction**, because of scar and stricture. It cannot diagnose epididymal obstruction, the commonest non-vasectomy site.

ALGORITHM

1. High scrotal incision; find the vas at the straight–convoluted junction; hemivasotomy at 25×.
2. **Examine vasal fluid first.** If motile sperm, then barbotage and cryopreserve **before any dye**.
3. Pass a 24-G sheath into the abdominal end and inject 1 mL saline. If flow is easy, then it is patent.
4. Rarely, give dilute indigo carmine; blue urine confirms.
5. Use formal contrast only for obstruction above the internal ring.
6. Close with 10-0 mucosa and 8-0 muscularis.

Vasal fluid / vasogram	Meaning and action
No sperm, abdominal end patent	Epididymal obstruction: vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO: locate it (vasogram, or 2-0 Prolene threaded down)
Block at ejaculatory ducts; both vasa fill from one side	EDO or midline cyst: transurethral resection of the ejaculatory ducts (TURED)
Vas ends blindly	Partial vasal agenesis: cystic fibrosis gene (CFTR) testing; retrieve and freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- Use **indigo carmine, not methylene blue**, which kills sperm (use it only after banking).

- Complications are **stricture** (repeated punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma and granuloma.
- **Transrectal seminal vesiculography** aspirates dilated vesicles under transrectal ultrasound (TRUS); sperm there show a patent vas and epididymis, allowing TURED without scrotal exploration.

5 · Vasovasostomy (vasectomy reversal)

Evaluation

- Prior fertility proves spermatogenesis. Check follicle-stimulating hormone (FSH) if testes are small or soft; raised FSH means poorer prognosis.
- An indurated epididymis predicts vasoepididymostomy. A **sperm granuloma at the testicular end gives a better prognosis**, because it vents pressure, whatever the interval.
- Operate only if able to do vasoepididymostomy.

Approach and preparation

- The approach is scrotal for most and inguinal for hernia or orchidopexy obstruction. **Never strip the perivasal sheath**.
- **Length**, in escalating steps: dissect to the external then internal ring; free the convoluted vas (4–6 cm; **do not unravel**); reroute beneath the inguinal floor; take the epididymis off the testis. This bridges gaps up to **10 cm**.
- A healthy cut end has a white mucosal ring, soft muscularis and a bleeding edge. **If the muscularis is gritty or bleeding poor, recut.**
- At the testicular end **aspirate, never flush**. Flush saline into the abdominal end to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

> **KEY POINT**
> Clear watery fluid still gets vasovasostomy: the epididymis has not blown out, and sperm usually return. Toothpaste fluid means secondary epididymal blow-out; a discrete level (dilated above, collapsed below) confirms it. When in doubt or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen is 2–5× the abdominal one

Principle	Reason
Leakproof	Sperm are antigenic; fluid lacks clotting factors, so the seal depends on mucosal sutures
Tension-free	Tension gives early sperm, then falling counts, then azoospermia
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six mapped dots guide **six 10-0 mucosal sutures**, then 9-0 deep muscularis, 9-0 adventitia and 8-0 sheath. The two-layer alternative is 10-0 mucosa and 8-0 full thickness.
- Reinforcing vasal sheath sutures (ReVas) relieve tension and raise patency.
- **Crossed vasovasostomy** suits unilateral inguinal obstruction with a contralateral atrophic testis, or one unreconstructable side.
- **Varicocele with reversal**: vasal vessels are compromised, so stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Check semen at **6 weeks**, then 6 months if no pregnancy. Poor motility gets a 4-week prednisone course. Azoospermia at 6 months means redo.
- An anastomotic granuloma heralds obstruction. **Late stricture** shows as falling motility, then count. **Bank sperm once motile sperm appear.**
- With sperm in vasal fluid on at least one side, sperm return in virtually all. Pregnancy is about half, depending on vasectomy interval and female age.

6 · Vasoepididymostomy

- The epididymis is a single tubule ~200 µm wide and 12–15 feet long, with its thinnest wall at the caput. So a **more distal anastomosis is easier and gives better pregnancy**.
- **Indications**: reversal by the fluid criteria; or non-vasectomy OA with proven spermatogenesis, no vasal sperm and a patent vas and ejaculatory ducts.

ALGORITHM

1. Hemitransect the vas at the straight–convoluted junction; sample fluid; confirm patency.
2. Open the tunica vaginalis; choose a site **above** the obstruction where tubules are dilated. If unclear, then puncture distal to proximal until sperm appear.
3. Make a 3–4 mm tunic window.
4. Bring the vas through the tunica vaginalis, **fixed with 5-0 Prolene to the albuginea (tension relief)**.
5. Place posterior tunic–vas 8-0, then intussusception sutures, then outer tunic–vas adventitia.

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid not checked before sutures set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule

Technique	For	Against
End-to-end	Rapid level finding; bridges gap	Haemostasis hard; inferior epididymal supply lost

- **No simultaneous varicocelectomy**; stage it after 6 months.
- Late shutdown occurs, so **bank sperm during surgery and once they appear**. If it fails, redo or retrieve sperm for ICSI.

7 · TURED

P3 Ejaculatory duct obstruction — congenital (midline cyst, atresia) vs acquired; low volume, acidic, low fructose, palpable vas, normal FSH; TRUS; TURED only after couple declines IVF; sperm return in ~half; cold-knife reobstructs; reflux, epididymitis, retrograde ejaculation; vesiculoscopy

Not in the paper answer:

- For retrograde ejaculation give pseudoephedrine 60 mg 90 min before, or use alkalinised urinary sperm.

ALGORITHM

1. Use a 24-Fr resectoscope with a finger in the rectum.
2. Resect the **verumontanum** to open the dilated orifices or cyst, sparing bladder neck, sphincter and rectum.
3. **Blue efflux confirms** adequacy; coagulate minimally.
4. Catheter overnight; 7 days of antibiotics.

8 · Sperm retrieval

P2 Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction) — ICSI essential; PESA/MESA/TESA/TESE/micro-TESE table; MESA by capillary action; micro-TESE ~half of NOA; retrieval by diagnosis (AZFa/b **0% — do not operate**); frozen = fresh except SCI

Not in the paper answer:

- Microsurgical epididymal sperm aspiration (MESA) gives most motile sperm and multiple vials. **Microdissection TESE (micro-TESE) is best in NOA**.

KEY POINT

Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

ALGORITHM

1. Micro-TESE: general or regional anaesthesia.
2. At 10×, find the avascular plane on the anterior albuginea; make a wide equatorial incision and bivalve the testis.
3. Excise larger tubules; an andrologist in theatre examines them; map positives.
4. If negative, then explore the other testis.
5. Close the albuginea with 5-0 and the tunica vaginalis with 4-0.

9 · Cryopreservation

- Freeze at diagnostic biopsy, at every reconstruction (before any dye), and from the ejaculate once motile sperm return after reversal.

10 · Varicocelectomy

P2 P3 Surgical management of bilateral varicocele; Varicocele: controversies — indications; approach table (microsurgical recurrence <1%, hydrocele ~0); micro-Doppler, testis delivery, impulse test; complications

Not in the paper answer:

- Conventional inguinal repair has hydrocele 3–15% and recurrence 5–15%, because it misses scrotal and periarterial veins.
- Ligate vasal veins only if >2.5 mm, keeping one set: **one intact deferential set guarantees venous return**.

ALGORITHM

1. Local anaesthesia with sedation; 2–3 cm incision over the external ring.
2. Deliver the cord on a Penrose, excluding the ilioinguinal nerve and genital branch.
3. **Map arteries with micro-Doppler before handling.**
4. Open the spermatic fasciae; work at 10–25× with 1% papaverine; loop the artery.
5. Ligate **all veins except vasal**; preserve lymphatics.
6. Deliver the testis; ligate external spermatic and gubernacular veins.
7. Do the impulse test. If there is no impulse, then a vein was missed.

- **End point**: only testicular and cremasteric arteries, lymphatics, and the vas with its vessels remain.

KEY POINT

Recurrence means a vein was missed; hydrocele, a lymphatic tied; atrophy, an artery tied. Microsurgery with testicular delivery addresses all three.

11 · Landmark trials

Trial	Design	Result	Take-home
Madgar (1995)	Randomised controlled trial (RCT): varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · Definition and key concepts

- **Erectile dysfunction (ED)** is the consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (American Urological Association (AUA) 2018).
- Erection needs **sinusoidal (trabecular) smooth muscle relaxation, arterial dilation and venous compression (veno-occlusion)**. Rigidity comes from trapping blood, not from inflow alone.
- Milestones MCQ: nitric oxide (NO) was identified in the 1990s, and oral phosphodiesterase type 5 inhibitors (PDE5Is) arrived in 1998.

2 · Functional anatomy

Corpora

- There are paired dorsal **corpora cavernosa** and a ventral **corpus spongiosum**, which surrounds the urethra and expands as the glans.
- The septum is **incomplete** in humans, so one injection fills both corpora.
- Sinusoidal oxygen tension (Po2) is venous when flaccid (**~35 mm Hg**) and arterial when erect (**~90 mm Hg**).

Tunica albuginea

- The **inner circular layer** gives off **intracavernous pillars** (absent at 5–7 o'clock). The **outer longitudinal layer** is thicker dorsally and absent in the ventral groove.
- The ventral tunica is ~0.8 mm against ~2.2 mm dorsally. So 5–7 o'clock is the weak zone for **penile fracture** and implant crossover or perforation.
- **Veno-occlusion**: emissary veins run obliquely between the layers and are compressed as the sinusoids stretch the tunica. Cavernous arteries escape because they lie in a periarterial sheath.
- The spongiosal tunica is thin, single-layered and absent in the glans, so spongiosum and glans are **low-pressure arteriovenous shunts** and never become rigid.

Arteries

- The internal iliac gives the **internal pudendal**, which becomes the common penile artery with three branches.
- The **cavernous** artery is the main corporal supply. Its **helicine arteries** are tortuous when flaccid and straight when erect, and open into sinusoids without capillaries.
- The **dorsal** artery fills the glans, and the **bulbourethral** artery supplies the bulb and spongiosum.
- **Accessory pudendal arteries** occur in ~one-third and may be dominant. Preserving them at radical prostatectomy speeds erectile recovery.

Veins

Sinusoids drain into the **subtunical venous plexus**, which exits through **emissary veins** in the tunica.

System	Drains	Destination
Deep dorsal vein (with or without circumflex and periurethral veins)	Glans and distal two-thirds of corpora	Periprostatic (Santorini) plexus
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

KEY POINT

Vascular anatomy varies widely between men, so the same "nerve-sparing" operation gives different erectile outcomes.

3 · Haemodynamics: phases of erection and detumescence

P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — sequence of events and phase table (flaccid, latent, tumescence, full, rigid, detumescence)

Not in the paper answer:

- Full-erection intracavernous pressure (ICP) plateaus at **~100 mm Hg**; the rigid phase is suprasystolic, driven by the pudendal nerve via ischiocavernosus.
- **Trap**: arterial inflow peaks in the latent phase, not at rigidity. This is why peak systolic velocity (PSV) on Doppler is measured early.
- Detumescence has three phases: initial (transient ICP rise), slow (veins gradually reopen) and fast (outflow restored).

4 · Neuroanatomy

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**
Route	Sympathetic chain, superior hypogastric plexus, hypogastric nerves, then pelvic plexus	Pelvic nerves (nervi erigentes) to the pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**; emission	**Erection** (cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** arise from the pelvic plexus, carry both autonomic types and run posterolateral to the prostate. Pelvic ganglia lie along the bladder–prostate junction and seminal vesicles, so they are vulnerable in radical pelvic surgery.
- The **bulbocavernosus reflex** (glans squeeze contracts bulbocavernosus) has limited value in ED work-up.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Reflexogenic	Genital touch	Sacral reflex arc to the cavernous nerves	Preserved in upper cord lesions
Nocturnal	Rapid eye movement (REM) sleep	Cholinergic pontine tegmentum on; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- The **medial preoptic area (MPOA)** is dopaminergic and integrates hormonal and sensory cues. The **paraventricular nucleus (PVN)** sends pro-erectile **oxytocin** neurons to the lumbosacral nuclei, and the nucleus paragigantocellularis is the **serotonergic brake**.

Central neurotransmitters

P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — promoters (dopamine, oxytocin, NO, noradrenaline, melanocortins, 5-HT2C) vs inhibitors (5-HT1A/2, GABA, opioids, cannabinoids, prolactin)

Not in the paper answer:

- Apomorphine is a D1/D2 agonist acting in the PVN, which releases oxytocin and then NO. Melanocortins act on the MC4 receptor.
- Serotonin 5-HT1A and 5-HT2 receptors inhibit erection but facilitate ejaculation.

> **KEY POINT**
> NO synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections, so NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · Molecular mechanisms of smooth muscle tone

Every erectogenic drug acts on one of these pathways.

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest** through myogenic tone, adrenergic tone and endothelial constrictors.
- **Calcium pathway**: noradrenaline (α1), endothelin-1 (ETA), angiotensin II (AT1), PGF2α and thromboxane A2 act via Gq and phospholipase C to form inositol trisphosphate (IP3), raising Ca^{2+}. Ca^{2+}–calmodulin then activates **myosin light chain (MLC) kinase**, which phosphorylates MLC20.
- **Ca^{2+} sensitisation by RhoA/Rho-kinase**: it inhibits **MLC phosphatase (MLCP)**, giving sustained contraction without a further Ca^{2+} rise.
- Rho-kinase is raised in diabetes mellitus (DM), hypertension, hypercholesterolaemia, hypogonadism, ageing and nerve injury, and it also drives fibrosis.

Relaxation (erection)

- **NO–cGMP**: NOS converts L-arginine and O2 into NO and citrulline. **Neuronal NOS initiates and endothelial NOS (eNOS) sustains** erection.
- NO activates soluble guanylyl cyclase to make cyclic GMP (cGMP), which activates **protein kinase G (PKG-I)**. PKG lowers intracellular Ca^{2+}, inhibits Rho-kinase and activates MLCP, so the muscle relaxes. PKG-I knockout mice cannot erect, so PKG is the bottleneck.

- **cAMP pathway**: PGE1, vasoactive intestinal peptide (VIP) and calcitonin gene-related peptide (CGRP) act via Gs to raise cyclic AMP (cAMP), which activates protein kinase A (PKA). It is **NO-independent**, which is why intracavernosal alprostadil works when nerves or endothelium fail.
- **PDE5 (cGMP-specific) is the principal cavernous PDE**. PDE6 is retinal (visual side effects); PDE2, 3 and 4 degrade cAMP. Papaverine is nonspecific.
- **Gap junctions (connexin-43)** synchronise relaxation, so a local input gives a uniform erection.
- Oxygen acts as a switch: erect Po2 >50–60 mm Hg permits full NOS activity, while low flaccid Po2 suppresses it.
- Excess **adenosine** with increased A2B signalling causes prolonged erection in adenosine deaminase-deficient and sickle cell mice, the **priapism models**.
- Acetylcholine acts indirectly (endothelial NO, presynaptic adrenergic inhibition), so **atropine does not abolish** neural erection MCQ.

Second messenger	Activators	Kinase	Degraded by	Drug
cGMP	NO, CO, H2S, natriuretic peptides	PKG-I	PDE5	PDE5Is
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil; papaverine (both)

Detumescence

NO release stops and PDE5 hydrolyses cGMP. Sympathetic noradrenaline (α1; the principal transmitter of flaccidity), endothelin-1 and Rho-kinase then contract the trabeculae, so the veins reopen.

> **KEY POINT**
> Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" means the phosphatase.

6 · Classification of ED

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · to **fill** (arterial) · to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component.

7 · Risk factors and endothelial dysfunction

ED often signals vascular disease before the heart does.

- **Age** is the strongest demographic predictor, but DM, heart disease and hypertension carry more risk (Massachusetts Male Aging Study (MMAS)).

- **Endothelial dysfunction**: oxidative stress causes **NOS uncoupling**, while arginase and asymmetric dimethylarginine (ADMA) further lower NO.
- **Artery-size hypothesis**: smaller penile arteries become symptomatic first. ED preceded coronary artery disease (CAD) by a mean of **~2 years**, so ED is a **sentinel** and needs cardiac risk stratification (see Ch 72).
- The common end-path is raised Rho-kinase and transforming growth factor-β (TGF-β), which lower the smooth muscle-to-collagen ratio and cause veno-occlusive failure.

8 · Pathophysiology by type

Psychogenic

- The brain inhibits the spinal centres, and excess sympathetic tone contracts the muscle. Causes include performance anxiety, partner conflict, depression and post-traumatic stress disorder (PTSD).

Neurogenic

- **Brain** causes include Parkinson disease, stroke, tumours and dementia. **Cord** causes include SCI (commonest), spina bifida, disc disease and multiple sclerosis (MS).

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (thoracolumbar pathway); marginal tumescence

- **Iatrogenic**: radical prostatectomy, cystoprostatectomy, abdominoperineal resection (APR) and pelvic radiation. Most men have at least a transient decline despite nerve sparing, worse with baseline ED and age >60.
- Cavernous nerve injury causes apoptosis and fibrosis, leading to venous leak. This is the rationale for early penile rehabilitation.
- **Pelvic fracture** injures the cavernous nerves or pudendal artery. With posterior urethral injury, early realignment preserves potency better than delayed anastomosis.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics, but it maintains NOS, PDE5 and smooth muscle. Low libido is the commonest sexual symptom of low T.
- AUA: measure **morning, fasting** T in all men with ED. Low T is **<300 ng/dL**; treat only with ≥2 low levels and symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal erections are androgen-dependent, but erections to visual stimuli are largely independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics) suppresses gonadotrophin-releasing hormone (GnRH), causing low T and low libido.
- A low T:oestradiol ratio (obesity), hyperthyroidism and hypothyroidism also cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic or traumatic occlusion of the hypogastric–pudendal–cavernous tree lowers perfusion pressure, giving a slower, softer erection.

- Risks are hypertension, hyperlipidaemia, smoking, DM, obesity, pelvic irradiation and blunt perineal trauma. Trauma causes focal stenosis in young men, who are revascularisation candidates.
- Chronic hypoxia raises TGF-β1 and collagen, so "inflow failure becomes storage failure".
- In hypertension, angiotensin receptor blockers (ARBs) and nebivolol protect penile tissue.

Vasculogenic – veno-occlusive (venous leak)

- Causes are **tunical** (ageing, DM, Peyronie disease, penile fracture), **fibroelastic trabecular change** preventing sinusoidal expansion, **insufficient relaxation** (anxiety, adrenergic tone, endothelial dysfunction), and **acquired shunts** after priapism shunt surgery. Congenital ectopic or crural veins cause primary ED.

> **KEY POINT**
> Veno-occlusion is an emergent property of healthy tissue (elastic tunica and fully relaxed trabeculae). So most "venous leak" is really cavernosal smooth muscle disease, which is why venous ligation surgery fails and is investigational.

Metabolic and systemic disease

- **DM** causes the earliest, most severe ED through arteriopathy, nitrergic neuropathy, endothelial dysfunction, advanced glycation end products (AGEs), raised Rho-kinase and hypogonadism. **Key**: the NO source is damaged but the response to exogenous NO is preserved, so PDE5Is work until fibrosis sets in.
- **Metabolic syndrome and obesity** improve with weight loss, Mediterranean diet, exercise and bariatric surgery. In **hyperlipidaemia**, oxidised low-density lipoprotein (LDL) inhibits NO, and statins are linked to better International Index of Erectile Function (IIEF) scores.
- **Chronic kidney disease**: up to **80%** of men with end-stage renal disease have ED, and it persists in ~half after transplant.
- Other associations include HIV, cirrhosis, chronic obstructive pulmonary disease, obstructive sleep apnoea and lower urinary tract symptoms.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers**, clonidine	Thiazide – unknown; β-blocker – prejunctional α2	ACE inhibitors, ARBs, nebivolol, α1-blockers (ejaculatory dysfunction), calcium channel blockers
Antidepressants	Selective serotonin reuptake inhibitors (SSRIs; paroxetine worst), tricyclics	Spinal 5-HT2/5-HT3 inhibition, less dopamine and NO; post-SSRI dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add a PDE5I
Antipsychotics	Typical > atypical	Dopamine block causes hyperprolactinaemia	Newer agents
Antiandrogens	LHRH agonists/antagonists, androgen receptor antagonists, cyproterone, **5α-reductase inhibitors (5ARIs)**	5ARI: least T effect yet persistent ED after stopping	None

Class	Cause ED	Mechanism / note	Alternative
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids	Endothelial damage, neuropathy, hypogonadism	Stop or moderate
Others	Cimetidine, spironolactone, carvedilol	—	—

Trazodone causes priapism, and among anticonvulsants carbamazepine and phenytoin are worse than lamotrigine.

Primary (lifelong) ED

- It is usually psychogenic, but exclude hypoplastic cavernous arteries, congenital venous leak, corporal atresia and micropenis.

> **KEY POINT**
> Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction and TGF-β-driven fibrosis.

72 Evaluation and Management of Erectile Dysfunction

1 · Definition and significance

- **Erectile dysfunction (ED)** (American Urological Association, AUA): a consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction. Minimum **3 months**, except after trauma or surgery.
- About **20%** of adult men have ED (up to 15% at 40–49 y; up to 40% at 60–69 y).
- **ED predicts cardiovascular disease (CVD)** through shared endothelial dysfunction, as strongly as smoking or a family history of myocardial infarction (MI). Younger onset means greater cardiovascular (CV) risk.

Independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	Diabetes mellitus (DM), the strongest disease; obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, body mass index (BMI) ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- Drugs cause up to 25% of ED (see Ch 71); opioids double the risk.

2 · Management principles

- **Shared decision making** (AUA 2018): all non-contraindicated options are offered **in parallel, not ranked**, and the patient may choose any first. They are phosphodiesterase type 5 inhibitors (PDE5Is), the vacuum erection device (VED), intraurethral (IU) alprostadil, intracavernosal injection (ICI) and prosthesis.
- In testosterone (T) deficiency (symptoms and total T below 300 ng/dL), restoring T probably improves the other ED treatments.

Cardiac risk assessment – Princeton Consensus

P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — Princeton high (defer sex and ED treatment) / indeterminate (stress test, restratify) / low (treat)

Not in the paper answer:

- Every man with ED is at potential CV risk; sex equals moderate exertion. **Refer** after failed therapy, for a young trauma patient, for deformity, in medicolegal cases, and for complex endocrine, psychiatric or vascular problems.

3 · Basic evaluation

History

- Ask about onset, pattern, morning erections, libido, ejaculation, curvature and prior treatment.
- Ask about DM, CVD, neurological disease, pelvic surgery or radiation, trauma and drugs. Screen for depression.

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

KEY POINT

Preserved nocturnal/morning erections prove an intact neurovascular axis, the strongest historical pointer to psychogenic ED.

Questionnaires

- The **International Index of Erectile Function (IIEF)** has 15 items in 5 domains: erectile function, orgasm, desire, intercourse satisfaction and overall satisfaction.

P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — IIEF-5/SHIM bands: 5–7 severe … 22–25 no ED

- The Sexual Health Inventory for Men (SHIM) has 5 items scored 1–5 over the past 6 months. Questionnaires grade severity and response but do **not** identify the cause.

Physical examination

- Look for hypogonadism (testes, gynaecomastia), vascular disease (blood pressure (BP), obesity, pulses), neuropathy (genital sensation, bulbocavernosus reflex) and structural problems (plaque, micropenis, chordee).

Laboratory tests

- **Recommended**: fasting glucose or glycated haemoglobin (HbA1c), lipids, serum chemistry, full blood count and **morning (7–11 am) total testosterone**.
- Repeat a low T, because about one-third of mildly low values normalise, and add luteinising hormone (LH) and prolactin. Check prostate-specific antigen (PSA) if prostate disease is suspected or before T therapy.

KEY POINT

The basic work-up finds the disease behind the ED (undiagnosed DM, dyslipidaemia, hypogonadism, CV risk); it does not prove ED.

4 · Hormonal evaluation

P4 Male menopause (andropause, late-onset hypogonadism) — bioavailable T = free + albumin-bound; SHBG rises with age; LH separates primary/secondary; prolactin if LH low; pituitary MRI if T <150 ng/dL

Not in the paper answer:

- The AUA threshold is **300 ng/dL (10.4 nmol/L)**; above about 230 ng/dL, sexual function correlates only weakly with T. Order a pituitary magnetic resonance imaging (MRI) scan if T is **<150 ng/dL**.
- Hyperprolactinaemia is caused by antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism and renal failure.

5 · Specialised testing

These tests are for young trauma patients, primary ED, surgical candidates, medicolegal cases and failed therapy.

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can the vessels produce a rigid erection?	Screening
Penile duplex Doppler	2nd	Arterial or veno-occlusive? Asymmetry?	Most reliable, least invasive
DICC (dynamic infusion cavernosometry/cavernosography)	3rd	Is there a venous leak, and where?	Trauma, lifelong ED, before venous surgery
Selective internal pudendal arteriography	3rd	Site of arterial lesion; is the inferior epigastric donor usable?	Young trauma patient before revascularisation

CIS test

- Inject alprostadil 10–20 µg or Bimix/Trimix 0.3 mL into the lateral base, then compress, stimulate and rate rigidity. A rigid, sustained erection means normal haemodynamics; anxiety or a low dose gives false negatives, so redose.
- If there is no detumescence by about 1 h, inject diluted phenylephrine (500 µg/mL) every 3–5 min.

Penile duplex Doppler

- Measure cavernous artery peak systolic velocity (PSV) and end-diastolic velocity (EDV) at the base every 5 min up to 20 min after ICI.

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Stays >25 cm/s
EDV	—	**>5 cm/s**
Resistive index (RI) = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full relaxation)

KEY POINT

Venous leak can be diagnosed only with adequate inflow and full smooth muscle relaxation; an anxious patient mimics veno-occlusive dysfunction, so redose first. Anatomical variants can falsely suggest arteriogenic ED.

DICC, NPTR and neurological tests

- DICC measures flow-to-maintain, pressure decay and cavernous artery systolic occlusion pressure after a vasodilator; cavernosography localises the leak.
- Nocturnal penile tumescence and rigidity (NPTR) testing in a sleep lab is the standard. The home **RigiScan** measures **radial** rigidity; normal is 4–5 erections a night, ≥60% rigid for ≥10 min. It suits obscure cause, medicolegal cases and suspected psychogenic ED.

6 · Treatment

ALGORITHM

1. Treat reversible causes (lifestyle, drugs, hormones, psychosocial) and stratify CV risk.
2. Shared decision: PDE5I, VED, IU alprostadil or ICI. Low-intensity extracorporeal shock wave therapy (Li-ESWT) is investigational.
3. If the response is inadequate, then optimise, combine or switch.
4. If still inadequate, then offer a penile prosthesis (see Ch 75).

Lifestyle and risk-factor modification

- Stopping smoking, exercise, weight loss and a Mediterranean diet improve function; control DM, lipids and BP.
- **Change medication**: replace thiazides or β-blockers with calcium channel blockers, angiotensin-converting enzyme inhibitors or angiotensin receptor blockers. For a selective serotonin reuptake inhibitor (SSRI), switch to bupropion or mirtazapine, or add a PDE5I.
- **Psychosexual therapy** works best combined with medical therapy.

Hormonal therapy

- Give T only for confirmed deficiency, as a ~3-month trial; it improves libido more than erection and is synergistic with PDE5Is.
- Monitor digital rectal examination, PSA, haematocrit, liver tests and lipids at baseline, 3–6 months and yearly. The main adverse effect (AE) is erythrocytosis; randomised trials show no causal CV link.
- Preparations include gels (transfer risk), intramuscular enanthate/cypionate (200–250 mg every 2–3 weeks) and undecanoate depot (750 mg at 0 and 4 weeks, then every 10 weeks). Avoid methyltestosterone because it is hepatotoxic.
- T does not correct hyperprolactinaemia: stop the drug, give bromocriptine for prolactinoma, and operate for failure or visual compromise.

PDE5 inhibitors (first-line oral)

They block cyclic guanosine monophosphate (cGMP) breakdown, so they **enhance, not initiate**, erection; sexual stimulation is obligatory.

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
Time to peak, T (h) max	0.8 (~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3-0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5
Fatty food	Lowers absorption	Lowers absorption	**No effect**	Lowers absorption
Characteristic AE	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation, so avoid with class 1A/3 antiarrhythmics** and congenital long QT	**Back pain/myalgia** (PDE11)	Fewest visual effects

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Nitrates	**Contraindicated (all)**			

- Headache is the commonest AE. Efficacy is equal across agents: about 70% achieve intercourse, and 40–50% in DM and after prostatectomy.
- **Nitrates** in any form, including amyl nitrite "poppers", cause profound hypotension. If angina occurs, give no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). There is no antidote; treat hypotension with head-down tilt, fluids and phenylephrine.
- Use caution with α-blockers (stabilise first, start low) and aortic stenosis; CYP3A4 inhibitors need a lower dose.
- They are not studied after MI or stroke within 6 months, with BP <90/50 or >170/100, or in retinitis pigmentosa. For sudden visual loss, stop the drug and seek care.
- **Before labelling failure**, check stimulation, timing, food, maximum dose and **up to 9–10 attempts**; correct hypogonadism, glycaemia and lipids; then switch agent or try daily tadalafil.
- **Penile rehabilitation** after prostatectomy is unproven, since REINVENT and other nightly-dosing trials were negative.

Other oral agents (yohimbine, apomorphine, trazodone) have little or no proven benefit.

Vacuum erection device

- Negative pressure fills the corpora and a base ring traps the blood; **remove it within 30 min**. The erection is hinged at the base, cool and cyanotic.
- Up to 90% get an erection but only 30–70% are satisfied; it suits older men in stable relationships. Take care with anticoagulants.

Intraurethral alprostadil (MUSE)

- A prostaglandin E1 (PGE1) pellet of 125–1000 µg is inserted into the urethra after voiding. About 50% respond, so it is a niche option: PDE5I failure after nerve injury, add-on to a PDE5I, or soft glans after prosthesis.
- Urethral pain affects about one-third; use a condom if the partner is pregnant.

Intracavernosal injection

ICI acts directly on smooth muscle, **independent of nerves and nitric oxide (NO)**, so it works when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1) 5–40 µg	Adenylate cyclase raises cyclic AMP (cAMP)	Only FDA-approved ICI; ~70% efficacy; metabolised locally; **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; Ca^{2+} channel block	Cheap; raises liver enzymes, causes priapism and fibrosis, so not used alone
Phentolamine	α-antagonist	Adjunct only; hypotension, reflex tachycardia
Trimix (alprostadil 10 µg, papaverine 30 mg and phentolamine 1 mg per mL)	Synergistic	≈90%; rescues monotherapy failures; less pain

Agent	Mechanism	Notes
Bimix	Papaverine with phentolamine, or alprostadil with phentolamine	≈90%

- Titrate in the office from a low dose, since neurogenic ED is very sensitive, and inject the lateral shaft.
- **Priapism** is highest with papaverine mixes and in neurogenic or young men; the patient must seek care beyond 4 h. Long-term use causes fibrosis and curvature.
- Contraindications: psychological instability, priapism risk (sickle cell), severe coagulopathy, unstable CVD, poor dexterity (the partner may inject), and monoamine oxidase (MAO) inhibitors, because phenylephrine reversal risks a hypertensive crisis.

Topical agents (alprostadil cream, the over-the-counter MD3000 gel) give modest benefit.

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Li-ESWT	New vessels; nerve, endothelial, smooth muscle regeneration	RCT meta-analysis: IIEF better than sham in **mild–moderate vasculogenic** ED, the best evidence; still investigational (Sexual Medicine Society of North America: research only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline	Short-term benefit in refractory ED

Combination therapy

Monotherapy fails in nearly 40%. Useful pairs are daily tadalafil with on-demand sildenafil, a PDE5I with T, IU, ICI or VED, and a prosthesis with IU for a soft glans. PDE5I with ICI is off-label, so test it in the office.

Surgery

- A **penile prosthesis** suits men whose other therapy is contraindicated, ineffective or unwanted, and those with corporal fibrosis or Peyronie disease (see Ch 75).

P2 Surgical management of erectile dysfunction and penile prosthesis — arterial revascularisation (age <55, nonsmoker, nondiabetic, focal post-traumatic pudendal stenosis; glans hyperaemia) and investigational venous ligation

7 · Landmark trials

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED gave 45% more cardiac events at 5 years	ED is a CV sentinel; stratify risk
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I is unproven

73 Priapism

1 · Definition and classification

- **Priapism** is a full or partial erection lasting **≥4 h** beyond, or unrelated to, sexual stimulation. Untreated ischaemic priapism causes fibrosis and permanent erectile dysfunction (ED).
- It is classified by **oxygenation, not duration**: failed outflow causes time-critical ischaemia, while excess arterial inflow gives an oxygenated, painless erection that can wait.

Feature	Ischaemic (low-flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, like a compartment syndrome	Repeated self-limiting episodes, often nocturnal	Fistula between cavernosal artery and sinusoids
Typical cause	Drugs, sickle cell disease (SCD), blood disorders	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas / Doppler	Low oxygen, high CO_2, acidotic; absent or low flow	Ischaemic in each episode	Arterial gas; high flow, fistula or blush
Management	Emergency decompression	Treat episode as ischaemic, plus prophylaxis	Observe, then embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

KEY POINT

Trap: stuttering priapism is **ischaemic**, so any episode lasting ≥4 h needs the same first-line treatment.

2 · Aetiology

P3 Priapism: Diagnosis and Management — causes: ICI, α-blockers, trazodone, antipsychotics, cocaine, anticoagulant withdrawal, SCD, leukaemia, metastases, venom; non-ischaemic trauma with delayed onset

Not in the paper answer:

- Metastases obstruct venous outflow.

Ischaemic – extra points

- Many other drugs, including oral phosphodiesterase type 5 inhibitors (PDE5Is), can cause it, so take a full drug history.
- In leukaemia, give cytotoxics and leukapheresis alongside penile treatment.
- **Recurrence after a shunt** may mean shunt thrombosis. Give **aspirin 325 mg + heparin 5000 U subcutaneously** before surgery, then aspirin 81 mg daily for 2 weeks.

Sickle cell disease

- Deoxygenated haemoglobin S polymerises, causing haemolysis and low nitric oxide (NO), which obstruct outflow.

- Lifetime risk is **29–42%**; SCD causes 23% of adult and **63% of paediatric** ischaemic priapism. Stuttering is commoner than major episodes.

Non-ischaemic

- Onset is often delayed ≥1 day after trauma, because the clot over the arterial tear disperses in a nocturnal erection.
- Transient post-ischaemic hyperaemia after treatment mimics it; colour Doppler distinguishes them.

3 · Molecular pathophysiology

- Low basal NO reduces cyclic guanosine monophosphate (cGMP) feedback, so **PDE5 is down-regulated**. A neural NO surge then produces cGMP that cannot be broken down, so relaxation is unopposed.
- Paradox: **a daily PDE5I restores PDE5 and prevents** recurrent episodes.

4 · Evaluation

P3 P1 Priapism: Diagnosis and Management; Doppler in urology — history, examination, corporal blood gas table (ischaemic pO_2 <30, pCO_2 >60, pH <7.25; flaccid 40/50/7.35; arterial >90/<40/7.4), PDUS (PSV <50 vs >50 cm/s), parallel labs

Not in the paper answer:

- Record baseline erectile function, because it is needed for prosthesis counselling.
- The glans and spongiosum stay soft in both types; their involvement suggests another disease.

> **ALGORITHM**
>
> 1. History, examination and corporal blood gas.
> 2. If ischaemic, then acute ischaemic algorithm.
> 3. If indeterminate, then penile/perineal PDUS.
> 4. If non-ischaemic, then expectant algorithm.

5 · Acute ischaemic priapism (AUA/SMSNA)

- Irreversible damage begins 4–12 h after presentation, so counsel about ED risk first. Conservative measures are **adjuncts only; never delay first-line treatment**. Oral terbutaline or pseudoephedrine is not recommended.

P3 Priapism: Diagnosis and Management — ED by duration (SCD series): ≤12 h 0%, 12–24 h 22%, 24–36 h 56%, ≥36 h **100%**

> **ALGORITHM**
>
> 1. If <36 h, then aspirate ± irrigate and inject intracavernosal phenylephrine (BP and cardiac monitoring).
> 2. If that fails, then distal shunt ± tunnelling; tunnel if not done.
> 3. If still erect, then proximal shunt or penoscrotal decompression; if refractory, prosthesis.
> 4. If >36 h, then counsel high ED risk with analgesia/follow-up, OR early penile prosthesis.

Aspiration / irrigation

- After a penile block, insert a **19-G needle** at the penoscrotal junction at **3 or 9 o'clock**; one side usually suffices. Aspirate, repeat and irrigate with saline.

- Aspiration alone works in fewer than 1 in 4. Cold (10 °C) irrigation gave 96% vs 60% in a randomised controlled trial (RCT), but **avoid chilled saline in SCD**.

Phenylephrine

- It is a selective α1-agonist with the least cardiac effect.
- Dilute to **100–500 µg/mL** and give **1-mL aliquots every 3–5 min** for up to **1 h** before declaring failure; the maximum is about 1 mg in 1 h (European Association of Urology).
- Toxicity includes hypertension, headache, arrhythmia and **reflex bradycardia**. Monitor, especially with cardiovascular disease or monoamine oxidase (MAO) inhibitors.

Shunts

P3 P2 Priapism: Diagnosis and Management; Management of priapism — Winter, Ebbehoj, T-shunt, Al-Ghorab, snake tunnelling (7/8 Hegar), Quackels/Sacher (stagger ≥1 cm), Grayhack/Barry, penoscrotal decompression

Not in the paper answer:

- AUA: proximal shunts only after a failed distal shunt with tunnelling. Complications include urethral injury or fistula, necrosis, infection and pulmonary embolism.

Early penile prosthesis

- It is used for priapism **>36 h** or failed shunts, because oedematous corpora dilate easily, while delayed surgery meets dense fibrosis.
- Early vs delayed implantation: shortening 3% vs 40% and satisfaction 96% vs 60%.
- Prior distal shunts raise the risk of distal perforation (see Ch 75).

6 · Non-ischaemic priapism

*It is **not an emergency**.*

- About 60% resolve with observation, compression and ice.

ALGORITHM

1. Observe and reassess at 4 weeks.
2. If it persists, then PDUS (fistula site, size), counsel, and selectively embolise the internal pudendal/cavernosal artery.
3. If it recurs, then repeat embolisation with a non-resorbable agent.
4. Only after repeated failure, ligate surgically.

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation succeeds in ≥75%, with 30–40% recurrence. Open ligation causes ED in about 50%.

7 · Stuttering priapism — prevention

- An episode ≥4 h is treated as acute ischaemic priapism. Short episodes are managed with fluids, walking and analgesia, and home self-injected phenylephrine for breakthrough.
- No prophylactic drug has robust evidence.

Agent	Dose	Note
PDE5I	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; start when flaccid; the only RCT missed its primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Suppress libido and erections; avoid before puberty is complete
Finasteride	5 mg, tapered to 3 mg then 1 mg	One SCD series
Digoxin	0.25–0.5 mg	Inhibits Na^+/K^+-ATPase
Baclofen	Oral 40 mg at night; intrathecal	Neurogenic/reflex erections

SCD specifics

- First-line aspiration and phenylephrine are the same; give IV hydration and analgesia **concurrently, never instead**. Exchange transfusion is not a primary treatment (AUA); involve haematology and avoid chilled irrigation.

8 · Landmark trials

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

74 Disorders of Male Orgasm and Ejaculation

1 · Spectrum

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**. It includes premature ejaculation (PE), delayed ejaculation (DE), anejaculation and retrograde ejaculation (RE). Painful ejaculation, anhedonia and postorgasmic illness syndrome (POIS) are rarer.
- Orgasm is a separate **cortical** event, so ejaculation and orgasm can dissociate.

2 · Physiology of ejaculation

- Lumbar interneurons at **L3–L5** form the **spinal ejaculation generator**.

Phase	Events	Mediation
Emission	Vas, seminal vesicles and prostate deliver semen to the posterior urethra	Autonomic, **T10–L2**
Ejection	Bulbospongiosus, ischiocavernosus, levator ani contract rhythmically; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** and a sympathetic reflex
Orgasm	Cerebral processing of pudendal input	Cortical

- **Dopamine (D2) promotes** and **serotonin (5-HT) inhibits** ejaculation.
- 5-HT2C stimulation delays and 5-HT1A shortens ejaculation, so lifelong PE is thought to be **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT**
> Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · Premature ejaculation — definition and classification

Short latency alone is not PE; loss of control and distress are needed.

P1 P3 Premature Ejaculation — ISSM 2014 three criteria; lifelong/acquired/variable/subjective; AUA 2 min; aetiology, evaluation (IELT, PEDT, IIEF-5), drug table

Not in the paper answer:

- DSM-5 duration (≥6 months, 75–100% of occasions) and severity bands are in the table below.

	Lifelong	Acquired	Variable	Subjective
Onset	From first encounters, every partner	After previously normal latency	—	—
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural
Disorder?	Yes	Yes	No	No

- The International Society for Sexual Medicine (ISSM) requires ejaculation within **about 1 min** (lifelong) or a bothersome fall, often to **≤ about 3 min** (acquired). It is the first

evidence-based definition, for vaginal intercourse only. **Latency alone never suffices**.

	ISSM	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (2–3 min or ≥50%)
Duration/frequency	Always or nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- The AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2-min line trades specificity for sensitivity.
- Community stopwatch intravaginal ejaculatory latency time (IELT) is positively skewed, median **5.4 min**; ≤1 min is statistically abnormal.

4 · Premature ejaculation — aetiology and evaluation

- Lifelong PE affects younger men with few comorbidities; acquired PE affects older men with metabolic syndrome or diabetes.
- **PE with erectile dysfunction (ED)** affects up to half of men with ED, who rush before a partial erection is lost.
- PE is the main sexual symptom in **chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)**, and it improves when **hyperthyroidism** is treated, though routine thyroid tests are not indicated.
- Examination is desirable in lifelong and **mandatory in acquired PE**, with digital rectal examination (DRE) for prostatitis.

5 · Premature ejaculation — treatment

ALGORITHM

1. Confirm PE.
2. If it is secondary to ED or another dysfunction, then treat that primary cause.
3. If lifelong, then SSRI ± behavioural therapy (patient preference).
4. If acquired, then treat the cause ± behavioural therapy/SSRI; try withdrawal after 6–8 weeks.
5. If variable or subjective, then reassurance, education, psychotherapy.

- Behavioural therapy is mainly an adjunct.

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not FDA); short half-life; works from the first dose; no PDE5I interaction; many stop it
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label

Drug	Dose / timing	Notes
Clomipramine (a tricyclic antidepressant, TCA)	12.5–50 mg daily or 3–4 h before	Off-label
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
Phosphodiesterase type 5 inhibitors (PDE5Is)	30–50 min before	Best with coexisting ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- **Daily selective serotonin reuptake inhibitors (SSRIs)** act in 5–10 days, fully at **2–3 weeks**.
- Use caution in bipolar disorder, with aspirin or NSAIDs, and under 18 years. **Never stop them abruptly.**
- Meta-analyses show that a PDE5I beats placebo and that a PDE5I with an SSRI beats an SSRI alone.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; established relationships

KEY POINT

The real limiter in PE is adherence, not efficacy: most stop SSRIs and topicals (cost, lost spontaneity, "antidepressant" stigma).

6 · Delayed ejaculation, anejaculation and anorgasmia

- DE (DSM-5) is a marked delay or absence of ejaculation despite adequate stimulation, on ~75–100% of occasions, with distress. There is **no validated cut-off**, but >25–30 min with distress suggests DE.
- **Anorgasmia** is absent perceived orgasm. DE is mostly acquired and rises with age; its hallmark is normal erections with **low subjective arousal**.

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism causes PE), hyperprolactinaemia
Neurogenic disease	Spinal cord injury (SCI), then diabetic autonomic neuropathy, then multiple sclerosis (MS)
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, retroperitoneal lymph node dissection (RPLND)
Anatomical (causes RE)	Transurethral resection of prostate (TURP), bladder neck incision

Category	Causes
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, tuberculosis (TB), schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), TCAs, antipsychotics, α-methyldopa, thiazides, alcohol

- **Superior hypogastric plexus** injury at RPLND causes dry ejaculation, which **nerve-sparing RPLND** preserves in most men.
- Ask whether masturbation works and whether he has orgasm without semen (RE or obstruction). Examine the vasa and do a DRE; check morning testosterone (T), HbA1c and lipids.

ALGORITHM

1. Orgasm? If never or sometimes, then inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique).
2. If always but no ejaculate, then look for **sperm in post-orgasm urine**; if present, RE.
3. If absent (aspermia), then failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO).

	Bilateral EDO	RE	Failure of emission
Antegrade volume	Very low	Low or absent	Absent
pH, fructose	Low (seminal vesicle fluid, ~80% of volume, is missing)	—	—
Post-orgasm urine	No sperm	**Sperm and fructose**	No sperm

Treatment

- Treat the cause, and counsel about anejaculation and sperm retrieval before pelvic surgery. Refer to a psychosexual therapist once organic and drug causes are excluded (AUA).
- Normalise T, change the offending drug and change masturbatory style. Drugs are all off-label, with about 40% success.

Drug	Action	Dose
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once or twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced DE); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

7 · Retrograde ejaculation

P3 Retrograde ejaculation — bladder neck failure; TURP 42–100%, TUIP 5–45%, diabetic neuropathy; post-ejaculate urine; pseudoephedrine/imipramine doses; urinary sperm retrieval for IUI/IVF/ICSI

Not in the paper answer:

- Diagnosis: **>5–10 sperm/high-power field** in urine passed after masturbation confirms RE.
- Aquablation preserves ejaculation better than TURP in randomised controlled trials.

8 · Sperm retrieval in SCI

- Reflex erection needs the sacral arc, but ejaculation needs thoracolumbar outflow, so it is rare (<5%) with complete upper motor neuron lesions; RE is common.

ALGORITHM

1. **Penile vibratory stimulation (PVS)** is first line, with up to ~70% success; it needs an intact reflex at the T12–S1 level.
2. If PVS fails, then **electroejaculation (EEJ)**: a rectal probe stimulates hypogastric sympathetic efferents; effective and safe.
3. If EEJ fails, then prostate massage, then surgical retrieval (TESE/TESA/MESA/PESA); testicular sperm extraction succeeds in most.

- Lesions above T6 risk **autonomic dysreflexia**: pretreat with **nifedipine** and monitor BP.

9 · Other disorders

Disorder	Key points
Painful ejaculation	Prostatitis/CPPS, urethritis, benign prostatic hyperplasia (BPH), seminal vesicle stones, EDO, α-blockers (least with alfuzosin); often none. Surgery (TURP, transurethral resection of ejaculatory ducts (TURED), pudendal neurolysis) weakly supported
Haematospermia	Usually benign (see pointer below). Seminal vesiculoscopy if refractory (see Ch 70); evaluate if >40 y or persistent
Climacturia	Leak at orgasm after radical prostatectomy. Steps: void first, condom, pelvic floor training, constriction ring, male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital numbness, low desire, ED, anhedonia persisting months to years
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral ache rising with excitement; sudden, intense at orgasm

P1 Haematospermia — causes, who needs work-up (≥40 y, persistent, haematuria), PSA/TRUS/MRI, treatment

75 Surgery for Erectile Dysfunction

1 · Place of surgery and devices

A prosthesis is the last step for erectile dysfunction (ED).

P2 P3 Surgical management of erectile dysfunction and penile prosthesis; Types of penile prosthesis — last-line but most effective (satisfaction 80–90%; repair 15% at 5 y, 30% at 10 y); Lue goal-directed approach; malleable vs 2-piece vs 3-piece; device names, expansion, coatings; consent; pre-op preparation; revascularisation

Not in the paper answer:

- Coloplast cylinders are polyurethane (Bioflex), so keep cautery **<35 W**; Boston and Rigicon silicone cylinders tolerate cautery.

2 · Selection and counselling

- **Consent is expectation management**: the device restores rigidity, not the former erection.
- The commonest disappointment is a **shorter erection**, because the pseudocapsule limits expansion (worse in Peyronie disease). Document it.
- With **phimosis**, circumcise first and implant about 6 weeks later.
- Raised glucose on the day needs intravenous (IV) insulin. Nasal methicillin-resistant Staphylococcus aureus (MRSA) carriers get chlorhexidine bathing and mupirocin 2% for 5 days.

3 · Surgical approaches

Approach	Advantages	Cautions
Penoscrotal	Majority choice; secure subdartos pump; good crural exposure	Urethra at risk, so Foley first; reservoir via inguinal canal
Infrapubic (Scott)	Reservoir under vision	Dorsal neurovascular bundle: stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise at the same time, or lymphoedema follows
Ventral penile	Rods without bending; foreskin kept	—

4 · Operative technique (three-piece)

ALGORITHM

1. Catheter; expose the tunica.
2. Two 2-0 polydioxanone (PDS) stay sutures per corpus; longitudinal corporotomy.
3. Dilate proximal, then distal (Metzenbaum, Hegar/Brooks 7→13 mm).
4. Goalpost test; measure (Furlow).
5. Cylinders in; test inflation 1. Close corporotomies; test inflation 2.
6. Reservoir, then pump; connect; test inflation 3.
7. Irrigate and close.

- Dilators point **dorsolaterally**, away from urethra and septum. The **goalpost test** (equal height and symmetry) detects crossover.

- The cylinder is equal to or 1 cm shorter than measured, because oversizing causes buckling and S-deformity; rear tip extenders adjust length.
- The pump sits in a dependent subdartos pouch, anterolateral to the testis.

Reservoir placement

- The **Retzius** space is entered with the bladder empty; the iliac vessels lie about 1 inch lateral.
- In a hostile pelvis (radiation, cystectomy, robotic prostatectomy, transplant) use an **ectopic** site, made safe by the **lockout valve**.
- **Submuscular suprafascial** is the commonest ectopic site. Fill it nearly full, because underfilling leads to migration.

5 · Intra-operative problems

- **Fibrotic corpora** follow priapism or explant (pelvic radiation does not add difficulty). Use cavernotomes or scar excision and **narrow cylinders**, always two.
- **Crossover** gives a lopsided erection and unequal measurements. Leave a large dilator in the common cavity and redilate the other side laterally; the Foley should lie midline between cylinders.
- **Crural perforation** (one dilator deeper, lengths differ by >1 cm) is repaired with a **suture sling** (2-0 Prolene through the rear tip, over the corporotomy).
- **Urethral perforation** at the corporotomy is closed and the operation proceeds. If distal (fossa navicularis, irrigant at the meatus), **abandon that cylinder**, place the contralateral one, leave no catheter and replace at 5–6 months.
- A **hypermobile glans** drooping ventrally is the supersonic transporter (**SST**) deformity. It is fixed with glans sutures (3-0 non-absorbable).
- **Residual curvature** is treated with high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X) and **modelling** (clamp tubing, bend 90° away for 90 s), then plication or incision with or without graft. Less than 20° is acceptable (see Ch 76).

> **KEY POINT**
> No catheter after distal urethral perforation: the urinary stream draws fluid into the urethra (Bernoulli), so urine does not enter the defect. Wait 5–6 months for scar to obliterate the false tract. Most intra-operative errors are **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

> **KEY POINT**
> After robotic prostatectomy the peritoneum is not replaced and bowel lies near the internal ring, so a blind inguinal Retzius reservoir can injure bowel. Choose an ectopic site or place it under vision.

6 · Complications

Infection usually costs the whole device.

Infection (1–3%)

- Risks include hyperglycaemia, revision, MRSA and scarred corpora.
- American Urological Association (AUA) prophylaxis is IV vancomycin, or a 3rd-generation cephalosporin plus gentamicin, 1 h before incision and for ≤24 h. Extended oral courses are unproven; in high-risk men add an antifungal, with or without piperacillin–tazobactam.

- Coatings (InhibiZone rifampin–minocycline; hydrophilic Titan/Infla) halve infection, and a **no-touch** technique gives <0.5%.
- Fever and erythema suggest infection; pus, an exposed device, progressive pain or a **pump fixed to the scrotal wall** are definitive. **Remove ALL components**, as retained parts harbour biofilm.

Option	Technique	Caveats
Explant, delayed replacement	Antibiotic irrigation drains for 3 days; reimplant at 2–6 months	Fibrosis, loss of several cm; vacuum erection device (VED) meanwhile
Mulcahy salvage	Remove all, antiseptic washes, new drapes, immediate new device	~84%; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin and gentamicin, salvage at 3 days	Aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Washes use dilute povidone-iodine (0.35–3.5%) for 3 min, then saline or antibiotic; hydrogen peroxide is dropped as cytotoxic.
- An uninfected revision carries ~10% infection, falling to ~3% with antiseptic washout of all cavities.

Mechanical failure

- This is the **commonest complication**: leak, pump malfunction, autoinflation, kinking or cylinder aneurysm.
- If the device is >3 years old, replace **all** parts; check any retained part with the **ohmmeter test**. A rod that loses rigidity has a core fracture and is replaced.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover, catheterisation; rods in insensate men (spinal cord injury)
Management	**Distal corporoplasty** (new dorsomedial plane behind pseudocapsule)	Remove all, washout, contralateral spacer, replace at 5–6 months

- **Reservoir erosion** into the bladder needs reservoir exchange and repair; into bowel, remove the whole device.
- **Penile necrosis** is linked to diabetes, vascular disease, smoking, prior implant and radiation. Avoid circumferential incisions and tight wraps, and remove cylinders promptly if it occurs.
- An early basal bulge is corporotomy disruption (repair); a late one is cylinder aneurysm (replace).

7 · Special situations

- After **priapism** >36 h, an **immediate implant** preserves length and relieves pain; avoid distally expanding cylinders (see Ch 73).
- In **Peyronie disease** length loss is greater, so use non-expanding high-rigidity cylinders with modelling.

- Transplant and HIV-positive men (adequate CD4) have no excess infection.

8 · Post-operative care

- It is day surgery. Leave cylinders **partially inflated** so the pseudocapsule forms large, remove the drain within 24 h and keep the penis upward for 4–6 weeks.
- Activation is at **4–6 weeks**, then the device is cycled daily. Dissatisfaction is mainly about length and rigidity.

9 · Penile revascularisation †

- †Only young men (<55) with a focal post-traumatic pudendal occlusion qualify (inferior epigastric to dorsal artery). It is not recommended for generalised atherosclerosis or venous leak.

76 Diagnosis and Management of Peyronie Disease

1 · Definition

- **PD** (American Urological Association, AUA 2015) is an acquired penile abnormality with **fibrosis of the tunica albuginea**, with or without pain, deformity, erectile dysfunction (ED) and distress.

> **KEY POINT**
> Myths corrected: PD is **not rare** (3–20%; higher in diabetes and ED). **Spontaneous resolution is rare** and up to half worsen untreated, so "wait 6–12 months" is a myth. **Calcification is not a marker of chronic disease**.

2 · Anatomy and pathophysiology

- The outer longitudinal tunica layer is **absent at 5–7 o'clock**, so the dorsum buckles; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma** from bending during intercourse causes **delamination at the septal insertions**.
- **Transforming growth factor-β1 (TGF-β1)** is central. Collagen becomes disorganised and elastin decreases, so the penis **curves toward the plaque**.
- A **susceptible host** is needed, and PD is associated with **Dupuytren and Ledderhose** disease; only 16–40% recall trauma.

3 · Clinical features and phases

- Features are pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), plaque and ED. Hourglass or hinge gives an **unstable penis**.
- Kelami grading: ≤30° mild, 31–60° moderate, >60° severe. Ventral and lateral curves are harder to manage than dorsal.

Feature	Active (acute)	Stable (chronic)
Pain	Present (flaccid, erection, intercourse)	Usually resolved
Deformity	Progressive	No progression
Aim	Prevent progression; non-steroidal anti-inflammatory drugs (NSAIDs) for pain; traction	Correct deformity (injection, surgery)
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- The differential includes congenital curvature, chordee, post-traumatic fibrosis and tumours (epithelioid sarcoma mimics PD).

4 · Evaluation

- The **key question** is: "If straight with current rigidity, would it be adequate for penetration?" If **no**, offer a prosthesis with straightening; the answer also predicts post-operative erectile function.
- Measure **stretched penile length (SPL)** from pubis to corona dorsally; this is essential, as patients overestimate loss.

- Extensive **calcification** on ultrasound predicts poor intralesional response and more surgery.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **Intracavernosal injection (ICI) with or without penile duplex Doppler ultrasonography (PDDUS)** is required by the AUA **before any invasive treatment**. It measures curvature and gives peak systolic velocity, end-diastolic velocity and resistive index (RI).

5 · Non-surgical management

P1 Management of Peyronie's disease — phases, evaluation, non-surgical options table, surgical algorithm

Not in the paper answer:

- **Oral**: the AUA "should not offer" any except **NSAIDs for pain**; vitamin E, tamoxifen, colchicine and carnitine give no benefit.
- **Extracorporeal shock wave therapy (ESWT)** helps pain, not curvature; radiotherapy is harmful.

Collagenase Clostridium histolyticum (CCH, Xiaflex)

- It is the only PD drug approved by the Food and Drug Administration (FDA, 2013). It degrades collagen **types I and III** and spares elastin, vessels and nerves.
- **Protocol (IMPRESS)**: 2 injections of **0.58 mg** 24–72 h apart, then **model** the penis 24–72 h after the second. Give up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential.
- **AUA criteria** are stable disease, curvature **>30° and <90°** and intact erections (with or without medication). IMPRESS excluded ventral plaques, extensive calcification and disease <12 months.
- Response is poor with calcification, curve ≥60°, severe indentation or hinge.
- Bruising and pain are common; the serious risk is **corporal rupture** (0–4.9%), so there is no intercourse until modelling ends.

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	Calcium channel blocker (CCB): reduces fibroblasts and extracellular matrix, increases collagenase	Randomised trials conflicting; stabilises; inferior to CCH; cheap	May offer
Interferon α-2b	Reduces fibroblasts and collagen, increases collagenase	Randomised controlled trial (RCT) vs saline: curvature reduced 27% vs 9%; flu-like symptoms (pre-treat with NSAIDs)	May administer
Nicardipine	CCB	Reduces pain and plaque; no curvature benefit	—

- Verapamil is given as 10 mg diluted (10–20 mL).

Mechanical therapy

- **Penile traction therapy (PTT)** in the acute phase reduces curvature, pain and the need for surgery, and gains length; it needs at least 3 h/day.
- The AUA makes no recommendation for traction, vacuum or combinations.

6 · Surgery

Indications

- Disease is **≥12 months from onset, with deformity stable ≥6 months**.
- Deformity or poor rigidity prevents intercourse, and conservative therapy has failed.

Consent

- The goal is a **functionally straight** penis (≤20°; European Association of Urology, EAU, ≤15°). Residual or recurrent curvature occurs in up to 16%.
- **Length loss** follows every procedure (plication more than grafting); 70–80% have already lost length, so document SPL.
- **New ED** is commoner after grafting than plication (~30% if pre-op rigidity is suboptimal); pre-op erection quality predicts it best.

Choice of procedure — algorithm

ALGORITHM

1. Ask: is the erection adequate (with or without a PDE5I)?
2. If **no**, place an inflatable prosthesis with modelling. If the curve persists, incise with or without a graft (graft if defect >2 cm).
3. If **yes**, do PDDUS. If the curve is simple and <60–70°, with no hourglass or hinge and acceptable length loss, **plicate** (if length loss is unacceptable, graft).
4. If **yes** but the curve is complex/multiplanar >60–70°, or there is severe hourglass, hinge or a short penis (<9 cm stretched), do **plaque incision/partial excision with graft** if the ED risk is accepted (otherwise prosthesis).

Plication (shortening)

- It shortens the **convex side opposite the plaque**. It is simple, with the least new ED, but does not fix hourglass, hinge or shortening.

Technique	Method	Point
Nesbit (1965)	Transverse elliptical full-thickness excision opposite the curve, closed transversely	Original; for a ventral curve, lift Buck fascia and take dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow the shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First incisionless plication
16-dot	Permanent extended Lembert, 4 dots per plication	Popular; no incision
Tunica albuginea plication (TAP, Levine)	Paired partial-thickness transverse incisions (outer layer only); buried permanent suture and 3-0 polydioxanone (PDS)	Cavernosal tissue undisturbed

Plaque incision/partial excision and graft (lengthening)

- It suits complex deformity with **strong erections** and no veno-occlusive dysfunction, and restores calibre and 0.5–3 cm of length.
- ED predictors are age >55, veno-occlusive dysfunction (RI <0.80), a large graft, a ventral curve and a curve >60°.
- **Key steps**: elevate Buck fascia with the neurovascular bundle, make an H/double-Y incision **crossing the septum**, and sew in the graft (pericardium or small intestinal submucosa) with running 4-0 PDS.

Prosthesis with straightening

- PD with PDE5I-refractory ED is treated in one operation; an **inflatable** device is preferred.
- **Manual modelling** (Wilson–Delk): inflate fully, clamp the tubing and bend opposite the curve for 60–90 s, with the hand on the shaft, not the glans. It straightens 86–100%.
- Residual curvature >30° needs a relaxing incision, with a graft if the defect is >2 cm. Length loss is the commonest complaint (see Ch 75).

7 · Landmark trials

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature reduced 34% (17°) vs 18% (9.3°); bother improved	Basis of FDA approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional interferon (IFN) vs saline	Curvature reduced 27% vs 9%	First RCT of intralesional therapy

★ Putting it together: andrology decisions

A · Infertile man: work-up

ALGORITHM

1. **History and examination**: testosterone use, mumps, cryptorchidism; testis volume, vasa, varicocele (see Ch 69).
2. **Two semen analyses** (abstinence 2–7 days). Low volume with acid pH suggests duct obstruction or congenital bilateral absent vas (CBAVD).
3. **Hormones**: follicle-stimulating hormone (FSH) and testosterone; if testosterone is low, add luteinising hormone (LH) and prolactin.
4. **Genetics** (azoospermia, severe oligozoospermia): karyotype, Y microdeletions; CFTR in CBAVD.

B · Azoospermia: obstructive or not

	Obstructive (OA)	Non-obstructive (NOA)
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low volume, vasectomy or hernia repair	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy
Treatment	Vasovasostomy (VV) or vasoepididymostomy (VE), resection of ejaculatory ducts, or sperm retrieval for ICSI (see Ch 70)	Micro-TESE with ICSI; fix a clinical varicocele first; no retrieval if complete AZFa/b deletion

KEY POINT

† Reversal suits a younger partner wanting >1 child; ICSI suits an older partner or female factor. Copious vas fluid or sperm means VV; toothpaste-like fluid or none means VE (see Ch 70).

C · Erectile dysfunction ladder

ALGORITHM

1. **Assess:** IIEF-5 score, HbA1c, lipids, morning testosterone. ED is a cardiovascular sentinel, so stratify cardiac risk (see Ch 72).
2. **Modify** risk factors; treating hypogonadism rescues PDE5 inhibitor (PDE5I) failures.
3. **Shared decision:** PDE5I (never with nitrates), vacuum, intraurethral alprostadil or intracavernosal injection.
4. **Penile prosthesis** (inflatable three-piece) if others fail; highest satisfaction. (see Ch 75).
5. **Young, focal post-traumatic arterial occlusion** → revascularisation.

D · Priapism: ischaemic vs non-ischaemic

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Semi-rigid, painless; after perineal trauma

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Like arterial blood: pO_2 >90, pCO_2 <40
Doppler	Little or no flow	High flow, fistula or pseudoaneurysm
Treatment	Aspirate with or without irrigation, then phenylephrine (100–500 µg every 3–5 min, up to 1 h), then distal shunt with or without tunnelling, then proximal shunt. >36 h or refractory: early prosthesis	Observe; selective embolisation if persistent

KEY POINT

Sickle cell: treat as above without waiting for transfusion. Stuttering priapism: daily PDE5I, nightly pseudoephedrine, or hormonal suppression after puberty (see Ch 73).

E · Peyronie disease

ALGORITHM

1. **Active**: analgesia, traction. Collagenase (CCH): stable dorsal/lateral 30–90°, intact erections; else verapamil, interferon α-2b.
2. **Surgery**: ≥12 months from onset, stable ≥6 months (see Ch 76).
3. If **erections are good**: simple <60–70°, no hourglass/hinge, adequate length → plication; complex, hourglass/hinge or short → incision/partial excision + graft (more ED).
4. If **erections are poor** despite PDE5I → inflatable prosthesis with modelling.

F · Premature ejaculation (PE)

KEY POINT

Diagnosis needs short latency (lifelong ≈1 min or less), poor control and distress. Treat ED first. Options: stop–start or squeeze, lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily SSRI (paroxetine best), tramadol (off-label, addiction risk) (see Ch 74).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 10

Testis, Penis & Urethra

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 78	Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum
Ch 79	Neoplasms of the Testis
Ch 80	Surgery of Testicular Tumours
Ch 81	Minimally Invasive Retroperitoneal Lymph Node Dissection
Ch 82	Tumours of the Penis
Ch 84	Inguinal Node Dissection
Ch 83	Tumours of the Urethra
Ch 85	Benign Disorders of the Penis and Urethra
Ch 86	Urethral Stricture Disease
Ch 87	Surgery of the Scrotum and Seminal Vesicles
★	Putting it together

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

1 · Definition, boundaries and organs

P1 Describe in brief the anatomy of the retroperitoneum… retroperitoneal laparoscopic nephrectomy — boundaries, primary vs secondary organs, flank layers, lumbodorsal fascia at 12th rib tip, three spaces, lateroconal fascia, perirenal cone open below, fat colour

Not in the paper answer:

- Primary organs lie in the perirenal space; secondary organs lie in the anterior pararenal space.
- Secondary organs exclude the first part of the duodenum (**D1**) and the pancreatic tail; the **anal canal** is primarily retroperitoneal.

> **KEY POINT**
> Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · Posterior abdominal wall and surface landmarks

- The upper pole is at the 11th rib (left) and 12th (right, lower). The hilum is at **L1**; the lower pole at L3–L4.
- **Psoas major** (T12–L5 to lesser trochanter) flexes the hip with iliacus. **Quadratus lumborum** lies posterolateral to psoas and stabilises the 12th rib; psoas minor is inconstant.
- Dividing the **costovertebral (lumbocostal) ligament**, which tethers the 12th rib to L1–L2, frees the rib in posterior approaches.
- The **intercostal bundle** lies under the rib above (vein, artery, nerve from top down), so the nerve is least protected. Supracostal access may cross the pleura and has more complications than subcostal.
- **Dorsal lumbotomy** incises the lumbodorsal fascia vertically, lateral to erector spinae and quadratus, so **no muscle is cut**.

3 · Retroperitoneal fasciae and spaces

Collections spread along these spaces.

Three fetal strata form three fasciae and spaces: outer (transversalis; posterior pararenal), intermediate (Gerota; perirenal, genitourinary) and inner (fusion fascia; anterior pararenal, gastrointestinal).

- The posterior Gerota lamina fuses medially with transversalis; this **must be incised to reach the hilum**.
- The Toldt (anterior) lamina is thinner than the Zuckerkandl (posterior) lamina.
- The perirenal cone may cross the midline below the hilum. Urinoma, haematoma and abscess are contained laterally but **track caudally**.
- The **anterior pararenal** space holds colon, pancreas and D2–D3, and is entered from the **white line of Toldt**.

4 · Gastrointestinal relations (operative hazards)

- **Right**: the pancreatic head lies in front of the inferior vena cava (IVC), and D2 lies on the right hilum, hence the **Kocher manoeuvre**. D3 crosses between the superior mesenteric artery (SMA) and the aorta.
- **Left**: pancreatic tail, spleen and stomach lie near the upper pole, and splenic vessels are at risk when the pancreas is lifted off renal fascia.

5 · Great vessels

Aorta and branches

The aorta enters at **T12**, left of the IVC, and bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac	Anterior	T12	Foregut
Middle adrenal	Paired lateral	SMA level	Adrenal
SMA	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
Inferior mesenteric (IMA)	Anterior	L3–L4	Hindgut

- Supernumerary renal arteries occur in ≥ ¼ (see Ch 84).
- **Ureteric supply**: proximal from renal; middle from aorta, common iliac and gonadal; distal from internal iliac. The abdominal ureter is supplied **medially**, the pelvic ureter **laterally**.
- IMA ligation is safe because the marginal artery of **Drummond** feeds the colon. **SMA ligation is not safe**: it causes severe bowel ischaemia.

IVC and veins

- The IVC passes the diaphragm at **T8**.
- The **left renal vein** crosses in front of the aorta below the SMA. It receives gonadal (right-angle entry), adrenal and ≥ 1 **lumbar vein** (avulsion risk). The **right** is short, without tributaries, and duplicated in one-sixth. Both enter the IVC at L1.
- The **right gonadal vein** enters the anterolateral IVC; the short **right adrenal vein** enters the posterolateral IVC.
- Left varicocele predominates (longer vein, perpendicular entry). A **new right varicocele** needs imaging for a retroperitoneal mass, e.g. renal cell carcinoma with venous thrombus.
- Dissect the right ureter **lateral** to the gonadal vein; medial dissection risks IVC injury and gonadal avulsion.
- The inferior mesenteric vein (**IMV**) lies in the left mesocolon and at reoperation is mistaken for the gonadal vein.

Variants (essential)

P1 P3 IVC anomalous development and retrocaval ureter; Retrocaval ureter — renal collar embryology, circumaortic/retroaortic vein, retrocaval ureter types and repair

- † A retroaortic left renal vein (≈ 2–3%) is injured when lumbars are divided or the aorta rolled in retroperitoneal lymph node dissection (RPLND).
- † A circumaortic vein (≈ 5–7%) has a posterior limb that mimics a lumbar vein, so check the pre-operative CT.
- † Retrocaval ureter is a **venous** anomaly with a "reverse-J" on pyelogram, repaired by dismembered pyeloplasty in front of the IVC.

6 · Lymphatics

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — right- to-left cephalad flow, cisterna chyli L1–L2, three nodal regions, landing zones (right interaortocaval, left para-aortic), right-to-left crossover only

- Lumbar nodes are the primary drainage of kidney, adrenal, ureter and gonad.

> **KEY POINT**
> The testis is embryologically retroperitoneal, so it drains to where it began. Crossover is one-way (right to left), so right templates include interaortocaval and some para-aortic tissue. See Ch 80, 81.

7 · Nerves

Autonomic

- **Parasympathetic** outflow is craniosacral (III, VII, IX, X, S2–S4). **Sympathetic** outflow is thoracolumbar, **T1–L2**.
- **Sympathetic trunks** lie medial to psoas behind the great vessels, and are met when lumbar vessels are divided.
- **Ejaculation**: † postganglionic fibres from the **T12–L3** ganglia cross in front of the aorta and IVC and converge in the hypogastric plexus below the IMA. They drive **emission** and **bladder neck closure**, so extensive dissection causes anejaculation or retrograde ejaculation, while erection is spared. See Ch 81.

Lumbar (lumbosacral) plexus

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, penile root, anterior scrotum
Genitofemoral	L1–L2	**On anterior psoas**; genital branch to cremaster	Anterior scrotum; upper anterior thigh
Lateral cutaneous thigh	L2–L3	Sensory only	Lateral thigh
Obturator	L2–L4	Medial to psoas, to obturator canal; adductors	Medial thigh
Femoral	L2–L4	Lateral to psoas; quadriceps	Anterior thigh, medial leg

Nerve	Setting	Result of injury
Genitofemoral	Psoas hitch, laparoscopic varicocelectomy, RPLND	Anterior scrotal/thigh numbness
Obturator	Lateral-wall TURBT (bladder tumour resection); pelvic node dissection	Adductor jerk causing bladder perforation
Femoral	Retractor blade on psoas; lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · Imaging anatomy

- On axial CT the **duodenum lies against the right hilum** and the left renal vein in front of the aorta.
- † Before RPLND or nephrectomy, check accessory renal arteries, retro-/circumaortic left renal vein and nodal size by zone.

79 Neoplasms of the Testis

1 · Epidemiology and risk factors

- Over 95% are **germ cell tumours (GCTs)**: seminoma or non-seminomatous GCT (NSGCT); 5% are extragonadal. It is the commonest solid malignancy in men aged **20–40 y**.
- Cisplatin raised metastatic cure to **80–90%**, so the aim is **minimum toxicity without losing curability**.
- Bilateral in **2%**, mostly metachronous.
- **Risk factors**: White race, cryptorchidism, family history, personal history, germ cell neoplasia in situ (GCNIS) and infertility. Undescended testis (UDT) has a relative risk (RR) of **4–6**, falling to 2–3 with orchidopexy before puberty. A previous GCT gives a 12-fold contralateral risk, yet the 15-y cumulative incidence is only 2%.
- **GCNIS** is an arrested fetal gonocyte formed **before birth**. It lies beside invasive GCT in 80–90% and invades in **50% at 5 y** and 70% at 7 y. Contralateral GCNIS is found in 5–9%, and in ≈36% with atrophy, UDT or age <40.
- Microlithiasis alone needs no work-up; it matters only after a previous GCT.

> **KEY POINT**
> GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes risk.

2 · Pathology and classification

P3 Classify testicular tumors and describe management of seminoma — WHO 2016 groups, i(12p), any NSGCT element = NSGCT, seminoma IHC

Not in the paper answer:

- Isochromosome 12p, i(12p), proves GCT origin in a carcinoma of unknown primary and in somatic-type malignancy from teratoma.

Subtype	Markers	IHC / histology	Hallmark
Seminoma	Human chorionic gonadotrophin (hCG) ≈15%; **never alpha-fetoprotein (AFP)**	CD117+, CD30−, placental alkaline phosphatase (PLAP)+	4th–5th decade; can transform to NSGCT
Spermatocytic	None	OCT3/4−, PLAP−; three cell types	6th decade; never mixed, not with UDT; **benign** unless sarcomatous/anaplastic
Embryonal carcinoma (EC)	AFP/hCG or normal	AE1/AE3+, OCT3/4+, CD30+†, c-KIT−	Totipotent; spreads with normal markers; predicts occult spread
Yolk sac	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG	Syncytio- and cytotrophoblasts	Spreads by blood (lung, liver, brain); bleeds; poor risk
Teratoma	Normal/mild AFP	≥2 germ layers; maturity irrelevant after puberty	Benign histology, malignant genetics; **chemoresistant**

- Teratoma is present in ≈half of mixed GCTs. Unresected, it causes growing teratoma, somatic transformation or late relapse, so surgery is integral to cure.
- hCG shares its α-subunit with TSH and LH, so it can cause hyperthyroidism and gynaecomastia.

> **KEY POINT**
> Management follows the most chemoresistant element: EC drives occult metastasis, choriocarcinoma haemorrhage, teratoma surgery.

3 · Presentation

- A **painless mass** is commonest. Metastases are present at diagnosis in ≈2/3 of NSGCT and 15% of seminoma; 10–20% present with metastatic symptoms such as back pain or a left supraclavicular node.
- About 1/3 are first misdiagnosed, so re-examine epididymo-orchitis 2–4 weeks after antibiotics.

4 · Diagnosis and initial management

Scrotal ultrasound (US)

- GCT is **hypoechoic**: homogeneous suggests seminoma, heterogeneous suggests NSGCT.
- In metastatic GCT with normal palpation, US seeks a **burned-out primary**, which still needs orchidectomy; normal testes mean extragonadal GCT.
- An impalpable <10 mm lesion with normal markers is GCT in 20–50%: options are orchidectomy, inguinal exploration with frozen section, or serial US.

Serum tumour markers

P1 Tumour markers in urology — sources, never-raised-in, half-lives, S stage, raised AFP = NSGCT

Not in the paper answer:

- Half-lives: AFP **5–7 days**, hCG **24–36 h**, LDH 24 h.
- After orchidectomy, a slow decline with negative imaging is monitored, but **rising markers mean chemotherapy**.
- An unstable patient with very high AFP/hCG gets chemotherapy first and orchidectomy later.

False positives (hepatic disease, hypogonadal LH cross-reactivity, marijuana) occur, and miR-371a-3p is a GCT-specific marker not raised by teratoma.

Radical inguinal orchidectomy

- Remove testis and cord to the internal ring within 1–2 weeks. **Trans-scrotal approach or biopsy is contraindicated**. Orchidectomy alone cures 80–85% of clinical stage (CS) I seminoma and 70–80% of CS I NSGCT.
- **Testis-sparing surgery** suits synchronous bilateral or solitary testis (<2–3 cm, adequate androgen) or a suspected benign lesion, with frozen section; GCNIS then gets ≥20 Gy.
- Biopsy the contralateral testis if atrophy, UDT, age <40 or suspicious US.

5 · Staging

Spread and imaging

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — retroperitoneum first site 70–80%, landing zones, crossover

Not in the paper answer:

- CT abdomen–pelvis for all misses disease in **25–35% of CS I NSGCT**, so landing-zone nodes of 5–9 mm are suspicious. Equivocal nodes with normal markers are re-imaged in 6–8 weeks.
- Brain imaging if hCG >10,000 IU/L; no routine bone scan or FDG-PET at diagnosis.
- Chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC.

Base decisions on CT and markers within 4 weeks of treatment, and offer **sperm cryopreservation** before RPLND, chemotherapy or radiotherapy.

TNM staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	LVI, or tunica vaginalis
pT3 / pT4	Spermatic cord / scrotum
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN1 ≤5 nodes, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N and hCG <5000 mIU/mL and AFP <1000 ng/mL
S2	LDH 1.5–10×N or hCG 5000–50,000 or AFP 1000–10,000
S3	LDH >10×N or hCG >50,000 or AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3** (raised markers, no visible disease)
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

IGCCCG risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no NPVM, AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	Any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	**NPVM** (10%; 72%)
Poor	**Mediastinal primary**, NPVM, AFP >10,000, hCG >50,000 or LDH >10×N (16%; 48%, now ≈71%)	**None**

NPVM = non-pulmonary visceral metastases.

KEY POINT

IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is invalid at relapse. Seminoma risk depends on NPVM alone; mediastinal NSGCT is always poor risk.

6 · Treatment principles

- Centralise care, give full-dose cisplatin and resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent, so surgery needed
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT in 10–15%, so biopsy non-responders.

GCNIS management

- Options are orchidectomy, radiotherapy **≥20 Gy** (lower doses allow recurrence) or observation. Up to 40% later need testosterone after radiotherapy.
- With a normal contralateral testis and a wish for paternity, orchidectomy is preferred because scatter harms the other testis. Cisplatin only partly clears GCNIS.

7 · NSGCT management

Every CS I option cures ≈100%; the choice is about toxicity.

CS I NSGCT

- Adjuvant therapy overtreats 70–80%.
- **Risk factors** are **LVI** (strongest) and **EC predominance**. High-risk tumours relapse in ≈50%, low-risk in <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y and mostly retroperitoneal; relapses are chemo-naive and curable.
- **Primary RPLND** removes teratoma and finds pathological stage II in 19–28%, mostly cured by surgery alone. pN1 is observed; pN2–3 (relapse >50%) gets **2 cycles adjuvant** (relapse ≤1%) or observation.
- **Bleomycin–etoposide–cisplatin (BEP) ×1** gives the lowest relapse (≈2–3%) but leaves teratoma, exposes all to late toxicity, and relapses are chemoresistant.
- Full bilateral nerve-sparing template (ejaculation >90%) is standard; a modified template is acceptable for clinically negative nodes.

ALGORITHM

1. Assess CS I NSGCT for LVI and EC predominance.
2. If low risk (neither), then surveillance.
3. If high risk, then surveillance, BEP×1 or expert nerve-sparing RPLND, chosen by preference and toxicity.

> **KEY POINT**
> Surveillance maximises quality-adjusted survival when relapse risk is <≈1/3. All options cure ≈100%; they differ in overtreatment versus difficulty of salvage.

CS IS, IIA–B

- **CS IS** is treated with induction chemotherapy.
- **CS IIA–B** with raised markers or nodes **>3 cm** gets chemotherapy, with or without post-chemotherapy RPLND (PC-RPLND). With normal markers and nodes <3 cm, **primary RPLND** is used: 13–35% are pN0, ≈30% have teratoma, and half avoid chemotherapy.

CS IIC–III: induction by IGCCCG

P2 CRE & BEP — BEP×3/EP×4 good, BEP×4 (VIP if lungs poor) intermediate/poor, toxicities

Not in the paper answer:

- BEP×3 equals BEP×4; etoposide–cisplatin (EP) ×3 and carboplatin are inferior.
- Second line is TIP×4 (paclitaxel, ifosfamide, cisplatin), VeIP×4 or VIP; TIP is best for late relapse.
- High-dose chemotherapy (HDCT) failed in three randomised trials, so it is for specialised centres.

Post-chemotherapy residual masses (NSGCT)

- With normal markers and a residual mass **>1 cm**, do **PC-RPLND** with a full bilateral template. Histology is necrosis in ≈40%, teratoma in ≈45% and viable tumour in ≈15%.
- No model, imaging or **FDG-PET** predicts necrosis (teratoma is not avid), so always resect.
- After a complete response (≤1 cm), observe, or resect if pre-chemotherapy nodes were >3 cm (controversial).
- With multiple sites, resect all, retroperitoneum first, because its necrosis predicts necrosis elsewhere. Somatic transformation (6–8%) needs complete resection.
- After viable tumour, adverse factors are intermediate/poor risk, incomplete resection and >10% viable cells.

Relapsed NSGCT

P1 Growing teratoma syndrome — enlarging mass, normal markers: stop chemotherapy, resect

- **Chemo-naive relapse** gets induction by IGCCCG, with cure >95%.
- **Early relapse (≤2 y) after chemotherapy** gets TIP/VeIP/VIP×4 or HDCT, then post-salvage surgery for residual masses.
- **Desperation surgery** is for rising markers with a single resectable site; survival is 33–57%.
- **Late relapse (>2 y)** occurs in ≈3%. Yolk sac tumour is commonest and chemoresistant, so **complete resection** is the treatment.

8 · Seminoma management

P3 P2 Classify testicular tumors and describe management of seminoma; Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis. — CS I options table, risk factors, Calvert, stage II–III, residual mass, follow-up

Not in the paper answer:

- CS I risk factors are tumour **>4 cm** and rete testis invasion (relapse 12/16/32% with 0/1/2), but they are not validated, so surveillance is preferred for all.

- IIA–B nonbulky (≤3 cm): dog-leg radiotherapy 25–30 Gy with a 5–10 Gy boost, BEP×3/EP×4, or primary RPLND (SEMS, COTRIMS). Bulky or multiple nodes get chemotherapy.
- Carboplatin ×2 gives relapse ≈1.5%. Seminoma relapses late (median 14 months), and radiotherapy carries second malignancy risk.

Residual seminoma masses are necrosis in ≈90%, and many regress over 12–18 months.

ALGORITHM

1. If the residual seminoma mass is <3 cm, then observe.
2. If >3 cm, then FDG-PET ≥6 weeks after chemotherapy (earlier scans give false positives).
3. If PET-negative, then observe.
4. If PET-positive, then template PC-RPLND (or repeat PET/observe). Mass-only excision gives 33% local relapse.

- Relapse after radiotherapy gets first-line chemotherapy; after chemotherapy, VeIP×4 or HDCT.

9 · Brain metastases

- Present in ≈1% at diagnosis, with **choriocarcinoma or very high hCG**; they can bleed fatally. Give BEP×4, then resect residuals.

10 · Treatment sequelae and follow-up

- RPLND can cause ejaculatory loss (>80% after full template without nerve sparing). Orchidectomy alone causes hypogonadism in 10–20%.
- Chemotherapy toxic death is 0–2.4% (good risk) and 3–4.4% (intermediate/poor). Azoospermia is near universal; ≈half recover by 2 y, 80% by 5 y.
- Late effects: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%, **cardiovascular disease** and **second malignancy** (60–100% excess after cisplatin or radiotherapy, 200% after both).
- Follow-up is most intensive in years 1–2 (NSGCT) or 1–3 (seminoma).

KEY POINT

Because cure is expected, late cardiovascular disease and second cancers, which rise with each added modality, decide between equally curative options.

11 · Non-germ-cell tumours†

- †**Leydig cell**: the commonest sex cord–stromal tumour, causing precocious puberty or gynaecomastia. ≈10% are malignant; these resist chemo- and radiotherapy, so nodes need RPLND. A small marker-negative lesion suits testis-sparing excision.
- †**Sertoli**: mostly benign; the large-cell calcifying variant occurs with Carney complex and Peutz–Jeghers.
- †**Lymphoma**: the commonest testicular tumour in men **>50–60 y** and the commonest bilateral one. It is treated by orchidectomy, then R-CHOP with CNS prophylaxis and contralateral testis radiotherapy.

12 · Landmark trials

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced	Equal survival, less neurotoxicity	BEP standard
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3
VIP vs BEP×4	RCT, intermediate/poor	Equal, more toxicity	VIP only if lungs at risk
MRC TE19/EORTC 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y RFS 94.7% vs 96%	Equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

RCT = randomised controlled trial; RFS = relapse-free survival; PFS = progression-free survival; PA = para-aortic.

80 Surgery of Testicular Tumours

1 · Evaluation of the testis mass

It is urgent, because delay escalates curative therapy.

- Examine supraclavicular nodes, breasts, abdomen and scrotum. Scrotal ultrasound (US) mainly checks the **contralateral testis** (bilateral synchronous ≈ **1%**). Markers (AFP, β-hCG, LDH) give diagnosis, stage and post-orchidectomy baseline.
- Discuss a prosthesis and † offer sperm banking.

2 · Radical inguinal orchidectomy

Treatment of choice: **early cord control** before handling the testis, removing testis, epididymis and cord to the internal ring. † A scrotal approach breaches a second (inguinal) lymphatic field and leads to local recurrence.

ALGORITHM

1. Make a 3–5 cm incision over the canal and open the external oblique.
2. **Preserve the ilioinguinal nerve**.
3. Mobilise the cord at the pubic tubercle and apply a **Penrose early clamp**.
4. Deliver the testis and divide the gubernaculum.
5. Dissect the cord to the internal ring.

- **High ligation**: ligate vas and gonadal vessels **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail to mark the stump for excision at retroperitoneal lymph node dissection (RPLND).

KEY POINT

† **Scrotal violation**: inguinal exploration, high cord ligation and excision of the scrotal scar. Routine hemiscrotectomy is unnecessary; chemotherapy-era outcomes are not compromised.

Partial orchidectomy (testis-sparing surgery)

P4 Testis sparing surgeries for CA testis — indications (solitary/bilateral, small, polar), intraoperative US, cold ischaemia <30 min, frozen section, GCNIS 18–20 Gy, recurrence: completion orchidectomy

Not in the paper answer:

- Needs **all three**: polar, **≤2 cm**, contralateral testis compromised/absent. **Never elective with a normal contralateral testis.**
- Masses **<5 mm** with negative markers have low malignancy, so selective surveillance is an option. Cold ischaemia can be omitted if resection takes **<30 min**.
- GCNIS options are completion orchidectomy, **18–20 Gy** or observation (local recurrence ≈ 7.5%). **Local recurrence always needs completion orchidectomy**.

Delayed orchidectomy

Life-threatening metastatic germ cell tumour (GCT) gets chemotherapy first, then **delayed orchidectomy in all**, because the testis is a sanctuary with **discordant response**. An apparent extragonadal retroperitoneal primary shows intratubular neoplasia in up to 42%, so lateralising nodes mean orchidectomy on that side.

3 · RPLND — terminology and indications

Term	Indication
Primary RPLND	High-risk clinical stage (CS) I or low-volume stage II non-seminomatous GCT (NSGCT), **normal markers**
Post-chemotherapy (PC-RPLND)	After induction; residual **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy
Desperation	Markers still elevated; resectable
Redo	Retroperitoneal recurrence after prior RPLND
Late relapse	**>24 months** after complete response

Each step down means more chemoresistant disease and harder surgery.

Preoperative planning

- CT within **6 weeks** and markers within **7–10 days**; note vascular variants; no bowel prep.
- After **bleomycin**, test lung function and ask anaesthesia to use **low FiO_2, restrictive fluids**.
- Total IVC occlusion by thrombus needs inferior vena cava (IVC) resection; partial needs reconstruction.

4 · Templates

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — landing zones, bilateral infrahilar and modified templates, omit suprahilar/interiliac, 3–23% out-of-template, PC-RPLND full bilateral; modified only if ≤5 cm in landing zone + normal markers + good/intermediate risk; L1–L4 fibres to hypogastric plexus

- The bilateral template is bounded by renal vessels, ureters and common iliac arteries. A modified post-chemotherapy template needs a lesion **≤5 cm**.

5 · Open RPLND technique

- The approach is **transperitoneal** or **extraperitoneal** (less ileus).
- Incise from the caecum to the inferior mesenteric vein (IMV) and along the line of Toldt to the foramen of Winslow, freeing duodenum and pancreas. Use **liberal clips** to prevent chyle leak.
- **Split and roll**: split at 12 o'clock on the aorta from the left renal vein down to the inferior mesenteric artery (IMA), roll packets off and ligate lumbars.
- Lumbar arteries are usually 3 pairs, with the **2nd pair at IMA level**. The **right sympathetic trunk**, posterolateral to the IVC, is mistaken for lymphatic tissue.

Nerve sparing

- Left fibres run along the lateral aortic border near the IMA. Right fibres emerge **beneath the IVC** and cross the aorta obliquely, so they are at risk when lumbar veins are ligated.

> **KEY POINT**
> Identify fibres **prospectively, before dividing anything**; loop them and pass the packet beneath. Avoid traction, which causes neuropraxia and transient anejaculation.

Postoperative care

- No routine drain; if one is used, remove it at **<100 mL/24 h** on diet. No nasogastric (NG) tube, low-fat diet early, and home on days 3–5.

6 · Auxiliary procedures (PC-RPLND 23–45%)

- **Nephrectomy** is commonest, usually **left**. The strongest predictor is a residual mass **>10 cm**.
- **Vascular** repair is planned in every PC-RPLND, using a patch or graft.
- **IVC**: narrowing **<25%** is repaired by venorrhaphy, >25% by patch or graft; ligation is acceptable. **The aorta is never ligated**.
- **Liver**: histology is discordant in ≈half; resect, or ablate if **<3 cm** and unresectable. **Thorax**: do RPLND first, then resect lesions >1 cm unless the retroperitoneum shows fibrosis only (thoracic necrosis then in ≈89%).

7 · Decision making

- **Complete remission (≈70%)**: generally observe, though ≤1 cm residuals hold teratoma in ≈ 1/5 and viable tumour in ≈ 1/20.
- **Necrosis is predicted by** no teratoma in the primary, normal pre-chemotherapy AFP/β-hCG, raised LDH, >50% shrinkage and residual <2 cm.
- **After primary RPLND**: pN0, teratoma only or pN1 are observed; pN2 gets 2 cycles or observation. Observation cures ≈70%.

8 · Histology of residual mass after chemotherapy

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	Relapse-free (RFS) and cancer-specific survival (CSS) ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

> **KEY POINT**
> **Resect every NSGCT residual >1 cm**: no imaging (including PET †) or marker separates necrosis from teratoma, and teratoma is chemoresistant, grows, relapses late and transforms. Surgery is both staging and the only treatment.

- **Fizazi** model for viable GCT: incomplete resection, ≥10% viable cells and IGCCCG intermediate/poor risk. With 0, 1 or ≥2 factors, 5-y overall survival (OS) is 100/83/51%.
- Adjuvant chemotherapy: NCCN for incomplete resection or intermediate/poor risk; EAU for incomplete resection only. Not after salvage chemotherapy.

9 · High-risk PC-RPLND

- **Salvage**: viable GCT is at least doubled; OS 60–75%.
- **Desperation**: OS ≈54%. Operate if disease is confined and completely resectable, with markers declining or plateaued, or chemotherapy exhausted. Do **not** operate with rapidly rising markers or multisite progression. **Postoperative marker normalisation** is the only OS predictor.

- **Redo** is mostly for technical failure (retroaortic/retrocaval tissue), so do **complete great-vessel mobilisation the first time**.
- **Late relapse** (yolk sac, somatic-type malignancy) is treated by **surgery first**.

10 · Unique situations

- **Seminoma residual**: dense desmoplasia, viable in ≈10%. If **<3 cm**, observe; if **>3 cm**, do FDG-PET (≥6 weeks †). If avid, choose selected PC-RPLND, salvage chemotherapy or HDCT.
- **Sex cord stromal**: malignant features include >4 cm, necrosis, mitoses and lymphovascular invasion; RPLND is of unproven benefit.

11 · Outcomes, fertility and complications

- **Ejaculation**: template or nerve sparing preserves antegrade ejaculation in **>90%** (primary) at no oncological cost. † If lost, options include sympathomimetics and sperm retrieval.
- Complications: primary 10.6–24%, PC up to 56%. Mortality: primary ≈0, PC <1%.

Complication	Key facts / management
Pulmonary (bleomycin)	PC 3–5%, acute respiratory distress syndrome. Low FiO_2, restrictive fluids; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest); extraperitoneal approach, no NG tube, early feeding
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; linked to suprahilar dissection (cisterna chyli L1–L2)
Lymphocele	Symptomatic ≤1.7%; drain if infected
Venous thromboembolism (VTE)	Primary <1%, PC 0.1–3.1%; drug prophylaxis if prior chemotherapy, VTE or obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease

P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — milky tap, triglyceride 2–8× serum, protein >3 g/dL; stepwise management

ALGORITHM

1. Paracentesis; drain if it reaccumulates.
2. Very-low-fat, high-protein diet with medium-chain triglycerides (MCT) and octreotide.
3. If no response, nil by mouth with total parenteral nutrition (TPN).
4. If refractory, lymphangiography with or without embolisation, peritoneovenous shunt or exploration.
5. Continue a low-fat diet ≥ **4 weeks** after resolution.

12 · Landmark trials

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	RCT, pathological stage II: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; equal survival	Observe pN1; individualise pN2
SEMPET	FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm PPV 37%	PET for >3 cm only

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

P3 P4 Minimal invasive retroperitoneal lymph node dissection; Robotic RPLND — indications, templates, nerve sparing, robotic steps, lap vs robot, outcomes, AUA/EAU positions, miR-371a-3p

Not in the paper answer:

- The benchmark recurrence-free rate is ≈**85%** for limited pN+ without adjuvant chemotherapy; low MI recurrence is **confounded** by high pN0 and adjuvant rates.

1 · Indications and rationale

- Goals in early-stage disease are to preserve cure, avoid **dual therapy** (surgery and chemotherapy) and minimise lifelong toxicity.
- **Primary** RPLND (retroperitoneal lymph node dissection) is for marker-negative clinical stage (CS) I non-seminomatous germ cell tumour (NSGCT), and small-volume stage II NSGCT and seminoma. It spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **Post-chemotherapy (PC-RPLND)** is for NSGCT residual **>1 cm** and select seminomas.
- The robot adds fourth-arm retraction and **circumferential great-vessel dissection with lumbar control**.

> **KEY POINT**
> Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach, so the benefit of access alone is smaller than it looks.

2 · Guidelines and controversy

- The AUA (early stage) says it **may be offered** by a surgeon expert in germ cell tumours and minimally invasive surgery (MIS), with the patient told long-term data are lacking. The EAU restricts it to **specialist centres** expert in open and MIS.
- With no noninferiority evidence, it is highly controversial. Concerns are staging-only dissection without lumbar division, excess adjuvant chemotherapy, operating on the wrong patients, conversion in **≈5%**, and short follow-up with reported peritoneal carcinomatosis.

3 · Patient selection

- There are no absolute contraindications; relative ones are extensive prior surgery and morbid obesity.
- **Suited to MIS**: chemo-naive high-risk IB; small-volume II (1–2 landing-zone nodes, none **>3 cm**); a post-chemotherapy solitary landing-zone mass with little desmoplasia.
- **Favour open**: most PC-RPLND, a large mass, large chemotherapy shrinkage, possible vessel reconstruction, somatic malignancy or retrocrural disease.

4 · Unchanging operative principles

- Imaging within **≈4 weeks** and markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND**: template, completeness and nerve-sparing are unchanged.

ALGORITHM

1. Safe entry and exposure.
2. **Split-and-roll** nodal removal.
3. Prospective nerve sparing.
4. **Lumbar vessel division**.
5. Spermatic cord removal, then closure.

- Chemo-naive patients may have an extended unilateral nerve-sparing template, converted to bilateral if enlarged nodes are found. Post-chemotherapy, the **bilateral full template** is standard.
- Incomplete RPLND or mass-only excision worsens survival.

5 · Preparation, positioning and ports

- Plan for haemorrhage with an **open set in theatre**; after bleomycin, manage fluids and oxygen carefully. A high-fat meal the day before highlights lymphatics.
- Positioning is lateral decubitus, laparoscopic supine, or robotic steep Trendelenburg with a **peritoneal hammock** that allows any template.

6 · Operative steps

- **Right**: incise the Toldt line to the foramen of Winslow and **Kocherise**; divide the gonadal vein at the inferior vena cava (IVC); split over the IVC; clip lymphatics; roll the paracaval packet, spare fibres and divide lumbars. **Do not ligate the right renal artery** in the interaortocaval space.
- **Left**: divide the gonadal vein at the left renal vein and control the **lumbar vein entering the left renal vein**.
- Lower the pneumoperitoneum to check haemostasis; **no routine drain**; discharge at **1–2 days**.

7 · Complications and management

ALGORITHM

1. Major vascular injury: apply pressure and raise the pneumoperitoneum.
2. Inform anaesthesia, get blood and call vascular help.
3. If uncontrolled, keep pressure, undock and convert.

- For MIS repair, **identify the injury before acting: no blind clipping**.
- Prevent chylous ascites with liberal clips rather than energy devices; rates match open, and it is treated by low-fat diet, paracentesis, octreotide and parenteral nutrition.
- Antegrade ejaculation is kept in **≥90%** after primary nerve-sparing.

8 · Outcomes vs open

- There is **no randomised trial**. MIS gives shorter stay and less blood loss but longer operative time; brisk bleeding is the unique risk.
- Robotic node yields approach the open benchmark of **≈30**, but adjuvant chemotherapy is still too frequent. Post-chemotherapy data are too small.

9 · Landmark trials

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: BEP×3 vs EP×4	Most relapses/deaths without or with incomplete RPLND	Complete RPLND matters

82 Tumours of the Penis

1 · Overview

- Squamous cell carcinoma (SCC) is **~95%** of penile malignancy, and **0.4–0.6%** of male cancers in North America and Europe, but up to **10%** in parts of Asia, Africa and South America.
- Prognosis rests on **primary stage and histology** and on **nodal metastasis**.

2 · Premalignant lesions: penile intraepithelial neoplasia (PeIN)

P1 Premalignant conditions in urology (including premalignant lesions of penis) — PeIN definition, WHO 2022 all high grade, HPV-associated vs differentiated, E6/E7/p16, EQ/Bowen/Bowenoid table, LS 2.3–9%, Buschke–Löwenstein, organ-preserving treatment

Not in the paper answer:

- Erythroplasia of Queyrat (EQ) is a red velvety glans lesion in uncircumcised men that mimics balanitis; it progresses in **10–33%**.
- Bowen disease is a scaly shaft plaque mimicking psoriasis or eczema; it progresses in **~5%**. Bowenoid papulosis (HPV-linked shaft papules in young men) **does not progress**.

3 · Aetiology and risk factors of SCC

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer; it protects against invasive SCC, **not CIS** (carcinoma in situ). Adult circumcision gives little or no protection
Phimosis	Present in **25–75%**; **7–10×** risk. It mediates the non-circumcision risk, as smegma causes inflammation and fibrosis
HPV	Low risk 6, 11; high risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest**. Present in **~50%**; wives have ~3× cervical cancer risk
Human immunodeficiency virus (HIV)	~4× risk of penile cancer death; PeIN progresses ~6 years earlier
Tobacco	All forms; dose-dependent and independent of phimosis
Lichen sclerosus (LS)	Cancer in **2.3–9%**; HPV-**independent** SCC
Trauma, obesity	Scarred shaft; obesity causes buried penis and functional phimosis
†**Psoralen and ultraviolet A (PUVA)**	† Genital exposure markedly increases SCC, so shield the genitalia

- **Prevention** is by neonatal circumcision (AAP 2012: benefits outweigh risks), hygiene, no tobacco and HPV vaccination (nonavalent adds 31/33/45/52/58).

> **KEY POINT**
> Two routes converge on the preputial sac: HPV-driven (E6/E7, p16-positive, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none.

4 · Natural history and spread

- Most arise on the glans or prepuce; **flat ulcerative tumours metastasise earlier**.
- **Buck fascia** is a temporary barrier; once it and the tunica are breached, tumour enters the corpora and spreads by blood.
- Distant metastases are uncommon (**1–10%**) and late. Untreated, death comes within **2 years** from femoral erosion haemorrhage or sepsis.

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — penile superficial/deep lymphatics, stepwise bilateral spread, no skips

Not in the paper answer:

- Pelvic stations are external iliac, internal iliac and obturator; distant sites (lung, liver, bone, brain) are rare without nodal disease.

5 · Presentation and diagnosis

- It presents as induration, a warty growth or an ulcer, often hidden by phimosis. Examine the lesion and **palpate both groins**.
- **Biopsy is mandatory before definitive therapy**, because depth and grade predict occult nodes and superficial biopsies mislead.
- **Hypercalcaemia without bone metastases** comes from PTH-related substances, tracks bulky inguinal disease and may resolve after node excision. Treat with saline, then bisphosphonate.

6 · Histology

Subtype (WHO 2022)		HPV	Features	Nodal risk
Usual (keratinising)	**45–65%**	Independent	Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓	25–40%
Warty		Associated	Cauliflower, fibrovascular cores; slow	17–25%
Basaloid		Associated	Small cells, necrosis, deep	**>½ node positive**
Papillary		Independent	Low grade, no koilocytosis	≤12%
Verrucous		Independent	**Broad pushing border**	**No metastasis**; recurs if under-excised
Sarcomatoid		Independent	Spindle cells	**Most aggressive**
Rare: pseudoglandular, adenosquamous; clear cell, lymphoepithelioma-like (HPV)		—	Deep invasion	Pseudoglandular, clear cell worst

- **Any anaplastic cells make it grade III**. **LVI** predicts nodes in every study, and **PNI** does so strongly.
- **High-risk HPV or p16 positivity means better disease-specific survival (DSS)**, not better nodal status.

7 · Imaging

A negative scan never clears the groin.

- **Primary**: examination is reliable; **MRI** (with or without artificial erection) helps if equivocal and organ sparing is planned.
- **cN0 groin**: **no imaging detects micrometastases**, so stage surgically; use CT only if obese or after prior groin surgery.
- **cN+ or T3–T4**: CT chest, abdomen and pelvis. **FDG-PET/CT** is best for pelvic and distant disease.

8 · Staging: TNM 8th edition (2017)

T1a versus T1b decides who needs groin staging.

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (incl. pure verrucous without destructive invasion)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**
T2	**Corpus spongiosum**, with or without urethra
T3	**Corpora cavernosa** (incl. tunica albuginea), with or without urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral**, no extranodal extension (ENE)
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed mass or pelvic nodes	**ENE or pelvic nodes**

Stage	TNM
I	T1a N0
IIA	T1b, T2 N0
IIB	T3 N0
IIIA / IIIB	T1–3 N1 / N2
IV	T4, any N3, any M1

- In the 8th edition, **urethral invasion no longer upstages**.

> **KEY POINT**
> Traps: T1b and T2 share stage IIA; any N3 or T4 is stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery.

9 · Management of the primary tumour

- **Penectomy** is the oncological gold standard (local recurrence **0–8%**), but over half of patients are ≤60, so preserve where safe.
- **Organ preservation** suits PeIN, Ta, T1 and grade I–II in **compliant** men. It is **contraindicated** by size **≥4 cm**, grade III, deep cavernosal invasion, or unfitness for salvage.
- Local recurrence is higher, but **early recurrence is salvaged without survival cost**, except in pT3 or high grade.
- **Margins**: microscopic spread is **~5 mm** for grade I–II and **~10 mm** for grade III. Only margins **<1 mm** raised recurrence, so a **negative frozen-section margin (>1 mm) is enough**; the 2-cm dogma is abandoned.

Stage / site	Treatment
PeIN, glans	Laser, resurfacing; topical alternative
Ta / PeIN — foreskin, shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 glans	Excision, resurfacing, glansectomy, radiotherapy (RT; not Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** with or without distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy, then surgery in responders
Recurrence after conservation	Re-excision or penectomy; superficial low-grade re-treated conservatively

- **Glans resurfacing** strips the epithelium and adds a split-skin graft, keeping erections.
- **Laser**: CO_2 reaches only ~0.1 mm (recurrence up to 50%), so **Nd:YAG** (up to 6 mm) is preferred, with base biopsies; recurrence is <10% for CIS and 10–25% for T1.
- † Partial penectomy leaves a stump for standing micturition; total penectomy needs a **perineal urethrostomy**.

> **KEY POINT**
> Margin width depends on grade, not a fixed 2 cm. Low-grade tumours spread millimetres and their recurrences are salvageable, so preservation is safe; in high-grade or cavernosal disease local relapse is lethal and amputation regains its place.

10 · Inguinal nodes (see Ch 84)

P2 P3 Importance and management of Inguinal lymph nodes in CA penis; Prognostic index score in carcinoma penis — nodes = strongest prognostic factor, Kroon early vs delayed, risk groups, DSNB, FNAC without antibiotic trial, PLND >2 nodes/ENE, neoadjuvant TIP, ILND morbidity

Not in the paper answer:

- Cure is ~**80%** with ≤2 unilateral nodes and no ENE or pelvic nodes; with positive pelvic nodes, 5-year survival after surgery alone is ~**14%**.
- Occult node rates are **0–18%** for PeIN/Ta/T1a, one-third to over half for T1b and above, and ~60% with corporal invasion.

ALGORITHM

1. **cN0, low risk**: surveillance. If a node appears, do fine-needle aspiration cytology (FNAC); if positive, inguinal lymph node dissection (ILND) and contralateral staging.
2. **cN0, high risk (T1b+)**: bilateral dynamic sentinel node biopsy (**DSNB**) or superficial/fascia-sparing/minimally invasive ILND. If positive, completion ILND.
3. **cN1–2, mobile, <4 cm**: biopsy, then ILND and contralateral staging. If >2 nodes or ENE, add pelvic lymph node dissection (PLND).
4. **Fixed, ≥4 cm or pelvic**: neoadjuvant TIP (cisplatin-eligible) or chemoradiation, then resect responders.
5. **Adjuvant therapy** if >2 nodes, bilateral nodes, ENE or pelvic nodes.

- DSNB refinements (pre-operative ultrasound with or without FNAC, serial sectioning) cut false negatives from ~18% to ~7%. Morbidity is greatest with radical ILND, then superficial or fascia-sparing, then minimally invasive, and least with DSNB.
- No pelvic skip metastases occur, so PLND is mainly staging.

11 · Radiotherapy

- **Circumcise first** to expose the lesion and prevent oedema and phimosis. Surgical salvage remains curative.
- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks** with **bolus**; ~55% are recurrence-free at 5 years.
- **Brachytherapy** (Ir-192) suits **T1–T2 glans tumours <4 cm**: ~60 Gy over ~5 days gives ~79% relapse-free survival and ~85% preservation at 5 years, with better erections than EBRT.
- **Late effects** are **meatal stenosis** (10–45%, at 18–24 months) and ulceration; **1 in 5 local recurrences occur after 5 years**.
- For the groins, give **no prophylactic RT** for cN0. Adjuvant RT for **≥2 nodes or ENE** is ≥50 Gy, with **54 Gy for ENE and 57–60 Gy for gross residual**. Chemoradiation (cisplatin if GFR >50 mL/min) treats inoperable nodes.

12 · Systemic therapy

- No randomised trials are completed. **TIP** (paclitaxel, ifosfamide, cisplatin) is most active; **Bleomycin regimens are discouraged**.
- **Neoadjuvant TIP ×4** for bulky N2–N3 or pelvic nodes gives ~50% response, and ~⅔ of responders stay disease-free, so **response selects for surgery**.
- Adjuvant chemotherapy has no proven survival gain but may be considered for pN3 or pelvic nodes.
- Palliative platinum is first line; checkpoint inhibitors help ~1 in 6 in trials.

13 · Follow-up

Groin relapse after observation is usually fatal.

- After organ preservation, use close follow-up and self-examination.
- Observed groins are checked 2–3-monthly for 2 years, then less often. Only ~⅓ of cN0 men with groin relapse survive 5 years.

14 · Non-squamous penile tumours (~5%)

Tumour	Key features / treatment
BCC	Proximal shaft; almost never metastasises. Treat by excision or Mohs
Melanoma	<1%; glans; spreads by blood and lymph; AJCC cutaneous staging. Bilateral modified ILND if Breslow ≥1 mm, ulceration or Clark IV–V (DSNB reasonable); adjuvant nivolumab for stage IIIB–IV
Sarcoma	Malignant are **proximal**; vascular commonest; **local recurrence characteristic**. Superficial: wide excision or partial; deep corporal: total. No ILND unless palpable
Extramammary Paget	Penoscrotal eczematoid plaque; CK7+, CEA+, S-100−; ~½ have another cancer, so screen. Wide excision (up to 3 cm) guided by **mapping biopsies**
Lymphoma	Usually systemic: R-CHOP even if isolated
Metastases	From **bladder, prostate, rectum**; **priapism** is the commonest sign; most die within a year; palliate

15 · Landmark trials

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; TIP ×4, cN2–N3 M0	Response 50%; median OS 17.1 months; durable mainly in responders	Standard before surgery for bulky/pelvic nodes
Netherlands Cancer Institute early vs delayed	Early DSNB vs dissection after surveillance relapse	3-y survival 84% vs 35%; ENE near-universal if delayed	Stage high-risk cN0 groins early

84 Inguinal Node Dissection

1 · Why the groin matters in penile cancer

Groin surgery alone can cure node-positive disease.

- Penile squamous cell carcinoma (SCC) has a **locoregional phase**, so the groin is a **curable station**.
- **Nodal status is the strongest prognostic factor**. Extranodal extension (ENE), even in one node, makes it pN3.
- Early versus delayed inguinal lymph node dissection (ILND) for impalpable nodes gave ENE ≈20% vs ≈95% and 3-year disease-specific survival ≈84% vs ≈35%. So manage the groins within **3 months** of diagnosis.
- An adequate dissection removes ≥**8 nodes per side**.

2 · Indications and risk groups

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

NCCN = National Comprehensive Cancer Network; EAU-ASCO = European Association of Urology–American Society of Clinical Oncology; DSNB = dynamic sentinel node biopsy; G = grade. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- For the impalpable high-risk groin, **NCCN leads with ILND** and **EAU-ASCO leads with DSNB**. Select on **primary pathology only**, not HPV or nomograms.
- **cN+**: about half of palpable nodes are metastatic. Do fine-needle aspiration cytology (FNAC), then excisional biopsy if negative. If any is positive, do **bilateral radical ILND**. Give no antibiotic trial.
- With unilateral palpable nodes, the other side has micrometastasis in ≈**30%**, so stage it.
- **Bulky or fixed (cN3)**: give cisplatin–taxane chemotherapy first, then surgery. Resect upfront only if chemotherapy-ineligible or vessel invasion is imminent; involved skin needs a myocutaneous flap.

3 · Staging work-up

- Palpable nodes or high risk need 18FDG-PET/CT or CT chest–abdomen–pelvis. Imaging misses deposits <1 cm, so it has no role in the impalpable groin.
- **FNAC** is specific but insensitive: if positive, do ILND; if negative, do DSNB or ILND and never discharge.
- **DSNB**: do ultrasound, with or without FNAC, first, because a **tumour-replaced node** reroutes tracer and gives a false negative. Then use ^{99m}Tc nanocolloid and blue dye with serial sectioning. Positive: ILND †; negative: surveillance.

KEY POINT

Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · Lymphatic anatomy

P2 Importance and management of Inguinal lymph nodes in CA penis. — superficial/deep systems, bilateral stepwise no-skip drainage, Daseler's five zones (superomedial = sentinel), superficial 4–25 and deep 1–3 nodes, Cloquet

Not in the paper answer:

- The **femoral triangle** is bounded by the inguinal ligament, **adductor longus** medially and **sartorius** laterally.
- Raise flaps deep to **Camper's fascia**, which holds the capillary network; too thin causes necrosis and too thick leaves nodes.
- In the fossa ovalis the order, lateral to medial, is **nerve, artery, vein**, so **never dissect lateral to the femoral artery**.

5 · Templates

Template	Extent	Features
Radical	Whole femoral triangle, superficial and deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND nodes go to frozen section; if **positive, convert to radical** †.

6 · Open radical ILND

ALGORITHM

1. Frog-leg position; incision ≈2 cm below and parallel to the inguinal ligament.
2. Raise flaps 3 cm above the ligament to the apex; divide the saphenous at the apex (spare if disease minimal).
3. Open fascia lata over sartorius and adductor longus; dissect cephalad on the femoral vessels.
4. Divide the saphenous at the junction; clear deep nodes up to the femoral canal.
5. **Sartorius transposition**; close the femoral canal.
6. Suction drains; tack the flaps.

7 · Pelvic lymph node dissection (PLND)

- If the inguinal frozen section is positive, do PLND at the same sitting; otherwise stage it.
- † EAU: with ≥2 positive inguinal nodes or ENE (pN2–3), do ipsilateral PLND, open or minimally invasive.

8 · Video-endoscopic ILND (VEIL)

P4 Videoendoscopic inguinal lymphadenectomy — same template beneath Scarpa's, 15 mm Hg, frozen section gates deep dissection; similar yield, fewer wound complications; non-bulky nodes only

- The commonest robotic complication is lymphocele.

9 · Complications and post-op care

Fear of complications must not cause omission.

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, tack flaps, incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, modified template, VEIL; debride, then flap or negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices; drain until **<30 mL/day**; drain symptomatic or infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month
Others	Phlebitis, neurapraxia, pulmonary embolism (PE), femoral bleeding	Thromboprophylaxis; stay medial to artery

Mobilise early. NCCN advises antibiotic prophylaxis, though the authors do not use it routinely.

10 · Follow-up

Most groin relapses occur within 2–3 years (median ≈1 year). EAU-ASCO advises examination 3-monthly for 2 years, then 6-monthly to 5 years, with imaging only if node-positive; NCCN images everyone.

83 Tumours of the Urethra

1 · Benign urethral tumours

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm suburothelial mass; grows in pregnancy	Excision; may recur
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding	Fulguration/laser; open excision if extensive
Fibroepithelial polyp	Boys; bulbar/posterior	Polyp on a stalk; filling defect	Transurethral resection

2 · Anatomy, histology and lymphatics

Tumour site predicts its type and its nodal basin.

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Urothelium in the prostatic urethra, columnar in the anterior urethra, squamous in the fossa navicularis	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior drains to **inguinal** nodes (bulbar also external iliac); posterior drains to **pelvic** nodes	

KEY POINT

Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women it is also the **continence boundary**: the distal third can be excised without incontinence.

3 · Staging: AJCC 8th edition

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral): **number, not size**	

- **Evaluation** is by examination under anaesthesia with cystourethroscopy, bimanual palpation and biopsy, plus **MRI** for local extent and CT chest if invasive.

- **Palpable inguinal nodes (20–30%) are almost always metastatic**, so, unlike penile cancer, no antibiotic trial is given.

4 · Male primary urethral cancer (PUC)

Distal tumours are curable by surgery; bulbar tumours need multimodal therapy.

- Patients are >60 years. **Urethral stricture** is the commonest risk factor (up to 50%); others are lichen sclerosus, pelvic radiotherapy (RT) and HPV 16.
- † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis** depends on stage and location: distal tumours do 2–3-fold better, and bulbar survival is as low as 20–30%.

Pendulous / fossa navicularis (distal)

- Surgery alone controls low-stage disease, ranging from transurethral resection to **distal urethrectomy with or without partial penectomy**, to total urethrectomy with perineal urethrostomy, with or without total penectomy.
- **Penile preservation** is endorsed by EAU and NCCN, using options such as topical 5-fluorouracil (5-FU) for distal CIS, distal urethrectomy, or glansectomy. A positive margin needs re-excision or adjuvant RT.
- A **margin of 5 mm** is now standard (1 cm traditionally).
- **Primary RT** is only for early anterior lesions in men refusing surgery, and it is ineffective for bulbar PUC.

Bulbar (proximal anterior)

- The few low-stage tumours are treated by transurethral resection, or segmental excision and anastomosis.
- **Advanced** disease needs radical cystoprostatectomy, pelvic lymphadenectomy, total urethrectomy and penectomy. Surgery alone gives ~¼ 5-year survival, so use **multimodal therapy**.

Inguinal nodes (both sexes)

- There is **no proven benefit of prophylactic inguinal dissection in cN0**. For cN1, give neoadjuvant chemotherapy or chemoradiotherapy, then lymphadenectomy.

5 · Male posterior (prostatic) urethral cancer

Only stromal invasion needs cystectomy.

- It is usually **secondary to bladder UC**, found in 20–40% at radical cystectomy (RC), so always evaluate the bladder and upper tracts.
- Diagnose by TURP biopsy; it is often CIS. BCG for high-grade superficial disease gives ≥70% response.

Involvement	Treatment	5-year survival
Mucosal	TURP + BCG	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC with or without urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy**, with or without neoadjuvant chemotherapy	~36%; follows bladder stage

KEY POINT

The only question is "stroma breached?" TURP before BCG debulks, improves contact and stages. Ductal disease is not T2, so do not jump to cystectomy.

6 · Female urethral cancer

- It is ~⅓ of PUC, in women >60 years. Risks include chronic irritation, HPV and **urethral diverticulum (6–10% harbour malignancy**, mostly adenocarcinoma).
- Distal tumours are SCC; proximal ones are UC or adenocarcinoma, which is **mucinous** or **clear cell** (in diverticula).
- **Location decides continence-sparing excision**; stage with MRI pelvis.
- **Stage dominates prognosis**: distal tumours do better than proximal, which do better than pan-urethral.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection or laser, with or without BCG
Low-stage distal	**Distal urethrectomy** (transvaginal) **or RT** (brachytherapy with or without external beam, ~60 Gy): equal, ≥70% cured. Stress incontinence (>40%), meatal stenosis; ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC, total urethrectomy, extended pelvic lymphadenectomy and anterior vaginal wall, with or without hysterectomy and pubic rami. Alone, most die, so use **multimodal** therapy
Palpable groin nodes	Chemoradiotherapy, chemotherapy or lymphadenectomy; often curable. Pelvic nodes are poor

- RT local control depends on the **length of urethra involved**, not histology or size.

7 · Multimodal therapy (both sexes)

- For advanced or proximal disease, give **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery. Response predicts survival.
- **Chemoradiotherapy** is 45–55 Gy to the genitalia, perineum and groins with 5-FU and mitomycin C. It is preferred for advanced SCC, though every responder develops a stricture.

Histology	Regimen
UC	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin); gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection, with or without intraurethral BCG or chemotherapy
Male T2 pendulous	Partial urethrectomy with or without partial penectomy; if margin positive, chemoradiotherapy, surgery or RT

Stage (NCCN, anterior male / female)	First line
Male T2 bulbar	Urethrectomy with or without RC; if ≥pT3/N1, adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy with or without surgery; or neoadjuvant chemotherapy, then surgery or RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy, with or without consolidative surgery

- Recurrence after resection of invasive disease is ~50%, so survey the urethra, groins and body.

8 · Urethral recurrence (UR) after radical cystectomy

Urethral risk decides whether a neobladder is safe.

Male

- Risk is **1–15%** (~5%), mostly within 2 years.
- **Risk factors** are **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS and multifocality. An orthotopic neobladder (ONB) is protective (~5% vs 9%).
- Routine prophylactic urethrectomy is abandoned, because 64–100% of specimens are negative.
- **Before ONB**, confirm a negative intraoperative frozen section of the urethral margin.
- **Urethrectomy** is absolute for a positive margin or gross urethral tumour, and relative for diffuse CIS, prostatic urethral involvement or multifocal disease. Delayed urethrectomy is as good as simultaneous.

Female

- The main risk is **bladder neck involvement**, but only 30–50% of these have urethral tumour, so it is not an absolute contraindication.
- ONB is allowed if the bladder neck, urethra and anterior vaginal wall are uninvolved and frozen section is negative; UR after ONB is <5%.

Surveillance and treatment

- Urethral wash cytology has no proven survival benefit; if positive, do urethroscopy and biopsy.
- **Total urethrectomy is treatment of choice**, including the meatus; after ONB, convert to cutaneous diversion. **Intraurethral BCG treats CIS** (not papillary or invasive). Invasive UR has median survival ~17 months.

85 Benign Disorders of the Penis and Urethra

1 · Reconstructive surgical technique

- **Cautery**: use **bipolar on the urethra** to limit spongiosal damage, and monopolar on subcutaneous tissue and muscle.
- A **bougie à boule** calibrates. **Goodwin sounds** dilate over a guidewire, so they are safer than van Buren sounds.
- **Sutures** are always **absorbable**, because permanent ones form a stone nidus; monofilament polydioxanone (PDS) is preferred. Support is needed for **30–60 days**, so avoid chromic, which lasts 14 days. Use **5-0** for most urethral work and **6-0** for a single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA).

2 · Tissue transfer: grafts and flaps

P1 Enlist various grafts used in urology and role of BMG in urology — graft vs flap; take by imbibition then inosculation (~48 h each), immobilise 5–10 days; STSG vs FTSG; other grafts; BMG properties, buccal = lingual

Not in the paper answer:

- **Extragenital full-thickness skin graft (FTSG)** fails in the urethra because its greater mass makes take ischaemic. The exception is the **posterior auricular (Wolfe) graft**, which behaves like a split-thickness skin graft (STSG).
- Oral mucosa graft (OMG) has a **panlaminar plexus** and sacculates less than skin. Buccal onlay failures occur in the **first 12 months**.

> **KEY POINT**
> Graft trade-off: a dense fine plexus and low mass favour take, while carried dermis or lamina propria gives durability and resists contraction. This explains STSG vs FTSG, the failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps

- Flaps are classified by **composition** (fasciocutaneous, fascial, musculocutaneous, muscle or visceral), **blood supply** (axial on a named vessel, or random as in a Z-plasty), **movement** (advancement, rotation or transposition) and **location** (local, distant pedicled or free).

3 · Surgical anatomy

Fascial layers decide flap planes and how blood spreads.

- Layers, deep to superficial, are **tunica albuginea, Buck's fascia, dartos, skin**. Dartos is fat-free, carries the skin vessels and fuses with Colles fascia.
- **Skin supply**: the superficial external pudendal artery feeds a subdermal plexus, which is the basis of the McAninch circular flap.
- **Deep supply**: the internal pudendal artery continues as the common penile artery, giving the **bulbourethral, dorsal and cavernosal** arteries.

Situation	Spread
Buck's intact (penile fracture)	Confined to the shaft, giving an **eggplant deformity**

Situation	Spread
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on the thigh by fascia lata
Fournier gangrene	From perineum and scrotum up the abdominal wall via Scarpa

KEY POINT

Two envelopes, two vascular worlds: dartos carries the skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery.

4 · Lichen sclerosus (LS)

P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — meatus sentinel, Littre-gland stricture mechanism, spares proximal urethra, biopsy features, RUG first, clobetasol, oral mucosa never skin, perineal urethrostomy

Not in the paper answer:

- Malignant transformation (mostly SCC) reaches up to **8%**.
- **Steroid**: clobetasol propionate or betamethasone 0.05% once or twice daily for 1–2 months, then on alternate days for 3–4 months.
- Meatal stenosis is treated by extended meatotomy or BMG onlay, and early repair may avert a panurethral stricture.

KEY POINT

LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, retrograde urethrography (RUG) before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · Other urethral lesions

Lesion	Key features	Management
Urethral haemangioma	Bloody discharge; often extends beyond view, so image with **MRI**; benign	Observe; small: laser or triamcinolone; large: excision, staged repair
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; HLA-B27 in ~2/3; **circinate balanitis**	Urethritis is self-limiting
Amyloidosis	Urethral mass, obstruction, haematuria; needs biopsy	Observe; dilate or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias, phalloplasty)	Treat the **underlying process** as well as the tract
Male urethral diverticulum	Congenital (all layers) or acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, UTI, stone	Cystourethroscopy and RUG; repair and treat the cause
Enlarged utricle	Müllerian remnant, with proximal hypospadias	A catheter may enter the utricle; UTI, post-void dribble

6 · Prepuce and meatus

- **Paraphimosis (emergency)**: a trapped foreskin causes oedema and can lead to glans necrosis. Reduce by steady compression, with the thumbs pushing the glans and the fingers pulling the foreskin. If irreducible, do an emergency **dorsal slit**; offer elective circumcision for recurrence.
- **Phimosis** is physiological in infants (resolving at 1–8 years); in adults it leads to balanitis and cancer. Treat with steroid, dorsal slit or circumcision.
- **Balanitis**: **Candida** is commonest. Treat with hygiene and antifungal, and **check blood glucose**.

P2 Management of Urethral meatal stenosis. — adult causes, RUG first, ventral meatotomy with 3 sutures, steroid self-dilation, urethroplasty if fossa involved

Adult circumcision (sleeve)

ALGORITHM

1. Penile block. Foreskin retracted: mark the cuff incision across the frenular base; incise to Buck's.
2. Foreskin advanced: mark the coronal incision.
3. Excise the sleeve using **bipolar**, because adult vessels do not seal by pressure.
4. Quadrant dartos sutures, then skin closure.

- **Human immunodeficiency virus (HIV)**: the foreskin has many CD4+ cells and microtears. Circumcision lowers female-to-male HIV by ~**60%**, and also lowers herpes simplex and HPV.

7 · Penile curvature

P1 Congenital Penile curvature.. — chordee without hypospadias vs congenital curvature (young man, large erection); never divide a urethra reaching the glans tip

Not in the paper answer:

- Congenital curvature is a smooth whole-shaft curve, whereas chordee without hypospadias shows a hooded prepuce and ventral inelasticity.

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient (chordee without hypospadias, I–III)
II	Dysgenetic Buck's and dartos; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric tunica = **congenital curvature**
V	Congenital short urethra; its existence is doubted

ALGORITHM

1. Excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Artificial erection; ventral midline septotomy.
3. If still curved, do dorsal tunical ellipse excision or plication under the elevated neurovascular bundle.

- **Acquired non-Peyronie curvature** (see Ch 76) follows deep urethrotomy, neglected or **subclinical fracture**, or a septal tear. Erections are normal, with no shortening. Treat by **scar excision and graft**, not contralateral plication.

8 · Penile transplantation

A composite allotransplant restores appearance, erection and voiding after penile loss, but needs lifelong immunosuppression; the alternative is phalloplasty.

9 · Adult acquired buried penis (AABP)

It is self-perpetuating and usually needs surgery.

- Risks are **obesity**, aggressive circumcision, scrotal lymphoedema and LS.
- It is a **vicious cycle**: the escutcheon tethers the penis, and trapped urine causes inflammation, LS and scarring, which trap it further. So **weight loss alone rarely cures**.
- The **Pariser** classification grades the repair needed: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- **Evaluate the urethra**, because a severe stricture needs urethroplasty **before** grafting.

ALGORITHM

1. Slit the phimotic ring and place glans traction.
2. Circumcising incision; excise non-viable skin.
3. Do escutcheonectomy (harvest the graft from it; keep 2–3 cm of skin at the base).
4. Tack Scarpa to the rectus fascia or pubis; place drains.
5. Cover with STSG (0.012–0.020 inch), fibrin glue and a bolster for 5–7 days.

- **Scrotal lymphoedema** may be congenital, infective (**Wuchereria bancrofti**, lymphogranuloma venereum), malignant or after node dissection or radiation. Resect **all** involved tissue and cover with spared perineal or lateral scrotal skin.
- **Foreign body injection** (silicone, paraffin) causes fibrosis, necrosis and abscess. Excise it all and graft, and screen for **body dysmorphic disorder**.

10 · Landmark trials

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three randomised controlled trials (RCTs), adult male circumcision vs none	~60% reduction in female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

86 Urethral Stricture Disease

1 · Definitions, anatomy, aetiology

P2 P3 Urethral stricture disease: etiology, presentation and management (including recent advances); Urethral stricture: classification, DVIU and surgery after failed VIU — anterior-only definition, spongiofibrosis, area ∝ diameter², Jordan spectrum, dorsal eccentric bulb, dual supply, aetiology table (idiopathic ≈ iatrogenic), LS via Littre glands

Not in the paper answer:

- Straddle strictures often present late; prostate cancer treatment causes posterior stenosis.

Site	Correct term
Anterior urethra	Stricture (epithelial scar with or without spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

- The common penile artery supplies the corpus spongiosum. The bulbar urethra lies dorsally, so a deep 12 o'clock urethrotomy in the bulb is dangerous.
- In lichen sclerosus (LS), microabscesses form in the glands of Littre; antibiotics relieve symptoms, not the stricture.

> **KEY POINT**
> Retrograde glans flow allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation causes ischaemia and PFUI repair failure.

> **KEY POINT**
> Aetiology predicts biology: iatrogenic and straddle strictures are focal scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

2 · Presentation and evaluation

- **Acute retention**: if a catheter fails, do a retrograde urethrogram (RUG) and place a **suprapubic cystostomy**, not blind dilation. Pass any guidewire under flexible endoscopic vision.
- Four determinants guide treatment: **location, length, depth, density**.
- The treating surgeon does a dynamic RUG and voiding cystourethrogram (VCUG). Do not dilate at first urethroscopy. MRI is for tumours and PFUI only.
- **Hydraulic trap**: voiding pressure holds the urethra above the stricture open, so a segment that cones down is scarred.
- **Urethral rest** with a suprapubic tube for **6–8 weeks** reveals the true length.

3 · Decision making

- First agree on **palliation or cure**. The **reconstructive ladder is archaic**: go to the procedure most likely to cure.

> **ALGORITHM**
>
> 1. If bulbar, <1.5 cm, without dense spongiofibrosis and untreated, then dilate or do one internal urethrotomy; if it recurs, then urethroplasty.
> 2. If bulbar 1–2 cm (up to 3–5 cm proximally), then excision and primary anastomosis (EPA).
> 3. If longer bulbar, then oral mucosa graft (OMG) onlay, with or without augmented anastomosis.
> 4. If penile, then onlay (graft or flap); if hostile bed, LS or failed hypospadias, then staged OMG.
> 5. If panurethral, options exhausted or elderly, then perineal urethrostomy.

4 · Dilation and internal urethrotomy (DVIU)

P3 Urethral stricture: classification, DVIU and surgery after failed VIU — secondary-intention race, 10–2 o'clock thin spongiosum, complications, second urethrotomy futile, self-dilation, mitomycin C, laser = cold knife, catheter 3–7 days

Not in the paper answer:

- DVIU does worse when luminal narrowing exceeds **74%**; the safest dilation is a **balloon** over an endoscopically placed guidewire.

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

- Self-catheterisation tapers over **3–6 months**; the stricture recurs when it stops. A 6-week catheter is no better than 3–7 days.

> **KEY POINT**
>
> No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; each repetition spends the patient's reconstructive capital.

5 · Excision and primary anastomosis (EPA)

- EPA is the **gold standard** for short bulbar strictures; recurrence is rare. Rules: **excise all fibrosis, spatulate widely, keep it tension-free**.
- Length comes from spongiosal mobilisation, freeing Buck's fascia and the intracrural space, and detaching bulbospongiosus from the perineal body. **The more proximal the stricture, the longer the excisable segment.**

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved

	Transecting	Vessel-sparing transecting	Non-transecting
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy, excise scar from within, close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); its other benefits are unproven.

6 · Substitution urethroplasty

- **Buccal mucosa graft (BMG)** is the workhorse, one-stage or staged. Skin grafts are not used in LS. Grafts do best in the **bulb**, which has a thick vascular bed.
- Graft position may be ventral (with spongioplasty), lateral, or **dorsal (Barbagli)**, spread-fixed on the corpora.
- **Augmented anastomosis** often beats pure onlay. **Kulkarni** is a one-stage panurethral dorsal onlay via the perineum with **penile invagination**.
- **Staged**: stage 1 lays the urethra open and quilts the graft to dartos. Stage 2 tubularises it to about 28 Fr after at least 6 months. It is used for LS and failed hypospadias.

> **KEY POINT**
> A graft lives on its bed: it succeeds where the bed is thick and vascular (bulb, dorsal corpora), and is staged when the bed, or skin disease, cannot be trusted in one step.

- **Flaps** use hairless genital skin on dartos, such as the **Orandi** ventral longitudinal flap or a **circular island** flap up to **15 cm**; fossa navicularis strictures use a Jordan or De Sy island.
- Grafts equal flaps; flaps suit irradiated or multiply operated urethras. **Onlay beats tube**.
- **LS**: skin repairs fail, so use **oral mucosa only**, often staged.
- **Perineal urethrostomy** opens the urethra dorsally to healthy mucosa, sutures it to skin, and can add a BMG to complete the dorsal wall.

7 · Meatal stenosis

- **Child**: post-circumcision meatitis needs ventral meatotomy and parental dilation; early meatitis responds to 0.05% clobetasol.
- **Adult**: the fossa is usually involved, so evaluate it as a stricture before ventral meatotomy or dorsal Y-V plasty.

8 · Pelvic fracture urethral injury (PFUI)

P3 Describe early and late management of PFUD injuries — bulbomembranous site, primary realignment debate, cystogram + RUG, bladder neck not predictive, MRI, delayed perineal repair after 3 months, progressive steps, voiding study 21–28 days

Not in the paper answer:

- PFUI occurs in about **10%** of pelvic fractures. Repair at **3–6 months**; endoscopic core-through is not curative.
- Failure is mostly **ischaemic stenosis** of the mobilised bulb; internal pudendal status decides the order of surgery (table below).

ALGORITHM

1. On-table endoscopy; exaggerated lithotomy; perineal incision.
2. Mobilise the bulb; pass a sound via the suprapubic tract; excise fibrosis.
3. Spatulate to **≥30 Fr**; place 8–10 tension-free 4-0 PDS sutures over a silicone stent.
4. If tension remains, then add steps in order: mobilisation, intracrural separation, infrapubectomy, corporal rerouting.

- Success is in the high **90s %**.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularise for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men with **duplex ultrasound, then angiography if abnormal**.

9 · Vesicourethral stenosis and fistula

- Stenosis after prostate cancer treatment is incised deep at 3 and 9 o'clock, adding mitomycin C if recalcitrant. Obliteration needs robotic or perineal repair, often followed by an artificial urinary sphincter.
- **Rectourethral fistula**: a simple one (<2 cm, non-irradiated) is repaired perineally or by York-Mason. A complex (irradiated) one needs faecal diversion and **gracilis interposition, with or without BMG**.

10 · Complications and follow-up

- Recurrence is early after EPA and within 12 months after grafts. ED is uncommon, higher with long repairs and straddle injuries, and often improves.
- Follow up with symptoms, uroflow and post-void residual, and urethroscopy if suspicious.

11 · Landmark trials

Trial	Design	Result	Take-home
ROBUST III	Phase III RCT, Optilume (paclitaxel balloon) vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

87 Surgery of the Scrotum and Seminal Vesicles

1 · Surgical anatomy of the scrotum

- Lax, rugated skin gives **poor tamponade**, so haematomas keep enlarging and haemostasis must be meticulous.
- Dartos keeps necrotising fasciitis off the testes. **Transdermal absorption** is about **40×** that of other skin, and the skin is rich in 5α-reductase.

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical, from internal iliac
Testis, secondary	Cremasteric	Inferior epigastric, from external iliac

KEY POINT

Three trunks (aorta, internal iliac, external iliac) anastomose in the cord and at the cauda. So the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal.

2 · Peri-operative principles and access

- Shave with a **razor on the day of surgery**. The American Urological Association (AUA) advises a single IV cephalosporin or clindamycin **only with risk factors**.
- Benign disease uses a scrotal incision. **Possible malignancy** needs an inguinal incision, with the cord clamped before the testis is handled.
- **Always close the dartos**, because most haematomas come from inadequate dartos closure.

3 · Scrotal wall: excision, scrotectomy, reconstruction

- **Fournier gangrene** is the commonest indication for partial scrotectomy. Give fluids and broad-spectrum antibiotics, and **debride urgently**, because it spreads **2–3 cm/h**; mortality is about **20%**. Re-debride every **24–48 h**.
- Scrotal violation in non-seminomatous germ cell tumour (NSGCT) calls for partial scrotectomy of the scar.
- If more than **50–60%** of skin is lost, do not reconstruct at once; protect the testes in thigh pockets or under a vacuum dressing (VAC).

ALGORITHM

1. If <50% and tension-free, then close primarily.
2. If <50% with tension, then local scrotal advancement or secondary intention.
3. If >50%, then VAC, then stretched scrotal advancement or delayed primary closure.
4. If no local tissue, then fasciocutaneous thigh flap; meshed split-thickness skin graft (STSG) last.

KEY POINT

The scrotum works because it is lax. Graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · Hydrocele

- **Congenital** hydrocele comes from a patent processus; about **80%** resolve by 18 months. **Acquired** hydrocele reflects a production–absorption imbalance.
- **Colour Doppler ultrasound first** excludes tumour and decides the scrotal or inguinal route.

Technique	Steps	Best for
Lord plication	Small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim to 2-cm cuff, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick, large, multiloculated; lowest recurrence

KEY POINT

The more tunica excised, the lower the recurrence but the more haematoma. Recurrence after any successful repair is **0–6%**.

- **Aspiration and sclerotherapy** suit the unfit; aspiration alone recurs, so add a sclerosant such as doxycycline. With **retreatment**, resolution is **85–96%**, but results are poor if multiloculated or more than **750 mL**.

5 · Spermatocele and epididymal surgery

- Operate on a spermatocele only for symptoms, and **counsel about obstructive azoospermia**.
- A suspected malignant epididymal tumour is approached inguinally; a proven benign one by scrotal microsurgical excision.
- **Epididymectomy** gives **poor results for chronic epididymitis** (>6 weeks), so it is a last resort; it works better with a **palpable lesion**.

6 · Varicocele repair

P2 Surgical management of bilateral varicocele — ~15% of men; Palomo, laparoscopic, microsurgical subinguinal (lowest recurrence and hydrocele, gold standard) and embolisation compared; microsurgical steps

Not in the paper answer:

- †Embolisation suits recurrence after surgery. For orchialgia, **microsurgical repair** gives more complete relief.

7 · Vasectomy

- **Counsel**: the man is not sterile at once, so contraception continues until post-vasectomy semen analysis (PVSA). Repeat vasectomy is needed in <**1%**; pregnancy occurs in about **1 in 2000** even after an azoospermic PVSA; chronic pain affects about **1–2%**.

- Give **no routine antibiotics**. **Isolation** determines morbidity; **occlusion** determines efficacy.

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm	≤ **10 mm**	Puncture <10 mm
Status	Difficult cases	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974)

- Open on the **straight vas**, because the convoluted vas is harder now and at reversal.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) and fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC and FI, testicular end open	Less congestive epididymal pain; granuloma linked to better reversal

KEY POINT

Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division is required; segment excision is optional (**1 cm** is adequate).

ALGORITHM

1. Do PVSA at **8–16 weeks**.
2. If azoospermia or rare non-motile sperm (RNMS, ≤ **100,000/mL**), then stop contraception.
3. If motile sperm, then repeat at 6 months; if still motile, then it is a failure, so repeat the vasectomy.

- **Early failure** is technical, such as a missed vas; **late failure** is recanalisation after a successful PVSA.
- **Sperm granuloma** is expected rather than a complication. Post-vasectomy pain affects about **2%**; if a cord block helps, then denervation or reversal follows.

8 · Orchidectomy and prosthesis

- **Bilateral simple orchidectomy** treats hormone-sensitive metastatic prostate cancer. It is cheaper than androgen deprivation therapy but has no proven survival benefit.
- **Unilateral** orchidectomy is for trauma or non-salvageable torsion, and for orchialgia only after all conservative measures. **Epididymis-sparing** orchidectomy gives a better contour.
- **For orchialgia, use the inguinal route**, because it removes more of the proximal ilioinguinal, iliohypogastric and genitofemoral nerves.
- **Prosthesis**: offer it to every man lacking a testis, before cancer orchidectomy. Pierce the fill port ≤ **5 times**, never after insertion. A **high scrotal subdartos** pouch has the least infection and erosion.

9 · Testicular torsion and fixation

- †**Acute torsion**: if suspicion is high, explore without waiting for imaging. Detorse and assess, then fix or remove; **always fix the contralateral testis**.
- Intermittent torsion (pain <2 h) and postpubertal retractile testis need elective orchidopexy.

Fixation	Method	Note
Suture fixation	Albuginea to dartos (3-0 absorbable)	Re-torsion still possible; transparenchymal sutures damage spermatogenesis
Dartos pouch	Subdartos pouch with a purse-string at the cremaster exit	No transparenchymal suture, so **preferred when fertility matters**

10 · Chronic scrotal pain (>3 months)

ALGORITHM

1. History, examination, duplex ultrasound; if a cause is found, then treat it.
2. If none, then exclude extrascrotal causes (abdomen, pelvis, spine, hip, prostatitis, pelvic floor).
3. If non-specific, then give support, NSAIDs, tricyclics or gabapentin for ≥3 months.
4. If epididymal, then epididymectomy.
5. If testicular or post-vasectomy, then do a cord block: if relief >**50%**, then microsurgical denervation (or reversal); if <50%, then pain clinic, with inguinal orchidectomy last.
6. If post-hernia, then ilioinguinal block; if relief >**90%**, then neurolysis.

- **Microsurgical denervation of the spermatic cord (MDSC)** is done through a subinguinal incision. A micro-Doppler helps **preserve testicular and cremasteric arteries, lymphatics, and the vas with its vessels**; everything else, including the trifecta nerve complex, is divided.

11 · Complications of scrotal surgery

- **Haematoma** is commonest; drain it if large, infected or enlarging. A falling haematocrit after orchidectomy suggests a slipped cord ligature with a **retroperitoneal bleed**, which needs re-ligation through a retroperitoneal extension.

12 · Seminal vesicles

- The seminal vesicles (SV) are Wolffian derivatives that bud at about **12 weeks** and make about **60%** of seminal fluid. The **neurovascular bundles lie lateral to their tips**.
- **MRI is most accurate**. On transrectal ultrasound (TRUS), suspect obstruction if the SV is >**1.5 cm** or the ejaculatory duct >**2.3 mm**; confirm by **aspiration showing sperm**.
- **Zinner syndrome** is an SV cyst with **ipsilateral renal agenesis** and ejaculatory duct obstruction (EDO), often with an ectopic ureter into the cyst.
- Observe or aspirate small cysts. **Transurethral unroofing** is preferred for cysts, EDO and stones; large symptomatic cysts need laparoscopic or robotic excision.

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal, phyllodes, sarcoma, carcinoid, seminoma, yolk sac

- Image tumours with MRI and biopsy by TRUS if inconclusive. Secondary invasion is far commoner. Adenocarcinoma needs radical excision and lymphadenectomy.

13 · Surgical approaches to the seminal vesicles

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical	Children, large cyst with or without nephroureterectomy	Ureter, vas
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; dominant, because it opens neither organ	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle
Transcoccygeal	Hostile abdomen/perineum	Rectum

- A rectal injury needs two-layer repair; a ureteric injury needs reimplantation over a stent.

★ Putting it together

A · Testicular mass: the first week

ALGORITHM

1. Ultrasound.
2. Tumour markers AFP, hCG and LDH **before** orchidectomy; sperm bank.
3. **Radical inguinal orchidectomy** (never trans-scrotal).
4. Repeat markers (AFP half-life 5–7 days, hCG 1–1.5 days); CT chest, abdomen and pelvis.

B · Germ cell tumour by stage

Stage	Seminoma	Non-seminoma (NSGCT)
CS I	Surveillance (occult spread 10–15%); single-dose carboplatin	Low risk: surveillance. High risk (lymphovascular invasion, embryonal-predominant): surveillance, BEP ×1 or nerve-sparing RPLND
II	≤3 cm: radiotherapy, chemo or RPLND; >3 cm: chemo	Normal markers and small nodes: RPLND; otherwise chemo
Advanced (IGCCCG)	Full-dose cisplatin. Good: BEP ×3 or EP ×4. Intermediate/poor: BEP ×4 (VIP ×4 if lung concern).	
Residual mass	>3 cm: FDG-PET (≥6 weeks post-chemotherapy)	Any >1 cm: post-chemotherapy RPLND (teratoma)

C · Penile cancer: primary and groin

ALGORITHM

1. **Primary:** if PeIN, then topical 5-FU/imiquimod, laser or glans resurfacing. If invasive, then organ-sparing surgery, glansectomy, or partial/total penectomy; margins <1 cm are acceptable.
2. **cN0:** if low risk (pTis/pTa/pT1a), then surveillance; if intermediate/high, then DSNB or modified ILND; if positive, then radical.
3. **cN+:** if FNA positive, then bilateral radical ILND; if cN3, then neoadjuvant TIP first. Add pelvic dissection if ≥2 inguinal nodes or extranodal extension.

D · Anterior urethral stricture

Stricture	Treatment
Bulbar <1.5 cm, first time	Dilation or DVIU once; if it recurs, urethroplasty
Bulbar 1–2 cm (3–5 cm proximally)	EPA
Longer bulbar	BMG onlay or augmented anastomosis
Penile	Onlay graft/flap; staged if hostile bed, LS, failed hypospadias
Panurethral, frail	Perineal urethrostomy
Pelvic fracture urethral injury	Suprapubic catheter, then delayed perineal anastomosis

In retention, place a suprapubic catheter; never blind-dilate.

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 11

Stones, Obstruction & Trauma

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 93	Management of Upper Urinary Tract Obstruction
Ch 94	Upper Urinary Tract Trauma
Ch 95	Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis
Ch 96	Acute Care of Urinary Lithiasis
Ch 97	Surgical Management of Ureteral Calculi
Ch 98	Surgical Management of Upper Urinary Tract Calculi
Ch 99	Prevention of Urinary Stone Disease
Ch 100	Lower Urinary Tract Calculi
Ch 54	Management of Paediatric Kidney Stone Disease
Ch 125	Genital and Lower Urinary Tract Trauma
Ch 62	Paediatric Genitourinary Trauma
★	Putting it together

93 Management of Upper Urinary Tract Obstruction

1 · Ureteropelvic junction obstruction (UPJO): presentation

- Trauma: see Ch 94, Ch 125; stones: see Ch 95–100.
- UPJO is usually congenital. **Neonates** show prenatal hydronephrosis on ultrasonography, with azotaemia if bilateral or solitary.
- **Older children and adults** have intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI) and, rarely, hypertension.

2 · Diagnosis

Prove impaired drainage before operating.

KEY POINT

Dilation is not obstruction. Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows where; functional tests (renography, Whitaker) show whether it matters. Combine one of each.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula (unlike multicystic kidney); ureter not seen
Non-contrast CT (acute pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study
Retrograde pyelography	**At the time of repair** (avoids introducing infection)
Nephrostomy and antegrade study	When retrograde fails; tests recoverability; permits Whitaker

P1 Radionuclide studies and diuretic renography in PUJ obstruction — tracers, well-tempered protocol, T½ cut-offs, P40, O'Reilly curves

- **Renography**: hydrate, then give furosemide **0.3–0.5 mg/kg** at **20 min**; mercaptoacetyltriglycine (MAG3) is preferred.
- **Obstruction is a half-time (T½) > 40 min**. After successful pyeloplasty, drainage occurs even before furosemide.

ALGORITHM

1. **Whitaker test**
2. Place a nephrostomy and a bladder catheter; perfuse at **10 mL/min**.
3. If pressure stays ≤12–15 cm H2O, there is no obstruction; if it exceeds **15–20 cm H2O**, there is.
4. It is invasive, so use it only when other tests are non-diagnostic.

3 · Indications and choice of intervention

- **Indications**: symptoms; impaired or falling function; stones or infection; rarely hypertension.

- **Neonates** are selectively observed; operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults** with indeterminate results have serial renography.
- The **procedure of choice** is dismembered pyeloplasty, now robotic at most major centres; endopyelotomy fails more.
- Drain before repair only for infection, azotaemia (solitary or bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- A crossing vessel is **prognostic, not a contraindication**, but it pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO: cross over.** Failed endopyelotomy is treated by pyeloplasty; failed pyeloplasty by endourology first.
- **Nephrectomy** suits a thin shell with a normal contralateral kidney, stones with chronic infection, or multiple failed repairs. Adults with **<15%** differential function are non-salvageable; if uncertain, stent or drain and re-scan.

4 · Endopyelotomy

- **Principle**: a full-thickness **lateral** incision into peripelvic fat heals over a stent; lateral is **free of crossing vessels**.
- Pyeloplasty (≈**95%**) wins because it makes a wide dependent funnel; endourology wins only on recovery.
- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy. Urine must be sterile first.

Percutaneous antegrade

- Best for **UPJO with pelvicalyceal stones**. Puncture the **mid-posterior or superolateral calyx**, not the lower pole, whose acute infundibular angle blocks passage.
- Stent with a 14/7-Fr endopyelotomy stent, traditionally for **6 weeks**; 2 and 4 weeks are equal.
- **Clear stones before incising**, because fragments migrate periureterally and cause fibrosis.

Retrograde ureteroscopic (standard)

P1 Retrograde endopyelotomy — indications, holmium lateral-incision steps, results (≈90% without crossing vessel >4 mm), complications, balloon/cautery-wire alternatives

Not in the paper answer:

- Control bleeding with haemostatic laser settings or 10-min balloon tamponade.

ALGORITHM

1. General anaesthesia; working and safety wires; flexible ureteroscope via a sheath.
2. With a 200-µm holmium fibre, incise **laterally** into fat and into normal ureter.
3. Leave a stent and Foley; do a renogram 4 weeks after stent removal.

- Percutaneous endopyeloplasty (endoscopic Heineke–Mikulicz) fails in secondary UPJO.

Complications and follow-up

ALGORITHM

1. **Percutaneous bleeding**
2. Bed rest and transfusion.
3. **Do not irrigate** the nephrostomy, so clot tamponades.
4. If bleeding persists, do **selective embolisation**.

- Renogram at 1 and 6 months, then yearly for **2–3 years**. Pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- After failure, get CT angiography; a significant crossing vessel needs **pyeloplasty, not repeat incision**.

5 · Pyeloplasty

KEY POINT

Principles (open = minimally invasive): widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** transition.

P3 P1 Laparoscopic pyeloplasty; Management of failed pyeloplasty — Anderson–Hynes steps, approaches, contraindications, stent 4–6 weeks, vascular hitch, late failures, redo options

Not in the paper answer:

- Dismembered (Anderson–Hynes 1949) repair suits any insertion, allows pelvic reduction and **transposition over crossing vessels**, and alone excises the abnormal UPJ.
- Open approach is flank extraperitoneal by default; anterior for horseshoe or pelvic kidney.

ALGORITHM

1. Dissect the ureter with periureteral tissue; place a lateral marking stitch.
2. Excise the UPJ; spatulate **laterally**; bring the apex to the inferior pelvis, anterior to any crossing vessel.
3. Do a watertight absorbable anastomosis, with or without reduction; leave a stent and drain.

- Remove the **stent** at **4–6 weeks**. Reserve an added nephrostomy for secondary UPJO.
- Laparoscopic and robotic success is >**90–95%**, equal to open; the robot's advantage is ergonomic.

ALGORITHM

1. Remove the Foley at 24–36 h, then the drain if output is negligible.
2. If output rises after Foley removal (reflux along the stent), **replace the Foley for 7 days**.
3. Remove the stent at 4–6 weeks; then renography.

6 · Other UPJ reconstructions

KEY POINT

Every UPJ repair asks **what tissue bridges the gap while keeping blood supply**. Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width ≤ **3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- Outline flaps longer than needed; trim the apex, never the base.

P2 Uretero-calycostomy — resect lower-pole parenchyma (not nephrotomy), mucosa-to-mucosa, omental cover; cortex <5 mm predicts failure

7 · Retrocaval ureter

P3 Retrocaval ureter — venous anomaly, always right, fish-hook/S-shape, CT; transpose ureter anterior to IVC

8 · Ureteral stricture — evaluation and endourology

Exclude cancer; endoscopy suits only short, well-vascularised strictures.

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis, endoscopy, infection (tuberculosis, schistosomiasis).
- **Exclude malignancy first** with CT, with or without ureteroscopy and biopsy.
- Stricture follows ureteroscopy (URS) in ≈ **3%** vs 1.5% after shock-wave lithotripsy; an impacted stone >2 months gives ≈1 in 4.
- Endourology needs ipsilateral function ≥ **25%** (the algorithm uses 20%).
- **Stents** suit intrinsic strictures; extrinsic compression needs tandem or metallic stents. Chronic stenting suits unfit or cancer patients.
- **Balloon dilation** is inflated until the **waist disappears**, then stented 2–4 weeks. Contraindications: infection, stricture > **2 cm**. It works best (≈85%) in short, recent, non-anastomotic, well-vascularised strictures; anastomotic ≈50%.
- **Endoureterotomy** (retrograde holmium) is first line under 2 cm. Cut full thickness into **2–3 mm of normal ureter** each side, over a safety wire.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels
Proximal ureter	**Lateral/posterolateral**	Great vessels

- Complete obliteration in high-risk patients can be crossed by **cutting toward the light**, then stenting 8–10 weeks.

ALGORITHM

1. If <2 cm and function >20%, then endoscopic incision.
2. If >2 cm and function >20%, then reconstruction.
3. If function <20%, then nephrectomy (children may still merit repair).
4. Transplant ureter: consider balloon. Follow with renography for 2 years.

KEY POINT

Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · Ureteral stricture — reconstruction ladder

Defect length and level choose the repair.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy** suits short defects only, because tension causes stricture. Preserve adventitia, spatulate 180° apart, stent and cover with omentum; success is >90%.

ALGORITHM

1. **Persistent leak**
2. X-ray the stent position.
3. Pull the drain off the anastomosis and stop suction.
4. Place a Foley and give an anticholinergic; if it persists, place a nephrostomy.

P3 P2 Boari flap; Trans Uretero Ureteric anastomoses — contracted bladder contraindicates; recurrent stricture commonest; eGFR fell in 44%; TUU donor above IMA, contraindications, correct recipient reflux

- In adults, refluxing and non-refluxing reimplants are equal. Treat outlet obstruction first.
- **Renal descensus** rotates the kidney inferomedially and fixes its lower pole; the renal vein limits descent.
- **Ileal ureter** is isoperistaltic. Contraindications: creatinine > **2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Start endoscopic surveillance from year 3.
- **Buccal mucosa onlay** suits dense scar. **Autotransplantation** suits an absent or poor contralateral kidney, or when other repairs are impossible.

KEY POINT

Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · Ureteroenteric anastomotic stricture

- Strictures occur in 4–9% of conduits, so **upper-tract surveillance is lifelong**.

- **The cause is ischaemia**. The **left is commoner**, because it needs extra length, passes under the sigmoid mesentery and angulates at the inferior mesenteric artery.
- Risks: **non-refluxing anastomosis** (≈13% vs 2–3%), carcinoma in situ at the margin, obesity and major complications.
- Conduit dilation is often non-obstructive, so confirm with renography; rapid progression with a mass suggests recurrent cancer.
- **Open revision is the gold standard** (≈91% vs 26% endoscopic); robotic is equal. Endourology is **antegrade**. Endoscopic success is ≈50% under 1 cm and better on the right.

ALGORITHM

1. If <2 cm and function >20%: right side, endoscopic incision; **left side, formal repair** (sigmoid mesentery bleeding).
2. If >2 cm, then formal repair.
3. If function <20%, then nephrectomy or observation.

11 · Retroperitoneal fibrosis

P3 P1 Retroperitoneal fibrosis — idiopathic 70% vs secondary (drugs, lymphoma); plaque at L4–L5, medial deviation; PET for activity; deep biopsies; drain, then prednisone (80% respond), then bilateral ureterolysis and omental wrap

ALGORITHM

1. If uraemic, decompress with a **nephrostomy** or stents and manage the **post-obstructive diuresis**.
2. Stop any culprit drug; biopsy.
3. Give steroids; if they fail, do ureterolysis.

- Steroids work while the lesion is cellular; second line is immunosuppressants or **tamoxifen**. Ureterolysis with an omental wrap is most definitive.

94 Upper Urinary Tract Trauma

P2 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — mechanism, haematuria, AAST 2018, imaging indications, CT red flags, non-operative care, exploration, vascular control, renovascular injury, complications

Not in the paper answer:

- Polar injury is treated by partial nephrectomy with omental cover.

1 · Renal injury — mechanism and presentation

- The kidney is the commonest urological organ injured. Blunt trauma causes 80–90%; most are grade I–III and heal conservatively.
- **Deceleration** injures the fixation points (hilum and ureteropelvic junction, UPJ), causing artery thrombosis, vein disruption, and pedicle or UPJ avulsion.
- **Penetrating** injury causes more bleeding, nephrectomy and complications; high-velocity missiles cause delayed necrosis.
- Signs: flank ecchymosis and **ipsilateral rib 9–12 fractures** (strongest association).

2 · Haematuria

- Significant haematuria is **gross**, or **microhaematuria (>5 red blood cells per high-power field) with systolic blood pressure (SBP) <90 mm Hg**.
- Degree does not match severity; haematuria is absent in ~⅓ of blunt renovascular injuries and **~½ of UPJ injuries**. Test the first aliquot.

> **KEY POINT**
> Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · AAST kidney injury scale (2018)

The AAST grade, set on CT, predicts intervention and nephrectomy.

Grade	AAST 2018
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration **≤1 cm**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's**
IV	Collecting system laceration with extravasation; pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's; segmental/complete infarction without active bleeding
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for bilateral or multiple injuries, up to III; grade by the highest finding. MCQ Main-vessel injury was grade IV in 1989 and is now V; segmental pseudoaneurysm or arteriovenous fistula (AVF) is now IV.

4 · Indications for imaging (AUA = EAU)

- **Penetrating** trauma with **any haematuria** needs CT, if the patient is stable.
- **Blunt** trauma needs CT with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 at any time; otherwise blunt microhaematuria is **observed without imaging**.
- **Children** hold blood pressure until ~50% blood loss, so image liberally.

5 · Imaging studies

Without a delayed phase, urinary extravasation is missed.

- Gold standard: **contrast CT with arterial/nephrographic and delayed (~10 min) phases**. A medial haematoma with normal arteries suggests vein injury.
- Focused assessment with sonography for trauma (FAST) cannot separate haematoma from urine.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Contrast blush	Brisk bleeding; strongly predicts need for angioembolisation (AE)
≥2 of: rim distance >4 cm, medial laceration, blush	High-risk grade IV

- The **MiGUTS nomogram** predicts bleeding intervention; its largest contributor is **haematoma rim distance**.

ALGORITHM

1. **One-shot intravenous pyelogram (IVP)** (unstable, no CT): **2 mL/kg contrast → single film at 10 min**. It confirms a working contralateral kidney.

Perinephric collection	Hounsfield units (HU)	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · Nonoperative management

- Stable, staged **grade I–IV and selected V** injuries are observed, whatever the mechanism. Success is >90% in blunt injury, even with extravasation; early mobilisation is safe.
- **Failure triggers**: instability; transfusion of **2–3 units**; new pain or guarding; enlarging urinoma with fever. Most failures need only stent, nephrostomy or AE.
- **Penetrating** injury breaches Gerota's fascia, losing tamponade, so more fail; stabs do better than gunshots.
- **Reimaging** (AUA): CT at ~48 h for **grade IV–V deep lacerations** or deterioration, not for uncomplicated I–III.

7 · Urinary extravasation

- Parenchymal leaks **heal in 76–90%**. **Drain early** when urine cannot pass antegrade: pelvis or UPJ avulsion, ureteric clot, UPJ obstruction.
- A symptomatic or enlarging urinoma gets a retrograde stent and catheter; if that is impossible, a nephrostomy.

> **KEY POINT**
> Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · Intervention

Needless exploration releases tamponade and costs kidneys.

Absolute indications	Relative (increasingly observed)
Instability, no or transient response	Extravasation with significant devascularisation
Expanding or pulsatile haematoma	Colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

- **AE** is superselective, raises salvage and can be repeated. **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria.
- At laparotomy, exploration is **not mandatory** for staged grade I–III. The only absolute indication is a **pulsatile or expanding haematoma**.

> **ALGORITHM**
> 1. **Vascular control first**
> 2. Eviscerate the small bowel; incise peritoneum over the aorta above the inferior mesenteric artery (IMA), up to the ligament of Treitz.
> 3. Follow the aorta to the left renal vein; loop and retract it; loop both renal arteries.
> 4. Reflect the colon and open Gerota's fascia; occlude the vessels if bleeding.

- **Damage control** (hypothermia, acidosis, coagulopathy): pack, close temporarily, resuscitate and return at 24–48 h.

> **ALGORITHM**
> 1. Expose and gain vascular control; debride only devitalised tissue.
> 2. Suture-ligate vessels; close the collecting system **watertight**.
> 3. Approximate parenchyma over a bolster; cover with Gerota's fascia or **pedicled omentum**; drain.

- Colon or pancreatic injury no longer mandates nephrectomy.
- **Revascularise** only a solitary kidney or bilateral injury, with warm ischaemia **<120 min**; unilateral repair salvages only ~⅓.
- The **left renal vein may be ligated; the right cannot**.
- Hypertension follows thrombosis repair in ≥43%, so monitor blood pressure for a year.

9 · Complications and follow-up

- Delayed bleeding comes within 21 days and is treated by AE.

- **Page kidney** (subcapsular compression) is treated with an ACE inhibitor or receptor blocker, then drainage if refractory; post-traumatic AVF needs AE.
- EAU: no imaging for grade I–II; salvaged grade III–V get a dimercaptosuccinic acid (DMSA) scan.
- Bleeding risk peaks at **2–6 weeks**; a solitary kidney does not bar contact sport.

10 · Ureteral injury

Often silent; late diagnosis means fistula or kidney loss.

P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — aetiology, risk sites, prevention, CT with delays/retrograde, damage control, repair by level (hitch 95–100%, Boari up to 12 cm, TUU, ileal ureter), follow-up

Not in the paper answer:

- Delayed upper-ureter repair may use appendix (right) or buccal graft.

AAST ureter grade	Description
I	Contusion/haematoma without devascularisation
II	Laceration **<50%** transection
III	Laceration ≥50% transection
IV	Complete transection, **<2 cm** devascularisation
V	Avulsion, >2 cm devascularisation

- Iatrogenic injury is commoner (distal); external is mostly gunshot (proximal). **Haematuria is absent in 25–45%**.
- The **distal ureter** is the commonest iatrogenic site; prolapse surgery is highest-risk, and minimally invasive hysterectomy is riskier than open.
- **Ureteroscopy**: stricture ~3%, avulsion <1%. At a tear, stop basketing and stent.
- Preoperative stents aid **recognition**, not proven prevention. Check efflux at cystoscopy after minimally invasive hysterectomy and prolapse surgery. **Methylene blue** is a monoamine oxidase inhibitor; avoid it in pregnancy and G6PD deficiency.

Modality	Role
CT urography, **10-min delays**	Best in a stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic; never rely on it

- CT signs: **no contrast distal to the injury**, medial extravasation (UPJ avulsion), delayed nephrogram, urinoma.

> **KEY POINT**
> Ureteral injury is diagnosed by absent contrast beyond it, so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

ALGORITHM

1. If stable at laparotomy, repair now; if unstable, do damage control and repair later.
2. If contusion, stent, or excise and repair.
3. If found late or partial, stent; if stenting fails, place a nephrostomy and repair later.
4. If ureterovaginal fistula, try a stent first.

- **Damage control**: ligate the proximal stump with long silk and place a nephrostomy once stable, or exteriorise a single-J stent.
- Principles: spare adventitia; debride to bleeding edges; make it **spatulated, tension-free, stented and watertight**; drain. **No bowel or autotransplantation acutely.**
- In trauma, reimplant **non-tunnelled**, because tunnelled reimplants stricture more.
- Psoas hitch beats distal ureteroureterostomy, because the distal stump's supply is tenuous; spare the genitofemoral and femoral nerves.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

1 · Overview and epidemiology

- **Removal is surgical, but recurrence is metabolic**. Prophylaxis: see Ch 99.
- Lifetime prevalence is **1–15%**; in the United States ≈ **10%** and rising.
- **Sex**: males outnumber females 2–3 : 1, but the gap is narrowing. **Oestrogen is protective**, hence a second female peak at menopause.
- **Race**: highest in White, lowest in Asian people. **Age**: peak in the 4th–6th decades; paediatric stones are rising.
- **Climate and occupation**: heat lowers urine volume and pH, sunlight causes vitamin D-mediated hypercalciuria, and sedentary work causes hypercalciuria.
- **Obesity**, **metabolic syndrome** and **diabetes** raise risk. **Insulin resistance** unifies these: it lowers urine pH, causing uric acid stones, and hyperinsulinaemia causes hypercalciuria.
- Hypertension, cardiovascular disease and chronic kidney disease (CKD) are associated. Fluid intake is inversely related (proven by randomised controlled trial, RCT).

2 · Physicochemistry and stone formation

Prevention lowers supersaturation or aids inhibition.

P2 Stone formation (etiopathogenesis) and medical management of urolithiasis — saturation states, heterogeneous nucleation, steps, Randall plaque vs ductal plugs, inhibitor list, matrix 2.5%

Not in the paper answer:

- Free-particle theory (crystals trapped as plugs) vs fixed-particle theory (anchored on plaque).
- Stones form by **supersaturation, then nucleation, growth and aggregation, then retention**. The concentration product (CP) multiplies ion concentrations; the solubility product (Ksp) is CP at saturation; the formation product (Kf) is CP at spontaneous crystallisation.

State	CP	What happens
Undersaturated	< Ksp	No crystals; stones may dissolve
Metastable	Ksp–Kf (most urinary salts)	No homogeneous (spontaneous) nucleation; growth and heterogeneous nucleation possible; **inhibitors effective, so this is the zone of therapy**
Unstable	> Kf	Spontaneous nucleation; inhibitors ineffective

- The **relative saturation ratio (RSR)** is CP ÷ Ksp. Below 1, stones dissolve; above 1, they grow.
- Changing pH lowers the saturation of **phosphate, uric acid and cystine, but not oxalate**. **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- Urine holds calcium oxalate at ≈ 4× its solubility in water; with inhibitors it precipitates only at **7–11×**.
- Nephron transit is too short for one crystal to block a tubule, so aggregation occludes.
- **Randall plaque** (MCQ 1937) is **calcium apatite**. It starts in the **thin-limb basement membrane**, spreads through the interstitium and erodes the papilla; calcium oxalate then

grows on it.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present, as the anchor	Uncommon, mild
Intestinal bypass	Absent	Apatite plugs in inner medullary collecting ducts; injury, fibrosis
Brushite	Present	Apatite plugs; interstitial fibrosis
Distal renal tubular acidosis (RTA), hyperparathyroidism, cystinuria	—	Present

KEY POINT

Plaque vs plug: a plaque is an idiopathic interstitial apatite anchor; a plug is intratubular obstruction with injury, caused by systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Infection or shock-wave lithotripsy may convert a calcium oxalate former into a brushite former.

3 · Inhibitors and promoters

- Inhibitors raise the supersaturation needed for nucleation. **Uric acid and cystine have no inhibitor**; only pH and concentration matter. Promoters: low volume and citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits precipitation, agglomeration and urate-induced nucleation	Most important small-molecule inhibitor
Magnesium	Complexes oxalate	Synergy with citrate, phytate
Glycosaminoglycans	Bind crystal surface	Lower in stone formers
Nephrocalcin	Inhibits calcium oxalate monohydrate nucleation, growth, aggregation	Formers excrete isoforms lacking γ-carboxyglutamic acid
Tamm-Horsfall protein	Inhibits aggregation, not growth	Most abundant urinary protein, made in the thick ascending limb (TAL); **a promoter in acid urine**
Osteopontin	Inhibits nucleation, growth, aggregation, epithelial binding	Inducible

- **Matrix** is ≈ **2.5%** of stone weight.

4 · Mineral metabolism

- **Calcium**: 30–40% is absorbed in the small bowel, transcellularly (calcitriol-dependent) at low intake and paracellularly at high intake.
- **Oxalate-rich food lowers calcium absorption, and luminal calcium lowers oxalate absorption**.
- **Calcitriol**, made in the proximal tubule under **parathyroid hormone (PTH) and hypophosphataemia**, is the most potent stimulator of gut calcium absorption.

- **PTH** causes bone resorption, raises renal calcium reabsorption and calcitriol, and lowers phosphate reabsorption; it has **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive; claudin-16/19; the calcium-sensing receptor (CaSR) acts through claudin-14 to block it
Distal convoluted tubule (DCT)	8–10% (fine control)	Transcellular: entry by TRPV5 channel, then carried by calbindin-D28k

- **Oxalate**: 6–14% is absorbed, unregulated, and excreted almost entirely in urine; half is dietary, half hepatic. **Ascorbic acid** is the largest endogenous source; Oxalobacter formigenes degrades oxalate.

5 · Classification

Stone	Frequency
Calcium oxalate (pure or with apatite)	**60%**
Hydroxyapatite	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite	≈ 7% (falling)
Cystine	1–3% (up to 10% children)
Triamterene, silica, 2,8-dihydroxyadenine (2,8-DHA)	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. phosphate leak)	Increased gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism (PHPT)	3–5
Hyperuricosuric calcium	Purine excess	10–40
Hypocitraturic calcium	Idiopathic, diarrhoea, distal RTA, thiazide	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	Reduced gut magnesium absorption	5–10
Idiopathic low urine pH	Low pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease; cystine transport	1–5; < 1
No abnormality	—	< 3

KEY POINT

Classify by the **urinary abnormality, not the stone**: composition starts the work-up. Fewer than 3% have no abnormality. Work-up: see Ch 99.

6 · Hypercalciuria

Fasting urine and PTH reveal the source.

- It is the commonest abnormality (35–65%).
- **Definition**: > **200 mg/day** after 1 week on a 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day.
- The **Pak classification** (gut, kidney, bone) is an exam framework, not a therapy guide.

Feature	Absorptive	Renal	Resorptive
Defect	Increased gut absorption	Renal leak	PHPT
Serum calcium	Normal	Normal	**High**
PTH	**Suppressed**/normal	**High** (secondary)	High
Fasting urinary calcium	**Normal** (< 0.11 mg/dL glomerular filtrate)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption suppresses PTH and raises the filtered load	Renal loss raises PTH, which raises calcitriol	PTH drives bone resorption and calcitriol, causing hypercalcaemia

- Absorptive **type I** persists on calcium restriction; **type II** normalises.
- **CYP24A1** loss stops calcitriol inactivation, causing hypercalciuria and nephrocalcinosis. In **renal phosphate leak**, hypophosphataemia raises calcitriol and gut calcium absorption.

Disorder	Gene	Features
Dent disease (X-linked)	ClC-5	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)	Claudin-16, -19	TAL magnesium/calcium loss
Bartter	NKCC2, ROMK, ClC-Kb, barttin; type V activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria

KEY POINT

Fasting urine and PTH separate gut from kidney: in renal leak, fasting calcium and PTH are high; in absorptive hypercalciuria, fasting calcium is normal and PTH suppressed.

- **PHPT** causes ≈ 5% of stones, and stones occur in < 5% of PHPT. Suspect it if serum calcium is > **10.1 mg/dL**; repeat with intact PTH, and use ionised calcium for normocalcaemic PHPT. A **thiazide challenge** unmasks it.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High/inappropriately normal	Commonest in outpatients

Cause	Mechanism	PTH	Clue
Sarcoidosis	Macrophage 1α-hydroxylase	Suppressed	Corrects with **corticosteroids**
Malignancy	PTH-related protein (PTHrP), osteolysis (RANKL)	Suppressed	Commonest in inpatients
Glucocorticoid excess	Increased bone resorption	Raised	Cushing syndrome

7 · Hyperoxaluria

- **Definition**: > **40 mg/day**. Types: primary, enteric, dietary and idiopathic.

P2 Primary Hyperoxaluria — PH1–3 enzymes, markers, course; oxalate ≥100 mg/day, oxalosis; pyridoxine (Gly170Arg); lumasiran; liver–kidney transplant in PH1

ALGORITHM

1. **Enteric hyperoxaluria**
2. Fat malabsorption leaves fatty acids in the gut lumen.
3. Fatty acids saponify calcium and magnesium, so less calcium binds oxalate and more oxalate is free.
4. Free oxalate undergoes **colonic hyperabsorption**, helped by bile salts.

- **Enteric** is the commonest acquired form: small-bowel resection or Crohn disease, jejunoileal bypass and malabsorptive bariatric surgery (not band or sleeve).
- It **needs an intact colon**; with an ileostomy, uric acid or ammonium urate stones form instead. Diarrhoea also causes hypocitraturia, hypomagnesiuria and low urine pH.
- **Dietary**: nuts, chocolate, tea, spinach, rhubarb and vitamin C. **Severe calcium restriction is counter-productive**, as less calcium binds gut oxalate.

8 · Hyperuricosuria, hypocitraturia, hypomagnesiuria

- **Hyperuricosuria** is > **600 mg/day**. Urate promotes calcium oxalate stones by **adsorbing inhibitors, "salting out" and epitaxy**; allopurinol cut recurrence (trial below).
- **Hypocitraturia** is < **320 mg/day**; **acid–base status is the main determinant**. Causes: distal RTA, chronic diarrhoea, animal protein, thiazides (through hypokalaemia), ACE inhibitors.

KEY POINT

Citrate excretion reads the proximal-tubular **intracellular pH**: acidosis, hypokalaemia or an acid diet make the tubule reclaim citrate. Hence potassium citrate is the treatment (see Ch 99).

- Hypomagnesiuria: **two RCTs of magnesium showed no benefit**. **Low urine pH** (< 5.5) causes uric acid and calcium oxalate stones.

9 · Renal tubular acidosis

P2 Renal tubular acidosis — type 1/2/4 comparison (urine pH, K^+, citrate, stones up to 70%), incomplete dRTA, ammonium chloride load, topiramate/acetazolamide

Not in the paper answer:

- Genetics: SLC4A1 (dominant); ATP6V1B1/ATP6V0A4 (recessive, with **sensorineural hearing loss**).

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**	< 5.5 at steady state	Low
Urinary citrate	**Profoundly low**	Near normal	—
Stones	**Up to 70%**: calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Associations	Sjögren, lupus, obstruction; genetic	Fanconi	Diabetic nephropathy

- **Incomplete distal RTA**: electrolytes are normal, but urine pH cannot fall below 5.5 after an ammonium chloride load.
- In distal RTA, **profound hypocitraturia** is probably the most important stone factor.

KEY POINT

Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria, the recipe for **calcium phosphate** stones.

10 · Uric acid stones

P2 Uric acid calculi: etiopathogenesis, medical management and allopurinol — no uricase, pKa 5.35, three determinants (low pH dominant), insulin resistance/ammoniagenesis, idiopathic vs hyperuricosuric calcium

Not in the paper answer:

- Uric acid forms about one-third of stones in type 2 diabetes.

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, diarrhoea, gout	**Single most important**
Low volume	Diarrhoea, heat	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative disease, uricosurics	Least often the driver

- Humans lack hepatic **uricase**, so uric acid is not converted to soluble allantoin.
- The pKa of uric acid is **5.35**. Undissociated uric acid dissolves only to ≈ 100 mg/L, but at pH 6.5 > 1200 mg/L stays soluble.
- Stones are radiolucent on plain film but visible on non-contrast CT.

ALGORITHM

1. **Why the urine is acid in insulin resistance**
2. Insulin resistance and lipotoxicity lower proximal ammoniagenesis and sodium–hydrogen exchanger 3 (NHE3) activity.
3. H^+ is excreted as titratable acid instead of ammonium.
4. Urine pH stays **persistently low**, uric acid stays undissociated, and stones form.

KEY POINT

Uric acid stones are a disease of **urine pH, not uric acid**. No inhibitor exists, so raising pH above the pKa removes the driver; alkalinisation can dissolve stones (see Ch 99).

11 · Cystine and struvite stones

Both recur aggressively without lifelong prevention or complete clearance.

P1 P4 Cystinuria; Struvite stones — SLC3A1/SLC7A9 types, solubility by pH, hexagonal crystals, CKD; urease cascade, biofilm, Proteus/Ureaplasma, discordant cultures

Not in the paper answer:

- **Cystinuria genes**: SLC3A1 (type A), SLC7A9 (type B), or both (type AB). It is autosomal recessive, and with **no inhibitor** cystine is soluble only at alkaline pH (≈ 1000 mg/L at pH 9).
- **Struvite**: **Proteus mirabilis** is commonest; **Ureaplasma** is easily missed. **Less than half of staghorns are infection stones**, and bacteria within the stone demand complete clearance.

ALGORITHM

1. **Urease cascade**
2. Urease splits urea into ammonia (NH_3) and CO_2.
3. Ammonia forms NH_4^+ and OH^-, raising urine pH to 7.2–8.0.
4. With magnesium, phosphate forms **struvite**; with calcium, carbonate apatite.

12 · Miscellaneous and drug-related stones

	Xanthine	2,8-DHA
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Radiology	Both radiolucent, so mistaken for uric acid	
Treatment	—	Allopurinol or febuxostat (more effective)

- **Ammonium acid urate** stones are radiolucent and follow diarrhoea, ileostomy, **laxative abuse** and infection.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	Insoluble at physiological pH; stones 4–13%; radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicates	Drug stones
Indirect	Acetazolamide, **topiramate**, zonisamide	Mimic distal RTA, causing calcium phosphate stones
Indirect	Loop diuretics	Hypercalciuria (up to 64% of low-birth-weight infants on furosemide)
Indirect	Steroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, urate

13 · Anatomic predisposition and pregnancy

Anomaly	Stones	Metabolic findings
Ureteropelvic junction obstruction (UPJO)	≈ 20%; children ~70× risk	Common (hypercalciuria first); recurrence after repair, often contralateral

Anomaly	Stones	Metabolic findings
Horseshoe kidney	≈ 36% adults	All tested abnormal; **hypocitraturia**; calcium stones ≈ 89%
Calyceal diverticulum	Up to 40%	Abnormal in half; UTI up to 40%
Medullary sponge kidney	Frequent, with nephrocalcinosis	Hypercalciuria, hypocitraturia; distal RTA on acid loading

KEY POINT

Anatomy sets the stage but metabolism writes the stone: anatomical stone formers still need **metabolic evaluation**, and pyeloplasty alone does not prevent recurrence (see Ch 99).

- **Pregnancy**: stones occur in the 2nd–3rd trimester. Physiological hydronephrosis affects up to 90%, **right more than left**.
- Raised GFR and placental calcitriol promote stones, while raised citrate and magnesium inhibit them; the **net risk is unchanged**, but stones skew to **calcium phosphate**.

14 · Landmark trials

Trial	Design	Result	Take-home
Borghi water trial	RCT, high fluid vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier (see Ch 99)
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones (see Ch 99)

96 Acute Care of Urinary Lithiasis

1 · Presentation and diagnosis

Fever with obstruction is an emergency.

- **Renal colic** is flank-to-groin pain; a UVJ stone causes penile-tip pain and frequency.
- Haematuria occurs in ~**90%**, so it is absent in >10%.
- **Fever means an infected obstructed system**, needing urgent intervention.
- Send bloods, urinalysis and a **pregnancy test** in women of reproductive age. Urine >50 WBC/hpf or nitrites suggest infection.

Modality	Role	Limitation
Non-contrast CT (NCCT)	Gold standard (sens 97%, spec 95%); shows alternative diagnoses	8–16 mSv; misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm and in obesity
Ultrasound (US)	First line in pregnancy, children, known formers	Up to 40% still need CT
Plain X-ray (KUB)	**Avoid** for acute diagnosis	Lowest accuracy
MRI	Second line in pregnancy and children	Poor stone detection

KEY POINT

Unknown diagnosis: NCCT answers everything at once. Known stone: US showing hydronephrosis suffices.

2 · Pain management

- **NSAIDs** block prostaglandin release and match opioids with less vomiting.

Agent	Place	Note
Ketorolac 30 mg	**First line** (EAU, NICE)	Avoid in renal failure, GI bleed, pregnancy
IV paracetamol	Second line	NSAID barred or insufficient
Opioids	Intractable pain	**Continue the NSAID**: morphine plus ketorolac beats either
Antispasmodics; forced fluids	**Ineffective**	Fluids only for vomiting

3 · Medical expulsive therapy (MET)

P2 Medical expulsive therapy (MET) in urinary calculi — α1 mechanism, tamsulosin/silodosin, passage-by-width table, SUSPEND/STONE, benefit in distal 5–10 mm, intervene after 4–6 weeks

Not in the paper answer:

- Measure **transverse width**, not length; a 6 mm wide stone passes in ≈ 33%.
- Trial of passage, with or without MET, lasts **4–6 weeks**.

4 · Urgent intervention

- **AUA indications**: obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Stent and nephrostomy (PCN)** are equal for sepsis; stents cause dysuria and haematuria, PCN a longer stay.
- **Stent failure** is predicted by P-CUT (ureteric diameter at the stone minus stone width) ≥ **7.5 mm**, stone ≥9.5 mm and multiple stones; prefer PCN then.
- Acute SWL (proximal) or URS (distal <10 mm) is done **only without infection**.

ALGORITHM

1. Give analgesia; get NCCT.
2. If infection or an AUA criterion, then drain only (stent/PCN); treat the stone later.
3. If not and stone <10 mm, then trial with or without MET for 4–6 weeks.
4. If it fails, then URS or SWL.

5 · Special populations

P2 Stones (urolithiasis) in pregnancy — US, then MRI, then low-dose CT; conservative first, stent/PCN encrustation, URS any trimester

Not in the paper answer:

- Resistive index difference ≥ **0.06** means obstruction.
- About 30% need intervention; change stent/PCN every **4–6 weeks**. URS is safe in any trimester (ACOG 2019).

Population	Pitfall	Key point
Elderly	Atypical pain; first stone after 60 unusual	Always CT to exclude aortic/cardiac mimics; drain early
Transplant kidney	Denervated, so **painless**	Fever or renal failure; endoscopic, else percutaneous

6 · Landmark trials

Trial	Design	Result	Take-home
SUSPEND	RCT, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm

97 Surgical Management of Ureteral Calculi

P2 Management of upper ureteric stones — passage rates, EAU choice by size, SWL vs URS table, SWL failure factors, flexible URS, antegrade and laparoscopic routes, complication rates

Not in the paper answer:

- SWL **within 48 h** of an acute obstructing stone beats delayed SWL.
- Little is gained beyond 2 SWL sessions; **forced diuresis is useless**.

1 · Background and indications

- Non-contrast CT (NCCT) is the gold standard; ultrasound misses mid-ureteric stones. US with a plain film (KUB) suits follow-up.
- Medical expulsive therapy is controversial after SUSPEND; intervene after **4–6 weeks**.
- The ladder is SWL, then ureteroscopy (URS), then percutaneous nephrolithotomy (PCNL) with antegrade URS.

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis or sepsis: **drain only**, never manipulate the stone

- Work-up: creatinine and **urine culture**; exclude coagulopathy and untreated UTI.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high risk
SWL	Stop if feasible	Not recommended; choose another modality
Laser URS	**Need not stop**	None; more haematuria and stent symptoms

KEY POINT

Only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen; laser URS works inside the lumen under vision.

2 · Choosing treatment

- Upper ureter ≤2 cm and distal ureter: stone-free rate (SFR) is similar, but SWL needs more sessions.
- **Impaction** (thick wall around the stone on NCCT) defeats SWL, so treat endoscopically.
- A large hard proximal stone needs PCNL with antegrade URS; if unfit, staged URS.

Modality	Size guidance	
SWL	Historical limit 2 cm; SFR falls above	**10–15 mm**
URS	No definite limit	
PCNL with or without antegrade URS	>15–20 mm, large hard proximal	

KEY POINT

Constant trade-off: SWL means fewer complications but more sessions; URS means one session with slightly more morbidity.

3 · Ureteroscopes

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered	Smallest tip	~8.5 Fr	7.7 Fr
Cost	Cheap	Repair ~every **15** cases		Low volume

- Semirigid scopes suit the distal and mid ureter; flexible scopes suit proximal stones.
- Digital scopes are **not more durable**; keep a fibreoptic scope for a narrow ureter.

4 · Access

Most injuries happen at access, so be safe before fast.

P3 Accessories for RIRS; Ureteric stents and ureteroscopy with or without stenting — guidewire types, 12/14 Fr sheath, safety wire (Segura), pre-stent 2–4 weeks, stent indications, Denstedt stentless trial

Not in the paper answer:

- **α-Blockers** ~1 week pre-URS give easier access, higher SFR and less stent pain.
- Reserve the sheath for stones **above the iliac vessels**.
- For a difficult ureter, place a stent and return after **2–4 weeks**.

> **ALGORITHM**
> 1. If the wire will not pass, place an open-ended catheter 1–2 mm short of the stone.
> 2. Do a retrograde pyelogram.
> 3. Use contrast hydrodissection with a hydrophilic wire.
> 4. Advance the catheter past the stone, then exchange for a stiff wire.

- If the catheter abuts the stone, the wire exits at 90° and causes **perforation**. If the ureter is inaccessible, **stent and stage**.
- If the scope will not enter the orifice, use **railroading** between two wires, a dual-lumen catheter or scope rotation.

Dilation method	Best for
Indwelling stent	Whole ureter narrow; gentlest, but commits to a second procedure
Serial/coaxial dilators	Long narrow segments
Balloon	Focal stenosis; inflate until the waist resolves, hold 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting is not needed** for ureteral stones; it helps only renal stones.
- The access **sheath** helps with large burden, dusting and high power, lowering intrarenal pressure (IRP) and heat.

5 · Lithotripsy

- **Basketing**: keep the stone **5–10 mm from the tip**, **never basket blind**, and when in doubt **fragment first**.

Source	Key points
Electrohydraulic	High ureteral trauma; abandoned

Source	Key points
Pneumatic	~12 Hz; **rigid/semirigid only**; best below brim; retropulsion
Holmium laser (Ho:YAG)	Short pulse fragments; long or modulated pulse lowers retropulsion; dusting is low energy, high frequency
Thulium fibre laser (TFL)	Low peak power and long pulse give fine dust; minimal retropulsion; >40 Hz misses

P4 Thalium laser in urology — TFL physics, settings (0.6 J × 6 Hz; dust 0.2 J × 40 Hz), "paint, do not drill", RCTs vs holmium

Not in the paper answer:

- Impacted stone: target the centre first, so the sides fall away from the wall.

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	0.6 J × 6 Hz short pulse (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- **Settings, not only the laser, decide outcome**.
- **Thermal** damage begins above **43 °C**, and **heat tracks power, not laser type**.
- Cap at **≤10 W in ureter** and ≤25 W in kidney, with continuous irrigation, because stopping it causes thermal stricture.

6 · After URS

Most need neither long antibiotics nor a stent.

- Give a **single perioperative intravenous antibiotic dose**; extend only after urosepsis this episode.
- Stents cause symptoms in ~4/5; stentless uncomplicated URS is safe.
- Stent for perforation, stricture, tight ureter, infection or a second stage.

7 · SWL for ureteral calculi

- SWL is an option at all levels.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (for treatment and 48 h); distal obstruction; untreated UTI; uncontrolled hypertension	Skeletal deformity or obesity preventing focusing; adjacent aneurysm

- **Predictors**: size matters most, and URS is better ≥10 mm. Impaction (CT "ring" sign) is possibly the strongest.
- Cystine, brushite and calcium oxalate monohydrate resist SWL. Density >1000 HU and skin-to-stone distance >10 cm are proven for renal stones only.
- **No routine pre-stenting**. Hold aspirin, NSAIDs and anticoagulants for treatment and 48 h. Give antibiotics only with stents, struvite or recent UTI.
- Position the upper ureter supine, the mid-ureter **prone** and the distal ureter supine; **60–90 shocks/min** beats 120. α-Blockers raise SFR.

8 · Special situations

- **Pregnancy**: treat conservatively first; **SWL is contraindicated**; URS is safe with shielded fluoroscopy. Stents encrust, so change them **every 4 weeks**.
- **Diversion or reimplant**: use SWL (prone) or **antegrade flexible URS**. Transplant: both routes are safe.

9 · Complications

- **Perforation** (0–4%): stop, stent for ~4 weeks and drain any urinoma.
- **Avulsion** (0.06–0.5%) follows forceful extraction, mostly in the **proximal third**; place a **nephrostomy and repair later**.
- **Stricture** occurs in 3–6% and is silent, so image every patient.
- **Sepsis** (~5%) is linked to positive culture, pre-stenting, diabetes and long surgery.

> **KEY POINT**
> Nearly every serious URS complication is force without vision: blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

10 · Landmark trials

Trial	Design	Result	Take-home
Stentless URS	RCT, stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

98 Surgical Management of Upper Urinary Tract Calculi

1 · Choosing the approach

- The framework is the 2016 AUA/Endourological Society guideline, built on index patients and shared decision making.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction or stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe or ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure or transplant, solitary kidney, diversion, pregnancy

KEY POINT

Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude a modality, not surgery.

Asymptomatic renal stones

- The natural history is not benign: **~50%** have a stone event within **5 years**. **Size predicts intervention**; risk rises above 5 mm and above 10 mm.
- With **recurrent UTI and a non-obstructing stone**, remove it whatever its composition, and **culture the stone**.
- Treat residual fragments **> 4 mm** after ureteroscopy (URS).
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side.

Preoperative evaluation

- AUA: non-contrast CT (NCCT) **should** precede percutaneous nephrolithotomy (PCNL), and **may** guide the choice of shock wave lithotripsy (SWL) versus URS.
- SWL works best when density is **< 900 HU** and skin-to-stone distance (SSD) is **< 10 cm**.
- Everyone needs urine culture within 1 month.

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL has fewer complications; URS has higher stone-free rate (SFR)
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line, because fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- EAU: for >20 mm use PCNL, then URS or SWL; for **10–20 mm, SWL and endourology are in equipoise**; for <10 mm use SWL or URS, then PCNL.

Stone morphometry scores

P1 Stone complexity scores — Guy's, STONE, CROES, S-ReSC predict SFR, not complications; Triple D predicts SWL

2 · Shock wave lithotripsy

P2 ESWL: mechanism, indications, factors affecting outcome, complications and current status — peak **30–100 MPa**; sources; mechanisms (dynamic squeezing strongest); **60–90 shocks/min**, ramping, coupling; haematoma, steinstrasse (no routine stent); refuted effects; burst wave

Not in the paper answer:

- **Absolute contraindications** are pregnancy, uncorrected coagulopathy or antithrombotics, untreated UTI, severe untreated hypertension, aortic or renal artery aneurysm (aorta > **4 cm**), obstruction distal to the stone, and inability to target the stone.
- No prophylactic antibiotic is needed if urinalysis is negative and there are no risk factors (AUA, EAU).
- The electrohydraulic HM3 (Dornier, 1980) remains the efficacy benchmark MCQ.

3 · Ureteroscopy (retrograde intrarenal surgery, RIRS)

URS is the workhorse for small stones and antithrombotic patients.

- Pre-treat bacteriuria and give a single antibiotic dose.
- **Antithrombotics**: URS is the modality for those who must continue them; antiplatelets increase admissions.

Instruments and principles

- **Safety wire**: AUA and EAU recommend it, although a multicentre study found no outcome difference. Never force a wire past an impacted stone.
- The **ureteral access sheath (UAS)** lowers intrarenal pressure but gives **no clear fall in infection**. It injures the ureter in nearly half, less if **pre-stented**.
- **Anti-retropulsion** devices reduce stone migration.
- AUA allows **stent omission** only when all four apply: normal anatomy, normal renal function, no ureteral injury and no planned second stage.

Success

"Clinically insignificant fragments" cause reintervention in up to a fifth, so report zero-fragment rates.

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes; treat the urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (rate underestimated)

Complication	Rate	Risk factors / management
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter; immediate nephrostomy, delayed reconstruction

4 · Percutaneous nephrolithotomy

PCNL clears the most stone; safe access is the key skill.

- Landmarks MCQ: Fernström and Johansson performed the first percutaneous stone removal in 1976.
- For 1–2 cm stones PCNL gives higher SFR than SWL or URS. It is the treatment of choice for **staghorn** and most stones >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture, injury or reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative

- Stop anticoagulants with the prescriber (bridge if high risk); **low-dose aspirin is safe**.
- **NCCT is the gold standard**, showing calyceal anatomy, SSD, HU and the surrounding organs.
- For a poorly functioning kidney, get a renogram; below 10–20% function, nephrectomy may suit.
- Give single-dose, culture-directed prophylaxis. **Tranexamic acid** reduces transfusion without more thromboembolism.

Positioning

P1 P2 Prone vs Supine PCNL / patient positions in PCNL — prone: upper pole access, ocular/brachial plexus risk; supine: airway, gravity drainage, ECIRS

Access: the most important step

- The **ideal** target is a posterior calyx with a short, wide infundibulum.
- **Supracostal (above the 12th rib)** upper pole access reaches most of the system but raises chest complications. Lower pole access risks colonic injury.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle seen end-on	Mediolateral in the oblique view, cephalocaudal in the anteroposterior (AP) view
Depth	Rotate back past AP	Short advances, alternating views

- Confirm entry by urine return, not contrast.
- **Ultrasound access** avoids radiation and shows adjacent organs.
- In **endoscopic combined intrarenal surgery (ECIRS)**, a ureteroscope watches the puncture, giving fewer tracts and higher clearance.

Dilatation and tract size

- Dilate with Alken metal, Amplatz serial or **balloon** dilators (one pass, 20–30 atm), keeping a safety wire.
- Multiple tracts raise bleeding and reduce function, especially in chronic kidney disease.

P4 Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques — standard 24–34, mini 15–22, ultra-mini 11–14, micro <10 Fr; mini: equal SFR, less bleeding

Completion and exit

- Finish with **flexible nephroscopy or URS**.

ALGORITHM

1. Confirm the collecting system and ureter are clear.
2. Withdraw the sheath under vision, with the safety wire out last.
3. Compress the tract.
4. If bleeding persists, re-inflate the balloon or place a **Kaye tamponade catheter**, and alert the anaesthetist.
5. If it still persists, arrange **selective angioembolisation**.

- Drain with a nephrostomy (≤10 Fr means less pain) or a double-J stent, or go tubeless.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications**, with fewer secondary procedures.
- Complications include haemorrhage (delayed bleeding from arteriovenous fistula or pseudoaneurysm needs angioembolisation), visceral injury, hydrothorax or pneumothorax, and sepsis.

5 · Special anatomy

Anatomy	Key adaptation
Horseshoe (stones in ~1 in 5)	SWL: long SSD. PCNL: access is medial and caudal; the upper pole posterior calyx is usually subcostal; more colonic injury; UPJO is frequent
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper or interpolar (laser-incise or dilate the ostium); lower pole: PCNL. Fulguration controversial
Pelvic kidney	SWL or URS <2 cm; PCNL with ultrasound, with or without CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable
Transplant	Always remove stones. The denervated graft presents with fever, oliguria and haematuria, not colic. Ultrasound-guided PCNL
Spinal deformity	SWL poor; URS good; PCNL equal clearance but more infection

- **Laparoscopic or robotic** surgery is only for when SWL, URS and PCNL fail or are expected to.

6 · Landmark trials

Trial	Design	Result	Take-home
Lower Pole I	Randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

99 Prevention of Urinary Stone Disease

1 · Stone disease as a chronic condition

Recurrence is the disease, so prevention is lifelong.

- Up to **50%** recur within 5–10 years. The **ROKS** nomogram predicts a second stone.
- **Low urine volume** is the single most important risk; others are young age, male sex, obesity, family history and low citrate.

2 · Screening and metabolic evaluation

P1 P2 Metabolic Evaluation in Stone Disease — screening (history, blood, urine, stone analysis, imaging) for every first-time former; high risk: one or two 24-hour urines; fast/calcium-load testing discouraged

Not in the paper answer:

- A 24-hour urine is **not** part of screening.
- Lithogenic drugs include calcium, vitamins C and D, acetazolamide and steroids.

3 · General measures (all formers)

- **Fluid** is the cornerstone: aim for urine ≥ **2.5 L/day**.
- **Limit nondairy animal protein and sodium**, which raise urinary calcium and lower citrate.
- Keep **calcium intake normal**, because restriction increases stones.

> **KEY POINT**
> Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · Hypercalciuria: thiazides

- The AUA makes thiazides the **preferred drug** for hypercalciuria with recurrent calcium stones, **and** for recurrent calcium stones with a normal evaluation.
- **Hypokalaemia causes intracellular acidosis, which leads to hypocitraturia**. So add **potassium citrate**, and check potassium 1–2 weeks after starting or changing the dose.
- With amiloride, **never add potassium**. Avoid triamterene, as it forms stones.
- Side effects include gout, diabetes and sexual dysfunction.

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	Gastrointestinal (GI) upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea (magnesium); ataxia, numbness (pyridoxine)
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

5 · Hyperoxaluria

- Stone risk rises continuously with urinary oxalate. Avoid large portions of spinach, nuts, chocolate and tea.
- **Enteric** hyperoxaluria follows malabsorption or Roux-en-Y gastric bypass; urine oxalate is >40 mg/day. Unabsorbed fat binds calcium, which frees oxalate for colonic absorption.
- Treat with **calcium up to 1 g/day with meals**, very high fluid, **liquid** potassium citrate, magnesium and a low-oxalate diet.

P2 Primary Hyperoxaluria — oxalate often > **100 mg/day**; PH1 (~70%, AGXT, pyridoxine-responsive), PH2 (GRHPR), PH3 (HOGA1); lumasiran silences glycolate oxidase (ILLUMINATE-A)

Not in the paper answer:

- Lumasiran is approved for PH1.

6 · Hypocitraturia

- AUA offers **potassium citrate** 30–60 mEq/day for recurrent calcium stones with low citrate, **and** for those with no abnormality.
- Sodium citrate does not lower urinary calcium. Use caution in renal impairment.
- Over-alkalinising to **pH ≥7** creates **calcium phosphate** risk.
- **Alkali citrate, not total citrate, matters**.

7 · Uric acid stones: a pH disease

P2 Uric acid calculi: etiopathogenesis, medical management and allopurinol — pKa **5.5**; low pH the main cause; potassium citrate to **6.0–6.5**, avoid >7.0; allopurinol not first-line; hypersensitivity (fatal up to 27%)

- Give allopurinol only for hyperuricaemia with hyperuricosuria, or failed alkalinisation.

8 · Cystine stones

ALGORITHM

1. **Dilute**: urine ≥ **3 L/day**, including at night; sodium ≤2300 mg/day; moderate protein (not in children).
2. Aim for cystine < **250 mg/L**.
3. If not enough, **alkalinise** with potassium citrate to pH **7.0–7.5**.
4. If still forming stones, add a **thiol** (tiopronin).

- pH >7.5 risks calcium phosphate, and sodium alkali raises cystine excretion.
- Thiols form a soluble cystine–drug disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop it	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

9 · Infection (struvite) stones

- **Clear the stone completely**, because every fragment harbours bacteria.

- **Acetohydroxamic acid (AHA)** is a urease inhibitor. AUA: use it only for residual or recurrent struvite after surgical options are exhausted.
- AHA toxicity is ~30%, with **deep venous thrombosis** the most severe. It is contraindicated if creatinine is >**2.5 mg/dL** or in pregnancy.
- Most struvite formers are also hypercalciuric or hypocitraturic, so do a 24-hour urine.

10 · Monitoring

ALGORITHM

1. Start therapy.
2. Repeat the 24-hour urine within **6 months**.
3. Then check urine and blood (for side effects) at least yearly.
4. If a new stone forms, **re-analyse** it.
5. Image periodically.

11 · Landmark trials

Trial	Design	Result	Take-home
Borghi 1996	Randomised controlled trial (RCT), high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein and salt with normal calcium halved events vs low calcium	Never restrict calcium
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

100 Lower Urinary Tract Calculi

1 · Bladder calculi

- Bladder stones are ~5% of urinary stones. Types are **migrant**, **primary** (endemic) and **secondary**.

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; boys far more than girls	Men >60 with outlet obstruction
Cause	Cereal diet low in animal protein and phosphate	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless the cause is removed

- Foreign bodies act as a nidus: a migrated intrauterine device, sutures, balloon fragments or hair during clean intermittent catheterisation (CIC).

> **KEY POINT**
> Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

- Adults present with **terminal haematuria** and lower urinary tract symptoms.

2 · Management of bladder stones

The route, not the stone, dictates modality.

Option	Indication / key point
Chemodissolution	Adjunct only; for catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; works poorly in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice; risks urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction or closure); spares the urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

3 · Special situations

Treat the cause along with the stone.

- **BOO** is present in 45–79%. **A bladder stone secondary to BOO is an absolute indication for BPH surgery**, but **prove BOO** first.
- **Diversion or augmentation** stones arise from acidosis-driven hypercalciuria, enteric hyperoxaluria, urease producers and **mucus**.
- A **continent pouch: go percutaneous**, which protects continence. A conduit with stomal stenosis needs stone clearance and stomal revision.
- **Spinal cord injury**: risk peaks **3 months** after injury; **CIC beats indwelling catheters**. Stones ≤4 cm get transurethral laser; larger ones percutaneous or open.

- **Transplant**: **remove all exposed suture**. **Children**: use percutaneous cystolithotripsy for large stones.

4 · Urethral calculi

- Urethral stones are 0.3–1% of stones, in males. Most are **migratory**, and the commonest site is the posterior urethra.
- They cause **acute painful retention**; anterior stones give a palpable mass. ~98% are radio-opaque.

Site	Treatment
Posterior	**Push back** into the bladder, then laser; else treat in situ
Anterior	**No pushback**; in situ holmium laser; urethrotomy for large stones
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction with diverticulectomy or stricture repair, same sitting or staged

5 · Unusual sites

- **Preputial** stones follow phimosis; treat by circumcision and send the **excised skin to histopathology**.
- **Prostatic** stones are linked to BPH, **not cancer**.
- **Seminal vesicle** stones cause haematospermia; **TRUS is most accurate**.

54 Management of Paediatric Kidney Stone Disease

1 · Epidemiology

Children carry stone disease for life.

- Early onset means stones before **24 years**. About 50% recur within 3–5 years.

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%**, mostly idiopathic, on Randall's plaques	75–80%
Calcium phosphate (CaP) / uric acid	CaP commoner and uric acid rarer, because urine is more alkaline	≈ 5% each
Lifetime stakes	Chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); cumulative radiation	Shorter

- **High-risk** groups include bladder augmentation (≈ 1%/year) and malabsorption.

KEY POINT

Childhood is the bone-accrual window, so seek low BMD early.

2 · Diagnosis and imaging

ALGORITHM

1. If a stone is suspected, do grey-scale **ultrasound (US)** with colour Doppler.
2. If US is negative but suspicion is high, or there is hydronephrosis, then do **low-dose non-contrast computed tomography (NCCT)**.
3. If US is negative and suspicion is low, stop.

- US shows the **twinkle artefact** (sensitivity > 70%, specificity > 95%) but overestimates size by 1–2 mm. NCCT is ≈ 100% accurate.

3 · Metabolic evaluation: every child

An abnormality raises recurrence up to **fivefold**, so do a **full evaluation after the first stone**.

- **History**: ketogenic diet, vitamin D, furosemide, topiramate, prematurity.
- **Examination clues**: dysmorphism suggests Williams syndrome, rickets suggests Dent disease or HHRH, and tetany suggests FHHNC.
- Do **stone analysis** first: cystine, struvite, 2,8-dihydroxyadenine and xanthine are each diagnostic.
- With hypercalcaemia and hypercalciuria, check for **CYP24A1** loss.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m²/day
Citrate (lower limit)	Hypocitraturia is < 128 mg/g creatinine (male), < 300 (female)
Cystine	< 50–60 mg/1.73 m²/day
Uric acid	< 815 mg/1.73 m²/day; spot < 0.56 mg/dL glomerular filtration rate

KEY POINT

Never use adult limits: infants excrete more calcium, so the school-age 0.21 ratio overcalls them. Citrate's abnormality is too little.

4 · Urinary abnormalities

Hypercalciuria: commonest (30–50%)

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (calcium-sensing receptor gain)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH; if none, **idiopathic**

Hyperoxaluria

P2 Primary Hyperoxaluria — urinary glycolate/L-glycerate, then genetics; pyridoxine

Not in the paper answer:

- Only 10–30% of primary hyperoxaluria type 1 (PH1) respond to pyridoxine.
- Thiols and lumasiran are the only drugs approved for paediatric stone prevention.

Others

- **Hypocitraturia**: incomplete dRTA needs an acid-load test. **Cystinuria** may present in infancy with a staghorn.

Monogenic disease	Inheritance	Signature
Dent (CLCN5)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (CLDN16/19)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · Management principles

- Tamsulosin is off-label (≤ 6 weeks).
- The residual fragment threshold is **4 mm on US**: 2 mm on CT plus the ≈ 2 mm US overestimate.
- Give antibiotic prophylaxis for all URS and PCNL, but for SWL only with infection risk; limit it to ≤ 24 h.

Scenario (AUA/Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation with or without an α-blocker
Ureteric, failed or unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting
Renal ≤ **20 mm**	SWL or URS
Renal > 20 mm	PCNL or SWL (with stent or nephrostomy); authors prefer PCNL

Scenario (AUA/Endourological Society 2016)	Recommendation
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · Ureteroscopy

The small ureter is the main limit, so never force access.

- Clearance is ≈ 90%, but worse under 6 years and in the lower pole.

ALGORITHM

1. Confirm a sterile culture; place in lithotomy; do cystoscopy.
2. Do a retrograde pyelogram and place a safety wire.
3. Calibrate the orifice with an 8/10 Fr coaxial dilator.
4. **If it will not pass, then place a 4.6–4.8 Fr stent for ≥ 14 days and return later.**
5. If it passes, do URS.
6. Stent if the case was long, needed many passes, or caused trauma or oedema.

- **No balloon or active orifice dilatation**; rely on passive stent dilatation.

7 · Shockwave lithotripsy

- SWL equals URS for small stones.
- Favourable: < 10 mm, non-lower-pole, < 1000 HU. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- Use < 20 kV and ≤ 2000–3000 shocks (< 2000 in infants); **80/min is better than 120/min**.
- Absolute contraindications are bleeding diathesis, untreated UTI and pregnancy.

8 · Percutaneous nephrolithotomy

- PCNL is first-line for stones > 20 mm. A 24–30 Fr tract in a child equals ≈ 72 Fr in an adult, so **miniaturise**.

ALGORITHM

1. Prevent hypothermia; place a ureteric catheter.
2. Position prone (latex precautions in myelomeningocele).
3. Puncture a posterior calyx under US or fluoroscopy.
4. Dilate serially to an 18 Fr sheath (24 Fr for staghorn).
5. Fragment and check with a flexible scope.
6. Place a nephrostomy (7 Fr infant to 16 Fr adolescent).

9 · Secondary prevention

- **Fluid**: at least maintenance; > 2–2.5 L/day in adolescents; 1.5–2 L/m² in cystinuria or PH.
- **Sodium** < 2–3 mEq/kg/day. **Calcium is not restricted**, because it binds gut oxalate.
- Target urine pH > 6.5 for uric acid and > 7 for cystine; avoid > 6.5 in calcium formers.

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; for hypokalaemia add potassium citrate
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP risk if pH > 6.5
Tiopronin	Cystinuria failing fluid and alkali	Fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; can cause xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

PH1 with end-stage disease needs liver–kidney transplantation.

10 · Landmark trials

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate lowered by 65%	RNA interference therapy for PH1

125 Genital and Lower Urinary Tract Trauma

1 · Principles

- Key decision: **immediate reconstruction vs deliberate delay**. Testicular rupture is repaired at once; pelvic fracture urethral injury (PFUI) gets a suprapubic tube now and urethroplasty later.
- Imaging: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · Penile fracture

P2 Fracture penis — tunica thinnest ventrolaterally; snap, then detumescence, then eggplant deformity; deviates away from tear; urethral injury ~20% West vs ~3% Middle East; clinical diagnosis, ultrasound, then magnetic resonance imaging (MRI) Intraoperative flexible cystoscopy; delay ≤7 days harmless; complications ~20%

Not in the paper answer:

- Bilateral tears (~10%) carry more urethral injury, and absence of blood at the meatus does not exclude it.
- Collagenase ruptures mostly occur after the 14-day observation window; manage them conservatively unless the urethra is injured.

ALGORITHM

1. Explore promptly. Through a ventral penoscrotal (or small lateral) incision, evacuate haematoma and find the tear (artificial erection).
2. Irrigate; close the tunica with interrupted **2-0/3-0 absorbable** sutures.
3. If the urethra is injured, repair it.

- A breached Buck's fascia gives a butterfly haematoma, and a negative MRI reliably excludes a fracture. Avoid deep corporal ligation and excessive debridement.

3 · Penetrating, bite, amputation, zipper and strangulation

- **Gunshot**: explore immediately without waiting for imaging. Debride, give antibiotics and close the tunica.
- Urethral injury occurs in **15–50%** of penile gunshots, so check with RUG or intraoperative dye. Viable tissue is repaired primarily; high-velocity tissue loss needs suprapubic diversion and staged repair.
- **Dog bites**: irrigate, debride and close **primarily**, with amoxicillin–clavulanate and tetanus and rabies cover. **Human bites** present late and infected, so **leave them open** and give amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- It is usually self-mutilation, so **always get psychiatric consultation**. **Microsurgical replantation** is preferred; otherwise repair urethra and corpora macroscopically.
- Transport the part **"double bagged"**: saline gauze in a sterile bag, inside an outer bag of ice slush.
- Erections return in >50% with either technique, but microvascular repair gives less stricture and sensory loss. If the part is lost, formalise the stump; total loss needs phalloplasty.

ALGORITHM

1. Suprapubic cystostomy; debride.
2. Tunica with 2-0 absorbable; urethra over a catheter with 4-0.
3. Anastomose dorsal artery with **11-0**, dorsal vein with **9-0** and dorsal nerve with **10-0** nylon.
4. Provide skin cover.

- **Zipper** injury (children): under penile block, try mineral oil once; if that fails, cut the slider's **median bar** with bone cutters.
- **Strangulation**: in children think of hair tourniquet and **child abuse**. Escalate from lubrication to the **string technique**, then to cutting tools with saline cooling in theatre.

4 · Testicular injury

- About 80% of penetrating scrotal wounds also involve neighbouring structures, including the **femoral vessels**.
- Consider rupture in every blunt injury; haematoma size does not predict it. **Ultrasonography** sensitivity is only ~64–75%, so an **equivocal scan must not delay exploration**.
- A non-palpable testis after trauma suggests **dislocation**, commonest to the superficial inguinal pouch (50%). Trauma does not exclude torsion.

AAST testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM

1. Explore; remove necrotic and extruded tubules.
2. Close the tunica with small absorbable sutures.
3. If the defect is large, use a tunica vaginalis flap or graft (never synthetic); drain.

- Explore a **significant haematocele** (up to 80% hide rupture) and an **intratesticular haematoma** (delayed orchidectomy 15%).
- Always explore penetrating injury; ligate an injured vas and reconstruct later.
- Salvage is **>90%** if repaired within **72 h**; non-operative care leads to infection, atrophy and late orchidectomy.

5 · Genital skin loss

- Fournier gangrene debridement is the commonest cause. Treat it with antibiotics, resuscitation and **repeated debridement**, then negative pressure wound therapy; hyperbaric oxygen is unproven.

	Penis	Scrotum
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch, because meshed graft contracts	**Meshed** STSG after local flaps (exudate escapes, so better take)

	Penis	Scrotum
Key step	Excise residual subcoronal skin, which otherwise causes lymphoedema; never reuse avulsed skin	Suture the testes together to avoid a bifid scrotum; ≤50% loss closes primarily
Note	Graft is insensate, but sexual function is preserved	Thigh pouches not recommended

- Graft 5–7 days after the last debridement, once the wound is granulating, under a bolster or vacuum dressing.

6 · Bladder injury

Intraperitoneal tears need surgery; most extraperitoneal tears heal on a catheter.

P2 Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries — pelvic fracture in 83–95% of bladder injuries, bladder in 5–10% of fractures; cystography for gross haematuria and pelvic fracture; retrograde fill **350 mL** , drainage film (plain only); flame-shaped extraperitoneal; 22-Fr catheter for uncomplicated extraperitoneal; intraperitoneal needs immediate repair; suprapubic tube adds nothing

Not in the paper answer:

- Underfilling the bladder gives false-negative cystograms.
- Bone spicules in the wall prevent healing, so repair them; a ureteric orifice injury needs a stented reimplant.

AAST bladder grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Intraperitoneal leakage causes chemical peritonitis and, once reabsorbed, **acidosis and azotaemia**.

> **KEY POINT**
> Immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, laparotomy for other reasons.

7 · Posterior urethral injury (PFUI)

Drain now and reconstruct later.

P2 P3 Lower urinary tract trauma in pelvic fracture / early and late management of PFUD — bulbomembranous junction; triad blood at meatus, retention, palpable bladder; RUG technique; suprapubic tube standard, brief realignment optional (no difference); delayed excision and primary anastomosis at **2–3 months** (corporal split, then pubectomy, then rerouting); erectile dysfunction (ED) up to 82%, restenosis 5–15%, incontinence <4%

Not in the paper answer:

- PFUI complicates ~10% of male and up to 6% of female pelvic fractures, mostly **straddle (four-rami)** and unstable fractures.

- ED is predicted by symphyseal diastasis, lateral displacement and a long gap. Endoscopic "cut-to-the-light" fails for gaps ≥1 cm.

AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

ALGORITHM

1. If there is blood at the meatus, do RUG before any catheter.
2. Place a 16-Fr catheter in the fossa navicularis and inflate the balloon with 3 mL.
3. Apply penile traction, position oblique, inject 20 mL contrast and image during injection.

- In females, examine the vagina in every pelvic fracture. A suprapubic tube is safe with pelvic hardware if tunnelled high in the midline.

8 · Anterior urethral injury

- Most are **straddle injuries** crushing the bulbar urethra against the pubis. Contusion or incomplete tear needs a catheter; major injury a suprapubic tube.
- Later, the 1.5–2 cm obliterated bulbar segment is treated by **excision and primary anastomosis** (>95% success). Urethrotomy through an obliteration is futile.

9 · Landmark trials

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

62 Paediatric Genitourinary Trauma

1 · Renal injury: how the child differs

- The kidney is the commonest site of paediatric genitourinary (GU) trauma. Most injuries are **blunt** and grade 1–3; half have other injuries.
- Congenital anomalies such as ureteropelvic junction (UPJ) obstruction are commoner than in adults.

Feature	Child	Adult
Protection	Less perirenal fat, weaker wall, kidney below the ribs	Better cushioned
Blood pressure	Maintained until ≈ **25–30% volume loss** ; hypotension is late	Falls earlier
Haematuria	May be absent (up to 2/3 of grade ≥ 2 injuries)	Classic trigger
Imaging decision	**Mechanism-led**; keep radiation as low as reasonably achievable (ALARA)	Led by haematuria and hypotension

2 · Indications for imaging

Setting	Indication (American Urological Association [AUA])
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria with systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest, even without haematuria

- Triggers: fall > **10 feet (3 m)**, high-speed collision, flank blow; rib, spine, femur or pelvic fracture. All four markers together (haematuria, hypotension, associated injury, mechanism) give > **98% sensitivity**.
- Paediatric hypotension (1–10 years) is SBP < 70 + (2 × age) mm Hg.

> **KEY POINT**
> Trap: "no haematuria, so no imaging". In children both adult markers fail, so mechanism drives the decision.

3 · Imaging studies

- Renal ultrasound (US) detects grade 3–5 without radiation and suits screening, but **contrast computed tomography (CT)** is preferred. Add a **delayed (excretory) phase** if a leak is suspected.

4 · Grading

American Association for the Surgery of Trauma (AAST): grades 1–3 are **low** and 4–5 are **high**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion

Grade	1989 AAST	2011 revision
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

KEY POINT

Grades 1–3 differ only in depth (1 cm) and never involve the collecting system; entry makes grade 4. Shattered kidney is grade 5 (1989) but grade 4 (2011).

High-risk CT criteria (grade 4–5)

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation with no contrast in distal ureter	Urine leak needing intervention

With 0–1 factor, < 10% need intervention; with 2–3, > 50% do (grade 4a vs 4b).

5 · Non-operative management (NOM)

- NOM is the **default at every grade**, including grade 4–5 and stable penetrating injury, provided the child is stable.

ALGORITHM

1. Confirm stability; grade the injury.
2. Bed rest, with serial haemoglobin and abdominal examination.
3. When gross haematuria resolves and counts are stable, mobilise.

- Grade 1–2 needs only **23-hour observation**. Repeat imaging: AUA CT at 48 h (grade 4–5), EAU US at 2–3 days; but **symptoms, not imaging, predict complications**.
- Grade 3 (devitalised) to 5: ≈ **25%** need embolisation, ≈ **15%** urinoma intervention, ≈ 5% exploration.

6 · Complications during NOM

- **Urine leaks** occur only in grade 4–5, and up to **85%** resolve. A urinoma shows **flank pain, ileus and low-grade fever**.
- If no contrast reaches the distal ureter, do retrograde pyelography (RPG). Escalate from **stent and Foley catheter** (AUA first line, ≈ 90%; out at 4–6 weeks) to percutaneous drain, then nephrostomy, then open repair.

Bleeding	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Fistula or pseudoaneurysm ruptures as the haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild
Unstable	Urgent angioembolisation	

- Superselective embolisation succeeds in > 80%.

7 · Operative management

- Operate for **instability from renal bleeding**, pulsatile or expanding haematoma, or failed embolisation. A stable non-pulsatile haematoma found at laparotomy is **left alone**.
- A **one-shot pyelogram** (2 mL/kg, film at 10–15 min) confirms a **contralateral kidney**.

P2 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — vascular control, renorrhaphy, nephrectomy; revascularise only solitary/bilateral; Page kidney

Not in the paper answer:

- Early vascular control did not lower nephrectomy in the one randomised trial.

ALGORITHM

1. Confirm a contralateral kidney.
2. Open Gerota fascia; control the vessels; debride; close the collecting system, then parenchyma over a bolster.
3. Drain. If the injury is penetrating, inspect the ureter.

- **UPJ disruption** follows deceleration, often without haematuria; CT shows **medial extravasation, empty distal ureter and often no laceration**. Repair early if stable; otherwise nephrostomy and repair at 2–3 months.

8 · Follow-up

- Scarring: grade 1–2 none, grade 3 ≈ 50–60%, grade 4–5 ≈ 100%, so do **US at 2–4 months**.
- **Hypertension** is rare and renin-mediated, so use renin–angiotensin blockade. Check blood pressure yearly after grade 3–5.

9 · Ureteral trauma

- Most are penetrating. **Haematuria is absent in 25–50%** and ≈ 10% are missed at laparotomy, so diagnose by **delayed-phase CT** or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

- Found within 1 week: repair if stable; ligate and place a nephrostomy if unstable. Found after 5–7 days: stent first; if that fails, nephrostomy and reconstruction at 6–12 weeks.

P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — repair by level: ureteroureterostomy, Boari, reimplant ± psoas hitch; ileal ureter, autotransplant

Not in the paper answer:

- **Avoid transureteroureterostomy in children**, because a long life puts the other kidney at risk.

10 · Bladder injury

Feature	Child	Adult
Bladder position	**More intraperitoneal**, so isolated injury is commoner (≈ half have no fracture)	Pelvic organ
Pelvic fracture	Half as frequent, so it implies higher energy and more urethral injury	Commoner
Posterior urethral injury	Prostatic; **bladder neck injury commoner**	Bulbomembranous

- **Gross haematuria or retention** calls for **retrograde cystography** with retrograde urethrography (RUG); delayed-phase CT may miss a rupture.
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V bladder neck or trigone.
- Extraperitoneal tears get a catheter, but are repaired for bone fragments, hardware, rectal or vaginal tear, or bladder neck injury. Intraperitoneal tears are **always repaired**. Keep the catheter 1–2 weeks, then do a cystogram.

KEY POINT

Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor and many need secondary procedures.

11 · Urethral injury

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- Do **RUG before any catheter**; if a catheter will not pass, place a **suprapubic tube**.
- Realignment is a brief attempt only; EAU advises against immediate urethroplasty. **Bladder neck injury is repaired at once.**

ALGORITHM

1. Divert with a suprapubic tube and wait several months.
2. Assess with cystogram and RUG, with or without endoscopy.
3. Do transperineal excision and primary anastomosis; if there is tension, add pubectomy.
4. Catheter for 2–3 weeks; cystogram; remove the suprapubic tube.

- Restricture ≈ 10%, incontinence ≤ 20%, and erectile dysfunction up to half (gap > 2.5 cm).

12 · Genital injury

> **KEY POINT**
> Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Involve social services before discharge.

- Penis AAST: I skin; II Buck fascia; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or ≥ 2 cm; V total penectomy.
- **Circumcision** is the commonest cause; the **Mogen clamp** causes the worst (glans amputation).
- Testis AAST: I contusion; II subclinical tunica tear; III < 50% loss; IV ≥ 50% loss; V destruction. If US shows tunica discontinuity, haematocele or focal flow loss, **explore and close the tunica**.
- Hymenal, vaginal or perianal injury means **penetration: evaluate for abuse**; a **normal examination does not exclude abuse**.

★ Putting it together

A · Renal colic

ALGORITHM

1. Give a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless there is renal failure, bleeding or pregnancy. Confirm with non-contrast CT, or ultrasound in pregnancy and children.
2. With infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration or refractory pain, **decompress now** (stent or nephrostomy; equal). Never lithotripsy while infected.
3. **Otherwise**, if the stone is <10 mm, try passage with or without an α-blocker (distal 5–10 mm) for 4–6 weeks; if that fails, do ureteroscopy (URS) or shock wave lithotripsy (SWL).

B · Stone operation

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS clears more, SWL has fewer complications; hard proximal >15–20 mm: consider percutaneous nephrolithotomy (PCNL)
Renal <2 cm, not lower pole	SWL or URS	SWL works best <900 Hounsfield units and skin-to-stone <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line (Lower Pole I)
Burden >2 cm, staghorn	PCNL	Staged URS an option

KEY POINT

SWL absolute contraindications: pregnancy, coagulopathy or antithrombotics, untreated infection, uncontrolled hypertension, arterial aneurysm, and obstruction distal to the stone. No routine stent after SWL or uncomplicated URS.

C · Prevention

ALGORITHM

1. **All patients:** urine ≥2.5 L/day, normal calcium, low sodium and less animal protein.
2. **If high risk**, give therapy guided by 24-hour urines, repeated within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide and potassium citrate (hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate
Uric acid	Alkalinise to pH 6.5–7; allopurinol only for hyperuricosuria/gout or failed alkali
Cystine	≥3 L/day fluid, pH 7.0–7.5, then tiopronin
Struvite	Complete clearance; check metabolism

D · Renal trauma

ALGORITHM

1. **Image** with delayed-phase CT for gross haematuria, microhaematuria with shock, deceleration or penetrating injury.
2. If stable with grade I–IV (or selected V), manage **non-operatively** (>90% succeed). Embolise a contrast blush; stent a symptomatic leak.
3. If unstable or expanding, **explore**; a one-shot pyelogram (2 mL/kg, 10 min) confirms the other kidney.

E · Bladder and urethra

Injury	Management
Extraperitoneal bladder rupture	22 Fr catheter, cystogram at 2–3 weeks; repair if at laparotomy, bone spicule, or unhealed at 4 weeks
Intraperitoneal bladder rupture	Surgical repair
Pelvic fracture urethral injury	Suprapubic tube (or brief realignment) now, delayed urethroplasty later
Penile fracture	Prompt tunical repair; check the urethra
Testicular rupture	Immediate repair

Retrograde cystography must fill to ≥350 mL.

F · Ureteric defects

Defect	Repair
2–3 cm, upper/mid	Spatulated ureteroureterostomy
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplant, nephrectomy

G · Children

KEY POINT

Stones: ultrasound first; full metabolic evaluation after one stone. Renal ≤20 mm: SWL or URS; >20 mm: PCNL. **Trauma:** image by mechanism (fall >3 m, deceleration); a perirenal haematoma >2.2 cm predicts embolisation.

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 12

Bladder Cancer & Diversion

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 129	Tumours of the Urinary Bladder
Ch 130	Management Strategies for Non–Muscle-Invasive Bladder Cancer
Ch 131	Management of Muscle-Invasive and Metastatic Bladder Cancer
Ch 132	Surgical Management of Bladder Cancer: Transurethral, Open and Robotic
Ch 133	Use of Intestinal Segments in Urinary Diversion
Ch 134	Continent Urinary Diversion
Ch 135	Minimally Invasive Urinary Diversion
Ch 124	Bladder Surgery for Benign Disease
★	Putting it together

129 Tumours of the Urinary Bladder

1 · Epidemiology (headline)

- It is the **fourth most prevalent** cancer and the **costliest per patient over a lifetime**.
- The median age at diagnosis is **73**, and the **male:female ratio is about 3:1**.
- **Women present later and do worse stage for stage**, as haematuria is mistaken for infection.

2 · Risk factors

The exposure predicts the histology.

P2 CA bladder, etiology, newer investigations and treatment — risk-factor table (occupation, arsenic, Lynch/MSH2, obesity, diet)

Not in the paper answer:

- **Cyclophosphamide** is the only proven chemotherapeutic carcinogen, via **phosphoramide mustard** (acrolein causes haemorrhagic cystitis, not cancer). Risk is ~4.5× above 20 g, with latency peaking at 10–14 years.
- **Radiation**: external beam radiotherapy for cervical cancer raises risk 2–4×, and for prostate cancer 1.5–2×.
- **Exstrophy** carries a malignancy risk of 3.3–7.5%, and **>90% is adenocarcinoma**.
- **Smoking** causes **30–40%** of urothelial carcinoma (UC) and raises risk **2–3×**.
- **N-acetyltransferase 2 (NAT2) slow acetylators** carry the highest risk, because acetylation detoxifies aromatic amines. Risk falls to near that of never-smokers **>15 years** after quitting.
- **Schistosoma haematobium** causes chronic inflammation and fibrosis, which leads to **squamous cell carcinoma (SCC)**. Long-term catheters in spinal cord injury and stones also cause SCC.
- Germline variants (MSH2, BRCA1/2) occur in 13–24% of UC, so test selected high-grade patients.

3 · Pathology

Benign lesions (reactive mimics)

Lesion	Hallmark	Potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recur; resect
Urothelial papilloma	Delicate stalks; FGFR3 mutant, TP53/RB wild type	None; resect
Nephrogenic adenoma	Tubule-like glands after injury	None; re-resect if it recurs
Leukoplakia	Keratinising squamous metaplasia; floating white flakes	**Not premalignant**, so no follow-up
Cystitis cystica/glandularis	From proliferating von Brunn nests	Glandularis: low adenocarcinoma risk, so resect, treat cause and survey

Flat (precursor) lesions

Loss of polarity means dysplasia; loss of umbrella cells means carcinoma in situ (CIS).

Lesion	Key features	Significance
Hyperplasia	≥10 layers, no atypia	May precede low-grade papillary tumour (FGFR3)
Reactive atypia	Atypia proportionate to inflammation; polarity kept	Coexists with dysplasia/CIS, so investigate
Atypia of unknown significance	Atypia disproportionate to inflammation	Limited utility
Dysplasia	Loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** progress to CIS if there was prior cancer
CIS	Flat, high grade, noncohesive pleomorphic cells; umbrella cells lost	Precursor of invasive high-grade disease

UC grade (2004 WHO) – noninvasive spectrum

Non-muscle-invasive bladder cancer (NMIBC) makes up about **70%** of UC and includes Ta, CIS and T1.

	PUNLMP*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chromosome 9 loss; FGFR3, PI3K, RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20% at 5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

About 80% have mixed (mostly squamous) differentiation. **Altered growth patterns behave worse than their stage.**

Variant	Hallmark	Chemosensitive?	Management
Micropapillary (0.7–2.2%)	Tight clusters with retraction artefact; **any amount counts**; ERBB2 amplification	No	**Early radical cystectomy (RC) even for NMIBC**
Sarcomatoid (0.3–0.6%)	Large infiltrative mass	Poor	Immediate RC
Plasmacytoid (<1%; includes signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, but relapse is common	Neoadjuvant chemotherapy (NAC) whenever possible
Nested (M:F 6:1)	Mimics von Brunn nests and inverted papilloma	No	Often upstaged; stage-matched outcome as UC
Squamous (16–22%, up to 60%)	Basal subtype	NAC active in mixed tumours	Linked to recurrence and progression

Variant	Hallmark	Chemosensitive?	Management
Glandular (~10%)	Gland-like spaces	–	Outcome as conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC	2–5% in West; majority where schistosomiasis is endemic	**Secondary is commoner than primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; from the allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic syndromes	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude a primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; 5-fluorouracil–cisplatin by analogy	**En bloc dome with urachal ligament and umbilicus**

KEY POINT

Small cell is systemic disease: primary cystectomy gives poor disease-free survival that adjuvant chemotherapy does not rescue, whereas neoadjuvant chemotherapy markedly improves survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation, showing that inflammation drives it.

Urachal carcinoma (2016 WHO criteria): the epicentre is in the dome or anterior wall, there is no widespread cystitis glandularis, and a secondary source is excluded. TNM staging is unsuitable.

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum
IV	Metastatic

4 · Molecular biology

	Papillary pathway	Flat pathway
Origin and product	Hyperplasia gives low-grade papillary tumour	Dysplasia gives CIS, then high-grade invasive tumour
Genetics	**FGFR3** (up to 80% of low-grade Ta vs 10–20% of MIBC), chromosome 9 deletion, PIK3CA	**TP53** (up to 76% of MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS mutations are **mutually exclusive with FGFR3**, because they act in the same pathway.

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3 and FOXA1
Behaviour	Less aggressive; papillary	Aggressive; squamous; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; checkpoint inhibitors; EGFR

- The **consensus classification (2020) has six classes**. Luminal papillary has the best survival, **basal/squamous (35%) is the commonest**, and neuroendocrine-like (3%) is the worst. Luminal-infiltrated tumours respond poorly to cisplatin NAC.

5 · Clinical presentation and evaluation

About one in five visible haematurias hides a cancer.

- **Painless gross haematuria occurs in 85%**. Irritative symptoms suggest CIS.
- Evaluate every gross haematuria, even on anticoagulants. Check prostate-specific antigen, because ~10% of recurrent gross haematuria is prostate cancer.
- After a negative work-up, repeat it at 3–5 years only if haematuria persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	Computed tomography urogram (CTU)	CTU (AUA 2012); ultrasound may replace it
Cytology/markers	Reasonable	Avoid routinely

The dipstick has sensitivity 91–100% but specificity only ~65%, so **confirm with microscopy**.

Cystoscopy and enhanced imaging

Cystoscopy with biopsy is the gold standard; flexible equals rigid. White light misses small tumours and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate (HAL) makes porphyrins build up in tumour, which fluoresce red under blue light	No dye; 415 and 540 nm light is absorbed by haemoglobin, so vessels stand out
Benefit	Finds more Ta and CIS; CIS is seen only on BLC in ~¼	Better CIS detection; lower 12-month recurrence in low risk
Status	Adjunct; only HAL is FDA approved	Adjunct

Urine cytology and biomarkers

P1 P3 Urine cytology; Asymptomatic haematuria and urinary biomarkers — good for high grade only; Paris System (HGUC N:C ≥0.7); NMP22, BTA, FISH, ImmunoCyt; none replaces cystoscopy

ALGORITHM

1. Cytology shows high-grade cells but cystoscopy is negative.
2. Take random bladder biopsies and prostatic urethral biopsies.
3. If these are negative, take selective upper-tract washings for cytology, with or without ureteroscopy.

6 · Staging and risk groups

Stage	TNM, American Joint Committee on Cancer (AJCC) 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)
I	T1 – lamina propria (subepithelial connective tissue)
II	T2a inner / T2b outer half of muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **The 2017 change**: regional nodes without distant metastasis are now stage III (previously IV). MCQ Common iliac nodes are N3, not M1a.
- **T1a/T1b substaging** is recommended, because ~50% of T1 are upstaged to ≥T2 at cystectomy.
- **Prostatic urethra**: mucosal CIS is not adverse in itself but escapes intravesical therapy. Only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta that is multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

1 · Definition and natural history

- **Non-muscle-invasive bladder cancer (NMIBC)** is urothelial carcinoma that has not invaded the detrusor: Ta, T1 and carcinoma in situ (CIS). It makes up **70–80%** of new bladder cancers.
- Recurrence occurs in **40–60%**; progression ranges from ~6% in low-grade (LG) Ta to 17% in high-grade (HG) T1.
- **Grade, not stage, is the strongest predictor of progression**, and CIS comes second. Primary CIS progresses in **>50%**.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in **one-third to one-half**.

> **KEY POINT**
> NMIBC is two diseases. LG disease is a **recurrence** problem; HG disease is a **progression** problem. Chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) treats progression, and cystectomy is for progression that cannot be controlled.

2 · Risk stratification

*Risk is **dynamic**, so reassess it at every recurrence.*

American Urological Association/Society of Urologic Oncology (AUA/SUO) 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single postoperative chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction of chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG and maintenance to **3 yr**. If BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene or gemcitabine–docetaxel

EORTC and CUETO scoring models

Factor	EORTC	CUETO
Number, prior recurrence rate, T stage, concurrent CIS, grade (WHO 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC **overestimates** risk, because its patients had little maintenance BCG. CUETO, derived in BCG-treated patients, fits better. MCQ Both use **WHO 1973** grading.

3 · Endoscopic management

TURBT: principles

- Transurethral resection of bladder tumour (TURBT) quality depends on **complete resection** and **detrusor in the specimen**; missing muscle means higher mortality.
- Image the upper tracts at diagnosis. Upper tract tumours occur in **<5%**, more often with trigonal tumour, CIS and high risk.
- **Diverticular tumour**: there is no detrusor, so invasion beyond the lamina propria is **T3a**. For HG, strongly consider partial cystectomy or RC.
- **En bloc resection** gives fewer perforations, more detrusor and better staging, but the **same recurrence**.

Repeat (second-look) TURBT

P4 Re-look (second-look) TUR bladder tumour — indications (all T1, incomplete, no muscle; consider HG Ta) Within 6 weeks; residual ≥40%, HG T1 upstaged 25–30% (~50% without muscle); residual T1 means 80% progression

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Random biopsies are **not** needed in low risk; consider them with positive cytology or high risk. Biopsy the prostatic urethra if there is CIS or a bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences can be fulgurated in the office, but this needs a **prior LG diagnosis and negative cytology**. **Active surveillance** also suits selected patients.

Enhanced cystoscopy

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins fluoresce red under blue light; needs intravesical instillation	415 and 540 nm light is absorbed by haemoglobin; no agent
Gain	~**one-third more CIS**; smaller papillary tumours	More papillary tumours and CIS
Weakness	Cost; no 3-yr recurrence benefit in a pragmatic trial	False positives ~60%

4 · Single post-operative instillation

P3 Immediate post TURBT intravesical Therapy for TCC — anti-implantation; within 6 h (useless at 24 h); MMC, epirubicin, gemcitabine 2 g (SWOG S0337); best in solitary LG; withhold if perforation; never BCG

- Give mitomycin C (MMC) within **6 h** (level 1 evidence). After perforation it risks chemical cystitis, contracture and rare deaths.

5 · Intravesical chemotherapy

P4 Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy — MMC optimisation (Au 2001); Gem/Doce regimen; device-assisted; BCG mechanism, schedule, toxicity

Not in the paper answer:

- Chemotherapy reduces **recurrence**, with **no proven effect on progression**. In intermediate risk, MMC or gemcitabine equals BCG. Give a 6-week induction, then monthly for 1 yr.

- A quinolone for 24–48 h after each BCG dose reduces cystitis without loss of efficacy.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	FDA approved for BCG-refractory CIS; complete response (CR) ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- **Gemcitabine–docetaxel (Gem/Doce)** is increasingly used in BCG-naive HG disease.

6 · Bacille Calmette-Guérin (BCG)

BCG prevents progression, but only with maintenance.

Mechanism

BCG binds fibronectin, releasing cytokines that recruit immune cells and produce a **T-helper 1 (Th1)** cytotoxic response. No strain is superior.

Administration

ALGORITHM

1. Wait **2–4 weeks** after TURBT (re-epithelialisation).
2. Check urinalysis before each dose to exclude infection and gross bleeding.
3. Instil 1 vial in 50 mL saline by catheter and retain it for 1–2 h, restricting fluids beforehand.
4. If catheterisation is traumatic, delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	Urinary tract infection; liver disease (isoniazid cannot then be given for sepsis); previous tuberculosis; anti-TNF drugs; poor performance status; advanced age

- Most absolute contraindications mean a **breached urothelium, allowing intravasation**. *Transplant and steroid patients may receive it, with weaker response.

Efficacy

- BCG is the **only intravesical agent that reduces progression**, and only with **maintenance**.
- In CIS, the initial CR is up to **84%**, and about half are durable.
- A second induction for persistent CIS gives a further 30–50% CR, but beyond 2 courses progression risk rises to 20–50%. **The response at 6 months predicts prognosis.**

Schedule

ALGORITHM

1. **SWOG (Southwest Oncology Group; Lamm 2000)**: give induction weekly ×6.
2. Then give 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**.
3. If high risk, give full dose for 3 yr. If intermediate risk, give 1 yr (EORTC 30962).

Toxicity management

Grade	Features	Management
1	Cystitis, mild haematuria, fever <38.5°C, lasting <48 h	Symptomatic; consider culture
2	Severe symptoms or >48 h	Culture, chest X-ray, liver tests; isoniazid, rifampicin and pyridoxine; resume at ½–⅓ dose
3	Haemodynamic change, persistent fever; allergic (arthralgia, rash); organ involvement (liver, lung, kidney)	**Stop BCG**; isoniazid, rifampicin and ethambutol; prednisone only **with** antimycobacterial cover; infectious disease input

- MCQ BCG is resistant to **pyrazinamide**. Asymptomatic granulomatous prostatitis needs no treatment.

BCG shortage

ALGORITHM

1. Prioritise high-risk patients.
2. Reduce the dose, especially in maintenance, and split vials.
3. Use chemotherapy: single agent for intermediate, Gem/Doce for high risk. Early RC for the highest risk.

7 · Recurrence after BCG

P2 BCG failure — definitions table: adequate BCG (≥5/6 induction + ≥2/3 maintenance; T1 induction alone), unresponsive (HG within 6 months Ta/T1, 12 months CIS), resistant (reinduce, CR 30–50%), intolerant, delayed relapse

Not in the paper answer:

- After failure, check the prostatic urethra and upper tracts, because recurrence there develops in about half on long follow-up.

KEY POINT

Trap: HG T1 at 3 months after induction alone counts as unresponsive. CIS or HG Ta at 3 months is "resistant", so give more BCG.

Bladder-sparing options (unfit for or declining RC)

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-PD-1; 200 mg intravenously every 3 weeks up to 2 yr	41%, then 19%	63%; grade 3–4 toxicity 13% (immune)
Nadofaragene firadenovec*	Adenoviral interferon-α2b gene; intravesical 3-monthly	53%, then 24%	71%; grade 3–4 4%

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
N-803 + BCG	Interleukin-15 superagonist	71% (any time), then 45%	~85%
Gem/Doce	Weekly ×6, monthly ×2 yr	—, then 60% (43% at 2 yr)	84%; no level 1 evidence

*FDA-approved. CR roughly halves by 12 months, and none matches cystectomy.

8 · Early radical cystectomy in high-risk NMIBC

- T1 failing BCG at 3 months progresses in **82%**, and CIS in ~50%.
- Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%** (64% if understaged). Even a 12-week delay worsens survival.
- **Patients who progress to muscle invasion under surveillance do worse than those with de novo muscle-invasive disease.**

ALGORITHM

1. **Strongly consider RC** if: BCG-unresponsive HG; persistent HG T1 on re-resection; T1 with CIS, LVI or variant histology; deep lamina propria invasion; diffuse CIS.
2. Also if: diverticular HG tumour; distal ureteric or prostatic ductal involvement; incompletely resectable tumour; patient preference.

9 · Surveillance

Cystoscopy with cytology is the gold standard. **Tumour at the 3-month cystoscopy** and number at first resection best predict recurrence.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	At 3 months, then annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly for 1–2 yr, then 6–12-monthly for 2 yr, then annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr**, then 6-monthly for 2 yr, then annually **for life**	Same schedule	Annually for 2 yr, then lengthen

- Restart the clock at each recurrence, using the risk of the **latest resected** tumour.
- Positive cytology with negative cystoscopy needs biopsies and upper-tract work-up.
- Urinary markers (UroVysion FISH is the most specific) do not replace cytology.
- **Extravesical recurrence**: upper tract UC occurs in <1% at low risk and ~10% at high risk. The prostatic urethra is involved in 20–40% of high-risk patients at 10 yr; HG ductal disease needs cystoprostatectomy.

10 · Prevention

- **Advise stopping smoking** at every visit, although its effect on recurrence is uncertain.

11 · Landmark trials

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline, suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine is the standard single instillation
SWOG maintenance (Lamm 2000)	Induction with or without maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance is essential
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

1 · Natural history and histology

MIBC kills through micrometastases, so cure needs systemic treatment too.

- **20–30%** of new bladder cancers are muscle-invasive (MIBC), and **~12%** are node-positive or metastatic.
- **Secondary** MIBC (**~20%**), which has progressed from non-muscle-invasive disease, does worse than **primary** MIBC, with more understaging and less response to neoadjuvant chemotherapy (NAC).
- Most deaths after local therapy occur within **2 years**, from micrometastases present at diagnosis.
- Urothelial carcinoma (UC) is 90%. **Radical cystectomy (RC) is standard whatever the histology**; subtype changes only the use of NAC and radiation.

Variant	Key points	Treatment
Squamous (5%)	Schistosomiasis, chronic cystitis	Poor chemo- and radiosensitivity, so up-front RC with lymphadenectomy
Adenocarcinoma (2%)	Commonest in exstrophy; urachal; exclude colorectal or breast metastasis	RC (urachal: partial cystectomy)
Small cell	High stage; ectopic ACTH, hypercalcaemia	**Chemosensitive**, so NAC with etoposide and cisplatin, then RC
Micropapillary	Frequent lymphovascular invasion (LVI)	If predominant, up-front RC
Mixed UC + variant	Quantify the variant	UC-predominant: NAC; variant-predominant: RC

2 · Clinical staging

Stage by resection depth, examination under anaesthesia (EUA), computed tomography (CT) with excretory phase, then chest imaging. In cT2, pathology upstages **40–54%**; LVI, variant histology, female sex and secondary MIBC predict upstaging.

TURBT

- Resect all visible tumour if safe, with a separate deep specimen; this is needed for bladder preservation, and pT0 at RC improves survival.
- If high-grade T1 or no muscle, repeat resection within **6 weeks**.
- **Prostatic urethral biopsy** (trigone or bladder-neck tumour, carcinoma in situ (CIS), multifocal disease) is a full-thickness loop from mid-prostate to verumontanum at **5 and 7 o'clock**.
- **Frozen urethral margin at cystoprostatectomy** is mandatory if the prostatic biopsy was positive, and advisable whenever a neobladder is planned; a neobladder is made only if it is negative.

Bimanual EUA

Examine before and after resection with the bladder empty; EUA more often understages than overstages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no three-dimensional mass
T3a	Mobile three-dimensional mass
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before resection**, as post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**; hydronephrosis suggests extravesical disease.
- Magnetic resonance imaging stages T best. The **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the muscularis line: intact in 1–3, interrupted in 4, into fat in 5. A score **≥3** is suspicious.
- FDG PET/CT is reserved for equivocal imaging.
- Chest CT is needed for all; bone or brain imaging only with symptoms or raised alkaline phosphatase. Renal function decides NAC and orthotopic diversion.

3 · Pathological staging (AJCC 8th) and prognostic markers

Category (American Joint Committee on Cancer)	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac is **N3, not M1**; T4b N0 is stage IVA. pT4a needs **stromal** invasion, since urethral or ductal CIS does not upstage.
- **Subtypes**: luminal survives best, **basal gains most from NAC**, and p53-like is chemoresistant.
- **Circulating tumour DNA**: over a third stay positive after RC and do badly; in IMvigor010, adjuvant atezolizumab helped only them.

4 · Radical cystectomy: principles and extent

- **RC with bilateral pelvic lymph node dissection (PLND) is standard for resectable MIBC**, ideally after cisplatin NAC. Delay > **12 weeks** worsens survival.
- **Men**: bladder, perivesical tissue, prostate and seminal vesicles are removed. **Women** (anterior exenteration): bladder, urethra, uterus, cervix, ovaries, tubes and anterior vaginal wall.

- **Male organ-sparing** needs no bladder-neck disease and negative prostatic biopsies; incidental prostate cancer is found in ¼–½.
- **Female organ-sparing** needs no base or trigone tumour. Keeping the uterus and anterior vagina supports a neobladder; keep ovaries premenopause unless there is breast or ovarian cancer risk.

> **KEY POINT**
> Spare an organ only where it is oncologically silent. The gatekeepers are, in men, the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall.

5 · Pelvic lymph node dissection

Nodal status is the strongest predictor of survival.

P4 Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — templates; German trial and S1011 negative, so standard PLND; extranodal extension

Not in the paper answer:

- About **25%** are node-positive at RC. Recurrence is **70–80%** if node-positive vs ~30% if node-negative.
- Standard PLND reaches the common iliac bifurcation; extended reaches the aortic bifurcation. About 10 nodes is the usual cut-off for pelvic recurrence risk.

6 · Intraoperative decision making

- If **clinically node-positive**, give cisplatin chemotherapy first, then **consolidative RC** in responders.
- **Unexpected nodes**: proceed only if completely resectable within the true pelvis. **Abort RC** if nodes are unresectable, the bladder is fixed to the sidewall or the rectosigmoid is invaded.
- **Ureteric margin** is involved in 6–8% and raises upper-tract recurrence (4–10%, mostly at 2–4 years). Frozen section is not mandated, but send one for gross tumour.
- **Male urethral recurrence** is 1–8% and highest with **prostatic stromal invasion** (~30%), so consider urethrectomy then. A positive pre-op prostatic biopsy does **not** bar a neobladder; the **frozen urethral margin decides**.
- **Female urethra** is involved in 2–12%. Urethrectomy is mandatory only for T4 disease involving urethra or vagina; send a proximal urethral frozen section in every neobladder candidate.

7 · Outcomes after RC

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** ultimately die of bladder cancer, mostly from metastases within 2 years.
- **Pelvic recurrence**: low risk (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**.

8 · Neoadjuvant therapy

Cisplatin NAC gives a level 1 survival gain.

P2 Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types — cisplatin NAC (ddMVAC 3–6/GC 4), rationale, pT0 ~90% RFS; adjuvant chemo (ABC 2022 +6%), nivolumab, Cairo radiotherapy

Not in the paper answer:

- About half of patients are cisplatin-ineligible, and they go straight to local therapy.
- Neoadjuvant immunotherapy (IO) phase II trials: **ABACUS** (atezolizumab) complete response 31%; **PURE-01** (pembrolizumab) pT0 42%.

Criterion	Cisplatin-ineligible (any one; ~50%)	Platinum-ineligible
Renal	Glomerular filtration rate **<60 mL/min**	Creatinine clearance <30 mL/min
ECOG performance status	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / ≥2	— / grade ≥3
Heart failure	NYHA III–IV	NYHA III–IV; multiple high-grade comorbidities

- The NCCN prefers dose-dense methotrexate, vinblastine, doxorubicin and cisplatin (ddMVAC).
- **Neoadjuvant IO** blocks the PD-1/PD-L1 checkpoint and was not yet recommended in the source. †NIAGARA has since shown that perioperative durvalumab with gemcitabine–cisplatin (GC) improves event-free and overall survival (OS).
- Neoadjuvant radiation downstages but gives no survival gain.

9 · Adjuvant therapy

- The target is **pT3–4 or N+**. Post-RC morbidity prevents delivery in ~¼, so neoadjuvant treatment is preferred.
- **Chemotherapy** is not standard, but discuss ddMVAC or GC in **cisplatin-naive** pT3–4/N+ patients.
- **Nivolumab for 12 months** is given for ≥pT3/N+ after NAC, or if cisplatin-ineligible or declining. IMvigor010 (atezolizumab) was negative.
- **Radiation** may be considered for pT3/N+, where local failure is 20–40%.

10 · Bladder preservation

Selected patients keep their bladder with similar survival.

Trimodality therapy (TMT)

P4 Discuss bladder preservation surgery for TCC of urinary bladder — selection (cT2, no hydronephrosis/CIS; 6–19%), sensitisers, schedules, outcomes, salvage RC 10–30%, follow-up; partial cystectomy

Not in the paper answer:

- **Dose**: a continuous course to **60–66 Gy** with restaging at 3–4 months is commonest; **55 Gy in 20 fractions** is non-inferior.
- Concurrent chemotherapy beats radiation alone in two randomised trials.

> **KEY POINT**
> TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopy decide success. Complete resection cuts the need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months)
Radiation alone (64–66 Gy/32 or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging; ~25% have occult nodes
Partial cystectomy with PLND	<3 cm solitary, no CIS, margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Late recurrence, so lifelong follow-up; convert to RC if the frozen margin is positive

11 · Metastatic urothelial carcinoma

First-line choice hinges on platinum eligibility.

- Bajorin adverse factors are Karnofsky status <80% and visceral metastasis.
- **Cisplatin-eligible**: GC or ddMVAC. **GC equals MVAC** and is less toxic (von der Maase), and dose-dense beat standard MVAC (EORTC 30924).
- **Cisplatin-ineligible, platinum-eligible**: gemcitabine–carboplatin (EORTC 30986) or enfortumab vedotin (EV) with pembrolizumab. †EV-302 later showed EV with pembrolizumab roughly doubled OS vs platinum, making it the preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052) or atezolizumab if PD-L1-positive. Trap: **first-line IO alone is only for platinum-ineligible patients.**
- **Maintenance avelumab** is given if there is no progression after platinum (JAVELIN Bladder 100). IO combined with chemotherapy is not recommended.
- **Second line** is **pembrolizumab** (KEYNOTE-045, regardless of PD-L1); atezolizumab failed (IMvigor211). Platinum can be rechallenged if the last dose was ≥12 months earlier.
- **Antibody–drug conjugates**: **EV** (Nectin-4; neuropathy, rash) beat chemotherapy after platinum and IO (EV-301). **Sacituzumab govitecan** (Trop-2; neutropenia) is third line.
- **Erdafitinib**, an oral FGFR inhibitor, treats FGFR3/FGFR2 alterations after platinum, with 40% response; it causes **hyperphosphataemia**.
- **IO toxicity** includes rash, colitis, hepatitis, pneumonitis, myocarditis and endocrinopathies, which are often **irreversible**.

ALGORITHM

1. If platinum-eligible, give GC or ddMVAC (or gemcitabine–carboplatin, or EV with pembrolizumab).
2. If there is response or stable disease, give avelumab maintenance.
3. On progression, give pembrolizumab, or erdafitinib if FGFR-altered.
4. Then give EV, then sacituzumab govitecan.
5. If platinum-ineligible, give pembrolizumab, then EV.

12 · Follow-up

- **After RC**: CT with excretory phase and chest CT 6-monthly for 2–3 years, then yearly to 5 (EAU, AUA). Check renal function at each visit and vitamin B12 yearly from year 2.
- Upper-tract recurrence (4–10%) is the commonest **late** one; pelvic and distant recurrences come mostly within 2 years.
- **After bladder preservation**: cystoscopy 3–4-monthly in years 1–2, 6-monthly in years 3–4, then yearly to **10 years**, with CT urogram and chest CT 3–6-monthly for 2 years, then yearly to 5.

13 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3, then RC, vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 is a surrogate; OS P = .06
BA06 30894 (EORTC/MRC)	Cisplatin, methotrexate, vinblastine (CMV) ×3, then RC or radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS hazard ratio (HR) 0.86; +5% at 5 years	Combination platinum only
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year progression-free survival 66% vs 56%	ddMVAC better but more toxic
SWOG S1011	Standard vs extended PLND	No difference in disease-free survival (DFS) or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	DFS 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation with or without mitomycin C/5-fluorouracil	Locoregional DFS 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (not significant)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine–carboplatin vs MCAVI, cisplatin-ineligible	Equal OS, less toxicity	Gem–carbo standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard

Trial	Design	Result	Take-home
EV-301	EV vs chemotherapy after platinum and IO	OS 12.9 vs 9.0 months	ADC third line

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

1 · Framing

- Transurethral resection (TUR) stages every patient and cures low-grade non-muscle-invasive bladder cancer (NMIBC); muscle-invasive disease (MIBC) needs radical cystectomy (RC) and pelvic lymph node dissection (PLND), after cisplatin neoadjuvant chemotherapy (NAC) if eligible.

2 · Transurethral resection of bladder tumour (TURBT)

Complete resection with muscle in the specimen makes staging correct.

Pre-operative evaluation

- **Continue aspirin**; hold other anticoagulants and P2Y12 inhibitors by half-life. **Treat active infection before resection**.
- Image the upper tracts by CT urogram, MRI, or ultrasound with retrograde study. Iodinated contrast needs a glomerular filtration rate (GFR) **≥60** mL/min; gadolinium needs GFR **>30**.

Set-up and inspection

ALGORITHM

1. Antibiotic within 1 h (avoid fluoroquinolones).
2. Inspect with 30° and 70° lenses (120° if incomplete).
3. Map tumours; add blue-light or narrow-band imaging if available (more tumours and carcinoma in situ (CIS) found, less recurrence).

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Remove all visible tumour with **detrusor in the specimen** and a **~2 cm** margin of normal tissue where possible.
- Then examine under anaesthesia (EUA) with the bladder empty: a **mass palpable after resection is cT3b**.
- **En bloc resection** keeps orientation and detrusor, and improves T1 staging.
- Maximal resection is therapeutic in NMIBC, but before RC for MIBC it gives no survival gain; residual tumour is only prognostic.

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk**: nerve stimulation makes the leg adduct and the loop perforates	Low distension; bipolar; muscle relaxant (general anaesthesia)

Site	Hazard	Countermeasure
Ureteric orifice	Coagulation causes stricture and hydronephrosis	**Pure cutting current**; routine stent not needed; short-term stent after large resection or heavy cautery is reasonable
Dome	Intraperitoneal perforation; heat injury to bowel	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site / frequency	Away from dome; commoner	Dome, posterior wall; rare, large ones morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at same sitting	
Management	**Foley drainage and observation**	**Laparotomy**, inspect bowel, repair; catheter and drain

> **KEY POINT**
> Perforation can follow adequate resection of advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak outside the bladder.

Post-operative intravesical agents

P3 Immediate post TURBT intravesical Therapy for TCC — rationale, agents, contraindications

Not in the paper answer:

- Single mitomycin C or epirubicin **within 24 h** gives a ~**13%** absolute fall in recurrence (meta-analysis); gemcitabine (SWOG S0337) cut recurrence by 12% at 4 years. They reduce recurrence, not progression.
- **BCG (bacille Calmette–Guérin) is never given immediately**: wait **2–4 weeks** with no gross haematuria, because of the risk of systemic absorption.

3 · Radical cystectomy: preparation

- **Mark a stoma for every patient** (sitting and standing) and consent for a conduit.
- **Avoid routine mechanical bowel prep** and oral antibiotics; omitting them speeds recovery and cuts wound complications.
- **Alvimopan** (peripheral μ-opioid antagonist), given 30–90 min pre-op and then twice daily, lessens ileus and shortens stay.
- Give an IV antibiotic within 1 h covering skin and bowel flora, e.g. **cefoxitin**.
- Start **venous thromboembolism (VTE) prophylaxis** (compression and drug) **before induction** and extend it to **4 weeks**: 28 rather than 8 days gives fewer major VTE without extra bleeding. Enoxaparin is standard; apixaban is equal, with better compliance.

4 · Open radical cystectomy: technique

ALGORITHM

1. Lower midline; Retzius space; divide urachus and vas or round ligament.
2. Mobilise caecum and sigmoid; make the **retrosigmoid window** (left ureter).
3. Free and divide ureters with adventitia.
4. Retractor (check vena cava); PLND; staple, clip or seal vesical pedicles.

- **Distal ureteric frozen section** is controversial: a positive margin predicts upper-tract recurrence, but clearing it has no proven benefit.

Male

ALGORITHM

1. Incise peritoneum over seminal vesicles; separate rectum; incise Denonvilliers fascia.
2. Control posterior pedicles; open endopelvic fascia.
3. Ligate dorsal venous complex (DVC); divide urethra.

- **Rectal injury is repaired primarily**, with or without faecal diversion.
- **Urethral frozen section** before a neobladder has a very high negative predictive value. **A positive result rules out orthotopic neobladder**, so warn the patient beforehand.
- **Nerve sparing** leaves few men potent at 1 year, and local recurrence occurs even in pT2. Avoid it in advanced disease or pre-existing erectile dysfunction.
- **Prostate sparing** improves potency and continence, but occult prostate cancer is common. It needs normal DRE and prostate-specific antigen (PSA), a TURP and a negative frozen section.

Female

- The standard is **anterior exenteration**, essential if EUA is positive or cT4a. Discuss **ovarian sparing** if disease is not locally advanced.

ALGORITHM

1. Ligate ovarian vessels; incise posterior fornix; control pedicles; excise urethra.
2. Watertight **clamshell vaginal closure** (keeps girth, loses length).

- **Vagina and urethra sparing for neobladder** needs ≤cT2, an uninvolved bladder neck and a negative urethral frozen section; preserve the striated sphincter.

5 · Pelvic lymphadenectomy

Template	Boundaries	Evidence
Standard	Lateral genitofemoral nerve; medial internal iliac artery; distal Cooper ligament; proximal ureter crossing common iliac	Reference
Extended	Standard plus common iliac and presacral	Better staging; possible survival gain
Superextended	Up to inferior mesenteric artery (preaortic)	No benefit over extended

- **≥10 nodes** removed goes with better survival (NMIBC at RC: >20).
- A randomised trial found **extended not superior to limited** for recurrence-free or cancer-specific survival, but its staging value is certain.

6 · Partial cystectomy

- Indications are a **solitary small MIBC without CIS** and **urachal adenocarcinoma** (en bloc umbilicus, urachus and dome with frozen margins).
- Oncology is as for RC: PLND, NAC if eligible, and consent for RC. Muscle-invasive recurrence needs salvage RC.

ALGORITHM

1. PLND; open the bladder away from tumour.
2. Full-thickness excision with fat and a 1–2 cm frozen-section margin; reimplant ureter if needed.
3. Watertight closure; warm water irrigation (against seeding); drain.
4. Cystogram day 7, then remove catheter.

7 · Robotic radical cystectomy (RARC)

KEY POINT

No effective salvage exists: urine spillage, poor PLND or a positive soft-tissue margin leads to near-uniformly fatal recurrence, so robotics is judged on oncology first.

- Indications match open surgery; prior surgery and obesity are relative contraindications. A **15-mm left assistant port** allows stapling and bagging.

ALGORITHM

1. Release sigmoid; develop presacral plane; open cul-de-sac to prostate apex.
2. Open Retzius; control anterior pedicle; free ureters (**never grasp**).
3. **Extended PLND with ureters intact**.
4. Clip ureters, send margin, pass left ureter under sigmoid.
5. Posterior pedicles, DVC, urethra (clip proximally only if neobladder).

- PLND before ureteric division keeps urine output as a resuscitation guide. Bag each packet whole; the obturator nerve is **not** the posterior limit, so go down to the hypogastric vein.

8 · Complications and post-operative care

- Gastrointestinal complications lead, then infection. For **ileus**, use alvimopan, early feeding and opioid-sparing analgesia, with **no routine nasogastric tube**.
- Ureteric devascularisation causes ischaemic stricture (keep adventitia).

9 · Landmark trials

Trial	Design	Result	Take-home
Grossman 2003	NAC and RC vs RC	Median survival 46 vs 77 months	Offer cisplatin NAC
SWOG S0337	Single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low grade
RAZOR (2018)	Non-inferiority RARC vs open; open diversion	2-year progression-free survival 72.3% vs 71.6%; less blood loss, shorter stay	Robotic oncologically equivalent

Trial	Design	Result	Take-home
iROC	RARC with intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE, wound complications	Recovery advantage

133 Use of Intestinal Segments in Urinary Diversion

1 · Surgical anatomy

A diversion depends on its pedicle's blood supply.

- **Stomach** is supplied by the right gastroepiploic (gastroduodenal, from hepatic) and left gastroepiploic (from splenic) along the greater curve. The graft is antrum or a fundal wedge, which must **never reach the pylorus or lesser curve**.
- **Small bowel** is ~22 ft: the proximal 2/5 is jejunum (wider, single arcades) and the distal 3/5 ileum (multiple arcades). Bowel > **8 cm** from a straight vessel may not survive, so clear ≤8 cm of mesentery at a segment end.
- **Avoid pelvic small bowel after radiotherapy**, including the fixed terminal ileum. A right colic arising from the ileocolic lets the ascending colon reach the pelvis.
- **Colon** is supplied by the superior mesenteric artery (SMA: ileocolic, right colic, middle colic) and inferior mesenteric artery (IMA: left colic, sigmoid and superior haemorrhoidal). The internal iliac gives the middle haemorrhoidal; all link through the **marginal artery of Drummond**.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic and ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid and superior haemorrhoidal

> **KEY POINT**
> Watershed zones get the smallest end branches of two arteries, so they go ischaemic first in hypotension or embolism; site anastomoses to one side.

2 · Segment selection

P2 P1 Use of intestinal segments in urology / Intestinal segments in urology — segment table (electrolytes, pros, cons, when to choose), NH4Cl mechanism, alkali therapy

Not in the paper answer:

- Losing **ileum** causes B12, bile-salt and fat malabsorption; losing the **ileocaecal valve** causes bacterial overgrowth, diarrhoea and bicarbonate loss. **B12** liver stores last **3–5 yr**, so give empirical parenteral B12 after 5–7 yr of ileal diversion.
- **Stomach** suits renal failure, short gut and dense adhesions, but causes the **haematuria–dysuria syndrome** (treat with H2 blockers). Use **jejunum** only if nothing else works.

3 · Bowel preparation and enhanced recovery

P3 Methods of preoperative bowel preparation — mechanical lowers amount, antibiotics lower concentration; oral regimens and harms; IV prophylaxis most important; ERAS; who still needs full prep

Not in the paper answer:

- The descending colon holds **$10^{10–12}$/g** bacteria; oral antibiotics cut this to ~10^2/g. The **IV antibiotic**, given 1–2 h before incision and covering anaerobes, matters most.
- **Alvimopan** (peripheral μ-opioid antagonist) is given as 12 mg 30 min–5 h pre-op, then twice daily, limited to 15 doses for inpatients.

- **Enhanced recovery after surgery (ERAS)** uses **no routine bowel prep**; postoperative ileus is the commonest cystectomy complication.
- Salvage or benign cystectomy (radiation, fistula, neurogenic) still needs full mechanical and antibiotic prep.

4 · Intestinal anastomoses

1. **Get exposure** and wall off the field; check the **blood supply** (transilluminate, bleeding edges, no tension).
2. **Prevent spillage** with non-crushing clamps, and irrigate the segment.
3. Join **serosa to serosa, inverted** and watertight, with **no strangulating sutures** and the **mesentery aligned**.

- Breakdown follows ischaemia, faecal sepsis, a **drain on the anastomosis** or irradiated bowel (most lethal). Sutured and stapled are equal. Leak presents late (2nd week); a neobladder–vaginal fistula causes continuous incontinence.
- **Obstruction** occurs in ~10% with ileum or stomach and ~5% with colon, mostly from adhesions. Prevent it with unirradiated bowel, closed mesenteric apertures, omentum and **reperitonealisation**.
- **Ogilvie syndrome** is a dilated caecum; exclude obstruction with a water-soluble enema. If the caecum is **>12 cm** or white count rises, give neostigmine or decompress by colonoscopy; operate for ischaemia or peritonitis.

5 · Ureterointestinal anastomoses

- Mobilise minimally, keep the periadventitia, spatulate, and sew mucosa to mucosa. **Bring the bowel to the ureter**, retroperitonealise, and use soft stents (they cut leak from 3–5% to near zero).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm)	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel made from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%
Wallace	Refluxing end-to-end	Ureters joined together (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

KEY POINT

Conduit pressure is damped by ureteric peristalsis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop colonisation, so a stented refluxing anastomosis is the default; reflux also allows a loopogram.

- **Stricture** follows ischaemia, leak, radiation or infection, so surveillance is lifelong. The classic site is the **left ureter under the IMA**.
- Open reanastomosis is most durable; endoincision works in ~50–60% at 2 yr. Endourology fails more if the stricture appears <1 yr post-op, is **≥1.5 cm**, or is left-sided.

6 · Conduit diversion

- The **ileal conduit** is simplest, with fewest early complications; avoid it in short bowel, inflammatory small-bowel disease and irradiated ileum.

ALGORITHM

1. Take **10–15 cm** of ileum, 10–15 cm from the ileocaecal valve, keeping a major arcade.
2. Restore continuity by ileoileostomy; close the mesenteric window; flush the segment.
3. Pass the left ureter behind the sigmoid mesentery; make stented anastomoses.
4. Fix the base retroperitoneally in the right lower quadrant (RLQ); make the stoma.

- Intractable stomal bleeding suggests portal-hypertensive varices, treated by TIPS or embolisation.
- A **jejunal conduit** (15–25 cm from Treitz, left upper quadrant stoma) is used only if ileum and colon are unusable.

	Transverse	Sigmoid	Ileocaecal
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy	Long ureteric replacement
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- Avoid colon conduits in inflammatory bowel disease and chronic diarrhoea. An **ileal vesicostomy** suits neurogenic bladder and is reversible.
- Remove stents at days 4–6; drain retroperitoneally, **never intraperitoneally**.

Complication (%)	Ileal conduit	Colon conduit	Continent
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27
Renal deterioration	18	15	—

A gastric conduit causes no acidosis (0%).

7 · Stomas

- Mark sitting and supine; go **through the rectus** (lateral stomas herniate); stay ≥5 cm from the incision and avoid creases, scars and irradiated skin.
- **Keep the subcutaneous fat**, cut the fascia in a cross, and make the aperture admit two fingers.
- A **nipple (rosebud)** stoma suits an appliance (2–3 cm spout, less stenosis); a **flush** stoma suits a continent catheterisable channel.

	End stoma	Loop (Turnbull)
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%
Best for	Standard	**Obese** (thick, short mesentery)

- Parastomal hernia occurs in 2–6.6% of ileal conduits. Stomal stenosis occurs in 20–24% of ileal and 10–20% of colon conduits, and in >50% of paediatric catheterisable stomas.

8 · Renal function

- Renal deterioration after conduits is 10–74% (ileal conduit 18%), and ~6% die of renal failure. After ureterosigmoidostomy it is the leading cause of death beyond 15 yr.
- Continent diversions demand more of the kidney than conduits. **Tubular acidification matters as much as glomerular filtration rate (GFR).**

Test	Threshold for continent diversion
Creatinine and urine protein	**<2.0 mg/dL**, no proteinuria
GFR	**>35 mL/min**
Ammonium chloride load	Urine pH **≤5.8**
Water deprivation	Osmolality **≥600 mOsm/kg**

- **Bowel invalidates** urine osmolality and pH and serum urea and creatinine, because they are reabsorbed.

9 · Metabolic complications

Absorption depends on the **segment, surface area, dwell time, solute concentration, renal function and pH**.

Segment	Na	K	Cl	pH	Associated	Treatment
Stomach	—	Low	Low	High	Raised aldosterone; lethargy, seizures, arrhythmia	H2 blocker, PPI; arginine HCl; remove segment
Jejunum	Low	High	Low	Low	Raised renin, angiotensin; vomiting, dehydration	IV saline, bicarbonate, thiazide
Ileum/colon	—	Low	High	Low	Total-body K depletion; hypocalcaemia	Alkali; chlorpromazine, nicotinic acid

KEY POINT

Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na and Cl, high K); ileum and colon absorb NH4Cl (high Cl). **Only jejunum is hyperkalaemic**.

- **Gastric alkalosis** is usually trivial, but severe with renal failure or dehydration. Above a **gastrin** of **120 ng/L**, small gastrin rises cause large bicarbonate rises. Treat stepwise: rehydration and H2 blocker, arginine HCl if life-threatening, omeprazole, then segment removal.
- In the **jejunal syndrome**, NaCl loss causes hypovolaemia and activates renin–aldosterone, so the kidney keeps losing K-rich urine. It is worse with proximal segments, can be fatal, and needs long-term oral salt and thiazide.
- **Hyperchloraemic acidosis** is worst with **ureterosigmoidostomy** and colonic pouches; $NH4^+$ also enters via K^+ channels. Mucosal villous atrophy occurs, but transporters persist, so the risk persists.
- Alkali options are sodium bicarbonate, citrate solutions, or potassium citrate when K is needed. **Chlorpromazine 25–50 mg tds** or **nicotinic acid 400 mg 3–4×/day** inhibit cyclic AMP; they limit, but do not correct, acidosis.

- **Hypokalaemia** is commoner with colon. **Trap**: correcting acidosis alone causes profound hypokalaemia and flaccid paralysis, so give K with bicarbonate. Mg and Ca are also low.
- **Altered sensorium** is usually **ammonia** (liver dysfunction plus ureterosigmoidostomy). Treat by draining the diversion, oral neomycin, protein restriction, IV arginine glutamate and lactulose.
- **Drugs are reabsorbed**, notably **phenytoin, some antibiotics and methotrexate**, so **drain continent pouches during chemotherapy**.
- **Osteomalacia** is commonest with ureterosigmoidostomy, from acidosis, vitamin D resistance and renal Ca loss. Alkaline phosphatase is raised. Correcting a base deficit >2.5 mEq/L prevents bone loss; treat with alkali and calcium, then active vitamin D.
- **Infection**: ~3/4 of conduit urines are bacteriuric; **treat pure Proteus or Pseudomonas** and observe asymptomatic mixed cultures.
- **Stones** (struvite, Ca phosphate) follow acidosis, urease organisms, **staples** and mucus: colon conduit 3–4%, ileal conduit 10–12%, continent caecal reservoir ~20%.

Cancer

P4 Secondary tumours after urinary diversion — Kälble risk table by diversion, surveillance, pathogenesis, adenocarcinoma at ureterocolic junction

Not in the paper answer:

- After **ureterosigmoidostomy**, cancer develops in 6–29% after **10–20 yr**. Defunctioned anastomoses still develop cancer, so **excise them, do not ligate**.

10 · Neuromechanical aspects

- Detubularising bowel into a sphere disrupts coordinated peristalsis and, by Laplace's law, increases volume and lowers pressure; this is the basis of reservoirs.

11 · Landmark trials

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

134 Continent Urinary Diversion

1 · Definition and basic principles

P4 P2 Continent urinary diversion: orthotopic vs non-orthotopic / Principles of continence in continent diversion — definition, three principles (reliable outlet, low pressure by detubularisation + sphere/Laplace, capacity ≥500 mL; 44 cm ileum <200 to 500–600 mL in a year), ileum stretches more than colon

Not in the paper answer:

- Orthotopic is now preferred (natural voiding, no appliance, fewer revisions); cutaneous suits pre-existing incontinence or urethral disease.

KEY POINT

Continence is a pressure balance: low reservoir pressure (from detubularisation) against adequate outlet resistance (sphincter or efferent valve). If either fails, the patient leaks. Camey I (tubular ileum) failed because its pressure was too high.

2 · Patient selection

Selection means excluding a bad urethra, kidneys or hands.

P4 Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion — eligibility 80–90% men / ~75% women, counsel all options (AUA/EAU), absolute and relative contraindications table, renal thresholds, ~10% intraoperative change, age/obesity not bars

Not in the paper answer:

- **Locally advanced or node-positive (N+) disease found at surgery is not a contraindication**: pelvic recurrence is ~7% (13% if extravesical or N+) and rarely disables the neobladder. The exception is a grossly positive margin at pubis, rectum or side wall.
- eGFR 40–59 is safe, as decline is no faster than with a conduit.
- A short or fatty mesentery is the commonest technical reason to abandon orthotopic diversion.

ALGORITHM

1. Estimated glomerular filtration rate (eGFR) >40.
2. Intact external sphincter and adequate bowel.
3. Not frail; no severe liver, cardiac or psychological disease.
4. Men: no distal prostatic urethral cancer. Women: no palpable anterior vaginal tumour or cervical invasion.
5. If all are met and the patient will self-catheterise, offer an orthotopic neobladder.

- Mark a conduit site in everyone, through mid-rectus and away from creases, umbilicus and scars.

Urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually N+, so give neoadjuvant chemotherapy)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op	Transurethral resection (TUR) biopsy at 5 and 7 o'clock beside the verumontanum — optional	Examination under anaesthesia (EUA); bladder-neck biopsy

Feature	Men	Women
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; the American Urological Association (AUA) requires a negative margin)	
Bladder carcinoma in situ (CIS)	Not predictive	
Skip lesions	—	None: a negative bladder neck means a negative urethra

- Risk rises with depth: none < CIS/mucosal/ductal < stromal. **The neobladder itself is protective** (hazard ratio (HR) 0.44).
- Prostatic involvement needs urethral surveillance (wash cytology, urethroscopy).
- In women, keep the urethra if bladder-neck biopsy and proximal urethral frozen section are negative.

Patient-related factors

- **Renal function** is the key contraindication. Proceed if creatinine < **1.7–2.2 mg/dL** or eGFR > **35–40 mL/min**, with hydration and oral bicarbonate and no nephrotoxins.
- **Self-catheterisation** is needed by 10–20% of men and ~30% of women with a neobladder, and by all with cutaneous pouches. The need may appear years later, so all must be willing; multiple sclerosis, quadriplegia or frailty favour a conduit.
- **Sphincter weakness** is managed with an artificial urinary sphincter (AUS) in men, and Burch or sling in women (who will likely catheterise).
- **Prior radiation** is acceptable without incontinence, stricture, fistula or severe bowel damage; the final decision is made intraoperatively.
- After **prior prostatectomy**, pre-op continence predicts outcome. After **prior bowel resection** with short bowel, use a sigmoid neobladder.

3 · Preserving continence at cystectomy

The **rhabdosphincter** (pudendal nerve) is critical. It lies **anterior and lateral** to the urethra in both sexes.

Men

- Never dissect posteriorly beyond the apex. Incise endopelvic fascia and puboprostatic ligaments minimally; control the dorsal venous complex (DVC) with **absorbable** suture, as clips or staples erode into the anastomosis.

ALGORITHM

1. Open the endopelvic fascia; incise the puboprostatic ligaments slightly.
2. Ligate the DVC close to the apex.
3. Incise the anterior urethra and place sutures.
4. Place posterior sutures through the rectourethralis, then divide the posterior urethra.
5. Tag the sutures until the ureteroenteric anastomosis is done.

- Place sutures **before** full division, as the male urethra retracts. Apical frozen section with tumour means urethrectomy and cutaneous diversion or conduit; atypia or focal CIS means a second distal margin.

- Nerve sparing has a controversial continence benefit; it may help night continence by keeping urethral sensation.

Women

- Sparing uterus and anterior vagina in selected low-stage disease avoids fistula, improves sexual function and reduces retention. A palpable bladder-neck or vaginal tumour needs en bloc urethrectomy and cutaneous diversion.

ALGORITHM

1. Divide the posterior pedicles and develop the plane off the anterior vagina.
2. Use the Foley balloon to mark the bladder–urethra junction.
3. Free the bladder neck and the first 0.5 cm of urethra.
4. Clamp the bladder neck, divide just distal to it, and send a frozen section.

- Leave endopelvic fascia untouched; the bladder neck is not needed for female continence.
- Always interpose an **omental flap** between vagina and pouch, and keep vaginal and urethral suture lines apart.

4 · Choice of bowel and ureteric reflux

Ileum gives lowest pressure; obstruction, not reflux, loses kidneys.

- **Ileum is the default**, being most distensible; its mucosa atrophies, lowering mucus and absorption. Colon has pressure waves; stomach has poor compliance and haematuria–dysuria. Close with absorbable suture, as staples cause stones.
- **Antireflux is unnecessary**: Valsalva raises pressure equally throughout, and every valve can obstruct. A direct (Wallace or Leadbetter) anastomosis strictures in **3–6%**, against 7–29% for Le Duc. **Obstruction, not reflux, destroys kidneys.** Kock nipple failures are late (>10 yr) and silent.

5 · Orthotopic neobladder designs

P2 P4 Use of intestinal segments in urology, and orthotopic neobladder — design table (Studer, Hautmann, T pouch, Abol-Enein, Camey II, Hemi-Kock, Padua, Mainz III, sigmoid); Studer steps; Hautmann vs Studer

Not in the paper answer:

- Hautmann became refluxing in 1997, cutting stenosis from 9.5% to 1%. Studer is the most popular design; the sigmoid neobladder has the worst night continence.

ALGORITHM

1. **Studer:**
2. Isolate ileum ending 15–20 cm proximal to the ileocaecal valve; keep a 40–44 cm reservoir (two 20–22 cm limbs) and an afferent limb.
3. Open the reservoir, oversew the back wall, and fold it the opposite way.
4. Make stented end-to-side ureteroileal anastomoses.
5. Join the dependent point to the urethra.

KEY POINT

Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · Continent cutaneous reservoirs

P2 Principles of continence in continent urinary diversion; Mitrofanoff principle — nipple vs flap vs imbricated valve, appendix ≥5–6 cm/≥12 Fr, Mitchell extension, Yang–Monti, tapered ileum, Indiana

Not in the paper answer:

- **Indiana** uses the right colon and ~10 cm of terminal ileum, with a double-imbricated ileocaecal valve and tapered ileum; the whole colon is opened to abolish caecal peristalsis.
- An **umbilical stoma** has less stenosis and is easiest for paraplegics; the appendix takes only 14–16 Fr catheters.

7 · Perioperative care

- **Enhanced recovery after surgery (ERAS)** means prehabilitation, carbohydrate loading, restrictive fluids, no nasogastric (NG) tube, early diet and opioid-sparing analgesia. **Alvimopan** cut ileus from ~23% to ~7%; ileus is the commonest gastrointestinal complication.
- **Drainage**: single-J stents and a 24 Fr urethral catheter irrigated for mucus; remove the catheter at **3 weeks**. Then use timed voids, lengthened progressively, with pelvic floor exercises.

8 · Results and complications

Men tend to leak, women to retain.

Early (<90 days) morbidity is no higher than with an ileal conduit.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; if undrained, percutaneous drain with or without nephrostomies; **avoid early open repair**
Bacteriuria	Very common	Treat only if symptomatic or Proteus/Pseudomonas; in febrile urinary tract infection (UTI), look for obstruction, stones, residual; if recurrent, nitrofurantoin or methenamine
Ureteroileal stricture	Can appear after 10 yr, often silent	Suspect on CT or ultrasound (US); confirm by pouchogram or antegrade study; endoscopy (retreatment >50%) inferior to open
Pouch stones	Kock and stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention	Cystogram; catheter with or without drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; beyond the first weeks it won't heal, so repair transvaginally or transabdominally, or convert to cutaneous
Stomal stenosis (cutaneous)	Up to ~20%; after 90 days, radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Endoscopy for level; explore if below fascia
Incisional hernia	Common; impairs emptying	Repair

Continence

- Day continence comes over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2, and nocturnal incontinence persists in **20–50%**. Risks are age >65, colon, diabetes, no nerve sparing and hysterectomy.
- Night leakage reflects loss of the sphincter–detrusor reflex; advise voiding at bedtime and an alarm 2–3 times a night.
- Evaluate persistent leakage after the 12–18-month plateau, excluding fistula and using urodynamics.

Stress incontinence	Men	Women
Bulking agent	Marginal	Mild only; <50%, poorly durable
Definitive	AUS or male sling	Pubovaginal sling — most then catheterise
Safety rule		No blind retropubic needle passage; use bone anchors or the prepubic route

Retention

- It presents as infection, overflow, palpable pouch or new hydronephrosis. It affects men in 4–10% (~21% by 10 yr) and **women in 20–30%**.
- In women the cause is **mechanical**: the pouch falls back and angulates the pouch–urethral junction. Prevent with omentum behind the pouch, vaginal suspension, sacrocolpopexy and uterine preservation.
- Treat with clean intermittent self-catheterisation (CISC), as drugs are useless; exclude stricture or recurrence.

> **KEY POINT**
> Men fail by **leaking** (sphincter without reflexes), so augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks), so support the pouch.

9 · Follow-up

- Stricture occurs in the first 4 months, recurrence from 4 months to 3 yr, and stones, late obstruction and urethral or upper-tract tumour after 3 yr.
- Review 4-monthly in year 1, 6-monthly to 3 yr, then annually (examination, metabolic panel, blood count), with CT and chest imaging to 5 yr.
- Check B12 yearly. Do urethral wash cytology in men, and bimanual examination and voided cytology in women. After 5 yr continue lifelong upper-tract US, metabolic panel and blood count.
- No diversion has proven superior quality of life; the neobladder gives better body image and the conduit better urinary function.

10 · Landmark trials

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 radical cystectomy (RC) patients	No difference in renal function or infection at 3 yr; T pouch more secondary procedures	Antireflux adds cost without benefit

Trial	Design	Result	Take-home
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

135 Minimally Invasive Urinary Diversion

1 · Context

- Robot-assisted radical cystectomy (RARC) equals open oncologically in trials (RAZOR), but with **extracorporeal** diversion (ECUD); intracorporeal diversion (ICUD) is untested.
- In the International Robotic Cystectomy Consortium (IRCC), ICUD rose ≈ **11%/year**, driven by the **ileal conduit**, the commonest robotic diversion.

2 · Patient selection

A bar to one step is not a bar to all.

Obesity, age, advanced disease and prior surgery no longer exclude.

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary tolerance of steep Trendelenburg (relative; eased by Xi and AirSeal)
Use of ileum	Short bowel; severe inflammatory bowel disease; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function (as for open)

ALGORITHM

1. Pasadena Consensus learning pathway:
2. RARC with extracorporeal diversion.
3. Intracorporeal conduit (≈30–100 cases).
4. Continent diversion (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · Perioperative bundle

Enhanced recovery (ERAS) uses alvimopan, goal-directed fluids, immediate oral diet and opioid-sparing analgesia, giving less ileus and a shorter stay. Caution: a stay **<5 days** about doubles 90-day major complications.

Element	Recommendation
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, and no oral antibiotics (electrolyte upset, liquid spillage, no benefit)
Thromboprophylaxis	Pneumatic compression and stockings; low-molecular-weight heparin before and up to **4 weeks** after
Antibiotics	Broad-spectrum intravenous within **1 h**

4 · Position and ports

- Use lithotomy and steep Trendelenburg, with Hasson entry after prior surgery.
- **Six to seven ports sit more cephalad** to reach over small bowel and take nodes to the aortic bifurcation.
- A neobladder adds a left stapling port; a suprapubic 12-mm port serves bowel reanastomosis and extends for specimen extraction.

5 · Nerve sparing

One plexus serves sexual function and neobladder continence.

	Male	Female
Danger points	Presacral and medial common iliac; high pedicle ligation; seminal vesicles and Denonvilliers fascia; prostatic pedicles and apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus)
Manoeuvre	Athermal, intrafascial, traction-free pedicle control	Preserve tissue around proximal urethra and anterior vaginal wall; extract via Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by dorsal venous complex; dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third, so continence survives loss of bladder neck

6 · Intracorporeal ileal conduit

ALGORITHM

1. Pass the left ureter under the sigmoid mesentery to the right.
2. Choose 12 cm ileum, 15–20 cm from the ileocaecal valve; place the marionette stitch.
3. Isolate with a 45-mm stapler.
4. Do ureteroileal anastomoses (left first) and stent.
5. Reanastomose bowel, then form the stoma.

- The **marionette stitch** (1-0 silk through the stoma site and conduit end) is held **untied**; it replaces the missing hand and positions the conduit.
- Avoid a redundant segment (See Ch 133), and tag cut ends to prevent malrotation.
- The **enterotomy** is an athermal cold-scissors buttonhole, because thermal slits gave more strictures.
- Single-J stents are used, and the **guidewire comes out only after anchoring**. Use a Bricker (separate end-to-side) or Wallace (Y, one anastomosis) technique.

7 · Intracorporeal neobladder

The key change is **early neobladder–urethral anastomosis** (Karolinska): dependent ileum is anchored to the urethra first, then the pouch is built in place. Design principles: See Ch 134.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded
UCLH Pyramid	50 cm	Koie-derived cross-fold
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	One trough at a time; chimneys

ALGORITHM

1. Urethral anastomosis (Van Velthoven barbed suture).
2. Detubularise distal 40 cm (cold scissors); form the posterior plate.
3. Join ureters to afferent limb or chimney, stented.
4. Close the anterior wall once the 22-Fr catheter is confirmed.

- A **tension-free urethral anastomosis** lacks tactile feedback. It is helped by urethral length, less Trendelenburg, perineal pressure and Penrose traction. Incising mesenteric peritoneum risks pouch ischaemia.

Intracorporeal Indiana pouch

ALGORITHM

1. Mobilise the right colon.
2. Divide ileum 15 cm from the ileocaecal junction and transverse colon proximal to the middle colic; restore continuity.
3. Detubularise along the anterior taenia and fold.
4. Bring ureters through the posterior plate on double-J; place a caecostomy.
5. Taper ileum over 14 Fr, imbricate the valve, and form an umbilical stoma.

KEY POINT

Capacity comes from detubularised right colon; continence from tapered ileum and the imbricated ileocaecal valve.

8 · Postoperative care

Avoid overhydration; there is no routine nasogastric tube (Cochrane). Irrigate the neobladder **8-hourly** for mucus.

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Neobladder catheter	**3–4 weeks** , after leak-free pouchogram
Externalised stents / double-J	≈1 week / cystoscopically at 3–4 weeks
Feeding-tube stents tied to Foley	With the catheter

9 · Outcomes

Most problems come from the diversion, and many appear late.

- **Operative time** depends most on diversion type (conduit ≈70 min shorter); ICUD and ECUD take similar time. Proficiency needs ≥**66 RARCs**.
- **Complications** are mostly diversion-related, infection commonest. Two-thirds of reoperations occur after 3 months, so 90-day reporting underestimates them.
- **Parastomal hernia** reaches ≈one-third by 3 years (risks: long operation, fascial defect ≥30 mm, low eGFR).
- ICUD versus ECUD (IRCC) is neutral for high-grade complications and readmission after adjustment, with less transfusion.
- **Reoperation** reaches ≈half by 5 years; predictors are neoadjuvant chemotherapy and any high-grade complication.

- **Uretero-enteric stricture** rises over 5 years; robotic revision equals open.

> **KEY POINT**
> Night continence is worse because the urethrosphincteric guarding reflex is inactive in deep sleep, so continence depends on reservoir capacity. Hence detubularisation, nerve and sphincter preservation, a night alarm and limiting night output. Assess at 6–12 months.

- Good ECOG status and retained filling sensation predict continence. Up to 25% of men need clean intermittent catheterisation (CIC) after open neobladder.
- Intracorporeal series report day continence 68–100%, night 55–85%, and CIC up to 11%.

10 · Landmark trials

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD
iROC	Multicentre RCT, intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early and recovery-related

124 Bladder Surgery for Benign Disease

1 · Surgical anatomy essentials

- The bladder is an extraperitoneal organ (mucosa, lamina propria, detrusor, perivesical fat) derived from the endodermal urogenital sinus.
- **Anteriorly** lies the space of Retzius, the usual access. **Posteriorly**, peritoneum reflects near the prostatorectal junction (men) or vesicouterine space (women), and Denonvilliers fascia separates bladder and prostate from rectum. **Laterally** lie the pelvic sidewall, obturator space, and vas or round ligament.
- The **medial umbilical ligaments** are the obliterated umbilical arteries; the **median umbilical ligament** is the urachus, which can persist as a patent urachus, sinus or cyst.
- The **uterine artery** crosses over the ureter near the internal os, and the **vas** crosses it near the seminal vesicle tip. The distal ureter runs medial to the medial umbilical ligament.
- Superior and inferior vesical arteries form the posterolateral pedicles; lymph drains mainly to external iliac nodes. Nerves are sympathetic T10–L2 and parasympathetic S2–S4; the pudendal nerve (Onuf's nucleus) supplies the sphincter.

KEY POINT

Three landmarks organise every benign bladder operation: Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety. They decide whether surgery stays extraperitoneal.

KEY POINT

Common aftercare: watertight **two-layer 2-0 polyglactin** closure (mucosa, then seromuscular), leak test, large-bore Foley for about **1 week**, drain out at about **48 h** if output is low, and a **cystogram before catheter removal**. With high drain output, drain creatinine above serum means a urine leak.

2 · Bladder diverticula

A sac without muscle cannot empty: stasis, infection, stones, cancer.

P2 P3 Bladder diverticulum / Hutch diverticulum — definition (no muscularis propria), congenital vs acquired table, VCUG/cystoscopy/CT/urodynamics work-up, indications, tumour inside means cystectomy

Not in the paper answer:

- Voiding cystourethrogram (VCUG) shows hidden diverticula and the outlet obstruction, and decides between a preoperative stent and simultaneous reimplantation.
- A fibrous **pseudocapsule** gives the surgical plane. Most acquired diverticula are asymptomatic and incidental.
- CT urogram shows hydronephrosis, ureteric displacement and stones. Send cytology in all, add flow rate or urodynamics, and obtain a **negative urine culture** before surgery.
- At cystoscopy, enter every sac and note neck size and distance from the orifice; treat stricture or bladder neck contracture before or with the diverticulectomy.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection

Situation	Management
Large with residual urine, recurrent urinary tract infection (UTI), stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy with or without reimplantation
Cancer in diverticulum	Partial or radical cystectomy, not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**, and augmentation for neurogenic bladder. Do **endoscopic outlet surgery first**, because later irrigation disrupts fresh suture lines.
- Localise the sac by selective filling or flexible cystoscopic transillumination, and place a ureteral catheter if it lies near the ureter.
- Open extraperitoneal surgery remains the gold standard, even for posterior sacs; laparoscopic and robotic surgery is reasonable in experienced hands.

ALGORITHM

1. **Open intravesical diverticulectomy**
2. Fill the bladder with about 200 mL via an 18 Fr Foley.
3. Lower midline incision; split recti; incise transversalis fascia.
4. Sweep peritoneum off the dome and open a vertical cystotomy.
5. Identify the orifices (ureteral catheter if concerned) and excise the sac.
6. Place a 20–22 Fr Foley, close in two layers, leak-test, and drain.

Step	Intravesical	Combined intra-/extravesical
Delivery	Sac everted or dissected from the pseudocapsule from inside; mucosal incision about **0.5 cm** round the neck	A finger in the sac pushes it out; the neck is incised from outside
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if the inner and outer incisions do not meet
Closure	Two layers 2-0 polyglactin with a secure seromuscular layer, which prevents recurrence	

Laparoscopic / robotic

- Use steep Trendelenburg and cystoscopy first, with pneumoperitoneum at **15 mm Hg**. Place the assistant port opposite the sac, circumscribe the neck, and excise and bag the sac with the ureter in view; staplers risk stones forming on the staples.
- If the sac sits at the ureteral hiatus, amputate the neck, make a new hiatus and reimplant over a double-J. Extraperitoneal, single-site and single-port variants exist.

Complications

- **Urinary extravasation is commonest**. It shows as high drain output, infected urinoma, or ileus and urinary ascites after transperitoneal surgery. Check drain creatinine and do CT cystography.

ALGORITHM

1. **Managing a leak**
2. If asymptomatic, keep Foley and drain.
3. If a symptomatic collection forms, drain it percutaneously.
4. Do a cystogram before removal; if the leak persists, choose nephrostomies or exploration and closure.

- Other complications are UTI, fistula, bladder spasm, wound infection and bowel injury.
- Unrecognised ureteric injury presents with acute kidney injury or flank pain; try a retrograde stent, and if it fails, place a nephrostomy and repair later.

KEY POINT

Diverticulectomy is a dissection problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely, and fix the outlet first.

3 · Bladder stones, foreign bodies and fistulae

A bladder stone signals stasis, so treat the cause too.

- The cause is **stasis**: bladder outlet obstruction, neurogenic bladder, or augmented and substituted bladders. Foreign bodies are mostly iatrogenic (mesh, migrated intrauterine device).
- **Non-contrast CT** is best for stone burden, prostate volume and hydronephrosis. Cystoscopy is needed only for haematuria, suspected cancer or prostate assessment.
- Treat the stone **and** its cause; the concomitant operation often picks the route (transvesical prostatectomy, diverticulectomy or enterocystoplasty).

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation risks urethral stricture
Percutaneous cystolithotripsy	Large stones	Nephroscope, short stay, no stricture	Contraindicated after prior surgery, diversion or augmentation (bowel injury)
Open cystolithotomy	Very large burden; with prostatectomy, diverticulectomy, augmentation	Historic gold standard	Incision, longer catheter
Laparoscopic/robotic	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technically demanding

- **Percutaneous** access two fingerbreadths above the symphysis is dilated to a **30 Fr** sheath for a 24–26 Fr nephroscope. Stones under 1 cm are grasped whole and larger ones fragmented; the suprapubic catheter is clamped at 24–48 h. After prior surgery, image first to exclude interposed bowel.
- **Transvesical laparoscopy** uses a pneumovesicum of **8–12 mm Hg**, avoids the peritoneum and suits eroded mesh. Transperitoneal laparoscopic or robotic cystolithotomy allows simultaneous prostatectomy or diverticulectomy.

- **Open** cystolithotomy opens the bladder vertically, removes the stones, inspects for tumour and intravesical lobe, and closes in two layers with a drain.
- Refractory fistulae (colovesical, vesicovaginal) may need partial cystectomy, and a refractory urinary fistula may justify simple cystectomy.

4 · Partial cystectomy for benign disease and urachal surgery

- Indications are leiomyoma, paraganglioma, endometriosis, urachal anomalies and fistulae; most benign tumours need only transurethral resection (TUR).
- **Avoid it with small capacity or severe dysfunctional voiding.** Exclude malignancy first, because cancer needs wide margins and lymphadenectomy. Lesions near the orifice need reimplanting.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; endovesical, intramural or extravesical	Transvaginal ultrasound (US), MRI	Small endovesical: TUR; large or intramural: wide local resection
Paraganglioma	Episodic hypertension, hypotension or syncope on **micturition or sex**; or painless haematuria	Plasma metanephrines; MIBG scan; yellow raised mucosa	**Alpha-blockade before any instrumentation**; partial cystectomy preferred over TUR (recurrence)
Endometriosis	Cyclical haematuria with menses	US, MRI (bright on T1); cystoscopy limited	Partial cystectomy (disease is extravesical)
Urachal anomaly	Umbilical discharge, mucinuria, infected mass; malignant risk	US/CT; VCUG or sinogram; cystoscopy	Culture and antibiotics, with or without drainage; if recurrent, excise urachus with a dome cuff

- Dome and anterior lesions are approached extraperitoneally, posterior ones transperitoneally, with no node dissection. **Aspirate before opening to avoid spillage**.
- Excise full thickness with about a **2-cm margin** if capacity allows, with frozen-section margins, and reimplant if the orifice is involved.
- **Urachal excision** takes the whole tract with peritoneal wings and a small dome cuff, with **no wide margin or frozen section**; the umbilicus may be kept if benign. Open excision uses a periumbilical midline incision down to a transmural dome cuff.
- Complications are extravasation (commonest), ureteric injury and reduced capacity.

> **KEY POINT**
> Leiomyoma, paraganglioma and endometriosis infiltrate and recur if incompletely excised, so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, a margin). Urachal surgery is the exception: the whole tract matters, not a wide margin.

5 · Simple (supratrigonal) cystectomy

A last resort that helps only if the bladder is the source.

- It treats end-stage refractory dysfunction: radiation cystitis, neurogenic bladder, refractory incontinence, fistula, bladder neck contracture, refractory anastomotic stricture, and interstitial cystitis/bladder pain syndrome (IC/BPS).
- Prove the bladder is the source, since non-vesical pain persists. In IC/BPS, success is predicted by a **small fibrotic bladder, small capacity under anaesthesia and Hunner**

lesions.

- Diversion alone risks pyocystis, bleeding, pain and secondary cancer.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed
Lymphadenectomy	None	Performed
Bladder opened	Often, to find the trigone and bladder neck	Avoided
Excision	Bivalve and amputate near the trigone; stump oversewn and residual mucosa cauterised	En bloc

Steps are to divide the ureters, bivalve the bladder, divide the pedicles, amputate each half and cover the stump with posterior peritoneum. An ileal conduit usually follows, and robotic surgery may reduce complications. Drains stay 5–7 days, until drain creatinine equals serum, and complications are common.

6 · Ureteral reimplantation in adults

Adults need reimplantation mainly for injury and stricture.

- Indications are distal ureteral trauma, iatrogenic injury, **stricture** and late-presenting injury, and reimplantation alongside diverticulectomy or partial cystectomy. Primary VUR is rare.
- In thermal or gunshot injury, debride even normal-looking ureter; assess partial injury with on-table retrograde pyelogram or methylene blue.
- Strictures follow impacted stone, radiation or chronic inflammation. An atrophic kidney needs decompression, then a renogram.
- Measure the gap with antegrade and retrograde imaging and exclude malignancy. A **poorly functioning kidney needs nephrectomy, not reimplantation**. **Persistent extravasation despite adequate stenting** is an indication.

> **KEY POINT**
> Refluxing reimplantation is acceptable in adults. Refluxing anastomoses show no faster renal decline, more stenosis or more infection than tunnelled ones, while needing less ureter and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / mobilisation	More	Less; reaches the posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical	Cephalad-medial to native hiatus	Submucosal tunnel; right-angle clamp draws ureter through

Technique	Access	Neo-hiatus	Signature step
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over a Kelly clamp	Clamp pulls ureter in

- Divide the ureter proximal to disease, with a frozen-section margin in stricture. Mobilise the bladder off peritoneum, dividing the ipsilateral medial umbilical ligament. Test reach on the **decompressed** dome.
- **Lich-Gregoir**: a small cystotomy forms the neo-orifice, detrusor is incised toward the bladder neck sparing mucosa, a spatulated 4-0 anastomosis is stented, and detrusor is closed over the ureter.
- **Politano-Leadbetter**: through an anterior cystotomy, a neo-orifice about 1.5 cm superior-medial to the native one leads into a submucosal tunnel.
- For a poorly vascularised ureter, use a non-transecting side-to-side refluxing anastomosis.
- Preserve periureteric adventitia, spatulate, and keep knots outside the lumen. The stent is a **6 Fr × 26–28 cm double-J**.
- Primary reimplantation suits defects up to **4 cm**; longer gaps need a psoas hitch or Boari flap, and renal descent adds up to **5 cm**. Further reach comes from dividing the contralateral medial umbilical ligament and pedicle.
- Enter **superior-medially, toward the bladder neck**, because lateral sites kink on filling.
- Open surgery is preferred for trauma; laparoscopic and robotic surgery give less blood loss and shorter stay with equal patency.
- Aftercare is a Foley for 1 week with a cystogram and a stent for **≈6 weeks**. A **leak** is managed by draining the urinoma and keeping the Foley and stent; if refractory, place a nephrostomy.
- **Late obstruction** usually follows stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site); short strictures are treated endoscopically and high-grade ones by redo.

> **KEY POINT**
> Reimplantation fails from ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other.

7 · Bridging distal ureteral defects

P3 Boari flap — reconstructive ladder, Boari dimensions (base ≥4 cm, apex ≥3 cm, ≤3:1, measured empty, reach up to 15 cm), psoas hitch nerves, contraindications, advancement flap

Not in the paper answer:

- Nonrefluxing Boari length is the defect plus about 3 cm; longer flaps need a wider base.
- Beyond reach, options are transureteroureterostomy, ileal ureter and autotransplantation.

Option	Reach	Rule
Primary reimplant	≤4 cm	Toward bladder neck
Psoas hitch	Gains about 5 cm	Sutures superficial, parallel to the fibres; reimplant before tying
Boari flap	Up to **15 cm**	Base ≥ **4 cm** , apex ≥ **3 cm** , length:base ≤ **3:1**

- Hitch plus flap shortens the flap and preserves capacity. The hitch sutures go into the psoas minor tendon, avoiding the **genitofemoral** (surface) and **femoral** (deep) nerves; reimplant on the superolateral dome.
- The Boari flap is based on the diseased side, anchored to psoas and tubularised in two layers.
- A bladder advancement flap gives comparable reach and is simpler. A **small-capacity, poorly compliant bladder** contraindicates both hitch and flap.
- Complications are femoral or genitofemoral neurapraxia, iliac vessel injury, ischaemic stricture of a narrow-based flap, and reduced capacity.

KEY POINT

The bladder is the donor tissue: hitch first, flap second. Every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and bladder capacity.

8 · Enterocystoplasty (augmentation)

Augmentation protects the kidneys from high bladder pressure.

- Indication: neurogenic bladder with poor compliance **despite** clean intermittent catheterisation (CIC), anticholinergics, botulinum toxin and sacral neuromodulation, especially with upper tract deterioration or incontinence with skin breakdown.
- Bowel lowers pressure only if **detubularised**, which abolishes peristalsis.
- Work-up includes video-urodynamics, serum chemistry, B12 and bowel history; **confirm the patient can catheterise**. Metabolic effects: See Ch 133.
- Add a sling, artificial urinary sphincter, bladder neck closure, catheterisable stoma or nonrefluxing reimplant as needed.

Segment	Note
Ileum (commonest)	25 cm taken **≈20 cm** proximal to the ileocaecal valve (spares B12 and bile-salt absorption); acidosis risk
Ileocaecal	When a catheterisable channel is wanted
Sigmoid	Colonoscopy first
Stomach	Possible; needs a higher incision

ALGORITHM

1. Isolate ileum with a broad mesenteric base and check reach to the dome.
2. Restore bowel side-to-side; close the window.
3. **Detubularise** on the antimesenteric border and fold into a U.
4. **Bivalve the bladder widely** and make a wide-mouthed anastomosis.
5. Suprapubic tube with or without Foley; leak-test; drain.

- A narrow mouth creates a poorly emptying pseudodiverticulum.
- Early complications are extravasation (preserve the arcade), UTI, bowel injury, fistula and intra-abdominal infection.
- **Perforation** follows missed catheterisation and is often masked by sensory loss. Diagnose with CT cystogram and proceed to **emergency exploration**.
- Stones are the commonest long-term complication. Neoplasia risk means surveillance endoscopy from **10 years**, with routine metabolic and vitamin bloods.

- If the urethra is not catheterisable, do a continent ileocaecocystoplasty; if the patient cannot catheterise, an incontinent ileovesicostomy.

KEY POINT

Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem. Detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, bloods and endoscopic surveillance are the price.

9 · Continent catheterisable channels

P2 Principles of continence in continent diversion; Mitrofanoff principle — submucosal tunnel flap valve, appendix first choice, Yang-Monti alternative, channel catheter 2–4 weeks, continence >95%, stomal stenosis commonest

Not in the paper answer:

- Channels suit adequate-capacity bladders after failed urethral CIC; augment if compliance is poor.
- The bladder is drained for 2–4 weeks before channel catheterisation starts. Always have a fallback if the appendix is absent or short.
- A Monti uses 2–4 cm of ileum; a double Monti risks channel diverticula; vas, colon and stomach channels do poorly.

ALGORITHM

1. **Minimally invasive appendicovesicostomy**
2. Detach the appendix with a caecal cuff and spatulate it.
3. Make a ≈4-cm submucosal detrusor tunnel and anastomose.
4. Bring the stoma to the right lower quadrant or umbilicus.
5. Drain the bladder for **2–4 weeks**. Continence is **>95%**.

KEY POINT

Use the appendix when present and long enough; reconfigure ileum when not. Continence comes from tunnel length through detrusor, and missed catheterisation endangers the bladder, not just the channel.

10 · Vesicovaginal fistula (VVF) †

P3 P2 VVF: management options, early vs delayed repair — causes, double-dye test, CT urography, catheter trial, timing, vaginal (Latzko, flap-split) vs abdominal (O'Conor) routes, interposition flaps

Not in the paper answer:

- Leakage usually starts 1–3 weeks after hysterectomy, and a coexisting ureteric injury is present in about 10%.
- A ureterovaginal fistula is treated with a ureteric stent first, and by ureteroneocystostomy if that fails.

★ Putting it together

A · NMIBC: risk and treatment

AUA/SUO risk is assigned after complete TURBT.

Risk	Features	Adjuvant	Cystoscopy
Low	Low-grade (LG) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation	3 mo, then yearly
Intermediate	LG Ta recurring <1 yr/>3 cm/multifocal; LG T1; high-grade (HG) Ta ≤3 cm	Induction MMC/gemcitabine or BCG; maintain 1 yr	3–6-monthly for 1–2 yr
High	HG T1; HG Ta >3 cm/multifocal/recurrent; CIS; variants; LVI; HG prostatic urethra	Re-TURBT (T1); BCG with **maintenance 3 yr** (SWOG); early RC if highest-risk	3–4-monthly for 2 yr, 6-monthly for 2 yr, then yearly; upper tracts yearly for 2 yr
BCG-unresponsive	HG despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC; if unfit: pembrolizumab (CIS), nadofaragene, gem–docetaxel	—

B · MIBC: cystectomy vs preservation See Ch 131, 132

	NAC + RC + pelvic node dissection	Trimodality therapy
Who	Default if fit; NAC (gem–cis or ddMVAC) if cisplatin-eligible	Solitary T2, complete TURBT, **no CIS/hydronephrosis**, good function; or unfit
Cisplatin-ineligible	GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, or NYHA III–IV: local therapy	
After	Adjuvant nivolumab (CheckMate 274): ≥pT3/N+ after NAC, or no NAC	Lifelong cystoscopy; salvage RC 10–30%

C · Choosing the diversion See Ch 133–135

	Ileal conduit	Orthotopic neobladder	Continent cutaneous
Best for	Frail, poor renal function, cannot catheterise	Default continent option, either sex	Unusable urethra or sphincter
Must have	Usable (non-irradiated) ileum	eGFR **>35–40** ; intact sphincter; negative urethral frozen section; will catheterise	eGFR >35–40; dexterity; efferent limb
Main problems	Stoma problems, ureteroileal stricture	Night leakage; late retention (women); acidosis	Stomal stenosis, catheterisation difficulty, stones

D · Metabolic complications by segment

Segment	Disturbance	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis**	Bicarbonate, potassium citrate

Segment	Disturbance	Treatment
Jejunum	**Hyponatraemic hypochloraemic hyperkalaemic acidosis**	Sodium chloride; avoid jejunum
Stomach	**Hypochloraemic hypokalaemic alkalosis**; haematuria–dysuria syndrome	H2 blocker/PPI

Losing terminal ileum lowers B12 and causes bile-salt diarrhoea; buffering acidosis in bone causes osteomalacia. See Ch 133

E · Metastatic urothelial carcinoma

ALGORITHM

1. **Fitness**: cisplatin-, platinum-only or platinum-ineligible.
2. **First line**: enfortumab vedotin (EV) with pembrolizumab (EV-302); otherwise gem–cis (gem–carbo if cisplatin-ineligible).
3. **If no progression on platinum**, give avelumab maintenance (JAVELIN Bladder 100).
4. **Later lines**: pembrolizumab; erdafitinib (FGFR-altered); EV; sacituzumab govitecan. See Ch 131

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 13

Kidney & UTUC

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 136	Benign Renal Tumours
Ch 137	Malignant Renal Tumours
Ch 138	Urothelial Tumours of the Upper Urinary Tract and Ureter
Ch 139	Surgical Management of Upper Urinary Tract Urothelial Tumours
Ch 141	Contemporary Open Surgery of the Kidney
Ch 142	Laparoscopic and Robotic Surgery of the Kidney
Ch 143	Focal Therapy for Renal Tumours
Ch 144	Treatment of Advanced Renal Cell Carcinoma
★	Putting it together

136 Benign Renal Tumours

1 · The benign small renal mass

Many resected "cancers" are benign.

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are imaging diagnoses; size, sex and age are unreliable.
- Management follows the risk of renal cell carcinoma (RCC); the SRM algorithm is in Ch 137.

2 · Renal cysts and the Bosniak classification

- A **simple cyst** (any size, homogeneous, **≤20 Hounsfield units (HU)**) needs no further imaging. Otherwise use Bosniak.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall or septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to **5 yr**
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall or septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT

Ladder: wall ≤2 mm, 3 mm, ≥4 mm, irregular, then nodule. Traps: any calcification fits II, a class I wall may enhance, and growth of I/II is not malignancy.

KEY POINT

Why surveil small III/IV? Cystic RCC is indolent and mostly fluid, so size predicts poorly (proposed **pT1c**) and the number needed to treat is very high.

- European Association of Urology (EAU): treat Bosniak IV as a solid mass, and biopsy a solid component only if it changes management.

Diffuse cystic disease

P2 ADPKD (Adult polycystic kidney disease); Acquired renal cystic disease — genetics, extrarenal cysts, US criteria, tolvaptan, cyst complications; ACKD prevalence, ADPKD vs ACKD

Not in the paper answer:

- Acquired cystic kidney disease (ACKD) affects **80%** by 10 yr of dialysis, with RCC risk **≥10×** (multifocal, bilateral, often papillary).
- PKD1 relatives aged 15–29 need **≥3 cysts** for diagnosis; a normal scan excludes disease only from age 40.

3 · Oncocytoma

Biopsy cannot reliably separate it from chromophobe RCC.

- It is the **commonest benign enhancing renal mass**, up to half of benign SRMs <4 cm.
- **Birt–Hogg–Dubé** (folliculin, chr 17) gives fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC).
- A **stellate scar** is non-specific, whereas **segmental enhancement inversion** favours oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

KEY POINT

Why can a core not clear oncocytoma? chRCC has oncocytic areas, so some biopsy "oncocytomas" are RCC. EAU: no definitive core diagnosis.

4 · Angiomyolipoma

AML can bleed catastrophically, so treatment follows bleeding risk.

P3 P2 Management of AML; Renal angiomyolipoma — PEComa, TSC, mTOR, fat on CT, epithelioid variant, triggers, SAE, NSS, everolimus

Not in the paper answer:

- An observed AML ≥4 cm needs CT angiography to look for an aneurysm.
- EAU (sporadic AML): use a mammalian target of rapamycin (mTOR) inhibitor only if unfit for embolisation or surgery.

ALGORITHM

1. **Spontaneous retroperitoneal haemorrhage** (Wunderlich): resuscitate, then selective arterial embolisation (**SAE**). Nephrectomy only if SAE is unavailable or unsuitable.
2. **Symptomatic** (pain, haematuria, barriers to care): SAE, then nephron-sparing surgery (NSS) or RN; ablation in selected cases.
3. **Asymptomatic sporadic**: if >**6 cm**, aneurysm **≥5 mm**, growth >**0.25 cm/yr**, or childbearing age/pregnancy, then treat; otherwise surveil.
4. **Tuberous sclerosis (TSC)**: if >**3 cm** or growing, then everolimus first line; SAE or kidney-sparing resection second.

KEY POINT

Why abandon 4 cm? It was poorly specific and caused overtreatment. "Barriers to care" counts because surveillance needs access to imaging and emergency care.

5 · Other benign epithelial and mesenchymal tumours

- **Papillary adenoma** is **≤15 mm** and cannot metastasise, so it is acceptable in a donor kidney.
- **Metanephric adenoma** (women in their 50s, polycythaemia) is **WT1 positive, CK7 negative**; excise it by NSS.
- **Mixed epithelial and stromal tumour and adult cystic nephroma** form one family and are managed by Bosniak.
- **Leiomyoma** is often really a lipid-poor AML; **Melan-A** is positive only in AML.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	Von Hippel–Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
Reninoma (juxtaglomerular cell tumour)	Young women; refractory hypertension, hypokalaemia, high renin; excision cures
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · Landmark trial

Trial	Design	Result		Take-home
EXIST-2	Phase III randomised trial, everolimus vs placebo, AML in TSC or lymphangioleiomyomatosis	≥50% volume reduction in **42%** **vs 0%**; no bleeding needing embolisation		Approved; mTOR inhibition first line in TSC

137 Malignant Renal Tumours

1 · Overview, epidemiology & aetiology

Most kidney cancers are incidental, so overtreatment is the modern risk.

- Renal cell carcinoma (RCC) makes up 2–3% of adult cancers and is the most lethal common urological cancer. Male to female ratio is **2:1**, peak age 55–75 y, and **4–6% are familial**.
- Over 60% are incidental and ~70% are organ-confined.
- Risk factors: **tobacco** (best accepted; relative risk ~1.4, dose-dependent); **obesity** (the "obesity paradox" of higher BMI but lower stage); **hypertension** (the disease, not its drugs); chronic kidney disease (CKD) and end-stage renal disease (ESRD); family history (2–4×).
- In children and young adults the commonest RCC is **Xp11/TFE3 translocation RCC**. These patients do well with surgery even when node-positive, so do a formal lymph node dissection (LND) at radical nephrectomy (RN).

2 · Classification & pathology

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified (NOS)
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

KEY POINT

Trap: epithelioid angiomyolipoma (AML) is malignant, whereas classic AML, papillary adenoma and oncocytoma are benign. The papillary type 1/2 split was **abandoned** because "type 2" was several diseases (MET-altered, MiT translocation, FH-deficient), so WHO moved to molecular definitions.

Gross features

- RCC has a **pseudocapsule** of compressed parenchyma and is not infiltrative, except collecting duct and sarcomatoid tumours. Cystic degeneration occurs in 10–25% (better prognosis).
- Venous involvement occurs in ~10% and can reach the right atrium.
- Sporadic RCC is bilateral in 2–4% and multicentric in 10–20%; occult satellites are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Hypervascular; clear cytoplasm; carbonic anhydrase IX (CA-IX) positive; sarcomatoid 5–8%	Most venous invasion; worse when localised, better when metastatic

Subtype (share)	Genetics	Hallmarks	Behaviour
Papillary (10–15%)	**Trisomy 7, 17**, loss of Y; MET altered 81%	Hypovascular; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival (CSS) 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in Birt–Hogg–Dubé (BHD)	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	From intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	Seen in ESRD and VHL; now benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8 positive; diagnosis of exclusion	Poor; urothelial-like, so may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative	Dismal, median 8–13 mo
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells with papillae; FISH confirms	40% of paediatric RCC; late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before calling a tumour "unclassified"	Responds to ALK inhibitors in some
Unclassified (1–5%)	—	Shrinking catch-all	Poor

- **Indolent types:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC and clear cell papillary. **Lethal types:** medullary, FH-deficient, collecting duct, unclassified and sarcomatoid.
- Because chromophobe and oncocytoma are hard to separate, WHO 2022 added "oncocytic renal neoplasm of low malignant potential".
- **Sarcomatoid** change (1–5%) is dedifferentiation of any subtype, with median survival <1 y. Sarcomatoid or rhabdoid features mean automatic grade 4 and must be reported in every RCC.

> **KEY POINT**
> Localised clear cell RCC does worse than papillary or chromophobe because it is intrinsically aggressive. Yet metastatic clear cell does better, because VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/ISUP grading: replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100

Grade	Criterion
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Nuclear grade is an independent prognostic factor.

3 · Molecular biology & hereditary syndromes

The VHL pathway explains clear cell vascularity and drug targets.

P2 P1 VHL gene and von Hippel–Lindau disease; Familial malignancies in urology — VHL–HIF pathway, two-hit, VHL lesions, genotype–phenotype, VHL screening, 3-cm rule, belzutifan; VHL/HPRC/HLRCC/BHD/TSC genes and features

Not in the paper answer:

- BAP1 mutation (up to 14% of sporadic clear cell) gives high grade and rapid growth, so these tumours are unsuitable for active surveillance (AS).
- Loss of **CDKN2A** (9p) carries a poor prognosis in all subtypes.

ALGORITHM

1. Normoxia: the VHL protein joins elongin B/C, cullin-2 and RBX1 to form an E3 ligase, which degrades hypoxia-inducible factor (HIF)-1α/2α.
2. If VHL is lost (or in hypoxia), **HIF-2α** accumulates and pairs with ARNT. This raises VEGF, PDGF, TGF-α, GLUT1, erythropoietin and CA-IX → hypervascular clear cell RCC.

Drug targets: VEGF ligand (bevacizumab); VEGF-receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib); mTOR (temsirolimus, everolimus); HIF-2α (**belzutifan**, FDA-approved in 2021 for VHL, ~50% response); CTLA-4 (ipilimumab); PD-1 (nivolumab, pembrolizumab); PD-L1 (atezolizumab, avelumab).

Syndrome (beyond topics)	Gene (chr)	Renal tumour	Other / management
Hereditary leiomyomatosis and RCC (HLRCC)	FH (1q42–43)	Aggressive, often **solitary**	Uterine leiomyosarcoma; **prompt excision at any size, wide margins**; PN only if a wide margin is feasible; FDG-PET finds small metastases
SDH-RCC	SDHA/B/C/D, SDHAF2	Onset ≤40 y	Paraganglioma/phaeochromocytoma, GIST; excision, PN preferred
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast, thyroid; 3-cm rule
Tuberous sclerosis	TSC1/TSC2	AMLs; clear cell 1–3%	AML: surveil <3 cm, everolimus 3–5 cm, embolise or excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma; nephron-sparing excision
MiTF-associated	MiTF p.E318K	RCC >5×	Melanoma; management undefined

4 · Clinical presentation

- The **classic triad** (flank pain, gross haematuria, mass) is rare and means advanced disease. Other clues are weight loss, bone pain, a **non-reducing or right-sided varicocele**, and leg

oedema from inferior vena cava (IVC) obstruction.

- Over 50% of spontaneous perirenal haematomas are due to tumour, so repeat CT later.

Paraneoplastic syndromes (10–20%)

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Marker of advanced disease
Hypercalcaemia	Up to 13%; parathyroid hormone-related protein (PTHrP) or osteolytic metastases	**The only one managed medically**: hydration, then furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy with or without metastasectomy
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3–20%; interleukin-6	Non-metastatic liver dysfunction (ALP raised almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence means viable tumour and poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders

- Paraneoplastic syndromes resolve only with excision or systemic therapy, so persistence after nephrectomy means residual disease.

5 · Diagnosis

Imaging sorts a mass into cyst, fat or enhancing tumour.

P4 Radiology for solid renal masses — CT phases and HU thresholds, MRI, CEUS, sestamibi, girentuximab, FDG-PET, biopsy yield, venous thrombus CT sensitivity, R.E.N.A.L./PADUA

Not in the paper answer:

- The strongest predictors of malignancy are **male sex (~3×)** and **size**: 20–30% of masses ≤4 cm and ~40% of ≤2 cm are benign, while few ≥7 cm are benign.
- A cT1 mass is ~20% benign, ~60% indolent and ~20% potentially aggressive.
- Renal mass biopsy (RMB): **multiple cores over fine-needle aspiration**; haematoma ~5%, specificity 94%. A non-malignant result does not prove the mass is benign.
- **Multiphase CT** is the single most important test: enhancement **>15–20 Hounsfield units (HU)** suggests RCC, and fat **<−20 HU** means AML. MRI is best for IVC thrombus and organ invasion.

ALGORITHM

1. If simple cyst, then observe.
2. If complex or solid: eGFR >45 → CT or MRI with/without contrast; eGFR 30–45 → hydrate, then the same; eGFR <30 → non-contrast CT or gadolinium MRI; contrast allergy → ultrasound (USG), contrast USG or MRI.
3. Then: fat → AML; complex cyst → Bosniak; solid enhancing → management options, with or without RMB.

- **Bosniak** (see Ch 136): malignancy is 3–10% in IIF, ~50% in III and 75–90% in IV. The key discriminator is unequivocal **measurable enhancement**. Avoid ablation and biopsy of cystic masses; AS is acceptable for III/IV.
- Do RMB when the result may change management, or when lymphoma, metastasis or infection is suspected. Do not biopsy patients who will be managed the same way regardless. Biopsy before ablation.
- **Screening:** there is none for the population or for autosomal dominant polycystic kidney disease. In ESRD, screen with USG or CT from the 3rd year of dialysis if fit, and continue after transplant.
- Offer genetic counselling for RCC at **≤46 y**, multifocal or bilateral tumours, or a suggestive history or pathology.

6 · Staging & prognosis

Pathological stage is the strongest predictor of outcome.

TNM (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage groups: I is T1N0; II is T2N0; III is T1–2N1 or T3 any N; IV is T4 or M1. The 8th edition moved pelvicalyceal invasion to T3a.

KEY POINT

TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus a supradiaphragmatic thrombus) are lost. Record all of them and use integrated nomograms.

- **Work-up:** abdominal CT and chest X-ray (chest CT if symptoms or high risk). Bone scan only with raised ALP, bone pain or poor performance status (PS); no PET. Treat an abnormal adrenal as involved.

- **Pathological stage** is the most important factor, then grade, size, subtype, necrosis and sarcomatoid change. Node-positive 5-y survival is 5–30%, and with a thrombus but otherwise confined disease it is 45–69%.
- Venous wall invasion is as important as thrombus level, and collecting-system invasion is adverse.
- Nomograms: **UISS** uses TNM, grade and ECOG PS; **SSIGN** (Mayo) uses Stage, Size, Grade and Necrosis; **Leibovich** predicts recurrence in clear cell RCC.

7 · Management of localised RCC

Every option trades cancer control against kidney function.

P1 P2 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy; Nephron-sparing surgery / partial nephrectomy in RCC — imperative/relative/elective PN, AUA RN rule (GFR >45), surgical vs medical CKD, PN steps, enucleation, solitary kidney, AS/TA criteria, 3-cm rule, EORTC 30904, DISSRM

Not in the paper answer:

- AUA 2021: refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30. Do LND for clinically positive nodes and adrenalectomy only if suspicious.
- CKD follows RN in 65% vs PN in 20%; surgical CKD progresses ~0.7%/y vs medical CKD ~4.7%/y. **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25**.
- **Triggers to intervene on AS**: >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms.
- On AS, small masses grow 0.1–0.3 cm/y and metastasise in <2%; image every 3–6 mo.
- PN function depends first on parenchymal quality, then on the **amount of vascularised parenchyma preserved (the main modifiable factor)**, and only secondarily on ischaemia.
- Leak and bleeding are commoner after PN than RN (steps: Ch 142).
- In a solitary kidney at least 20–30% of the kidney is needed. If bilateral tumours make PN impossible, options are RN with dialysis or neoadjuvant TKI.
- Enucleation is controversial in sporadic RCC because 40–50% infiltrate the pseudocapsule.

Radical nephrectomy

- Robson's principles (early vascular ligation, removal outside Gerota's, adrenalectomy, extended LND) are now modified: perifascial dissection is still essential (~25% of cT1b/T2 invade fat), but there is **no routine adrenalectomy** (unless enlarged on imaging or extensive tumour) and **no routine extended LND**.
- Laparoscopic RN is standard without gross invasion, major venous involvement or bulky nodes; open surgery uses a subcostal or thoracoabdominal approach (see Ch 141).

Thermal ablation (see Ch 143)

- Thermal ablation (TA) works best <2.5–3 cm and gives no pathological staging. Cryoablation is lethal at ≤−20 °C with a double freeze–thaw; neither cryoablation nor radiofrequency is superior.
- Recurrence is mostly salvaged by repeat TA, and nodular enhancement means recurrence.

Familial RCC: 3-cm rule (National Cancer Institute)

- VHL tumours <3 cm had no metastases at ~5 y, while ~25% metastasise when larger. **HLRCC is the exception**.

- Recurrence is near-universal, so surveillance is lifelong. **Exclude phaeochromocytoma** and central nervous system haemangioblastoma before surgery.

8 · Locally advanced RCC

Even with a caval thrombus, complete surgery can cure.

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Clamp infrarenal IVC, then contralateral renal vein, then suprarenal IVC; occlude lumbars
III	Intrahepatic, below diaphragm	T3b	Mobilise liver and milk thrombus below hepatic veins; or suprahepatic clamp with Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass with or without hypothermic arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins; graft/reconstruct

- Image close to surgery, because the thrombus grows. **No routine renal artery embolisation** (more deaths). Do coronary angiography before bypass (levels III–IV) and use intraoperative transoesophageal echocardiography.
- Level and wall invasion predict survival, and level IV without metastasis is curable. Mortality is 5–10% for supradiaphragmatic thrombus.

Locally invasive (T4)

- Only ~40% of suspected invasion is confirmed. En bloc resection is the only cure, but prognosis is poor.

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant. With ≥2 risk factors (≥10 cm, cT3/T4, grade 3–4, sarcomatoid change, necrosis), >40% are node-positive.
- LND gave no survival benefit (EORTC 30881), so it is done for suspicious nodes, mainly for staging.

9 · Recurrence, adjuvant therapy & follow-up

A third of recurrences come late; one adjuvant drug works.

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, radiofrequency ablation 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone

Local recurrence	After RN	After PN	After TA
Management	Only 20–40% isolated, so do a full work-up; en bloc resection gives 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- A positive PN margin is usually surveilled; completion nephrectomy is only for gross involvement.
- About 20–35% metastasise after "curative" surgery, but cytokines, vaccines and most TKIs failed as adjuvants. **Adjuvant pembrolizumab** is FDA-approved for high risk.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging starts at 3–6 mo and intensifies with risk; very high risk is followed for up to 10 y. A microscopic positive margin moves the patient up one level. **30% of recurrences occur after 5 y**.
- Benign pathology needs no imaging. After TA, do contrast imaging within 6 mo, then follow the intermediate schedule.

10 · Other renal malignancies

Lymphoma and metastases need biopsy and drugs, not nephrectomy.

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins; the first resection is the best chance; margin and grade give prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%); massive adenopathy, splenomegaly; prior lymphoma means recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP with or without radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung primary commonest; multifocal, 5–30 HU	Biopsy; systemic therapy; nephrectomy only for refractory bleeding
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery and platinum
Primitive neuroectodermal tumour	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than in children

- In a suspected sarcoma, any RCC feature excludes it (the differential is sarcomatoid RCC).

> **KEY POINT**
> Non-RCC malignancies are almost all **hypovascular**. The key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis: biopsy, don't resect). RCC nodes are interaortocaval (right) or para-aortic (left), so nodes elsewhere suggest lymphoma.

11 · Landmark trials

Trial	Design	Result	Take-home
EORTC 30904	Randomised trial (RCT), RN vs elective PN, ≤5 cm, normal contralateral kidney	PN better function; 10-y overall survival (OS) better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT, RN with or without LND	No survival benefit	Underpowered, low-risk (4% node-positive); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival (DFS) 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS hazard ratio 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry, AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

1 · Epidemiology & risk factors

UTUC is often invasive at diagnosis, and its causes act as a field effect.

P3 P2 Upper Tract Urothelial Carcinoma: Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management; Etiopathogenesis and management of Upper tract TCC — epidemiology, tobacco/aristolochic acid/Lynch/occupation/phenacetin/arsenic, seeding vs field effect, CTU, cytology, URS biopsy, TNM, EAU risk groups, management

Not in the paper answer:

- Upper tract urothelial carcinoma (UTUC) is **5–10%** of urothelial cancers, but **40%** of renal neoplasms in the Balkans. Concurrent bladder cancer occurs in **17%**, ~60% are invasive at diagnosis, and multifocality is **10–20%**.
- After UTUC, the bladder recurs in **22–47%**.
- **EAU: screen for Lynch syndrome (LS) at the first interview** if age <60, a personal LS-spectrum cancer, a first-degree relative (FDR) <50 with LS cancer, or two FDRs with LS cancer.

> **KEY POINT**
> **Why UTUC presents invasive:** the muscle layer is thin, so tumours reach muscle, parenchyma or periureteric fat earlier than bladder tumours.

- **Smoking** carries a relative risk (RR) of **2.5–7**. Chronic infection or stones (RR 1.5–2) favour squamous cell carcinoma (SCC). Cyclophosphamide, long-term anthranoid laxatives and pelvic radiotherapy are iatrogenic causes.

> **KEY POINT**
> Aristolochic acid, arsenic, phenacetin and Lynch all behave as a **field disease** (multiple, bilateral recurrences). This favours kidney-sparing treatment, but Balkan endemic nephropathy patients often have renal insufficiency.

- **Lynch**: UTUC risk is ≈**14×**, and ≈**75×** in MSH2 carriers. If mismatch repair proteins are lost on immunohistochemistry, do germline testing.

2 · Pathology & molecular biology

- Over 90% are urothelial carcinoma. Variant histology (≈**25%**: squamous, glandular, sarcomatoid, micropapillary, neuroendocrine) is aggressive, and **micropapillary** is independently adverse.
- Ureteric inverted papilloma harbours malignancy in ≈**18%**, so follow it for ≥2 yr.
- SCC (0.7–7%) is commoner in the pelvis and invasive at presentation; adenocarcinoma is rare and linked to stones.
- Microsatellite instability distinguishes UTUC from bladder cancer, is prognostic and flags LS; the **RAS-mutated** subtype does worst.

3 · Diagnosis

Biopsy gives grade but not stage.

- Haematuria is the commonest presentation, then flank pain. **Always do cystoscopy**; positive cytology with normal cystoscopy (including urethra) means the upper tracts must be evaluated.
- **CT urography (CTU)** is first line (EAU prefers it to MR urography): sensitivity ≈**92%**, specificity ≈95%, but flat lesions are missed. Hydronephrosis means an invasive ureteric tumour in up to 80%.
- For MR urography, avoid gadolinium if creatinine clearance <30 mL/min. Do a split-function scan before radical surgery.
- Cytology detects ≈65% overall (G1 20%, G3 75%); selective washing beats voided urine, and it should be **collected before contrast**.
- Flexible ureteroscopy (URS) with biopsy raises accuracy to ≈90%; biopsy grade is ≈90% accurate, but visual grading only ≈70%.

> **KEY POINT**
> **Why URS biopsy cannot stage:** samples are small and shallow, so a Ta biopsy is often T3 at resection. Stage is **inferred** from grade, appearance, hydronephrosis and cytology, which is why kidney-sparing patients need intensive follow-up.

- **Diagnostic URS indications:** uncertainty, planned kidney-sparing treatment, solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence without changing survival, so avoid it when CTU and cytology already justify RNU.

4 · Staging & grading

Pelvic T3 vs T4 is a classic trap.

TNM 8th	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal; pT3b = macroscopic or peripelvic fat (worse)
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single node ≤2 cm / single >2 cm or multiple

- AJCC stages: I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**.
- Pelvic and upper ureteric tumours drain to hilar and retroperitoneal nodes, and lower ureteric tumours to pelvic nodes. WHO grades are PUNLMP (managed as low grade), low and high grade.

5 · Prognosis & risk stratification

The risk group decides between kidney-sparing treatment and RNU.

- **Stage matters most**, then grade. SEER 5-yr cancer-specific survival after RNU: ≤T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.

- **Ureteric and multifocal tumours do worse** than pelvic ones at the same stage, because there is no parenchymal barrier and the periureteric lymphovascular plexus is rich.
- Age alone is never a reason to deny RNU. Do **RNU within 12 wk** for invasive disease. Size **2 cm** is the best cut-off for ≥pT2.
- Post-op adverse factors include nodal disease, lymphovascular invasion (never seen on biopsy, so report it on every specimen), positive margins, necrosis, sessile growth, CIS and variant histology.

Low risk — ALL	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
<2 cm	>2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

KEY POINT

Every pre-op factor is a stand-in for invasion, so one adverse feature means high risk. **Trap:** a small, low-grade, unifocal tumour **with hydronephrosis is high risk**.

6 · Recurrence patterns

- Bladder recurrence after UTUC is 15–75% at 5 yr, usually within 2 yr.
- Upper tract tumours after bladder cancer occur in 2–4%, late (median >3 yr). About **30%** of bacillus Calmette–Guérin (BCG)-treated bladder CIS (or cystectomy for BCG-refractory disease) develop UTUC.

KEY POINT

Asymmetric recurrence: antegrade seeding makes bladder recurrences common, early and monoclonal, while the field effect gives multifocal tumours of different clones. Both operate.

- Intravesical recurrence after RNU is linked to male sex, prior bladder cancer, ureteric or multifocal tumours, laparoscopic or extravesical cuff technique, positive margins and diagnostic URS.
- **Pan-urothelial disease** (bladder plus ≥2 extravesical sites) has no evidence-based algorithm; note that cystectomy before RNU may lose cisplatin eligibility.

7 · Management overview & follow-up

ALGORITHM

1. Work-up: cystoscopy and CTU, with or without URS biopsy and cytology; assess contralateral function; screen for LS; then assign the risk group.
2. If **low risk**, then kidney-sparing treatment (endoscopic or segmental) with intensive surveillance.
3. If **high risk**, then RNU with bladder cuff, with or without LND, within 12 wk.
4. If **cN+**, then systemic chemotherapy first, and surgery only if there is a response.
5. After RNU, give a single intravesical chemotherapy dose; all patients have cystoscopic follow-up.

- Pre-op nomograms (≥pT2 or non-organ-confined) select patients for neoadjuvant chemotherapy, given before RNU because nephrectomy often removes cisplatin eligibility.
- Lymph node dissection (LND) gives the best staging, but its curative role is debated (technique and POUT: Ch 139).
- Follow-up: cystoscopy for all. After kidney-sparing treatment, add URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory), do yearly upper-tract CT. Surveillance is prolonged because the risk is constant.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Epidemiology, risk factors, pathology, staging and imaging: Ch 138

1 · Principles & patient selection

Tumour risk and renal reserve together decide between removing and saving the kidney.

P3 Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management — AUA/SUO favourable/unfavourable groups and ablation role, elective vs imperative KSM, URS and percutaneous steps, lasers, segmental options, RNU principles, abandoned cuff techniques, topical BCG/UGN-101, follow-up

Not in the paper answer:

- Endoscopy carries ~3× more reinterventions than RNU; ipsilateral recurrence after segmental resection is 33–55%; no routine adrenalectomy.
- **Standard** treatment of upper tract urothelial carcinoma (UTUC) is radical nephroureterectomy (RNU) with bladder cuff, if the other kidney is normal and the patient fit.
- UTUC is **unforgiving of spillage**, so every approach keeps the collecting system closed.
- Work-up is CT or MR urogram, cystoscopy, then ureteroscopy (URS) with biopsy and selective cytology; URS raises accuracy from ≈75% to 85–90%.
- Keep diagnostic endoscopy brief and low-pressure to avoid seeding.
- Low-grade (LG) tumours are mostly Ta/T1; high-grade (HG) tumours are mostly invasive.

*AUA/SUO risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG with obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, invasion on imaging, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

*American Urological Association / Society of Urologic Oncology.

KEY POINT
Low grade does not mean automatic ablation: obstruction or multifocality makes an LG tumour unfavourable, where RNU keeps a role.

KEY POINT
Two axes: tumour risk and renal reserve. High risk with adequate reserve means extirpation; low risk means preservation. Imperative indications are where they conflict.

ALGORITHM

1. Imaging suspicion: CT or MR urogram.
2. Endoscopy with biopsy and cytology; assign risk group; shared decision.
3. If high risk with adequate glomerular filtration rate (GFR), then RNU with or without neoadjuvant chemotherapy.
4. If low risk or poor GFR, then kidney-sparing management (KSM).

- Imperative KSM must not significantly compromise cancer control; dialysis risk is weighed against oncological risk.
- Unfavourable lower ureteric tumours need distal ureterectomy, not endoscopy.
- **Expectant management** (AUA/SUO option) is only for major comorbidity or high risk of end-stage renal disease (ESRD); untreated UTUC progresses.

2 · Endoscopic (kidney-sparing) management

- **Grade** is the key selector and the strongest predictor of recurrence. AUA/SUO: tumours **<1.5 cm**; caution at ≥1.5 cm.
- For a bulky or suspected HG tumour, use a **staged approach**: biopsy and partial ablation only, avoiding perforation, and complete it once pathology confirms suitability.
- Versus RNU, survival is similar but there are **~3× more reinterventions** and lifelong surveillance.

	Retrograde ureteroscopic	Antegrade percutaneous
System	**Closed**	**Open** – tract seeding ≈0.3%, all HG
Best for	Low-volume ureteric, small renal	Large pelvic/upper ureteric, lower-pole calyx, diversion
Staging	Not possible	Possible – base sent separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical therapy	Repeat URS; catheter or stent reflux	Via retained nephrostomy

- **URS:** saline barbotage improves cytology, and a tight ureter is pre-stented rather than forcefully dilated. The usual method is a **3-Fr biopsy, then laser**; resection never goes beyond the lamina propria.
- A tumour protruding from the orifice needs ablation or resection of the whole intramural ureter.
- **Never fulgurate circumferentially**, because a stricture follows.
- Lasers: holmium:YAG <0.5 mm, ideal for the ureter; neodymium:YAG 5–6 mm, with transmural risk; thulium <0.2 mm, with strong coagulation.
- **Percutaneous:** puncture the calyx distal to a peripheral tumour, never through it, and dilate to 30 Fr. Resect with cold-cup forceps or a **bipolar** resectoscope, since monopolar risks hypo-osmotic absorption.
- A **nephrostomy is always left** for second look and topical therapy. HG or invasive histology means RNU.
- In a diversion, use flexible or antegrade endoscopy.

KEY POINT

Does URS spread tumour? Prior ureteroscopy increases bladder recurrence (seeding), but pre-RNU URS has not increased metastasis. A single post-op intravesical instillation mitigates this.

3 · Segmental ureterectomy

- It saves nephrons **and** gives a staging specimen; in selected patients it equals RNU.
- **URS of the entire tract above is mandatory**. For HG, add ipsilateral pelvic lymph node dissection (LND).
- Ipsilateral recurrence is **33–55%**, mostly distal and lifelong.
- Segmental resection with uretero-ureterostomy is rare and needs 1–2 cm margins with frozen section.
- Distal ureterectomy with reimplant suits LG low-stage tumours, or selected HG distal ones when the kidney must be kept.
- **Boari flap:** contraindicated by a small or irradiated bladder. The base is ≥2 cm wider than the apex, and the width ≥3× the ureteric diameter.
- **Ileal ureter:** 20–25 cm of ileum ≥15 cm from the ileocaecal valve, isoperistaltic.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap with hitch	Further 10–15 cm	Adequate capacity
Renal descensus with nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · Positive cytology & carcinoma in situ (CIS)

Trace a positive cytology before removing any kidney.

ALGORITHM

1. Positive voided cytology: repeat it.
2. CT urogram and bladder biopsies. If the bladder is positive, then treat the bladder.
3. If still positive with a negative bladder (or after BCG "6+1"), then selective upper-tract cytology, retrograde pyelography (RPG), prostatic urethral biopsy (men) and URS.

- Upper-tract CIS is a diagnosis of exclusion: persistently positive selective cytology with normal URS and imaging.
- **No RNU for cytology alone**, because of false positives and later contralateral disease.
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy; RNU if more than surface disease is proven.
- CIS with HG papillary UTUC needs radical or multimodal treatment.

5 · Topical therapy in the upper tract

- Indications are solitary kidney, bilateral UTUC and ESRD risk. Routes are nephrostomy (≤25 cm H_2O), ureteric catheter or stent reflux.
- Always use **low pressure and sterile urine**, as sepsis deaths are reported.

- Adjuvant instillation after papillary ablation has **no proven benefit**. BCG is acceptable after complete ablation of HG favourable Ta/T1; its commonest complication is renal granulomas.
- **UGN-101 (Jelmyto)** is mitomycin C (MMC) in a reverse-thermal gel, FDA-approved in 2020 for LG UTUC **not endoscopically manageable**. Stricture occurs in ≈44%.

6 · Radical nephroureterectomy

Remove the whole ureter and cuff without spillage.

- **Indications:** non-metastatic high-risk pelvic or proximal ureteric tumour (HG, obstruction, parenchymal invasion), with cuff and LND; or LG with unfavourable features or failed or infeasible endoscopy.
- **Kidney first, cuff last. Clip the ureter early**, control the hilum before manipulation and stay outside Gerota's fascia. **Adrenalectomy is not routine.**
- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%** and cannot be surveyed. After bowel diversion, excise the ureteroenteric anastomosis.
- Extravesical or transvesical excision is acceptable; lasso, pluck, stripping and stapling are abandoned because of spillage and positive margins. Close in two layers.
- Laparoscopic or robotic: wide perihilar margin and extract **intact in a bag**.
- Ureteric blood supply is anteromedial in the upper third, medial in the middle and **lateral** in the lower third.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

> **KEY POINT**
> Principle beats approach: port-site seeding and carcinomatosis follow spillage, not the approach.

Lymphadenectomy

P4 Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — UTUC templates (right interaortocaval), ≥8 nodes, chylous leak; neoadjuvant rationale, EA8141, AUA/SUO CrCl <50, POUT

Not in the paper answer:

- Outcome ranks pN0 better than pNx, better than pN+; the prognostic cliff is T2 to T3.
- LND has value in **T2–T4**, especially for pelvis and proximal ureter, and the surgeon, not the approach, sets the yield.
- Caudal limits: renal pelvis to the inferior mesenteric artery; upper two-thirds of ureter to the aortic bifurcation; lower third to the pelvis.

7 · Perioperative adjuncts

- Bladder tumours follow in 30–50% after RNU or endoscopy, mostly within 2 yr. **AUA/SUO strongly recommend** a single dose after RNU (optional after endoscopy).

- Post-op (day 3–7), do a cystogram, then give MMC or gemcitabine with a 1–2 h dwell. An **intraoperative** instillation, clamped until the ureter is clipped, needs no cystogram.
- **Renal function is the hinge**, so give platinum **before** nephrectomy. Neoadjuvant is favoured if post-op creatinine clearance (CrCl) is expected **<50** mL/min.
- In EA8141, patients with CrCl >50 received accelerated MVAC ×4.
- Adjuvant (POUT): gemcitabine–cisplatin (CrCl >50) or gemcitabine–carboplatin (30–49). cN+ disease receives chemotherapy, then consolidative surgery.

8 · Complications

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; no circumferential fulguration
Percutaneous	Bleeding, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · Follow-up

- Every visit includes **cystoscopy**, cytology and a metabolic panel; after KSM add ipsilateral URS.
- Use CT or MR urography (RPG if no contrast) and chest imaging.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	With cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · Landmark trials

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% vs 16% at 1 yr	Single-dose instillation standard
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU	Bladder recurrence 42% vs 17% at 2 yr	Confirms single post-RNU instillation
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable

Trial	Design	Result	Take-home
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

141 Contemporary Open Surgery of the Kidney

1 · Role & preoperative work-up

- Open surgery offers **exposure and early great-vessel control**. It suits renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours and a hostile abdomen (minimally invasive surgery: Ch 142).
- **Renal function** decides partial (PN) versus radical nephrectomy (RN). Estimate glomerular filtration rate (GFR) with the **CKD-EPI 2021 race-free** equation, which beats MDRD.
- If GFR is **<60** or there is significant proteinuria, ask for a perioperative nephrology consult.
- In advanced RCC, check blood count and metabolic panel (MSKCC and IMDC risk models), liver tests (**Stauffer syndrome**) and cardiopulmonary fitness.
- Contrast is generally safe if GFR is **>30**. If thrombus is suspected, MR or CT venography must be **≤1 month old**, because thrombus propagates rapidly.
- Give cefazolin ≤1 h pre-incision, pneumatic compression and subcutaneous heparin 5000 IU; no routine bowel preparation.
- **Renal artery embolisation** lets the vein be tied first when the hilum is hidden, but it is no longer routine because of postinfarction syndrome and harder surgery if delayed >48 h.

2 · Surgical anatomy & approaches

Most disasters come from misread anatomy.

- At risk on the **right** are liver, duodenum and colon; on the **left**, spleen, pancreatic tail, colon and superior mesenteric artery (SMA).
- The **right renal artery** lies behind the renal vein and IVC and can be reached in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- The **SMA** leaves the front of the aorta **1–2 cm above LRV** and veers left, so it can be mistaken for the left renal artery.
- The right gonadal and adrenal veins drain into the IVC; the left ones drain into the LRV.
- Anterior exposure reflects the colon along the **white line of Toldt**. On the right, add a **Kocher** manoeuvre without cautery, because duodenal injury is devastating.
- **Cattell-Braasch** (right colon and midgut mobilisation) and partial **Mattox** (left medial visceral rotation) expose the great vessels.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour or IVC thrombus (early aortic access); bilateral	Bowel mobilisation needed
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN with thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN or PN; 8-cm mini-flank above the 11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at an old rib tip	Rib division

Incision	Best for	Limitations
Thoracoabdominal, dorsal lumbotomy	Largely historical; thoracoabdominal is still an option for level III–IV thrombus	

- The **flank** position lowers venous return and lung function, so it is poor in cardiopulmonary disease; pad against rhabdomyolysis.
- In the **supracostal** approach, push the pleura up and divide the intercostals close to the **upper border of the rib** to spare the bundle. Spare the **subcostal nerve**.

3 · Simple nephrectomy

- Stay **within Gerota's fascia**, on the capsule, for a benign nonfunctioning kidney (obstruction, infection, stones, reflux), and spare the adrenal.
- Benign PN suits focal atrophy, a duplex pole or a symptomatic angiomyolipoma.

4 · Radical nephrectomy

- Robson's operation included adrenal and extended nodes; now the **normal adrenal is spared** and lymph node dissection (LND) is selective.
- **Indications:** tumours not amenable to PN, such as a nonfunctioning kidney, tumour replacing the kidney, nodes, cytoreduction or venous thrombus.
- Staging is abdominal CT or MRI and **chest CT**; a bone scan only for symptoms or raised alkaline phosphatase.
- Biopsy only if it changes the plan, such as an infiltrative mass or before systemic therapy.
- T4 invasion means altered enhancement of the organ, not just contact.

ALGORITHM

1. **Right:** survey the abdomen; open Toldt's line down to the iliac vessels; take down the hepatic flexure; Kocher; expose the IVC.
2. Loop and palpate the renal vein; find the artery behind it; tie **artery, then vein**; remove the kidney outside Gerota's.
3. **Difficult hilum:** ligate the artery in the interaortocaval space, so the kidney decompresses; then double-ligate the flaccid vein.
4. Alternative: start at the lower pole and lift the kidney from posterior to anterior to show the artery.
5. **Left:** mobilise the splenic flexure down to the iliacs; divide the renocolic ligament (pad the pancreas).
6. Follow the aorta or gonadal vein to the LRV. Tie the artery above and behind it, confirm occlusion, then the vein.

KEY POINT

Why artery first? Tying the vein first leaves the kidney engorged, so collaterals bleed. Arterial ligation shrinks the kidney and softens the vein, which can then be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or the SMA has been tied.

- **Lumbar veins:** use silk ties and **no clips on the IVC side**, because clips block emergency caval clamping.

- **Lost pedicle control:** tell the anaesthetist, compress, use two suckers and **never clamp blindly**.
- **Adrenalectomy** only for an upper-pole tumour with a lost plane, an adrenal mass or diffuse involvement. Metastasis occurs in **<5%**, and contralateral metastasis is as common as ipsilateral.
- **LND:** nodes are positive in ≈5%, and the EORTC RCT showed no overall survival benefit, though it included many low-stage tumours.
- **Do LND if nodes are enlarged**: a short axis **≥1.5 cm ≈ 50% positive**. Consider it in high risk (>10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- The template runs from crus to bifurcation. **Clip lymphatics above the renal vein** to prevent chylous ascites.
- **Closure:** fill with saline and give a big breath; bubbles mean a pleural tear, which is sutured over a 14-Fr catheter. Take a **chest X-ray in all**.

5 · IVC tumour thrombectomy

Surgery is the only cure; thrombus level decides the plan.

- RCC is the commonest cause (**5–10%** of RCC). Untreated, most die within 1 yr, so surgery (RN and LND, with or without systemic therapy) is the backbone.
- **Tumour thrombus** differs from **bland thrombus** (clot below the obstruction). **MR venography** best shows extent, wall invasion and bland thrombus.
- Anticoagulate for venous thromboembolism, bland thrombus or IVC occlusion. Use **no suprarenal filters**, because thrombus grows onto the device.
- Use **transoesophageal echo (TEE)** for levels II–IV, and involve a **cardiothoracic surgeon for III–IV**.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At or above major hepatic veins, below diaphragm	Liver mobilisation with or without Pringle; occlusive thrombus: suprahepatic clamp; nonocclusive: venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) with deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip or staple); do not extract
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

1. **Level I (right):** control the IVC before handling the tumour; ligate the right artery interaortocavally.
2. Loop the LRV and the suprarenal and infrarenal IVC with **Rummel tourniquets**.
3. Milk the thrombus toward the ostium; place a Satinsky round the ostium with the thrombus inside the jaws.
4. Do an ostial cavotomy keeping ≥half the IVC width, and extract.
5. **Level I–II:** ligate the renal artery early. Tighten tourniquets in order: **infrarenal IVC, contralateral renal vein, suprarenal IVC**.
6. Open the cava, extract, flush and inspect the wall. Close with 4-0 Prolene.
7. Release the **caudal clamp first**, flush on a held breath, then release the cranial clamp.

- A **left tumour (level II)** is harder: tie the left artery first, then Kocher, **split-and-roll** the IVC and excise the LRV ostium en bloc. Do not clamp the right renal artery.
- **Level III:** keep bypass on standby and mobilise the liver as in transplantation. The **major hepatic veins are never divided**, and **Budd-Chiari syndrome is a contraindication**.
- The **Pringle** manoeuvre is needed if clamping above the hepatic veins. Limit it to ≤60 min, ideally **≤20 min**.
- **Level III sequence:** clamp infrarenal IVC, contralateral renal vein, porta hepatis, then suprahepatic IVC; open and extract. Unscrapable wall invasion needs IVC resection.
- Do a 2–5 min **clamp trial**. If cardiac output falls by >50% or mean arterial pressure by >30%, use bypass. Occlusive thrombus tolerates clamping because of azygos collaterals; free-floating thrombus does not.
- Incomplete tumour removal is universally fatal, so never compromise IVC control to save time.

	VVB	CPB with DHCA
Access	Femoral to jugular, percutaneous	Sternotomy; cool to 16–18 °C and drain 95% of blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar and intercostal back-bleeding	Cardiac and neurological events
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- In **VVB**, place cannula tips far from the thrombus, because dislodgement causes massive pulmonary embolism (PE).

P3 Vascular grafts and vascular surgery in urology — caval repair: ≥50% lumen primary closure, else pericardial patch (Leibovich–Bower width), circumferential: PTFE 16–20 mm + lifelong aspirin 81 mg; left tumour needs right-kidney drainage

Not in the paper answer:

- Suprarenal IVC ligation with a right tumour is acceptable if the LRV is tied caval to its tributaries, which act as collaterals.

KEY POINT

"Shortcuts" that become "longcuts": opening the IVC before all inflow is controlled; side-biting a level I; clips on the IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping the test clamp.

6 · Partial nephrectomy

P2 Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches — PN standard for T1; imperative/relative/elective; relative contraindications (<20% nephron mass, pedicle encasement, thrombus, cT4, nodes); hereditary observe to 3 cm except HLRCC

Not in the paper answer:

- Hyperfiltration injury (focal segmental glomerulosclerosis, proteinuria) is rare unless nephron mass falls by >80%.
- Multifocality is ≈2% in clear cell and ≈10% in papillary RCC, so inspect the whole kidney with intraoperative ultrasound.
- PN is size-independent if it is safe; collecting-system invasion is a relative contraindication.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss and stay

KEY POINT

PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes. Hypothermia and mannitol are optional.

- Always **control the hilum first**, even if not clamping, so a clamp can go on at once.
- **Enucleation** follows the pseudocapsule plane; **resection** takes a 5–10 mm rim and is usually clamped.
- **Renorrhaphy:** the inner layer closes vessels and collecting system with absorbable sutures (permanent sutures become a stone nidus); the outer layer is capsular over a bolster.

7 · Complications & management

Injury	Management
SMA	Injured when mistaken for the left renal artery or by Cattell-Braasch traction. Potentially lethal, so call a vascular surgeon for primary repair or graft
Inferior mesenteric artery or vein (IMA/IMV)	Ligation safe if the marginal artery of Drummond is patent
Superior mesenteric vein (SMV)	Ligation causes bowel ischaemia, splanchnic hypertension and hypotension. **Never ligate**; repair with 6-0 Prolene venorrhaphy. If transected, reanastomose or use autologous vein; leave the abdomen open
Duodenum	Haematoma narrowing the lumen: incise serosa and muscularis, drain. Tear: debride, 2-layer closure, omentum, drain, nasogastric tube
Pancreas	Glue and drain (check amylase); ductal injury: general surgeon; late collection: percutaneous drain, nil by mouth, octreotide
Pleura	Purse-string over a 14-Fr catheter, evacuate on a positive-pressure breath; large tear: chest tube

- In thrombectomy, air embolism is prevented by releasing the caudal clamp first. Use **no cell saver**, because it spreads tumour.
- A **urine leak** mostly appears within 72 h. If drain output is >50 mL/24 h, a drain creatinine above serum means urine; a persisting urinoma needs a double-J stent and Foley, and most resolve in 4–6 wk.
- A significant **pseudoaneurysm** (flank pain, haematuria or shock) needs **angiography** with **selective embolisation**, because CT angiography can miss it and cannot treat it.
- An **arteriovenous fistula** is observed, then coil-embolised, with surgery if complex.
- **Renal insufficiency** is usually mild and transient; exclude obstruction, drug toxicity and thrombosis.

142 Laparoscopic and Robotic Surgery of the Kidney

1 · Principles & patient selection

Fitness here means fitness for the gas.

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ; first laparoscopic partial nephrectomy (PN): Winfield, 1993.
- Minimally invasive surgery (MIS) gives less pain, blood loss and convalescence than open, with **equivalent long-term outcomes** and the same indications.
- **General anaesthesia is mandatory.** The CO_2 pneumoperitoneum impairs ventilation, venous return and cardiac output, so risk is high in chronic obstructive pulmonary disease (COPD) and cardiac disease.
- In severe COPD with hypercarbia, lower the pressure, then use helium, then convert.
- With the robot, **docking and undocking delay emergency access**, so the team must communicate closely.
- In obesity, **visceral adipose tissue**, not body mass index (BMI), predicts difficulty. Prior surgery is **not a contraindication**.

KEY POINT

The patient must tolerate the **pneumoperitoneum**, not the renal operation: cardiopulmonary reserve, not age or BMI, decides fitness.

2 · Access & port placement

Routes are transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted (HAL) and single-site (LESS).

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	Reached after colon reflection	Direct, especially the posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP** uses a 30–45° modified flank position with 3–5 trocars.
- **Robotic** arms are spaced **8 cm (Si) / 6 cm (Xi)**, forming a **broad-based triangle** with the camera at the apex.
- **RP** uses the full flank position; the space is made by finger or balloon below the 12th rib in the posterior axillary line.

KEY POINT

RP pitfalls: a balloon inflated within muscle causes a flank hernia; too anterior injures the peritoneum or colon; too posterior makes the quadratus or psoas bleed. Bail-out: open the peritoneum and continue transperitoneally.

- **HAL** gives tactile feedback and helps novices, donors and scarred kidneys; a hand through the port can compress a bleeding vessel.
- **LESS and the single-port robot** improve cosmesis but lose triangulation, with unchanged outcomes.

3 · Nephrectomy — operative steps

ALGORITHM

1. Reflect colon; find ureter and gonadal vessels on the psoas.
2. Lift kidney to expose the hilum; artery, then vein.
3. Upper pole (adrenal spared or taken); ureter and lateral attachments; bag; extract.

- Keep the **lateral attachments** until late, or the kidney rolls over the hilum. On the right, dissect cold near the duodenum and do a **Kocher**.
- The **ureter** lies just behind the gonadal vessels and is a useful handle.
- **Hilum:** divide the **artery first, then the vein**, with an endo-GIA stapler. **Never leave clips in the staple line.**
- **En bloc hilar stapling** is safe (no arteriovenous fistula, AVF) and saves ~45 min.
- **Extraction:** intact in a bag, or morcellated in an impermeable **LapSac**, which keeps type and grade but loses size and stage.

4 · Simple vs radical nephrectomy

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple** nephrectomy is for benign nonfunctioning kidneys, such as chronic pyelonephritis, xanthogranulomatous pyelonephritis, obstruction, reflux or tuberculosis; after pyonephrosis or stones, use hand assistance or a subcapsular approach.
- **Radical nephrectomy (RN):** MIS equals open oncologically at 5 and 10 yr. In RCTs of RP versus TP, the only consistent RP advantage is **shorter operative time**.
- **Lymph node dissection (LND)** has staging value only, with no overall survival benefit; the AUA advises removing clinically positive nodes.
- Tumours >10 cm are feasible, though landmarks are lost; conversion is higher laparoscopic than robotic.
- Selected **inferior vena cava (IVC) thrombus** cases can be done by experts, with early arterial ligation that may retract the thrombus (Ch 141). Laparoscopic **cytoreductive** nephrectomy allows earlier systemic therapy.

KEY POINT

Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery.

5 · Partial nephrectomy

P2 P1 Nephron-sparing surgery / partial nephrectomy in RCC; Role of renal nephrometry in management of RCC — PN default where feasible; RENAL score predicts complexity, WIT, conversion; enucleation; TP anterior vs RP posterior

Not in the paper answer:

- In cT1b/T2 tumours, PN gives better estimated GFR with equal oncology, but **tumours >5 cm upstaged to pT3a recur more after PN**.
- The collecting system is entered in about 60%, so always drain.

ALGORITHM

1. Intraoperative ultrasound (size, depth); open Gerota's away from the lesion; score the capsule.
2. Clamp the hilum (Satinsky en bloc or bulldogs); cold-scissor excision.
3. Close vessels and collecting system; renorrhaphy; unclamp; drain.

- **Trifecta:** negative margin, no major complication within 30 days, and **WIT ≤25 min**.
- **Enucleation** gives less blood and function loss but more positive margins, with the same survival. It suits familial, multifocal, solitary-kidney and severe chronic kidney disease (CKD) cases.
- **Renorrhaphy:** sliding-clip and barbed sutures shorten warm ischaemia time (WIT), and **single-layer** closure gives better function and fewer pseudoaneurysms.
- The **collecting system** is entered in **~60%**; repair it with absorbable suture and **always drain**.
- **Ischaemia:** the historical warm limit is **30 min**, but **preserved parenchymal volume predicts function better than WIT**. Use cold ischaemia for a solitary kidney, CKD or a long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery and vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

KEY POINT

No clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped.

- Robotic PN has fewer conversions and shorter WIT than laparoscopic, and less blood loss than open.
- **Positive margins (0–7%)** matter only in high-risk tumours, so the answer is **surveillance, not reoperation**.
- **Port-site recurrence is ~0.1%** and follows spillage, not the gas.

- A **hilar** tumour touches the renal artery or vein; consider enucleation with superficial renorrhaphy bites.
- Salvage surgery after PN or ablation often ends in RN, because the planes are lost.

6 · Other renal procedures

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first proves the cyst causes the pain; excise the wall, biopsy, fulgurate the base
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex
Nephropexy	Descent >5 cm or 2 vertebrae from supine to erect, with pain, reduced flow or obstruction	Usually right side, slim young women; 20% asymptomatic. Fix the capsule to psoas or quadratus fascia
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Suture the neck, obliterate the cavity with fat
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation with or without omental wrap
Ablation	≤3 cm, comorbid, solitary kidney, hereditary RCC	Percutaneous preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour. Radiofrequency ablation (RFA): 60–100°C (>105°C vaporises), 1 cm margin, no real-time imaging; more retreatment

7 · Complications & management

Dangerous complications happen out of view, so seek them.

P1 Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy — vascular injury commonest (<1/1000, fatal 8–17%); venous bleed control; gas embolism (left side down, head down, aspirate); pneumothorax; thermal bowel injury, CT with oral contrast; rhabdomyolysis

Not in the paper answer:

- The overall complication rate is **~15%** (PN > RN > simple), with a learning curve of about 20 nephrectomies.
- **In left RN, the superior mesenteric artery (SMA) can be mistaken for the renal artery**, leading to bowel necrosis.

ALGORITHM

1. Major bleed: locate; temporary control (grasper or pressure); inform the anaesthetist (blood, open set).
2. Clear the field: raise pneumoperitoneum or add a trocar; then definitive clip or suture.
3. If that fails, then convert and call a vascular surgeon.

- Vascular injury happens at **access** or at the hilum. **Never clip or suture blindly.**
- At the end of the case, **lower the insufflation pressure** to unmask venous bleeding.
- **Haematuria at ~day 12 after PN** suggests a pseudoaneurysm or AVF and is treated by selective embolisation.

- A **urine leak** after PN (0.8–10%) is drained percutaneously, with a stent if it persists.
- **Bowel injury** (<1%, mortality ~3%) is thermal in about half. It presents with **pain at a single trocar site**, distension, diarrhoea and a **normal or low white count**.
- Pancreatic injury is insidious, so inspect at the end. A diaphragm injury shows as a sudden rise in airway pressure and is closed with a purse-string, followed by a chest X-ray.
- Bowel injury is repaired laparoscopically if stable, or by laparotomy if unstable or with gross spillage.
- **Neuromuscular** injury (2.7%) includes flank rhabdomyolysis, treated by hydration.

> **KEY POINT**
> MIS complications are the open ones in altered form and timing: the lethal ones (access vessel injury, thermal bowel injury) happen out of view and present late or atypically.

8 · Laparoscopic live donor nephrectomy †

P4 P1 Laparoscopic donor nephrectomy — left preferred (longer vein), donor keeps better kidney; gonadal–ureter packet; artery then vein; no Hem-o-lok on donor artery; mortality 3/10 000; lifetime ESRD risk

Not in the paper answer:

- † Donors need lifelong annual blood pressure and renal function checks.

143 Focal Therapy for Renal Tumours

1 · Rationale & indications

Ablation suits patients whom surgery harms more than the tumour.

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do (Ch 137).
- **Partial nephrectomy (PN) is the gold standard** for cT1a. Thermal ablation (TA) is the alternative: cryoablation (CA), radiofrequency ablation (RFA) or microwave ablation (MWA).
- **Indications:** high surgical risk, solitary kidney, bilateral tumours, or hereditary syndromes (von Hippel–Lindau, VHL) needing repeat treatment. Recovery is faster and renal function equal or better than PN.

KEY POINT

Trap: TA is never the default for a fit patient.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a **≤3 cm**, especially comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	For frail/comorbid; TA not routinely **>3 cm**; CA not **>4 cm**
National Comprehensive Cancer Network (NCCN) 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · Selection & work-up

The tumour stays in place, so biopsy gives the only histology.

ALGORITHM

1. Counsel: age, comorbidity, CKD risk, size and location.
2. Biopsy, then re-counsel with histology.
3. Choose PN, surveillance or percutaneous ablation.

- **The AUA advises biopsy of all**, ideally at a **separate earlier session**. Post-ablation, biopsy only a new enhancing lesion.
- Treat urinary infection first; colonised urine (ileal conduit) risks abscess.
- Complication predictors are the RENAL score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (about 3×) and $(MC)^2$ (CA only). Day-case suits ≤3 cm, low complexity and no anticoagulation.

3 · Mechanisms

P2 Biological effects of cryosurgery — freeze/thaw cell injury, vascular thrombosis, apoptosis, immunomodulation; argon cools/helium warms; kidney dies at −19.4 °C; <−20 °C only 3 mm inside iceball edge

Not in the paper answer:

- Cryo target is **≤−40 °C**; the iceball extends **5–10 mm beyond tumour**, pushing the −40 °C isotherm past the tumour edge.
- Heat: **≥60 °C** causes coagulative necrosis; ≥100 °C causes vaporisation and charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**.
- **Heat sink:** large vessels carry heat or cold away, so perivascular tumour survives. Counter with colder or longer freezes, more probes, or MWA.

4 · Choice of energy

- RFA electrodes vary: the cool-tip gives the largest zone, the umbrella the most spherical, and a single monopolar tip suits only ≤2 cm.

	Cryoablation	RFA	MWA
Physics	Argon cools (**Joule–Thomson**); helium warms	450–1200 kHz current causes ionic friction	915 MHz/2.45 GHz causes **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No**; inferred from impedance and temperature	Gas bubbles on ultrasound (US)
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

KEY POINT

Why RFA is self-limiting: charring raises impedance and halts current (cap 105 °C, electrodes cooled). MWA radiates energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- In CA the second freeze enlarges necrosis. MWA for cT1 uses 45–65 W for 5–15 min.

5 · Technique

- **Percutaneous is standard.** Laparoscopic is kept for prior scarring or failed hydrodisplacement. General anaesthesia with jet ventilation reduces renal motion.

ALGORITHM

1. Give intravenous (IV) antibiotics.
2. Position prone or flank on the computed tomography (CT) gantry.
3. Place probes under image guidance; confirm on non-contrast CT.
4. Ablate, then contrast CT for completeness and haematoma.

- **Hydrodisplacement** pushes organs away (colon >1 cm) with saline or 5% dextrose (**D5W for RFA**, as saline conducts current). Use caution in cardiac failure.
- **Pyeloperfusion** (central tumours) irrigates a ureteral catheter at the **opposite temperature**: cold for RFA/MWA, warm for CA.
- **CA:** cryoprobes sit **1–2 cm apart** and come out only after full thaw. Bleeding rises with size and probe number (>4 cm). **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; overlap ablations for larger lesions.
- **MWA:** antennas go tangentially, never into the collecting system: 1 for ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm.

6 · Follow-up, success & salvage

With no specimen, imaging alone shows success.

- **Success is complete loss of contrast enhancement** on CT/MRI.
- **Incomplete ablation** enhances on the first scan and is re-ablated. **Local recurrence** later enlarges or enhances and needs biopsy. **Enlargement after any modality is recurrence until proved otherwise.**
- CA zones shrink >50% in year 1; RFA zones barely shrink (**fibrotic halo**).
- **AUA:** CT/MRI with and without contrast plus chest imaging at **6, 12, 24, 36, 48, 60 months**, then shared decision. US is not routine unless a contrast-enhanced (CEUS) protocol exists.
- Salvage is re-ablation, PN, radical nephrectomy (RN), or stereotactic body radiotherapy (SBRT) if comorbid.

7 · Oncological outcomes vs PN

- There is no randomised trial. Local recurrence is commoner after TA but **equal once re-ablation is counted**.
- **Metastasis-free and cancer-specific survival are equal**; overall survival is lower in these older, comorbid patients.
- **Size predicts failure:** RFA does worse at ≥3 cm, and T1b worse than T1a. No energy is superior.

> KEY POINT
> **Why comparisons are biased:** benign and under-graded tumours flatter TA; occult pT3a penalises it; non-surgical candidates die of other causes first. Few RCC deaths occur, so trials need huge cohorts.

8 · Complications & management

- Major complications (Clavien ≥3) are **≤5%**, fewer than PN; percutaneous beats laparoscopic, and MWA is lowest.

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Ablate (RFA/MWA coagulate); otherwise selective embolisation
Urinoma, fistula, pelviureteric junction (PUJ) or ureteric stricture	Pyeloperfusion; CA for central tumours	Conservative or stent; drain a large urinoma

Complication	Prevention	Management
Bowel or other organ injury (tumour <1 cm away)	Hydrodisplacement, positioning	Fistula: stent, low-residue diet; diversion if persistent
Pneumothorax (access above 12th rib)	Recognise during procedure	Aspirate; chest tube rarely
Nerve injury / RFA pad burn	Pads level on posterior thighs	Neuralgia is self-limiting
Infection <1%, may be lethal; abscess at 1–6 months	Urine culture, prophylaxis	†Drain and antibiotics

9 · Newer modalities (investigational)

- **High-intensity focused ultrasound (HIFU)** ablates incompletely; **histotripsy** (non-thermal) has no clinical data yet.
- **SBRT** needs no anaesthesia but leaves viable tumour on biopsy; it may have a salvage role.
- **Irreversible electroporation** spares vessels but controls cancer poorly.

144 Treatment of Advanced Renal Cell Carcinoma

1 · Overview

In metastatic RCC, drug choice rests on risk group and mechanism.

P2 Metastatic RCC (mRCC): assessment and management, including recent concepts — work-up, MSKCC/IMDC, first-line by risk, switch-mechanism later lines, CARMENA/SURTIME, metastasectomy 35–50% 5-yr, KEYNOTE-564, nccRCC

Not in the paper answer:

- About **1/3** present with metastases; a further **20–40%** of localised tumours metastasise later. mRCC 5-yr survival is about **12%**; 10-yr <5%.

Standard care is now **immune checkpoint (immuno-oncology, IO) combinations**.

2 · Prognostic models

- **MSKCC (Motzer)** came from the cytokine era; **IMDC (Heng)** came from first-line VEGF-targeted therapy.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	**>10 mg/dL**	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** from diagnosis to systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

KEY POINT

Three factors are shared. IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and short interval. The IMDC risk group **decides the first-line regimen**.

3 · Biology and drug targets

- The **VHL** protein (3p) tags **hypoxia-inducible factor (HIF)-α** for degradation; when VHL is lost, HIF drives VEGF, PDGF, TGF-α, GLUT-1 and erythropoietin. Drugs block the ligand (bevacizumab), receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mTOR inhibitors) or HIF-2α (belzutifan). MET/AXL/FGFR escape explains broader later TKIs.
- **CTLA-4** and **PD-1/PD-L1** switch off T cells. Blocking them gives autoimmune toxicity, and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory, the rationale for IO plus TKI. Clear cell RCC (ccRCC) is chemoresistant.

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (raised ALT)**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, HFS; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, hyperglycaemia, hyperlipidaemia, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · Surgery in metastatic disease

Surgery now serves systemic therapy; the question is who and when.

- **Cytoreductive nephrectomy (CN) rationale:** spontaneous regression of metastases is <1%, and a bulky primary suppresses T cells. CN helps only as a platform for systemic therapy.
- Fewer than 1/4 reach systemic therapy with ECOG 2, multiple sites, or brain or liver metastases.
- **CARMENA caveats:** about 45% were poor risk, about 25% of the CN arm were off-protocol, and it stopped early underpowered. It does **not** show harm in fit, low-burden, favourable patients.

ALGORITHM

1. Assess mRCC: histology, IMDC risk, PS and metastatic burden.
2. If fit, with low burden, favourable features and a resectable primary, then consider upfront CN followed by systemic therapy.
3. If intermediate/poor risk or high burden, then give systemic therapy first and offer deferred CN to responders.

KEY POINT

CN helps only patients fit enough, with disease slow enough, to reach systemic therapy; operating first on poor-risk patients delays it.

- **Metastasectomy** does best with complete resection, a solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm and metachronous disease, all markers of indolent biology.
- **Palliative** nephrectomy treats pain and paraneoplastic syndromes; try **angioembolisation for haematuria** first. Brain and spinal metastases get surgery and radiation.

5 · Systemic therapy for ccRCC

First line follows IMDC risk; later lines switch mechanism.

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, multiorgan failure; early mortality 2–5%

- IL-2 is now kept for carefully selected fit patients with clear cell histology and prior nephrectomy; IFN-α is not used first line.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT

Choosing: IO/TKI gives higher response and PFS and is better tolerated; IO/IO gives deeper durable responses (CR 9%) with autoimmune toxicity; never compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk (favourable did better on sunitinib in CheckMate 214). PD-L1 does not select. Use a TKI or temsirolimus if IO is contraindicated.

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Tivozanib is used after ≥2 prior therapies. TKI + mTOR and nivolumab + sunitinib/pazopanib combinations failed.

- **Ipilimumab + nivolumab toxicity:** grade 3–4 events **46%**, high-dose steroids 35%, discontinuation 22%. Dose: nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4.
- † For immune-related events, hold IO and give steroids for grade ≥2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥4 weeks. Grade 4 means permanent stopping.
- TKI toxicity is managed by dose reduction or interruption; mTOR inhibitors need glucose, lipid and phosphate monitoring.

6 · Non-clear cell RCC

- Non-clear cell RCC (nccRCC) is 15–25%. No agent is proven, so offer a **trial to all**.
- Papillary tumours respond to sunitinib in only 5–10%; temsirolimus beat IFN in the nccRCC subgroup.
- **Hereditary papillary RCC (HPRC)** carries a germline activating **MET** mutation (chromosome 7) and gives type 1 papillary RCC. Sporadic papillary has **trisomy 7 in >2/3**. MET inhibitors (savolitinib, crizotinib) are active here.
- **Hereditary leiomyomatosis and RCC (HLRCC)**: fumarate hydratase loss gives type 2 papillary RCC. The cells depend on glycolysis, so **bevacizumab + erlotinib** restricts glucose (partial response about 2/3).
- **Collecting duct** carcinoma is treated with **gemcitabine + carboplatin**; **sarcomatoid** tumours with gemcitabine-based therapy.

7 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Later comparator
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; patients preferred pazopanib (PISCES)
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS better, OS not	Little used
Temsirolimus phase III	vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer	Lenvatinib + everolimus

Trial	Design	Result	Take-home
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS better; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS better; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

★ Putting it together

A · Small renal mass

	Active surveillance (AS)	Ablation (Ch 143)	Partial nephrectomy (PN)	Radical nephrectomy (RN)
Best for	Frail, life expectancy <5 y; <2 cm or cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard; imperative if solitary, bilateral, hereditary or CKD	Only if high complexity, no CKD/proteinuria, and new eGFR **>45**
Numbers	Growth 0.1–0.3 cm/y; treat if >3 cm or >5 mm/y	Cryo: 2 freeze–thaw cycles; RFA/MWA >60–70 °C	Trifecta: R0, no complication, warm ischaemia ≤25 min	New GFR ≈ GFR × contralateral split × 1.25
Weakness	Imaging adherence	More local recurrence; no pathology	Leak, bleeding	CKD 65% vs 20% after PN

B · Renal lesions

Lesion	Imaging	Genetics / IHC	Action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Fat ≤−20 HU	TSC2/TSC1; HMB-45, Melan-A+	Embolise if >6 cm, aneurysm >5 mm or bleeding; everolimus in TSC
Oncocytoma	Central scar	KIT+, CK7 single cells; Birt–Hogg–Dubé	Surveil or PN (chromophobe mimic)
Clear cell RCC	Hypervascular	3p/VHL loss activates HIF-2α; CA-IX	PN/RN/ablation/AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple losses; diffuse CK7	As RCC; good prognosis
Lymphoma/metastasis	Infiltrative	—	**Biopsy; don't operate**
UTUC	Filling defect, ~46 HU, cytology+	MSI suggests Lynch	Risk-stratify, then KSM or RNU

C · UTUC: kidney-sparing (KSM) vs radical nephroureterectomy (RNU)

	KSM (endoscopic or segmental)	RNU + bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low grade, no invasion on CT urogram; or imperative (solitary, CKD, bilateral)	ANY of multifocal, >2 cm, high grade, hydronephrosis, prior cystectomy, variant
Extras	Surveillance ureteroscopy; UGN-101 mitomycin gel	≤12 wk; template node dissection; single post-op intravesical chemo

	KSM (endoscopic or segmental)	RNU + bladder cuff
Systemic	—	Neoadjuvant platinum if post-op CrCl <50; adjuvant (POUT) ≥pT2/N+

D · Open vs minimally invasive

- **Open (Ch 141)** is kept for IVC thrombus needing bypass, complex anatomy, prior surgery and trauma.
- **Laparoscopic/robotic (Ch 142)** is standard for organ-confined RN and most PN, with faster recovery and the same oncology.

E · Metastatic RCC

ALGORITHM

1. **Score IMDC (6 factors, Ch 144)**: 0 is favourable, 1–2 intermediate, ≥3 poor.
2. **First line**: IO + TKI (pembro + axitinib, nivo + cabozantinib, pembro + lenvatinib) at any risk; ipi + nivo only if intermediate/poor.
3. **Cytoreductive nephrectomy**: upfront only if fit and low burden; otherwise deferred (CARMENA, SURTIME).
4. **Later lines**: switch mechanism; belzutifan after IO + TKI.

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 14

Adrenal & Retroperitoneum

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 145	Surgical and Radiographic Anatomy of the Adrenals
Ch 146	Pathophysiology, Evaluation and Medical Management of Adrenal Disorders
Ch 147	Surgery of the Adrenal Glands
Ch 140	Retroperitoneal Tumours
★	Putting it together: adrenal and retroperitoneal decisions

145 Surgical and Radiographic Anatomy of the Adrenals

P1 Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — relations, arteries, veins, zones, embryology

Not in the paper answer:

- Neonatal gland weight and adrenal rests (§1).

1 · Embryology

- The cortex and gonad share the adrenogonadal ridge, which explains adrenal rests; neural crest forms the medulla.
- At birth the gland is **2×** adult weight. The **fetal zone involutes in year 1**.
- In **renal agenesis** the adrenal is **discoid**.
- **Adrenal rests** occur in ≈1% of adults and in 7.5–15% of neonates along gonadal descent; in congenital adrenal hyperplasia (CAH) they form **testicular masses**.

2 · Zones and hormones

Cortex is ≈90% and medulla ≈10%. †Outside inwards: **GFR = salt, sugar, sex**.

- Glomerulosa ~15% makes aldosterone; fasciculata **~80%** makes cortisol (lipid-laden "clear cells"); reticularis 5–7% makes DHEAS; the medulla (~10%) holds chromaffin cells.
- The medulla is a **modified sympathetic ganglion** of axonless postganglionic neurones.

3 · Relations and blood supply

- Each gland weighs ≈5 g. The right is **retrocaval**, directly above the upper pole; the left lies medial to the upper pole.
- **Exposure**: on the right, liver above, inferior vena cava (IVC) medial, duodenum and kidney below. On the left, pancreas and spleen above, splenic flexure medial.
- Three arteries enter via many twigs, but **one vein drains**. Cortical blood drains into the medulla, so cortisol drives **epinephrine synthesis**.
- The **right vein** is short and enters the **posterior IVC**, so it can avulse. The **left** enters the left renal vein, joined by the inferior phrenic.

> **KEY POINT**
> The right side is riskier: high, retrocaval, with a short vein whose avulsion tears the cava.

5 · Imaging appearance

- **CT** shows an inverted **V/Y** with limbs of 3–6 mm. Unenhanced **≤10 HU** means lipid-rich.
- **MRI** has the best contrast; on **T2 the gland resembles fat**. Ultrasound separates solid from cystic.
- Adrenal venous sampling (AVS) is the only **functional** study; cavography shows intraluminal tumour.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

1 · Applied anatomy and physiology

P1 Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — veins, arteries, zones, PNMT, COMT forms metanephrines

Not in the paper answer:

- Anomalous venous drainage occurs in up to **9%**, which matters for adrenal vein sampling (AVS).
- **Pituitary failure or exogenous steroids spare aldosterone**, because the glomerulosa obeys angiotensin II and potassium; the inner zones atrophy without ACTH.

2 · Evaluation of an adrenal mass – principles

KEY POINT

Two questions: **(1) Is it malignant?** (imaging). **(2) Is it hormonally active?** (biochemistry). Yes to either means treat, usually by resection. Imaging cannot identify functional adenomas, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal or not visualised; thrombus to adrenal vein level
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · Imaging

P2 P4 Algorithm for management of an incidentaloma; Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — APW/RPW formulas, size–malignancy risk, ESE–ENSAT follow-up

Not in the paper answer:

- Up to **7%** of metastases are <10 HU, so every lesion still needs biochemistry; RCC and hepatoma metastases can also wash out.
- **Unenhanced CT**: **≤10 Hounsfield units (HU)** means a lipid-rich adenoma, which covers >70% of adenomas. Chemical-shift MRI drop-out is about as accurate.
- **CT washout** is the gold standard for lipid-poor lesions, with phases at 0, 60 s and 15 min. It is an adenoma if the absolute percentage washout (APW) is **>60%** or the relative washout (RPW) **>40%**.

ALGORITHM

1. Unenhanced CT: if ≤10 HU or MRI drop-out, then adenoma.
2. If >10 HU, then do a 15-min washout CT.
3. If APW >60% / RPW >40%, then it is an adenoma; if below, manage as indeterminate.

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% adrenocortical carcinoma (ACC)
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** is the consensus threshold (sensitive, poorly specific); 6 cm is more specific.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series.
- Heterogeneity, irregular margins and invasion are suspicious. **Macroscopic fat** (<−30 HU) means myelolipoma. **Calcification is not reassuring**, as it occurs in up to 30% of ACC.
- Classic follow-up re-images at 6 and 12 months; growth of ≥1 cm means consider resection.

Functional imaging

P2 MIBG scan — isotopes; best in MEN-2; cold in SDHB/VHL

Not in the paper answer:

- For phaeochromocytoma >5 cm, do MIBG or FDG PET (best in SDHB/VHL) pre-op to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · Hormonal work-up (all lesions >1 cm)

KEY POINT

Every incidentaloma gets a **1-mg dexamethasone suppression test (DST) and metanephrines**. Add an **aldosterone-to-renin ratio (ARR)** if there is hypertension or unexplained hypokalaemia. **Never biopsy, sample or excise before excluding phaeochromocytoma**.

Incidentalomas comprise cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5% and metastasis 2.5%, so **≈18%** are potentially surgical.

Cortisol

P2 Cushing syndrome and the dexamethasone suppression test — DST cut-offs and pitfalls, MACS

- 1-mg DST: **≤1.8 µg/dL (50 nmol/L)** excludes autonomy; 1.8–5.0 means possible autonomy; **>5.0 µg/dL (140 nmol/L)** is >95% specific but misses up to 18% of Cushing syndrome.
- Urinary free cortisol (UFC) is insensitive for mild excess and unreliable if GFR is <60 mL/min.

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Sensitivity 97–99% – best rule-out	>97% by tandem mass spectrometry
Conditions	**Supine after 20 min rest**, fasting	Normal renal function
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× upper limit of normal)	—

- **False positives** come from paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness and seated sampling.
- If equivocal, repeat supine off β-blockers, check chromogranin A, or do a **clonidine suppression** test.

5 · Biopsy and incidentaloma management

- Biopsy answers only "is this a metastasis?", because cytology cannot separate adenoma from carcinoma. ESE–ENSAT advises **no biopsy without a history of malignancy**; complications occur in 3–12%, with seeding in ACC.

> **ALGORITHM**
> 1. If imaging characterises the lesion, then do not biopsy.
> 2. If it is functional, then do not biopsy.
> 3. If ACC is suspected, then do not biopsy.
> 4. If the result will change management (e.g., known primary) and justify risk, then biopsy, **after excluding phaeochromocytoma**.

- Incidentalomas appear on ≈5% of scans; ≈93% are inactive and need no treatment.

6 · Primary aldosteronism (PA)

P2 How will you evaluate and manage 38 yr male with uncontrolled hypertension — ARR, saline, CT, AVS; cure ≈94%

Not in the paper answer:

- PA causes 3–13% of hypertension; most patients (63–91%) are **normokalaemic**.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – mineralocorticoid receptor antagonist (MRA)
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial hyperaldosteronism (FH)	<1 each	Carcinoma/ectopic yes; FH no

- **FH-I** (glucocorticoid-remediable) results from a chimeric gene that puts aldosterone under ACTH control; it causes early severe hypertension and is treated with glucocorticoid.
- **Screen (Endocrine Society 2016)**: resistant hypertension (≥3 drugs including a diuretic) or control on ≥4 drugs; BP ≥150/100 on three days; hypertension and hypokalaemia; **incidentaloma with hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.
- **ARR** is measured in the morning, seated, as plasma aldosterone concentration (PAC, ng/dL) divided by plasma renin activity (PRA, ng/mL/h). It is positive if **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2 and correct K^+ first.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound	Stop ≥4–6 weeks; use α1-blocker or calcium channel blocker (CCB)

Drug	Effect on ARR	Action
β-blockers	Lower renin, so **false positive**	May continue (or stop)
ACE inhibitors / angiotensin receptor blockers	Raise renin, so false negative	Continue; undetectable renin is suspicious
Calcium channel blockers	Neutral	Continue

Confirmatory test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg, when salt load is contraindicated	PAC stays >15 ng/dL

- **Skip confirmation** if there is spontaneous hypokalaemia, undetectable renin and PAC >20 ng/dL.
- CT all surgical candidates to exclude ACC, but ≈20% of aldosteronomas are <1 cm, so CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well) is done in the morning with K^+ corrected, sampling both adrenal veins and the IVC for aldosterone and cortisol.
- **Selectivity** proves the catheter is in the vein (adrenal-to-IVC cortisol >1.1–5:1). **Lateralisation**: a cortisol-corrected aldosterone ratio **>2–4:1** indicates ipsilateral adrenalectomy.
- **Omit AVS** if age <40 with a clear unilateral adenoma >1 cm and a normal contralateral gland, or if ACC is suspected.

ALGORITHM

1. If there is a screening indication, measure the ARR.
2. If the ARR is positive, confirm with a suppression test. If there is a family history or onset <20 y, test for FH-I.
3. If not a surgical candidate, then MRA.
4. If a surgical candidate, do CT. If age <40 with a unilateral >1 cm nodule, then adrenalectomy.
5. Otherwise do AVS: if lateralised, laparoscopic adrenalectomy; if not lateralised, MRA.

- After surgery, check PAC and ARR; † stop K^+ and MRA, and watch for **hyperkalaemia** for weeks because the contralateral glomerulosa is suppressed. BP settles over 6 months; residual hypertension is likelier in older, obese men with long-standing disease.
- MRA: spironolactone 25–50 mg, up to a maximum of 400 mg (gynaecomastia, erectile dysfunction); eplerenone 25 to 100 mg (selective).

7 · Cushing syndrome and cortisol excess

P2 Cushing syndrome and the dexamethasone suppression test — causes, IPSS, Nelson, pituitary cure

Not in the paper answer:

- Post-op insufficiency follows ≈65% of MACS resections; the HPA axis recovers in a median of 6.5 (MACS) to 11 (Cushing) months.
- **Urological**: hypogonadotropic hypogonadism and **urolithiasis in up to 50%**, so test cushingoid stone formers.

ALGORITHM

1. Exclude exogenous steroid.
2. Screen with 1-mg DST, late-night salivary cortisol (LNSC) ×2 or UFC ×2.
3. If abnormal, endocrinologist confirms; measure **morning ACTH**.
4. If ACTH low: adrenal, do CT.
5. If normal/high: pituitary MRI and **inferior petrosal sinus sampling after CRH**. A central-to-peripheral gradient means Cushing disease; no gradient means ectopic.

- **Bilateral adrenalectomy** is for failed pituitary treatment, life-threatening hypercortisolism, or unresectable or occult ectopic ACTH. The cost is lifelong replacement, crisis risk and **Nelson syndrome** (chiasmal compression, oculomotor palsies).

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	Primary pigmented nodular adrenocortical disease (PPNAD)
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex
Association	McCune–Albright	Carney complex (≈half)
Clue	Looks ACTH-driven – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of the larger gland has worked in AIMAH	

- **Mild autonomous cortisol secretion (MACS)**: individualise surgery for hypertension, obesity, dysglycaemia or osteopenia; progression to overt Cushing is very unlikely.
- **Post-op adrenal insufficiency is expected** after overt Cushing (≈99.7%), because the contralateral gland is atrophic.
- Give hydrocortisone 100 mg IV on the day of surgery, then an oral taper (25–50 mg/day). Alternatively give dexamethasone 0.5 mg 6-hourly, where a morning cortisol <50 nmol/L confirms cure. Maintenance is 10–12 mg/m²/day in 2–3 doses.

8 · Phaeochromocytoma and paraganglioma (PPGL)

These can kill on the table, so block before surgery.

P2 Phaeochromocytoma: overview, pre-operative and intra-operative management — genetics, blockade doses, targets, post-op care, lifelong metanephrines

Not in the paper answer:

- ≈24% of apparently sporadic cases carry germline mutations.
- 75–90% are adrenal; paraganglioma is extra-adrenal. Headache occurs in 60–90%.

KEY POINT

The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

Syndrome	Gene	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2	Medullary thyroid cancer (MTC), hyperparathyroidism	50% / ≈3%; adrenal, bilateral
MEN-2B	RET	MTC, mucosal neuromas, marfanoid	50% / ≈3%

Syndrome	Gene	Features	PPGL risk / malignancy
VHL type 2	VHL 3p25	RCC, haemangioblastoma, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
Neurofibromatosis 1 (NF1)	NF1 17q11	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

All are autosomal dominant. VHL type 1 has no phaeochromocytoma, whereas type 2 does.

- On CT the mean is ≈35 HU; the T2 "light bulb" sign is unreliable.
- **Block all with abnormal biochemistry, even if normotensive.** Phenoxybenzamine causes hypotension after resection; α1-selective drugs do so less. CCBs can be sole therapy in mild cases.
- **Metyrosine** 250 mg 2–3 times a day, increased to 1.5–2 g, is added to α-blockade for large or metastatic tumours; it causes sedation and **extrapyramidal** effects.

KEY POINT

Why never β before α? β-blockade removes β2-mediated arteriolar dilatation. Catecholamines then act unopposed on α1 receptors and cause a hypertensive crisis. Hence use selective β1 agents.

- **Targets**: seated BP 120–130/80; standing systolic >80–90 mmHg; † heart rate 60–70/min. **Volume repletion** is key.
- In **MEN-2 and VHL**, do **cortical-sparing partial adrenalectomy**.
- Malignancy is defined **only by metastases**; risk factors are SDHB, extra-adrenal site, >5 cm and dopamine secretion.
- Check metanephrines at ≈6 months, then annually for life, because recurrence occurs even >15 years later.

9 · Non-functional benign lesions

Lesion	Key features	Management
Myelolipoma	Fat and haematopoietic tissue; macroscopic fat diagnostic; may exceed 10 cm	Exception to 4-cm rule; still do hormonal work-up. Excise if symptomatic or large; embolise a bleed
Cysts	Endothelial 45%, pseudocyst 39% (harbours malignancy ≈7%); >30 HU means haemorrhage	Endocrine work-up. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**
Ganglioneuroma	Young; encases vessels without narrowing them	Benign; diagnosed at resection
Oncocytoma	Rare; large; ≈10% functional	Resect; >30% malignant (unlike renal)

10 · Adrenocortical carcinoma (ACC)

P4 Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — syndromes, CT, Weiss, ENSAT, R0 surgery, mitotane, trials, surveillance

Not in the paper answer:

- Overall 5-year survival is 20–47% (stage IV <5%), and 60–80% recur after resection.

ENSAT stage	Definition	At diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous thrombus	11–19%
IV	Distant metastasis only	39–49%

- Resect recurrences if low grade with a long disease-free interval.
- † Mitotane often needs higher glucocorticoid doses, because it raises binding globulins and speeds clearance.
- **Paediatric ACC** is >90% functional (virilisation), warrants TP53 testing and has a better prognosis.

11 · Adrenal metastases

- With known malignancy, **>50%** of new adrenal lesions are metastases. RCC and hepatoma deposits may wash out, so growth and biopsy decide. **Still exclude phaeochromocytoma**.
- Resect an isolated metastasis in selected patients: non-small cell lung cancer (level 1 evidence) and melanoma, and RCC as part of cytoreduction.

12 · Adrenal insufficiency

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (anti-21-hydroxylase), **tuberculosis**, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH)	Pallor
Electrolytes	Low Na^+, **high K^+**; salt craving	Low Na^+, normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

- A **crisis** mimics an acute abdomen. Those at risk have a cortisol-secreting tumour, bilateral adrenalectomy or a **previous contralateral nephrectomy/adrenalectomy**, so review old notes.
- Diagnose with morning cortisol and ACTH, then ACTH stimulation. Give ½–⅔ of the hydrocortisone dose on waking.

KEY POINT

† **Adrenal crisis**: do not wait for results. Draw cortisol and ACTH, give **hydrocortisone 100 mg IV bolus**, then 200 mg/24 h (infusion or 50 mg 6-hourly). Give 0.9% saline (1 L in the first hour) with dextrose, treat the precipitant, and taper over 1–3 days. Fludrocortisone is unnecessary while hydrocortisone is >50 mg/day. Teach sick-day rules and issue a steroid card.

Surgical stress (chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Potency	Replacement (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36
Dexamethasone	25–50	0.5	36–54

13 · Congenital adrenal hyperplasia (CAH)

- CAH is autosomal recessive, and **21-hydroxylase deficiency is >95%**; low cortisol raises ACTH, causing hyperplasia and androgen excess (see Ch 59).
- Non-classic CAH may present as an adult mass, and testicular rests mimic tumour, so consider them before orchidectomy.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + etoposide, doxorubicin, cisplatin (EDP) vs mitotane + streptozotocin, advanced ACC	Better response and progression-free survival; overall survival similar	M-EDP first-line
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No difference in recurrence	No routine adjuvant mitotane in low risk

15 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest ectopic ACTH source: bronchial carcinoid.

147 Surgery of the Adrenal Glands

P2 Adrenal Sx. — indications, contraindications, preparation, approaches, steps, complications

Not in the paper answer:

- Size and imaging criteria for incidentaloma (§2).

1 · Surgical anatomy (essentials)

- The **right** gland lies behind the bare area of the liver. Its vein is **<1 cm** long and enters the **posterior IVC at 45°**. The **left** gland is more mobile.

2 · Indications

Two gates: any functional tumour, or malignant potential.

Category	Indication
Functional (any size)	Aldosteronoma; cortisol adenoma; Cushing disease after failed trans-sphenoidal surgery; phaeochromocytoma (PCC); virilising/feminising tumour
Non-functional	Adrenocortical carcinoma (ACC); symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm** ; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast CT **>10 HU**; washout <50% with delayed >35 HU; chemical-shift MRI not adenoma; FDG-PET uptake equal to or above liver

KEY POINT

Trap: bilateral hyperplasia causing aldosteronism is **not** surgical, so adrenal venous sampling (AVS) comes before adrenalectomy for aldosteronism.

Incidentaloma work-up

- ACC is found in ~2% and autonomous secretion in up to 20%. Measure sex steroids if ≥4 cm or virilised.
- All get a 1-mg dexamethasone suppression test (never spot cortisol) and metanephrines (>2× upper limit is significant).
- If hypertensive, check the aldosterone-to-renin ratio (ARR >20 with aldosterone >20 ng/dL), then salt loading, then AVS; up to 40% are normokalaemic.
- Inert, ≤4 cm, ≤10 HU lesions get CT at 12 months. Operate on myelolipoma (≤ −20 HU) only if **>7 cm**, symptomatic or atypical.

3 · Contraindications

- Correct physiology first. Low-dose aspirin may continue; necessary surgery in pregnancy can be done in any trimester. Confirm that a prior nephrectomy has not removed the ipsilateral adrenal before contralateral adrenalectomy.

4 · Pre-operative preparation

Pathology	Preparation	Why
PCC	α-blockade for **2 weeks** (phenoxybenzamine; doxazosin, calcium channel blocker, metyrosine alternatives); β **only after α**; volume expansion	Contracted patients collapse once vein tied

Pathology	Preparation	Why
Cushing	†Hydrocortisone 100 mg at induction, tapered over months; thromboprophylaxis	Contralateral gland is suppressed
Aldosteronoma	†Correct K^+; control BP	†Hypokalaemia causes arrhythmia

5 · Approaches

- **Minimally invasive surgery is standard** (since 1992).
- The **retroperitoneoscopic** route (lateral or **prone**) avoids bowel, so it suits prior abdominal surgery; bilateral small tumours can be done prone without repositioning.
- **Open** surgery is for suspected **ACC**, large tumours, or invasion of the IVC or organs.

6 · Operative steps

Control the vein, then free the gland.

ALGORITHM

1. **Right**:
2. Elevate the liver; Kocher the duodenum; expose the IVC.
3. Dissect up from the renal vein to the short adrenal vein.
4. **Warn anaesthesia**, then clip early.
5. Dissect superiorly and laterally down to psoas; bag the gland intact.

ALGORITHM

1. **Left**:
2. Reflect the colon at Toldt, outside Gerota; lift spleen and pancreas.
3. Follow the left renal vein to the adrenal vein; clip after warning.
4. Protect pancreas and splenic artery; divide superomedial tissue; bag.

- **Retroperitoneoscopic**: enter below the 12th rib tip with a balloon. Along psoas the **renal artery is met first**; the right vein enters the posterolateral IVC above the hilum.

KEY POINT

Ligate the vein early in PCC to limit the catecholamine surge; hypotension follows, so tell anaesthesia first.

7 · Special pathologies

- **PCC intra-op**: treat a surge with nitroprusside, nitroglycerin or phentolamine, and arrhythmia with esmolol.
- **ACC** needs a high-volume surgeon (>6, ideally ≥20/year) and an **open, intact, en bloc** resection.
- **Metastasis** (two-thirds from lung or kidney) is resected when the adrenal is the only disease; renal origin and <5 cm do best.
- Avoid ablation of PCC or ACC outside trials.

8 · Partial (cortical-sparing) adrenalectomy

- **Indications**: bilateral or hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma and cysts.
- Arteries enter superiorly, medially and inferiorly, so **lateral dissection is safe**; the main vein may be divided. †Retain about one-third of a gland, guided by intra-op ultrasound.

9 · Complications

- For bleeding, apply pressure, raise the pneumoperitoneum, side-bite the IVC, and convert if uncontrolled.
- †Check glucose hourly for 24–48 h after PCC. Treat insufficiency with IV hydrocortisone 100 mg and saline. For hyperkalaemia after aldosteronoma, stop K^+ and spironolactone, occasionally adding fludrocortisone.

10 · Outcomes

- PCC recurs in up to **16%** over life, so check BP and metanephrines annually.
- †Aldosteronoma surgery corrects hypokalaemia in nearly all and cures hypertension in about half. Advanced-stage ACC has a 5-year survival **<15%**.

140 Retroperitoneal Tumours

1 · Anatomy of the retroperitoneum

P1 Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. — boundaries, contents, spaces, vessels, nerves at risk

Not in the paper answer:

- **Depth of psoas invasion predicts nerve injury**: genitofemoral injury causes thigh paraesthesia, while femoral injury weakens hip flexion and knee extension.

KEY POINT

The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · Classification

The group decides first treatment.

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma, sex cord stromal	Platinum chemotherapy (germ cell tumour, GCT); RP lymph node dissection, RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, undifferentiated pleomorphic sarcoma (UPS), synovial, Ewing, desmoplastic small round cell tumour (DSRCT), unclassified	Surgery; chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, solitary fibrous tumour (SFT), PEComa, desmoid, myelolipoma, GIST	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, malignant peripheral nerve sheath tumour (MPNST), ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP)	Lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision
Unknown primary	Poorly differentiated carcinoma encasing vessels	Histology-led

3 · Clinical evaluation

Tissue only if it changes treatment.

- Most are painless or incidental. The sequence is **imaging, then serology, then tissue**.
- **CT chest/abdomen/pelvis** comes first (NCCN). **Negative HU (fat)** suggests liposarcoma, angiomyolipoma or myelolipoma, but CT cannot tell origin or malignancy. **MRI** shows vascular involvement; PET-CT is selective.

- In a young man check **AFP and hCG**; in an older man, PSA. Add blood count and electrophoresis; † metanephrines if paraganglioma is possible.
- **Biopsy** only when it **changes first treatment**: lymphoma, GCT, Ewing, or high-grade sarcoma for neoadjuvant therapy. Use a **core (14–16 G coaxial) rather than FNA**; † take a retroperitoneal route, with the tract inside the future resection field.
- Avoid biopsy of purely cystic lesions, † an unblocked paraganglioma, or a young man with raised AFP/hCG.

4 · Mesenchymal tumours (sarcoma)

- Soft-tissue sarcoma is <1% of adult cancers; **15–20%** arise in the RP.

Entity	Key features	Treatment
Well-/de-differentiated (WD/DD) liposarcoma (commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical compartmental resection
Leiomyosarcoma (**~20%**)	From vessel media (IVC); no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant
UPS (ex-MFH)	5th–6th decade; no fat, peripheral nodular enhancement	Wide resection
Unclassified (≤20%)	Exclusion; prior RT in a quarter	Screen MDM2
Synovial	Calcification; TLE1; **t(X;18)**	Chemosensitive
SFT	**NAB2-STAT6**; Doege-Potter hypoglycaemia (IGF)	En bloc; radiosensitive
Ewing	EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based
DSRCT	Young men; peritoneal implants; WT1-EWSR1 t(11;22)	Multimodal; poor
PEComa	HMB-45 + muscle markers; TSC2	Excision; mTOR inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

- **Liposarcoma**: **dense nodules in a fatty mass mean de-differentiation** and are the biopsy target. WD recurs locally in up to 40% but rarely metastasises; DD spreads distantly in up to 20%. Mean time to recurrence is ~**2 years**, so surveillance is lifelong.
- If unresectable, anthracycline or trabectedin offers modest benefit. **Pleomorphic liposarcoma** is MDM2-negative and metastasises early to lung.
- **Leiomyosarcoma** obstructs veins, urine and bowel; biopsy underestimates its aggressiveness.

> **KEY POINT**
> MDM2 amplified (WD/DD) means local disease, won or lost at the margin; MDM2 absent (pleomorphic) means systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable.

- † **FNCLCC grade** adds differentiation (1–3), mitoses per 10 high-power fields (1: 0–9, 2: 10–19, 3: ≥20) and necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology, not size, predict distant spread.

5 · Neurogenic tumours

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated; S100+; slow	Local resection curative (often enucleable)
Neurofibroma	NF-1 or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; H3K27me3 loss; spreads along roots	Resect with **parent nerve**; 5-year ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision
Neuroblastoma	MYCN adverse; MIBG staging	Paediatric chemotherapy with or without surgery
Paraganglioma	Commonest at organ of Zuckerkandl (below IMA); T2-bright; secretory ~40%; metastases 20–50%; NF-1, MEN-2A, SDHB	Surgery after α-blockade† (see Ch 145)

6 · Germ cell, haematological, cystic, unknown primary

- **GCT**: an RP mass in a young man is GCT until proven otherwise; extragonadal disease is often a **burnt-out testicular primary**. Give platinum first, then resect all nonseminomatous residual masses, because teratoma is chemoresistant and marker-silent (see Ch 79–80).
- **Sex cord stromal** tumours resist chemo and radiotherapy, so do early RPLND in high-risk node-negative primaries.
- **Lymphoma** forms confluent nodal masses: biopsy, never resect.
- **Cystic** lesions (lymphangioma, mucinous cystadenoma, cystic mesothelioma) need **complete excision**, because aspiration or marsupialisation recur.
- **Unknown primary**: if midline or i(12p)-positive, give platinum; a single deposit is excised or irradiated.

> **KEY POINT**
> Histology decides first treatment: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning sarcoma resection.

7 · Surgery for RP sarcoma

The first operation decides cure.

- **Complete en bloc (R0) resection at the first operation** in an experienced centre is the best chance.
- **Compartmental resection** removes adherent uninvolved neighbours (kidney, colon, psoas fascia), which lowers local recurrence. Preserve duodenum, pancreatic head, major vessels and femoral nerve unless invaded. † **Never enucleate**, because the pseudocapsule contains tumour.
- Nephrectomy is needed in up to **55%** but rarely leads to dialysis; † get a split renogram first.
- After **bleomycin**, keep FiO_2 low and fluids restrictive.
- Through a midline laparotomy, **split and roll** over the great vessels and ligate lumbar vessels and retrocrural lymphatics to prevent chylous ascites.
- The **aorta is always reconstructed**. A chronically occluded **IVC** tolerates ligation; IVC grafts thrombose in 10–40%, so anticoagulate for ≥3 months.

8 · RPLND in germ cell tumours

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — landing zones, templates, PC-RPLND, nerve-sparing

- Primary RPLND gives survival ~100% with either template. In post-chemotherapy RPLND, a full bilateral infrahilar template is standard; unilateral only if the mass is **≤5 cm** in the landing zone, markers are normal, and IGCCCG risk is good or intermediate.
- Sparing the L1–L4 fibres and the superior hypogastric plexus gives antegrade ejaculation in **90–100%**.

9 · Complications

P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — risk, diagnosis, stepwise treatment

Not in the paper answer:

- Chylous ascites occurs in 2–7% after chemotherapy. Other risks are pancreatic leak (check drain amylase), VTE after caval reconstruction and femoral neuropraxia.

ALGORITHM

1. **Chylous ascites**:
2. Paracentesis and a medium-chain triglyceride (MCT) diet, with or without octreotide.
3. If it reaccumulates, place a drain.
4. Then nil by mouth and total parenteral nutrition.
5. If output stays >100 mL/day, then peritoneovenous (LeVeen) shunt or ligation.
6. Continue the MCT diet for 1–3 months after resolution.

10 · Adjuvant and neoadjuvant therapy

Neither is routine; surgery is the mainstay.

- **Radiotherapy**: pre-op is better tolerated than post-op. **STRASS** showed no gain, so it is **not standard**; † an exploratory benefit in low-grade liposarcoma keeps it an option.
- **Chemotherapy**: WD liposarcoma and leiomyosarcoma are largely resistant; myxoid liposarcoma, synovial sarcoma and Ewing respond. **Standard neoadjuvant chemotherapy is only for Ewing**.

11 · Recurrence and follow-up

- † Local recurrence kills in liposarcoma, whereas leiomyosarcoma fails distantly (lung, liver).
- † CT every 3–6 months for 2–3 years, 6-monthly to 5 years, then annually; lifelong for liposarcoma.
- Re-resect slow-growing resectable recurrence (WD liposarcoma).

12 · Landmark trials

Trial	Design	Result	Take-home
EORTC-62092 STRASS	Phase III, 266 patients: pre-op RT + surgery vs surgery	No abdominal recurrence-free survival difference (HR 1.01); more complications	Pre-op RT not standard

Trial	Design	Result	Take-home
STRASS II (ongoing)	Neoadjuvant doxorubicin + ifosfamide/dacarbazine + surgery vs surgery	Awaited	Neoadjuvant chemotherapy unproven

★ Putting it together: adrenal and retroperitoneal decisions

A · Adrenal incidentaloma: two questions

ALGORITHM

1. **Malignant?** If unenhanced CT is ≤10 HU or MRI shows drop-out, then adenoma. If >10 HU, do washout CT (APW >60% / RPW >40% = adenoma). If >4 cm, suspicious or growing, consider resection.
2. **Functioning?** Test all with 1-mg DST and metanephrines, plus ARR if hypertensive or hypokalaemic.
3. If **silent, <4 cm and benign**, then no follow-up.

B · Functional tumours compared

	Aldosteronism	Cushing	Phaeo
Screen	ARR	1-mg DST, salivary or urinary cortisol	Metanephrines
Next	Confirm, then CT, then AVS	ACTH: low = adrenal; normal/high = pituitary/ectopic	CT/MRI, functional imaging; genetics
Pre-op	Correct K^+	Steroid cover	α then β; fluids, salt
Post-op	Stop K^+/spironolactone; hyperkalaemia	Steroids until axis recovers	Hypotension, hypoglycaemia
Medical	Mineralocorticoid antagonist (bilateral)	Steroidogenesis inhibitors	Long-term α-blockade

C · Adrenalectomy approach

KEY POINT

Use laparoscopy for benign tumours. Operate open for large suspected carcinoma or invasion, because rupture causes recurrence. On the right, control the short caval vein carefully. Partial adrenalectomy suits bilateral or hereditary disease.

D · Retroperitoneal mass

ALGORITHM

1. **Image**, then **exclude** paraganglioma (metanephrines), germ cell tumour (testis, markers) and lymphoma.
2. If it changes treatment, do a **core biopsy** by the retroperitoneal route.
3. **Sarcoma:** en bloc R0 resection at the first operation in a specialist centre; never enucleate. Follow liposarcoma with lifelong CT.

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 15

Renal, Nephrology & Transplant

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 88	Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter
Ch 90	Fundamentals of Nephrology
Ch 91	Renal Transplantation
★	Putting it together: kidney function and transplant decisions

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

1 · Kidney: position and relations

The kidney's neighbours are what you injure at surgery.

- The kidney is ≈**11 cm** long, 125–170 g (male), at T12–L3. The **right lies 1–2 cm lower**; both descend ≈3 cm on inspiration or standing.

P1 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal mass — relations, coverings, Gerota open below, hilum order

Not in the paper answer:

- Traction on the hepatorenal or splenorenal ligament tears the liver or splenic capsule.

P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — access hazards: pleura, lung, liver/spleen, retrorenal colon

Not in the paper answer:

- Upper-pole excision through the diaphragm risks pneumothorax.

2 · Coverings and compartments

- Each perirenal space is separate, so extravasation rarely crosses the midline. The **anterior pararenal space** (colon, duodenum, pancreas) does cross it; the posterior pararenal space holds fat only.
- Transperitoneally, open the **white line of Toldt**. Reach the right hilum by a **Kocher manoeuvre**; on the left, reflect the pancreatic tail, spleen and splenic vessels medially.

3 · Parenchyma and its imaging

- Cortex is ≈1 cm. **Columns of Bertin** carry interlobar arteries, so the safe track is **through pyramid and papilla**.
- Glomerular filtration rate (GFR) is ≈**125 mL/min**; ≈99% of filtrate is reabsorbed.

Modality	Normal kidney
Ultrasound	Cortex iso-/hypoechoic to liver (hyperechoic in newborn)
Computed tomography (CT)	Unenhanced 30–60 Hounsfield units (HU), cortex = medulla; enhanced 80–120 HU
Magnetic resonance (MRI)	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**
Excretory	>3 min	Collecting system, ureter

- On intravenous urography, no excretion at 24 h means a non-functioning kidney.
- **Corticomedullary differentiation** is lost in chronic kidney disease and acute kidney injury and returns with recovery.

Congenital anomalies

- **Malrotation** shows a central pelvis with calyces medial to it. In **ectopia**, poor drainage causes stasis, infection and stones.
- **Horseshoe**: lower poles fuse and ascent is arrested by the inferior mesenteric artery (**IMA**); it brings ureteropelvic junction obstruction (UPJO), stones, infection and multiple arteries.

4 · Renal vessels, lymphatics and nerves

End arteries make partial nephrectomy possible.

P2 Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment — segmental end arteries, accessory 25–28%, Brödel line, veins; left renal vein **6–10 cm**, nutcracker, circumaortic

Not in the paper answer:

- **CT angiography** is the gold standard for vascular anatomy but misses vessels <2 mm.
- Right-sided lymph drains to interaortocaval/paracaval nodes but **may cross to left para-aortic** nodes.

5 · Percutaneous puncture and vessels

P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — Sampaio puncture table: upper infundibulum 67%, fornix no arterial injury

Not in the paper answer:

- A through-and-through puncture hits anterior segmental vessels and causes marked haemorrhage.

> **KEY POINT**
> Why the fornix? Arteries run on both faces of the infundibula, and a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Vascular injury causes haemorrhage, arteriovenous fistula, pseudoaneurysm and loss of function.

6 · Pelvicalyceal system

P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — Sampaio types, duplex pelvis, perpendicular and crossed calyces

Not in the paper answer:

- Dilatation is mimicked by an extrarenal pelvis, **megacalycosis** and a calyceal diverticulum.

7 · Ureter: course and relations

Injuries occur at crossings; stones stick at narrowings.

- The adult ureter is 22–30 cm. Stones impact at three **narrowings**: ureteropelvic junction (UPJ), iliac crossing and ureterovesical junction (UVJ).
- It lies on psoas and crosses the **common iliac bifurcation** (landmark), with gonadal vessels in front. The **pelvic brim** is a commonest injury site.
- In females it passes under the **uterine artery 1.5–2 cm** from the cervix ("water under the bridge"). In males the vas loops over it.
- The **intramural** segment is 1.2–2.5 cm long. Bladder pressure closes this oblique tunnel, which prevents reflux.

8 · Radiologic anatomy of the ureter

- Because of **peristalsis**, the whole ureter is rarely opacified; a continuous column suggests obstruction or atony.
- A stricture needs **proximal dilatation**; crossing vessels and iliac indentation can mimic it.
- Incomplete duplication (one bud, one orifice) is commoner than complete duplication (two buds, two orifices).

KEY POINT

Weigert–Meyer rule: the upper-moiety orifice is **inferior and medial** and obstructs (ureterocele); the lower moiety refluxes (vesicoureteral reflux, VUR).

- Bilateral medial displacement suggests retroperitoneal fibrosis or pelvic lipomatosis.

9 · Ureteric blood supply, nerves, histology

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- The longitudinal anastomosis is fragile, so preserve adventitia and avoid lateral retraction to prevent ischaemic stricture.
- Peristalsis is **myogenic**, so denervation does not cause reflux. Pain (T11–L1) refers to flank, groin, scrotum or labia.
- In **UPJO**, the junction has excess longitudinal muscle and collagen.

10 · Endoscopic anatomy

- The trigone is the bladder's most vascular part. Orifices lie 1–2 cm from the midline and move laterally on filling.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- A higher grade means more reflux. The **intramural ureter is narrowest** (3–4 mm) and may need dilatation.
- The lower-pole infundibulum needs ≈ **140°** deflection.

ALGORITHM

1. Infundibulotomy: incise superior, then inferior, then posterior quadrants.
2. **Never anterior**.

90 Fundamentals of Nephrology

1 · Why a urologist needs nephrology

- Each kidney has ≈**1 million** **nephrons**, fixed at birth. When nephron mass falls (nephrectomy, agenesis), the rest **hyperfilter**.
- Interstitial fibroblasts make **erythropoietin**; the proximal tubule makes **calcitriol** by 1α-hydroxylation.

> **KEY POINT**
> The glomerulus is a pressure filter set by arteriolar tone; the tubule is energy-hungry and near-hypoxic. Most disorders fail one or the other.

2 · Measuring kidney function

Serum creatinine (SCr) often misleads in surgical patients.

P2 GFR estimation: clinical assessment and in the prospective transplant donor — inulin/iohexol gold standard; creatinine secreted (24-h clearance overestimates); CKD-EPI 2021 race-free; cystatin C; combined equation most accurate

Not in the paper answer:

- SCr is falsely low with small muscle mass, and falsely raised by **trimethoprim and cimetidine**, which block secretion. Cystatin C is raised by hyperthyroidism, steroids and inflammation.

> **KEY POINT**
> Trap: after cystectomy or bladder repair, a rising SCr with ascites may be an **intraperitoneal urine leak** reabsorbed across the peritoneum, not kidney injury.

P1 Proteinuria — normal <**150 mg/day**, nephrotic >3.5 g/day; spot early-morning ACR first, PCR for non-albumin protein; dipstick misses light chains

Not in the paper answer:

- **eGFR is invalid when function is changing**, so acute kidney injury (AKI) is defined by SCr change and urine output.

3 · Acute kidney injury

Glomerular haemodynamics

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32.

Arteriolar change	Renal blood flow	GFR
Afferent constriction	Falls	Falls
Afferent dilatation	Rises	Rises
Efferent constriction	Falls	Rises
Efferent dilatation	Rises	Falls

Angiotensin II blockade therefore drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥ **0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

P2 Acute renal failure (AKI) and acute tubular necrosis — causes, ATN mechanism, AEIOU dialysis indications

Not in the paper answer:

- **3 months** separates acute kidney disease (AKD) from CKD.

Prerenal AKI

- Urine is bland, fractional excretion of sodium (FeNa) is low, and function **recovers immediately** on correction.
- **Autoregulation**: the **myogenic reflex** dilates the afferent arteriole via prostaglandins, so it is **disabled by NSAIDs** (non-steroidal anti-inflammatory drugs).
- In **tubuloglomerular feedback (TGF)**, angiotensin II constricts the efferent arteriole, so it is **disabled by renin–angiotensin–aldosterone system (RAAS) blockade**.
- **Modifiable** factors are hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis and hypercalcaemia.

P2 Acute renal failure (AKI) and acute tubular necrosis — prerenal vs ATN indices: urine Na 20, FeNa < **1%**, FeUrea < **35%**, SG, muddy-brown casts, fluid response

Not in the paper answer:

- High FeNa after diuretics does not prove ATN; use fractional excretion of urea (FeUrea). **Response to correction is the single most important test.**
- Keep mean arterial pressure (MAP) > **65 mm Hg**. **Fluids** (mOsm/L): plasma and 0.9% saline 308; Plasmalyte 294; Ringer's lactate 273; 0.45% saline 154; 5% dextrose (D5) 0, so D5 is free water.

Intrinsic AKI: the cast localises the compartment

	Acute tubular necrosis (ATN)	Acute glomerulonephritis (GN)	Acute interstitial nephritis (AIN)
Cause	Ischaemia, toxins	Immune complex, anti-glomerular basement membrane (anti-GBM), cell-mediated	Drug, infection, systemic
Urine	Granular casts	Dysmorphic red cells, **red cell casts**, proteinuria	White cells, **white cell casts**, with or without eosinophils

Acute tubular necrosis

- **Rhabdomyolysis** (obese patient, long flexed lateral robotic case) needs **early aggressive isotonic fluid**; alkalinisation is unproven. †Prevent it with padding and limited flexion time.
- **Give diuretics only for symptomatic overload; they do not treat ATN.**

Postcontrast AKI

- This is an SCr rise within **48 h** of intravenous (IV) iodinated contrast. Risks include CKD, diabetes and the arterial route.

- **IV fluid before contrast is the only effective prevention** (†for eGFR <30 or AKI); use the lowest dose.
- **Gadolinium** groups I and III cause nephrogenic systemic fibrosis (**NSF**), so avoid them in AKI, eGFR <**30** and dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) is allowed. **Group III (gadoxetate) is treated as group I.**

Acute interstitial nephritis

- **Drugs cause most**, notably β-lactams, NSAIDs, proton pump inhibitors and **checkpoint inhibitors**.
- Suspect it in unexplained hospital AKI with **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ.
- **Stop the drug**; early steroids may help, and about half recover fully.

Glomerulonephritis

- In microhaematuria, a rising SCr, proteinuria, red cell casts, hypertension or purpura call for **urgent nephrology and biopsy**, not cystoscopy first.

Complications of AKI

- **Overload** is urgent only with respiratory distress. Give a high-dose IV loop diuretic and add metolazone or acetazolamide (**sequential blockade**).
- For uraemic bleeding, give desmopressin (**DDAVP**) **0.3 µg/kg** IV/IM 30–60 min before the procedure.

Renal replacement therapy

- AEIOU (acidosis, electrolytes, ingestions, overload, uraemia) means failed medical management, not a biochemical threshold. **Early dialysis gives no benefit.** Continuous therapy is not superior.
- Non-recovery for ≥**90 days** means CKD or end-stage kidney disease (ESKD).

4 · Drug dosing in reduced kidney function

- In AKI, eGFR overestimates function, so reassess doses daily.
- †**Cockcroft–Gault** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers in dehydration or perioperative hypotension ("sick-day" rule).
- †Avoid nitrofurantoin if eGFR <30 (UK guidance allows short courses at 30–44). †Stop metformin at eGFR <30, and withhold it before iodinated contrast in AKI or eGFR <30.

5 · Chronic kidney disease

- **KDIGO** (Kidney Disease: Improving Global Outcomes) defines CKD as abnormal structure or function for ≥**3 months**. Classify by **cause, GFR and albuminuria**.

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 count as CKD only with albuminuria or structural damage. Albuminuria can shift risk two bands: eGFR 43 with ACR 1100 mg/g is **G3bA3, very high**.

- **Each perioperative AKI is a modifiable progression factor**.

P2 Metabolic abnormalities in CKD — complications table: anaemia (Hb **10–11.5 g/dL**, >13 harmful), secondary/tertiary HPT, acidosis (bicarbonate ≥**22**), hyperkalaemia (binders, keep RAAS blockade), hypertension (chlorthalidone), ED

Not in the paper answer:

- Tertiary hyperparathyroidism (HPT) is treated with a calcimimetic, then subtotal parathyroidectomy.

Slowing progression

- Target glycated haemoglobin <7.5% and blood pressure <**130/80**; the agent matters only in proteinuric disease (RAAS blockade).
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors** protect kidney and heart and cut mortality, with or without diabetes. Add **finerenone** in CKD with type 2 diabetes.

6 · Haematuria and proteinuria: when to involve a nephrologist

- †Microhaematuria is ≥**3** red blood cells per high-power field (RBC/HPF) on microscopy; a positive dipstick needs confirmation.
- **Glomerular clues** are dysmorphic red cells, red cell casts, raised ACR or protein-to-creatinine ratio (PCR), rising SCr, hypertension and systemic disease.

ALGORITHM

1. Haematuria with or without proteinuria: send urine microscopy, ACR, SCr/eGFR.
2. If there are casts, dysmorphic cells, AKI or systemic features, refer for **urgent nephrology**.
3. If isolated non-glomerular haematuria, do a urological evaluation.
4. If isolated albuminuria, stage CKD and refer per criteria.

- †KDIGO referral criteria: AKI or an abrupt GFR fall; GFR <30; ACR ≥300 mg/g (≥30 mg/mmol); a fall >5 mL/min/1.73 m²/year; red cell casts or haematuria >20 RBC/HPF; BP refractory to ≥4 drugs; persistent potassium abnormality; recurrent stones; hereditary disease.

7 · Electrolyte and acid–base disorders

Water and sodium

- **Sodium disorders are water disorders**. **Volume depletion overrides hypo-osmolality** in driving antidiuretic hormone (ADH).
- Of 1 L, D5 leaves **83 mL** in the circulation, 0.9% saline 250 mL and 5% albumin 1000 mL.

Hyponatraemia

- †Transurethral resection (TUR) syndrome is absorption of hypotonic glycine or sorbitol during monopolar resection.

Volume	Urine	Causes
Hypovolaemic	Na >20	Diuretics, mineralocorticoid deficiency, cerebral salt wasting
Hypovolaemic	Na <20	Vomiting, diarrhoea
Euvolaemic	Osm > serum	Syndrome of inappropriate ADH (SIADH), cortisol deficiency, hypothyroidism, pain, drugs
Euvolaemic	Osm low	Polydipsia, low solute ("tea and toast", beer potomania)
Hypervolaemic	Na >20	Kidney failure
Hypervolaemic	Na <20	Heart failure, cirrhosis, nephrotic

- Treat SIADH with water restriction, then vaptans, salt with furosemide, or urea.
- **Rate**: in chronic hyponatraemia, raise sodium ≤ **8–12 mEq/L per 24 h**, or **osmotic demyelination** follows. Treat overcorrection with DDAVP and free water.
- With severe symptoms, give **3% saline to raise sodium 4–6 mEq/L**, then slow down.

Hypernatraemia

- Nephrogenic diabetes insipidus (DI) follows lithium, hypercalcaemia, hypokalaemia and **relief of obstruction**.
- Give free water. In chronic cases, lower sodium ≤ **0.5 mEq/L/h**, or cerebral oedema follows.

Potassium

P1 Management of acute hyperkalaemia — 98% intracellular; causes incl. obstructive uropathy, jejunal conduit, pseudohyperkalaemia; ECG sequence; > **6.0** urgent; binders (SZC ≈1 h, patiromer ≈7 h)

ALGORITHM

1. **Stabilise** if there is any ECG change: calcium gluconate 10%, 10 mL IV.
2. **Shift**: insulin 10 U with 50 mL of 50% dextrose; nebulised salbutamol 20 mg.
3. **Remove**: haemodialysis if oliguric or ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; a binder.
4. Stop culprit drugs.

- **Sodium polystyrene sulfonate (SPS)** is slow and causes **colonic necrosis**, so it is not first-line.
- **Hypokalaemia** from **vomiting** is caused by kidney loss, not gastric loss. Replace potassium chloride, **oral preferred**, 20–40 mEq repeated.

Calcium

- Saline and loop diuretics raise urine calcium; **thiazides lower it**.
- **Hypercalcaemia in a stone former**: non-suppressed PTH means primary HPT; if suppressed, check PTH-related peptide (renal, bladder, squamous cancer) and 1,25-vitamin D (sarcoid, lymphoma). Low urine calcium suggests familial hypocalciuric hypercalcaemia.
- Treat if symptomatic or ≥ **14 mg/dL**: **isotonic saline** 200 mL/h to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin; IV bisphosphonate; denosumab.

Acid–base

- Anion gap (AG) = Na − (Cl + HCO_3); it is raised if >12. If the AG rises less than the bicarbonate falls, a non-AG acidosis coexists.
- **Urine AG** = (Na + K) − Cl. Negative means ammonium excretion, so gut loss; positive means a renal cause (renal tubular acidosis, RTA).

P2 Renal tubular acidosis — type 1/2/4 table: defect, K, urine pH, stones; obstruction causes type 4; UAG

Not in the paper answer:

- **Topiramate and zonisamide** cause calcium phosphate stones. MCQ Laxative abuse leads to **ammonium acid urate stones**.
- †Bowel conduits and neobladders absorb ammonium chloride, causing hyperchloraemic non-AG acidosis.

8 · Urinary tract obstruction and post-obstructive diuresis

Relief of obstruction can cause dangerous fluid loss.

Term	Meaning
Hydronephrosis	Pelvicalyceal dilatation, with or without obstruction
Obstructive uropathy (OU)	Obstruction at any level
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **curable**, so exclude it (†by ultrasound) in any unexplained AKI or CKD.

P1 Post-obstructive diuresis — after bilateral obstruction relief; up to **300 mL/h**, ≈2 days; 3 mechanisms (AQP2/Na-K-ATPase loss, volume expansion/ANP, urea); gradual decompression no role

ALGORITHM

1. Relieve the obstruction, then allow free oral fluids; do not give routine IV fluid.
2. If output stays high, or there is hypovolaemia, or the patient is obtunded or nil by mouth, then check a panel every **4–6 h** (Na, K, phosphate, Ca, Mg).
3. Replace ≈**75%** of the output with **0.45% saline** plus electrolytes.

KEY POINT

Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete unilateral ureteral obstruction	100% at 7 days; 70% at 14 days; 30% at 4 weeks; **none at 6 weeks**

- Duration, not dilatation, decides the injury. A small stone with low-grade obstruction can be observed safely for **2–4 weeks**; beyond 4 weeks the outcome is unpredictable.
- Keep a **low threshold for a stent or nephrostomy**. †Infection with obstruction needs urgent drainage.

9 · Landmark trials

Trial	Design	Result	Take-home
†PRESERVE	Bicarbonate vs saline; N-acetylcysteine (NAC) vs placebo	No difference	Saline; no NAC
†STARRT-AKI	Accelerated vs standard renal replacement therapy	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	SGLT2 inhibitor, CKD with or without diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	Finerenone, CKD with type 2 diabetes	Fewer kidney and cardiovascular events	Add to RAAS blockade
†CHOIR / TREAT	High vs lower haemoglobin target	Harm with high target	Avoid Hb >13 g/dL

91 Renal Transplantation

1 · End-stage disease and why transplant

Transplant beats dialysis, so plan it early.

- KDIGO stage **G5 (eGFR < 15 mL/min/1.73 m²)** is end-stage renal disease (ESRD).
- **Transplant is optimal therapy**, and **pre-emptive** transplant (before dialysis) is best. It may triple life expectancy, but death risk is **higher for the first 3–6 months**, then much lower (Wolfe).

2 · Recipient evaluation

> **KEY POINT**
> Two questions: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Deceased-donor listing is allowed at eGFR ≤ **20**.

Recurrence risk	Diseases	Implication
Very low	Autosomal dominant polycystic kidney disease (ADPKD), dysplasia, Alport, **genetic** focal segmental glomerulosclerosis (FSGS)	None
Higher	**Primary FSGS**, membranoproliferative GN, membranous	Counsel; monitor
Histological often, clinical rarely	IgA nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Disease coded in native-kidney genes cannot recur in the graft; disease from circulating or immune factors can.

- **Cardiac**: coronary artery disease is the main cause of death; screen high-risk patients.
- **Vascular**: heavy iliac calcification may prevent anastomosis (non-contrast CT). ADPKD with a family history of intracranial bleed needs an aneurysm screen.
- **Obesity** is a relative contraindication that increases wound complications, delayed graft function (DGF) and rejection.
- **Infection**: active tuberculosis contraindicates until treated. Screen for hepatitis B, C and HIV, and **vaccinate before** transplant, because live vaccines are barred afterwards.

Malignancy	Transplant
Prostate Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Without prior treatment
Basal/squamous skin, melanoma in situ, renal cell carcinoma <3 cm, small low-grade thyroid, superficial bladder cancer	No wait after cure
All other	After treatment, individualised wait

P2 Pre transplant nephrectomy — indications: ADPKD space/infection/bleeding/pain, chronic infection, VUR with UTI, heavy proteinuria

3 · Immunological compatibility

P3 Discuss the role of tissue typing and cross match in live and cadaveric — ABO rules (A2 donors), HLA class I/II, sensitisation, crossmatch, Luminex virtual crossmatch, paired exchange

4 · Deceased donation

P2 P1 Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval; Kidney retrieval techniques for organ donation — DBD vs DCD, contraindications, KDPI/EPTS, en bloc < **15 kg**, retrieval steps, UW/HTK, machine perfusion ↓DGF

Not in the paper answer:

- En bloc kidneys go to an adult with body mass index <30 and low immunological risk; thrombosis is 10–15%.

	Donation after brain death (DBD)	Donation after circulatory death (DCD)
Warm ischaemia	Minimal	Present, so rapid aortic cannulation and cooling are needed
Preservation	Static cold storage	Usually machine perfusion (reduces DGF)

ALGORITHM

1. Midline incision and sternotomy; mobilise bowel off the retroperitoneum.
2. Control the aorta and inferior vena cava (IVC).
3. Find aberrant or accessory arteries **before cannulation**.
4. Perfuse; ice via opened Gerota fascia.
5. Remove kidneys en bloc with **periureteric tissue preserved**.
6. Separate on the back table; store cold.

5 · Living donation

P2 Pre-operative evaluation of the living (related and unrelated) renal donor — donor safety, exclusions (bilateral stones), GFR by 2 methods, CT angiography, split function; function 60–80%, ESKD 90 vs 14/10,000

Not in the paper answer:

- Donor mortality is **3/10,000**. An independent donor advocate is mandatory.
- **Laparoscopic donor nephrectomy** is standard.

6 · The operation

- On the back table, ligate hilar lymphatics (lymphocele). **Never dissect the triangle between lower pole and ureter.** Extend a short right vein with donor IVC.
- **Multiple arteries** (18–43%) mean more DGF but similar survival; use a Carrel patch, conjoining or separate anastomoses.
- **Always keep the lower pole artery**, because it supplies the ureter; upper pole branches ≤2 mm may be ligated.
- The graft goes in the retroperitoneal iliac fossa. **Contralateral placement** brings the pelvis and ureter anterior.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near the anterior superior iliac spine (ASIS)	Midline suprapubic, up the rectus edge
Access	Iliac vessels	Better to common iliac, IVC
Hernia	Lower	Higher; rectus denervation

ALGORITHM

1. **Vein first**: end-to-side to the external iliac vein, 5-0/6-0 polypropylene.
2. Artery end-to-side to the external iliac artery (punch arteriotomy; avoid calcified clamp sites).
3. Reperfuse, then position the kidney without kink or stretch.

- A flaccid graft suggests poor inflow and a tense, engorged graft poor outflow; check with intraoperative ultrasound.

ALGORITHM

1. **Lich–Gregoir** extravesical: fill the bladder; incise the detrusor down to mucosa.
2. Open the mucosa distally and spatulate the ureter.
3. Suture mucosa to mucosa with 5-0/6-0 absorbable, with or without a stent.
4. Close the detrusor loosely as a short tunnel.

- Document stent removal, because forgotten stents encrust.

7 · Surgical complications

Early vascular problems lose grafts within hours.

P1 Vascular complications after renal transplantation, including transplant renal artery stenosis — arterial/venous thrombosis (reversed diastolic flow), TRAS (parvus-tardus, angioplasty first), thrombosis risks, torsion, pseudoaneurysm, biopsy AV fistula

Not in the paper answer:

- Bleeding (brisk drain, falling haematocrit) needs transfusion and re-exploration.

P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — risks incl. mTOR inhibitors; lymph creatinine = serum; treat if symptomatic; laparoscopic fenestration best, aspiration worst

Transplant nephrectomy

- Indications are **graft intolerance syndrome** (pain, swelling, haematuria, fever; steroids first), infection and tumour, not an asymptomatic failed graft.
- **Early (<6–8 weeks), go extracapsular**; later, go **subcapsular**, except for tumour.

8 · Rejection

Chronic antibody injury now limits graft survival.

- The usual regimen is induction, then tacrolimus and mycophenolate, with or without steroids.

P1 Renal allograft rejection. — hyperacute/acute/subclinical/chronic; acute <10%, chronic AMR limits survival; biopsy (Banff TCMR/AMR); risks; de novo DSA, dd-cfDNA NPV >90%; TCMR steroids, then ATG; AMR plasmapheresis, IVIG, rituximab, eculizumab, anti-IL-6

Not in the paper answer:

- De novo donor-specific antibody (DSA) is linked to class II mismatch and under-immunosuppression.

- Hyperacute rejection is †irreversible, so the graft is removed.

9 · Landmark trials

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

★ Putting it together: kidney function and transplant decisions

A · Rising creatinine after urological surgery

ALGORITHM

1. **Postrenal first:** check the catheter and scan. After cystectomy, a rise with ascites may be a urine leak (drain creatinine ≫ serum).
2. **Prerenal vs intrinsic:** FeNa <1% with bland urine means volume depletion; >1% with granular casts means acute tubular necrosis. Stop nephrotoxins, RAAS blockers and metformin.
3. **Dialysis** only for refractory "AEIOU": acidosis, electrolytes (K), intoxication, overload, uraemia.

B · Post-obstructive diuresis

KEY POINT

After bilateral or solitary-kidney relief, monitor output and electrolytes. Diuresis is mostly physiological; †it is pathological if it persists after euvolaemia. †Do not replace every millilitre; if the patient cannot drink, replace ≈75% with 0.45% saline.

C · Transplant ureteric complications

Problem	Clue	Management
Urine leak (early)	High drain output; fluid creatinine ≫ serum	Foley and nephrostomy ± stent; explore if large or persistent
Ureteric stricture	Falling function ± mild hydronephrosis	Nephrostomy, then dilation or endoureterotomy; open repair is most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter or vessels	Treat only if symptomatic (<15%); laparoscopic window is best
Stent infection	Colonised stent	Remove at 14–21 days (earlier risks leak; later risks infection)

Prevent them with a stent and by sparing the lower pole–ureter triangle.

D · Rejection at a glance

Type	Cause	Point
Hyperacute	Preformed donor-specific antibody (DSA)	Crossmatch prevents it; †irreversible, so remove graft
Acute	T-cell-mediated (TCMR) or antibody-mediated (AMR), Banff	Biopsy. TCMR: IV steroids, then antithymocyte globulin (ATG). AMR: steroids, IV immunoglobulin, plasmapheresis, rituximab
Chronic	Mainly antibody-mediated	Main late loss; subclinical

E · Child going for transplant

Fix the bladder before transplant.

ALGORITHM

1. Most have congenital anomalies (CAKUT), chiefly valves.
2. Make the bladder safe: age-appropriate capacity at detrusor pressure <40 cm H_2O (antimuscarinics, catheterisation, augmentation).
3. Remove refluxing ureters or kidneys causing infection, polyuria or hypertension.
4. A poor bladder destroys the graft.

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 16

BPH

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 148	Anatomy and Development of the Prostate
Ch 149	Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History
Ch 150	Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia
Ch 151	Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia
Ch 152	Simple Prostatectomy: Open, Laparoscopic and Robotic
★	Putting it together: the BPH pathway

148 Anatomy and Development of the Prostate

1 · Development (embryology)

The androgen pathways that build the prostate also drive BPH.

- **Origin**: the urogenital sinus (UGS), endoderm in mesoderm. Incomplete cloacal division (imperforate anus, urorectal fistula) impairs development.
- **Dihydrotestosterone (DHT)**: †5α-reductase type 2 in UGS mesenchyme converts testosterone to DHT, the more potent androgen receptor (AR) ligand. **SRD5A2** loss leaves the prostate incomplete; castration or inactive AR leaves it absent. †The same enzyme is the target of 5α-reductase inhibitors (5-ARIs) in BPH.
- **Androgen is not rate-limiting**: testosterone is present at week 9 but budding starts at week 10, gated by **AR demethylation** in the UGS.

Week	Event
6–7	Cloaca divides into the ventral UGS and the dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR, whose andromedins instruct the epithelium
10	Epithelial buds invade the mesenchyme; this fixes the adult ductal pattern

Mesenchyme instructs, epithelium responds

- Budding and branching need **mesenchymal (stromal)** AR, not epithelial AR; UGS mesenchyme can even turn bladder epithelium into prostate (Cunha).
- **Andromedins** are androgen-induced stromal signals, chiefly **fibroblast growth factor 10 (FGF10)**; Nkx3-1 specifies the epithelium.

> **KEY POINT**
> McNeal: **re-awakening of embryonic induction** in ageing transition-zone stroma may initiate new glands, one theory of BPH.

- Gestational diethylstilboestrol, phthalates and dioxin disrupt development; dioxin-induced hyperinnervation may explain variable α-blocker response.

2 · Zonal anatomy (McNeal)

P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — zones (PZ 70% glands/70% cancers, CZ 25%, TZ 5%/BPH, AFMS), 35° urethral bend, surgical capsule, fasciae, NVB at 5/7 o'clock, urethral/capsular arteries, Santorini plexus, lymph nodes

Not in the paper answer:

- The urethra bends **≈35°** at the verumontanum, where 13–15 ducts open. This is the distal limit of transurethral resection (TURP) and enucleation.
- The transition zone (TZ) has the shortest ducts and gives †≈20% of cancers; the peripheral zone (PZ) has the longest.

> **KEY POINT**
> Trap: the central zone (CZ), around the ejaculatory ducts, is not the TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). The enlarging TZ compresses PZ into the **surgical (pseudo)capsule**, the enucleation plane (see Ch 152).

3 · Fascia, vessels †

- †The capsule is incomplete anteriorly and apically. The neurovascular bundles lie outside it, so enucleation spares them.
- †Prostatic veins join the vertebral (**Batson**) plexus, a route to spinal metastases.

4 · Histology and cell types

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	Prostate-specific antigen (PSA), NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, prostatic acid phosphatase (PAP)	Almost none

- The urethra and proximal ducts carry club, hillock and neuroendocrine cells; BPH stroma contains prominent lymphocytes.

5 · Secretions and clinical markers

- Prostatic fluid is **37–44%** of semen and acidic (pH ≈6.3); seminal vesicle fluid is the alkaline majority.
- Kallikreins liquefy semen within **20–30 min**; incomplete liquefaction at 60 min is nonliquefaction.
- **Ion trapping**: macrolides concentrate in acidic fluid; inflammation alkalinises it.
- PSA, PSMA and PAP all occur outside the prostate, so "prostate-specific" is a misnomer.

Product	Key facts
PSA (KLK3)	Serine protease (19q13.4); liquefies semen
PSMA	Folate hydrolase; rarely deleted in cancer, hence PSMA-PET and ^{177}Lu-PSMA
PAP	First tumour marker (Huggins and Hodges, 1941) MCQ
Zinc	Highest of any tissue; inhibits mitochondrial aconitase, so citrate is secreted

KEY POINT

Benign cells: high zinc blocks aconitase, so citrate is exported. Cancer: **zinc is lost** and citrate is oxidised, so citrate is low (†the basis of MR spectroscopy).

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

1 · Definitions and the four overlapping entities

Histology, size, obstruction and symptoms differ; treatment depends on which.

P2 Pathophysiology, classification and management of LUTS. — BPH vs BPE vs BPO vs BOO, ICS symptom phases, static/dynamic (α1A) components, capsule/TUIP, bladder hypertrophy then decompensation, one-third keep storage LUTS after surgery

Not in the paper answer:

- †BOOI = PdetQmax − 2Qmax: **>40 obstructed, 20–40 equivocal, <20 unobstructed**.
- "Hypertrophy" is wrong because BPH raises cell **number**, not size; "prostatism" is obsolete (Abrams coined "LUTS") MCQ.

Term	How measured
Benign prostatic hyperplasia (BPH): a histological increase in epithelial and stromal cell number in the periurethral/transition zone (TZ)	Histology
Benign prostatic enlargement (BPE): enlargement caused by BPH	Digital rectal examination (DRE), transrectal ultrasound (TRUS), MRI
Bladder outlet obstruction (BOO): obstruction from any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow (best); flow rate (indirect)
Benign prostatic obstruction (BPO): BOO caused by BPE	Pressure–flow
Lower urinary tract symptoms (LUTS): storage, voiding and post-micturition; not prostate-specific	International Prostate Symptom Score (IPSS)

- The four entities overlap only **partially**; BOO may exist without BPE (for example, a stricture). Detrusor ageing, polyuria, sleep disorders and impaired contractility can all mimic BPH.

> **KEY POINT**
> Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are separate; obstruction is diagnosed only by pressure–flow.

2 · Epidemiology

- At autopsy there is **no BPH under 30**; prevalence is ≈ **50% in the 50s**† and **88% in the 80s**. It is **similar across all ethnic and geographic groups**.
- No clinical definition is globally accepted. BPH is described by **symptoms** (IPSS), **enlargement** and **obstruction**, plus bother. The proposed volume cut-off is **>30 mL**; a community composite is IPSS >7, maximum flow rate (Qmax) <15 mL/s and volume >20 mL.

P3 P1 IPSS score; Methods to estimate prostate size — IPSS items, bands 0–7/8–19/20–35, QoL item, no incontinence/pain items, LURN; DRE underestimates, TRUS ellipsoid, MRI, CT

Not in the paper answer:

- Probability of obstruction by Qmax: <10 mL/s high, 10–15 intermediate, >15 low. Qmax falls almost linearly with age (Olmsted median 20.3 mL/s at 40–44 versus 11.3 at 75–79).
- DRE underestimates size by 25% to ≥50%.

ALGORITHM

1. Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**.
2. Transition zone volume (TZV) ≈ 15 → 25 mL.

3 · Aetiology

KEY POINT

BPH is an **accumulation** disease: volume = proliferation minus death. Established BPH shows equal or reduced replication and raised bcl-2. Androgens act chiefly by **suppressing cell death**, so withdrawal shrinks the gland.

- New glands otherwise form only in the fetus, hence the theory of **embryonic reawakening** of stromal induction.

3.1 Androgens: permissive, not causal

- Prepubertal castrates and men with androgen defects **never develop BPH**; withdrawal gives only **partial** involution. The two established determinants are **age** and **functioning testes**.
- Serum testosterone falls with age, but **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. DHT is **maintained, not elevated**; old "raised DHT" reports used cadaveric controls MCQ.
- Serum testosterone does not predict volume or PSA. Androgens act via growth factors rather than as direct mitogens, and they are anti-inflammatory.

ALGORITHM

1. Testosterone enters the stromal cell.
2. **Type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) converts it to DHT.
3. DHT acts on the stroma (autocrine) or diffuses to the epithelium (paracrine).
4. The DHT–AR complex binds DNA and switches on growth factors.
5. Growth factors drive epithelial survival and growth.

- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% is adrenal. DHT has higher AR affinity and forms a more stable complex.

Feature	Type 1	Type 2
Site	Skin, liver	Prostate **stromal fibroblasts**, some basal cells
5α-reductase deficiency	Normal	**Mutated**
Finasteride / dutasteride	Not inhibited / inhibited	Inhibited / inhibited
Role	Possibly cancer	Development and BPH growth

- Finasteride shrinks the gland as much as dual inhibition, so type 1 is unimportant in BPH.
- SRD5A2 lies on **chromosome 2p23** MCQ; its polymorphisms link inconsistently to BPH.

3.2 5α-reductase inhibitor (5-ARI) resistance

- 5-ARIs are the **only medical therapy that reduces volume**.
- Resistance is **not** failure to lower DHT but focal lack of luminal atrophy in nodules, or stromal hyperplasia, often linked to inflammation (NF-κB-driven AR variants, SRD5A2 silencing). The proposed definition is unchanged volume and no atrophy despite lower DHT.

3.3 Oestrogens

- In dogs, androgen plus oestradiol produces BPH; oestrogen induces AR and increases stroma. Oestrogen rises with age relative to testosterone, and adipose tissue aromatises.
- **Oestrogen receptor α (ERα) is stromal** and proliferative; **ERβ is epithelial** and antiproliferative. An ERβ agonist failed in a trial for LUTS.

3.4 Cell death; stromal–epithelial interaction

- Apoptosis after castration is active and probably mediated by transforming growth factor-β (TGF-β).
- Hypothesis: a **stromal defect removes a brake** on epithelial proliferation. Periurethral interstitial fibroblasts (carrying type 2 5α-reductase) increase in BPH and expand the TZ; **no drug targets them**.
- Chemokines such as interleukin-8 (IL-8) turn fibroblasts into collagen-producing myofibroblasts, and the resulting fibrosis contributes to LUTS.

3.5 Growth factors

Androgen permits; growth factors signal and are the likelier primary defect. The first prostatic growth factor found was basic fibroblast growth factor (FGF-2) MCQ.

Factor	Source and target	Role in BPH
FGF-2	Stroma (autocrine)	Raised 2–3-fold; induced by TGF-β and hypoxia
FGF-7 (keratinocyte growth factor, KGF)	Made by stroma; receptor **only on epithelium**	Leading stromal mediator of hormonal control
FGF-10	Smooth muscle to epithelium	Raised by androgens
TGF-β1	Stroma	**Inhibits epithelium** but is **mitogenic for fibroblasts**; BPH stroma shows a blunted response
Insulin-like growth factors (IGF-1, -2)	—	Transgenic IGF-1 causes enlargement; serum IGF-1 is unrelated to volume
Others	—	Epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), FGF-17

- **Hypoxia hypothesis**: hypoxia induces HIF-1, which raises FGF-2 and FGF-7, so growing nodules become more hypoxic; vascular disease such as diabetes may add risk.

3.6 Inflammation

- **Activated T cells** secrete growth factors, and cytokines drive the stroma. Autoimmune disease increases BPH, whereas TNF antagonists are associated with less.
- **Inflammation correlates with enlargement and 5-ARI resistance**, but causality is unproven.

3.7 Metabolic syndrome, lifestyle

- Metabolic syndrome causes prostatic inflammation and LUTS via aromatisation, **hyperinsulinaemia driving sympathetic overactivity**, and low HDL.
- **Waist circumference** predicts prostatectomy and LUTS independently of BMI. BMI >35 versus <25 gives an odds ratio (OR) of **3.5** for a prostate ≥40 mL.

Factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (LUTS)	Increases risk
Physical activity; moderate alcohol	Lowers risk (alcohol lowers BPH, **not LUTS**)
Smoking, diet, vasectomy, sexual activity	No consistent effect (**vasectomy is not a risk factor**)
Jewish religion	More prostatectomy, not more BPH (access to care) MCQ

- **Depression** is strongly linked to LUTS, and LUTS severity correlates with sexual dysfunction.
- **Drugs that worsen LUTS**: α-sympathomimetic cold remedies, antidepressants, antihistamines and bronchodilators; anticholinergic and adrenergic drugs can precipitate retention.

3.8 Genetics and others

- In young men with large glands (>37 g), first-degree relatives have ≈ **4-fold** risk of surgery. Inheritance is **autosomal dominant**; ≈ **50%** of men operated on before 60 have heritable disease, versus ≈9% after 60.
- **Familial BPH** shows early onset, a **large gland**, normal androgens and a normal 5-ARI response. Only **dogs and humans** get BPH.

4 · Pathology

- **Every nodule arises in the TZ or the periurethral region**. The TZ lies outside the preprostatic sphincter; the periurethral zone lies inside it, with glands parallel to the urethra.
- The **middle lobe** is **periurethral**, because there is no TZ at the bladder neck MCQ. Patterns are middle, lateral, trilobar and posterior commissural (median bar).

> **KEY POINT**
> Only humans have a **capsule**, transmitting expansion onto the urethra; dogs get BPH but rarely obstruct. Transurethral incision of the prostate (TUIP) relieves obstruction without changing volume. Hence **size does not correlate with obstruction**.

Feature	Periurethral nodules	TZ nodules
Composition	Purely stromal (early), like **embryonic mesenchyme**	Glandular, tall epithelium
Origin	Interstitial fibroblasts	New duct branches forming a new ductal system

- **Two phases**: for ≈20 years nodule number rises; then glandular nodules grow.
- Small glands are mainly stroma and large glands mainly epithelial nodules. This variable stromal:epithelial ratio explains variable drug response, since α-blockers target smooth muscle and 5-ARIs target epithelium.

5 · Pathophysiology of obstruction

Many symptoms come from the bladder, so some persist after surgery.

ALGORITHM

1. Hyperplasia raises urethral resistance (static and dynamic components).
2. The detrusor compensates, at a cost to storage function.
3. Storage LUTS appear, with or without impaired emptying.
4. In a minority, the detrusor decompensates.

- The static component (bulk under the capsule, elastic ECM) is **not** reduced by α-blockade. The dynamic component is smooth muscle tone via **α1a** receptors (the most abundant subtype).
- Other regulators include endothelin and **phosphodiesterases 4 and 5 (PDE4/5)**, which is why PDE5 inhibitors help LUTS.
- Many LUTS arise from **obstruction-induced bladder change**. ≈ **One-third** of men keep voiding dysfunction (mostly storage) after relief.
- Storage changes are detrusor overactivity (DO) and lower compliance; emptying changes are reduced contractility and, in a minority, detrusor failure.

ALGORITHM

1. **Three-stage remodelling**:
2. Smooth muscle hypertrophy.
3. Compensation.
4. Decompensation (underactive bladder).

- Stretch, pressure and **hypoxia/ischaemia** switch the detrusor to a **secretory phenotype** that lays down ECM.
- **Trabeculation** reflects increased detrusor **collagen** and progresses to cellules, diverticula and failure; there is no accepted grading.

KEY POINT

Storage LUTS persist after transurethral resection of the prostate (TURP) because the collagen-rich, overactive detrusor does not reverse, so counsel men with DO before surgery. Acute urinary retention (AUR) is **not** the inevitable end: many men in AUR have an adequate detrusor and a precipitant.

6 · Natural history

Who progresses decides watchful waiting versus treatment.

- **Regression to the mean**: entry thresholds select men at a temporary extreme, so they seem to improve. It is part of every placebo effect.
- Within trials, IPSS, Qmax and volume barely correlate (restriction of range); only PSA and volume still correlate.
- ≈ **40%** improve without treatment (watchful waiting behaves like placebo); active drug gives about twice the placebo improvement.
- **Olmsted County**: IPSS rises ≈ **0.2–0.34 points/year**; Qmax falls ≈ **2.1%/year** (faster if age ≥70 or Qmax <10); volume grows ≈ **0.6 mL/year** (faster with higher PSA or volume). Volume >50 mL gives ≈3.5-fold odds of moderate–severe LUTS.
- A perceptible improvement is ≈ −2 IPSS points at low baseline versus −6 at high baseline.

Predictor of progression	Predicts
Age	Everything; AUR rises linearly from age 40 to 80+
PSA ≥1.4–1.5 ng/mL	Growth, symptoms and flow, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, new LUTS
IPSS ≥8 or high bother	AUR, surgery
Qmax <10–12 mL/s	AUR, flow decline, treatment
Raised post-void residual (PVR)	Treatment failure, AUR

- **MTOPS** (Medical Therapy of Prostatic Symptoms) placebo arm: progression **4.5 per 100 patient-years** (17% at 4 yr), mostly an **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.
- **PLESS** (Proscar Long-term Efficacy and Safety Study) placebo arm: symptoms drifted back to baseline. Surgery was 10% at 4 yr, rising from ≈6% to ≈15% across PSA and volume tertiles.
- **REDUCE** placebo arm: volume 40–80 mL and Qmax <10 each gave a hazard ratio of 1.67 for new LUTS.

> **KEY POINT**
> PSA is a cheap surrogate for volume, the main driver of AUR and surgery; models with >100 predictors barely beat PSA alone.

7 · Complications

These are the absolute indications for surgery.

> **KEY POINT**
> Death, renal failure, stones and infection are now rare. The common events are **AUR and surgery**, predicted by age, symptoms, low flow and a large gland (or high PSA).

7.1 Acute urinary retention

- Precipitants include infection, overdistension, fluid load, alcohol and bed rest; prostatic infarction is **not** proven.

	Spontaneous AUR	Precipitated AUR
Trigger	None	Surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- AUR incidence is ≈ **4.5–6.8/1000 person-years** in the community, 18 in PLESS placebo and 6 in MTOPS. The hazard is constant over time.
- **Age and symptoms multiply risk**: in Olmsted, men in their 70s with IPSS >7 had 34.7/1000 person-years. Relative risks: IPSS ≥8 3.2; Qmax <12 3.9; volume >30 mL 3.0; PSA >1.4 several-fold.

7.2 Other complications

- **Renal**: "**silent prostatism**" is asymptomatic renal failure from BOO; MTOPS had none in >3000 men. †Catheterise first, watch for post-obstructive diuresis, let creatinine stabilise, then operate.
- **Bladder stones** are ≈ **8×** commoner because of stasis; renal stones are not.
- **Haematuria** reflects increased microvessel density; **finasteride** is first-line because it lowers VEGF. †Exclude cancer and stones first.
- **UTI** is rare (MTOPS 0.1/100 patient-years). **Incontinence** is overflow or urge (DO in up to half).
- **Surgery** is a soft end point and AUR a hard one. In Olmsted, moderate–severe symptoms, Qmax <12, volume >30 and PSA ≥1.4 each gave ≈4× treatment.

8 · Landmark trials

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure RR 0.48; 5-yr 10% vs 21%; ≈36% crossed over; delayed TURP smaller gains	Watchful waiting safe; delay may reduce benefit
PLESS	Finasteride vs placebo, 4 yr	Placebo returns to baseline; AUR/surgery rise with PSA/volume	PSA/volume predict progression
MTOPS	Placebo vs doxazosin vs finasteride vs combination	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4; no renal failure	Defined "clinical progression"
Olmsted County Study	Population cohort, 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural history; age, IPSS, Qmax, volume predict AUR

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

1 · Definitions and the diagnostic problem

Not every man with urinary symptoms is obstructed; find the cause first.

- Benign prostatic hyperplasia (BPH) is **histological** proliferation in the **transition zone (TZ)**. It can lead to benign prostatic enlargement (BPE), benign prostatic obstruction (BPO, a form of bladder outlet obstruction, BOO) and lower urinary tract symptoms (LUTS).
- LUTS are multifactorial, including detrusor overactivity (DO) or underactivity (DUA), stricture and infection (UTI).

> **KEY POINT**
> Bothersome LUTS are not BPH-related BOO until assessed. Aims: **grade severity and bother** and **identify the cause**. Symptoms, flow, post-void residual (PVR), volume and cystoscopy cannot separate obstruction from bladder dysfunction; only the **pressure–flow study** can (gold standard, reserved for when it changes surgery).

Guidelines: American Urological Association (AUA) BPH (2021); European Association of Urology (EAU) non-neurogenic male LUTS (2022).

2 · History and symptom assessment

- **Comorbidities**: diabetes mellitus or insipidus, renal disease, heart failure, sleep apnoea, neurological disease, haematuria and prior lower-tract surgery.
- **Drugs**: loop diuretics, β-blockers, excess fluid and caffeine worsen storage; anticholinergics impair contractility.
- Assess **sexual function**. LUTS do **not** raise prostate cancer (PCa) risk.
- Storage symptoms (frequency, urgency, urge incontinence, nocturia) form **overactive bladder (OAB)**.

P3 IPSS score — AUASI 7 items + QoL (0–6, WHO 1993), subscores not validated, does not diagnose BOO, VPSS/BII/OAB-q alternatives

Not in the paper answer:

- The AUA Symptom Index (AUASI) and International Prostate Symptom Score (IPSS) total **0–35**: mild **0–7**, moderate **8–19**, severe **20–35**.
- The minimum perceptible change is **3 points**; with a severe baseline, **percentage** reduction is more reliable.

3 · Examination and DRE

- Look for a palpable bladder, meatal stenosis and phimosis, and check **anal tone** and lower-limb neurology.
- **Digital rectal examination (DRE)** estimates volume and screens for PCa. It **underestimates prostate volume (PV) above 30 mL**; a flat gland suggests <30 mL.
- PV does not correlate with symptoms or obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · Frequency–volume charts and nocturia

- A frequency–volume chart (FVC) records times and volumes; a bladder diary adds intake and urgency. **Indication**: predominant **storage LUTS or nocturia**. **Duration**: **3 days**, since adherence falls by 7 days.
- **Nocturia** is waking ≥1 time to void. **Nocturnal polyuria** is night output **>20%** of 24-h output (young) or **>33%** (over 65).

ALGORITHM

1. **Nocturia**
2. Complete an FVC.
3. If †global polyuria (>40 mL/kg/24 h), look for diabetes or excess intake.
4. If nocturnal polyuria, look for heart failure, sleep apnoea, evening fluids or diuretic timing.
5. If reduced capacity, think of OAB or BOO.
6. Treat the cause. If nocturia persists, consider **sleep apnoea** (AUA).

Nocturnal polyuria: no fluid 2 h before bed, no evening alcohol or caffeine, retimed diuretics, with or without **desmopressin**. †It risks **hyponatraemia**, so check sodium at baseline, 1 week and 1 month.

5 · Laboratory tests

- **Urinalysis is mandatory** for UTI, microhaematuria, glycosuria and proteinuria. Gross haematuria needs exclusion of bladder cancer; **cytology** is for severe storage symptoms with dysuria, or smokers.
- **Prostate-specific antigen (PSA)** gives PCa risk, a volume surrogate and progression risk. Test only if life expectancy is **>10 years** and it changes management.
- PSA predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s). Baseline PSA also predicts acute urinary retention (AUR), surgery, and change in symptoms and maximum flow rate (Qmax).
- **5α-reductase inhibitors (5-ARIs) lower PSA by 40–50% by 12 months**, so take a baseline first. †Double the PSA after ≥6 months; any rise from the nadir needs evaluation.
- **Creatinine and eGFR** are checked (EAU) if renal impairment or retention is suspected.

6 · Instrumental investigations

PVR

- A PVR **<30 mL** is insignificant; persistently **>50 mL** is important; **200–300 mL** may mean DUA and a poorer response; **>300 mL** is significant chronic retention.
- High PVR means BOO **or DUA**; it predicts symptom deterioration (MTOPS >39 mL; ALTESS >93 mL), **not surgery**.

Uroflowmetry

- Record **Qmax**, voided volume and pattern. Repeat if voided volume is **<150 mL**.
- It **cannot distinguish BOO from DUA** (DUA in 9–48%). Qmax **10 mL/s** is more specific; **15 mL/s** is more sensitive.
- Optional at first; **recommended before active treatment** and for monitoring.

Urodynamics

P2 Role of UDS before prostate surgery — cystometry/PFS, BOOI and BCI, BOO = high pressure + low flow vs DUA, EAU indications, UPSTREAM: not routine

Not in the paper answer:

- Pressure–flow study (PFS) indications (EAU): failed invasive treatment; cannot void **>150 mL**; PVR **>300 mL**; age **>80** or **<50 years** with voiding LUTS; neurological disease or pelvic surgery. AUA: before surgery if **Qmax >10 mL/s**.
- DUA predicts long-term surgical failure.

Imaging and endoscopy

P1 Methods to estimate prostate size — TAUS/TRUS ellipsoid, planimetry, MRI, CT; IPP grading

Not in the paper answer:

- **Intravesical prostatic protrusion (IPP) ≥10 mm** is high grade; it predicts need for therapy and a **failed trial without catheter (TWOC)**. Bladder wall thickness and other non-invasive tests are accurate, but **none replaces PFS**.
- **Transrectal ultrasound (TRUS)** is the method of choice for PV, needed before a 5-ARI and **any BPH surgery**; a median lobe guides the choice of minimally invasive therapy.
- **Cystoscopy** neither diagnoses BOO nor decides treatment. Indications include haematuria, recurrent UTI, suspected stricture, and **excluding a median lobe before prostatic urethral lift (PUL)**.

7 · Evaluation algorithms

ALGORITHM

1. **EAU (men ≥40 yr)**
2. History (including sexual), symptom score, examination with DRE, urinalysis, PVR, and PSA if life expectancy >10 yr.
3. If DRE, PSA or urinalysis is abnormal, or neurological disease, follow the relevant guideline.
4. If not bothersome, no further work-up.
5. If PVR is significant, renal ultrasound with or without renal function.
6. If bothersome, FVC (storage or nocturia), then prostate ultrasound and uroflow if treating, then the medical algorithm.
7. Before surgery, endoscopy if it alters the choice; PFS for specific indications.

ALGORITHM

1. **AUA**
2. History, examination, IPSS, urinalysis, with or without PVR and flow.
3. If **PVR >300 mL**, manage as non-neurogenic chronic retention.
4. Counsel: behavioural, medical, procedural.
5. Reassess at **4–12 weeks**. If improved, follow up. If nocturia predominates, FVC and a drug. If incomplete, FVC, PVR and flow, urodynamics and cystoscopy, then the OAB pathway, surgery, or both.

8 · Watchful waiting (WW)

- **Indication**: mild–moderate uncomplicated LUTS, minimal bother. **Excluded**: gross haematuria, UTI, raised PSA or PVR, and neurological disease.
- WW is reassurance, education, lifestyle and **yearly** review; in the VA trial ≈a quarter crossed over because of **bother**.
- Lifestyle: fluids **1.5–2 L/day**, none 2 h before bed; less caffeine and evening alcohol; retime diuretics; **double voiding** and **urethral milking**.

KEY POINT

Treatment is driven by **bother and complicating features**, not by score or size alone.

9 · Medical therapy: choice

- Drugs are first line without an **absolute surgical indication** (refractory or recurrent AUR, recurrent UTI, renal insufficiency, stones, recurrent haematuria; see Ch 151).

P2 BPH: medical management and surgical modalities — drug classes and doses, MTOPS/CombAT/SMART, IFIS, ALFAUR, desmopressin, surgical options by size

Not in the paper answer:

- Antimuscarinics need PVR <200 mL; avoid oxybutynin in the elderly; hexane-extract Serenoa is an EAU option.

ALGORITHM

1. **EAU**
2. Not bothersome: WW.
3. Nocturnal polyuria: lifestyle with or without desmopressin.
4. Storage predominant: antimuscarinic or β3-agonist.
5. If **PV >40 mL and long-term treatment** intended: 5-ARI with or without an α1-blocker or phosphodiesterase-5 inhibitor (PDE5I).
6. Otherwise α1-blocker or PDE5I; if storage symptoms remain, add an antimuscarinic or β3-agonist. **AUA**: α-blocker (PDE5I if erectile dysfunction, ED) → **prostate >30 mL** → add 5-ARI → PDE5I trial → surgery.

KEY POINT

Reading trials: placebo improves scores and flow, so judge by a 3-point or percentage threshold; the goal includes **preventing progression**.

10 · α1-Blockers

P2 Tamsulosin; Silodosin. — α1a ≈70% of prostatic α1, subtype table, dynamic component, onset, ejaculatory dysfunction, IFIS, silodosin renal dosing, greatest BOOI fall

Not in the paper answer:

- **No α1-blocker is clearly more effective**; they differ in adverse effects (AEs). Doxazosin cut progression 39% (MTOPS), **not AUR or surgery**.
- **ALLHAT**: the doxazosin arm stopped for heart failure, so never use it as sole antihypertensive; **never use two α-blockers**.

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; abandoned

Class	Drug	Dose	Notes
Short-acting	Prazosin, indoramin	bd	Not used for LUTS
Long-acting	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin immediate release (IR)	2–8 mg od	Start 2 mg at bedtime, titrate
	Doxazosin GITS (sustained release)	4–8 mg od	Start **4 mg**, **no titration**
Uroselective	Alfuzosin extended release (ER)	10 mg od	Clinically uroselective
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d
	Silodosin	8 mg od	Most α1a-selective
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia

KEY POINT

Titration: α1b blockade causes **first-dose hypotension** and syncope; a bedtime start and titration avoid it. Uroselective drugs are fixed-dose; doxazosin GITS starts at 4 mg without titration.

- They double the placebo symptom gain with **durable** benefit but do **not** shrink the gland; tamsulosin plateaus at **0.4 mg**.
- **AEs**: dizziness, asthenia, orthostatic hypotension and anejaculation. In the **elderly**, dizziness causes falls; severe hypotension is mostly in the **first 8 weeks**.
- **Sexual**: ED equals placebo. Ejaculatory dysfunction is mostly **anejaculation** (emission failure): silodosin (odds ratio ≈32) > tamsulosin (≈9) > alfuzosin.

KEY POINT

Intraoperative floppy iris syndrome (IFIS): **billowing iris, iris prolapse, progressive miosis**. α1a blockade causes dilator atrophy, so IFIS occurs **years after stopping**; stopping pre-op does **not** help. Highest with **tamsulosin**, then alfuzosin, doxazosin and naftopidil. †If cataract surgery is awaited, defer the α1-blocker.

Drug	Vascular	Ejaculatory	IFIS
Terazosin	High	Low	Reported
Doxazosin	High (GITS lower)	Low	Lower
Alfuzosin	Intermediate	Lowest uroselective	Intermediate
Tamsulosin	Low	Dose-dependent	**Highest**
Silodosin	Lowest	**Highest**	Not well studied

KEY POINT

One efficacy, three toxicities: **α1b blockade gives vascular effects** (terazosin, doxazosin); **α1a blockade affects ejaculation and causes IFIS** (silodosin, tamsulosin); **α1d blockade gives storage benefit** (naftopidil).

11 · 5α-Reductase inhibitors

The only drugs that prevent retention and surgery.

P2 Dutasteride. — type 1+2, DHT ↓90%, 0.5 mg, PV ↓25%, AUR ↓57%/surgery ↓48%, PSA doubling, REDUCE, teratogenic

Not in the paper answer:

- **Finasteride 5 mg** (type 2) lowers dihydrotestosterone (DHT) by †~70% and PV by 18–20%; in PLESS, AUR fell 57% and surgery 55%. It performs the same as dutasteride head to head.
- Indication: moderate–severe LUTS with PV >40 mL (EAU) or >30 mL (AUA).
- They relieve the **static component**. Benefit is **slow** (maximum shrinkage at **6 months**) and **volume dependent**, best at **>40 mL**.
- They are the **only class altering natural history**.
- PCPT and REDUCE showed fewer cancers but slightly more Gleason ≥8, so they are **not licensed for prevention**.
- **AEs**: ED, lower libido and lower ejaculate volume (≈2× placebo, waning after 1–2 yr), **depression** (FDA 2011) and rarely gynaecomastia. **Post-finasteride syndrome** is persistence of symptoms after stopping.

> **KEY POINT**
> 5-ARIs treat the gland: slow, volume-dependent and modest for symptoms, but they prevent AUR and surgery, and they halve PSA (both a trap and a tool).

12 · Antimuscarinics

- They block detrusor **M2/M3** receptors, mainly on **afferent** pathways.
- **Indication**: predominant storage LUTS, **PVR <200 mL** and no prior AUR.
- Agents: darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine ER 4 mg and trospium.
- In BOO, PVR rises slightly and AUR equals placebo (trials excluded PVR ≥200).
- AEs: **dry mouth**, constipation and cognitive impairment. **Oxybutynin** has most brain penetration, so avoid it in the elderly; trospium least†.

> **KEY POINT**
> Antimuscarinics treat the bladder, so every recommendation is anchored to a PVR ceiling (200 mL).

13 · β3-Agonists

- β3 stimulation gives **storage relaxation without weakening voiding** (Qmax unchanged), with less dry mouth than antimuscarinics.
- **Mirabegron** starts at **25 mg** and rises to **50 mg**; in BOO, PVR rose only at 100 mg.
- AEs: nasopharyngitis, headache, blood pressure rise ≤1 mm Hg. **Contraindicated** in uncontrolled hypertension **≥180/110 mm Hg**; it is a moderate CYP2D6 inhibitor†.
- **Vibegron** 75 mg has no CYP2D6 inhibition; data are mainly in women.

14 · PDE5 inhibitors

- **Tadalafil 5 mg daily** is the only licensed agent (2.5 mg is ineffective). It lowers IPSS by ≈6 (placebo 3.6), **independent of ED**, and reduces nocturia. **Qmax is unchanged**.
- First choice for LUTS with ED (AUA). AEs: headache, back pain, dyspepsia, flushing. †Contraindicated with nitrates and riociguat.

15 · Combination therapy

- **MTOPS** (Medical Therapy of Prostatic Symptoms): progression = AUASI rise **≥4**, AUR, UTI or urosepsis, incontinence, or creatinine rise ≥1.5 mg/dL. Placebo 17%. Number needed to treat (NNT) for combination 8.4 (≈5 if PSA >4 and PV >40).
- **CombAT** (PV ≥30, PSA ≥1.5): AUR or surgery fell two-thirds versus tamsulosin but not significantly (NS) versus dutasteride.
- **Indication**: high risk, PV **>30–40 mL** and PSA **>1.5 ng/mL**. **SMART**: stop the α-blocker after 6 months.

MTOPS reduction vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

KEY POINT

α-blockers prevent symptomatic progression only; 5-ARIs prevent AUR and surgery.
The α-blocker gives fast dynamic relief; the 5-ARI gives slow static shrinkage.

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS (tolterodine 4-arm; NEPTUNE)	PVR <150–200 mL only; AUR <1–3%
α1-blocker + β3-agonist	Persistent storage (MATCH)	PVR rise; limited data
α1-blocker + PDE5I	LUTS with or without ED; small gain	Hypotension; not recommended; start lowest dose
PDE5I + 5-ARI	Large prostate with or without ED	Single trial; FDA capsule 2021

16 · Phytotherapy

- Evidence is product-specific. Most **Serenoa repens** extracts show no benefit, but the **hexane extract (Permixon) 320 mg daily with food** reduced nocturia and raised Qmax and is an EAU option. AUA/EAU make **no general recommendation** for phytotherapy.

17 · Follow-up

- Worsening symptoms and rising PVR predict AUR.

ALGORITHM

1. Review at **4–12 weeks** (α-blocker response, AEs).
2. Review at **6 months** (5-ARI).
3. Then **annually**: score, flow, PVR; PSA on a 5-ARI if life expectancy >10 yr. WW: yearly.

18 · Acute urinary retention

P2 Catheter free trial — spontaneous vs precipitated AUR, α-blocker before TWOC (ALFAUR), predictors of failure, recurrence, 5-ARI prevention

Not in the paper answer:

- CombAT: AUR fell 68% versus tamsulosin but only 18% (NS) versus dutasteride; α-blockers do not prevent AUR (ALTESS, MTOPS).
- About two-thirds of AUR is spontaneous, with a worse TWOC outcome than precipitated AUR.

ALGORITHM

1. Catheterise (†suprapubic if urethral fails).
2. Start an **α1-blocker**.
3. Do a **TWOC at ~3–5 days**.
4. If he voids, continue (add a 5-ARI for a large gland).
5. If he fails, try up to **3 TWOCs**.
6. If AUR recurs or TWOC fails: PFS and surgery (see Ch 151).

- TWOC success is ≈ **60%**, and recurrence is mostly within **6 months**.
- Failure predictors: age, severe LUTS, drained volume **>1 L**, spontaneous AUR and high-grade IPP. 5-ARIs prevent AUR (PLESS, MTOPS).

KEY POINT

The α-blocker rescues the TWOC (dynamic component); **the 5-ARI prevents the next episode** (static component).

19 · Landmark trials

Trial	Design	Result	Take-home
VA transurethral resection (TURP) vs WW	Randomised trial, moderate LUTS	TURP better; most WW did not fail; crossover by bother	WW safe
PLESS	Finasteride vs placebo, 4 yr, PV ~55 mL	AUR 7% to 3%, surgery 10% to 5%; combined fall 51%	5-ARI prevents AUR/surgery
Dutasteride phase III	2 yr, PV ≥30 mL	DHT fell 90%, PV 26%, AUR 57%, surgery 48%	≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 yr	Progression fell 39/34/66%; AUR/surgery only with 5-ARI	Combination for high risk
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 yr	AUR/surgery fell 66% vs tamsulosin	Combination for large gland
SMART	α-blocker withdrawal	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI prevention	Fewer cancers, more high grade	Not licensed
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

1 · Scope and principles

- **American Urological Association (AUA) 2023 operations**: transurethral incision (TUIP), resection (TURP) and vaporisation (TUVP) of the prostate; photoselective vaporisation (PVP); prostatic urethral lift (PUL; UroLift); water vapour thermal therapy (WVTT; Rezūm); temporary implanted prostatic device (TIPD; iTind); robotic waterjet treatment (RWT; Aquablation); holmium or thulium laser enucleation (HoLEP/ThuLEP); simple prostatectomy; prostate artery embolisation (PAE).
- Transurethral microwave therapy (TUMT) was removed in 2023. Because the sham effect is large, new devices need sham-controlled trials.

KEY POINT

Volume predicts progression (acute urinary retention (AUR), surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County).

2 · Work-up and indications for surgery

- History, examination, IPSS and urinalysis at baseline (Statement (St) 1). International Prostate Symptom Score (IPSS) bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery, post-void residual (PVR) is **required** (St 6). **Consider** size and shape by transrectal ultrasound (TRUS), abdominal ultrasound, cystoscopy, MRI or CT (St 5), uroflowmetry (St 7), and pressure–flow study if the diagnosis is uncertain (St 8).
- A raised prostate-specific antigen (PSA) needs biopsy **before** BPH surgery. Counsel on failure and retreatment (St 9).

AUA Statement 26 (surgery recommended)	† European Association of Urology (EAU): absolute vs relative
Renal insufficiency from BPH; refractory retention; recurrent urinary tract infections (UTIs); recurrent bladder stones or gross haematuria from BPH; lower urinary tract symptoms (LUTS) refractory to, or unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation or renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5α-reductase inhibitor (5-ARI). **Relative**: moderate–severe bothersome LUTS failing or declining medical therapy

- **Antibiotics (AUA 2019)**: one dose suffices for most transurethral cases. Cover gram-negatives and enterococci (1st/2nd-generation cephalosporin or co-trimoxazole), guided by antibiogram and urine culture. **Defer surgery if UTI is active and symptomatic**.

3 · Choosing the operation

Volume / situation	AUA 2023 options	Restrictions
<30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes

Volume / situation	AUA 2023 options	Restrictions
>80–150 cc / >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc
Higher bleeding risk	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Preserve erection and ejaculation	PUL or WVTT (St 37)	Within 30–80 cc

KEY POINT

Only enucleation spans every size band, joined above 80 cc only by simple prostatectomy. Minimally invasive surgical therapies (MISTs) and RWT live in 30–80 cc, so volume and median lobe assessment come first.

4 · Monopolar TURP (m-TURP): the benchmark

Every newer operation is judged against TURP.

P1 P2 TURP instruments and techniques; Measures to decrease complications of TURP — resectoscope, Nesbit/floor-first resection, canoe chips, prevention of each complication

Not in the paper answer:

- AUA: TURP for <80 cc (St 28), monopolar or bipolar by expertise (St 29). Monopolar irrigant is non-ionic (water, **1.5% glycine**†, sorbitol) and **hypo-osmolar**, which causes TUR syndrome.

ALGORITHM

1. Inspect the bladder. If tumour, resect it and defer TURP.
2. Identify orifices, neck, trigone, verumontanum (veru), sphincter; if orifices hidden, give indigo carmine or methylene blue.
3. Resect median lobe, then lateral lobes, then anterior tissue and apex last.
4. Evacuate chips, secure haemostasis, insert a large three-way catheter.

- **Anterior and apex last**, because the anterior fossa perforates easily and the sphincter lies more proximally anteriorly.
- **Arterial bleeding** is bright and must be fulgurated. **Venous bleeding** is darker and vanishes with a full bladder; once arteries are secured, use balloon traction.
- Even routine TURP absorbs **800–1000 mL** irrigant. The catheter comes out at 24–48 h.
- Warn of dysuria for weeks and eschar bleeding at **1–4 weeks**. Results last about 10 years.

5 · TURP complications

Missed TUR syndrome can kill.

P1 P3 TUR Syndrome — mechanism, glycine toxicity, features, risk factors, 3% saline protocol

- Risk factors are a gland **>45 cc**, resection **>90 min**, perforation and irrigant height. Keep the bag at about **60 cm**, because prostatic venous pressure is only about **10 mm Hg**. Na^+ **<120 mEq/L** is severe.

ALGORITHM

1. Stop or expedite surgery, secure haemostasis, stop irrigation; send Na^+.
2. If overloaded, give a †loop diuretic.
3. If severe central nervous system signs: **3% saline 100 mL** (+2–3 mEq/L), repeat ×2 at 10 min (max 300 mL).
4. Target +4–6 mEq/L; ≤12 mEq/L in 24 h (≤8–10 if presentation >48 h). Monitor in high dependency.

KEY POINT

Acute post-TURP hyponatraemia is safer to correct than chronic: the brain has not yet shed its organic osmolytes, so osmotic demyelination is less likely. Bipolar saline TURP eliminates TUR syndrome.

- **Perforation** shows glistening fat. A limited extraperitoneal tear needs only a longer catheter, an extensive one needs drainage, and an intraperitoneal tear needs laparotomy. A persistent intra-operative erection responds to intracorporeal phenylephrine.

Late complication	Cause / features	Management
Bladder neck contracture (BNC)	Over-resection or over-fulguration (small glands); onset ~6 months, flow declines	Dilate, then incise over a wire into perivesical fat. Recurrent: add mitomycin C (10 mL of 0.4 mg/mL). Intractable: Y-V plasty
Stricture	Meatus or fossa navicularis	Dilate meatus 2 Fr above sheath; plenty of gel
Retrograde ejaculation	About **two-thirds**	Counsel; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost; apical over-resection causes total stress incontinence (SUI). Rare, usually transient	Veru is the distal limit; †pelvic floor exercises, then †sling or artificial sphincter
Bleeding	Early: incomplete haemostasis. Delayed (1–4 weeks): eschar	Early major: theatre. Delayed mild: fluids; †clot evacuation if retention
Erectile dysfunction (ED)	No conclusive causal evidence	—
Retreatment	Residual apical/anterior tissue, >80 cc, regrowth, BNC, stricture	Rises with size and pre-op retention

6 · Bipolar TURP (b-TURP)

P2 Monopolar v/s Bipolar TURP — saline, no TUR syndrome (NNT ≈50), less clot retention (NNT ≈20), equal efficacy to 5 y

- It is equally effective and safer, so it may replace m-TURP as gold standard.

7 · TUIP

- TUIP releases capsule and bladder neck without removing adenoma. It is for **<30 cc only** (St 31) without a median lobe (relative contraindication; EAU the same). It suits a young man with a small gland who wants ejaculation.

ALGORITHM

1. Use a Collins knife, loop, or holmium/thulium laser.
2. Incise at 5 and/or 7 o'clock from distal to the orifice, through the bladder neck, to proximal to the veru.
3. Deepen to capsule (with or without fat); cauterise.
4. Catheter out and home on day 0–1.

- Compared with TURP it is shorter, bleeds less and relieves symptoms similarly, but gains less Qmax and needs **more reoperation**; retrograde ejaculation is rarer and BNC does not occur.

8 · Bipolar TUVP

- A broad electrode (PlasmaButton) vaporises and coagulates together, so haemostasis is excellent, but there is **no specimen**. Only **bipolar** is recommended (St 32: <30 and 30–80 cc); monopolar was abandoned for irritation, SUI and poor durability.
- Compared with TURP it needs a shorter catheter and fewer transfusions, but causes more dysuria, AUR and retreatment; retrograde ejaculation is as common.

KEY POINT

Vaporisation trades haemostasis for a coagulated, specimen-free bed. The coagulum causes dysuria and retention; with no anatomical plane, durability is weaker. PVP makes the same trade.

9 · Laser physics

P3 Lasers in BPH, including HoLEP — wavelength/chromophore/penetration table (thulium **0.2 mm** shallowest; Nd:YAG abandoned)

Not in the paper answer:

- Deeper penetration leaves more coagulum, hence Nd:YAG's sloughing and retention.

10 · PVP (GreenLight)

PVP suits anticoagulated men with glands under 80 cc.

P4 Green light lasers — physics, technique, AUA <80 cc + bleeding risk, GOLIATH, complications

Not in the paper answer:

- AUA St 33 recommends 120 W or 180 W PVP for <30 and 30–80 cc; anticoagulated men improve equally without more transfusion.

11 · HoLEP

Size-independent, and the only endoscopic operation beating TURP.

P4 P1 HoLEP — size-independent (St 38/43), only endoscopic method superior to TURP; settings, steps, learning curve 20/50, BNC if <40 cc

Not in the paper answer:

- It is durable to 18 years because the whole transition zone (TZ) goes, whereas TURP leaves a peripheral rim.

ALGORITHM

1. Incise the bladder neck at 5 and 7 o'clock (at 6 o'clock if no median lobe); carry to the veru and enucleate the median lobe.
2. Make a perimontanal inverted-U incision; open the apical plane.
3. Sweep anteriorly to 11–1 o'clock; incise the apical mucosal strip (no sphincter traction).
4. Develop the anterior plane to the neck ("Baleen" sign = bladder); release lobes into the bladder.
5. Haemostasis with a defocused beam; morcellate via nephroscope; 22 Fr three-way catheter, ~60 mL balloon.

- The smooth reflective capsule is the depth reference. If the morcellator engages mucosa, stop cutting and suction.
- **Transient SUI** is common early, in proportion to volume removed; early apical release reduces it.
- Retrograde ejaculation occurs in about **three-quarters**, but median-lobe-only HoLEP preserves it in ~90%. Erections are unchanged or improved and retreatment is rare.

12 · Thulium (ThuLEP) and variants

- Variants include ThuVARP (vaporesection), ThuVEP (vapo-incision) and ThuLEP (blunt dissection); steps match HoLEP.
- Function about equals HoLEP, with less bleeding than TURP (St 38 and 43). Retrograde ejaculation is ~four-fifths; erections are preserved.

	HoLEP	ThuLEP / ThuVEP
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MISTs

The ideal MIST is office-based, single session, needs no general anaesthesia, spares sexual function and ends medication. **Trade-off**: retreatment within 5 years exceeds surgery, and durability beyond 5 years is uncertain.

P4 UroLift (prostatic urethral lift) — implant, 30–80 mL no median lobe, >1.5 cm from neck, L.I.F.T./BPH6, encrustation

Not in the paper answer:

- AUA keeps 30–80 cc and has not adopted the FDA ≤100 cc label.
- PUL typically uses **4–6 implants**, placed anterolaterally between dorsal veins and neurovascular bundles.
- **WVTT (Rezūm)** is **convective**: radiofrequency steam is injected into the TZ, and condensation releases latent heat, causing 1.5–2 cm of necrosis confined by the capsule.
- Injections are given every 1 cm from neck to veru, with extra ones for a **median lobe**, under sedation or prostatic block.
- **Rezūm II** showed ~50% IPSS gain to 5 years, retreatment 4.4% and **no de novo ED**; reduced ejaculate settles by 12 months. (St 36/37).

KEY POINT

PUL moves tissue; WVTT ablates it. So PUL cannot treat a median lobe but needs no catheter and causes little dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritation. Both preserve the bladder neck, and so antegrade ejaculation.

14 · Robotic waterjet (RWT, Aquablation)

- A non-thermal saline waterjet ablates tissue mapped on real-time biplanar TRUS. It **needs general or spinal anaesthesia, so it is not a MIST**. It suits 30–80 cc (St 39), treats median lobes and gives no specimen.
- **Haemostasis is the weakness**: traction alone led to bleeding and transfusion, so **focal bladder-neck cautery** via resectoscope is now used.
- **WATER** showed non-inferiority to TURP to 5 years, with retrograde ejaculation 10% vs 36%.

15 · Temporary devices

- **iTind (TIPD)** is a nitinol device left **5–7 days** to cause ischaemic remodelling, then removed. St 41: 25–75 cc, no obstructing median lobe. In its sham trial the responder end point was met, but the absolute IPSS difference was not significant.
- **Optilume** (paclitaxel balloon) has single-arm data only and is not on the AUA list.

16 · Prostate artery embolisation (PAE)

No surgery, but less effective.

ALGORITHM

1. Femoral access under local anaesthesia; pelvic angiography with the catheter balloon as landmark.
2. Pass a 1.9–2.4 Fr microcatheter into the prostatic arteries (mostly from internal pudendal; often duplicated).
3. Inject microspheres to stasis; bilateral is better.

- **Contraindications**: underactive, acontractile or neurogenic bladder; stricture or BNC; large diverticula or stones; contrast allergy; chronic kidney disease.
- AUA St 40: by trained clinicians; no size limit; a role in refractory BPH haematuria. EAU: for men accepting results inferior to TURP.
- Compared with TURP, IPSS, Qmax and volume reduction are inferior but the stay is shorter; most men reintervene by 10 years.
- Complications include non-target embolisation, post-embolisation syndrome and AUR.

17 · Overall comparison

This is the counselling sheet.

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/general (GA); 1–2 d	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
TUIP	<30 cc	Day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritation	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT

The "gold standard" for a given man is the operation that meets his goals safely: a shared decision on symptoms, size and shape, bleeding risk, ejaculation wishes and surgeon expertise.

18 · Landmark trials

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	Lesser-efficacy option

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

1 · Definition and surgical anatomy

- It enucleates only the benign prostatic hyperplasia (BPH) adenoma and is the gold standard for outlet obstruction with glands **>80 g**.
- The plane is the **surgical capsule (pseudocapsule)**: peripheral zone compressed by the enlarging transition zone (see Ch 148).
- **Urethral arteries enter posterolaterally just distal to the bladder neck**, so a figure-of-eight suture goes at **5 and 7 o'clock**.
- Eponyms MCQ : **Freyer** is suprapubic transvesical and **Millin (1945)** is retropubic.

2 · Indications

- Gland **>80–100 g** with bothersome lower urinary tract symptoms (LUTS) refractory to medical or endoscopic therapy.
- Retention, recurrent urinary tract infection (UTI), **bladder stones**, haematuria or obstructive nephropathy; a diverticulum needing surgery (relative); or inability to lie in lithotomy for transurethral resection (TURP).

3 · Pre-operative evaluation

- Check bleeding history, anticoagulants, prior pelvic surgery (which dictates the route), **urine culture** (treat first) and coagulation.

> **KEY POINT**
> Work-up asks three questions. Is the gland big enough (transrectal ultrasound)? Is there cancer (digital rectal examination, prostate-specific antigen, MRI)? Will the bladder empty (urodynamics)?

4 · Open techniques

Freyer opens the bladder; Millin opens the capsule.

P1 P2 Millin's prostatectomy — Retzius, transverse capsulotomy ~2 cm distal to bladder neck, bladder not opened, apex divided under vision, 5/7 o'clock sutures, watertight closure; Millin vs Freyer table

Not in the paper answer:

- **Freyer** uses a transverse **cystotomy 1–2 cm proximal to the bladder neck**. It treats **median lobe, diverticula and stones**, but the apex is blind.
- **Millin**'s drawback is **dorsal venous complex (DVC) bleeding**.

> **KEY POINT**
> Every open variant: enter, find the pseudocapsular plane, divide the urethra at the apex **proximal to the verumontanum** without pulling on the sphincter, then 5/7 o'clock haemostasis with bladder neck mucosal advancement.

5 · Haemostasis

Technique	When / how
Pack fossa 5 min	First
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule
Bladder neck mucosal advancement ("trigonisation")	Haemostasis; prevents contracture
11 and 1 o'clock sutures	Anterior branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC and lateral pedicles, sparing neurovascular bundles (NVBs)
O'Conor plication	Diffuse bleeding: plicate the posterior capsule
Malament purse-string	Persistent bleeding: purse-string round the neck, tied suprapubically, partitions fossa from bladder

ALGORITHM

1. Catheter traction (50 mL balloon), with or without continuous bladder irrigation (CBI); exclude coagulopathy.
2. If bleeding continues, evacuate clot endoscopically and fulgurate.
3. If severe, reopen the fossa.

6 · Postoperative care (open)

- On day 2, if urine is clear, remove the Foley for a trial of void. If drain output is high, check **drain creatinine**.

7 · Laparoscopic and robotic

P3 Laparoscopic Trans vesical Prostatectomy — steps, same function as open with less morbidity

Not in the paper answer:

- Mucosal advancement usually means **no CBI, drain or suprapubic tube**, and the catheter comes out on day 5–7.
- Avoid posterolateral cautery, because thermal NVB injury causes erectile dysfunction (ED).

ALGORITHM

1. Vertical cystotomy; identify orifices.
2. Incise the bladder neck 3→9 o'clock under the median lobe; develop the **posterior plane** to the apex, then the lobes.
3. Divide the anterior commissure, see the veru, and transect the urethra **just proximal to the veru**.
4. Place 5/7 barbed sutures; do **vesicourethral mucosal advancement**; close in two layers.

- **Single-port transvesical surgery** with **pneumovesicum 8 mm Hg** allows same-day discharge.

8 · Outcomes and comparisons

The route changes recovery, not function.

Comparison	Bottom line
Robotic vs open	Less bleeding, shorter catheter and stay, fewer complications; longer operation. Function **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, **no TUR syndrome**, treats stones and diverticula; but more bleeding, longer stay
†vs holmium laser enucleation (HoLEP)	†Equivalent outcomes; HoLEP bleeds less, but has a long learning curve and cannot treat diverticula

KEY POINT

Minimally invasive routes improve only perioperative morbidity; function is identical because enucleation is identical. Open transfusion is now **<5%**.

9 · Complications

Complication	Cause / rate	Management
Extravasation	Incomplete closure; capsular tear	If drain creatinine equals urine, keep drain and Foley
Ureteric orifice injury	Neck incision or sutures	Stent
Bladder neck contracture	Open **2–6%** ; minimally invasive <1%	Dilation or Collings knife incision
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare	Bulking (mild); artificial sphincter (severe)
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op
UTI, stricture, venous thromboembolism (VTE)	VTE <1%	Prophylaxis

KEY POINT

The adenoma lies inside the pseudocapsule and proximal to the veru. Removal loses the bladder neck (retrograde ejaculation) but spares sphincter and NVBs, so incontinence and ED are rare.

★ Putting it together: the BPH pathway

A · Man with LUTS

ALGORITHM

1. **1. Assess:** International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), rectal examination, urinalysis, PSA if it changes management; uroflow and post-void residual (PVR) before intervention. **2. If complications, operate:** refractory retention, recurrent UTI, stones, prostatic haematuria, renal impairment. **3. If mild**, watchful waiting. **4. If bothersome**, drugs (B); if they fail, a procedure (C).

B · Medical therapy

Drug	Use	Notes
α1-blocker	First line; works in days	Ejaculatory dysfunction; floppy iris
5α-reductase inhibitor (5-ARI)	Prostate >30–40 mL	Less retention and surgery; halves PSA
α-blocker and 5-ARI	Large gland, progression risk	Best (MTOPS, CombAT)
Add antimuscarinic or β3-agonist	Persistent storage LUTS	Check PVR first
Tadalafil 5 mg daily	LUTS with ED	No nitrates
Desmopressin	Nocturnal polyuria	Check sodium if ≥65

C · Choosing a procedure

Situation	Options
<30 mL	TUIP or TURP
30–80 mL	Bipolar TURP, PVP, HoLEP/ThuLEP, Aquablation; MISTs (urethral lift, water vapour)
>80 mL	HoLEP/ThuLEP; simple prostatectomy (stones/diverticulum)
Anticoagulation continued	HoLEP, ThuLEP, PVP
Preserve ejaculation	Urethral lift (no median lobe) or water vapour
Unfit	Long-term catheter or †prostatic artery embolisation

D · TURP complications

KEY POINT

TUR syndrome causes confusion and hyponatraemia. Stop, check sodium, give a diuretic if overloaded and 3% saline if severe; bipolar prevents it. Late problems are retrograde ejaculation (~two-thirds), stricture, bladder neck contracture and incontinence (see Ch 151).

II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 17

Prostate Cancer

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 153	Epidemiology, Aetiology and Prevention of Prostate Cancer
Ch 154	Prostate Cancer Biomarkers
Ch 155	Prostate Biopsy: Techniques and Imaging
Ch 156	Pathology of Prostatic Neoplasia
Ch 157	Diagnosis and Staging of Prostate Cancer
Ch 158	Active Management Strategies for Localised Prostate Cancer
Ch 159	Active Surveillance of Prostate Cancer
Ch 160	Open Radical Prostatectomy
Ch 161	Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy
Ch 162	Radiation Therapy for Prostate Cancer
Ch 163	Focal Therapy of Prostate Cancer
Ch 164	Treatment of Locally Advanced Prostate Cancer
Ch 165	Management of Recurrent and Newly Metastatic Prostate Cancer
Ch 166	Management of Castration-Resistant Prostate Cancer
★	Putting it together

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

1 · Epidemiology — the headline

- It is the **second commonest** non-skin cancer and the **fifth** cause of cancer death in men; 2020 saw ≈ **1.4 million cases, 375,000 deaths**.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest). US lifetime risk is ≈ **1 in 8**. 5-year survival is 97%, yet it is the second cause of cancer death in US men.
- **Incidence and mortality are decoupled.** Incidence follows screening and wealth, while mortality peaks where African ancestry meets poor access: the **Caribbean has the highest mortality in the world** MCQ.
- **Latent (autopsy) cancer** is ~5% under 30 y, 29% at 60–69 and **59% >79 y**, and higher in Black men at every age.

> **KEY POINT**
> In a prostate-specific antigen (PSA)-screened population, risk-factor studies may find determinants of being tested, so the preferred end point is **aggressive disease**. **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits falsely seem to raise incidence while lowering mortality.

2 · Established risk factors

Only three are unequivocal: **age, race (ancestry) and family history**.

- **Age**: mean age at diagnosis is 66 and it is rare under 40. Diagnosis at ≥70 carries a higher risk of cancer death.
- **Race**: Black men have the highest incidence (172.6/100,000 in the US) and **twice** the cancer-specific mortality of White men, presenting younger and more metastatic. Drivers are unequal access, comorbidity and **8q24** (and 17q21) alleles; with **equal access, outcome is equal** in non-metastatic disease.
- **Family history** is present in ~20%. Twin-study heritability is up to **58%**, so genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative (father or brother)	2–3×
Two first-degree relatives	~3.5×
Both father and brother	**9×**
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

3 · Germline genetics

The key inherited genes are **DNA-repair genes**, which predict **aggressive** cancer.

Gene	Pathway	Association	Implication
BRCA2, BRCA1	Homologous recombination (HR)	BRCA2 familial, aggressive	Early screening; poly(ADP-ribose) polymerase (PARP) inhibitor; caution in active surveillance (AS)
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (**Lynch**)	Mismatch repair (MMR)	2–5% of cases	Checkpoint inhibitor; very sensitive to androgen deprivation therapy (ADT)
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	Genome-wide association study (GWAS) loci	Commoner in African ancestry	Explain part of the racial excess

- **Polygenic risk scores** predict any cancer, not aggressive cancer, so they are not yet practice-changing.

P4 Indications of genomic testing in carcinoma prostate — germline testing: metastatic, high-risk, intraductal/cribriform, family history, Ashkenazi

Not in the paper answer:

- GWAS has found >200 single-nucleotide polymorphisms, each of small effect.

4 · Molecular aetiology and inflammation

- **Chronic inflammation** is in nearly every biopsy and may be causal; **proliferative inflammatory atrophy** is the proposed precursor, via high-grade prostatic intraepithelial neoplasia (HGPIN).
- **TMPRSS2-ERG** fusion puts the ERG oncogene under an androgen-regulated promoter. It is in ~half of Western tumours but uncommon in East Asians. See Ch 154

5 · Possible risk factors — body, metabolism, lifestyle

Most lifestyle factors act on **progression** rather than initiation; evidence is observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% higher per 10 cm	More high-grade cancer and death
Obesity	Mixed; less low-grade	**Consistently raised**
Smoking	Not raised (smokers screen less)	**Mortality raised ~40%**; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, cardiovascular disease (CVD)	Slightly raised	Metabolic syndrome: worse pathology
Diabetes	Lower (low testosterone, IGF-1 and PSA; metformin)	—
Vigorous activity	No clear effect	Less advanced disease

Factor	Total incidence	Aggressive / fatal
Alcohol	Unclear	Mortality raised at ≥45 g/day; binge drinking raises it

- Adiposity raises insulin, IGF-1, leptin, inflammation and oestrogen, which drives proliferation and blocks apoptosis.
- **Adult serum testosterone does not predict risk** MCQ.
- A **composite healthy lifestyle** means **30–50%** less lethal cancer; in the highest polygenic-risk quartile, lifetime lethal risk falls from 5.3% to 1.6%.
- **Diet**: World Cancer Research Fund 2018 found **no dietary factor with strong evidence**; dairy, calcium and lycopene were downgraded.
- **Vasectomy** is not a risk factor (detection bias), and the AUA says it need not be discussed.
- Transgender women on oestrogen have a median PSA ≈ **0.02 ng/mL** (50-fold lower), so cut-points must be lowered.

6 · Supplements

P4 Antioxidants in urological malignancy; Prostate cancer prevention trials — SELECT null, vitamin E harmful; VITAL null; lycopene, green tea, soy/pomegranate/curcumin

Not in the paper answer:

- Calcium supplements raise fatal cancer (relevant on ADT), and zinc ≥75 mg/day raises lethal cancer.
- No supplement benefits healthy men on a balanced diet.

> **KEY POINT**
> Vitamin E worked in the ATBC trial but failed in SELECT: ATBC used a tenfold lower dose in **heavy smokers** with little PSA screening (aggressive cancers); SELECT used 400 IU in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · Chemoprevention

P4 Prostate cancer prevention trials — 5ARI rationale, PCPT/REDUCE high-grade controversy, FDA warning, no mortality effect; aspirin, statins, metformin

Not in the paper answer:

- No preventive agent is FDA-approved. 5α-reductase inhibitors shrink the gland about 25%, so on a 5ARI you **double the PSA**. See Ch 154
- **Secondary prevention**: in the Prostate Cancer Lifestyle Trial (men on AS), plant-based diet, exercise and stress management meant **5% vs 27%** treated at 2 y.
- Activity (≥150 min/week moderate or ≥75 min vigorous) lowers cancer-specific mortality ~31%, while a Western diet and continued smoking raise cancer death.

> **KEY POINT**
> Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does, so lifestyle advice benefits both.

8 · PSA screening

P2 PSA screening for prostate cancer: evidence, problems and solutions — trials, corrected ERSPC benefit, overdiagnosis, USPSTF D to C, MRI-first

- Screening lowers cancer death ~**20–30%**, and the benefit builds with time.

9 · Landmark trials

Trial	Design	Result	Take-home
ERSPC (European screening study)	>182,000; PSA every 2–4 y vs none	Cancer death fell 16% (9 y), then ~20% (16 y); excess incidence fell 85% to 36%	Benefit is time-dependent
PLCO (US)	Annual PSA vs usual care	Control arm heavily screened	Contamination dilutes the effect
Göteborg-1	Swedish screening	Significantly fewer cancer deaths	Supports ERSPC
CAP (UK)	Single PSA invitation	No benefit at 10 y; tiny fall in death at 15 y	One-off screening is weak
PCPT	18,882, PSA ≤3; finasteride 5 mg 7 y	Cancer fell **24.8%** (18.4 vs 24.4%); Gleason 7–10 6.4 vs 5.1%; 18-y death fell 25% (not significant)	Fewer cancers, more high-grade
REDUCE	PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg 4 y	Cancer fell **22.8%**; Gleason 8–10 0.9 vs 0.6%	FDA warning
SELECT	35,533; selenium 200 µg with or without vitamin E 400 IU	Stopped for futility; vitamin E raised risk (P = 0.06)	No supplements
VITAL	>25,000; vitamin D3 2000 IU 5 y	No prostate effect	Not preventive

154 Prostate Cancer Biomarkers

1 · Why biomarkers beyond PSA

- Prostate-specific antigen (PSA) is **organ-specific, not cancer-specific**, which causes negative biopsies and overdiagnosis of Grade Group (GG) 1. Guidelines advise **secondary testing** (imaging or biomarker) before biopsy; the target is **clinically significant cancer (GG ≥2)**. See Ch 153
- Settings are before initial or repeat biopsy, favourable-risk cancer (active surveillance (AS) vs treatment), and after radical prostatectomy (RP).

2 · PSA biology

- PSA is **hK3** (gene KLK3 on **19q13**), an androgen-regulated serine protease that **liquefies semen**. Half-life is **2–3 days**.
- It is made as **proPSA**, and **hK2 cleaves the leader** to give active PSA.
- Serum PSA rises when it leaks through a lost basal layer; per cell, cancer makes less PSA than benign tissue.

Serum form	% of PSA	Measured?
PSA–α1-antichymotrypsin (ACT)	60–90	Yes (predominant, inactive)
PSA–α2-macroglobulin	10–20	**No**: epitopes masked
PSA–α1-protease inhibitor	1–5	Yes
Free PSA	5–45	Yes (proPSA, benign PSA (BPSA), intact PSA)

Confounders

- After biopsy the rise lasts ≥**4 weeks**; ejaculation raises it for 24 h, so repeat after abstinence.
- **5α-reductase inhibitors (5ARIs)** lower PSA ~**50%** by 12 months, so **double** it.
- Age-specific upper limits (ng/mL): 40–49, 2.5; 50–59, 3.5; 60–69, 4.5; 70–79, 6.5.

3 · PSA derivatives

Derivative	Definition / values		Use
PSA density	PSA ÷ volume; cut-off (0.08–0.15)	**0.15**	After negative/equivocal magnetic resonance imaging (MRI); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification
PSA velocity	Change per year		Minimal value; **never an isolated biopsy trigger**

Derivative	Definition / values	Use
% free PSA	Cancer PSA binds ACT, so **low %fPSA suggests cancer**; thresholds 15–25%	FDA: PSA 4–10, benign digital rectal examination (DRE); mainly **after negative biopsy**; unaffected by 5ARI

> **KEY POINT**
> %fPSA was validated against any cancer, not GG ≥2, so it overestimates significant risk.

- **[-2]proPSA** rises with Gleason and underpins the Prostate Health Index (PHI); **BPSA** tracks benign hyperplasia and has no prognostic value.

4 · Blood and urine tests (NCCN 2023 pre-biopsy)

Six tests estimate GG ≥2 risk; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Components	Threshold / key fact
PHI	Blood: **(p2PSA ÷ fPSA) × √tPSA**	24–28.6 proposed; ~30% biopsies avoided
4Kscore	Blood: tPSA, fPSA, iPSA and **hK2**, with age, DRE, prior biopsy	**7.5%**: sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis risk
IsoPSA	Blood: PSA isoform structure	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine DLX1, HOXC6 mRNA with clinical data	NPV ~95%; needs prostate volume
ExoDx (EPI)	Urine exosomes: PCA3, ERG, SPDEF	**No DRE**; 15.6 initial, 20 repeat
MyProstateScore	Post-DRE urine PCA3, TMPRSS2:ERG and PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

- **PCA3** is a long non-coding RNA. Score = (PCA3 ÷ KLK3 mRNA) × 1000 after an **attentive DRE**; thresholds 25/35. It is **independent of prostate size** and FDA-approved after a **previous negative biopsy**.
- **TMPRSS2:ERG** is **~100% specific** but in only ~half of tumours, so it is combined with PCA3.

> **KEY POINT**
> Urine tests follow a DRE because cancer RNA comes only from shed cells massaged into the urethra; ExoDx avoids this, as exosomes are shed spontaneously. Urinary markers are highly specific, but their coverage is limited, hence panels.

5 · Tissue-based biomarkers

P2 Molecular staging of CA prostate — Decipher, Oncotype GPS, Prolaris, PORTOS, ConfirmMDx; limits: heterogeneity, retrospective, G-MAJOR

Not in the paper answer:

- **ConfirmMDx** measures GSTP1, APC and RASSF1 methylation in negative cores (NPV ~88–90%), using the **field ("halo") effect** around a missed tumour.
- **Decipher ≥0.4 post-RP** suggests benefit from radiotherapy.

6 · Germline testing

Up to 40–50% of risk is familial, mostly via **DNA-damage-repair** genes.

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	CHEK2 odds ratio ~2–3.4	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract

Gene	Risk	Phenotype
HOXB13 G84E	4.5×	**Not** more aggressive

Testing is for BRCA1/2 alone or a panel (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM). Indications: see Ch 153.

7 · Clinical algorithm

ALGORITHM

1. **Raised PSA**: repeat it, excluding infection, instrumentation, ejaculation and 5ARI. Then do MRI and/or a biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density). If risk is high, biopsy.
2. **Negative biopsy, persistent suspicion**: MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx. If positive, targeted re-biopsy.
3. **Favourable-risk**: PSA density and Prolaris/Oncotype/Decipher decide AS vs treatment.
4. **After RP**: PSA should be undetectable. If ≥0.2 ng/mL and confirmed, this is biochemical recurrence; Decipher/PORTOS then guides adjuvant or salvage radiotherapy.

155 Prostate Biopsy: Techniques and Imaging

1 · Indications and contraindications

On transrectal ultrasound (TRUS) the peripheral (PZ) and central zones are homogeneous and echogenic; the transition zone (TZ) is heterogeneous. **39%** of cancers are isoechoic, so a normal grey scale never precludes biopsy.

P2 TRUS / prostate biopsy — indications, contraindications, preparation, technique, TR vs TP, complication rates

Not in the paper answer:

- **Repeat** biopsy follows rising prostate-specific antigen (PSA), atypical small acinar proliferation (ASAP; **≈50%** cancer risk), positive PCA3 or ConfirmMDx, suspicious MRI, or active surveillance (AS).
- National Comprehensive Cancer Network (NCCN): biopsy for abnormal digital rectal examination (DRE) with raised PSA, or repeat PSA >3 ng/mL with PI-RADS ≥3; PSA velocity alone, or abnormal DRE with normal PSA, is not enough. See Ch 154
- PSA density **>0.15 ng/mL/cc** suggests cancer. Volume = π/6 × transverse × anteroposterior × length.

2 · Pre-biopsy mpMRI

P1 P4 Prostate multiparametric MRI; PI-RADS and MRI–ultrasound fusion biopsy — sequences, PI-RADS scoring, zone rules, cancer rate per score, targeting methods

Not in the paper answer:

- Multiparametric MRI is a **triage test**: with no lesion biopsy may be avoided. A lesion **>1.5 cm** or definite extraprostatic extension scores PI-RADS 5.

Feature	Peripheral zone	Transition zone
Dominant sequence	**DWI**	**T2WI**
Dominant score 3	Positive DCE upgrades to 4	DWI 5 upgrades to 4
v2.1 (2019)	—	T2WI 2 with DWI ≥4 becomes 3

KEY POINT

DCE never creates or downgrades PI-RADS 4–5 and never upgrades a TZ lesion. mpMRI misses a minority of significant cancers (mostly 3+4), so systematic cores remain relevant.

- Targeting is cognitive, software fusion or in-bore. Fusion-targeted cores are ≈3× likelier than systematic ones to find intermediate/high-risk disease.
- Take **2–4 cores per target** (AUA/SUO 2023: 2–3, with or without systematic cores) whenever biopsy is indicated and a PI-RADS ≥3 lesion exists.

3 · Systematic templates and technique

- **Extended 12-core** biopsy is the AUA standard, replacing sextant. Going from 6 to 12 cores raises detection, but 18–21 do not: **geometry, not core number**.
- Add TZ/anterior cores for repeat biopsy or gland >50 mL, and seminal vesicle cores for a palpable abnormality.
- Use an 18-gauge gun, and never reposition with the needle on the rectal wall.

4 · Transrectal vs transperineal (TP)

- TP passes clean perineal skin, causes no rectal bleeding and reaches apex and anterior gland (anterior-only cancers 6–20%).
- European Association of Urology 2023: TP is **preferred** (strong), with non-inferior detection. Give skin-flora cover or none; omitting antibiotics did not change the very low sepsis rate.
- Access is by brachytherapy grid, freehand guide (PrecisionPoint) or two 14-gauge cannulas.

5 · Antibiotic prophylaxis

- Every transrectal patient gets prophylaxis. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin with or without aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before, for **≤24 h**.
- Single doses: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Aztreonam replaces aminoglycoside in renal impairment.
- The FDA 2016 fluoroquinolone warning covers tendon rupture, neuropathy, CNS effects and myasthenia.
- Infection risk factors are antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI, travel and rectal resistant E. coli. These men need augmented or swab-targeted prophylaxis, or TP.
- Endocarditis risk needs IV ampicillin (vancomycin if allergic) and gentamicin. An enema has no proven infection benefit.

6 · Anticoagulation and anaesthesia

Contraindications are coagulopathy, severe immunosuppression, and acute urinary tract infection (UTI) or prostatitis.

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when INR <1.5
Direct oral anticoagulants (DOACs)	Stop 2–5 days
Mechanical valve	Bridge with heparin

- Transrectal: periprostatic block, 1–2% lidocaine 5 mL each side at the seminal vesicle–prostate junction (beats intrarectal gel).
- Transperineal: local to the skin, then 5 mL into the **periapical triangle** and 3 mL along the capsule.

7 · Complications and management

Complication	Rate	Management
Haematuria	23–63%	Clot retention (0.7%): catheter, washout
Haematospermia	10–50%	Lasts ≈1 month
Rectal bleeding	2–22%	Ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics
Sepsis	**0.3–3.1%**	Carbapenem until cultures

Complication	Rate	Management
Vasovagal	1.4–5.3%	Trendelenburg, fluids
Retention	0.2–0.4%	Temporary catheter

ALGORITHM

1. **Post-biopsy fever**: assess for sepsis and take cultures.
2. Give **no empirical fluoroquinolone**, as E. coli is resistant. Start a carbapenem (or amikacin/3rd-generation cephalosporin by sensitivity) and resuscitate.

ALGORITHM

1. **Persistent rectal bleeding**: apply pressure.
2. If it continues, tamponade (sheath or Foley balloon). If still bleeding: endoscopy with adrenaline/sclerosant, then angioembolisation, then transrectal suture.

KEY POINT

Bleeding is the commonest complication overall; infection leads serious ones and admissions (1.0% rising to 4.1% in the resistance era). This explains universal prophylaxis and the move to the perineum.

8 · Repeat and saturation biopsy

P2 Saturation biopsy in CA prostate — >12 cores, template mapping, indications, no gain at initial biopsy

Not in the paper answer:

- Micro-ultrasound (29 MHz) approaches mpMRI without contrast (emerging).

ALGORITHM

1. Cancer yield on successive biopsies in ERSPC: **22% → 10% → 5% → 4%**.

- Take a third or fourth biopsy only for high suspicion, using mpMRI with targeting, ConfirmMDx, anterior/TZ cores and the TP route.

9 · Landmark trials

Trial	Design	Result	Take-home
PROMIS (2017)	576: mpMRI, TRUS, 5-mm template mapping (reference)	Sensitivity 93% vs 48%; negative predictive value 89% vs 74%; 27% avoid biopsy	mpMRI better triage
PRECISION (2018)	500 biopsy-naive: MRI with targeted biopsy vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior
MRI-FIRST (2019)	Paired systematic and targeted	Each alone missed cancers	Combine for maximal detection
PREVENT (2024)	TP without antibiotics vs transrectal targeted prophylaxis	Infection not significantly different; detection comparable	Both acceptable

10 · MCQ one-liners

MCQ ONE-LINERS

1. Absent seminal vesicle: ipsilateral renal agenesis in 79%.

2. Commonest false-positive hypoechoic lesion: BPH foci and prostatitis.
3. Müllerian cysts are midline; wolffian cysts are lateral.
4. Significant cancer in PROMIS: Grade Group ≥3 or core length ≥6 mm.

156 Pathology of Prostatic Neoplasia

1 · Precursor and intraductal lesions

P2 PIN — definition, histology, precursor evidence, LGPIN not reported, isolated HGPIN needs no repeat biopsy, cribriform PIN reclassified

Not in the paper answer:

- ~**20%** of high-grade prostatic intraepithelial neoplasia (HGPIN) carries the **TMPRSS2:ERG** fusion.
- **Intraductal carcinoma of the prostate (IDC-P)** is dense cribriform atypia with **basal cells retained**, usually **retrograde spread of high-grade invasive cancer**. It independently predicts early biochemical recurrence (BCR), and Grade Group (GG) 1 with IDC-P is upgraded at radical prostatectomy (RP) in **>80%**.

Lesion	Defining feature	Action on isolated finding
Low-grade PIN (LGPIN)	Mild atypia; irreproducible	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	No repeat biopsy
PINATYP	HGPIN with adjacent small atypical glands	Manage as atypical
Atypical intraductal proliferation (AIP)	Atypia beyond HGPIN, below IDC-P	Consider re-biopsy (if benign or GG1 elsewhere)
IDC-P	Dense cribriform, basal cells retained	Definitive therapy (or re-biopsy); **excludes active surveillance (AS)**

KEY POINT

HGPIN is a weak risk marker; IDC-P usually marks high-grade cancer already present.
Traps: re-biopsy for HGPIN; surveillance for GG1 with IDC-P.

2 · Adenocarcinoma: location, spread, staging

- **Zones (McNeal)**: the peripheral zone (PZ) holds ~70% of cancers, including **85%** of T1c; they are posterolateral and palpable.
- The transition zone (TZ) holds ~20% of cancers, which are anterior, impalpable, under-sampled and lower grade. The central zone holds ~5%, rare but aggressive.
- Tumours are **multifocal in >85%**. With no true capsule, say "**extraprostatic extension**" (EPE); tumour exits via **perineural spaces**, posterolaterally.
- **Seminal vesicle invasion (SVI)** means tumour in the SV muscular wall, usually reached from the base through periseminal tissue.
- EPE is uncommon below **0.5 cm³**; nodes and SVI are uncommon below **4 cm³**.

Stage	Definition
T1a / T1b	Incidental at TUR of prostate (TURP)/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, found on needle biopsy
pT2	Organ-confined at RP
pT2+ (pT2x)	Positive margin at an intraprostatic incision or ill-defined boundary (apex)

Stage	Definition
pT3a	EPE; **includes microscopic bladder neck invasion**
pT3b	SVI
pT4	Other adjacent structures (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck invasion is pT3a; T1a/b depend on the 5% cut-off, not grade.

3 · Grading: Gleason and grade groups

P2 Gleason's score and its significance — history, scoring rules, significance, limitations

Not in the paper answer:

- **Biopsy score = commonest plus highest pattern**. The lowest score in practice is **6**, which has **no nodal metastatic capacity**.

Pattern	Morphology			
3	Discrete well-formed glands			
4	**Cribriform (incl. glomeruloid)**, poorly formed or fused glands			
5	Sheets, cords, single cells, solid nests, **comedonecrosis**			
	GG (International Society of Urological Pathology (ISUP); World Health Organization 2016)	Gleason	Nodes at RP	5-yr BCR-free
1		3+3	0%	**96%**
2		3+4	0.6%	**88%**
3		4+3	4.3%	**63%**
4		4+4, 3+5, 5+3	—	**48%**
5		9–10	—	**26%**

KEY POINT

The steep drop is between GG2 and GG3: never lump them as "7". 3+5 and 5+3 are GG4; GG5 starts at 9. Pure GG1 cannot metastasise, so calling it "cancer" is debated.

- Report **% pattern 4** for GG2–3 (ISUP 2014). GG2 with limited pattern 4 may suit AS.
- **Tertiary pattern 5** (<5% at RP) raises BCR risk but **stays in its GG**, because nodal risk is not raised. On biopsy any higher pattern counts as secondary, however small.

Issue	ISUP 2019	Genitourinary Pathology Society (GUPS) 2019
% pattern 4 in GG2–3; cribriform	Report	Report
IDC-P with invasive cancer	**Incorporated into grade**	**Not graded**; comment
Targeted cores of one lesion	**Aggregate grade** (plus per-core)	Each core; aggregate optional

KEY POINT

Cribriform pattern and IDC-P mark aggressive disease and argue against AS even in GG2, so they must be reported. Aggregate grading of a targeted lesion avoids under- or over-grading.

Upgrading setting	Risk
Biopsy GG1, then RP	~⅓ upgraded
Biopsy GG2, then RP	~20% become GG3
GG1 with negative mpMRI (PI-RADS 1–2)	Still ~20% at 6 yr

Reclassification within 3 yr usually means missed disease rather than progression.

4 · Needle biopsy assessment

- Use **separate jars per site**. Pitfalls: **at the base, SV/central zone mimic HGPIN; at the apex, Cowper gland mimics cancer**.
- Report the **number of positive cores plus one other extent measure**, and report **perineural invasion** on biopsy (it raises EPE risk).
- Basal cell markers, **high-molecular-weight cytokeratin (HMWCK) and p63**, stain benign glands, so their **absence supports cancer**. **AMACR** is an adjunct only.
- **Adenosis**, in the TZ, is the chief mimic of low-grade cancer and carries **no increased cancer risk**.
- **Atypical small acinar proliferation (ASAP)**: cancer follows in ~**40%**. Get **expert review before re-biopsy**, then re-biopsy the atypical site.

> **KEY POINT**
> Sampling error is one-directional: a bad biopsy is believed, a good one is provisional, and an atypical one is re-examined before it is repeated.

5 · TUR and RP specimens

- At TUR, **grade each major nodule**. Nodes are uncommon with Gleason <8.
- **EPE**: focal vs nonfocal extent predicts progression. **SVI** carries **65%** 5-yr progression.
- **Margins** are positive in ~15% (robotic), but only ~50% progress. Report **extent (focal vs nonfocal, 3 mm) and grade at the margin**.
- Independent predictors of BCR are grade, EPE extent, margins, SVI and nodes. **Tumour volume is not independent**.

6 · Treatment effect

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade looks artefactually higher	**No** where effect is present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months)	Only if no treatment effect
Cryotherapy / high-intensity focused ultrasound (HIFU)	Infarction; residual cancer unaltered	**Yes**

After radiotherapy, cancer with treatment effect has a prognosis close to no cancer.

7 · Subtypes and other tumours

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	**Pattern 4** (PIN-like 3; comedonecrosis 5)	Exophytic at the verumontanum; PSA may be normal; behaves like Gleason 8; **visceral recurrence at low PSA**
Mucinous	As usual	As acinar
Neuroendocrine differentiation (~half of acinar)	As usual	Not prognostic
Small cell (~50% mixed)	**Not graded**	Survival <1 yr
Squamous	—	**Osteolytic** metastases; hormone-unresponsive
Sarcomatoid	—	Dismal

- **Urothelial carcinoma in the prostate**: suburethral invasion is pT1 and stromal pT2, but **transmural spread from the bladder is pT4**. Urothelial cancer stains **GATA3**; prostate stains PSA and NKX3.1.
- Sarcomas are **rhabdomyosarcoma in children and leiomyosarcoma in adults**.

157 Diagnosis and Staging of Prostate Cancer

1 · Clinical presentation

- Early cancer is **asymptomatic**; symptoms imply advanced disease.
- **Locally advanced** disease causes voiding or ureteric obstruction and haematuria; **metastatic** disease causes bone pain, fracture, leg oedema and cord compression.

2 · Screening in brief

P2 PSA screening for prostate cancer: evidence, problems and solutions — ERSPC vs PLCO, NNI 570 / NND 18, USPSTF D to C, AUA, NCCN, overdiagnosis

3 · DRE and PSA

- **Digital rectal examination (DRE)** has low screening value at PSA <3, but an abnormal DRE raises the risk of Grade Group (GG) 4–5 at every PSA level.
- **Prostate-specific antigen (PSA)** is organ-specific, not cancer-specific, and rises ≈4% per mL of volume.
- PSA is lowered by hypogonadism, obesity, statins and 5α-reductase inhibitors (5-ARIs). On a 5-ARI, **double PSA** after ≥12 months, or treat the nadir as the new baseline.
- Risk is **continuous**: the Prostate Cancer Prevention Trial (PCPT) found cancer in 15% of men with PSA ≤4.
- Thresholds are 4 or 2.5–3 (ERSPC used 3). Repeat a raised PSA before biopsy, and give no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Yield, normal DRE	Yield, suspicious DRE	High-grade (normal / suspicious)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

- Baseline PSA in the 40s predicts metastasis and death decades later (Malmö).
- Derivatives: PSA density ≥0.15; velocity >0.75 ng/mL/yr; free PSA ≤25% (at PSA 4–10). See Ch 154, 155

4 · Staging: TNM (AJCC 8th, 2017)

Pathological stage is more accurate and prognostic than clinical stage.

Category	Definition
T1a / T1b	Incidental at transurethral resection (TUR): ≤5% / >5% of tissue
T1c	Impalpable, found on needle biopsy
T2a / T2b / T2c	Palpable, confined: ≤½ of one side / >½ of one side / both sides
T3a / T3b	Extraprostatic extension (EPE) / seminal vesicle invasion (SVI)
T4	Fixed, or invading sphincter, rectum, bladder, levators or pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites with or without bone

AJCC prognostic groups add PSA and GG.

KEY POINT

T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b; non-regional nodes are M1a; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is set by DRE, not imaging.

5 · Predicting extent and risk

- **PSA** cannot stage an individual. Organ-confined disease is ≈80% at PSA <4 and <50% above 10; nodes are positive in **75% at >50**.
- **Biopsy grade** is the most important biopsy finding, and tertiary pattern 5 on biopsy is recorded as the secondary pattern. See Ch 156

D'Amico	Stage	PSA	Gleason	10-yr disease-free after RP
Low (all criteria)	T1–T2a	≤10	≤6	83%
Intermediate (any one)	T2b	>10–20	7	46%
High (any one)	T2c	>20	8–10	29%

National Comprehensive Cancer Network (NCCN) group	Criteria
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores, ≤50% of any core
Low	T1–T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4–5, or PSA >20
Very high	T3b–T4, primary pattern 5, >4 cores GG4–5, or ≥2 high-risk features

- CAPRA and **nomograms** predict better than groups for individuals; nodal nomograms decide pelvic lymph node dissection (PLND). Tissue genomic tests refine risk but await prospective validation.

6 · Staging imaging and nodes

Risk	Imaging
Low / intermediate, asymptomatic	No routine computed tomography (CT) or bone scan
High	Bone scan with CT or mpMRI; consider PSMA PET

- **Multiparametric MRI** is used for local staging (EPE, SVI), treatment planning and surveillance selection.
- **PSMA PET** (68Ga-PSMA-11, piflufolastat F-18) is FDA-approved for high-risk staging and is more sensitive and specific for nodes and metastases.
- **PLND** has no consistent oncological benefit; its value is **staging**. Do it when the nomogram nodal risk exceeds the guideline threshold.

KEY POINT

Imaging yield follows pre-test probability: in low-risk men most positive scans are false positives, leading to more tests, anxiety and wrongly denied cure.

7 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	European screening randomised trial, from 1993	**20% relative reduction** in cancer death at 16 y	Saves lives, at the cost of overdiagnosis
PLCO	US screening randomised trial, from 1993	No mortality difference	Contaminated control arm
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤4	No safe PSA level
proPSMA (2020)	PSMA PET-CT vs CT and bone scan, high-risk	Accuracy ≈92% vs 65%; less radiation	PSMA PET is the better first-line staging test

8 · MCQ one-liners

MCQ ONE-LINERS

1. Finasteride 1 mg lowers PSA as much as 5 mg.

158 Active Management Strategies for Localised Prostate Cancer

1 · The modern question: timing as well as modality

- In 15-year randomised data on favourable risk, surveillance gave **more metastasis but no difference in cancer mortality**.
- **Spectrum**: a man likely to die of competing risks gets **watchful waiting (WW)**; good longevity with unclear benefit means **active surveillance (AS)**; a man who needs therapy gets radical prostatectomy (RP), external-beam radiation therapy (EBRT), brachytherapy, or whole/partial-gland ablation.

> **KEY POINT**
> Cancer mortality is always weighed against competing mortality. A competing risk argues against treatment only when it will act before the cancer does.

2 · Host evaluation

- NCCN estimates life expectancy from Social Security Administration or WHO tables.
- In the Prostate Cancer Outcomes Study (PCOS), 10-year prostate cancer mortality was **3–18%** by risk, but other-cause mortality was **≥33%** with any comorbidity.
- The blunt **"10-year life expectancy" rule** undertreats older and overtreats younger men, so individualise.
- The **Charlson index** (1-year inpatient mortality) is reweighted as **PCCI** for 10-year mortality; NNT rises steeply with Charlson score.
- The automated **Care Assessment Needs (CAN) score** (Veterans Health Administration) best predicts 5-year survival MCQ.

Baseline function (IPSS, UCLA Prostate Cancer Index)

Baseline state	Favours	Reason
Obstructive lower urinary tract symptoms (LUTS), large gland	Surgery	RP relieves obstruction; brachytherapy increases retention
Severe irritative LUTS	Surgery	Worst after radiation
Gland >60 cm^3	Surgery (or neoadjuvant androgen-deprivation therapy, ADT)	Pubic arch interference; if >80 g, ADT (shrinks 25–40%) before radiotherapy (RT)
Chronic retention / neurogenic bladder	Surgery	RT causes bladder fibrosis; if RT is chosen, switch to intermittent self-catheterisation
Prior transurethral resection (TUR) defect	Surgery	Brachytherapy dosimetry is hard and incontinence increases. Do TURP before RT and delay EBRT **6–8 weeks**
Inflammatory bowel disease (IBD), bowel dysfunction, prior rectal surgery	Surgery	RT bowel toxicity

Baseline state	Favours	Reason
Poor sexual function	—	Recovery depends on age, baseline erection, PDE5 inhibitor use and vascular risk; poor baseline function means non-nerve-sparing surgery is used more.

Modality	Absolute contraindication	Relative contraindication
Radiotherapy	Previous radical pelvic irradiation	Bowel in field (prior surgery, hernia); uncontrolled diabetes; IBD; bowel dysfunction
Surgery	—	Previous pelvic RT; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · Risk stratification

NCCN group	Criteria	Staging imaging
Very low	**All**: T1c; grade group (GG) 1; prostate-specific antigen (PSA) <10; <3 positive cores, ≤50% each; PSA density < **0.15**	Not indicated
Low	**All**: T1–T2a; GG1; PSA <10	Not indicated
Favourable intermediate	Intermediate-risk factors (IRFs): T2b–c, GG2, PSA 10–20. **All**: 1 IRF; GG1–2; <50% cores positive	
Unfavourable intermediate	**Any**: 2–3 IRFs; GG3; ≥50% cores positive	Bone scan and CT/MRI, plus chest imaging; prostate-specific membrane antigen (PSMA) PET may replace these (NCCN 2023)
High	**Any**: T3a; GG4–5; PSA >20	
Very high	**Any**: T3b–T4; primary pattern 5; >4 cores GG4–5	

KEY POINT

Favourable groups need ALL criteria; adverse groups ANY one. PSA density appears only in very low risk; GG3 alone makes intermediate unfavourable. AUA/ASTRO/SUO use very low/low/intermediate/high without substratification.

- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% of any core.
- Nomograms: **Partin** predicts pathological stage (accurate 72.4%); **Kattan** predicts relapse after RP; with **UCSF-CAPRA**, risk jumps at score ≥6.
- **proPSMA** (302 high-risk men): PSMA PET had accuracy **92% vs 65%** and sensitivity 85% vs 38% over conventional imaging, but a negative scan cannot exclude small nodes.
- Genomic tests are not routine (AUA 2022); use them only where they change an AS decision.

4 · Observation vs active surveillance

P2 Expectant treatment of CA prostate — AS (curative, deferred) vs WW (palliative): candidate, monitoring, triggers

Not in the paper answer:

- AS is **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN); selected low-volume GG2 is acceptable.
- Limitations are **25–45%** occult high grade on systematic biopsy (mostly GG2), declining adherence and no agreed triggers. PSA density >0.15 predicts reclassification.

5 · Comparative oncological evidence

- Population studies favour RP (Swedish subdistribution HR ≈1.8; PCOS disease-specific HR 0.35, curves apart by 3 years), but this is largely **selection bias**.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RP vs WW; 695; pre-PSA; ~90% palpable	23.2 yr: cancer deaths 63 vs 99, relative risk (RR) **0.56**; absolute 11%; NNT **8**; most benefit <65 (RR 0.45), intermediate risk	Benefit in younger men with intermediate risk
PIVOT	RP vs observation; 731 (2000 planned); PSA era; older, comorbid	10 yr: no difference; benefit if PSA >10. 19.5 yr: all-cause ARR 5.5% and cancer 4.0% (both not significant); significant only in **intermediate risk** (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit in low risk; underpowered
ProtecT	1643 screen-detected, 50–69; monitoring vs RP vs RT with 3–6 mo ADT; 77% low risk	15-yr cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); more metastasis with monitoring; 60% of monitored treated by 15 yr	AS safe in favourable risk; RP equals RT

KEY POINT

Competing mortality and low-risk enrolment dilute trial benefit, which appears only in younger intermediate-risk men with long follow-up.

7 · Functional outcomes compared

Cancer control is similar, so side effects usually decide.

Domain (vs AS; Chen, JAMA 2017)	RP	EBRT	Brachytherapy
Sexual	**Worst**; persists at 24 mo	Mild early decline; near AS by 24 mo	Mild early decline
Incontinence	**Worst**; persists at 24 mo	No change	No change
Obstructive/irritative	Worse at 3 mo, then **better than AS**	Worse at 3 mo; resolves by 12	**Worst acute**; resolves by 24 mo
Bowel	No change	**Worse**; persists at 24 mo	No change

- **PCOS** found differences at 2 and 5 years but **none at 15 years**. **ProtecT** found rising bloody stools after RT, with no difference in anxiety, depression or quality of life.
- In **CEASAR**, the erectile disadvantage of RP was confined to the top quartile of baseline function. In open vs robotic RP, the surgeon matters more than the approach.

KEY POINT

Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow decline. Curves converge by 15 years.

8 · Shared decision-making

- No treatment is clearly superior, so **patient preference carries unusual weight** (AUA, ASTRO, SUO).
- The **urologist's recommendation is the most influential factor**, so beware your own bias. **Longevity assessment may be the most important part of counselling**.

9 · Stepwise decision algorithm

ALGORITHM

1. **Treat or not?** Weigh disease and host factors. If longevity is short, then WW; if very low/low or selected favourable intermediate, then AS.
2. **Partial or whole gland?** Weigh focality, size, grade, potency, LUTS. If limited-focality intermediate risk, then focal ablation, ideally in a trial.
3. **Surgery, radiation or whole-gland ablation?** Compare outcomes, toxicities, baseline function and risk of secondary treatment.
4. **Delivery:** RP approach and node dissection, or EBRT vs brachytherapy, fractionation and ADT.

Risk / longevity	Usual recommendation
Limited longevity, low lethal potential	WW; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies are options
Favourable intermediate	RP or RT; AS for select (low-volume GG2); focal therapy in trials
Unfavourable intermediate	RP, or RT with ADT; AS strongly discouraged
High / very high	RP, or RT with ADT; consider even with competing risks if the cancer threatens within 10 yr; no focal therapy

- **Surgery:** early incontinence and erectile dysfunction (ED) are the norm, so quote your own outcomes; LUTS may improve.
- **Radiation:** it spares short-term function but causes yearly decline; secondary rectal/bladder cancer at 15–20 years is controversial.
- **Whole-gland ablation:** long-term data are scant and fibrosis hampers salvage. Cryoablation almost always causes ED; HIFU preserves function better, but retreatment is **29–38%**.
- EBRT, brachytherapy, SBRT and protons give comparable 10-year cancer control.

159 Active Surveillance of Prostate Cancer

1 · Rationale: overdiagnosis

- **Latent reservoir**: autopsy cancer prevalence roughly equals age as a percentage. PSA testing and biopsy detect cancer in 15–25%, yet lifetime risk of cancer death is only **3%**; overdiagnosis is 87–94%.
- Active surveillance (AS) rests on **two premises**: progression takes a long time, and cure is still possible at the first sign of higher risk.

2 · AS versus watchful waiting

P2 Expectant treatment of CA prostate — AS curative/deferred vs WW palliative; candidates, monitoring, switch at life expectancy <5–10 yr

Not in the paper answer:

- The switch to watchful waiting (WW) is typically near age 80.
- **Pitfall**: elderly men with unrecognised hypogonadism may progress (neuroendocrine dedifferentiation) without a PSA rise, so measure testosterone and get interval imaging.

> **KEY POINT**
> AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 · Biology of grade group 1

- **Pattern 3 is molecularly near-normal.** Patterns 4–5 show EGFR, HER2 and BCL2 changes and **PTEN deletion in 70–90%**; **MYC amplification with PTEN deletion** drives metastasis.
- **Metastatic potential of pure grade group (GG) 1 is effectively zero**, though occasional extraprostatic extension occurs.
- True grade progression is ~**1–2% per year**, mostly in high-volume Gleason 6, so follow-up must be long term.
- Deaths in low-risk series come from **occult higher grade** missed at biopsy, present in ~**30%** after systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True progression
Frequency	**Commonest (25–30%)**	1–2.3%/year
Usual result	Mostly GG2	Many 3+4 stay indolent
Remedy	MRI, targeted biopsy, predictors	Long-term surveillance

> **KEY POINT**
> The main hazard of AS is **sampling error at entry**, not biology over time. First job: get the entry grade right.

4 · Patient selection

Most GG1 should be on AS (AUA, EAU, ASCO, NICE and Cancer Care Ontario agree).

	Very low risk (Epstein)	Low risk
Definition	GG1; **≤2 positive cores; no core >50%; PSA density <0.15**	Any GG1, PSA <10
Progression to treatment	Lower	Higher
Metastasis	Exceptionally low	Exceptionally low

High-volume GG1 is still a candidate, but check it with multiparametric MRI (mpMRI) and targeted biopsy, re-biopsy or a genomic assay.

- **Toronto** is inclusive (all Gleason 6 and selected intermediate), with 15-yr cancer mortality of **5%** actuarial; **Johns Hopkins** is restrictive (Epstein very low), with **0.5%**. So **inclusion criteria set mortality, while intervention stays constant (~25% at 5 yr)**.
- In Toronto, baseline Gleason 7 had ≥20% metastasis at 15 yr (HR 3.75). In PRIAS (n=5302), cancer mortality was <1% at 10 yr, and cardiovascular disease was the commonest cause of death.
- **Favourable intermediate**: offer AS selectively for small-volume GG2. Only NICE lists AS and radical treatment as equal for intermediate risk.
- Exclude GG ≥3; intraductal or cribriform histology raises concern.
- **Black men** are AS candidates, perhaps with closer surveillance. **Family history** rarely justifies treatment, except metastatic disease <60 yr in a first-degree relative, which calls for germline testing.
- **BRCA2** carries the highest upgrading risk; consider definitive therapy even for low grade, or AS only with very close follow-up.
- Risk factors for higher grade are **PSA density >0.15**, Black race and biopsy volume; restricted diffusion on MRI is a powerful predictor.

5 · Monitoring protocol

ALGORITHM

1. At diagnosis, obtain mpMRI.
2. Do a **confirmation biopsy** (systematic + targeted) within 1–3 yr.
3. Check PSA 6-monthly, DRE every 1–2 yr and MRI ≤ every 18–24 months.
4. Repeat biopsy every 3–5 yr while stable.
5. If a trigger appears, then give definitive therapy; if life expectancy falls, then switch to WW.

- Confirmation biopsy may be omitted if the first biopsy was MRI-targeted. **Compliance is paramount**: compliers had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo then 3–5-yearly; MRI if discordant
AUA/ASTRO 2022	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3–4-monthly in year 1, then 6-monthly; **MRI at enrolment**

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	More Gleason 6 volume: get MRI or a biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2: may continue AS
	PSA rise alone: repeat MRI or biopsy

KEY POINT

Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger.

6 · MRI in AS

- **At entry the evidence is strong**: mpMRI before confirmation biopsy, with **systematic and targeted cores**, upgrades about ⅓. NICE recommends it; NCCN lists it as an option.
- **Serial MRI alone is weak** (NPV ~0.8–0.9 for GG ≥2), so it must never be the only way to exclude progression. MRI-invisible cancers behave better.

7 · Biomarkers and genomics

- Decipher predicts upgrading in GG1, not GG2; Oncotype DX evidence is mixed and Prolaris has none specific to AS.
- **Bayesian problem**: with a 3% prior metastasis risk, a 90%-specific test labels 10% "significant", so test only higher-risk candidates (high volume, high PSA density, GG2).
- Use the tests to set surveillance intensity, **never as the sole basis for AS**; a favourable score does not override adverse features.

8 · Lifestyle, drugs and psychology

- **Stop smoking**, exercise vigorously (ERASE: lowered PSA), keep a healthy weight and eat a vegetable-based diet.
- Only vitamin D 1000–2000 IU/day is reasonable. **Selenium and vitamin E are contraindicated** (SELECT).
- 5α-reductase inhibitors are not used to prevent progression (REDEEM); metformin, apalutamide and enzalutamide are investigational.

9 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	695, pre-PSA, palpable; RP vs WW; 23 yr	Cancer death 19.6% vs 31.3% (relative risk 0.55)	Benefit at age <65 and GG ≥3; none in GG1–2
PIVOT	731 US veterans; RP vs observation; 18.6 yr	All-cause death 68% vs 73% (not significant); low-risk cancer death 2.7%	Modest benefit in intermediate risk
ProtecT	1643 PSA-detected; monitoring vs RP vs radiotherapy; 15 yr	Cancer mortality <3%, no difference; **metastases 9.4% vs 4.5% vs 5.0%**; 61% of monitoring arm treated	Active monitoring is not AS (no scheduled biopsy, >⅓ intermediate/high)

Trial	Design	Result	Take-home
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (not significant); ~50% less AS failure at 2 yr	MRI benefit shows over follow-up
REDEEM	Dutasteride vs placebo on AS	38% less "progression" (less GG1 volume), no upgrading difference	Cytoreduction, not disease modification

160 Open Radical Prostatectomy

1 · Principles

- Radical prostatectomy (RP) alone has been shown in a randomised controlled trial (RCT) to reduce metastasis and cancer death (**SPCG-4**). Most RPs are now robotic, without proven benefit.
- **Goals, in order**: cancer control, then avoiding technical complications, then continence, then potency.

> **KEY POINT**
> The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 · Surgical anatomy

P1 P4 Anatomy of prostate and neurovascular bundles; Prostate anatomy, newer surgical concepts and NVB preservation — fascial layers table, NVB of Walsh, inferior vesical urethral/capsular branches, accessory pudendals, Santorini plexus, nerve-sparing planes

Not in the paper answer:

- Inferior vesical urethral branches enter at the bladder neck (**5 and 7 o'clock** pedicles); capsular branches mark the cavernous nerves.
- The pelvic plexus combines parasympathetic **S2–S4** and hypogastric sympathetic fibres. It lies beside the rectum, **5–11 cm** from the anal verge, centred on the seminal vesicle tip.
- The **striated sphincter** is a vertical tubular sheath, not a flat diaphragm, so apical dissection decides continence; it and the levator ani are both mainly **pudendal**.

> **KEY POINT**
> Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · Selection and preoperative care

- **Harder, not contraindications**: prior pelvic surgery or radiotherapy, transurethral resection of the prostate (TURP), hernia mesh, large gland.
- **Defer surgery 6 weeks after biopsy and 12 weeks after TURP**, so haematoma settles.
- Inguinal hernia affects ~15% before and up to 20% after RP, so examine standing and repair it preperitoneally at RP.
- Give cephalosporin prophylaxis; the **fibreoptic headlight** is the most useful instrument.

4 · Retropubic RP — operative steps

> **ALGORITHM**
> 1. Lower midline extraperitoneal incision; pelvic lymph node dissection (PLND).
> 2. Open endopelvic fascia; divide puboprostatic ligaments.
> 3. Control and divide the DVC, then the urethra and posterior sphincter.
> 4. Release the NVBs; divide pedicles, bladder neck and seminal vesicles.
> 5. Reconstruct the bladder neck and make the anastomosis.

- **PLND** is for intermediate/high risk, omitted for Grade Group 1–2 low-volume disease.
- **Endopelvic fascia**: open at the sidewall reflection, because too medial causes Santorini bleeding. Divide the **puboprostatic ligaments** but spare the **pubourethral** part, which anchors the sphincter.

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if the catheter moves freely	3-0 figure-of-eight to pubic perichondrium, plus a back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- The **anterior apex is the commonest positive margin site**, so get absolute venous control before touching it.
- **Urethra**: transect close to the apex and place six 3-0 sutures through **mucosa and submucosa only**, because muscle bites delay continence.

Nerve sparing

- The **interfascial** plane (levator vs prostatic fascia) is correct; Walsh never uses intrafascial because of margin risk. Wide (extrafascial) excision is for induration, a fixed bundle or doubtful margin.
- Both NVBs are usually kept, because extracapsular extension rarely exceeds 1–2 mm. Decide in theatre: **if induration or a fixed bundle, excise that side**.
- **Technique**: release the prostate from the bundle, not the bundle from the prostate, starting **at the apex**. Use small clips and **never thermal energy**.
- **High anterior release** (before urethral division) means less traction and faster recovery.
- Avoid it if ≥2 **Tsuzuki criteria** are present: PSA >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal DRE.
- For **wide excision**, clamp **medial to lateral** on the rectum, since the reverse risks rectal injury.

Pedicles, bladder neck, reconstruction

- **Complete seminal vesicle excision is essential**; the pelvic plexus lies on their lateral surface.
- Close the bladder neck as a **tennis racquet** with mucosal eversion, and make a tension-free anastomosis; a peritoneal window lets haematoma decompress.

> **KEY POINT**
> Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 · Perineal RP (brief)

- Perineal RP reflects the fasciae off the prostate and spares the DVC, so it bleeds less, but PLND needs a separate approach.

6 · Intraoperative complications

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	**Divide DVC completely and oversew** (best measure)
Obturator nerve	PLND	Reanastomose with fine non-absorbable suture
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair and omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Typical loss is 300–1000 mL; partial DVC division bleeds more.
- **Rectal repair**: finish the prostatectomy, dilate the anus, close in 2 layers and place omentum between rectum and anastomosis to prevent a rectourethral fistula.

7 · Postoperative care

- Discharge on day 1–2. Give ketorolac only if creatinine <1.2 mg/dL and there is no haematuria.
- Keep the **catheter ~1 week**, because day-4 removal causes frequent retention; never remove it with significant extravasation.

8 · Postoperative complications

- Mortality is **0.2%**. **Delayed haemorrhage** needs **early exploration and evacuation**, which shortens stay.
- **Thromboembolism** is the major cause of death. It is highest with PLND and peaks at **14–28 days**. Use compression and early ambulation; treat with low-molecular-weight heparin (LMWH).
- **Bladder neck contracture** appears at ~3 months. Treat with dilatation, then cold-knife incision at 3, 6 and 9 o'clock with self-catheterisation and triamcinolone; reconstruction with an artificial sphincter is rarely needed.

P2 Post-prostatectomy incontinence — rhabdosphincter dependence, evaluation, PFMT, sling vs AUS

Not in the paper answer:

- Continence means **0–1 pad/24 h**. Prevent incontinence with a preserved apical sphincter, a precise tension-free mucosal anastomosis and a small supple neck.
- **Erectile dysfunction (ED)**: recovery is predicted by **age <65, preoperative potency, no diabetes and bilateral nerve sparing**, and continues up to 4 years. PDE5 inhibitor rehabilitation helps; nightly is not better than on-demand.

> **KEY POINT**
> Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs. Hence early re-exploration, clips near the bundles and a peritoneal window.

9 · Outcomes

Time	Continent	Potent (Walsh)
3 months	80%	38%
12 months	93%	73%
18–24 months	97%	86%

- These are single-expert figures; elsewhere only a third regained baseline potency by 2 years. RP improves lower urinary tract symptoms for up to 15 years.

10 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

1 · Background

P1 Open vs laparoscopic vs robotic radical prostatectomy; Robotic radical prostatectomy — history, comparison table, contraindications, ports, steps, complication rates

Not in the paper answer:

- Avoid relative contraindications (including neoadjuvant hormones) early in the learning curve.

> **KEY POINT**
> The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery.

2 · Patient selection

- The only absolute contraindications are uncorrectable bleeding diathesis and unfitness for general anaesthesia.
- After extraperitoneal mesh repair, use the **transperitoneal** route.
- **Obesity**: place longer trocars more cephalad and lower pneumoperitoneum to 10 mm Hg. A **large gland or median lobe** takes longer and bleeds more, but continence is comparable; **prior TURP** brings the ureteric orifices close to the bladder neck.
- Multiparametric MRI (mpMRI) for extracapsular extension (ECE) has specificity **0.88** and sensitivity **0.61**; it guides nerve sparing.

3 · Preparation and anaesthesia

- Consent for **conversion to open**; **general anaesthesia is mandatory**.
- **Positioning**: tuck and pad arms; use **no fixed shoulder rests** (brachial plexus injury) and avoid exaggerated hip flexion (femoral neuropraxia).
- CO2 pneumoperitoneum causes vagal response, oliguria and hypercarbia, so raise ventilation and lower pressure. Restrict fluids to 1.5–2 L, mostly after the anastomosis.
- **Steep Trendelenburg** causes facial, laryngeal and corneal oedema and raises intraocular pressure, so protect the eyes and consider delayed extubation if the face is very swollen.

4 · Access

- **Extraperitoneal** access gives a smaller space, protects bowel and needs less tilt (suits obesity and lung disease), but absorbs more CO2. **Transperitoneal** gives a larger space. Neither has a consistent advantage.
- Transperitoneal **anterior** (commonest; Retzius first) differs from **retrovesical** (seminal vesicles first, useful with a median lobe).
- Place the camera port **15–17 cm** from the symphysis; insufflate to 12–15 mm Hg.

5 · Operative steps (transperitoneal anterior RALRP)

ALGORITHM

1. Drop the bladder (Retzius); open the endopelvic fascia; ligate the DVC.
2. Divide the bladder neck; free the seminal vesicles and vasa.
3. Develop the posterior (Denonvilliers) plane; control pedicles and release the NVBs.
4. At the apex, divide the DVC and urethra; inspect the specimen.
5. Do pelvic lymph node dissection (PLND); entrap the specimen.
6. Posterior reconstruction, anastomosis, extraction.

- Open the endopelvic fascia with **no cautery**. Pneumoperitoneum tamponades DVC bleeding, so it can be sutured, stapled, or divided late and oversewn.

Bladder neck — no tactile feedback

- Divide anteriorly in the midline, lift the catheter, inspect for median lobe and orifices, then divide the posterior neck angled **45°** downward.
- After TURP the balloon falls into the defect, so division drifts too distal; a balloon off the midline suggests a median lobe. When in doubt, **divide more proximally**.

Posterior plane, pedicles, nerve sparing

P1 P4 Anatomy of prostate and NVB; NVB preservation in RP — nerve-sparing planes, Denonvilliers excision for posterolateral ECE, injury mechanisms, apex = commonest margin

Not in the paper answer:

- **Athermal (veil) release**: incise levator fascia anteromedially and sweep the NVB posterolaterally off the prostatic fascia, then Hem-o-lok the pedicle and divide sharply. The **intrafascial** plane is accepted in selected men.
- Pedicle clips near the anastomosis can migrate into the bladder, causing haematuria, infection and stones.

Apex and reconstruction

- Preserve urethral length; a small urethral rim on the apex lowers margins without harming continence.
- Reconstruct the bladder neck (tennis racquet) only for a large gland, median lobe or prior TURP, and protect the orifices.
- The **Rocco stitch** joins the Denonvilliers remnant and posterior bladder to the posterior rhabdosphincter. Its continence benefit is inconsistent; its agreed benefit is a **tension-free anastomosis**.
- The **Van Velthoven anastomosis** runs two arms anteriorly from 6 o'clock.

6 · Continence-preserving variants

P4 Newer surgical concepts in CA prostate — Retzius-sparing: earlier continence, more margins; Hood: early data

- Retzius-sparing leaves the detrusor apron, puboprostatic ligaments and DVC untouched, on the idea that preserving anterior support beats reconstructing it.

7 · Pelvic lymph node dissection

P4 P1 PLND templates; Lymphocele — templates (extended ~63%), staging not survival; lymphocele drain, then fenestration

Not in the paper answer:

- **Indications**: may omit in low risk (nodes <5%). AUA 2018: unfavourable intermediate or high risk. EAU 2021: nomogram risk **>5%**. AUA/ASTRO 2022: risk-stratified by nomogram (Briganti and MSKCC nomograms predate PSMA PET).
- The standard template misses most drainage, so **if PLND is done, use the extended template**: proximal 2 cm above the common iliac bifurcation (ureter crossing), distal node of Cloquet, lateral **genitofemoral nerve**, medial bladder wall.
- **Clipping lymphatics** means fewer lymphoceles; node count is a poor quality surrogate.

PLND complications (~1 in 7)

- For the obturator nerve, remove a clip or repair a transection primarily. Minimise heat proximally, where the ureter is at risk.
- **Lymphocele is commonest** (**2–5%** significant). **Transperitoneal access is not protective**, because the bladder walls off the fossa; a **peritoneal interposition flap** reduces it. It classically presents with ipsilateral deep vein thrombosis (DVT).

8 · Postoperative care

- Use enhanced recovery, with same-day discharge for selected men; a drain is not routine.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2–3 weeks	Strongly consider

- Removal at day 3–4 causes more retention. Extravasation delays but does not reduce long-term continence.
- **Penile rehabilitation**: PDE5 inhibitors help on-demand erections, but early rehabilitative use does not improve later response (3 RCTs). Use injections for non-responders.

9 · MIS-specific complications

Complication	Rate	Management
Transfusion	1–2%	Low, because pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation or incision; rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression; drugs only with multiple risk factors
Robotic failure	<0.5%	Cancel or convert
Recognised rectal injury	Rare; higher in salvage	Multilayer repair and flap, no colostomy
Unrecognised rectal injury	Rare	Presents as rectovesical fistula; colostomy, later repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

- **Ureteric injury** occurs behind the vas (posterior approach), at the posterior bladder neck after TURP or with a median lobe, and during proximal PLND.

- With DVT, image the pelvis for lymphocele or haematoma; with a significant lymphocele, do a duplex for occult DVT.

> **KEY POINT**
> Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 · Outcomes

- **Oncologic**: positive surgical margin (PSM) rates are equal open and robotic (~15%; ~9% organ-confined). PSM depends on tumour (Gleason ≥7, perineural invasion, stage) and surgeon volume, **not** on obesity, gland size or prior surgery.
- **Continence**: no-pad rates are in the mid-90s% at 1 year. Persistent mild leakage needs a sling or ProACT; severe (<2%) an artificial urinary sphincter.
- **Potency**: up to 90% at 1 year in expert hands; recovery continues to 5 years.
- **Salvage RALRP**: exclude metastases with PSMA PET and use full bowel prep and an air-bubble test. Continence is only 33–80%, and most men are impotent.

11 · Landmark trials

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

162 Radiation Therapy for Prostate Cancer

P2 P4 Radiotherapy and brachytherapy for CA prostate; Role of radiotherapy in focal and advanced CA prostate — α/β, schedules, brachytherapy, ADT by risk, salvage, Phoenix, toxicity

Not in the paper answer:

- A single 19 Gy high-dose-rate fraction fails (§5).
- RADAR and RTOG 99-10 on ADT duration (§6).
- Salvage prostatectomy incontinence ~63% (§11).

1 · Radiobiology and fractionation

- Conventional fractionation gives **1.8–2 gray (Gy)** daily.
- In the **linear–quadratic model**, the **α/β ratio** sets a tissue's response to fraction size: low (2–4 Gy) in late-responding normal tissue, high (≥8 Gy) in most tumours.
- Prostate cancer has few cycling cells, so its α/β is **≈1.5 Gy**, **lower than the rectum (~3 Gy)**. Fractions >2 Gy therefore favour tumour kill over late toxicity, which is the rationale for hypofractionation.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy in 39 fractions)
Moderate hypofractionation	2.25–4 Gy	~4 wk (**60 Gy/20**)
Ultrahypofractionation = stereotactic body RT (SBRT)	≥5 Gy	≤5–7 fractions (**36.25 Gy/5**)

2 · EBRT techniques

- MRI is fused with planning CT; PSMA PET changes management in up to half of higher-risk men (see Ch 152).
- **IMRT or VMAT** lowers gastrointestinal (GI) but not genitourinary (GU) toxicity versus 3D conformal RT, as the urethra lies inside the target.

Volume	Prostate application
Gross tumour volume (GTV)	Dominant intraprostatic lesion (DIL) on MRI
Clinical target volume (CTV)	Prostate, with or without proximal seminal vesicles and pelvic nodes
Planning target volume (PTV)	CTV plus a margin for motion and set-up: **3–7 mm** with image guidance

- **Image-guided RT (IGRT)** with daily gold fiducials or cone-beam CT is the NCCN standard; its smaller margins make large fractions safe.
- A **rectal spacer** (hydrogel or hyaluronic acid) is injected transperineally into Denonvilliers' fascia. Late rectal toxicity falls to **2% vs 7%** (hydrogel, 79.2 Gy).
- The MR-linac allows 2 mm margins, and protons have shown no randomised benefit over IMRT.

3 · Dose escalation

- Six randomised trials, including RTOG 0126, raised **biochemical control in all risk groups** with **no overall survival (OS) gain**, because salvage works and metastasis fell only slightly. Still, ≥~78 Gy is standard.

- **FLAME**: a focal boost to the MRI DIL improved biochemical progression-free survival (HR 0.45) without more toxicity.

4 · Hypofractionation

- **Moderate**: CHHiP 57 Gy/19 was not shown noninferior; PROFIT 60/20 was noninferior; RTOG 0415 70 Gy/28 was noninferior but more toxic late; HYPRO 64.6 Gy/19 was **not superior**.
- After prostatectomy, HYPORT 62.5 Gy/25 was noninferior to 66.6 Gy/37.
- **SBRT**: HYPO-RT-PC gave ~84% 5-yr failure-free survival, with more acute toxicity.
- **Dose ceiling**: there is no gain above **40 Gy/5**. Give it on alternate days with 3–5 mm margins.

5 · Brachytherapy (BT)

Urinary function decides eligibility.

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I, ^{103}Pd, ^{131}Cs)	Temporary ^{192}Ir (380 keV) in a remote afterloader
Technique	TRUS-guided transperineal template; **peripheral loading** spares urethra	Dose shaped by dwell times and positions; no post-implant dosimetry
Uses	Monotherapy for low/intermediate risk; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Oedema, seed migration	**19 Gy × 1 fails**: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	**59.6 d**	140–160 Gy	108–110 Gy
^{103}Pd	17 d	110–125 Gy	90–100 Gy
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- **Post-implant dosimetry** (CT at 14–30 days): **D90** >90% of prescription; **V100** >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. A low D90 predicts biochemical failure. MCQ D90 is a dose, V100 a volume.
- **Relative contraindications**: **IPSS >20**; **Qmax <10 mL/s**; large median lobe; large TURP defect or TURP within 3–6 mo; prior pelvic RT; inflammatory bowel or connective tissue disease; **pubic arch interference**; inability to adopt lithotomy.
- A large gland shrinks ~30% with ADT. Retention risk is ≈29% if IPSS >20 vs ≈2% if <10, so start α-blockers before the implant.
- **With EBRT**: 45 Gy plus an implant at 60–70% of monotherapy dose. **RTOG 0232**: BT alone equalled EBRT plus BT in intermediate risk with less toxicity, so BT alone is standard.
- **ASCENDE-RT**: no OS difference; LDR boost grade 3 GU toxicity **18.4% vs 5.2%**.

KEY POINT

Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 · ADT with RT

- **Synergy**: ADT causes apoptosis, shrinks the target and lowers **Ku70**, impairing non-homologous end-joining repair (see Ch 164).

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	**4–6 mo**	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 mo better than 3); RTOG 99-10: 9 mo no better than 4
High / very high	**24–36 mo** (≥18)	12–36 mo	EORTC 22863, RTOG 92-02, EORTC 22961 (36 mo better than 6); PCS IV: 18 mo = 36 mo

- Needed duration falls as RT dose rises: in **RADAR**, 18 mo beat 6 mo only without an HDR boost.
- **STAMPEDE**: adding 2 yr abiraterone in high-risk N0/N1 M0 improved metastasis-free survival and OS.

7 · Pelvic nodal RT

- RTOG 94-13 and GETUG-01 showed no durable benefit, mainly because a field border at L5/S1 misses the **common iliac nodes**; modern fields extend to **L4/L5**.
- **POP-RT** (node risk >20%, long ADT): whole-pelvis RT raised 5-yr biochemical failure-free survival (~95% vs ~81%), with no OS difference and more acute GI toxicity.

8 · Post-prostatectomy RT

- RADICALS-RT, GETUG-AFU 17 and RAVES, with their meta-analysis, make **early salvage preferred** over adjuvant RT, which may still suit very high risk (Gleason 8–10, pT3b) (see Ch 164).
- Dose is ≥64 Gy, and PSMA PET localises recurrence.

9 · Follow-up and biochemical failure

Do not mistake a benign bounce for failure.

- Benign glands persist, so PSA stays detectable. The nadir is ≤0.1 ng/mL after EBRT plus BT and higher after EBRT alone; it is reached at ~1.5 yr after EBRT and ~2.5 yr after BT.

	ASTRO 1996	Phoenix 2005 (current)
Failure	3 consecutive rises	PSA **≥ nadir + 2 ng/mL**
Date	Backdated to midway between nadir and first rise	When reached (or when salvage started)
Notes	Delays diagnosis ≥18 mo; invalid with ADT or BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Poor prognosis is signalled by failure <18 mo and especially **PSA doubling time <3 mo**, which predicts cancer death.
- A **bounce** (rise >0.2 ng/mL, then durable fall) occurs in ~1/3 after LDR, mostly in the first 3 yr, and predicts a better outcome.
- **Biopsy** is best at **30–36 mo**, since irradiated cells die slowly. Marked radiation effect is not graded and behaves as negative.

10 · Toxicity

Toxicity	Features	Management / prevention
Acute GU	Peaks ~2 mo, settles by 12 mo	α-blockers; catheter for retention
Late GU	Grade ≥2 ~20%; stricture; incontinence ~3%, more after TURP	IGRT halves late GU toxicity; peripheral loading
Late GI (proctitis)	Bleeding at 6–18 mo; IMRT 81 Gy grade ≥2 ~5%; BT 1–2%; colostomy ~0.3%	Steroid or sucralfate enemas, then **argon plasma coagulation** or hyperbaric oxygen, then colostomy
Erectile dysfunction	Vascular, over 24 mo; often due to ADT; best with BT alone (~76% potent at 5 yr)	PDE5 inhibitors work in ~¾
Second cancers	Small excess of bladder and rectal cancers	Investigate haematuria or rectal bleeding fully

11 · Salvage after local failure

- Salvage needs **biopsy-proven, local-only** recurrence. Options are salvage RP (incontinence ~63%), BT (preferred), SBRT re-irradiation, cryotherapy and HIFU.

12 · Landmark trials

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low–intermediate	8-yr biochemical failure-free 65% vs 80%; OS same	Dose escalation improves control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (plus 12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT with no, 3 or 6 mo ADT	6 mo improved OS; 3 mo did not	≥6 mo short-course
RTOG 94-08	66.6 Gy with or without 4 mo ADT, T1b–T2b	10-yr OS 57% vs 62%; intermediate risk only	ADT for intermediate, not low
EORTC 22863	RT with or without 3 yr goserelin, T3–T4	10-yr OS 39.8% vs 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk

Trial	Design	Result	Take-home
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	RT lowered cancer death and raised OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate-only RT	No durable benefit	Field height matters
SPPORT	Salvage bed RT, with or without ADT and nodes	Pelvic RT plus ADT best	Treat nodes in salvage

163 Focal Therapy of Prostate Cancer

1 · Definition and rationale

- **Focal therapy (FT)** is guided ablation of an image-defined, biopsy-confirmed lesion with a **safety margin**. Sparing the neurovascular bundles (NVBs), sphincter and rectum preserves function.
- FT rests on mpMRI, targeted biopsy and the fact that **Gleason pattern 4/5** is lethal.

Index lesion hypothesis

- Cancer is multifocal in **50–91%** of radical prostatectomy (RP) specimens. The **index lesion (IL)**, the largest or highest-grade focus, is held to drive natural history and metastasis.
- **For**: the IL holds **80–90%** of tumour volume, and satellites are mostly **< 0.5 mL** (Stamey "insignificant").
- **Against**: a satellite is the aggressive focus in 12–38%, and IL Gleason matches overall grade in only ~⅔.

> **KEY POINT**
> "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT and active surveillance (AS)** form one strategy: ablate grade 4 disease, watch the low-grade remainder, and retreat new foci.

2 · Patient selection

- The patient must **accept close surveillance**.
- **TRUS biopsy** misses almost half the disease relevant to hemiablation compared with **transperineal mapping biopsy (TMB, 5-mm grid)**.
- **mpMRI** detects high-grade and anterior tumours. In **PROMIS**, its sensitivity for clinically significant cancer (csPCa) was **93%** and specificity 41%, so MRI lesions need biopsy and MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates volume**, so a margin is needed: **5 mm** ablates >95% of csPCa. About **90%** of outfield csPCa lies within **1 cm** of the target.

> **KEY POINT**
> Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland. Every consensus has kept all three.

Consensus criteria (2016–2019)

Variable	Consensus
Work-up	mpMRI plus targeted and systematic biopsy
Risk group	**Intermediate**; Gleason **3+4 = "sweet spot"**; 4+3 maximum; no cribriform or ductal histology
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016, tightened from 5 mm)

Variable	Consensus
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	**Not** a contraindication
Not suitable	High risk (no long-term data); low risk usually goes to AS

- The goal is now to treat and retreat so as to **delay or avoid radical therapy**. FT remains investigational, ideally within a trial.

3 · Ablation patterns and energies

- FT means **lesion, quadrant or hemiablation**; "hockey-stick" (subtotal) ablation is partial-gland ablation, not FT.

P2 P4 HIFU; Cryotherapy for carcinoma prostate — HIFU ≥55 °C, posterior lesions; cryo Joule–Thomson, urethral warmer, fistula ≤0.8%

Not in the paper answer:

- Focal cryotherapy 5-yr biochemical progression-free survival (BPFS) ~62–70%; COLD registry: partial equals whole-gland control with ~2× better erections.

Modality	Mechanism / route	Pros	Cons
TULSA (MRI-guided transurethral ultrasound ablation)	Transurethral transducer with MR thermometry	Anterior gland; control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~⅓
Focal laser	Interstitial fibre ≥60 °C; MR thermometry	Most predictable zone	Narrow margin, so **highest infield failure** (~40% at 6 yr)
Irreversible electroporation (IRE)	Non-thermal pulses >1600 V/cm open membrane pores	Spares connective tissue; good function	Needs deep anaesthesia; NVB and urethral injury on histology
Vascular-targeted photodynamic therapy (VTP)	IV padeliporfin (TOOKAD) 4 mg/kg activated by 753-nm light occludes vessels	**Only randomised evidence** (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency	~100 °C	Fat limits conduction	Infield csPCa 20–30%
Focal brachytherapy	Low-dose-rate seeds or high-dose-rate (HDR) catheters	Mild early toxicity	HDR: stricture ≤8.2%, ED in half; biopsy only at 24 mo

KEY POINT

Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants all aim to make that edge lethal, so laser's precise but narrow zone gives the most infield recurrence.

4 · Outcomes

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7–39% overall (~¼ csPCa)		95–100%	47–100%
HIFU	17–54% (csPCa 0–22%)	7–43% (csPCa 2–8%)	90–95%	50–78%

Modality	Infield positive biopsy	Outfield	Continence	Erections
Laser	11–50%	17–60%	95–100%	Minimal change
IRE	csPCa 1–10%	csPCa 5–16%	~All	Some decline
VTP	19–37%	17–19%	Preserved	Nil to mild fall

- HEAT (HIFU): 7-yr failure-free survival was **69%** (68% intermediate, 65% high risk).

5 · Complications

- The commonest is **urinary retention** (0–9%); others are infection, sloughing and stricture (~4–5% after HIFU or IRE).
- **Rectourethral fistula** occurs in ≤2% and is the most feared; some heal with a catheter for 4–6 weeks.

6 · Follow-up and failure

Follow-up relies on MRI and biopsy, not PSA.

- **PSA** falls only **50–80%**, and the Phoenix/ASTRO definitions are not validated after FT. A rise to nadir + **1.0** ng/mL at 12 months, or + 1.5 at 24–36 months, predicts csPCa.

Success (ICUD–SIU 2015)	Infield	Outfield
Short term	Eradication of treated focus	No csPCa — any outfield cancer = **selection failure**
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ALGORITHM

1. At **3–6 months**: mpMRI and targeted biopsy (4–6 cores) of the treated zone.
2. At 1–2 and 5 years: mpMRI; if a new lesion appears, then targeted biopsy.
3. At 1–2 and 5 years: systematic biopsy of the untreated gland.

- **PI-FAB**: read contrast-enhanced (**DCE) images first**, as scar obscures T2 and diffusion. Score 1 (both negative) is monitored, score 2 (DCE positive only) is biopsied if PSA rises, and score 3 (both positive) is biopsied.

7 · Salvage

- After FT failure, options are **focal retreatment** or whole-gland RP or radiotherapy.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	Progression-free survival 60%, metastasis-free survival 91% (48 months)
Continence	87–100%	64–87.5%	93%
Signature complication	Rectourethral fistula ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%

- Focal salvage controls cancer as well as whole-gland salvage, with better function.

8 · Landmark trials

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; conversion to radical therapy at 4 yr **24% vs 53%**; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT

Open problems are leaving satellites, the cost of retreatment and adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy, whereas prostate FT relies on surveillance and retreatment.

164 Treatment of Locally Advanced Prostate Cancer

P2 P1 Staging and management of locally advanced prostate cancer; Staging and management of non-organ-confined CA prostate — definition, risk groups, work-up, RT + ADT, RP, salvage RT, pN1, algorithm

Not in the paper answer:

- pN1: hormones plus RT beat hormones alone (OS 74% vs 55%) (§3).
- Bicalutamide 150 mg raised death in localised disease (HR 1.23) (§5).

1 · Definition and natural history

- The working definition is **T3–4, N0 or N1, M0**, and it calls for **multimodal therapy**.
- NCCN high risk is cT3a, PSA >20 or grade group (GG) 4–5. Very high risk is cT3b–T4, primary pattern 5, ≥2 high-risk features, or >4 cores of GG 4–5.
- Because of stage migration, high risk now mostly means **high grade**, not a palpable T3.
- **Grade drives death**: untreated Gleason 8–10 had 15-yr cancer mortality of 60–87% before the PSA era.

2 · Staging and risk assessment

- Clinical staging and imaging both **understage**.
- NCCN orders a bone scan and CT or MRI for cT3–T4, PSA >20, GG 4–5 or primary pattern 5; PSMA PET may substitute.
- In **proPSMA**, PSMA PET accuracy was **92% vs 65%**.

3 · Radical prostatectomy in high-risk disease

- **Clinical cT3 is overstaged in 7–27%**, and >⅓ of high-risk specimens are confined, so do not exclude men on clinical stage alone.
- NCCN advises RP with extended pelvic node dissection (**ePLND**) only if the tumour is **completely excisable**; otherwise give androgen deprivation (ADT, "AD" in trial reports) with radiotherapy (RT).
- In cT3, DSS after RP is **~80–90% at 10 yr**.

> **KEY POINT**
> Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37), higher than extracapsular extension (ECE), seminal vesicle invasion (SVI) or nodes, so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	**4.37** (highest)

Neoadjuvant AD before RP

- **Clinical downstaging occurs in 32–90%, but pathological in only 8–31%**. It has **no** effect on PSA recurrence, DSS or overall survival (OS).

- **CALGB 90203** (AD plus docetaxel before RP) missed its primary endpoint but improved MFS and OS.

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911	pT3 and/or positive margin; immediate RT vs observation	Biochemical disease-free survival 74% vs 53%; **benefit only with positive margins**
SWOG 8794	Same	**MFS (HR 0.71) and OS (HR 0.72)** — the only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES	Adjuvant vs early salvage (64–66 Gy)	**Early salvage preferred**; 5-yr event-free 87–94% in both arms

- **Trap**: very high-risk men (Gleason 8–10, pT3b) were only ~20% of the timing trials, so adjuvant RT may still help them, as may extensive margins.
- Dose is **≥64 Gy**. **GETUG-AFU 16**: **3 months** of AD with salvage RT gave 10-yr progression-free survival 64% vs 49% (HR 0.54) (see Ch 162).

Adjuvant AD and pN1

- In **pT3N0**, adjuvant AD reduces progression but gives **no DSS or OS gain**.
- **ECOG 7887** (nodes found at RP): immediate AD improved OS, DSS and PFS.
- **AD plus RT beats AD alone** in pN1: 10-yr DSS 86% vs 70%, OS 74% vs 55%.

> **ALGORITHM**
>
> 1. **pN1 after RP**
> 2. If favourable (few nodes, Gleason ≤8), then observe with PSA monitoring.
> 3. If adverse (>3 nodes, Gleason >8), then give AD plus RT (best OS).
> 4. Avoid AD alone in a fit man.

4 · RT + ADT

- The standard is **dose-escalated RT (>72, ideally ~78–81 Gy) with ADT**, often with pelvic RT (see Ch 162).
- ADT shrinks the target and is synergistic with RT.

Trial	Question	Verdict
RTOG 86-10	4 mo neoadjuvant/concurrent AD, bulky T2–T4	Disease-specific mortality fell from 36% to 23%; **OS unchanged**
TROG 96.01	RT with no, 3 or 6 mo NAD	6 mo lowered all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control and MFS; OS gain only in Gleason 8–10
PCS IV	18 vs 36 mo	OS comparable; 18 mo better quality of life
RTOG 94-13	Pelvic vs prostate RT, and neoadjuvant vs adjuvant AD	Early PFS gain lost; whole pelvis with adjuvant AD had worst OS

- **Duration**: intermediate risk 4–6 mo; high risk (NCCN) **18–36 mo** with external beam RT (EBRT), 12–36 mo with a brachytherapy boost.

- **ASCENDE-RT**: a low-dose-rate boost halved biochemical recurrence vs an EBRT boost (HR 2.04), with equal OS.
- **Trap**: AD plus RT benefits often exclude OS. The OS gain is seen with long AD in high risk (EORTC 22863; RTOG 92-02 Gleason 8–10 OS 81% vs 71%).

Adding RT to ADT

- **SPCG-7**: cancer death relative risk **0.44**; overall mortality RR 0.68. **PR.3/PR07**: DSS HR 0.54, and 7-yr OS rose from 66% to 74%. So RT adds **8–10% OS**.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD and RT	OS HR 0.69
GETUG 12	Docetaxel and estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel, estramustine and etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	**Abiraterone and prednisolone for 2 yr**, added to 3 yr AD and RT	**MFS HR 0.53, OS HR 0.60**

- STAMPEDE eligibility: **N1, or two of Gleason ≥8, cT3–4 and PSA ≥40**. **NCCN 2023 advises RT, ADT and 2 yr abiraterone for very high risk.**

5 · ADT alone and other options

- **MRC**: immediate rather than deferred AD gave fewer cancer deaths, mainly in M0.
- **Bicalutamide 150 mg** improved PFS in locally advanced disease but **raised death in localised disease** (HR 1.23), so never use it at low risk.
- AD alone does not stop local progression, and over half need a TURP.
- **RP vs RT with ADT**: there is no completed randomised trial; **SPCG-15** (cT3N0M0) is ongoing.

ALGORITHM

1. **Locally advanced / high-risk M0**
2. Stage with PSMA PET, or CT and bone scan.
3. If fit and resectable, then RP with ePLND. If pathology is adverse, then early salvage RT (adjuvant if pT3b, Gleason 8–10 or positive margins). If pN1, then observe or give AD plus RT.
4. Alternative: dose-escalated RT (with or without brachytherapy boost and pelvic nodes) plus 18–36 mo ADT.
5. If very high risk or N1, then add 2 yr abiraterone.
6. If unfit for local therapy, then ADT alone.

KEY POINT

Viva: "Why not ADT alone for T3 disease?" SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-7	ADT with or without RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT with or without 3 yr goserelin	5-yr OS 62% vs 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain in Gleason 8–10	Prolong ADT for high grade
STAMPEDE (abiraterone)	ADT and RT, with or without 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS and OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional imaging	Accuracy 92% vs 65%	PSMA PET for staging

165 Management of Recurrent and Newly Metastatic Prostate Cancer

1 · Biochemical recurrence: concept and definitions

Biochemical recurrence (BCR) is a rise in prostate-specific antigen (PSA) after curative therapy, before clinical recurrence. Clinical recurrence is almost always preceded by it.

- After radical prostatectomy (RP), **¼–⅖** develop BCR within 10 years; only ≈half of higher-risk men with BCR fail clinically.
- At MSK, PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis.
- **Ultrasensitive PSA** is prognostic, not diagnostic; about half of early values fall on retesting, so repeat it.

Setting	Definition	Logic
Post-RP — AUA	PSA **≥0.2 ng/mL**	No gland remains, so any PSA is suspect
Post-RP — NCCN	Never undetectable, two rises, or PSA ≥0.1	Any of three triggers
Post-RP — EAU	No threshold; risk groups (below)	Kinetics plus grade
After external beam RT (EBRT) ± androgen deprivation therapy (ADT)	**Phoenix**: **nadir + 2 ng/mL**	Validated against metastasis and death
After low-dose-rate (LDR) brachytherapy	≥0.2 may suit better; **<0.2 at 4 years ≈ cure**	The very low nadir means Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After radiotherapy (RT)
Low risk (needs BOTH)	PSA doubling time (PSADT) **>1 year** AND grade group (GG) <4	Interval to failure **>18 months** AND GG <4
High risk (EITHER)	PSADT ≤1 year OR GG 4–5	Interval ≤18 months OR GG 4–5

KEY POINT

PSA bounce after RT is a transient rise of 0.1–0.5 ng/mL at a mean of ≈ **9 months**, followed by a fall. It is **favourable**, so monitor; never salvage on a bounce.

2 · Risk prediction and natural history

- A **positive margin** is the only modifiable BCR predictor.
- **CAPRA** is the pre-operative score and **CAPRA-S** the post-operative one MCQ. Only the MSK nomograms predict 15-year cancer-specific survival.
- **Pound**: untreated, a median **8 years** passes from BCR to metastasis, then **5 years** to death (P2-144).
- After Phoenix failure, metastasis follows at ≈5.4 years. Follow-up is lifelong, because ¼ recur after 5 years.

3 · Imaging at recurrence

P2 65 yr male underwent RP… PSA has risen — BCR definitions, PSMA PET/mpMRI work-up, salvage RT dose/timing, RTOG 9601, GETUG-AFU 16, SPPORT, STOMP/ORIOLE, EMBARK, Pound prognosis

- Bone scan and CT are rarely positive below PSA **10 ng/mL**.
- **Prostate-specific membrane antigen (PSMA) PET** is positive in ≈½ at PSA **<0.5**, and finds disease outside the salvage field in ≈1 in 5. Fluciclovine-guided salvage improved control (**EMPIRE-1**).
- Multiparametric MRI (mpMRI) is best for local recurrence. **Biopsy after RT** is mandatory before local salvage.

> **KEY POINT**
> Salvage RT works best at the lowest PSA, but imaging improves only as PSA rises, so risk features (grade, margins, kinetics) guide early action.

4 · Salvage after radical prostatectomy

- **Salvage RT (SRT)** is the only curative option (P2-144). Give **≥64–65 Gy** to the fossa.
- Delay costs control: ≈10% relapse-free survival is lost per 0.1 ng/mL in men with ≥2 adverse features.
- No visible lesion on mpMRI predicts SRT failure, because the disease is probably occult metastasis.

5 · Salvage after radiotherapy

Prerequisites are biopsy-proven local recurrence, no metastasis on imaging, excellent performance status, life expectancy ≥5–10 years and a shared decision. Every modality gives ≈**50–70%** control, so selection (local-only disease, low PSA, GG <4) matters more than modality.

Option	Selection / features	Signature morbidity
Salvage RP (open/robotic)	EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, N0M0 on imaging, initial cT1–T2.	Incontinence, bladder neck contracture, rectal injury, near-universal impotence
Salvage brachytherapy (LDR ≈ high-dose-rate)	Only after EBRT	Grade ≥3 urinary toxicity ≈1 in 6
Salvage stereotactic body RT (SBRT)	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA <10 (ideally <4), PSADT **≥16 months**; a post-cryo nadir <0.2 predicts success	**Urethrorectal fistula** (1–3%), erectile dysfunction; incontinence single-digit with **urethral warming**
Salvage high-intensity focused ultrasound (HIFU)	≥2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Recurrence after **focal therapy** is worked up like RT failure.

6 · Non-curative approach to BCR: observation vs ADT

- AUA: observe (or enrol in a trial) after maximal local therapy; give ADT only for higher metastatic risk (PSADT **<12 months** or high grade), preferably intermittent.
- EAU advises observing low-risk BCR, limited life expectancy, or men declining local therapy. Primary ADT for localised disease is **not** recommended.
- **Messing (ECOG)**: immediate ADT for node-positive (pN+) disease after RP improved overall survival (OS), but the trial was small and pre-dated modern imaging.
- Low-volume pN+ with undetectable PSA: **20–40%** are cured by surgery alone.
- Immediate ADT: TOAD OS hazard ratio (HR) 0.55; EORTC 30891 benefit only if PSA >50 or PSADT <12 months.
- **EMBARK** entry: PSADT **≤9 months**, with PSA ≥1 after RP or ≥2 above nadir after RT.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent was **non-inferior** (OS 8.8 vs 9.1 years).
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% CI 0.99–1.23). This was **inconclusive**; intermittent may be worse here.

7 · Androgen deprivation therapy: biology and agents

Castration removes the ligand, not the receptor, so castration-resistant prostate cancer (CRPC) keeps androgen receptor (AR) signalling alive through AR amplification, ligand-binding-domain mutations, splice variants (**AR-V7**, poor prognosis) and tumour steroid synthesis. **Castrate testosterone: <50 ng/dL**.

P4 P2 GnRH (LHRH) analogues; Maximum androgen blockade and bicalutamide — agonist flare/FSH rise, antagonists (degarelix, relugolix/HERO), antiandrogen toxicities, CAB, withdrawal response

Method	Key facts
Bilateral orchiectomy	Castrate within hours. Fewer cardiac events and fractures than agonists with equal survival, so offer it for lifelong ADT
LHRH agonists / antagonists	See P4-103. HERO: castrate by day 4 in 56% vs 0%
Cyproterone acetate	Inferior to goserelin; cardiovascular events up to 10%; used 50–100 mg/day for hot flushes
AR-targeting agents (ARTAs): enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Seizures with enzalutamide; rash and hypothyroidism with apalutamide; darolutamide has the fewest effects
Abiraterone (selective CYP17), ketoconazole (non-selective)	Give abiraterone with prednisone to prevent ACTH-driven mineralocorticoid excess. Ketoconazole castrates within 4 h: an emergency option

KEY POINT

Flare matters with spinal or weight-bearing bone metastases: give an antiandrogen with the agonist for **21–28 days**. For **impending cord compression**, use an **antagonist or orchiectomy** (or ketoconazole). **Antiandrogen withdrawal**: 15–30% respond with bicalutamide and 0–10% with ARTAs, with no OS gain.

8 · ADT side effects and their management

Effect	Key facts	Management
Osteoporosis / fracture	Loss of peripheral oestrogen; fracture risk rises 20–45%	Baseline DEXA and FRAX. For all: calcium 1000–1200 mg and vitamin D 400–1000 IU/day, exercise, stop smoking. Give a drug if T-score is −1 to −2.5 with 10-year hip fracture risk **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly. Repeat DEXA at 1 year
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Oestrogens best; megestrol, cyproterone, venlafaxine, gabapentin. Clonidine is ineffective
Metabolic	Fat rises with sarcopenia; diabetes (agonist HR 1.44)	Exercise; monitor glucose and lipids
Cardiovascular	Agonists raise coronary events, sudden death and thromboembolism	Risk-factor control; orchiectomy or antagonist favoured
Anaemia	Normocytic; pre-ADT Hb <12 g/dL predicts shorter survival	Reversible ≈1 year after stopping
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy or tamoxifen

Follow-up on ADT

Check PSA and bloods 3–6-monthly. **If PSA rises, check testosterone first.** A rising PSA with castrate testosterone means **CRPC** (see Ch 166), so restage. The germline HSD3B1 (1245A>C) variant predicts ADT resistance MCQ.

9 · Metastatic hormone-sensitive prostate cancer (mHSPC)

P3 Management of newly Metastatic prostate cancer — CHAARTED/LATITUDE definitions, doublets and triplets with HRs, prostate RT for low volume, MDT, genetic testing, follow-up

Not in the paper answer:

- **SWOG 9346 PSA nadir at 7 months**: ≤0.2 gives median OS **75** months; 0.2–4 gives 44; >4 gives 13.
- Metachronous low volume does best (median OS ≈92 vs ≈43 months for synchronous high volume).
- Genes to test: germline BRCA1/2, ATM, PALB2, CHEK2, HOXB13, mismatch repair; somatic CDK12, MSI-H.

Definition	Criteria	Use
CHAARTED high volume	**≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral metastasis (no grade criterion)	Docetaxel benefit
LATITUDE high risk	**≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral	Abiraterone entry

- **ADT is the floor, never the ceiling**; long-term combined blockade is not recommended.

- Docetaxel is **75 mg/m² 3-weekly × 6** (meta-analysis HR 0.77).
- Observe only if asymptomatic with life expectancy <5 years. Cytoreductive prostatectomy is for trials only.
- **Metastasis-directed therapy (MDT)**: oligometastases are **≤3 sites** on PET, treated by SBRT or salvage node dissection.

10 · Landmark trials

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2)	Equivalent; ≥60% spared RT	Early salvage preferred
RTOG 0534 (SPPORT)	Bed vs bed plus 4–6 months ADT vs pelvis plus ADT	5-year freedom from progression 70.9 / 81.3 / **87.4%**	Add ADT and pelvic nodes (not OS)
RTOG 9601	SRT ± 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%	OS benefit
GETUG-AFU 16	SRT ± 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); MFS HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide ± enzalutamide vs enzalutamide	MFS 87.3% (combination) / 80.0% (enzalutamide) / 71.4% (leuprolide); HR 0.42 and 0.63	Enzalutamide for high-risk non-metastatic HSPC
CHAARTED (ECOG 3805)	ADT ± docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months	High volume only
GETUG-AFU 15	ADT ± docetaxel	OS HR 0.88, not significant	Negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic and high-risk M0	Docetaxel OS HR 0.78; abiraterone OS HR 0.63, failure-free survival HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT ± abiraterone	OS HR 0.62; rPFS 33.0 vs 14.8 months	Abiraterone in high risk
ARCHES	ADT ± enzalutamide	rPFS HR 0.39	Volume-independent
ENZAMET	ADT plus enzalutamide vs ADT plus older antiandrogen	OS HR 0.67	Control was not placebo
TITAN	ADT ± apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2, ± abiraterone and prostate RT	All: rPFS HR 0.54, OS HR 0.82; with docetaxel: rPFS 0.50, OS 0.75	Triplet with abiraterone
ARASENS	ADT and docetaxel, ± darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel
STAMPEDE arm H	ADT ± prostate RT	Overall HR 0.92; low volume HR 0.68 (3-year OS 81 vs 73%)	RT to prostate in low volume

166 Management of Castration-Resistant Prostate Cancer

1 · Definition and biology

CRPC means progression (biochemical, radiographic or clinical) despite castrate testosterone **<50 ng/dL**. On androgen deprivation therapy (ADT) alone, metastatic overall survival (OS) was historically 24–36 months.

P2 Castration-resistant prostate cancer (CRPC): definition and treatment — EAU definition, AR mechanisms, evaluation, general principles, nmCRPC, mCRPC options table, sequencing, supportive care

Not in the paper answer:

- A prostate-specific antigen (PSA) doubling time (PSADT) **<3 months** predicts a rapid course, so consider aggressive therapy.
- Bone pain or visceral-predominant disease calls for **upfront chemotherapy**.
- **Castration resistance is not androgen independence.** The androgen receptor (AR) still drives growth through amplification, mutations, the splice variant **AR-V7** and tumour steroid synthesis, so AR-pathway inhibitors (ARPIs) work.
- Once AR dependence is lost, use taxanes, radium-223, PSMA therapy, PARP inhibitors (PARPIs) or immunotherapy.

2 · Clinical states and progression criteria

The **Prostate Cancer Clinical Trials Working Group 3 (PCWG3)** clinical states model classifies men by prior therapy, from rising PSA through non-metastatic (nmCRPC) to metastatic CRPC (mCRPC).

- PCWG3 PSA progression is a rise of ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.

Modality	Pitfall	PCWG3 rule
PSA	Early rise with chemotherapy; flare with radium-223 and ^{177}Lu-PSMA	Confirm any rise
Bone scan	Early new lesions may be **flare** of healing disease	Rescan at **12–16 weeks** as the new baseline; progression needs **≥2 new lesions** confirmed by ≥2 more ("2 + 2")
CT	Soft tissue is little of the burden	RECIST; PFS preferred over response rate
PSMA PET	Apparent progression at 4 weeks resolves by 3 months in ≈13%	Criteria not defined

> **KEY POINT**
> Every marker (PSA, bone scan, PSMA PET) can flare early, so call progression only on confirmed, combined evidence.

- **Continue ADT indefinitely** during every later therapy.

3 · Germline and somatic testing

- **Germline testing** suits all metastatic disease, relevant family history (BRCA, Lynch, HOXB13) and Ashkenazi ancestry.

- **Somatic** sequencing usually uses the archival primary. In liquid biopsy, **clonal haematopoiesis (CHIP)** can give false DNA-repair calls.
- Somatic homologous recombination repair (HRR) mutations occur in **20–25%** of mCRPC. Mismatch repair deficiency (dMMR/MSI-H) occurs in **2–5%** and should prompt Lynch counselling.

4 · Non-metastatic CRPC (nmCRPC)

nmCRPC is a rising PSA on castration with no metastases on **conventional** imaging. PSA kinetics are the strongest predictor. **Add an ARPI (apalutamide, enzalutamide or darolutamide) to ADT when PSADT is ≤10 months.** All three improve metastasis-free survival (MFS) and OS. Abiraterone is active but **not approved**.

5 · mCRPC: androgen-signalling inhibitors

Agent	Mechanism	Dose	Toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17α-hydroxylase and 17,20-lyase)	1000 mg **fasting** with prednisone (10 mg/day in mCRPC)	Mineralocorticoid excess, hepatotoxicity; correct potassium first. Food raises absorption up to 10× (alternative: 250 mg with a low-fat meal)
Enzalutamide	Pure AR antagonist; blocks translocation and DNA binding	160 mg, no steroid	**Seizures (≈1%)**, fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	Does not cross the blood–brain barrier (BBB)	240 mg	**Rash, hypothyroidism**, fractures
Darolutamide	Structurally distinct; does not cross the BBB	600 mg twice daily	Least CNS toxicity

KEY POINT

Prednisone with abiraterone: blocking CYP17 lowers cortisol, so ACTH rises and causes mineralocorticoid excess; prednisone suppresses ACTH. Enzalutamide needs no steroid. All ARPIs cause falls and fractures, so take care over 75.

6 · Cytotoxic chemotherapy

Cabazitaxel has poor P-glycoprotein affinity, so it works after docetaxel.

Agent	Dose / line	Signature toxicity
Docetaxel	**75 mg/m² 3-weekly** with prednisone 5 mg twice daily; first line	Neutropenia, **neuropathy**, oedema, hyperlacrimation, nail dystrophy
Cabazitaxel	**20 mg/m²** (25 if low neutropenic risk) 3-weekly with prednisone and G-CSF; after docetaxel	**Neutropenia** (grade 3–4 82%), **diarrhoea**
Mitoxantrone	12 mg/m² with prednisone; palliation	No survival benefit
Platinum (with etoposide or taxane)	Neuroendocrine variant; DNA-repair-deficient	—

7 · Radiopharmaceuticals

Feature	Radium-223	^{177}Lu-PSMA-617
Emitter	**Alpha** (path **<100 µm**, so marrow is spared); calcium mimetic	Beta; needs PSMA PET-positive disease
Eligible	**Symptomatic bone metastases, no visceral or bulky nodal disease**	PSMA-avid, any site, after ARPI and taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia	Fatigue, **dry mouth**, anaemia
Caution	**Never with abiraterone/prednisone**; bone agent with any ARPI	—

Strontium-89 and samarium-153 only palliate pain.

8 · PARP inhibitors

P4 PARP inhibitors in Prostate cancer — synthetic lethality, PARP trapping, AR–PARP synergy, BRCA2 vs ATM, approvals, toxicity, key trials

Not in the paper answer:

- PARPIs are ineffective in unselected men; haemoglobin can fall suddenly, so watch blood counts.

9 · Immunotherapy

- **Sipuleucel-T**, an autologous dendritic-cell vaccine, was the first approved therapeutic cancer vaccine.
- It suits **asymptomatic/minimally symptomatic mCRPC with no visceral metastases or opioid pain**. It prolongs OS **without** PSA or PFS change.
- **Pembrolizumab** 200 mg IV 3-weekly is approved for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase** after ≥1 line.

10 · Neuroendocrine/aggressive-variant CRPC

Feature	Usual mCRPC	Neuroendocrine variant
AR / PSA	Retained; PSA tracks disease	Lost; **PSA low despite progression**
Pattern	Osteoblastic bone ± nodes; visceral <20%	Bulky masses, liver, **osteolytic** lesions with hypercalcaemia, brain
Molecular	AR alterations	≥2 of PTEN loss, TP53 mutation, RB1 loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	**Platinum + etoposide** (or taxane + carboplatin) ± radiotherapy (RT)
Survival	Years	**<12 months**

Biopsy a metastasis if progression is rapid with a low PSA.

11 · Sequencing and cross-resistance

A second ARPI helps only **15–30%** because of cross-resistance, so **switch mechanism** (CARD). See the P2-140 sequencing table; add pembrolizumab for MSI-H/dMMR and a PARPI if HRR-mutated.

- Enzalutamide induces CYP3A4/2C9/2C19, so check anticoagulants and antiepileptics.
- AR-V7 in circulating tumour cells predicts resistance to abiraterone and enzalutamide, not to taxanes.

12 · Complications and palliative care

P2 Metastatic prostate cancer with bone metastases — cord compression pathway, Patchell, pain RT, obstruction, zoledronic acid vs denosumab, ONJ prevention

Not in the paper answer:

- Obstructive uropathy must be **relieved before chemotherapy**, but not every unilateral case needs drainage.

Every focal back pain in a man with bone metastases is cord compression until proven otherwise.

> **ALGORITHM**
>
> 1. Back pain with or without deficit or sphincter change: suspect compression.
> 2. Give **high-dose IV dexamethasone** (10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day).
> 3. Arrange **urgent whole-spine MRI**.
> 4. Give RT with or without decompressive surgery.
> 5. Taper steroids over 2–3 weeks.

- Operate for progression during RT, unstable fracture, or recurrence after RT.
- Treat neuropathic pain with tricyclics or gabapentinoids, and start hospice care in time. Erythropoiesis-stimulating agents are not recommended for anaemia.

13 · Bone-targeted therapy

Two uses: ADT osteoporosis, and **prevention of skeletal-related events (SREs)** in CRPC bone metastases. There is **no survival effect**.

Feature	Zoledronic acid	Denosumab
Dose	**4 mg IV 4-weekly**, renal adjustment	**120 mg subcutaneous (SC) 4-weekly**; no renal adjustment
SRE	Better than placebo	**Better than zoledronic acid** (20.7 vs 17.1 months)
Toxicity	Flu-like, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ)	Hypocalcaemia (grade ≥3 5%), ONJ **2–4%**

> **KEY POINT**
>
> **ONJ** follows dental extraction or poor teeth, so check the teeth first. Give calcium 1000 mg and vitamin D 800 IU daily with both agents. Bone agents are strongly recommended with radium-223.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤10 mo; apalutamide	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug

Trial	Design	Result	Take-home
PROSPER	nmCRPC; enzalutamide	MFS improved; death reduced 27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemo to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemo
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m²; docetaxel first
CARD	After docetaxel and 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone	OS 14.8 vs 10.9 mo (HR 0.65)	Post-chemo
COU-AA-302	Chemo-naive, asymptomatic	rPFS 16.5 vs 8.3 mo	Pre-chemo
AFFIRM	Post-docetaxel; enzalutamide	OS 18.4 vs 13.6 mo (HR 0.63)	Post-chemo
PREVAIL	Chemo-naive; enzalutamide	Death reduced 29%; radiographic progression reduced 81%	Pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI and taxane, PSMA-positive; ^{177}Lu-PSMA	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ^{177}Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3–4; OS equal	Lutetium ≥ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	BRCA/ATM rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	BRCA rPFS HR 0.50; no ATM benefit	BRCA only
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI with abiraterone / with enzalutamide	rPFS gains, largest in BRCA; PROpel OS 42.1 vs 34.7 mo (P = 0.054)	HRR/BRCA
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab for SRE

★ Putting it together

A · PSA to diagnosis

ALGORITHM

1. **Raised prostate-specific antigen (PSA) or abnormal DRE**: repeat it; exclude infection and instrumentation; double the PSA on 5α-reductase inhibitors (→ Ch 154).
2. **Multiparametric MRI (mpMRI) before biopsy**: PI-RADS (→ Ch 155).
3. **Biopsy** PI-RADS 3–5 targets plus systematic cores, preferably transperineal.
4. Assign **ISUP Grade Group (GG)** (→ Ch 156) and risk group (→ Ch 157). If unfavourable intermediate or worse, stage with PSMA PET (or CT and bone scan).

B · Localised disease: matching risk to treatment

NCCN favourable groups need ALL criteria, adverse groups ANY one.

Risk group	Defined by	Management
Very low	T1c, GG1, PSA <10, density <0.15, <3 positive cores (≤50% each)	Active surveillance (AS)
Low	T1–T2a, GG1, PSA <10	AS preferred
Favourable intermediate	One factor (T2b–c, GG2, PSA 10–20); <50% cores	Radical prostatectomy (RP) or radiotherapy (RT); AS if low-volume GG2
Unfavourable intermediate	2–3 factors, or GG3, or ≥50% cores	RP, or RT with 4–6 months ADT
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP with extended node dissection, or RT with ADT 18–36 months; add abiraterone 2 years if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT

SPCG-4, PIVOT and ProtecT show that radical treatment helps only younger men with ≥intermediate-risk disease. ProtecT: RP = RT for 15-year cancer survival.

C · Surgery vs radiotherapy vs focal therapy

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Main toxicity	Incontinence, ED	Rectal/urinary irritation	Least; retreatment common
Advantages	Pathological staging	Suits comorbid men	Preserves function
Failure	PSA ≥0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	Undefined; biopsy-based
Salvage	Bed RT early at PSA 0.1–0.2 ± ADT	RP, cryotherapy, HIFU, brachytherapy	Repeat focal, RP, RT

D · Advanced disease

ALGORITHM

1. **Biochemical recurrence** (Ch 165): PSMA PET. After RP, salvage RT; after RT, local salvage if biopsy-proven. If high risk (PSADT ≤9 months), give ADT and enzalutamide (EMBARK).
2. **mHSPC**: never ADT alone; add an ARPI, or a docetaxel triplet if fit and high volume. If low volume, add prostate RT (STAMPEDE arm H). Test genes in all.
3. **nmCRPC** with PSADT ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).
4. **mCRPC** (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); ^{177}Lu-PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); PARP inhibitor if BRCA/HRR mutation (PROfound); pembrolizumab if MSI-H; protect bone.